

Toledo

Equation Library of the Human–AI Readout Programme

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Part I

APPENDIX C (SM DOMAIN EQUATION STREAM)

1 EQ-001 — Primordial difference

domain: — • *tier:* Ax • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *none*

$\exists a, b : a \neq b$

EQ-001/B.01.v1 — retained distinction (master root)

domain: B • *tier:* *untagged* • *status:* *current* • *coq:* *not_formalisable* • *parents:* *EQ-001*

$a \backslash neq b$ is retained as a distinction; the distinction is asymmetric, giving an ordering direction. The same shared root as *relativity/* and *quantum/*, not a new axiom.

EQ-001/B.02.v1 — ordered transitions and retention

domain: B • *tier:* *untagged* • *status:* *current* • *coq:* *not_formalisable* • *parents:* *EQ-001*

The asymmetric ordering yields ordered transitions with a positive retention time $\tau_c > 0$.

EQ-001/B.03.v1 — discrete ticks / ordered tape carrier G^*

domain: B • *tier:* *untagged* • *status:* *current* • *coq:* *not_formalisable* • *parents:* *EQ-001*

Retention discretizes into ticks; the ordered-tape carrier $G \cdot$ keeps order ($ab \backslash neq ba$), giving the finite carrier the BIO-G1 gate operates on.

EQ-001/B.04.v1 — count-carrier N^G obstruction (finite diagnostic)

domain: B • *tier:* *finite_diagnostic* • *status:* *current* • *coq:* *not_formalisable* • *parents:* *EQ-001*

The count carrier N^G (multiset/count quotient of $G \cdot$) FAILS to keep dynamics well-defined: $\text{nu}(ab) = \text{nu}(ba) = (1, 1)$ but $\text{nu}(F(ab)) = (2, 0) \backslash neq \text{nu}(F(ba)) = (0, 2)$ under the future-response signature F . Exact witness in *biology_closure_v0_1.py* section BIO-G1.

EQ-001/B.05.v1 — Ordered Sequence Carrier — q_{seq} closes

domain: B • *tier:* *untagged* • *status:* *current* • *coq:* *not_formalisable* • *parents:* *EQ-001*

Because the count carrier N^G loses the distinction the future-response signature F needs, the ORDERED quotient q_{seq} (which keeps $ab \backslash neq ba$) closes exact: dynamics are unambiguous under q_{seq} where they are ambiguous under N^G . STRUCTURALLY_CLOSED. 'sequence' is a carrier property, not yet a real amino-acid or nucleotide chain.

EQ-001/B.06.v1 — Functional Quotient q_F — order $\neq$ function

domain: B • *tier:* *untagged* • *status:* *current* • *coq:* *not_formalisable* • *parents:* *EQ-001*

The intervention-response equivalence class $q_F(w) = 0(w)$ (order-of-positions readout) is invariant under the swap intervention for the passing case, while the coarser keep-first-letter quotient q_{first} breaks output factorization (a failing control). q_F CLOSES as function; order and function are demonstrably distinct quotients.

EQ-001/B.07.v1 — boundary-conditioned state $z_A=(x,e)$ under a repair contract

domain: B • tier: Definition • status: current • coq: definition • parents: EQ-001

A two-component state (integrity x , reserve e) updates under a no-free-repair ledger $x'+e'=x+e+j-d$ for contract (d,j) and repair policy a , setting up the fixed-point search for BI0-G3.

EQ-001/B.08.v1 — Living-Unit Closure – self-maintaining fixed point V_A

domain: B • tier: Definition • status: current • coq: definition • parents: EQ-001

A self-maintaining fixed point $V_A=\text{Gamma}(V_A)$ exists under the no-free-repair ledger: exact witness $(1,1)$ with contract $\{(0,0),(1,1)\}$, policy $a=d$, stays $(1,1)$ and remains in V_A (PASSING); a sustained deficit contract $(d,j)=(1,0)$ drives integrity to 0, exiting V_A (FAILING). CONDITIONALLY_CLOSED on the boundary contract.

EQ-001/B.09.v1 — mortality = irreversible exit from V_A (finite diagnostic)

domain: B • tier: Definition • status: current • coq: definition • parents: EQ-001

The failing control of BI0-G3 (sustained $(d,j)=(1,0)$) exhibits integrity $x \searrow 0$ with no return path in the finite fixture, licensing 'mortality' as a structural READOUT: irreversible exit from V_A . Exact witness $x_2=0$.

EQ-001/B.10.v1 — Heredity q_H via descendant signatures + lineage counts

domain: B • tier: untagged • status: current • coq: not_formalisable • parents: EQ-001

A heredity quotient q_H (descendant-signature equivalence) closes structurally: a coarse parent-quotient that merges classes with different licensed replication counts breaks the successor-count function (FAILING control, $2 \neq 1$), while descendant-signature q_H keeps successor counts well-defined (PASSING). NO fitness term is introduced as a root variable anywhere in this rule.

EQ-001/B.11.v1 — lineage counts $B \rightarrow$ frequency $p_{j,n+1}$ (finite diagnostic)

domain: B • tier: finite_diagnostic • status: current • coq: closed • parents: EQ-001

From licensed replication-count matrix B and initial counts N_0 , $N_1=B \cdot N_0$ gives a frequency readout $p_1=N_1/\text{sum}(N_1)$ purely from lineage counts. Exact witness $B=\text{diag}(2,1)$, $N_0=(1,1) \searrow N_1=(2,1) \searrow p_1=(2/3,1/3)$. No external evolution law used.

EQ-001/B.12.v1 — endogenous operator $L_R[\text{Theta}]$ – augmented closure

domain: B • tier: untagged • status: current • coq: not_formalisable • parents: EQ-001

The I-only quotient q_I fails to distinguish $z_1=(1,1,0)$ and $z_2=(1,1,1)$ BEFORE the endogenous step (q_I identical), but their images differ AFTER ($z_1'=(1,1,0)$, $z_2'=(2,0,1)$) – q_I loses a distinction the dynamics needs. The augmented quotient $q_{\text{aug}}=(I,\text{Theta})$ distinguishes z_1, z_2 exactly and CLOSES. CLOSED_BY_AUGMENTATION.

EQ-001/B.13.v1 — I-only quotient failure (finite diagnostic supporting T2)

domain: B • tier: finite_diagnostic • status: current • coq: not_formalisable • parents: EQ-001

Exact witness: $\text{tstep}(1,1,0)=(1,1,0)$, $\text{tstep}(1,1,1)=(2,0,1)$; $q_I(z_1)=q_I(z_2)=(1,1)$ before, but $(1,1,0)_{\{1:2\}} \neq (2,0,1)_{\{1:2\}}$ after – the I-only quotient loses the distinction that forces the T2 augmentation.

EQ-001/B.14.v1 — homeostasis substrate: decay/turnover/balance

domain: *B* • tier: *Th_coqc* • status: *unverified* • coq: *not_formalisable* • parents: *EQ-001*
source tags this reading *Th_coqc*, but no Coq identifier for it was located in the imported developments (*coq_map.json*); tier kept as the source states, status set to *unverified* until an identifier is found (v1.1 checker warning). v1.2 lane *S* upgrade (2026-09-07): the actual theorem identifiers were located by direct inspection of solver arc (private) *formal/InfoBioHomeostasis_attempt.v* (rule *B-SUB1*'s own *coq_backing.source* in *readout_genesis domains/biology/RULE_REGISTRY.json*, which already names this file): *decay_geometric*, *homeostasis_balance*, *turnover_is_production*. Still not present in this Toledo tree's *coq_map.json* / *coq/* mirror, so status stays *unverified* – this is a more precise statement of what is missing, not a confirmation. Toledo v1.5 Lane *C* re-check (2026-09-07): confirmed today, by direct re-reading of "solver arc (private)" *formal/InfoBioHomeostasis_attempt.v* at commit 961151db33b0491cba8fabade69f594238d33f84 (unchanged from the commit already anchored elsewhere in this tree), that the three named theorems (*decay_geometric* line 49, *homeostasis_balance* line 62, *turnover_is_production* line 68) are present exactly as previously located, and that this file is still absent from *coq/solver-arc/formal/* in this Toledo tree and from *registry/coq_map.json* (grepped, zero matches). The v1.2 lane *S* finding stands, re-verified rather than stale; status remains *unverified* pending an N5+ import of this file.

Geometric decay, turnover-is-production, and homeostasis-balance theorems compile axiom-clean and root-native, backing the BIO-G3 living-unit substrate with a discrete-algebra homeostasis model.

EQ-001/B.15.v1 — relaxation substrate: setpoint one-step / n-step error

domain: *B* • tier: *Th_coqc* • status: *unverified* • coq: *not_formalisable* • parents: *EQ-001*
source tags this reading *Th_coqc*, but no Coq identifier for it was located in the imported developments (*coq_map.json*); tier kept as the source states, status set to *unverified* until an identifier is found (v1.1 checker warning). v1.2 lane *S* upgrade (2026-09-07): the actual theorem identifiers were located by direct inspection of solver arc (private) *formal/InfoHealthCausalRelax_attempt.v* (rule *B-SUB2*'s own *coq_backing.source* in *readout_genesis domains/biology/RULE_REGISTRY.json*, which already names this file): *setpoint_is_fixed*, *one_step_error*, *n_step_error*. Still not present in this Toledo tree's *coq_map.json* / *coq/* mirror, so status stays *unverified* – this is a more precise statement of what is missing, not a confirmation. Toledo v1.5 Lane *C* re-check (2026-09-07): confirmed today, by direct re-reading of "solver arc (private)" *formal/InfoHealthCausalRelax_attempt.v* at commit 961151db33b0491cba8fabade69f594238d33f84, that the three named theorems (*setpoint_is_fixed* line 47, *one_step_error* line 61, *n_step_error* line 83) are present exactly as previously located, and that this file is still absent from *coq/solver-arc/formal/* in this Toledo tree and from *registry/coq_map.json* (grepped, zero matches). The v1.2 lane *S* finding stands, re-verified rather than stale; status remains *unverified* pending an N5+ import of this file.

Setpoint relaxation (one-step and n-step error decay toward a fixed setpoint) theorems compile axiom-clean and root-native, backing the BIO-G3 living-unit substrate's dynamic-response quality.

EQ-001/B.16.v1 — cusp-fold substrate: bistable window / critical slowing

domain: *B* • tier: *Th_coqc* • status: *unverified* • coq: *not_formalisable* • parents: *EQ-001*
source tags this reading *Th_coqc*, but no Coq identifier for it was located in the imported developments (*coq_map.json*); tier kept as the source states, status set to *unverified* until an identifier is found (v1.1 checker warning). v1.2 lane *S* upgrade (2026-09-07): the actual theorem identifiers were located by direct inspection of solver arc (private) *formal/InfoHealthCuspFold_attempt.v* (rule *B-SUB3*'s own *coq_backing.source* in *readout_genesis domains/biology/RULE_REGISTRY.json*, which already names this file): *disc_even*, *cusp_factor*, *fold_from_double_root*, *bistable_window_dec*, *critical_slowing_marginal*, *rest_iff_critical*. Still not present in this Toledo tree's *coq_map.json* / *coq/* mirror, so status stays *unverified* – this is a more precise statement of what is missing, not a confirmation. Toledo v1.5 Lane *C* re-check (2026-09-07): confirmed today, by direct re-reading of "solver arc (private)" *formal/InfoHealthCuspFold_attempt.v* at commit 961151db33b0491cba8fabade69f594238d33f84, that the six named theorems (*disc_even* line 73, *cusp_factor* line 79, *fold_from_double_root* line 87, *bistable_window_dec* line 105, *critical_slowing_marginal* line 109, *rest_iff_critical* line 117) are present exactly as previously located, and that this file is still absent from *coq/solver-arc/formal/* in this Toledo tree and from *registry/coq_map.json* (grepped, zero matches). The v1.2 lane *S* finding stands, re-verified rather than stale; status remains *unverified* pending an N5+ import of this file.

A bistable window and critical-slowness theorem compile axiom-clean and root-native, backing the BIO-G3 living-unit substrate's stability-boundary quality.

EQ-001/B.17.v1 — coupled EP-3 substrate: coupling energy / locked-iff-zero / one-way-breaks-conservation

domain: *B* • tier: *Th_coqc* • status: *unverified* • coq: *not_formalisable* • parents: *EQ-001*

source tags this reading *Th_coqc*, but no Coq identifier for it was located in the imported developments (*coq_map.json*); tier kept as the source states, status set to *unverified* until an identifier is found (v1.1 checker warning). v1.2 lane *S* upgrade (2026-09-07): the actual theorem identifiers were located by direct inspection of solver arc (private) *formal/InfoCoupledCuspEP3_attempt.v* (rule *B-SUB4*'s own *coq_backing.source* in *readout_genesis domains/biology/RULE_REGISTRY.json*, which already names this file): *coupling_energy_nonneg*, *locked_iff_energy_zero*, *two_way_conserve*s, *one_way_breaks_conservation*. Still not present in this Toledo tree's *coq_map.json* / *coq/* mirror, so status stays *unverified* – this is a more precise statement of what is missing, not a confirmation. Toledo v1.5 Lane *C* re-check (2026-09-07): confirmed today, by direct re-reading of "solver arc (private)" *formal/InfoCoupledCuspEP3_attempt.v* at commit 961151db33b0491cba8fabade69f594238d33f84, that the four named theorems (*coupling_energy_nonneg* line 70, *locked_iff_energy_zero* line 86, *two_way_conserve*s line 98, *one_way_breaks_conservation* line 104) are present exactly as previously located, and that this file is still absent from *coq/solver-arc/formal/* in this Toledo tree and from *registry/coq_map.json* (grepped, zero matches). The v1.2 lane *S* finding stands, re-verified rather than stale; status remains *unverified* pending an N5+ import of this file.

Coupling-energy, locked-iff-energy-zero, and one-way-coupling-breaks-conservation theorems compile axiom-clean and root-native, backing the T2 endogenous-operator substrate's coupling quality.

EQ-001/B.18.v1 — protein-semantic bridge (uncalibrated)

domain: *B* • tier: *untagged* • status: *unverified* • coq: *not_formalisable* • parents: *EQ-001*

source class (*readout_genesis domains/biology/RULE_REGISTRY*.json*, rule *B-Y1*) is "*PARTIAL_UNCALIBRATED*" – the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (*coq_status=not_yet_formalised*).

BI0-G2 closes *q_F* as an abstract functional quotient; whether *q_F* corresponds to a real protein, enzyme, or catalytic function requires a calibrated encoding from retained state to biochemical observables. No such encoding exists yet.

EQ-001/B.19.v1 — living-unit → real-cell semantic bridge (uncalibrated)

domain: *B* • tier: *untagged* • status: *unverified* • coq: *not_formalisable* • parents: *EQ-001*

source class (*readout_genesis domains/biology/RULE_REGISTRY*.json*, rule *B-Y2*) is "*PARTIAL_UNCALIBRATED*" – the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (*coq_status=not_yet_formalised*).

BI0-G3 closes *V_A* as an abstract self-maintaining fixed point; whether *V_A* corresponds to a real cell or organism requires a calibrated encoding. No such encoding exists yet.

EQ-001/B.20.v1 — reproduction R_κ branch (declared, not derived)

domain: *B* • tier: *untagged* • status: *unverified* • coq: *not_formalisable* • parents: *EQ-001*

source class (*readout_genesis domains/biology/RULE_REGISTRY*.json*, rule *B-Y3*) is "*YELLOW_BRIDGE_PARTIAL*" – the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (*coq_status=not_yet_formalised*).

A reproduction operator R_κ is architecturally sketched as a branch off the living-unit fixed point *V_A* (a self-maintaining unit producing a new self-maintaining unit), but its construction and closure are declared, not derived or witnessed in *biology_closure_v0_1.py*.

EQ-001/B.21.v1 — heredity -> biological-selection semantic bridge (uncalibrated)

domain: *B* • tier: *untagged* • status: *unverified* • coq: *not_formalisable* • parents: *EQ-001*

source class (*readout_genesis domains/biology/RULE_REGISTRY*.json*, rule *B-Y4*) is "*PARTIAL_UNCALIBRATED*" – the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (*coq_status=not_yet_formalised*).

BIO-G4 closes q_H and the frequency readout $p_{\{j,n+1\}}$ structurally; whether this frequency change corresponds to real biological selection or fitness requires a calibrated encoding. No fitness variable is ever introduced as a root quantity.

EQ-001/B.22.v1 — evolution readout (semantic partial)

domain: *B* • tier: *untagged* • status: *unverified* • coq: *not_formalisable* • parents: *EQ-001*

source class (*readout_genesis domains/biology/RULE_REGISTRY*.json*, rule *B-Y5*) is "*YELLOW_BRIDGE_PARTIAL*" – the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (*coq_status=not_yet_formalised*).

The lineage-frequency trend (B-LIN) can be READ as an 'evolution' trajectory, but this reading is a declared interpretation, not a derived biological-evolution law – 'evolution' here is a structural count trend, not natural selection.

EQ-001/B.23.v1 — homeostasis substrate -> real physiological bridge (Dr, clinical)

domain: *B* • tier: *untagged* • status: *unverified* • coq: *not_formalisable* • parents: *EQ-001*

source class (*readout_genesis domains/biology/RULE_REGISTRY*.json*, rule *B-Y6*) is "*YELLOW_BRIDGE_PARTIAL*" – the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (*coq_status=not_yet_formalised*).

B-SUB1's discrete-algebra homeostasis theorems are Coq-green as FORMAL claims; whether they describe real physiological homeostasis is a declared clinical reading (Dr, not root-derived), open pending calibration.

EQ-001/B.24.v1 — relaxation substrate -> real recovery/relapse bridge (Dr, clinical)

domain: *B* • tier: *untagged* • status: *unverified* • coq: *not_formalisable* • parents: *EQ-001*

source class (*readout_genesis domains/biology/RULE_REGISTRY*.json*, rule *B-Y7*) is "*YELLOW_BRIDGE_PARTIAL*" – the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (*coq_status=not_yet_formalised*).

B-SUB2's setpoint-relaxation theorems are Coq-green as FORMAL claims; whether they describe real clinical recovery/relapse dynamics is a declared reading (Dr), open pending calibration.

EQ-001/B.25.v1 — cusp-fold substrate -> real bistable disease-state bridge (Dr, clinical)

domain: *B* • tier: *untagged* • status: *unverified* • coq: *not_formalisable* • parents: *EQ-001*

source class (*readout_genesis domains/biology/RULE_REGISTRY*.json*, rule *B-Y8*) is "*YELLOW_BRIDGE_PARTIAL*" – the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (*coq_status=not_yet_formalised*).

B-SUB3's bistable-window theorem is Coq-green as a FORMAL claim; whether it describes a real bistable disease state is a declared reading (Dr), open pending calibration.

EQ-001/B.26.v1 — coupled EP-3 substrate -> real physiological coupling bridge (Dr, clinical)

domain: B • tier: untagged • status: unverified • coq: not_formalisable • parents: EQ-001

source class (readout_genesis domains/biology/RULE_REGISTRY*.json, rule B-Y9) is "YELLOW_BRIDGE_PARTIAL" – the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (coq_status=not_yet_formalised).

B-SUB4's coupling-energy theorems are Coq-green as FORMAL claims; whether they describe real physiological coupling (hormonal, neural, etc.) is a declared reading (Dr), open pending calibration.

EQ-001/B.27.v1 — domain-discovery battery -> real event-resolved bridge (synthetic only)

domain: B • tier: untagged • status: unverified • coq: not_formalisable • parents: EQ-001

source class (readout_genesis domains/biology/RULE_REGISTRY*.json, rule B-Y10) is "YELLOW_BRIDGE_PARTIAL" – the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (coq_status=not_yet_formalised).

The domain-discovery battery is real and runnable (PASS), but has been run only on SYNTHETIC tapes – no real event-resolved biological data has been passed through it. The bridge to real data is open.

EQ-001/B.28.v1 — bR lineage ledger -> real event-resolved run bridge (architecture-only)

domain: B • tier: untagged • status: unverified • coq: not_formalisable • parents: EQ-001

source class (readout_genesis domains/biology/RULE_REGISTRY*.json, rule B-Y11) is "YELLOW_BRIDGE_PARTIAL" – the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (coq_status=not_yet_formalised).

The bacteriorhodopsin (bR) lineage ledger and its CLI are narrated and architected in the core, but NO real event-resolved run has ever been executed. This is architecture-only, not evidence.

EQ-001/B.29.v1 — calibrated encoding retained-state -> biological observables (OPEN)

domain: B • tier: Open • status: unverified • coq: open_prop • parents: EQ-001

source class (readout_genesis domains/biology/RULE_REGISTRY*.json, rule B-X1) is "OPEN" – the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (coq_status=not_yet_formalised).

TARGET, not established: a calibrated encoding mapping retained root state to real biological observables (sequence identity, functional identity, unit identity, heredity identity). This is the single central open bottleneck licensing all real-domain semantics; none of BIO-G1..G4/T2 supply it.

EQ-001/B.30.v1 — end-to-end root -> real biology via event-resolved data (OPEN)

domain: B • tier: Open • status: unverified • coq: open_prop • parents: EQ-001

source class (readout_genesis domains/biology/RULE_REGISTRY*.json, rule B-X2) is "OPEN" – the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (coq_status=not_yet_formalised).

TARGET, not established: derivation of any real biological result from the root through real (measured, event-resolved) data. End-to-end root \to real-biology through event-resolved data = 0%.

EQ-001/B.31.v1 — DNA / genome carrier real semantics (OPEN, destination only)

domain: *B* • tier: *Open* • status: *unverified* • coq: *open_prop* • parents: *EQ-001*

source class (readout_genesis domains/biology/RULE_REGISTRY*.json, rule B-X3) is "OPEN" – the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (coq_status=not_yet_formalised).

DNA/genome as a real molecular carrier is a destination the ordered-sequence carrier (BIO-G1) might eventually be calibrated to reach – never a premise anywhere in this registry.

EQ-001/B.32.v1 — cell / organelle real semantics (OPEN, destination only)

domain: *B* • tier: *Open* • status: *unverified* • coq: *open_prop* • parents: *EQ-001*

source class (readout_genesis domains/biology/RULE_REGISTRY*.json, rule B-X4) is "OPEN" – the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (coq_status=not_yet_formalised).

A real cell or organelle is a destination the living-unit closure (BIO-G3) might eventually be calibrated to reach – never a premise anywhere in this registry.

EQ-001/B.33.v1 — enzyme / catalytic-kinetics real semantics (OPEN, destination only)

domain: *B* • tier: *Open* • status: *unverified* • coq: *open_prop* • parents: *EQ-001*

source class (readout_genesis domains/biology/RULE_REGISTRY*.json, rule B-X5) is "OPEN" – the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (coq_status=not_yet_formalised).

Real enzyme kinetics (e.g. Michaelis-Menten-shaped catalysis) is a destination the functional quotient (BIO-G2) might eventually be calibrated to reach – never a premise. Michaelis-Menten remains a prohibited active token (see DRIFT_CONTRACT.json).

EQ-001/B.34.v1 — fitness / natural-selection as root-derived quantity (OPEN, destination only)

domain: *B* • tier: *Open* • status: *unverified* • coq: *not_formalisable* • parents: *EQ-001*

source class (readout_genesis domains/biology/RULE_REGISTRY*.json, rule B-X6) is "OPEN" – the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (coq_status=not_yet_formalised).

Fitness / natural selection is explicitly FORBIDDEN as a root variable anywhere in BIO-G4/B-LIN; it remains a destination requiring a calibrated encoding, fully open.

EQ-001/B.35.v1 — ecology / population-level real dynamics (OPEN)

domain: *B* • tier: *Open* • status: *unverified* • coq: *not_formalisable* • parents: *EQ-001*

source class (readout_genesis domains/biology/RULE_REGISTRY*.json, rule B-X7) is "OPEN" – the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (coq_status=not_yet_formalised).

Real population-level ecological dynamics beyond the abstract lineage-count trend (B-LIN/B-Y5) are fully open – no root-native construction exists. Lotka-Volterra remains a prohibited active token.

EQ-001/B.36.v1 — LINE-2 textbook curriculum -> root-native promotion (OPEN, explicitly forbidden)

domain: *B* • tier: *Open* • status: *unverified* • coq: *not_formalisable* • parents: *EQ-001*

source class (readout_genesis domains/biology/RULE_REGISTRY*.json, rule B-X8) is "OPEN" – the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (coq_status=not_yet_formalised).

The LINE-2 textbook curriculum solver (45/45 checklist, tier textbook_closure/finite_diagnostic, no Th_coqc) sits outside this DAG entirely as a CHECKER. This node exists ONLY to make the boundary explicit and permanently red – it must never be promoted, folded in, or counted toward the 47.2%/62.5% LINE-1 score.

EQ-001/C.01.v1 — RT-LEDGER-001

domain: *C* • tier: *untagged* • status: *current* • coq: *closed* • parents: *EQ-001*

If the declared closed boundary preserves ledger $L(n)=A\ n$, then $A(n1-n0)=0$.

EQ-001/C.02.v1 — RT-KERNEL-002

domain: *C* • tier: *untagged* • status: *current* • coq: *closed* • parents: *EQ-001*

A change delta preserves the ledger iff delta is in $\ker(A)$.

EQ-001/C.03.v1 — RT-EXTENT-003

domain: *C* • tier: *untagged* • status: *current* • coq: *open_prop* • parents: *EQ-001*

If columns of N form a basis for the admitted kernel subspace, every admitted change has coordinates ξ with $\delta=N\ \xi$.

EQ-001/C.04.v1 — RT-ADMISSIBLE-004

domain: *C* • tier: *untagged* • status: *current* • coq: *open_prop* • parents: *EQ-001*

Admitted coordinates satisfy $n0+N\ \xi \geq 0$ and all declared capacity/boundary inequalities.

EQ-001/C.05.v1 — RT-REFINE-005

domain: *C* • tier: *untagged* • status: *current* • coq: *open_prop* • parents: *EQ-001*

For a finite state set, repeatedly splitting cells by declared readout and successor-cell signatures terminates in an exact closed refinement.

EQ-001/C.06.v1 — RT-ADD-COND-006

domain: *C* • tier: *untagged* • status: *current* • coq: *closed* • parents: *EQ-001*

If composition is a free commutative monoid and R is a monoid homomorphism, then $R(c)=\sum_i c_i R(e_i)$.

EQ-001/C.07.v1 — RT-STATIONARY-007

domain: *C* • tier: *untagged* • status: *current* • coq: *closed* • parents: *EQ-001*

A quotient fixed point can hide source-state cycles or currents; therefore fixed quotient readout alone establishes only observational stationarity.

EQ-001/C.08.v1 — PR-GRAMMAR-001

domain: C • tier: untagged • status: current • coq: not_formalisable • parents: EQ-001

Candidate mathematical languages must be frozen and reported before held-out evidence is opened.

EQ-001/C.09.v1 — NP-EXACT-001

domain: C • tier: untagged • status: current • coq: definition • parents: EQ-001

Use integer/Fraction arithmetic for the finite proof witnesses in v0.901.

EQ-001/C.10.v1 — PR-GENERATOR-002

domain: C • tier: untagged • status: current • coq: not_formalisable • parents: EQ-001

Freeze generator identities, aliases, source profiles, contexts, and permitted transformations before quotient admission.

EQ-001/C.11.v1 — RT-FCM-008

domain: C • tier: untagged • status: current • coq: open_prop • parents: EQ-001

For a frozen finite generator registry G , N^G with coordinatewise addition and zero is a free commutative monoid.

EQ-001/C.12.v1 — RT-COUNT-HOM-009

domain: C • tier: untagged • status: current • coq: closed • parents: EQ-001

Counting registered generators maps empty source word to zero and concatenation to coordinatewise addition.

EQ-001/C.13.v1 — RT-PERM-GATE-010

domain: C • tier: untagged • status: current • coq: closed • parents: EQ-001

If equal occupation vectors imply equal registered readout and successor signatures in every frozen profile, the count quotient exactly factors those profiles.

EQ-001/C.14.v1 — RT-DECOMP-011

domain: C • tier: untagged • status: current • coq: closed • parents: EQ-001

Within N^G and a frozen independent generator basis, occupation coordinates uniquely specify structural decomposition.

EQ-001/C.15.v1 — RT-REFINE-COMP-012

domain: C • tier: untagged • status: current • coq: open_prop • parents: EQ-001

A declared nonnegative-integer coarsening map P commutes with composition: $P(c+d)=Pc+Pd$.

EQ-001/C.16.v1 — RT-LINEAGE-013

domain: C • tier: untagged • status: current • coq: closed • parents: EQ-001

The marked quotient stores structural occupation separately from an append-only lineage sidecar; equal structural projections need not identify marked histories.

EQ-001/C.17.v1 — NP-FINITE-GATE-002

domain: C • tier: untagged • status: current • coq: definition • parents: EQ-001

Enumerate all registered same-count pairs and compare exact frozen signatures.

EQ-001/E.01.v1 — A mathematical theory of communication

domain: E • tier: untagged • status: current • coq: not_formalisable • parents: EQ-001

A mathematical theory of communication

Occurrences (1): ‘mass_note.tex’ §lit

EQ-001/E.02.v1 — Szilárd’s information-work engine

domain: E • tier: untagged • status: current • coq: not_formalisable • parents: EQ-001

Szilárd’s information-work engine

Occurrences (1): ‘mass_note.tex’ §lit

EQ-001/E.03.v1 — Landauer’s erasure principle

domain: E • tier: untagged • status: current • coq: not_formalisable • parents: EQ-001

Landauer’s erasure principle

Occurrences (1): ‘mass_note.tex’ §lit

EQ-001/E.04.v1 — It from bit

domain: E • tier: Dr • status: current • coq: not_formalisable • parents: EQ-001

It from bit

Occurrences (1): ‘mass_note.tex’ §reading (Reading 1-4); proposed staircase "step 0" (2026-07-05, not yet mechanized)

EQ-001/E.05.v1 — Stinespring dilation

domain: E • tier: untagged • status: current • coq: not_formalisable • parents: EQ-001

Stinespring dilation

Occurrences (1): ‘mass_note.tex’ §lit

EQ-001/E.06.v1 — Completely positive maps

domain: E • tier: untagged • status: current • coq: not_formalisable • parents: EQ-001

Completely positive maps

Occurrences (1): ‘mass_note.tex’ §lit

EQ-001/E.07.v1 — States, Effects, and Operations (Kraus)

domain: E • tier: untagged • status: current • coq: not_formalisable • parents: EQ-001

States, Effects, and Operations (Kraus)

Occurrences (1): ‘mass_note.tex’ §lit, ‘URCF_RD_All.v’ (‘kraus_completeness’)

EQ-001/P.01.v1 — retained distinction (master root)

domain: P • tier: *untagged* • status: *current* • coq: *definition* • parents: *EQ-001*

$a \backslash \text{neqb}$ is retained as a distinction; the distinction is asymmetric, giving an ordering direction. Identical to relativity's RB-P0 - this is the one shared root, not a new axiom.

EQ-001/P.02.v1 — ordered transitions and retention

domain: P • tier: *untagged* • status: *current* • coq: *definition* • parents: *EQ-001*

The asymmetric ordering yields ordered transitions with a positive retention time $\tau_c > 0$.

EQ-001/P.03.v1 — discrete ticks and finite causal graph

domain: P • tier: *untagged* • status: *current* • coq: *definition* • parents: *EQ-001*

Retention discretizes into ticks $t = n \cdot \Delta_\theta$; a finite causal graph is built on those ticks.

EQ-001/P.04.v1 — retained-distinction axioms FORCE the L_R operator form

domain: P • tier: *Th_coqc* • status: *current* • coq: *not_formalisable* • parents: *EQ-001*

The retained-distinction axioms FORCE the finite retention operator L_R 's form - not merely posit it. Machine-checked, axiom-free, re-compiled 2026-07-21 (coqc -q -R . DQG).

EQ-001/P.05.v1 — telegraph coarse-grain / spine PDE

domain: P • tier: *untagged* • status: *current* • coq: *definition* • parents: *EQ-001*

Coarse-graining L_R over the ticks gives the spine second-order stepper $M \cdot d^2 + D \cdot d + K$, shared verbatim with the relativity domain's RB-P2/RB-P3.

EQ-001/P.06.v1 — spine characteristic equation / dispersion relation

domain: P • tier: *untagged* • status: *current* • coq: *definition* • parents: *EQ-001*

The spine stepper's characteristic equation is $M \cdot s^2 + D \cdot s + K \cdot \lambda = 0$, exact-rational, parametrized by a mode label λ .

EQ-001/P.07.v1 — oscillatory-regime split (spine dispersion)

domain: P • tier: *finite_diagnostic* • status: *current* • coq: *closed* • parents: *EQ-001*

$\lambda_c = D^2 / (4 \cdot M \cdot K)$. The discriminant $\text{disc}(\lambda) = D^2 - 4 \cdot M \cdot K \cdot \lambda$ is negative (complex-conjugate root pair, oscillatory) for $\lambda > \lambda_c$, positive (overdamped) for $\lambda < \lambda_c$, and exactly zero at $\lambda = \lambda_c$ (critical). Exact rational witness: $M=D=K=1 \rightarrow \lambda_c=1/4$; $\lambda=1$ gives $\text{disc}=-3 < 0$ (oscillatory); $\lambda=1/8$ gives $\text{disc}=1/2 > 0$ (overdamped); $\lambda=\lambda_c$ gives $\text{disc}=0$ (critical).

EQ-001/P.08.v1 — asymmetric-seed trifurcation $R_0 = \text{Diag} + \text{SymOff} + \text{SkewOff}$

domain: P • tier: *Th_coqc* • status: *current* • coq: *closed* • parents: *EQ-001*

Any asymmetric seed matrix R_0 decomposes uniquely as $R_0 = \text{DiagPart} + \text{SymOff} + \text{SkewOff}$. Exact rational witness with a 3×3 R_0 reproduces R_0 exactly from the three parts; SkewOff is genuinely nonzero (torsion/circulation branch).

EQ-001/P.09.v1 — seed torsion is the skew-off part; group and rank-N structure

domain: P • tier: Th_coqc • status: *current* • coq: *closed* • parents: *EQ-001*

The seed torsion/circulation branch is exactly *SkewOff*; the set of admissible *SkewOff* parts forms a group and the construction extends to rank N , axiom-free.

EQ-001/P.10.v1 — seed torsion genuine mixing

domain: P • tier: Th_coqc • status: *current* • coq: *closed* • parents: *EQ-001*

The torsion/circulation branch produces genuine (non-trivial, non-decomposable-further) mixing among retained components - a finite, exact-rational witness with a nonzero *SkewOff* block.

EQ-001/P.11.v1 — complexification gate (i as a readout, not a premise)

domain: P • tier: *untagged* • status: *unverified* • coq: *closed* • parents: *EQ-001*

source class (*readout_genesis domains/quantum/RULE_REGISTRY*.json*, rule *Q-C1*) is "*PROPOSED_INTERNAL_GATE*" - the Domain Registration Standard's own class enum treats an item in this state as *not-yet-established*; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (*coq_status=not_yet_formalised*).

An orientation J on a closed real oriented mode-pair with $J^2 = -I$ and G -adjoint $J^{\dagger}G = -J$ (G -skew under the positive retention metric G) EARNs the complex coordinate $\psi = u + J \cdot v$ as a READOUT - i is a consequence of a closed real structure, never a premise. Exact rational witness: $J = [[0, -1], [1, 0]]$, $G = I_2$, $J^2 = -I_2$, $J^{\dagger} = -J$. FAILING control: $J_{\text{bad}} = I_2$ (identity) has $J_{\text{bad}}^2 = +I_2 \neq -I_2$, and is correctly REFUSED the complex coordinate (*NO_COMPLEX_QUANTUM_COORDINATE*).

EQ-001/P.12.v1 — positive quantum norm

domain: P • tier: *finite_diagnostic* • status: *current* • coq: *closed* • parents: *EQ-001*

$N_Q(\psi) = \langle \psi, G \psi \rangle$ is ≥ 0 for all ψ and $= 0$ iff $\psi = 0$, on the exact rational witness ($G = I_2$, $\psi = (3, 4)$ gives $N_Q = 25$; $\psi = 0$ gives $N_Q = 0$).

EQ-001/P.13.v1 — reversible G -orthogonal evolution

domain: P • tier: *finite_diagnostic* • status: *current* • coq: *closed* • parents: *EQ-001*

An evolution operator U satisfying $U^{\dagger} G U = G$ preserves N_Q exactly. Exact witness: $U = J$ (the complexification-gate orientation itself, reused as a rotation), $N_Q(U \cdot \psi) = N_Q(\psi) = 25$.

EQ-001/P.14.v1 — memory-before-mass (formal algebra, Coq-green)

domain: P • tier: Th_coqc • status: *current* • coq: *not_formalisable* • parents: *EQ-001*

Retained memory (discrete algebra on the retention operator) FORCES a mass-like readout, axiom-free, machine-checked. This rule establishes the formal/algebraic claim only; whether this readout IS physical mass is a separate, explicitly YELLOW bridge (see Q-Y5). NOT counted as an independent node in the 32-node Quantum DAG closure audit - it is the supporting formal citation folded into Q-Y5's yellow accounting (matching the verifier's own 'memory-before-mass' yellow-list item).

EQ-001/P.15.v1 — discrete metric/curvature chain (folded from relativity, root-native)

domain: P • tier: Th_coqc • status: *current* • coq: *not_formalisable* • parents: *EQ-001*

The discrete connection $U_{\{j<-i\}}=V_j^{-1}\cdot V_i$, the curvature certificate $K_C=U_C-I$ (two finite transports failing to commute, nonzero exact witness), and the supporting discrete-geometry chain (metric-compatible curvature, 2nd Bianchi, Gauss-Bonnet, Riemann pair symmetry, rational $SO(3)$ curvature) are folded in here from the relativity domain's OB-03/OB-04 as shared root-native structure, deduplicated (same underlying L_R operator, not re-derived).

EQ-001/P.16.v1 — QM/SR bounded algebraic identity (imported Minkowski caveat)

domain: P • tier: *untagged* • status: *unverified* • coq: *not_formalisable* • parents: *EQ-001*

source class (readout_genesis domains/quantum/RULE_REGISTRY*.json, rule Q-Y1) is "YELLOW_BRIDGE_PARTIAL"
– the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (coq_status=not_yet_formalised).

InfoQuantumRelativityUnification.v (box_quad_is_spine_residual, spine_dispersi on_iff_box_quad_vanishes) and InfoSeedFeedsQuantumRelativity_attempt.v compile axiom-free, BUT box_quad INHERITS the imported Minkowski signature from InfoLorentzInv ariance.box_quad. This is therefore a bounded ALGEBRAIC IDENTITY between two independently-positied constructions, never a root-native derivation of QM/SR unification.

EQ-001/P.17.v1 — telegraph crossover discriminant (spine's own, not node-8 lapse)

domain: P • tier: *untagged* • status: *unverified* • coq: *not_formalisable* • parents: *EQ-001*

source class (readout_genesis domains/quantum/RULE_REGISTRY*.json, rule Q-Y2) is "YELLOW_BRIDGE_PARTIAL"
– the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (coq_status=not_yet_formalised).

InfoTelegraphHorizonUnification_attempt.v / InfoTelegraphCrossover_attempt.v prove disc_is_spine_discr, lam_c_is_spine_lambda_c, quantum_iff_above_horizon – the spine's OWN classical/quantum crossover discriminant $\lambda_c=D^2/4MK$, axiom-free. This is a real, machine-checked result, but the BRIDGE from 'above/below this discriminant threshold' to 'this regime IS quantum' is not established – the theorem names a threshold, not a semantic identification.

EQ-001/P.18.v1 — $v^2=K/M$ conditional oscillator-parameter identification

domain: P • tier: *untagged* • status: *unverified* • coq: *definition* • parents: *EQ-001*

source class (readout_genesis domains/quantum/RULE_REGISTRY*.json, rule Q-Y3) is "YELLOW_BRIDGE_PARTIAL"
– the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (coq_status=not_yet_formalised).

Under the DECLARED (not derived) identification of the spine's own front-speed-squared with K/M , the oscillatory-regime split of Q-OS1 can be read as an oscillator-parameter statement. This identification is conditional and declared, exactly parallel to relativity's OC-01 (graph front-speed vs measured-speed) – it is NOT derived from the root.

EQ-001/P.19.v1 — metric/curvature-to-quantum-regime bridge

domain: P • tier: *untagged* • status: *unverified* • coq: *not_formalisable* • parents: *EQ-001*

source class (readout_genesis domains/quantum/RULE_REGISTRY*.json, rule Q-Y4) is "YELLOW_BRIDGE_PARTIAL"
– the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (coq_status=not_yet_formalised).

Q-G1's discrete metric/curvature chain is root-native and Coq-green as GEOMETRY, but its bridge toward a quantum-regime interpretation (e.g. curvature as any kind of quantum-gravitational coupling) is not established – naming it here only as a formal-closed / bridge-partial connective, matching relativity's own OB-01 caveat (operator-to-metric direction only, reverse forbidden).

EQ-001/P.20.v1 — memory-before-mass physical/semantic bridge

domain: *P* • tier: *untagged* • status: *unverified* • coq: *not_formalisable* • parents: *EQ-001*

source class (readout_genesis domains/quantum/RULE_REGISTRY*.json, rule Q-Y5) is "YELLOW_BRIDGE_PARTIAL" – the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (coq_status=not_yet_formalised).

Q-M1 establishes the FORMAL algebraic claim (retained memory forces a mass-like readout) as Coq-green. Whether that readout IS real physical mass/inertia is a DECLARED, not derived, correspondence – exactly parallel to relativity's OC-02 (mass-memory declared correspondence).

EQ-001/P.21.v1 — channel / CPTP formal card (bridge-from-root open)

domain: *P* • tier: *untagged* • status: *unverified* • coq: *not_formalisable* • parents: *EQ-001*

source class (readout_genesis domains/quantum/RULE_REGISTRY*.json, rule Q-Y6) is "YELLOW_BRIDGE_PARTIAL" – the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (coq_status=not_yet_formalised).

A formal CPTP (completely-positive, trace-preserving) channel CARD exists as a target definition (the abstract shape a quantum channel would need to satisfy), but the bridge FROM the root's own reversible G-orthogonal evolution (Q-N2) TO a general CPTP map (which must also cover irreversible/open-system dynamics) is OPEN – not yet constructed or gated.

EQ-001/P.22.v1 — general root-derived complex psi (OPEN)

domain: *P* • tier: *Open* • status: *unverified* • coq: *not_formalisable* • parents: *EQ-001*

source class (readout_genesis domains/quantum/RULE_REGISTRY*.json, rule Q-X1) is "OPEN" – the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (coq_status=not_yet_formalised).

Q-C1 establishes complexification for ONE closed real oriented mode-pair, exact witness. General existence and uniqueness of a root-derived complex psi across all admissible retained states is OPEN.

EQ-001/P.23.v1 — quantum quotient q_Q (architecture, commuting-square NOT proven) (OPEN)

domain: *P* • tier: *Open* • status: *unverified* • coq: *open_prop* • parents: *EQ-001*

source class (readout_genesis domains/quantum/RULE_REGISTRY*.json, rule Q-X2) is "OPEN" – the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (coq_status=not_yet_formalised).

A candidate architecture for a quantum quotient q_Q (mirroring relativity's q_g / chem's q_D pattern) is sketched, but the commuting-square condition $q_Q \circ F_{\text{full}} = F_Q \circ q_Q$ has NOT been proven or even witnessed on a finite fixture. This is the central open bottleneck for the whole domain.

EQ-001/P.24.v1 — mixed-state from hidden lineage (OPEN)

domain: *P* • tier: *Open* • status: *unverified* • coq: *not_formalisable* • parents: *EQ-001*

source class (readout_genesis domains/quantum/RULE_REGISTRY*.json, rule Q-X3) is "OPEN" – the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (coq_status=not_yet_formalised).

Whether a mixed-state description (density-matrix-shaped object) arises from tracing out a hidden lineage of the retained-distinction root is OPEN – no finite witness exists.

EQ-001/P.25.v1 — subsystem composition / (x)_kappa (OPEN)

domain: *P* • tier: *Open* • status: *unverified* • coq: *not_formalisable* • parents: *EQ-001*

source class (readout_genesis domains/quantum/RULE_REGISTRY*.json, rule Q-X4) is "OPEN" – the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (coq_status=not_yet_formalised).

Whether a composition operation (x)_kappa on retained subsystems, derived from the no-free-copy law, reproduces tensor-product-shaped behavior is OPEN. No finite witness exists; this is distinct from and does not assume any imported tensor product.

EQ-001/P.26.v1 — measurement / Born-rule uniqueness (TARGET, OPEN)

domain: *P* • tier: *Open* • status: *unverified* • coq: *not_formalisable* • parents: *EQ-001*

source class (readout_genesis domains/quantum/RULE_REGISTRY*.json, rule Q-X6) is "OPEN" – the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (coq_status=not_yet_formalised).

TARGET, not established: uniqueness of the Born weight $p=|\psi|^2$ across ALL admissible refinements. quantum_closure_v0_1.py section E is FINITE_DIAGNOSTIC evidence FOR the quadratic form only: on one exact refinement (25=9+16), the quadratic weight (p=2) is refinement-invariant while a linear weight (p=1) is not (5 $\neq$ 7, correctly failing). This is evidence, not a uniqueness proof across all refinements/bases.

EQ-001/P.27.v1 — conditional update / measurement collapse (OPEN)

domain: *P* • tier: *Open* • status: *unverified* • coq: *not_formalisable* • parents: *EQ-001*

source class (readout_genesis domains/quantum/RULE_REGISTRY*.json, rule Q-X7) is "OPEN" – the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (coq_status=not_yet_formalised).

Whether a root-native conditional-update rule (projection-postulate-shaped) exists is OPEN – no finite witness, no gate constructed.

EQ-001/P.28.v1 — uncertainty relation (formal guard, boundary not root-derivation) (OPEN)

domain: *P* • tier: *Open* • status: *unverified* • coq: *not_formalisable* • parents: *EQ-001*

source class (readout_genesis domains/quantum/RULE_REGISTRY*.json, rule Q-X8) is "OPEN" – the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (coq_status=not_yet_formalised).

An RDL_Heisenberg-shaped formal guard exists and SURVIVED the hollow-card audit (it correctly refuses to fire on non-quantum inputs), but it is a boundary/consistency check, not a root-derivation of an uncertainty principle. Kept OPEN pending a root-native construction.

EQ-001/P.29.v1 — correlation / CHSH / Tsirelson bound (formal guard, boundary not root-derivation) (OPEN)

domain: *P* • tier: *Open* • status: *unverified* • coq: *not_formalisable* • parents: *EQ-001*

source class (readout_genesis domains/quantum/RULE_REGISTRY*.json, rule Q-X9) is "OPEN" – the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (coq_status=not_yet_formalised).

A CHSH/Tsirelson-shaped formal guard exists as a boundary check (it can flag classical-vs-quantum-shaped correlation bounds), but it is not a root derivation of quantum correlation, entanglement, or the Tsirelson bound. Kept OPEN, depends on the still-open Q-X4 composition.

EQ-001/P.30.v1 — mass / particle layer (beyond formal memory-mass algebra) (OPEN)

domain: P • tier: Open • status: unverified • coq: not_formalisable • parents: EQ-001

source class (readout_genesis domains/quantum/RULE_REGISTRY*.json, rule Q-X10) is "OPEN" – the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (coq_status=not_yet_formalised).

A particle ontology (localized, countable excitations with mass) is OPEN – Q-M1/Q-Y5 establish only a formal mass-like readout from memory algebra, declared not derived as physical mass.

EQ-001/P.31.v1 — spin / spin-statistics (fully OPEN)

domain: P • tier: Open • status: unverified • coq: not_formalisable • parents: EQ-001

source class (readout_genesis domains/quantum/RULE_REGISTRY*.json, rule Q-X11) is "OPEN" – the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (coq_status=not_yet_formalised).

Spin and the spin-statistics theorem are fully OPEN. Q-S1's SkewOff torsion branch is NOT spin – no rotation-group representation, half-integer quantization, or statistics connection has been constructed.

EQ-001/P.32.v1 — QFT (needs $q_Q + q_{\text{relativity}} + q_{\text{geometry}}$ joint) (OPEN)

domain: P • tier: Open • status: unverified • coq: not_formalisable • parents: EQ-001

source class (readout_genesis domains/quantum/RULE_REGISTRY*.json, rule Q-X12) is "OPEN" – the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (coq_status=not_yet_formalised).

A full QFT / effective-field-theory closure requires the JOINT construction of the quantum quotient q_Q , the relativity domain's quotient q_g , and a geometry quotient q_{geometry} – none of which are individually complete, let alone jointly commuting. Fully OPEN.

EQ-001/P.33.v1 — superposition gate (conditional, fixed-operator-only) (OPEN beyond the narrow case)

domain: P • tier: Open • status: unverified • coq: closed • parents: EQ-001

source class (readout_genesis domains/quantum/RULE_REGISTRY*.json, rule Q-X13) is "OPEN" – the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (coq_status=not_yet_formalised).

A conditional superposition gate exists ONLY for a fixed, pre-specified pair of orthogonal readout operators on the single closed-pair witness of Q-C1 – linear combinations of the two witness components stay in the closed pair and N_Q is additive on them (Pythagorean, exact rational). This is NOT a general superposition principle: it does not extend to arbitrary bases, arbitrary numbers of components, or non-fixed operators without further construction.

EQ-001/P.34.v1 — distinguishability and asymmetry

domain: P • tier: untagged • status: current • coq: definition • parents: EQ-001

$a \neq b$ is retained as a distinction; the distinction is asymmetric, giving an ordering direction.

EQ-001/P.35.v1 — telegraph coarse-grain and finite front speed

domain: P • tier: untagged • status: current • coq: definition • parents: EQ-001

Coarse-graining the stepper gives a telegraph process with finite front speed $v = \sqrt{D/\tau_c}$, derived from finite retention, not inserted.

EQ-001/P.36.v1 — finite causal cone

domain: P • tier: *untagged* • status: *current* • coq: *definition* • parents: *EQ-001*

The finite front speed bounds a causal cone $|x| \leq t$.

EQ-001/P.37.v1 — null coordinates on the cone

domain: P • tier: *finite_diagnostic* • status: *current* • coq: *definition* • parents: *EQ-001*

Define null coordinates $n_+ = t + x$, $n_- = t - x$ from the cone's own two edges.

EQ-001/P.38.v1 — cone product invariant candidate

domain: P • tier: *finite_diagnostic* • status: *current* • coq: *closed* • parents: *EQ-001*

$Q_v = n_+ \cdot n_- = v^2 t^2 - x^2$, composed from the cone's own null edges, not an imported interval.

EQ-001/P.39.v1 — O1 path-additivity gate

domain: P • tier: *finite_diagnostic* • status: *current* • coq: *closed* • parents: *EQ-001*

$B(\delta_1 + \delta_2) = B(\delta_1) + B(\delta_2)$ is imposed and forces linearity of B over $\mathbb{Q}$ (derived, not separately assumed).

EQ-001/P.40.v1 — O2 cone-preservation gate

domain: P • tier: *finite_diagnostic* • status: *current* • coq: *definition* • parents: *EQ-001*

The observer map B must send the cone to itself (preserve $|x| \leq t$ under the map).

EQ-001/P.41.v1 — O3 orientation gate

domain: P • tier: *finite_diagnostic* • status: *current* • coq: *definition* • parents: *EQ-001*

The observer map B preserves the forward/backward orientation of the cone.

EQ-001/P.42.v1 — O4 RD-neutrality gate

domain: P • tier: *untagged* • status: *unverified* • coq: *definition* • parents: *EQ-001*

source class (readout_genesis domains/relativity/RULE_REGISTRY*.json, rule OA-06) is "PROPOSED_INTERNAL_GATE"
– the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (coq_status=not_yet_formalised).

A pure observer change preserves the RD causal-cell count $n_+ \cdot n_-$ (Q_v). This is the one new internal gate added by this closure; it is proposed, not derived from the root backbone.

EQ-001/P.43.v1 — forced null rescaling (kappa parametrization)

domain: P • tier: *untagged* • status: *unverified* • coq: *closed* • parents: *EQ-001*

source class (readout_genesis domains/relativity/RULE_REGISTRY*.json, rule OA-07) is "PROPOSED_INTERNAL_BRIDGE"
– the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (coq_status=not_yet_formalised).

O4 forces $n_+' = \kappa^{-1} \cdot n_+$, $n_-' = \kappa \cdot n_-$, $\kappa > 0$, so that $Q_v' = Q_v$ automatically.

EQ-001/P.44.v1 — Γ_R from kappa and relative velocity

domain: P • tier: *untagged* • status: *unverified* • coq: *closed* • parents: *EQ-001*

source class (readout_genesis domains/relativity/RULE_REGISTRY*.json, rule OA-08) is "PROPOSED_INTERNAL_BRIDGE"
– the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (coq_status=not_yet_formalised).

Relative velocity $\beta = u/v$; $\kappa^2 = (1+\beta)/(1-\beta)$; $\Gamma_R = (\kappa+1/\kappa)/2 = 1/\sqrt{1-\beta^2}$. Exact witness: $v=5/4, u=3/4 \rightarrow \beta=3/5, \kappa=2, \Gamma_R=5/4$.

EQ-001/P.45.v1 — observer coordinate transforms

domain: P • tier: *untagged* • status: *unverified* • coq: *closed* • parents: *EQ-001*

source class (readout_genesis domains/relativity/RULE_REGISTRY*.json, rule OA-09) is "PROPOSED_INTERNAL_BRIDGE"
– the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (coq_status=not_yet_formalised).

$x' = \Gamma_R (x - u \cdot t)$, $t' = \Gamma_R (t - u \cdot x/v^2)$. Exact witness: event
($t=8, x=5$) $\rightarrow$ ($t'=7, x'=-5/4$); $Q_v = Q_{v'} = 75$.

EQ-001/P.46.v1 — time dilation and length contraction (kinematic consequences)

domain: P • tier: *untagged* • status: *unverified* • coq: *definition* • parents: *EQ-001*

source class (readout_genesis domains/relativity/RULE_REGISTRY*.json, rule OA-10) is "PROPOSED_INTERNAL_BRIDGE"
– the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (coq_status=not_yet_formalised).

$t'/t = 1/\Gamma_R$ and $L = L_0/\Gamma_R$ follow from OA-09 with no new law inserted.

EQ-001/P.47.v1 — relative simultaneity

domain: P • tier: *untagged* • status: *unverified* • coq: *definition* • parents: *EQ-001*

source class (readout_genesis domains/relativity/RULE_REGISTRY*.json, rule OA-11) is "PROPOSED_INTERNAL_BRIDGE"
– the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (coq_status=not_yet_formalised).

$\Delta t' = -\Gamma_R \cdot u \cdot \Delta x / v^2$ follows from OA-09 with no new law inserted.

EQ-001/P.48.v1 — velocity composition

domain: P • tier: *untagged* • status: *unverified* • coq: *definition* • parents: *EQ-001*

source class (readout_genesis domains/relativity/RULE_REGISTRY*.json, rule OA-12) is "PROPOSED_INTERNAL_BRIDGE"
– the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (coq_status=not_yet_formalised).

$u_{21} = (u_1 + u_2)/(1 + u_1 \cdot u_2/v^2)$ follows from OA-09; exact witness stays sub-cone
($|u_{21}| < v$).

EQ-001/P.49.v1 — operator-to-metric readout

domain: P • tier: *Th_coqc* • status: *current* • coq: *not_formalisable* • parents: *EQ-001*

$g_{ij} \sim 1/2 \cdot \text{Hess}(\sigma_{LR})$, the retention operator's principal symbol read as a metric. This direction carries the master core's [Th_coqc] base tier (Face 8). The reverse direction (spectrum-alone $\rightarrow$ metric) is explicitly forbidden.

EQ-001/P.50.v1 — finite-basis covariance and minimal closed state

domain: P • tier: *finite_diagnostic* • status: *current* • coq: *definition* • parents: *EQ-001*

The geometry-only quotient Theta lacks the variables to close on its own (Theta_{n+1} depends on Phi, Psi, not Theta alone); the minimal closed state is Z_n=(Phi_n, Phi_{n-1}, Psi_n, Psi_{n-1}, Theta_n). Basis covariance under this minimal state is only PARTIALLY closed.

EQ-001/P.51.v1 — discrete connection

domain: P • tier: *Th_coqc* • status: *current* • coq: *definition* • parents: *EQ-001*

$U_{\{j<-i\}} = V_j^{-1} \cdot V_i$ from the basis-change identity; transport defect $o_{\{i\to j\}} = \phi_j - U_{\{j<-i\}} \cdot \phi_i$.

EQ-001/P.52.v1 — curvature certificate

domain: P • tier: *Th_coqc* • status: *current* • coq: *definition* • parents: *EQ-001*

Loop transport $U_C = U_C(\text{loop}) - I$ defines curvature K_C : local transports failing to commute (finite witness: $U_x = [[1,1],[0,1]]$, $U_y = [[1,0],[1,1]] \Rightarrow U_C = [[0,1],[-1,3]]$, $K_C = [[-1,1],[-1,2]] \setminus \neq 0$).

EQ-001/P.53.v1 — free-path obstruction minimizer

domain: P • tier: *finite_diagnostic* • status: *current* • coq: *definition* • parents: *EQ-001*

Free path selects $0(P) = \sum_e o_e^T G_j o_e$ minimal; witness $\phi_i = [1,0]$, $\phi_f = [1,1]$: path $x \setminus \text{to } y$ obstruction 0, path $y \setminus \text{to } x$ obstruction 1, minimizer selects $x \setminus \text{to } y$.

EQ-001/P.54.v1 — geometry-field feedback commuting square

domain: P • tier: *finite_diagnostic* • status: *current* • coq: *open_prop* • parents: *EQ-001*

The geometry-field feedback loop closes as a time-unrolled DAG (not circular), satisfying $q_{LG} \circ F_{\text{full}} = F_{LG} \circ q_{LG}$, with a shared singular fail-state perp_singular .

EQ-001/P.55.v1 — observer-normalization / redshift / lapse quotient

domain: P • tier: *untagged* • status: *unverified* • coq: *definition* • parents: *EQ-001*

source class (readout_genesis domains/relativity/RULE_REGISTRY*.json, rule OB-07) is "PROPOSED_INTERNAL_GATE" – the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (coq_status=not_yet_formalised).

CLOSED in v0.2 by the Null-Transport Factorization Gate: the local causal-cell rate relates to a remote record rate via the internal lapse $N = \sqrt{\det B}$ ($d\theta_i = N d\theta_o$, $\nu_o = N \nu_i$), with no named metric inserted as a premise.

EQ-001/P.56.v1 — null-transport factorization (internal lapse N and relative-observer factor chi)

domain: P • tier: *untagged* • status: *unverified* • coq: *closed* • parents: *EQ-001*

source class (readout_genesis domains/relativity/RULE_REGISTRY*.json, rule GC-01) is "PROPOSED_INTERNAL_GATE" – the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (coq_status=not_yet_formalised).

The diagonal cone map $B = \text{diag}(a,b)$ factors uniquely as $B = N \cdot \text{diag}(\chi^{-1}, \chi)$, with $N = \sqrt{ab} = \sqrt{\det B}$ (the internal LAPSE) and $\chi = \sqrt{b/a}$ (relative-observer factor). Pure observer change $\Rightarrow \det B = 1 \Rightarrow N = 1$; geometry-dependent readout $\Rightarrow N = \sqrt{\det B}$. Exact witness: $a=2, b=8 \Rightarrow \det B = 16, N=4, \chi=2$.

EQ-001/P.57.v1 — redshift as relative retained-step rate

domain: P • tier: *finite_diagnostic* • status: *current* • coq: *closed* • parents: *EQ-001*

On a stationary path, $d\theta_i = N_{\{i|o\}} d\theta_o$ (θ = retained transition count), so $\nu_o = N \nu_i$ and $d\theta_o = d\theta_i / N$. No external light/gravity/metric law is used; 'frequency shift' here means relative retained-step rate only.

EQ-001/P.58.v1 — lapse composition and normalization holonomy

domain: P • tier: *finite_diagnostic* • status: *current* • coq: *closed* • parents: *EQ-001*

Determinants multiply under composition of diagonal observer maps, so $N_{\{j|o\}} = N_{\{j|i\}} \cdot N_{\{i|o\}}$, and $N_P = \text{product of } N_e \text{ along a path } P$. Around a loop C , the normalization holonomy $H_N(C) = \text{product of } N_e \text{ over the loop}$ is the scalar side of curvature. Exact witness: $B_{\{i|o\}} = \text{diag}(2,8)$ ($N=4$), $B_{\{j|i\}} = \text{diag}(3,27)$ ($N=9$), composed $B_{\{j|o\}} = \text{diag}(6,216)$ ($\det=1296$, $N=36=9 \cdot 4$).

EQ-001/P.59.v1 — horizon as rank-loss boundary, $N=0$ (finite, not infinity)

domain: P • tier: *untagged* • status: *unverified* • coq: *closed* • parents: *EQ-001*

source class (readout_genesis domains/relativity/RULE_REGISTRY*.json, rule GC-04) is "PROPOSED_INTERNAL_GATE" – the Domain Registration Standard's own class enum treats an item in this state as not-yet-established; no per-rule status field exists in the source, this reading has also not yet passed Toledo's own Coq verification (coq_status=not_yet_formalised).

$N=0$ iff $\det B=0$, a finite rank-loss boundary of the observer map (not a statement involving infinity). At $N=0$ the translation B has no inverse; the exterior cannot reconstruct the interior. Clone/agency-horizon connects here as a READOUT equivalence only. Exact witness: $B_h = \text{diag}(0,5)$, $\det=0$, $N=0$, singular (no inverse).

EQ-001/P.60.v1 — Unruh-fix: κ_R as an internally normalized readout of a_{local}

domain: P • tier: *untagged* • status: *current* • coq: *closed* • parents: *EQ-001*

From the DeclaredFormula $\tau_c(U) = \pi \cdot c/a_{\text{local}}$, $a_{\text{local}} = \pi \cdot c/\tau_c(U)$. The observer reads $a_o = N_{\{i|o\}} \cdot a_{\text{local}}$; at the horizon, $\kappa_R := \lim_{i \rightarrow H} N_{\{i|o\}} a_{\text{local}}(i)$, if finite. CORRECT STATUS: $a_{\text{local}} \neq \kappa_R$ directly, but $\kappa_R = \text{Normalize}_N(a_{\text{local}})$; 'surface gravity' requires EXTERNAL semantic calibration – only the INTERNAL bridge (κ_R as a normalized readout of a_{local}) is repaired here, not the physical identification. Exact witness: $a_{\text{local}}=7/2$, $N=4 \Rightarrow \kappa_R=14$, finite, $\kappa_R \neq a_{\text{local}}$.

EQ-001/P.61.v1 — graph front-speed vs measured-speed correspondence

domain: P • tier: *untagged* • status: *current* • coq: *not_formalisable* • parents: *EQ-001*

The graph front-speed v is declared (not derived) as playing the role a measured propagation speed would play; the identification with any specific physical speed is not established.

EQ-001/P.62.v1 — mass-memory declared correspondence

domain: P • tier: *untagged* • status: *current* • coq: *not_formalisable* • parents: *EQ-001*

Retention memory is declared (not derived) as playing a mass-like role; no mass-energy law is derived from the root.

EQ-001/P.63.v1 — Schwarzschild calculator identity

domain: P • tier: *untagged* • status: *current* • coq: *definition* • parents: *EQ-001*

$r_s = 2 \cdot G \cdot E / c^4$, admitted only as a declared calculator identity, not a derived theorem of this domain.

EQ-001/P.64.v1 — Unruh calculator identity

domain: P • tier: *untagged* • status: *current* • coq: *definition* • parents: *EQ-001*

$\tau_c = \pi \cdot c / a$, admitted only as a declared calculator identity, not a derived theorem of this domain.

EQ-001/P.65.v1 — complete horizon/redshift bridge and dynamic metric-source closure

domain: P • tier: *finite_diagnostic* • status: *current* • coq: *not_formalisable* • parents: *EQ-001*

CLOSED in v0.2: the horizon half by GC-04 ($N=0 \wedge \det B=0$, finite rank-loss boundary) plus GC-05 (Unruh-fix internal bridge $\kappa_R = \text{Normalize}_N(a_{\text{local}})$); the metric-source half by the Geometry Stationarity Gate (GD-01..GD-04), a source/record field Ψ driving a Θ -evolution that passes a commuting square AND has a failing control case (exact fixture $\Theta_{n+1}=3/8$, $R_\Theta=0$; failing control $\Theta_{\text{wrong}}=7/8$, $R_\Theta=1 \neq 0$).

EQ-001/P.66.v1 — geometry source $S_\Theta = \Phi^T G_a \Psi$

domain: P • tier: *finite_diagnostic* • status: *current* • coq: *closed* • parents: *EQ-001*

Since $G[\Theta] = G_0 + \sum_a \Theta^a G_a$, $dG/d\Theta^a = G_a$. The geometry source is $S_{\Theta,n} = \Phi_n^T G_a \Psi_n$ - reader x record changing the operator geometry reads. This is NOT posited as energy/mass/stress. Exact witness: $G_1 = [[0,1],[1,0]]$, $\Phi_n = [1,0]$, $\Psi_n = [0,1] \Rightarrow S_\Theta = 1$.

EQ-001/P.67.v1 — geometry stationarity law and explicit Θ stepper

domain: P • tier: *finite_diagnostic* • status: *current* • coq: *closed* • parents: *EQ-001*

The living-geometry state Θ obeys the same retained-action grammar as the rest of the root: $S_{\text{closed}} = S_{\text{DRL}} + \sum_n [(1/2) dt) d\Theta_n^T M_\Theta d\Theta_n - dt U_\Theta(\Theta_n)]$. Stationarity in Θ gives $M_\Theta d^2_t \Theta_n + \text{grad } U_\Theta(\Theta_n) + K S_{\Theta,n} = 0$, with explicit stepper $\Theta_{n+1} = 2 \Theta_n - \Theta_{n-1} - dt^2 M_\Theta^{-1} [\text{grad } U_\Theta(\Theta_n) + K S_{\Theta,n}]$. Exact fixture ($\Theta_{n-1}=0$, $\Theta_n=1/2$, $M_\Theta=2$, $K=1$, $dt=1$, $U_\Theta=\Theta^2/4$) with $S_\Theta=1$ gives $\Theta_{n+1}=3/8$ and action residual $R_\Theta=0$. Then $G_{n+1}=G[\Theta_{n+1}]$ and $g_{n+1}^{ij}=(1/2)$ Hess $\sigma_{G[\Theta_{n+1}]}$ - operator\to metric only, spectrum-alone still forbidden.

EQ-001/P.68.v1 — minimal closed state Z_n and commuting square (time-unrolled DAG)

domain: P • tier: *finite_diagnostic* • status: *current* • coq: *definition* • parents: *EQ-001*

The minimal closed state is $Z_n = (\Phi_n, \Phi_{n-1}, \Psi_n, \Psi_{n-1}, \Theta_n, \Theta_{n-1})$; $Z_{n+1} = F_{\text{MSG}}(Z_n)$. The commuting square $q_{\text{MSG}} \circ F_{\text{full}} = F_{\text{MSG}} \circ q_{\text{MSG}}$ holds, but the metric-only quotient q_g LACKS the variables to close alone - two states with equal metric but different Φ , Ψ , Θ_{n-1} give different next metrics. Time-unrolled, this is a DAG, not circular.

EQ-001/P.69.v1 — geometry stationarity fail rules

domain: P • *tier:* *finite_diagnostic* • *status:* *current* • *coq:* *definition* • *parents:* *EQ-001*

Four named fail states for the geometry stationarity gate: FAIL_GEOMETRY_PIVOT (det M_Theta=0 / low pivot), FAIL_METRIC_ADMISSIBILITY (bad-signature/degenerate metric), FAIL_NULL_MIXING (null edges mixed into a stationary claim), HORIZON_BOUNDARY (lapse=0, ties to GC-04). No silent regularization; no inverse computed across a horizon.

Part II

APPENDIX C

2 EQ-002 — Distinguishability needs a distinguisher

domain: — • *tier:* Ax • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *EQ-001*

$\exists A : A \text{ discriminates } E_1 \neq E_2$

EQ-002/E.01.v1 — The human-domain instance of the readout operator: a subject's readout is conditioned on their history and context

domain: E • *tier:* *Definition* • *status:* *current* • *coq:* *definition* • *parents:* *EQ-002/M.03.v1*

$r_n = R_H(x_n \mid H_n, c_n)$; extended with an explicit question index in Before Meaning,
Before Choice: $\rho^H H_n = R_H(x_n \mid H_n, c_n, Q_n)$

Occurrences (4): 22410666:MBN-readout, 22410666:H2, 22357744:EMG-01, 22424434:(9)

EQ-002/E.02.v1 — A source label is a readout, not an oracle: what an evaluator receives is itself operator-conditioned

domain: E • *tier:* *Definition* • *status:* *current* • *coq:* *closed* • *parents:* *EQ-002/M.03.v1*

$$R_A = O_A(W; \Pi_A)(\neq W); m(A) \neq \rho(A)$$

Occurrences (3): 22301202:(RA), 22301202:(m-rho), 22307841:(unnumbered, Sec.1)

EQ-002/E.03.v1 — The observer as a bounded selection–encoding–translation pipeline

domain: E • *tier:* *Definition* • *status:* *current* • *coq:* *definition* • *parents:* *EQ-002/M.03.v1*

$$M_A(E) = (T_A \circ \Pi_A)(S_A(E))$$

Occurrences (2): 18925129:(1), 18943971:Canonical Formula

EQ-002/E.04.v1 — meaning decomposed into analytic modes

domain: E • *tier:* *Definition* • *status:* *current* • *coq:* *definition* • *parents:* *EQ-002/M.03.v1*

$$\mu_n = (\mu_n^{aff}, \mu_n^{prag}, \mu_n^{auto}, \mu_n^{conc}, \mu_n^{epi})$$

Occurrences (1): 22424434:(11)

EQ-002/E.05.v1 — naming as articulation and restructuring operator

domain: E • *tier:* *Definition* • *status:* *current* • *coq:* *definition* • *parents:* *EQ-002/M.03.v1*

$$\ell_n = L_H(E_n, \mu_n \mid H_n, c_n); \mu_n \rightarrow E_n \rightarrow \ell_n \rightarrow \mu_{n+1} \rightarrow E_{n+1}$$

Occurrences (2): 22424434:(14), 22424434:(15)

EQ-002/E.06.v1 — rhythm/momentum/attraction feeding history-shaped semantic accessibility

domain: E • tier: Definition • status: current • coq: definition • parents: EQ-002/M.03.v1

$$\mathbf{m}_{\{t+1\}}(\mathbf{e}) = \rho \mathbf{m}_t(\mathbf{e}) + 1[\mathbf{e}_t = \mathbf{e}]; \mathbf{a}_{\{Q,t\}}(\mathbf{e}) = [\Phi_{\{Q,t\}}(\mathbf{s}) - \Phi_{\{Q,t\}}(\mathbf{s}')]_{+}; \kappa^{\text{sem}}_{\{t+1\}}(\mathbf{e} | Q) \propto \kappa^{\text{sem}}(0)_t(\mathbf{e} | Q) \cdot \exp(\beta \mathbf{a} + \mu \mathbf{m} - \nu \mathbf{c} + \xi)$$

Occurrences (3): 22424434:(18), 22424434:(19), 22424434:(20)

EQ-002/E.07.v1 — Total epistemic error is never zero

domain: E • tier: Ax • status: current • coq: closed • parents: EQ-002/M.03.v1

$$\epsilon_{\text{tot}} > 0 \text{ (Genesis Constraint-First); decomposed independently as } \epsilon_{\text{tot}} = \epsilon_{\text{clock}} + \epsilon_{\text{cross}} + \epsilon_{\text{sel}} + \epsilon_{\text{map}} + \epsilon_{\text{self}} \text{ (Structural Error theorem, Mind as Information Horizon)}$$

Occurrences (2): 19205869:(4), 19640361:(12)

EQ-002/E.08.v1 — Epistemic Multiplication Factor k_{epi} and frontier-regime classification

domain: E • tier: Definition • status: current • coq: closed • parents: EQ-002/M.03.v1

$$k_{\text{epi}}(t) = |\text{New}_{t+1}| / \max(1, |F_t|); k_{\text{epi}} < 1 \text{ contractive}, k_{\text{epi}} \approx 1 \text{ critical}, k_{\text{epi}} > 1 \text{ expanding}$$

Occurrences (4): 22308072:(7), 22308072:(8), 22308072:(9), 22308072:(10)

EQ-002/E.09.v1 — An agent's realized exposure to a source, indexed by outlet/channel/timing/relation/context

domain: E • tier: Definition • status: current • coq: definition • parents: EQ-002/M.03.v1

$$x_{i,n} = \text{Access}(A_n; O_i, L_i, T_i, R_i, C_i)$$

Occurrences (1): 21529456:(11)

EQ-002/E.10.v1 — Selection-then-encoding decomposition of an observation into a translated record

domain: E • tier: Definition • status: current • coq: definition • parents: EQ-002/M.03.v1

$$O_A[n] = \Pi_A(E[n]); \text{enc}_A(O_A)[n] = T_A(O_A[n])$$

Occurrences (2): 19205869:(1), 19205869:(2)

EQ-002/E.11.v1 — A named taxonomy of readout-domain sources (first-person, behavioral, neural, world)

domain: E • tier: Dr • status: current • coq: definition • parents: EQ-002/M.03.v1

$$D_n = \{D_n^{\text{first}}, D_n^{\text{beh}}, D_n^{\text{neural}}, D_n^{\text{world}}\}$$

Occurrences (1): 21425420:(39)

EQ-002/E.12.v1 — Friction, not magic: institutional certification earns force through reliable epistemic friction, not status alone

domain: E • tier: Dr • status: current • coq: closed • parents: EQ-002/M.03.v1

Friction, not magic: institutional certification has epistemic force insofar as institutions produce reliable epistemic friction (criticism, validation, replication, robustness testing, archival continuity, correction, and answerability). Certification does not create truth by status alone.

Occurrences (1): 22301202:Prin-5

EQ-002/H.01.v1 — The causal chain from world-state through body, nervous system, sensory interface, internal agency state, policy, and executed output back to world-state

domain: H • tier: Definition • status: current • coq: definition • parents: EQ-002/M.03.v1

$B[n] \rightarrow H_body[n] \rightarrow N[n] \rightarrow S[n] \leftrightarrow A[n] \rightarrow \pi[n] \rightarrow U[n] \rightarrow B[n+1]$, with each arrow a discrete update Φ_* ; independently paralleled by $x_n \rightarrow \zeta_n \rightarrow v_n \rightarrow \rho_n \rightarrow U_n \rightarrow \beta_n \rightarrow I_{\{n+1\}} \rightarrow a_n \rightarrow S_{\{n+1\}}$ (contact $\rightarrow$ valence $\rightarrow$ drive $\rightarrow$ identity-binding $\rightarrow$ appropriated experience $\rightarrow$ action $\rightarrow$ renewed state)

Occurrences (13): 19640361:(1), 19640361:(2), 19640361:(3), 19640361:(4), 19640361:(5), 19640361:(6), 19640361:(7), 19640361:(8), 21529456:(39), 21529456:(40), 21529456:(41), 21529456:(42) (+1 more)

EQ-002/H.02.v1 — An epistemic agent is a query-relative quotient of a retained state, not an indivisible substance; identity is compositional lineage

domain: H • tier: Definition • status: current • coq: definition • parents: EQ-002/M.03.v1

$A_{\{i,n\}} = q_A(Z_{\{i,n\}}; Q_A, O_A, c_n)$; $Aut(F_A, O_A) = \{h: O_A \circ h = O_A, h \circ F_A = F_A \circ h\}$; $I_n = q_comp(M_n \oplus Bel_n \oplus \Theta_n \oplus Roles_n \oplus BodyTool_n \oplus SocialLineage_n)$

Occurrences (3): 21529456:(36), 21529456:(37), 21529456:(38)

EQ-002/H.03.v1 — RG-HCA’s own proactive-capability pipeline

domain: H • tier: Definition • status: current • coq: definition • parents: EQ-002/M.03.v1

Retained Difference $\rightarrow$ Human Readout $\rightarrow$ Live Problem $\rightarrow$ Barrier Readout $\rightarrow$ Candidate Routes $\rightarrow$ Human Endorsement $\rightarrow$ Adaptive Scaffold $\rightarrow$ Practice $\rightarrow$ Withdrawal $\rightarrow$ Human Return $\rightarrow$ Novel Transfer $\rightarrow$ World Feedback $\rightarrow$ Opportunity Conversion

Occurrences (2): 22498047:(5), 22498047:Human Capability Advancement Principle

EQ-002/H.04.v1 — Each pipeline operator stage is attributed to human, AI, or joint control

domain: H • tier: Definition • status: current • coq: definition • parents: EQ-002/M.03.v1

$$\alpha_t(o) \in \{HUMAN, AI, JOINT\} \text{ for } o \in \Omega = \{q_{sem}, \Pi_P, \Pi_Q, B, a, m, \kappa, \Gamma_Q, U_D, \Pi_R\}$$

Occurrences (3): 22307148:(55), 22307148:(56), 22307148:(57)

EQ-002/H.05.v1 — Agent-level network readout

domain: H • tier: Dr • status: current • coq: open_prop • parents: EQ-002/M.03.v1, weld/M.02.v1

$$z_{\{i,t\}} = R_{\{i,t\}} \setminus \text{left}(q_D(S_t), m_{\{-i,t\}}, c_{\{i,t\}} \setminus \text{right})$$

Occurrences (1): 22639311:Tunnel-v2.1:eq.2

EQ-002/H.06.v1 — Readout-of-readout recursion

domain: H • tier: Dr • status: current • coq: open_prop • parents: EQ-002/M.03.v1

$$z_{\{j,t+1\}} = R_{\{j,t+1\}} \setminus \text{left}(z_{\{i,t\}}, c_{\{j,t+1\}} \setminus \text{right})$$

Occurrences (1): 22639311:Tunnel-v2.1:eq.3

EQ-002/M.01.v1 — When may a downstream distinction be credited to a given readout: fiber constancy, factorization, and stochastic composition

domain: M • tier: Definition • status: current • coq: closed • parents: EQ-002/M.03.v1

$\Phi: X \rightarrow Z$ is readout-admissible relative to $R: X \rightarrow Y$ iff constant on every fiber of R : $R(x) = R(x') \implies \Phi(x) = \Phi(x')$; equivalently $\ker R \subseteq \ker \Phi$, or $\Phi = \bar{g} \circ R$ (Factorization Theorem). Stochastic form: $C = G \circ K^*$ (access admissibility via a Markov kernel); data-processing inequality $I(X; Y_{\{j+1\}}) \leq I(X; Y_j)$; non-collapse $R^* \neq R^{\sim}$, $K^* \neq K^{\sim}$ (actual source relation vs claimant’s model of it).

Occurrences (20): 22301318:(5), 22301318:(6), 22301318:Def-1, 22301318:(7), 22301318:Prop-1, 22301318:Princ-4, 22301318:(8), 22301318:(9), 22301318:Prop-2, 22301318:(10), 22301318:(11), 22301318:(12) (+8 more)

EQ-002/M.02.v1 — programme legibility and its coherence multiplier

domain: M • tier: *finite_diagnostic* • status: *current* • coq: *closed* • parents: *EQ-002/M.03.v1*

`Coh_effective=Coh_latent×L_g; L_g^proxy=(public assets with one-click programme path)/(public assets); K1→RelatedAssetDiscovery→LongerExposure→K2 opportunity`

Occurrences (3): 22163849:(34), 22163849:(35), 22163849:(36)

EQ-002/M.03.v1 — The root readout gate itself: a bounded, reader/operator-conditioned map that never returns the source unchanged

domain: M • tier: *Definition* • status: *current* • coq: *closed* • parents: *EQ-002*

$$\text{Readout}_{Q,O,c}(S) = z, z \neq S$$

Occurrences (4): 22481924:(2), 22639311:Tunnel-v2.0:eq.23, 22550491:Eq. 52, 22550491:Eq. 66

EQ-002/M.04.v1 — Simulation success does not entail world validation

domain: M • tier: *Dr* • status: *current* • coq: *open_prop* • parents: *EQ-002/M.03.v1*

`Simulation \ Success \not\rightarrow World \ Validation`

Occurrences (1): 22639311:Tunnel-v2.1:eq.20

EQ-002/W.01.v1 — distributional readouts of conversion

domain: W • tier: *Definition* • status: *split* • coq: *definition* • parents: *EQ-002/M.03.v1*

N4 split (founder ruling 22:20, 2026-09-06): (22) $C^{10}/C^{50}/C^{90}$ percentile-readout definition and (23) $I^H_{(j,t)}$ inequality-index definition are two distinct formulas; 'Proposition 4 [Open]' ('Broad human expansion requires that conversion gains reach a declared population distribution...') is a narrative Open-tier proposition, no relational operator.

$$C^{10}_{j,t}, C^{50}_{j,t}, C^{90}_{j,t}, I^H_{j,t} = C^{90}_{j,t} - C^{10}_{j,t}$$

Occurrences (3): 22481926:(22), 22481926:(23), 22481926:Proposition 4

EQ-002/W.02.v1 — Conversion percentile readouts $C^{10}/C^{50}/C^{90}$

domain: W • tier: *Definition* • status: *current* • coq: *definition* • parents: *EQ-002/M.03.v1*, *EQ-002/W.01.v1*

$$C^{10}_{(j,t)}, C^{50}_{(j,t)}, C^{90}_{(j,t)}$$

Occurrences (1): 22481926:(22)

EQ-002/W.03.v1 — Conversion inequality index $I^H_{(j,t)}$

domain: W • tier: *Definition* • status: *current* • coq: *definition* • parents: *EQ-002/M.03.v1*, *EQ-002/W.01.v1*

$$I^H_{(j,t)} = C^{90}_{(j,t)} - C^{10}_{(j,t)}$$

Occurrences (1): 22481926:(23)

3 EQ-003 — Asymmetry is forced

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *EQ-001*, *EQ-002*

`A→B ≠ B→A` (a directed admissible-transition relation on the retained graph, not material implication)

4 EQ-004 — Temporal ordering

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *EQ-003*

$$t(s) := \min \# \text{steps}(s_0 \rightarrow s) \in \mathbb{N}$$

5 EQ-005 — Persistence / causal memory

domain: — • tier: *Ax* • status: *current* • coq: *root_layer_unwired* • parents: *EQ-001*, *EQ-004*

$$\tau_c > 0$$

6 EQ-006 — Discrete world before continuum

domain: — • tier: *Ax* • status: *current* • coq: *root_layer_unwired* • parents: *EQ-005*

$$t = n\Delta\theta, n \in \mathbb{N}, \Delta\theta > 0 \text{ } (\Delta\theta=0 \text{ undefined; continuum is a controlled readout limit})$$

7 EQ-007 — Finite causal propagation speed

domain: — • tier: *Ax* • status: *current* • coq: *root_layer_unwired* • parents: *EQ-005*, *EQ-006*

$$v = \sqrt{(D/\tau_c)} < \infty$$

8 EQ-008 — Retained graph-Laplacian operator L_R

domain: — • tier: *Ax* • status: *current* • coq: *root_layer_unwired* • parents: *EQ-005*, *EQ-006*, *EQ-007*

$$L_R := D_W - W \text{ } (D_W = \text{weighted degree matrix, } W = \text{retained adjacency; full operator on a multimode retained state, not a scalar})$$

Occurrences (1): (4)

9 EQ-009 — RD unit definition

domain: — • tier: *Ax* • status: *current* • coq: *root_layer_unwired* • parents: *EQ-001*

$$1 \text{ RD} := \text{one retained-distinction record}$$

EQ-009/E.01.v1 — Append-only record branch/split step

domain: *E* • tier: *Definition* • status: *current* • coq: *definition* • parents: *EQ-009*

$$z^\sim = C z \text{ } \backslash \text{to } z' = \sqrt{1-\gamma} z^\sim, \backslash \rho = -\sqrt{\gamma} z^\sim \text{ } \backslash \text{to } \Psi_{i_n+1} = \Psi_{i_n} \backslash \text{oplus } \backslash \rho_n$$

EQ-009/E.02.v1 — Append-only record T1 conservation check

domain: *E* • tier: *finite_diagnostic* • status: *current* • coq: *definition* • parents: *EQ-009/E.01.v1*

T1 conservation [Cyrillic text omitted in the print catalogue - see the JSON/site entry for the source text]: $Q(z_n) + \Sigma Q(\rho_n) = Q(z_{\{0\}})$ -- rel. error $< 10^{-1}$,
decay ω [Thai text omitted in the print catalogue - see the JSON/site entry for the source text] $(1-\gamma)^n < 10^{-9}$

EQ-009/E.03.v1 — Append-only record / DRL bridge (T4)

domain: E • tier: *Open* • status: *current* • coq: *not_formalisable* • parents: *EQ-009/E.02.v1, EQ-015/M.13.v1*

$T4 = [\text{Thai text omitted in the print catalogue - see the JSON/site entry for the source text}]: [\text{Thai text omitted in the print catalogue - see the JSON/site entry for the source text}] \text{ envelope } [\text{Thai text omitted in the print catalogue - see the JSON/site entry for the source text}] \ \gamma \rightarrow D/M$

10 EQ-010 — Domain encoder

domain: — • tier: Ax • status: *current* • coq: *root_layer_unwired* • parents: *EQ-009*

$x_{\text{domain}} - [\text{Enc}_\Omega] \rightarrow x_{\text{RD}}$ (encoder: domain quantity $\rightarrow$ RD coordinate)

11 EQ-011 — Domain decoder

domain: — • tier: Ax • status: *current* • coq: *root_layer_unwired* • parents: *EQ-009*

$y_{\text{RD}} - [\text{Dec}_\Omega] \rightarrow y_{\text{domain}}$ (decoder: RD coordinate $\rightarrow$ calibrated domain quantity)

12 EQ-012 — Resource-logic retention judgment

domain: — • tier: Ax • status: *current* • coq: *root_layer_unwired* • parents: *EQ-009*

$\Gamma \vdash_{\{\alpha, \rho, \kappa\}} \phi$

13 EQ-013 — No unrestricted contraction

domain: — • tier: Ax • status: *current* • coq: *root_layer_unwired* • parents: *EQ-012*

$A \not\Rightarrow A \otimes A$

14 EQ-014 — Copy licence

domain: — • tier: Ax • status: *current* • coq: *root_layer_unwired* • parents: *EQ-012, EQ-013*

$!_{\kappa} A \vdash A^{\otimes m}, m \leq \kappa$

15 EQ-015 — Universal spine PDE (mixed-tier term breakdown)

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *SpinePDE*

$M \partial_t^2 \Phi + D \partial_t \Phi + K \cdot L_R \Phi + \nabla V(\Phi) = J - \eta [M \partial^2 \Phi: Dr \cdot D \partial \Phi: \text{finite_diagnostic} \cdot K \cdot L_R \Phi: Th_coqc (\text{admissibility skeleton}) \cdot \eta: \text{finite_diagnostic/Open}]$

Occurrences (1): (5)

EQ-015/B.01.v1 — Health-stream theorem: setpoint_is_fixed

domain: B • tier: *Th_coqc* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

$\forall \alpha, \beta, u, dt \in \mathbb{Q}, \beta \neq 0 \implies \text{step}(\alpha, \beta, u, dt, \text{setpoint}(\alpha, \beta, u)) = \text{setpoint}(\alpha, \beta, u),$
 $\text{quad} \text{ where } \text{step}(\alpha, \beta, u, dt, C) := C + dt(\alpha u - \beta C),$
 $\text{setpoint}(\alpha, \beta, u) := \frac{\alpha u}{\beta}$

Occurrences (1): setpoint_is_fixed

EQ-015/B.02.v1 — Health-stream theorem: one_step_error

domain: B • tier: Th_coqc • status: current • coq: wrapped_related • parents: EQ-015

$$\forall \alpha, \beta, u, dt, C \in \mathbb{Q}, \beta \neq 0 \implies \text{step}(\alpha, \beta, u, dt, C) - \text{setpoint}(\alpha, \beta, u) = (1 - dt) \cdot \beta \cdot (C - \text{setpoint}(\alpha, \beta, u))$$

Occurrences (1): one_step_error

EQ-015/B.03.v1 — Health-stream theorem: n_step_error

domain: B • tier: Th_coqc • status: current • coq: wrapped_related • parents: EQ-015

$$\forall \alpha, \beta, u, dt \in \mathbb{Q}, \beta \neq 0, \forall n \in \mathbb{N}, C \in \mathbb{Q} : \text{iter}_n(\alpha, \beta, u, dt, C) - \text{setpoint}(\alpha, \beta, u) = (1 - dt)^n \cdot \beta \cdot (C - \text{setpoint}(\alpha, \beta, u))$$

Occurrences (1): n_step_error

EQ-015/B.04.v1 — Health-stream theorem: disc_even

domain: B • tier: Th_coqc • status: current • coq: wrapped_related • parents: EQ-015

$$\forall h \in \mathbb{Q} : \text{disc}(-h) = \text{disc}(h), \text{disc}(h) := 4 - 27h^2$$

Occurrences (1): disc_even

EQ-015/B.05.v1 — Health-stream theorem: cusp_factor

domain: B • tier: Th_coqc • status: current • coq: wrapped_related • parents: EQ-015

$$\forall R, h \in \mathbb{Q} : h = R - R^3 \implies 27h^2 - 4 = (3R^2 - 1)(9R^4 - 15R^2 + 4)$$

Occurrences (1): cusp_factor

EQ-015/B.06.v1 — Health-stream theorem: fold_from_double_root

domain: B • tier: Th_coqc • status: current • coq: wrapped_related • parents: EQ-015

$$\forall R, h \in \mathbb{Q} : V_p(R, h) = 0 \wedge V_{pp}(R) = 0 \implies 27h^2 = 4, \\ V_p(R, h) := R^3 - R + h, V_{pp}(R) := 3R^2 - 1$$

Occurrences (1): fold_from_double_root

EQ-015/B.07.v1 — Health-stream theorem: bistable_window_dec

domain: B • tier: Th_coqc • status: current • coq: wrapped_related • parents: EQ-015

$$\forall h \in \mathbb{Q} : \text{disc}(h) > 0 \iff 27h^2 < 4$$

Occurrences (1): bistable_window_dec

EQ-015/B.08.v1 — Health-stream theorem: critical_slowing_marginal

domain: B • tier: Th_coqc • status: current • coq: wrapped_related • parents: EQ-015

$$\forall R, h \in \mathbb{Q} : V_p(R, h) = 0 \wedge V_{pp}(R) = 0 \implies V_{pp}(R) = 0$$

Occurrences (1): critical_slowing_marginal

EQ-015/B.09.v1 — Health-stream theorem: rest_iff_critical

domain: B • tier: Th_coqc • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

$$\backslash\text{forall}\backslash, R, h, dt \backslash\text{in} \backslash\text{mathbb{Q}}, \backslash dt \backslash\text{neq} 0 : \backslash\text{quad} \backslash\text{mathrm{repair_step}}(R, h, dt) = R \backslash\text{iff} \\ V_p(R, h) = 0, \backslash\text{quad} \backslash\text{mathrm{repair_step}}(R, h, dt) := R - dt \backslash, V_p(R, h)$$

Occurrences (1): *rest_iff_critical*

EQ-015/E.01.v1 — Meaning-giving: a reader-conditioned significance relation, typed into affective/pragmatic/autobiographical/conceptual/epistemic modes

domain: E • tier: *Definition* • status: *current* • coq: *definition* • parents: *EQ-015/M.03.v1*

$$\mu_n = \Psi_H(r_n, H_n, c_n, Q_n); \text{typed as } \mu_n = (\mu_n^{\text{aff}}, \mu_n^{\text{prag}}, \mu_n^{\text{auto}}, \mu_n^{\text{conc}}, \\ \mu_n^{\text{epi}})$$

Occurrences (4): 21529456:(13), 22410666:MBN-config, 22357744:EMG-02, 22357744:EMG-03

EQ-015/E.02.v1 — Approach/aversion/blindness can distort the meaning operator without raising epistemic status

domain: E • tier: *Definition* • status: *current* • coq: *closed* • parents: *EQ-015/M.03.v1*

$$\xi_n = (\xi_n^+, \xi_n^-, \xi_n^0); \Xi_n = \alpha_n^+ P_n^{++} + \alpha_n^- P_n^{-+} + \alpha_n^0 P_n^{00}; G_n\{\mu, n\} = G_n\{\mu, \\ n\} \circ (I + \Xi_n); \epsilon_n\{\Xi, n\} = d_0(G_n\{\mu, n\}(z_n), G_n\{\mu, n\}(z_n))$$

Occurrences (3): 21529456:(44), 21529456:(45), 21529456:(46)

EQ-015/E.03.v1 — meaning as significance relation $\mu_n = \Psi_H(r_n, H_n, c_n, Q_n)$

domain: E • tier: *Definition* • status: *current* • coq: *definition* • parents: *EQ-015/M.03.v1*

$$\mu_n = \Psi_H(r_n, H_n, c_n, Q_n)$$

Occurrences (3): 22456564:(1), 22424434:(10), 22639311:Tunnel-v2.0:eq.6

EQ-015/E.04.v1 — Experience as phenomenon-as-meaningfully-read

domain: E • tier: *Definition* • status: *current* • coq: *closed* • parents: *EQ-015/M.03.v1*

$$E_n = \Phi_E(x_n, \mu_n, \gamma_n^\mu, c_n); \text{central identity : } Experience = \text{phenomenon-as-meaningfully-read}$$

Occurrences (11): 21529456:(14), 22410666:MBN-experience, 22357744:EMG-04, 22357744:EMG-05, 21425420:(12), 21425420:(13), 21425420:(15), 21425420:(16), 21425420:(17), 22424434:(12), 22424434:(13)

EQ-015/E.05.v1 — Naming as articulation, optionally recursive, never a precondition of significance

domain: E • tier: *Definition* • status: *current* • coq: *definition* • parents: *EQ-015/M.03.v1*

$$\ell_n = L_H(E_n, \mu_n \mid H_n, c_n), \text{ with } E_n, \mu_n \text{ permitted before stable } \ell_n; \text{ recursive form:} \\ \mu_n \rightarrow E_n \rightarrow \ell_n \rightarrow \mu_{\{n+1\}} \rightarrow E_{\{n+1\}}$$

Occurrences (6): 22410666:MBN-naming, 22410666:H1, 22410666:H3, 22410666:H5, 22357744:EMG-06, 22357744:EMG-07

EQ-015/E.06.v1 — Resonance as external-internal experiential congruence (current definition supersedes an earlier accessibility-diagnostic reading)

domain: E • tier: *Dr* • status: *current* • coq: *definition* • parents: *EQ-015/M.03.v1*

$$I_{\{H, n\}} = \text{Retrieve}(M_{\{H, n\}} \mid c_n, Q_n); \text{Res}_H(n) = C_H(E_n^{\text{cur}}, I_{\{H, n\}} \mid c_n, Q_n) \in [0, 1]; \\ \text{Res} \neq \text{Identity}, \text{Res} \neq \text{Truth}, \text{Res} \neq \text{Retention}, \text{Res} \neq \text{Improvement}$$

Occurrences (5): 22357744:EMG-09, 22357744:EMG-10, 22357744:EMG-11, 22357744:EMG-12, 22410666:MBN-resonance-diagnostic

EQ-015/E.07.v1 — Accumulation, activation-barrier crossing, and release — a candidate transition topology, not a physical-energy law

domain: E • tier: Definition • status: current • coq: definition • parents: EQ-015/M.02.v1

$\Delta W_{\{j,N\}} = \Sigma_{\{n=1\}}^N \eta_{n\text{-eligibleGradient}_n}; W_{n\text{-eff}} = W_{\{\text{info},n\}} \cdot g(\text{Res}_H(n), \text{eligibility}, \text{context}); \Sigma_{\{k \leq n\}} W_{k\text{-eff}} \geq \Delta V_{\uparrow\{\text{old} \rightarrow \text{new}\}}$

Occurrences (4): 22357744:EMG-16, 22357744:EMG-17, 22357744:EMG-18, 22357744:EMG-19

EQ-015/E.08.v1 — Retention: only a selected residue of a readout updates the retained state

domain: E • tier: Definition • status: current • coq: definition • parents: EQ-015/M.03.v1

$H_{\{n+1\}} = U_H(H_n, \text{Retain}(E_n, \mu_n, r_n), \delta_{\text{world}\{n:n+1\}})$ -- restated with an added other-experience term in later Human-AI chapters as $H_{\{n+1\}} = U_H(H_t, E^{\{\text{AI}\}}_{\{H,t\}}, \delta_{\{\text{world}\}}, X_{\{\text{other}\}})$

Occurrences (14): 21529456:(15), 22410666:MBN-retention, 22410666:H4, 22357744:EMG-08, 22357744:EMG-23, 21425420:(14), 19205869:(3), 22456414:(2), 22456487:unlabeled (5, Reflexive Reconfiguration stage), 22456564:unlabeled (§6, Integration bullet), 22424434:(16), 22424434:(17) (+2 more)

EQ-015/E.09.v1 — Domain-indexed epistemic alignment and its expected improvement under disciplined reasoning

domain: E • tier: Definition • status: current • coq: definition • parents: EQ-015/M.03.v1

$$V_A[n] = \text{Align}(M_A[n], \theta_W|D); E[V_A[n+1]|Rsn_A, D] > E[V_A[n]|D]$$

Occurrences (2): 19205869:(5), 19205869:(6)

EQ-015/E.10.v1 — Under unbounded external generation, production ceases to index internal organization; accumulation gives way to discrimination

domain: E • tier: Dr • status: current • coq: definition • parents: EQ-015/M.02.v1

When external generation of coherent structure increases without bound, observable production ceases to serve as a unique indicator of internal organization; regime transition: accumulation $\rightarrow$ discrimination

Occurrences (2): 18711408:L1, 18711408:L2

EQ-015/E.11.v1 — context-indexed knower status and weighted alignment profile

domain: E • tier: Definition • status: current • coq: definition • parents: EQ-015/M.01.v1

$$K_A(D, t) := V_A^D(M_A(t), \theta_D); V_A^D = w1P + w2I + w3S + w4R + w5L, \Sigma w_i^D = 1$$

Occurrences (2): 19215748:unlabeled (§7, display 1), 19215748:unlabeled (§7, display 2)

EQ-015/E.12.v1 — The Mission/event-domain reading of the root stepper: an agent's realized readout as a gain on a control signal plus three named noise/error terms

domain: E • tier: Definition • status: current • coq: closed • parents: EQ-015/M.02.v1

$$M_A[n] = K_A \cdot \theta(E[n]) + \eta_{sel} + \eta_{map} + \eta_{self}$$

Occurrences (2): 22307148:(1), 22163849:(65)

EQ-015/E.13.v1 — The live semantic-field state as a declared quotient of the raw readout stream

domain: E • tier: Definition • status: current • coq: definition • parents: EQ-015/M.03.v1

$$S_{A,t} := q_{sem,A,\Omega_t}(Z_t)$$

Occurrences (1): 22307148:(6)

EQ-015/E.14.v1 — No Bare Pedigree: a source label alone is epistemically incomplete reporting

domain: E • tier: Dr • status: *current* • coq: *closed* • parents: $EQ-015/M.01.v1$

No Bare Pedigree: a source label is epistemically incomplete reporting. The relevant object is the tuple of production procedure, selection conditions, dependencies, checks, error model, inferential role, and answerability.

Occurrences (1): 22301202:Prin-3

EQ-015/E.15.v1 — From Problem to Hypothesis's own family of non-collapse identities guarding attraction/momentum/accessibility from being read as warrant or truth

domain: E • tier: Dr • status: *current* • coq: *closed* • parents: $EQ-015/M.02.v1$

$M_A[n] \neq \theta(E[n])$; $S_{\{A,t+1\}} \neq S_{\{A,t\}}$ is admissible and expected; $reachable(H)=1 \Rightarrow$
 $Acc(H|Q) \text{ high}; wholesome \neq true, unwholesome \neq false; a_{\{Q,t\}}(H_i) \text{ high} \Rightarrow W_t(H_i) \text{ high},$
 $m_t(H_i) \text{ high} \Rightarrow W_t(H_i) \text{ high}, Acc(H_i) \text{ high} \Rightarrow W_t(H_i) \text{ high}; Attraction \neq Warrant,$
 $Momentum \neq Truth; H_i \sim \{U_{\{D,t\}}\} H_j \Rightarrow H_i \sim \{U_{\{D,t+1\}}\} H_j; a_G \text{ up} \Rightarrow W_G(H) \text{ up}, m_G$
 $\text{up} \Rightarrow truth; Attraction \neq Accessibility \neq Warrant \neq Truth$

Occurrences (9): 22307148:(2), 22307148:(7), 22307148:(18), 22307148:(22), 22307148:(30)-(32), 22307148:(33), 22307148:(38), 22307148:(47), 22307148:(58)

EQ-015/E.16.v1 — Canonical symbol rename resolving a three-way kappa collision (meaning strength / semantic-route accessibility / live-policy accessibility)

domain: E • tier: *Definition* • status: *current* • coq: *definition* • parents: $EQ-015/M.08.v1$

$\gamma^{\mu}_n = \text{meaning strength}$, $\kappa^{\mathrm{sem}}_{\{A,t\}}(e \mid$
 $Q) = \text{semantic-route accessibility}$, $\lambda^{\mathrm{live}}_{\{A,t\}}(\pi \mid$
 $g) = \text{live-policy accessibility}$

Occurrences (1): 22550491:Eq. 43

EQ-015/H.01.v1 — Language as a bounded transport of a prior human state, not its container

domain: H • tier: *Definition* • status: *current* • coq: *definition* • parents: $EQ-015/M.01.v1$

$$H_t \xrightarrow{L_H} Q_t; Q_t \rightarrow AI_t \rightarrow Y_t \xrightarrow{R_H} E_{H,t}^{AI}; H_{t+1} = U_H(H_t, E_{H,t}^{AI}, \delta^{world}, X^{other})$$

Occurrences (4): 22410666:MBN-articulation, 22357744:EMG-21, 22357744:EMG-22, 22357744:EMG-24

EQ-015/H.02.v1 — pre-AI entry-state anchor tuple

domain: H • tier: *Definition* • status: *current* • coq: *definition* • parents: $EQ-015/M.01.v1$

$A0 = \langle P0, M0, U0, E0, F0, S0 \rangle$ (latest, DCP eq.12); supersedes $H0^* = \langle P0, M0, U0, E0, \Phi0, \kappa0 \rangle$ and
 $ENTRY_{\{H \rightarrow AI\}} = H0^* \wedge V0 \wedge C0 \wedge R^*$

Occurrences (5): 22331922:EF-01, 22331922:EF-02 (Repair 1), 22331922:EF-03 (Repair 1), 22481928:(12), 22481928:(13)

EQ-015/H.03.v1 — Pre-Prompt Human State Principle & Live-Field Update Loop

domain: H • tier: *Definition* • status: *current* • coq: *definition* • parents: $EQ-015/M.03.v1$

$H_t \xrightarrow{L_H} Q_t; Q_t \rightarrow AI_t \rightarrow Y_t \xrightarrow{R_H} E^{\{AI\}}_{\{H,t\}}; H_{t+1} = U_H(H_t, E^{\{AI\}}_{\{H,t\}},$
 $\delta^{\{world\}}, X^{\{other\}}); L_{\{A,t+1\}} \neq L_{\{A,t\}}$

Occurrences (5): 22357788:CBC-11, 22357788:CBC-12, 22357788:CBC-13, 22357788:CBC-14, 22550491:Eq. 46

EQ-015/H.04.v1 — AI response chain and re-entry as encounter

domain: H • tier: Definition • status: current • coq: definition • parents: EQ-015/M.02.v1

[print fallback: this entry's LaTeX transcription did not compile as standalone display math; showing the source ascii/latest text instead — the registry, JSON entry and docs site carry the LaTeX transcription unmodified]

$$Q_t \rightarrow AI_t \rightarrow Y_t \text{ -R}_H \rightarrow E^{AI}_{\{H,t\}}; H_{\{t+1\}} = U_H(H_t, E^{AI}_{\{H,t\}}, \delta^{\text{world}}, X^{\text{other}})$$

Occurrences (5): 22424434:(30), 22424434:(32), 22424434:(33), 22481928:(2), 22639311:Tunnel-v2.0:eq.10

EQ-015/H.05.v1 — finite dialogue state and reciprocal-lineage diagnostic

domain: H • tier: Definition • status: current • coq: definition • parents: EQ-015/M.02.v1

[print fallback: this entry's LaTeX transcription did not compile as standalone display math; showing the source ascii/latest text instead — the registry, JSON entry and docs site carry the LaTeX transcription unmodified]

$$Z_dlg[s,n+1] = F\#_dlg(Z_dlg[s,n], u_H[s,n], u_AI[s,n], c[s,n], T[s,n]); \chi_recip[s,n,L] = |D_recip[s,n,L]| / |\Sigma[s,n]|$$

Occurrences (7): 22456564:(2), 22331922:EF-13, 22331922:EF-14, 22331922:EF-15 (Repair 6), 22331922:EF-16, 22639311:Tunnel-v2.0:eq.11, 22550491:Eq. 32

EQ-015/H.06.v1 — Selective retention as a rank-bounded factorized update (Human LoRA) — split 2026-09-06, keeps only the rank-bounded retention mechanism

domain: H • tier: Definition • status: current • coq: closed • parents: EQ-015/M.03.v1

$$\Delta 0_n^{\text{fast}} = B_n A_n; g_n = \Gamma_n(s_n, r_n^{\text{err}}, q_n, v_n, c_n, p_n) \in \mathbb{Q} \cap [0,1]; \Delta 0_n^{\text{ret}} = g_n \cdot \Delta 0_n^{\text{fast}}; 0_H[n+1] = 0_H[n] + \Delta 0_n^{\text{ret}} + \epsilon_n; \text{finite-bottleneck postulate } 0 < m_n < d_n; \text{rank_Q}(B_n A_n) \leq m_n \text{ (proved)}$$

Occurrences (18): 21425420:(18), 21425420:(19), 21425420:(20), 21425420:(21), 21425420:(22), 21425420:(23)-(24), 21425420:(25), 21425420:(26), 21425420:(27), 21425420:(28), 21425420:(29), 21425420:(32) (+6 more)

EQ-015/H.07.v1 — live possibility nesting $\Pi^{\text{live}} \subseteq \Pi^{\text{feas}} \subseteq \Pi^{\text{phys}}$

domain: H • tier: Definition • status: current • coq: definition • parents: EQ-015/M.03.v1

$$\Pi_{A,t}^{\text{live}}(g) \subseteq \Pi_{A,t}^{\text{feas}}(g) \subseteq \Pi_t^{\text{phys}}(g)$$

Occurrences (4): 22456414:(1), 22424434:(21), 22424434:(43), 22481924:(32)

EQ-015/H.08.v1 — practical accessibility weight and operational live set

domain: H • tier: Definition • status: current • coq: definition • parents: EQ-015/M.02.v1

$$L_{\{A,t\}}(g) = \{(\pi, \lambda^{\text{live}}_{\{A,t\}}(\pi|g)) : \pi \in \Pi^{\text{feas}}_{\{A,t\}}(g)\}; \Pi^{\text{live}}_{\{A,t\}}(g) = \{\pi : \lambda^{\text{live}}_{\{A,t\}}(\pi|g) \geq \tau_{\text{live}}\}$$

Occurrences (2): 22424434:(22), 22424434:(23)

EQ-015/H.09.v1 — corrigible/effective agency potential over a witnessed set

domain: H • tier: Definition • status: current • coq: definition • parents: EQ-015/M.03.v1

$$p *_{A,g}(h, z; T, B, P) = \max_{\pi \in \Pi_A^{\text{wit}}(g; h, z, T, B)} Pr_P^{\pi}(Read_g \cap D_g \cap X_g \cap F_g)$$

Occurrences (1): 22424434:(27)

EQ-015/H.10.v1 — AI output is knowledge-like, not validated knowledge

domain: H • tier: Definition • status: current • coq: definition • parents: EQ-015/M.03.v1

$$AI(Q) = K_{\text{like}}, K_{\text{like}} \neq K_{\text{validated}}$$

Occurrences (8): 22456487:(2), 22456564:(3), 22331922:EF-04, 22339909:CTSA-02, 22481924:(4), 22424434:(31), 22636999:eq.2, 22639311:Tunnel-v2.0:eq.12

EQ-015/H.11.v1 — knowledge-like status as a state machine (not a fluency upgrade)

domain: H • tier: Dr • status: *current* • coq: *definition* • parents: *EQ-015/M.01.v1*

$K_like \rightarrow K_assumed$ (dangerous shortcut); repaired: $K_like \text{ -check-} \rightarrow K_checked \text{ -independent}$
 $support \rightarrow K_supported \text{ -warrant } V* \rightarrow K_validated|V*$

Occurrences (3): 22331922:EF-05, 22331922:EF-06 (Repair 2), 22639311:Tunnel-v2.0:eq.13

EQ-015/H.12.v1 — gain-side/tunnel-side directional functions and their difference

domain: H • tier: Dr • status: *current* • coq: *definition* • parents: *EQ-015/M.02.v1*

$G_s = g(k, d, v, p, r, 1-f, a | \Theta_s, \Pi_s)$; $T_s = h(c, f, b, o | \Theta_s, \Pi_s)$; $\Delta_s = G_s - T_s$; $\eta_s > 0, \Delta_s > 0 \Rightarrow \text{expansion}$; $\eta_s > 0, \Delta_s < 0 \Rightarrow \text{tunnel}$

Occurrences (6): 22331922:EF-20, 22331922:EF-21, 22331922:EF-22 (Repair 8), 22331922:EF-25, 22331922:EF-26 (Repair 10), 22550491:Eq. 39

EQ-015/H.13.v1 — revised ten-problem architecture (Epistemic Fusion v8.1)

domain: H • tier: Dr • status: *current* • coq: *definition* • parents: *EQ-015/M.03.v1*

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$H0* \rightarrow K_like \rightarrow D^{\text{eff}} \rightarrow R^{\text{eff}} \rightarrow H \leftrightarrow AI \rightarrow \chi_recip \rightarrow \eta \rightarrow (G-T | \Theta, \Pi) \rightarrow Y^{\text{return}} \rightarrow J*$

Occurrences (1): 22331922:EF-27

EQ-015/H.14.v1 — epistemic resistance quality vs its accessibility

domain: H • tier: Dr • status: *current* • coq: *closed* • parents: *EQ-015/M.03.v1*

$R_s^{\text{ep}} = \rho(I_s, V_s, Q_s)$; $U_s^{\text{R}} = u(C_s^{\text{v}}, T_s^{\text{v}}, A_s^{\text{v}})$; $R_s^{\text{ex}} = \psi(R_s^{\text{ep}}, U_s^{\text{R}})$; Resistance
quality $\neq$ resistance accessibility

Occurrences (4): 22331922:EF-08 (Repair 4), 22331922:EF-09 (Repair 4), 22331922:EF-10, 22639311:Tunnel-v2.0:eq.16

EQ-015/H.15.v1 — pre/post confidence-warrant calibration audit

domain: H • tier: Dr • status: *current* • coq: *definition* • parents: *EQ-015/M.02.v1*

$$K_s = (\kappa_0, \kappa_1, W_0, W_1); \text{Calibration error} \approx N^{-1} \sum_i (\kappa_i - y_i)^2$$

Occurrences (2): 22331922:EF-11 (Repair 5), 22331922:EF-12 (Repair 5)

EQ-015/H.16.v1 — Human Return audit tuple, six-field (CTSA v9.1)

domain: H • tier: *Definition* • status: *current* • coq: *definition* • parents: *EQ-015/M.01.v1*

$$H_{\text{return}} = \langle G_{CTSA}, L, M, P, W, \Delta_{\text{dir}} \rangle$$

Occurrences (5): 22339909:CTSA-04, 22424434:(38), 22339909:CTSA-01, 22639311:Tunnel-v2.0:eq.18, 22550491:Eq. 41

EQ-015/H.17.v1 — Human Return tuple, four-field $\langle C, T, S, A \rangle$

domain: H • tier: *Definition* • status: *current* • coq: *definition* • parents: *EQ-015/M.01.v1*

$$R_H^{\text{return}} = \langle C, T, S, A \rangle$$

Occurrences (3): 22481928:(25), 22498047:(24), 22636999:eq.26

EQ-015/H.18.v1 — Difference, Resistance, Agency constitutive necessities

domain: H • tier: Definition • status: current • coq: definition • parents: EQ-015/M.02.v1

$$D > 0, Resist > 0, A_H > 0$$

Occurrences (5): 22456487:unlabeled (§5.2, AVRH), 22456564:(5), 22331922:CN-1, 22424434:(35), 22639311:Tunnel-v2.0:eq.19

EQ-015/H.19.v1 — three-axis human-AI outcome vector (augmentation, synergy, return)

domain: H • tier: Definition • status: current • coq: definition • parents: EQ-015/M.01.v1

$$J_{*s} = (AUG_s, SYN_s, RET_s); AUG_s = P_s^{joint} - P_s^H; SYN_s = P_s^{joint} - \max(P_s^H, P^A I_s)$$

Occurrences (3): 22331922:EF-23, 22331922:EF-24 (Repair 9), 22550491:Eq. 40

EQ-015/H.20.v1 — CTSA v9.1's Open hypotheses on incremental validity, dissociation, and regulation

domain: H • tier: Open • status: current • coq: definition • parents: EQ-015/M.02.v1

$$H1 - H6[Open], §12$$

Occurrences (6): 22339909:H1, 22339909:H2, 22339909:H3, 22339909:H4, 22339909:H5, 22339909:H6

EQ-015/H.21.v1 — deployment/triage policy across DCP-Lite, DCP-Standard, DCP-Critical

domain: H • tier: Definition • status: current • coq: definition • parents: EQ-015/M.01.v1

$\pi^{\text{deploy}} = f(\text{Stakes}, \text{LearningNeed}, \text{Irreversibility}, \text{DependencyRisk}, \text{UserSkill}); \text{State} \rightarrow \text{Challenge} \rightarrow \text{Check} \rightarrow \text{Own}(\text{Lite}); \text{Topic Entry} \rightarrow \text{Triage} \rightarrow \text{Minimum Sufficient Dialogue} \rightarrow \text{Escalate} \rightarrow \text{Periodic Return}(\text{Critical}); \pi^{\text{dialogue}} = f(\text{TopicMode}, \text{Goal}, \text{Skill}, \text{Stakes}, \text{Domain}, \text{LearningNeed}, \text{Reversibility})$

Occurrences (5): 22481928:(8), 22481928:(9), 22481928:AFP, 22481928:(10), 22481928:(35)

EQ-015/H.22.v1 — Reset vs Removal

domain: H • tier: Definition • status: current • coq: definition • parents: EQ-015/M.02.v1

Reset : fresh framing/*session*/source route; *Removal* : absence of decisive *AI assistance*

Occurrences (2): 22481928:(23), 22481928:(24)

EQ-015/H.23.v1 — session-level Human Return conversion vector and sampling frequency

domain: H • tier: Definition • status: current • coq: definition • parents: EQ-015/M.01.v1

$\Delta H_s = \langle \Delta C_s, \Delta T_s, \Delta S_s, \Delta A_s, \Delta A^{\text{corr}}_{\{H,s\}}, \Delta \Lambda^{\text{live}}_{\{H,s\}} \rangle; F^{\text{return}} = f(\text{Criticality}, \text{LearningNeed}, \text{FailureCost}, \text{DependencyRisk})$

Occurrences (3): 22481928:(26), 22481928:(28), 22639311:Tunnel-v2.0:eq.20

EQ-015/H.24.v1 — Human Return closed through action and world feedback

domain: H • tier: Definition • status: current • coq: definition • parents: EQ-015/M.02.v1

$I_s \rightarrow a_s \rightarrow \delta^{\text{world}}_{\{s+1\}} \rightarrow H_{\{s+1,0\}}; \text{Live Problem} \rightarrow \text{Question} \rightarrow \text{Dialogue} \rightarrow \text{Human Return} \rightarrow \text{Action} \rightarrow \text{World Feedback} \rightarrow \text{Revision or New Problem}$

Occurrences (9): 22481928:(29), 22481928:(30), 22481928:WCP, 22481928:(44), 22498047:(27), 22639311:Tunnel-v2.0:eq.21, 22550491:Eq. 50, 22550491:Eq. 51, 22550491:Eq. 65

EQ-015/H.25.v1 — DCP’s remaining Open propositions (conversion, deployment, agenda)

domain: H • tier: $Open$ • status: $current$ • coq: $definition$ • parents: $EQ-015/M.02.v1$

$$DCPp, DEPp, AgP[Open]$$

Occurrences (3): 22481928:DCPp, 22481928:DEPp, 22481928:AgP

EQ-015/H.26.v1 — DCP’s ten primary empirical hypotheses

domain: H • tier: $Open$ • status: $current$ • coq: $definition$ • parents: $EQ-015/M.02.v1$

$$H1 - H10, §15.2$$

Occurrences (10): 22481928:H1, 22481928:H2, 22481928:H3, 22481928:H4, 22481928:H5, 22481928:H6, 22481928:H7, 22481928:H8, 22481928:H9, 22481928:H10

EQ-015/H.27.v1 — four decisive practical questions closing the paper

domain: H • tier: Dr • status: $current$ • coq: $definition$ • parents: $EQ-015/M.02.v1$

What unresolved difference deserves my attention? / How much AI/friction does this task require? / What remains with me after AI is removed? / What does the world return when I act?

Occurrences (1): 22481928:(46)

EQ-015/H.28.v1 — interaction efficiency as a qualitative construct

domain: H • tier: $Definition$ • status: $current$ • coq: $definition$ • parents: $EQ-015/M.02.v1$

$$\mathcal{E} := f(ConstraintPrecision, ContextRecall, SourceTraceability, ErrorRepair)$$

Occurrences (1): 18517054:(1)

EQ-015/H.29.v1 — context-access amplification, bounded expansion, and architectural doorway hypotheses

domain: H • tier: $Open$ • status: $current$ • coq: $definition$ • parents: $EQ-015/M.02.v1$

H1: AI increases access to/reorganization of human interpretive contexts, not event access; H3: bounded interpretive expansion under high $\mathcal{E}$ and preserved S; H4: AI as architectural doorway to interpretive diversity

Occurrences (3): 18517054:Finding-1 (H1), 18517054:Finding-2 (H3), 18517054:Finding-3 (H4)

EQ-015/H.30.v1 — candidate HCA domain state tuple

domain: H • tier: $Definition$ • status: $current$ • coq: $definition$ • parents: $EQ-015/M.01.v1$

$$Z_{\{HCA, i, n\}} = \langle P^{live}_{\{i, n\}}, K^{life}_{\{i, n\}}, B^{\bar{}}_{\{i, n\}}, C^{cand}_{\{i, n\}}, C^{live}_{\{i, n\}}, h_{\{i, n\}}, R^{return}_{\{H, i, n\}}, \Omega^{real}_{\{i, n\}} \rangle$$

Occurrences (1): 22498047:(6)

EQ-015/H.31.v1 — Life-Capital Context Vector

domain: H • tier: $Definition$ • status: $current$ • coq: $definition$ • parents: $EQ-015/M.01.v1$

$$K_{i, n}^{life} = \langle E^{econ}, F^{base}, L^{lang}, D^{digital}, T^{disc}, H^{health}, M^{mob}, N^{mentor}, C^{cred} \rangle$$

Occurrences (1): 22498047:(8)

EQ-015/H.32.v1 — adaptive scaffolding fading rule

domain: H • tier: *Open* • status: *current* • coq: *definition* • parents: *EQ-015/M.02.v1*

Stable Unaided Return $\uparrow \Rightarrow$ h $\hat{}$ decisive $\downarrow$; Attempt $\rightarrow$ Minimal Sufficient Scaffold $\rightarrow$ Feedb
ack $\rightarrow$ Reattempt $\rightarrow$ Fading $\rightarrow$ Unaided Execution $\rightarrow$ Novel Transfer; R $\hat{}$ return_H $\uparrow \Rightarrow$ h $\hat{}$ decisive $\downarrow$

Occurrences (3): 22498047:(22), 22498047:(23), 22498047:(26)

EQ-015/H.33.v1 — child and adult governance minimum bundles

domain: H • tier: *Definition* • status: *current* • coq: *definition* • parents: *EQ-015/M.02.v1*

Adult: Purpose+Transparency+Refusal+Revision+Data Minimization; Child: Child
Assent+Adult Oversight+Privacy+Safety+No Opaque Persuasion(+Age-Appropriate Design)

Occurrences (2): 22498047:(34), 22498047:(35)

EQ-015/H.34.v1 — Before Meaning, Before Choice minimal empirical predictions H1-H6

domain: H • tier: *Open* • status: *current* • coq: *definition* • parents: *EQ-015/M.02.v1*

$$H1 - H6[Open], \S 17.2$$

Occurrences (6): 22424434:H1, 22424434:H2, 22424434:H3, 22424434:H4, 22424434:H5, 22424434:H6

EQ-015/H.35.v1 — Digital Yonisomanasikāra minimal empirical predictions

domain: H • tier: *Open* • status: *current* • coq: *definition* • parents: *EQ-015/M.02.v1*

$P1$: framing residual; $P2$: iteration without integration is insufficient

Occurrences (2): 22456564:P1, 22456564:P2

EQ-015/H.36.v1 — Six rival hypotheses (M0-M5) for the mechanism behind the Human-LoRA update O_H

domain: H • tier: *Open* • status: *current* • coq: *wrapped_related* • parents: *EQ-015/M.03.v1*

M0: context-only state change; M1: sparse but not low-rank update; M2: low-rank
factorized update (= CAN-054's own $\Delta O_n = B_n A_n$); M3: regularized full-rank update; M4:
finite graph rewiring; M5: hybrid fast trace plus slow update

Occurrences (6): 21425420:(43), 21425420:(44), 21425420:(45), 21425420:(46), 21425420:(47), 21425420:(48)

EQ-015/H.37.v1 — Well-being state vector W_n and multiplicative maladaptive-cost index $C_{mal}[n]$

domain: H • tier: *Definition* • status: *current* • coq: *wrapped_related* • parents: *EQ-015/M.03.v1*

$W_n = (F_n, A_n^{\text{agency}}, M_n^{\text{meaning}}, R_n^{\text{relation}}, C_n^{\text{competence}}, Q_n^{\text{repair}})$; $C_{mal}[n] = \text{rig}(a_n) \cdot \text{gen}(a_n) \cdot \text{mis}(a_n) \cdot \text{loss}(a_n)$

Occurrences (2): 21425420:(36), 21425420:(37)

EQ-015/H.38.v1 — Effort v0.3 Eq.28: NEW DERIVATION

domain: H • tier: *Dr* • status: *current* • coq: *open_prop* • parents: *EQ-015/M.03.v1*

$$Y_{\{n+1\}^{\text{world}}} \rightarrow O_Q \rightarrow \mu_{\{n+1\}^Q} \rightarrow E_{\{n+1\}^Q} \rightarrow \text{Retain}_Q \rightarrow A_{\{n+1\}^Q}$$

Occurrences (1): 22622206:eq.28

EQ-015/H.39.v1 — Effort v0.3 Eq.29: NEW DOMAIN DEFINITION

domain: *H* • tier: *Definition* • status: *current* • coq: *definition* • parents: *EQ-015/M.03.v1*

$$\text{RB_Q}(n) = 1 \text{ \textit{iff} } A_{\{n+1\}}^{\sim Q} \not\sim_Q A_n^{\sim Q}$$

Occurrences (1): 22622206:eq.29

EQ-015/H.40.v1 — Recursive epistemic network state

domain: *H* • tier: *Definition* • status: *current* • coq: *definition* • parents: *EQ-015/H.05.v1*, *A.8/M.02.v1*

$$\mathcal{G}_t = (\mathcal{A}_t, \mathcal{E}_t, \mathcal{P}_t)$$

Occurrences (1): 22639311:Tunnel-v2.1:eq.1

EQ-015/H.41.v1 — Recursive Epistemic Reflection (RER) minimal cycle

domain: *H* • tier: *Definition* • status: *current* • coq: *definition* • parents: *EQ-015/H.05.v1*

$$a_i \rightarrow a_j \rightarrow a_i$$

Occurrences (1): 22639311:Tunnel-v2.1:eq.4

EQ-015/H.42.v1 — RET diagnostic state vector

domain: *H* • tier: *Dr* • status: *current* • coq: *open_prop* • parents: *EQ-015/H.05.v1*, *weld/H.08.v1*, *EQ-015/H.14.v1*, *A.8/M.02.v1*, *EQ-015/H.10.v1*, *EQ-015/H.11.v1*

$$\mathfrak{R}_t = \langle \chi_{\{\text{recip}\}_t}^{\mathcal{G}}, D_{\{\mathcal{G}, t\}^{\{\text{eff}\}}}, R_{\{\mathcal{G}, t\}^{\{\text{ex}\}}}, P_{\{\mathcal{G}, t\}^{\{\text{ind}\}}}, K_{\{\mathcal{G}, t\}^{\{\text{status}\}}}, \kappa_{\{\mathcal{G}, t\}} \rangle$$

Occurrences (1): 22639311:Tunnel-v2.1:eq.8

EQ-015/H.43.v1 — Tunnel-contraction regime

domain: *H* • tier: *Definition* • status: *current* • coq: *definition* • parents: *EQ-015/H.05.v1*, *weld/H.08.v1*, *EQ-015/H.14.v1*

$$\Delta \chi_{\{\text{recip}\}_t}^{\mathcal{G}} > 0, \Delta \kappa_{\{\mathcal{G}\}_t} > 0, \Delta D_{\{\mathcal{G}\}_t}^{\{\text{eff}\}} < 0, \Delta R_{\{\mathcal{G}\}_t}^{\{\text{ex}\}} \leq 0, \Delta P_{\{\mathcal{G}\}_t}^{\{\text{ind}\}} \leq 0$$

Occurrences (1): 22639311:Tunnel-v2.1:eq.10

EQ-015/H.44.v1 — Recursive Epistemic Tunnel (RET)

domain: *H* • tier: *Definition* • status: *current* • coq: *definition* • parents: *EQ-015/H.05.v1*, *weld/H.08.v1*, *EQ-015/H.14.v1*

$$\text{RET} \neq \text{Falsehood}; \quad \text{RET} \neq \text{Consensus}$$

Occurrences (1): 22639311:Tunnel-v2.1:eq.11

EQ-015/H.45.v1 — Collective Epistemic Hallucination (CEH) status-inflation chain

domain: *H* • tier: *Definition* • status: *current* • coq: *definition* • parents: *EQ-015/H.10.v1*, *EQ-015/H.11.v1*

$$K_{\{\text{like}\}} \overset{\text{forget}}{\longrightarrow} K_{\{\text{assumed}\}} \overset{\text{recursion}}{\longrightarrow} K_{\{\text{reinforced}\}} \overset{\text{misattribution}}{\longrightarrow} K_{\{\text{collectively warranted}\text{-looking}\}}$$

Occurrences (1): 22639311:Tunnel-v2.1:eq.12

EQ-015/H.46.v1 — Multi-model consensus readout

domain: H • tier: Dr • status: *current* • coq: *open_prop* • parents: *EQ-015/H.10.v1*

$$\text{Consensus}_{\{\mathcal{M}\}}(c) = \frac{1}{\sum_{j=1}^k \mathbf{1}_{[M_j]}} \text{ accepts } c$$

Occurrences (1): 22639311:Tunnel-v2.1:eq.13

EQ-015/H.47.v1 — Consensus does not entail validation

domain: H • tier: Dr • status: *current* • coq: *open_prop* • parents: *EQ-015/H.10.v1*, *EQ-015/H.11.v1*

$$\text{Consensus}_{\{\mathcal{M}\}}(c) \uparrow \not\rightarrow \text{Validation}(c) \uparrow$$

Occurrences (1): 22639311:Tunnel-v2.1:eq.14

EQ-015/H.48.v1 — RET network stop rule

domain: H • tier: Dr • status: *current* • coq: *open_prop* • parents: *EQ-015/H.05.v1*, *weld/H.08.v1*, *EQ-015/H.14.v1*, *A.8/M.02.v1*

$$\begin{aligned} & \Delta \chi_{\{\mathrm{recip}\}}^{\{\mathcal{G}\}} > 0, \Delta \kappa_{\{\mathcal{G}\}} > 0 \ \backslash \ \text{while} \ \backslash \\ & \Delta D_{\{\mathcal{G}\}}^{\{\text{eff}\}} \leq 0, \Delta R_{\{\mathcal{G}\}}^{\{\text{ex}\}} \leq 0, \Delta P_{\{\mathcal{G}\}}^{\{\text{ind}\}} \leq 0 \end{aligned}$$

Occurrences (1): 22639311:Tunnel-v2.1:eq.21

EQ-015/H.49.v1 — Session reset does not entail epistemic reset

domain: H • tier: Dr • status: *current* • coq: *open_prop* • parents: *EQ-015/H.05.v1*, *A.8/M.02.v1*

$$\text{SessionReset} \not\rightarrow \text{EpistemicReset}$$

Occurrences (1): 22639311:Tunnel-v2.1:eq.22

EQ-015/M.01.v1 — retained root state $S_n = (G_n, \Lambda_n, T_n)$

domain: M • tier: *Definition* • status: *current* • coq: *closed* • parents: *EQ-015*

$$S_n = (G_n, \Lambda_n, T_n)$$

Occurrences (3): 22498047:(1), 22424434:(3), 22639311:Tunnel-v2.0:eq.1

EQ-015/M.02.v1 — root transition $S_{n+1} = F(S_n, u_n, c_n, T_n)$

domain: M • tier: *Definition* • status: *current* • coq: *closed* • parents: *EQ-015*

$$S_{n+1} = F(S_n, u_n, c_n, T_n)$$

Occurrences (3): 22498047:(2), 22424434:(4), 22639311:Tunnel-v2.0:eq.2

EQ-015/M.03.v1 — Constitutional ordering of epistemic terms

domain: M • tier: Dr • status: *current* • coq: *closed* • parents: *EQ-015*

$$\begin{aligned} & \text{retention} \rightarrow \text{structure} \rightarrow \text{translation} \rightarrow \text{readout} \rightarrow \text{meaning} \rightarrow \text{experience} \rightarrow \text{memory} \rightarrow \\ & \text{belief} \rightarrow \text{claim} \rightarrow \text{checking} \rightarrow \text{status} \rightarrow \text{public report (forbidden order: starting from a} \\ & \text{familiar name and retrospectively presenting the result as knowledge)} \end{aligned}$$

Occurrences (2): 21529456:(9), 22550491:Eq. 44

EQ-015/M.04.v1 — readout-native admission order for semantic categories

domain: M • tier: Definition • status: current • coq: definition • parents: EQ-015/M.03.v1

Retention→Structure→Translation→Readout→Meaning→Report (compressed); Retention→Structure→Candidate State→Sufficiency→Quotient→Domain Dynamics→Readout→Meaning→Experience→Memory→Knowledge→Checked Report (MEMK extension)

Occurrences (4): 22424434:(1), 22424434:(2), 22424434:(49), 22424434:(50)

EQ-015/M.05.v1 — the Decorrelated Verification Protocol

domain: M • tier: finite_diagnostic • status: current • coq: definition • parents: EQ-015/M.02.v1

DVP*=ReduceCorrelatedError+ExposeResidualDependence+BottomOutWherePossible;
Disagreement=>Resolve v Declare; NoHumanAvailable=>DVP→K1→HumanAcquisition

Occurrences (3): 22163849:(27), 22163849:(30), 22163849:(105)

EQ-015/M.06.v1 — the Legitimacy Circulation Loop

domain: M • tier: Definition • status: current • coq: closed • parents: EQ-015/M.02.v1

[print fallback: this entry's LaTeX transcription did not compile as standalone display math; showing the source ascii/latest text instead — the registry, JSON entry and docs site carry the LaTeX transcription unmodified]

HorizontalGeneration→EpistemicFriction→VerticalStrengthening→ResourceReturn→HorizontalGrowth

Occurrences (1): 22163849:(64)

EQ-015/M.07.v1 — the integrated architecture and its endpoint constructs

domain: M • tier: Definition • status: current • coq: definition • parents: EQ-015/M.03.v1

Phenomenon→AIExploration→DVP→HumanMastery→Integrity→K1→{Global,Local Friction}→K2→Revision→K3; EpistemicPosition=RecognizableProblem+RepeatHumanNodes+CitableAssets+CorrectionHistory; CrediblePath=EarlyTimestamp+ExplicitProvisionality+Rapid Revision+ExternalFriction+EventualCertification

Occurrences (3): 22163849:(89), 22163849:(94), 22163849:(95)

EQ-015/M.08.v1 — Rhythm, recent-path momentum, and history-shaped accessibility are parallel descriptors, not a derivation chain

domain: M • tier: Definition • status: current • coq: definition • parents: EQ-015/M.02.v1

$T_n = \{(x_k, t_k, w_k)\}_{k \leq n}$, $Rhythm_n = \Omega(T_n, B_n)$; $m_{t+1}(e) = \rho m_t(e) + 1[e_t = e]$, $0 \leq \rho < 1$;
 $\kappa^{\{sem\}}_{t+1}(e|Q) \propto \kappa^{\{sem\}}_t(e|Q) \cdot \exp(\beta a_{\{Q, t\}}(e) + \mu m_t(e) - \nu c_{\{A, t, Q\}}(e) + \xi_t(e))$;
corrected non-derivation: $Rhythm_t, m_t, a_t, c_t \rightarrow \kappa^{\{sem\}}_{t+1}$

Occurrences (6): 22357744:EMG-13, 22357744:EMG-14, 22307148:(19), 22307148:(20), 22307148:(21), 22550491:Eq. 15

EQ-015/M.09.v1 — Experience arises from coupled brain-body and mind-core dynamics, neither alone

domain: M • tier: Definition • status: current • coq: definition • parents: EQ-015/M.03.v1

$$B[t+1] = F(B[t]) + C_H(H[t]); H[t+1] = G(H[t]) + C_B(B[t]); E[t] = R(B[t], H[t])$$

Occurrences (3): 19640361:(9), 19640361:(10), 19640361:(11)

EQ-015/M.10.v1 — protocol usability-burden vector

domain: M • tier: Definition • status: current • coq: closed • parents: EQ-015/M.02.v1

$B^{use} = \langle \text{Time}, \text{CognitiveLoad}, \text{VerificationCost}, \text{Interruption}, \text{LiteracyDemand} \rangle$; Epistemically optimal $\neq$ Behaviorally adoptable

Occurrences (3): 22481928:(37), 22481928:(38), 22636999:eq.30

EQ-015/M.11.v1 — The paper’s own four-part taxonomy of ways an epistemic basis can be augmented

domain: M • tier: Definition • status: current • coq: closed • parents: EQ-015/M.03.v1

Def-2 Access augmentation ($Y1 \rightarrow (Y1, Y2)$); Def-3 Contrast/relevance operation ($X \rightarrow X' \subseteq X$);
Def-4 Inferential commitment (a rule changes what can be inferred without new access);
Def-5 Decision-policy augmentation (a loss/threshold/rule maps epistemic state to action)

Occurrences (4): 22301318:Def-2, 22301318:Def-3, 22301318:Def-4, 22301318:Def-5

EQ-015/M.12.v1 — $M_A[n] \neq \theta(E)$

domain: M • tier: untagged • status: current • coq: closed • parents: EQ-015/M.02.v1

$$M_A[n] \neq \theta(E)$$

Occurrences (1): 22163849:(66)

EQ-015/M.13.v1 — Discrete Retention Lagrangian (single-field ring)

domain: M • tier: finite_diagnostic • status: current • coq: wrapped_related • parents: EQ-015

$$\mathbb{L}^n[X] = (M/2dt) \cdot \Delta X^T (G \otimes I) \Delta X + (D/2) \cdot X^T \{ \Omega \otimes I \} \Delta X - (dt/2) \cdot X^T (G \otimes (K \cdot L_R + k_2 I)) X$$

Occurrences (1): logic.md.DRL_Lagrangian_restatement

EQ-015/M.14.v1 — Discrete Retention Hamiltonian (conserved companion of the DRL Lagrangian)

domain: M • tier: finite_diagnostic • status: current • coq: wrapped_related • parents: EQ-015/M.13.v1

$$H = M \cdot v \cdot \Phi^T v \Psi + K \cdot \Phi^T L_R \Psi + k_2 \cdot \Phi^T \Psi \text{ (conserved quantity)}$$

EQ-015/M.15.v1 — General-N nonlinear Discrete Retention Hamiltonian

domain: M • tier: finite_diagnostic • status: current • coq: wrapped_related • parents: EQ-015/M.14.v1

$$H_{nl} = \sum_i M_i v \Phi_i v \Psi_i + K \Phi^T L_w \Psi + \Psi^T \nabla V(\Phi) - J^T \Psi$$

EQ-015/M.16.v1 — River-summary narrative closure: experience, the changed reader, possibility, choice, action, and the re-read world

domain: M • tier: Definition • status: current • coq: definition • parents: EQ-015/M.02.v1

$\text{\texttt{Experience changes the reader; the changed reader changes what can become possible; what becomes possible changes choice; choice changes action and the world; the changed world is read again.}}$

Occurrences (1): 22550491:Eq. 45

EQ-015/P.01.v1 — Planck–Einstein relation ‘ $E=\hbar\omega$ ’

domain: P • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Planck–Einstein relation $E=\hbar\omega$

Occurrences (1): ‘formal/URCF_RD_All.v’, ‘InfoSchrodinger’

EQ-015/P.02.v1 — Schrödinger equation / eigenvalue problem

domain: P • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Schrödinger equation / eigenvalue problem

Occurrences (1): ‘InfoSchrodinger’

EQ-015/P.03.v1 — Special relativity / Lorentz invariance

domain: P • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Special relativity / Lorentz invariance

Occurrences (1): ‘InfoLorentzInvariance.v’

EQ-015/P.04.v1 — General relativity field equations ‘ $G_{\mu\nu}+\Lambda g_{\mu\nu}=8\pi GT_{\mu\nu}$ ’

domain: P • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

General relativity field equations $G_{\mu\nu}+\Lambda g_{\mu\nu}=8\pi GT_{\mu\nu}$

Occurrences (1): ‘InfoQuantumGravityRootBridge_attempt.v’, ‘mass_note.tex’ §holo

EQ-015/P.05.v1 — Schwarzschild metric

domain: P • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Schwarzschild metric

Occurrences (1): ‘InfoAnalysisLift.v’

EQ-015/P.06.v1 — Regge–Wheeler equation

domain: P • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Regge–Wheeler equation

Occurrences (2): ‘InfoQuantumGravityRootBridge_attempt.v’, ‘InfoQuantumGravityRootBridge_attempt.v’, ‘scripts/verify_quantum_gravity.py’

EQ-015/P.07.v1 — Electroweak mixing: ‘ $A=\sin\theta W\cdot W^3+\cos\theta W\cdot B$ ’, ‘ $Z=\cos\theta W\cdot W^3-\sin\theta W\cdot B$ ’, ‘ $m_W^2=g^2v^2/4$ ’, ‘ $m_Z^2=(v^2/4)(g^2+g'^2)$ ’, ‘ $\sin^2\theta W=g'^2/(g^2+g'^2)$ ’

domain: P • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Electroweak mixing: $A=\sin\theta W\cdot W^3+\cos\theta W\cdot B$, $Z=\cos\theta W\cdot W^3-\sin\theta W\cdot B$, $m_W^2=g^2v^2/4$, $m_Z^2=(v^2/4)(g^2+g'^2)$, $\sin^2\theta W=g'^2/(g^2+g'^2)$

Occurrences (1): ‘domains/standard_model/electroweak_decoder_v0_3.py’, ‘formal/InfoElectroweakNullDirection_attempt.v’

EQ-015/P.08.v1 — Fermi coupling ‘ G_F ’ and ‘ $v=(\sqrt{2} G_F)^{-1/2}$ ’ (Higgs vev)

domain: P • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Fermi coupling G_F and $v=(\sqrt{2} G_F)^{-1/2}$ (Higgs vev)

Occurrences (1): ‘domains/standard_model/electroweak_decoder_v0_3.py’

EQ-015/P.09.v1 — Gauge-anomaly cancellation conditions ‘ $[\mathrm{SU}(3)]^2\mathrm{U}(1)$ ’, ‘ $[\mathrm{SU}(2)]^2\mathrm{U}(1)$ ’, ‘ $[\mathrm{grav}]^2\mathrm{U}(1)$ ’, ‘ $[\mathrm{U}(1)]^3$ ’ fixing the SM hypercharges

domain: P • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Gauge-anomaly cancellation conditions $[\mathrm{SU}(3)]^2\mathrm{U}(1)$, $[\mathrm{SU}(2)]^2\mathrm{U}(1)$, $[\mathrm{grav}]^2\mathrm{U}(1)$, $[\mathrm{U}(1)]^3$ fixing the SM hypercharges

Occurrences (1): ‘domains/standard_model/sm_discovery_pipeline_v0_4.py’; ‘formal/InfoHyperchargeAnomalyClosure_attempt.v’

EQ-015/P.10.v1 — Peter–Weyl theorem / character convolution (class-function convolution multiplies representation coefficients)

domain: P • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Peter–Weyl theorem / character convolution (class-function convolution multiplies representation coefficients)

Occurrences (1): ‘domains/standard_model/triality_spectral_flow_v0_6.py’; ‘formal/InfoTrialitySpectralFlow_attempt.v’

EQ-015/P.11.v1 — Wilson-loop area-law criterion for confinement ‘ $-\log\langle W(C) \rangle = \sigma \cdot A(C)$ ’, and the center ‘ Z_N ’ role in confinement|K. Wilson (1974); center-symmetry / center-vortex work (‘t Hooft 1978; Greensite)|1974; 1978+

domain: P • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Wilson-loop area-law criterion for confinement $-\log\langle W(C) \rangle = \sigma \cdot A(C)$, and the center (Z_N) role in confinement|K. Wilson (1974); center-symmetry / center-vortex work (‘t Hooft 1978; Greensite)|1974; 1978+

Occurrences (1): *u_p-1*

EQ-015/P.12.v1 — Ginsparg–Wilson relation ‘ $\{\Gamma, D\} = a D \Gamma D$ ’ (lattice chirality without naïve anticommutation) and the overlap/sign-operator realization ‘ $D = (1/\bar{a})(I - V)$ ’, ‘ $V = \Gamma \epsilon$ ’

domain: P • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Ginsparg–Wilson relation $\{\Gamma, D\} = a D \Gamma D$ (lattice chirality without naïve anticommutation) and the overlap/sign-operator realization $D = (1/\bar{a})(I - V)$, $V = \Gamma \epsilon$

Occurrences (1): ‘domains/standard_model/tape_kinetic_operator_v1_8.py’; ‘formal/InfoTapeKineticGW_attempt.v’

EQ-015/P.13.v1 — Nielsen–Ninomiya fermion-doubling no-go (locality + translation invariance + Hermiticity + exact naïve chiral anticommutation $\Rightarrow$ ‘ 2^d ’ species)

domain: P • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Nielsen–Ninomiya fermion-doubling no-go (locality + translation invariance + Hermiticity + exact naïve chiral anticommutation $\Rightarrow$ 2^d species)

Occurrences (1): ‘domains/standard_model/tape_kinetic_operator_v1_8.py’; ‘domains/standard_model/relation_channel_dimension_v1_9.py’

EQ-015/P.14.v1 — Clifford algebra ‘ $\{A_\mu, A_\nu\} = 2\delta_{\mu\nu} I$ ’ / irreducible Dirac-matrix dimension (2×2 Pauli, mutually-orthogonal directions in $\mathbb{R}^3$)

domain: P • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Clifford algebra $\{A_\mu, A_\nu\} = 2\delta_{\mu\nu} I$ / irreducible Dirac-matrix dimension (2×2 Pauli, mutually-orthogonal directions in $\mathbb{R}^3$)

Occurrences (1): ‘domains/standard_model/relation_channel_dimension_v1_9.py’; ‘formal/InfoDimensionFourClosure_attempt.v’

EQ-015/P.15.v1 — Unitary ' $N \times N$ ' quark-mixing-matrix physical-parameter counting (' N^2 ' real params, ' $2N-1$ ' removable by phase redefinition $\Rightarrow$ ' $(N-1)^2$ ' physical, split into ' $N(N-1)/2$ ' mixing angles + ' $(N-1)(N-2)/2$ ' CP-violating phases; nonzero phase count requires ' $N \geq 3$ ')

domain: P • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Unitary $N \times N$ quark-mixing-matrix physical-parameter counting (N^2 real params, $2N-1$ removable by phase redefinition $\Rightarrow$ $(N-1)^2$ physical, split into $N(N-1)/2$ mixing angles + $(N-1)(N-2)/2$ CP-violating phases; nonzero phase count requires $N \geq 3$)

Occurrences (1): 'domains/standard_model/item2_exploration/item2_family_index_v2_fit.py'

EQ-015/P.16.v1 — Gatto–Sartori–Tonin relation ' $\sin \theta_C \approx \sqrt{(m_d/m_s)}$ ' (Cabibbo angle from the down-type quark mass ratio)|R. Gatto, G. Sartori, M. Tonin|1968

domain: P • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Gatto-Sartori-Tonin relation $\sin \theta_C \approx \sqrt{(m_d/m_s)}$ (Cabibbo angle from the down-type quark mass ratio)|R. Gatto, G. Sartori, M. Tonin|1968

Occurrences (1): *Vus*

EQ-015/P.17.v1 — Group-averaging / Reynolds ("twirl") projection onto the invariant subspace ' $\Pi(X) = (1/|G|) \sum_{R \in G} R^T X R$ ', here for the signed-permutation (hyperoctahedral) group ' $B_4 = (\mathbb{Z}_2)^4 \rtimes S_4 \Rightarrow \Pi_4(X) = (\text{Tr } X/4)I$ '

domain: P • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Group-averaging / Reynolds ("twirl") projection onto the invariant subspace $\Pi(X) = (1/|G|) \sum_{R \in G} R^T X R$, here for the signed-permutation (hyperoctahedral) group $B_4 = (\mathbb{Z}_2)^4 \rtimes S_4 \Rightarrow \Pi_4(X) = (\text{Tr } X/4)I$

Occurrences (1): 'domains/standard_model/isotropic_fixed_point_v1_10.py'; 'formal/InfoIsotropicFixedPoint_attempt.v'

EQ-015/P.18.v1 — Osterwalder–Schrader reflection positivity via the Gram / transfer-matrix construction (a half-slab amplitude ' B ' gives a positive kernel ' $K = B^\dagger B \geq 0$ ', i.e. reflection positivity by construction; hidden-midpoint reconstruction)

domain: P • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Osterwalder-Schrader reflection positivity via the Gram / transfer-matrix construction (a half-slab amplitude B gives a positive kernel $K = B^\dagger B \geq 0$, i.e. reflection positivity by construction; hidden-midpoint reconstruction)

Occurrences (1): 'domains/standard_model/frame_mixing_from_action_v1_11.py'; 'formal/InfoFrameMixingAction_attempt.v' (also used in 'universal_rp_slab_v1_4.py')

EQ-015/P.19.v1 — Perron–Frobenius / doubly-stochastic consensus (a primitive doubly-stochastic ' P ' has a simple eigenvalue 1 with ' $P|_{1^\perp}$ ' spectral radius ' < 1 ' $\Rightarrow$ convergence to the mean)

domain: P • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Perron-Frobenius / doubly-stochastic consensus (a primitive doubly-stochastic P has a simple eigenvalue 1 with $P|_{1^\perp}$ spectral radius $< 1 \Rightarrow$ convergence to the mean)

Occurrences (1): 'domains/standard_model/isotropic_fixed_point_v1_10.py'; 'formal/InfoIsotropicFixedPoint_attempt.v'

EQ-015/P.20.v1 — Quasinormal-mode late-time power-law ("Price") tail

domain: P • tier: *finite_diagnostic* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Quasinormal-mode late-time power-law ("Price") tail

Occurrences (1): 'causal-quantum-gravity/scripts/price_tail.py'

EQ-015/P.21.v1 — Late-time tail dependence on initial data

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Late-time tail dependence on initial data

Occurrences (1): ‘causal-quantum-gravity/scripts/price_tail.py’ (T1 control, momentum-type vs. time-symmetric data)

EQ-015/P.22.v1 — Late-time tails, further numerical study

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Late-time tails, further numerical study

Occurrences (1): ‘causal-quantum-gravity/scripts/price_tail.py’

EQ-015/P.23.v1 — Quasinormal mode spectrum (continued-fraction method)

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Quasinormal mode spectrum (continued-fraction method)

Occurrences (1): ‘scripts/verify_quantum_gravity_root_bridge.py’

EQ-015/P.24.v1 — Raychaudhuri equation

domain: P • tier: Open • status: current • coq: wrapped_related • parents: EQ-015

Raychaudhuri equation

Occurrences (1): ‘InfoCutGrowth_attempt.v’

EQ-015/P.25.v1 — de Broglie’s internal clock

domain: P • tier: Dr • status: current • coq: wrapped_related • parents: EQ-015

de Broglie’s internal clock

Occurrences (1): ‘mass_note.tex’ §mass (“Mass and the internal clock”)

EQ-015/P.26.v1 — Dirac equation

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Dirac equation

Occurrences (1): ‘mass_note.tex’ §mass

EQ-015/P.27.v1 — Zitterbewegung (relativistic)

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Zitterbewegung (relativistic)

Occurrences (1): ‘mass_note.tex’ §mass, Reading 4

EQ-015/P.28.v1 — Zitterbewegung interpretation

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Zitterbewegung interpretation

Occurrences (1): ‘mass_note.tex’ §mass

EQ-015/P.29.v1 — Higgs mechanism

domain: P • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Higgs mechanism

Occurrences (1): ‘mass_note.tex’ §mass (scope-boundary note)

EQ-015/P.30.v1 — Electroweak Higgs mechanism / one-doublet EWSB: residual ‘ $Q=T_3+Y$ ’ stabilizer, ‘ $m_W^2=g^2v^2/4$ ’, ‘ $m_Z^2=(g^2+g'^2)v^2/4$ ’, ‘ $m_W=m_Z \cos\theta$ ’, and tree-level custodial ‘ $\rho=1$ ’

domain: P • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Electroweak Higgs mechanism / one-doublet EWSB: residual $Q=T_3+Y$ stabilizer, $m_W^2=g^2v^2/4$, $m_Z^2=(g^2+g'^2)v^2/4$, $m_W=m_Z \cos\theta$, and tree-level custodial $\rho=1$

Occurrences (1): ‘domains/standard_model/order_higgs_closure_v1_12.py’; ‘formal/InfoOrderHiggsClosure_attempt.v’

EQ-015/P.31.v1 — Elitzur’s theorem (a local gauge symmetry cannot break spontaneously; only gauge-invariant order parameters like ‘ $\langle H^\dagger H \rangle$ ’ are physical)

domain: P • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Elitzur’s theorem (a local gauge symmetry cannot break spontaneously; only gauge-invariant order parameters like $\langle H^\dagger H \rangle$ are physical)

Occurrences (1): ‘domains/standard_model/order_vacuum_selection_v1_13.py’; ‘formal/InfoOrderVacuumSelection_attempt.v’

EQ-015/P.32.v1 — Robb’s causal-order axiomatization

domain: P • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Robb’s causal-order axiomatization

Occurrences (1): ‘mass_note.tex’ §lit

EQ-015/P.33.v1 — Zeeman: causality determines Lorentz group

domain: P • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Zeeman: causality determines Lorentz group

Occurrences (1): ‘mass_note.tex’ §lit

EQ-015/P.34.v1 — Malament: causal order fixes topology

domain: P • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Malament: causal order fixes topology

Occurrences (1): ‘mass_note.tex’ §lit

EQ-015/P.35.v1 — One-loop non-abelian gauge β -function with matter, ‘ $b_a = (11/3)C_2(G_a) - (2/3)\Sigma T_a(R_{\text{fermion}}) - (1/3)\Sigma T_a(R_{\text{scalar}})$ ’ (the ‘11/3, 2/3, 1/3’ kinematic weights are the gauge-boson+ghost, Weyl-fermion, and complex-scalar one-loop vacuum-polarization contributions)

domain: P • tier: *finite_diagnostic* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

One-loop non-abelian gauge β -function with matter, $b_a = (11/3)C_2(G_a) - (2/3)\Sigma T_a(R_{\text{fermion}}) - (1/3)\Sigma T_a(R_{\text{scalar}})$ (the 11/3, 2/3, 1/3 kinematic weights are the gauge-boson+ghost, Weyl-fermion, and complex-scalar one-loop vacuum-polarization contributions)

Occurrences (1): ‘domains/standard_model/item25_exploration/beta_function_coefficients_v1.py’

EQ-015/P.36.v1 — GUT hypercharge normalization ‘ $T_1(R) = (3/5)Y^2$ ’ (SU(5) embedding factor needed to compare U(1)_Y on the same Dynkin-index footing as SU(2)/SU(3))

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

GUT hypercharge normalization $T_{\{1\}}(R) = (3/5)Y^{\{2\}}$ (SU(5) embedding factor needed to compare U(1)_Y on the same Dynkin-index footing as SU(2)/SU(3))

Occurrences (1): ‘domains/standard_model/item25_exploration/beta_function_coefficients_v1.py’

EQ-015/P.37.v1 — Schutz’s independent Minkowski axioms

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Schutz’s independent Minkowski axioms

Occurrences (1): ‘mass_note.tex’ §lit

EQ-015/P.38.v1 — Regge calculus

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Regge calculus

Occurrences (1): ‘SUPPLEMENT.md’ §novelty, ‘mass_note.tex’ §lit

EQ-015/P.39.v1 — Causal set theory

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Causal set theory

Occurrences (1): ‘SUPPLEMENT.md’ §novelty

EQ-015/P.40.v1 — Wolfram hypergraph rewriting project

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Wolfram hypergraph rewriting project

Occurrences (1): ‘SUPPLEMENT.md’ §novelty, ‘mass_note.tex’ §lit

EQ-015/P.41.v1 — Tree-level Higgs quartic self-coupling relation ‘ $\lambda = m_H^2/(2 v^2)$ ’ (from ‘ $V(H) = -\mu^2 H^\dagger H + \lambda (H^\dagger H)^2$ ’ minimized at ‘ v ’)

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Tree-level Higgs quartic self-coupling relation $\lambda = m_H^2/(2 v^2)$ (from $V(H) = -\mu^2 H^\dagger H + \lambda (H^\dagger H)^2$ minimized at v)

Occurrences (1): ‘domains/standard_model/item18_exploration/higgs_quartic_DOCUMENTED_NON_RESULT_v1.py’ (proven circular as a derivation of ‘ m_H ’); ‘domains/standard_model/item18_exploration/higgs_quartic_fit_calibrated_v1.py’ (registered as an explicit ‘fit_calibrated’ value, computed FROM measured ‘ $m_H=125.25$ ’ GeV, per ‘DRIFT_CONTRACT.json’ DEV-SM-004)

EQ-015/P.42.v1 — QCD running coupling ‘ $\alpha_s(M_Z)$ ’ (asymptotic freedom / one-loop-and-beyond running of the SU(3) gauge coupling)

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

QCD running coupling $\alpha_s(M_Z)$ (asymptotic freedom / one-loop-and-beyond running of the SU(3) gauge coupling)

Occurrences (1): ‘domains/standard_model/fit_calibrated_registry.py’ (‘ALPHA_S_MZ’); ‘domains/standard_model/item_gauge_couplings_

EQ-015/P.43.v1 — QCD vacuum angle ‘ θ_{QCD} ’ and the strong CP problem (CP-violating term ‘ $\theta (g^2/32 \pi^2) G \tilde{G}$ ’ in the QCD Lagrangian; neutron EDM as the experimental probe)|G. ’t Hooft (theta-vacuum structure, 1976); neutron EDM experimental collaborations (bound)|1976 (theta-vacuum); ongoing (EDM bound)

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

QCD vacuum angle θ_{QCD} and the strong CP problem (CP-violating term $\theta (g^2/32 \pi^2) G \tilde{G}$ in the QCD Lagrangian; neutron EDM as the experimental probe)|G. ’t Hooft (theta-vacuum structure, 1976); neutron EDM experimental collaborations (bound)|1976 (theta-vacuum); ongoing (EDM bound)

Occurrences (1): θ_{QCD} \

EQ-015/P.44.v1 — Neutrino oscillation mass-squared splittings ‘ Δm^2_{21} ’, ‘ Δm^2_{31} ’ (PMNS-driven flavor oscillation probability depends only on these differences, not absolute masses)

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Neutrino oscillation mass-squared splittings Δm^2_{21} , Δm^2_{31} (PMNS-driven flavor oscillation probability depends only on these differences, not absolute masses)

Occurrences (1): ‘domains/standard_model/fit_calibrated_registry.py’ (‘DELTA_M2_21_EV2’, ‘DELTA_M2_31_EV2’); ‘domains/standard_model/item23_exploration/neutrino_mass_fit_calibrated_v1.py’

EQ-015/P.45.v1 — Clausius relation ‘ $\delta Q = T \delta S$ ’

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Clausius relation $\delta Q = T \delta S$

Occurrences (1): ‘InfoStrainTensorBridge_attempt.v’ (‘clausius_form’), ‘mass_note.tex’ §holo

EQ-015/P.46.v1 — Bekenstein–Hawking area-entropy law

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Bekenstein–Hawking area-entropy law

Occurrences (1): ‘InfoStrainTensorBridge_attempt.v’ Ansatz H, ‘causal-quantum-gravity/paper/main.tex’ §arealaw

EQ-015/P.47.v1 — Unruh temperature

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Unruh temperature

Occurrences (1): ‘mass_note.tex’ §holo (Tolman–Unruh–Hawking direction, unchecked)

EQ-015/P.48.v1 — Jacobson’s thermodynamic derivation of Einstein’s equation

domain: P • tier: Ax • status: current • coq: wrapped_related • parents: EQ-015

Jacobson’s thermodynamic derivation of Einstein’s equation

Occurrences (1): ‘InfoStrainTensorBridge_attempt.v’ header, ‘SUPPLEMENT.md’ §5

EQ-015/P.49.v1 — Verlinde’s entropic gravity

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Verlinde’s entropic gravity

Occurrences (1): ‘SUPPLEMENT.md’ §novelty, ‘mass_note.tex’ §lit

EQ-015/P.50.v1 — Sakharov induced gravity

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

`Sakharov induced gravity`

Occurrences (1): ‘mass_note.tex’ §lit

EQ-015/P.51.v1 — Susskind holographic principle

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

`Susskind holographic principle`

Occurrences (1): ‘mass_note.tex’ §lit

EQ-015/P.52.v1 — Maldacena AdS/CFT

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

`Maldacena AdS/CFT`

Occurrences (1): ‘mass_note.tex’ §lit

EQ-015/P.53.v1 — Ryu–Takayanagi entanglement=area

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

`Ryu–Takayanagi entanglement=area`

Occurrences (1): ‘mass_note.tex’ §lit

EQ-015/P.54.v1 — Van Raamsdonk: spacetime from entanglement

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

`Van Raamsdonk: spacetime from entanglement`

Occurrences (1): ‘mass_note.tex’ §lit

EQ-015/P.55.v1 — Quasinormal-mode analytic representation

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

`Quasinormal-mode analytic representation`

Occurrences (1): ‘SUPPLEMENT.md’ §6

EQ-015/P.56.v1 — QNM review (literature target values)

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

`QNM review (literature target values)`

Occurrences (1): ‘SUPPLEMENT.md’ §6

EQ-015/P.57.v1 — Hyperboloidal-slicing method

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

`Hyperboloidal-slicing method`

Occurrences (1): ‘SUPPLEMENT.md’ §5.2 (attempt 6-7)

EQ-015/P.58.v1 — Perfectly Matched Layer absorbing boundary

domain: *P* • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Perfectly Matched Layer absorbing boundary

Occurrences (1): ‘SUPPLEMENT.md’ §5.2 (attempt 8), ‘scripts/verify_quantum_gravity_root_bridge.py’

EQ-015/P.59.v1 — Kepler’s third law ‘ $T^2 = 4 \pi^2 a^3 / (G M)$ ’

domain: *P* • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Kepler’s third law $T^2 = 4 \pi^2 a^3 / (G M)$

Occurrences (1): ‘engine/formulas.py’ (‘kepler_third_law’)

EQ-015/P.60.v1 — Bernoulli’s equation (steady incompressible streamline energy balance)

domain: *P* • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Bernoulli’s equation (steady incompressible streamline energy balance)

Occurrences (1): ‘engine/formulas.py’ (‘bernoulli_pressure’)

EQ-015/P.61.v1 — Continuity equation ‘ $A_1 v_1 = A_2 v_2$ ’ (incompressible 1D duct form)

domain: *P* • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Continuity equation $A_1 v_1 = A_2 v_2$ (incompressible 1D duct form)

Occurrences (1): ‘engine/formulas.py’ (‘continuity_flow’)

EQ-015/P.62.v1 — 2D truss method of joints (joint-equilibrium linear system, ‘ $m+r=2j$ ’ determinacy count)

domain: *P* • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

2D truss method of joints (joint-equilibrium linear system, $m+r=2j$ determinacy count)

Occurrences (1): ‘engine/systems.py’ (‘solve_truss’)

EQ-015/P.63.v1 — Euler–Bernoulli beam theory

domain: *P* • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Euler–Bernoulli beam theory

Occurrences (1): ‘engine/systems.py’ (‘solve_beam_fem’)

EQ-015/P.64.v1 — Cubic-Hermite beam FEM element (4-DOF stiffness matrix, consistent UDL fixed-end forces; exact nodal values for point/uniform loads)

domain: *P* • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Cubic-Hermite beam FEM element (4-DOF stiffness matrix, consistent UDL fixed-end forces; exact nodal values for point/uniform loads)

Occurrences (1): ‘engine/systems.py’ (‘_beam_element_k’)

EQ-015/P.65.v1 — 1D axial-bar FEM element (‘ $k_e = EA/L$ ’ two-node element)

domain: *P* • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

1D axial-bar FEM element ($k_e = EA/L$ two-node element)

Occurrences (1): ‘engine/systems.py’ (‘solve_bar_fem’)

EQ-015/P.66.v1 — Modal analysis closed forms for uniform spring-mass chains, ‘ $\omega_n^2 = (2k/m)(1-\cos(n\pi/(N+1)))$ ’, and the generalized eigenproblem ‘ $K\phi = \omega^2 M\phi$ ’

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Modal analysis closed forms for uniform spring-mass chains, $\omega_n^2 = (2k/m)(1-\cos(n\pi/(N+1)))$, and the generalized eigenproblem $K\phi = \omega^2 M\phi$

Occurrences (1): ‘engine/systems.py’ (‘solve_modal’), ‘tests/test_systems.py’ (chain closed form as test oracle)

EQ-015/P.67.v1 — 1D Ising model + transfer-matrix exact solution (‘ $f = -kT \ln \lambda_+$ ’, ‘ $\xi = -1/\ln \tanh(\beta J)$ ’ at ‘ $h=0$ ’)

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

1D Ising model + transfer-matrix exact solution ($f = -kT \ln \lambda_+$, $\xi = -1/\ln \tanh(\beta J)$ at $h=0$)

Occurrences (1): ‘engine/systems.py’ (‘solve_ising_1d’)

EQ-015/P.68.v1 — 2D square-lattice Ising critical temperature ‘ $T_c = 2J/(k_B \ln(1+\sqrt{2}))$ ’

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

2D square-lattice Ising critical temperature $T_c = 2J/(k_B \ln(1+\sqrt{2}))$

Occurrences (1): ‘engine/systems.py’ (‘ising_2d_onsager_Tc’)

EQ-015/P.69.v1 — 1D Ising real-space decimation RG, ‘ $K' = (1/2) \ln \cosh(2K)$ ’ (b=2 block-spin)

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

1D Ising real-space decimation RG, $K' = (1/2) \ln \cosh(2K)$ (b=2 block-spin)

Occurrences (1): ‘engine/systems.py’ (‘solve_ising_rg’)

EQ-015/P.70.v1 — SUVAT kinematics equations (‘ $v=u+at$ ’, ‘ $s=ut+\frac{1}{2}at^2$ ’, ‘ $v^2=u^2+2as$ ’)

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

SUVAT kinematics equations ($v=u+at$, $s=ut+\frac{1}{2}at^2$, $v^2=u^2+2as$)

Occurrences (1): ‘engine/formulas.py’ (‘suvat_final_velocity’, ‘suvat_displacement’, ‘suvat_velocity_from_displacement’, ‘suvat_average_velocity’)

EQ-015/P.71.v1 — Hooke’s law ‘ $F=kx$ ’ and spring elastic PE ‘ $\frac{1}{2}kx^2$ ’

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Hooke’s law $F=kx$ and spring elastic PE $\frac{1}{2}kx^2$

Occurrences (1): ‘engine/formulas.py’ (‘hooke_spring_force’, ‘spring_elastic_pe’)

EQ-015/P.72.v1 — Simple-pendulum and mass-spring SHM period (‘ $T=2\pi\sqrt{L/g}$ ’, ‘ $T=2\pi\sqrt{m/k}$ ’ and closed-form SHM motion (‘ $x=A \cos(\omega t+\phi)$ ’)

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Simple-pendulum and mass-spring SHM period ($T=2\pi\sqrt{L/g}$, $T=2\pi\sqrt{m/k}$) and closed-form SHM motion ($x=A \cos(\omega t+\phi)$)

Occurrences (1): ‘engine/formulas.py’ (‘shm_period_pendulum’, ‘shm_period_spring’, ‘shm_position’, ‘shm_max_velocity’, ‘shm_max_acceleration’)

EQ-015/P.73.v1 — Snell’s law of refraction ‘ $n_1 \sin \theta_1 = n_2 \sin \theta_2$ ’ and total-internal-reflection critical angle

domain: *P* • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Snell’s law of refraction $n_1 \sin \theta_1 = n_2 \sin \theta_2$ and total-internal-reflection critical angle

Occurrences (1): ‘engine/formulas.py’ (‘snell_refraction_angle’, ‘critical_angle’)

EQ-015/P.74.v1 — Young’s double-slit interference fringe spacing ‘ $\Delta y = \lambda L / d$ ’

domain: *P* • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Young’s double-slit interference fringe spacing $\Delta y = \lambda L / d$

Occurrences (1): ‘engine/formulas.py’ (‘double_slit_fringe_spacing’)

EQ-015/P.75.v1 — Diffraction grating equation ‘ $d \sin \theta = m \lambda$ ’

domain: *P* • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Diffraction grating equation $d \sin \theta = m \lambda$

Occurrences (1): ‘engine/formulas.py’ (‘diffraction_grating_angle’)

EQ-015/P.76.v1 — Malus’s law ‘ $I = I_0 \cos^2 \theta$ ’

domain: *P* • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Malus’s law $I = I_0 \cos^2 \theta$

Occurrences (1): ‘engine/formulas.py’ (‘malus_law’)

EQ-015/P.77.v1 — Radioactive decay law ‘ $N = N_0 e^{-\lambda t}$ ’ and half-life ‘ $t_{1/2} = \ln 2 / \lambda$ ’

domain: *P* • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Radioactive decay law $N = N_0 e^{-\lambda t}$ and half-life $t_{1/2} = \ln 2 / \lambda$

Occurrences (1): ‘engine/formulas.py’ (‘radioactive_decay_N’, ‘half_life_from_decay_constant’, ‘decay_constant_from_half_life’, ‘activity’)

EQ-015/P.78.v1 — Bohr model hydrogen energy levels ‘ $E_n = -13.6056931 / n^2$ eV’ (full-precision CODATA 2018 Rydberg energy)

domain: *P* • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Bohr model hydrogen energy levels $E_n = -13.6056931 / n^2$ eV (full-precision CODATA 2018 Rydberg energy)

Occurrences (1): ‘engine/formulas.py’ (‘bohr_energy_level’, ‘bohr_transition_wavelength’)

EQ-015/P.79.v1 — Planck-Einstein photon energy relation ‘ $E = hf = hc / \lambda$ ’

domain: *P* • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Planck-Einstein photon energy relation $E = hf = hc / \lambda$

Occurrences (1): ‘engine/formulas.py’ (‘photon_energy_freq’, ‘photon_energy_wavelength’)

EQ-015/P.80.v1 — Mass-energy equivalence ‘ $E = mc^2$ ’

domain: *P* • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Mass-energy equivalence $E = mc^2$

Occurrences (1): ‘engine/formulas.py’ (‘rest_mass_energy’)

EQ-015/P.81.v1 — Fourier’s law of heat conduction ‘ $Q/t=kA\Delta T/L$ ’

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Fourier’s law of heat conduction $Q/t=kA\Delta T/L$

Occurrences (1): ‘engine/formulas.py’ (‘conduction_heat_rate’)

EQ-015/P.82.v1 — Newton’s law of cooling (convective form) ‘ $Q/t=hA\Delta T$ ’

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Newton’s law of cooling (convective form) $Q/t=hA\Delta T$

Occurrences (1): ‘engine/formulas.py’ (‘convection_heat_rate’)

EQ-015/P.83.v1 — Stefan-Boltzmann law ‘ $P=\epsilon\sigma AT^4$ ’

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Stefan-Boltzmann law $P=\epsilon\sigma AT^4$

Occurrences (1): ‘engine/formulas.py’ (‘radiation_heat_rate’)

EQ-015/P.84.v1 — Carnot heat-engine efficiency ‘ $\eta=1-T_c/T_h$ ’

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Carnot heat-engine efficiency $\eta=1-T_c/T_h$

Occurrences (1): ‘engine/formulas.py’ (‘carnot_efficiency’)

EQ-015/P.85.v1 — Linear thermal expansion ‘ $\Delta L=\alpha L_0\Delta T$ ’

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Linear thermal expansion $\Delta L=\alpha L_0\Delta T$

Occurrences (1): ‘engine/formulas.py’ (‘thermal_expansion’)

EQ-015/P.86.v1 — Force on a current-carrying wire ‘ $F=BIL$ ’

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Force on a current-carrying wire $F=BIL$

Occurrences (1): ‘engine/formulas.py’ (‘force_on_wire’)

EQ-015/P.87.v1 — Solenoid magnetic field ‘ $B=\mu_0 nI$ ’ and straight-wire field ‘ $B=\mu_0 I/(2\pi r)$ ’ (Ampère’s law special cases)

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Solenoid magnetic field $B=\mu_0 nI$ and straight-wire field $B=\mu_0 I/(2\pi r)$ (Ampère’s law special cases)

Occurrences (1): ‘engine/formulas.py’ (‘solenoid_field’, ‘straight_wire_field’)

EQ-015/P.88.v1 — Faraday’s law of induction ‘ $\mathcal{E}=-N\Delta\Phi/\Delta t$ ’ and the motional-EMF special case ‘ $\mathcal{E}=BLv$ ’

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Faraday’s law of induction $\mathcal{E}=-N\Delta\Phi/\Delta t$ and the motional-EMF special case $\mathcal{E}=BLv$

Occurrences (1): ‘engine/formulas.py’ (‘faraday_emf’, ‘motional_emf’)

EQ-015/P.89.v1 — Coulomb’s law potential of a point charge ‘ $V=kq/r$ ’, uniform-field relation ‘ $E=V/d$ ’, and capacitor relations (‘ $Q=CV$ ’, ‘ $C=\epsilon_0 A/d$ ’, ‘ $E=\frac{1}{2}CV^2$ ’, series/parallel capacitance)

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Coulomb’s law potential of a point charge $V=kq/r$, uniform-field relation $E=V/d$, and capacitor relations ($Q=CV$, $C=\epsilon_0 A/d$, $E=\frac{1}{2}CV^2$, series/parallel capacitance)

Occurrences (1): ‘engine/formulas.py’ (‘electric_potential_point_charge’, ‘parallel_plate_field’, ‘capacitor_charge’, ‘parallel_plate_capacitance’, ‘capacitor_energy’, ‘series_capacitance’, ‘parallel_capacitance’)

EQ-015/P.90.v1 — Real-source terminal voltage with internal resistance ‘ $V=\epsilon-Ir$ ’

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Real-source terminal voltage with internal resistance $V=\epsilon-Ir$

Occurrences (1): ‘engine/formulas.py’ (‘terminal_voltage_internal_resistance’)

EQ-015/P.91.v1 — Rotational dynamics: ‘ $\tau=I\alpha$ ’, ‘ $L=I\omega$ ’, ‘ $KE=\frac{1}{2}I\omega^2$ ’, and standard moment-of-inertia forms (rod, disc, sphere, hoop)

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Rotational dynamics: $\tau=I\alpha$, $L=I\omega$, $KE=\frac{1}{2}I\omega^2$, and standard moment-of-inertia forms (rod, disc, sphere, hoop)

Occurrences (1): ‘engine/formulas.py’ (‘torque_inertia’, ‘angular_momentum’, ‘rotational_ke’, ‘moi_rod_center’, ‘moi_disc’, ‘moi_sphere_solid’, ‘moi_hoop’, ‘rolling_no_slip_velocity’)

EQ-015/P.92.v1 — Coulomb friction model ‘ $f=\mu N$ ’, incline force components, Atwood machine acceleration, Pascal’s principle hydraulic press

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Coulomb friction model $f=\mu N$, incline force components, Atwood machine acceleration, Pascal’s principle hydraulic press

Occurrences (1): ‘engine/formulas.py’ (‘friction_force’, ‘incline_gravity_component’, ‘incline_normal_force’, ‘atwood_acceleration’, ‘pascal_hydraulic_press’)

EQ-015/P.93.v1 — Torsion of circular shafts: shear stress ‘ $\tau=Tr/J$ ’, angle of twist ‘ $\theta=TL/GJ$ ’, polar moment of a solid/hollow circular section

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Torsion of circular shafts: shear stress $\tau=Tr/J$, angle of twist $\theta=TL/GJ$, polar moment of a solid/hollow circular section

Occurrences (1): ‘engine/formulas.py’ (‘shear_stress_torsion’, ‘twist_angle_torsion’, ‘polar_moment_solid_shaft’, ‘polar_moment_hollow_shaft’)

EQ-015/P.94.v1 — Fast Fourier Transform (Cooley–Tukey algorithm; used here via ‘numpy.fft.rfft’)

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Fast Fourier Transform (Cooley–Tukey algorithm; used here via `numpy.fft.rfft`)

Occurrences (1): ‘engine/systems.py’ (‘signal_spectrum’)

EQ-015/P.95.v1 — Periodogram / power spectral density estimation (one-sided PSD scaling used here follows the periodogram lineage; Hann window per J. von Hann, popularized by Blackman & Tukey)

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Periodogram / power spectral density estimation (one-sided PSD scaling used here follows the periodogram lineage; Hann window per J. von Hann, popularized by Blackman & Tukey)

Occurrences (1): 'engine/systems.py' ('signal_spectrum')

EQ-015/P.96.v1 — Total harmonic distortion (THD, ratio of harmonic to fundamental amplitude)

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Total harmonic distortion (THD, ratio of harmonic to fundamental amplitude)

Occurrences (1): 'engine/systems.py' ('signal_spectrum')

EQ-015/P.97.v1 — Fokker–Planck equation (and its overdamped/Smoluchowski stationary limit ' $p(x) \propto \exp(-V(x)/D)$ ')

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Fokker–Planck equation (and its overdamped/Smoluchowski stationary limit $p(x) \propto \exp(-V(x)/D)$)

Occurrences (1): 'engine/systems.py' ('fokker_planck_1d')

EQ-015/P.98.v1 — Boltzmann distribution ' $p \propto \exp(-E/k_B T)$ '

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Boltzmann distribution $p \propto \exp(-E/k_B T)$

Occurrences (1): 'engine/systems.py' ('fokker_planck_1d')

EQ-015/P.99.v1 — Hydrogen fine-structure energy shift ' $dE_{fs} = -(Ry \alpha^2/n^4)(n/(j+1/2) - 3/4)$ ' (relativistic kinetic + spin-orbit, combined first-order result)

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Hydrogen fine-structure energy shift $dE_{fs} = -(Ry \alpha^2/n^4)(n/(j+1/2) - 3/4)$
(relativistic kinetic + spin-orbit, combined first-order result)

Occurrences (1): 'engine/formulas.py' ('hydrogen_fine_structure_shift', PhD curriculum audit 2026-07-10)

EQ-015/P.100.v1 — Normal Zeeman effect ' $dE = m_l \mu_B B$ '

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Normal Zeeman effect $dE = m_l \mu_B B$

Occurrences (1): 'engine/formulas.py' ('zeeman_splitting_normal')

EQ-015/P.101.v1 — Rutherford differential scattering cross section ' $d\sigma/d\Omega = (Z_1 Z_2 k_e e^2/4E)^2 / \sin^4(\theta/2)$ '

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Rutherford differential scattering cross section $d\sigma/d\Omega = (Z_1 Z_2 k_e e^2/4E)^2 / \sin^4(\theta/2)$

Occurrences (1): 'engine/formulas.py' ('rutherford_cross_section')

EQ-015/P.102.v1 — Fermi-Dirac distribution

domain: *P* • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Fermi-Dirac distribution

Occurrences (1): 'engine/formulas.py' ('fermi_dirac_occupation')

EQ-015/P.103.v1 — Bose-Einstein distribution

domain: *P* • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Bose-Einstein distribution

Occurrences (1): 'engine/formulas.py' ('bose_einstein_occupation')

EQ-015/P.104.v1 — Quantum harmonic oscillator canonical partition function ' $Z = 1/(2 \sinh(\hbar \omega/2 k_B T))$ '

domain: *P* • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Quantum harmonic oscillator canonical partition function $Z = 1/(2 \sinh(\hbar \omega/2 k_B T))$

Occurrences (1): 'engine/formulas.py' ('qho_partition_function')

EQ-015/P.105.v1 — Planck's law of blackbody spectral radiance ' $B(\lambda, T)$ '

domain: *P* • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Planck's law of blackbody spectral radiance $B(\lambda, T)$

Occurrences (1): 'engine/formulas.py' ('planck_spectral_radiance')

EQ-015/P.106.v1 — Rectangular waveguide TE_{mn} cutoff frequency

domain: *P* • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Rectangular waveguide TE_{mn} cutoff frequency

Occurrences (1): 'engine/formulas.py' ('waveguide_te_cutoff_frequency')

EQ-015/P.107.v1 — Oscillating electric dipole radiated power ' $P = p_0^2 \omega^4/(12 \pi \epsilon_0 c^3)$ '

domain: *P* • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Oscillating electric dipole radiated power $P = p_0^2 \omega^4/(12 \pi \epsilon_0 c^3)$

Occurrences (1): 'engine/formulas.py' ('dipole_radiation_power')

EQ-015/P.108.v1 — Relativistic Doppler shift (radial) ' $f_{\text{obs}} = f_{\text{source}} \sqrt{(1-\beta)/(1+\beta)}$ '

domain: *P* • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Relativistic Doppler shift (radial) $f_{\text{obs}} = f_{\text{source}} \sqrt{(1-\beta)/(1+\beta)}$

Occurrences (1): 'engine/formulas.py' ('relativistic_doppler_shift')

EQ-015/P.109.v1 — Vis-viva equation ' $v^2 = GM(2/r - 1/a)$ '

domain: *P* • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Vis-viva equation $v^2 = GM(2/r - 1/a)$

Occurrences (1): 'engine/formulas.py' ('vis_viva_orbital_speed')

EQ-015/P.110.v1 — General-relativistic perihelion/apsidal precession ‘ $d_{\phi} = 6 \pi G M / (c^2 a (1-e^2))$ ’

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

General-relativistic perihelion/apsidal precession $d_{\phi} = 6 \pi G M / (c^2 a (1-e^2))$

Occurrences (1): ‘engine/formulas.py’ (‘gr_perihelion_precession’)

EQ-015/P.111.v1 — Schwarzschild radius ‘ $r_s = 2GM/c^2$ ’

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Schwarzschild radius $r_s = 2GM/c^2$

Occurrences (1): ‘engine/formulas.py’ (‘schwarzschild_radius’)

EQ-015/P.112.v1 — Schwarzschild static-observer gravitational time dilation ‘ $d_{\tau}/dt = \sqrt{1-r_s/r}$ ’

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Schwarzschild static-observer gravitational time dilation $d_{\tau}/dt = \sqrt{1-r_s/r}$

Occurrences (1): ‘engine/formulas.py’ (‘gravitational_time_dilation’)

EQ-015/P.113.v1 — Debye model heat capacity, low-temperature ‘ T^3 ’ limit ‘ $C_V = (12 \pi^4/5) N k_B (T/\theta_D)^3$ ’

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Debye model heat capacity, low-temperature T^3 limit $C_V = (12 \pi^4/5) N k_B (T/\theta_D)^3$

Occurrences (1): ‘engine/formulas.py’ (‘debye_heat_capacity_low_T’)

EQ-015/P.114.v1 — Resonance quality factor from damping ratio ‘ $Q = 1/(2 \zeta)$ ’

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Resonance quality factor from damping ratio $Q = 1/(2 \zeta)$

Occurrences (1): ‘engine/formulas.py’ (‘quality_factor_from_zeta’)

EQ-015/P.115.v1 — Ziegler-Nichols closed-loop PID tuning rule (‘ $K_p=0.6 K_u$ ’, ‘ $T_i=0.5 P_u$ ’, ‘ $T_d=0.125 P_u$ ’)

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Ziegler-Nichols closed-loop PID tuning rule ($K_p=0.6 K_u$, $T_i=0.5 P_u$, $T_d=0.125 P_u$)

Occurrences (1): ‘engine/formulas.py’ (‘ziegler_nichols_pid’)

EQ-015/P.116.v1 — Hagen-Poiseuille equation ‘ $Q = \pi r^4 dP/(8 \mu L)$ ’

domain: P • tier: untagged • status: current • coq: wrapped_related • parents: EQ-015

Hagen-Poiseuille equation $Q = \pi r^4 dP/(8 \mu L)$

Occurrences (1): ‘engine/formulas.py’ (‘hagen_poiseuille_flow’)

EQ-015/P.117.v1 — Blasius laminar flat-plate boundary layer similarity solution (‘delta_99 $\approx 5.0x/\sqrt{\text{Re}_x}$ ’, ‘Cf_x=0.664/ $\sqrt{\text{Re}_x}$ ’)

domain: P • tier: *untagged* • status: *current* • coq: *wrapped_related* • parents: *EQ-015*

Blasius laminar flat-plate boundary layer similarity solution (delta_99\appr
ox5.0x/sqrt(Re_x), Cf_x=0.664/sqrt(Re_x))

Occurrences (1): ‘engine/formulas.py’ (‘blasius_boundary_layer’)

EQ-015/S.01.v1 — Regime Structure & Update Laws (translation operator T_R , interaction operator I_R)

domain: S • tier: Ax • status: *split* • coq: *definition* • parents: *EQ-015/M.02.v1*

N_4 split (founder ruling 22:20, 2026-09-06): CE-05 ‘ $R := (T_R, I_R)$ ’ (Axiom III, regime structure), CE-06/CE-07 (the two update laws for M and A), and CE-08 ‘ $\text{Etic}(A;t) := \dots$ ’ (a derived definition) are four distinct formal objects, not restatements of one equation.

$$R := (T_R, I_R); M(t'+\Delta t') = T_R(M(t')); A(t'+\Delta t') \subseteq I_R(A(t'), M(t')); \text{Etic}(A;t') := \\ \exists R_A(t') \in R_{\text{adm}}(t'): A \subseteq M(t'), M(t'+\Delta t') = T_{\{R_A\}}(M(t')), A(t'+\Delta t') \subseteq I_{\{R_A\}}(A(t'), M(t'))$$

Occurrences (4): 18444260:CE-05, 18444260:CE-06, 18444260:CE-07, 18444260:CE-08

EQ-015/S.02.v1 — Ethical Load, Causal Memory & Spectral Margin — the Causal Ethics Inequality (locked core definition)

domain: S • tier: *Definition* • status: *split* • coq: *definition* • parents: *EQ-015/M.03.v1*

N_4 split (founder ruling 22:20, 2026-09-06): CE-09 ($V \geq 0$ Lyapunov potential), CE-10 (causal-memory positivity), CE-11 (spectral-margin positivity), CE-12 (Ethical(A) biconditional, ‘locked’), CE-13 (V -dot-plus notation), CE-19 (Eth_ind biconditional) are six distinct definitions/conditions.

$$\text{Ethical}(A) \Leftrightarrow \exists \text{Choice}(A \rightarrow R_A): d/dt' V_{\{A, R_A\}}(M(t')) \leq 0 \wedge \tau_{c'}(R_A) > 0 \wedge \Delta_{\text{spec}}(R_A) > 0$$

[author-labelled final, locked, will not be weakened or reinterpreted later]

Occurrences (6): 18444260:CE-09, 18444260:CE-10, 18444260:CE-11, 18444260:CE-12, 18444260:CE-13, 18444260:CE-19

EQ-015/S.03.v1 — Collective/Individual Conflict, Spectral Survival, and Structural Injustice

domain: S • tier: *Definition* • status: *split* • coq: *definition* • parents: *EQ-015/M.03.v1*

N_4 split (founder ruling 22:20, 2026-09-06): CE-14 (spectral decomposition of M), CE-15 (survival subspace), CE-16 (projection operator), CE-17/CE-18 (individual survival limsup/lim conditions), CE-20 (collective potential V_G), CE-21 (Eth_col), CE-22/CE-23 (the two conflict directions), CE-24 (regime incompatibility), CE-30 (structural injustice) are eleven distinct definitions.

$$\text{Eth_col}(G) \Leftrightarrow d/dt' V_G(M) \leq 0 \wedge \min_i \Delta_{\text{spec}}(R_i) > 0; \text{Conf}_{\{\text{ind} \rightarrow \text{col}\}} \Leftrightarrow \\ d/dt' V_{\{A_i\}} \leq 0 \wedge d/dt' V_G > 0; \text{Conf}_{\{\text{col} \rightarrow \text{ind}\}} \Leftrightarrow d/dt' V_G \leq 0 \wedge \exists i: d/dt' V_{\{A_i\}} > 0; \\ \Delta_{\text{spec}}(R_i \cup R_j) \leq 0 \text{ [regime incompatibility]}; \text{structural injustice} \Leftrightarrow \exists i, j: \\ C_{\{A_i, R_i\}} \gg C_{\{A_j, R_j\}}$$

Occurrences (11): 18444260:CE-14, 18444260:CE-15, 18444260:CE-16, 18444260:CE-17, 18444260:CE-18, 18444260:CE-20, 18444260:CE-21, 18444260:CE-22, 18444260:CE-23, 18444260:CE-24, 18444260:CE-30

EQ-015/S.04.v1 — Non-Collapse Constitutive List (the 17 role-separations the live-possibility architecture depends on)

domain: S • tier: *Definition* • status: *split* • coq: *definition* • parents: *EQ-015/M.03.v1*

N_4 split (founder ruling 22:20, 2026-09-06): CBC-16 (sec. 15, ‘Non-Collapse Laws’) states verbatim a semicolon-separated list of 17 distinct non-identity claims (‘objective possibility $\neq$ structural feasibility; feasibility $\neq$ live possibility; ...; coupled performance $\neq$ Human Return’) – each is its own claim (none derivable from another by renaming/scaling), i.e. 17 non-equivalent members bundled as one code; CBC-07 and CBC-15 restate/extend two more instances of the same non-collapse pattern from elsewhere in the paper.

objective possibility $\neq$ structural feasibility; feasibility $\neq$ live possibility; live possibility $\neq$ stated preference; stated preference $\neq$ free consent; choice $\neq$ enactment; enactment $\neq$ evaluator-visible agency; option count $\neq$ effective agency; accessibility $\neq$ warrant; resonance $\neq$ consent; repetition $\neq$ legitimacy; retention $\neq$ improvement; norm conformity $\neq$ absence of agency; resistance to norms $\neq$ agency by definition; meaning-shaping $\neq$ domination; live-field narrowing $\neq$ structural violence without causal and normative conditions; AI influence $\neq$ manipulation by definition; coupled performance $\neq$ Human Return

Occurrences (3): 22357788:CBC-16, 22357788:CBC-07, 22357788:CBC-15

EQ-015/S.05.v1 — Meaning-Shaped Practical Possibility: five core propositions (P1-P5)

domain: S • tier: Dr • status: $not_an_equation$ • coq: $definition$ • parents: $EQ-015/M.03.v1$

$N4$ split ruling (founder 22:20, 2026-09-06, registry/split_proposal_SW.json): $CBC-P1..CBC-P5$ (sec. 5, 'The Core Thesis: Choice Begins Before Choice') are five narrative propositions, e.g. $CBC-P1$: 'Objective possibility is not practical possibility. An action may be physically or legally available while remaining absent from the agent's practical field.' – none contains a relational operator; the row's own *CANONICAL* statement.latest ('P1 objective $\neq$ practical possibility; ...') is itself a compressed paraphrase, not a quote of an equation.

P1 objective $\neq$ practical possibility; P2 practical possibility is meaning-shaped; P3 meaning-shaping can alter agency, prior to preference; P4 choice can be real within a narrowed field; P5 expansion of the live field can also be harmful

Occurrences (5): 22357788:CBC-P1, 22357788:CBC-P2, 22357788:CBC-P3, 22357788:CBC-P4, 22357788:CBC-P5

EQ-015/S.06.v1 — Potential as a Witnessed Readout (corrigible agency potential p^* , over the witnessed set)

domain: S • tier: $Definition$ • status: $split$ • coq: $definition$ • parents: $EQ-015/M.03.v1$

$N4$ split (founder ruling 22:20, 2026-09-06): $CBC-08/(1)$ ($p^*_{-}\{A,g\}$ witnessed-readout definition, restated near-identically in both papers), (2) (p^*_A vector and threshold-set C^α_A definition), (3) ($A^\wedge corr_A$ aggregate definition), and 'unnumbered' (the $r*d*x*f$ chain-rule decomposition) are four distinct formal objects.

$p^*_{-}\{A,g\}(h,z;T,B,P) = \max\{\pi \in \Pi^\wedge\{wit\}_{-}A(g;h,z,T,B)\} \Pr^\wedge \pi_P(Read_g \cap D_g \cap X_g \cap F_g)$, with $p^*_{-}\{A,g\} := 0$ when the set is empty; $p^*_A = (p^*_{-}\{A,g\})_{-}\{g \in G\}$, $C^\alpha_A = \{g: p^*_{-}\{A,g\} \geq \alpha\}$; $A^\wedge\{corr\}_{-}A(h,z;T,B,P,w) := \sum_g w_g p^*_{-}\{A,g\} \in [0,1]$; chain-rule $p_{-}\{A,g\}(\pi) = r \cdot d \cdot x \cdot f$

Occurrences (5): 22357788:CBC-08, 22361830:(1), 22361830:(2), 22361830:(3), 22361830:unnumbered

EQ-015/S.07.v1 — Recoverable Envelope Gap (Layer-2 potential minus Layer-1, on the probability envelope)

domain: S • tier: $Definition$ • status: $split$ • coq: $definition$ • parents: $EQ-015/M.03.v1$

$N4$ split (founder ruling 22:20, 2026-09-06): (4) defines the Layer-2 potential $p^*(2)_{-}\{A,g\}$; (5) defines the recoverable envelope gap $L^\wedge recoverable_A$ as a difference of $A^\wedge corr_A$ values – two distinct quantities, the second built from but not equal to the first.

$$p^*(2)_{A,g} := \max_{z \in J_{feas}} p^*_{A,g}(h,z;T,B,P); L_A^{recoverable} = \max_{z \in J_{feas}} A_A^{corr}(h,z) - A_A^{corr}(h,z0)$$

Occurrences (2): 22361830:(4), 22361830:(5)

EQ-015/S.08.v1 — Recoverable Live-Field Gap (distance-based, on the possibility field)

domain: S • tier: $Definition$ • status: $current$ • coq: $closed$ • parents: $EQ-015/M.03.v1$

$$L_{A,g}^{live} = \max_{z \in J_{feas}} D_L(L_A^z(g), L_A^{z0}(g))$$

Occurrences (1): 22357788:CBC-10

EQ-015/S.09.v1 — Corrigibility / Spectral-Support Readout Condition

domain: S • tier: Definition • status: current • coq: closed • parents: EQ-015/M.03.v1

$$\Delta_{\text{spec}}(Ri) > 0 \iff \text{channel}_i = \text{open} \wedge \dot{R}_i \neq 0$$

Occurrences (1): 22361830:(6)

EQ-015/S.10.v1 — Pseudo-Peace Gate-Collapse Signature

domain: S • tier: Definition • status: split • coq: definition • parents: EQ-015/M.03.v1

N_4 split (founder ruling 22:20, 2026-09-06): (7) ' $Y_{\text{obs}} = \text{calm} \wedge p * \text{wit} = \text{low} \wedge F = \text{blocked}$ ' is the pseudo-peace signature definition (a formula); P-A ('Pseudo-peace signature: amplitude-suppressing interventions raise the calm an evaluator reads while $p * \text{wit}$ does not rise and F stays blocked. Defeated if...') is a falsifiable proposition stated in prose that TESTS the definition – not itself an equation.

$$Y_{\text{obs}} = \text{calm} \wedge p * \text{wit} = \text{low} \wedge F = \text{blocked}$$

Occurrences (2): 22361830:(7), 22361830:P-A

EQ-015/S.11.v1 — Seven Distinctions Anatomy of 'Potential' (D1-D7)

domain: S • tier: Definition • status: split • coq: definition • parents: EQ-015/M.03.v1

N_4 split (founder ruling 22:20, 2026-09-06): D1..D7 (sec. 3, 'An anatomy of potential: seven distinctions') are seven distinct $\neq$ claims, e.g. D1 'Potential $\neq$ exercised $\neq$ observed', D2 'Declared set $\neq$ witnessed set' – seven non-equivalent members under one code, matching the row's own name 'Seven Distinctions D1-D7'.

D1 Potential $\neq$ exercised $\neq$ observed; D2 Declared set $\neq$ witnessed set; D3 Layer1 $\neq$ layer2; D4 Task potential $\neq$ aggregate potential; D5 Feasible $\neq$ permitted; D6 Diagnosis of compression $\neq$ attribution of responsibility; D7 Recoverable gap $\neq$ accumulated loss

Occurrences (7): 22361830:D1 (NC-78), 22361830:D2, 22361830:D3, 22361830:D4, 22361830:D5, 22361830:D6 (NC-79), 22361830:D7

EQ-015/S.12.v1 — Falsifiable Propositions Testing the Potential-Readout Architecture (P-B, P-C, P-D)

domain: S • tier: Open • status: not_an_equation • coq: definition • parents: EQ-015/M.03.v1

N_4 split ruling (founder 22:20, 2026-09-06, registry/split_proposal_SW.json): P-B, P-C, P-D (sec. 8, 'Falsifiable propositions') are narrative falsifiable propositions, e.g. P-B: 'Channel shift: when F is blocked and C^α shrinks in task g , load rises in a coupled task g ' while the aggregate (3) stays flat. Defeated if profiles stay flat across tasks.' – descriptive hypotheses with a defeat condition, not standalone equations.

P-B channel shift (load displaces between coupled tasks while the aggregate stays flat);
P-C the excluded path (hidden cost concentration inside a rising mean); P-D layer-2 dominates the structural case

Occurrences (3): 22361830:P-B, 22361830:P-C, 22361830:P-D

EQ-015/S.13.v1 — Axiom III: Regime Structure

domain: S • tier: Ax • status: current • coq: definition • parents: EQ-015/M.02.v1, EQ-015/S.01.v1

$$R := (T_R, I_R)$$

Occurrences (1): 18444260:CE-05

EQ-015/S.14.v1 — Update Law (state M)

domain: S • tier: Definition • status: current • coq: definition • parents: EQ-015/M.02.v1, EQ-015/S.01.v1

$$M(t' + \Delta t') = T_R(M(t'))$$

Occurrences (1): 18444260:CE-06

EQ-015/S.15.v1 — Update Law (agent A)

domain: S • tier: Definition • status: current • coq: definition • parents: EQ-015/M.02.v1, EQ-015/S.01.v1

$$A(t' + \Delta t') \subseteq I_R(A(t'), M(t'))$$

Occurrences (1): 18444260:CE-07

EQ-015/S.16.v1 — Definition: Etic

domain: S • tier: Definition • status: current • coq: closed • parents: EQ-015/M.02.v1, EQ-015/S.01.v1

$$\begin{aligned} \text{Etic}(A; t') &:= \exists R_A(t') \in R_adm(t') : A \subseteq M(t'), M(t' + \Delta t') = T_{\{R_A\}}(M(t')), \\ A(t' + \Delta t') &\subseteq I_{\{R_A\}}(A(t'), M(t')) \end{aligned}$$

Occurrences (1): 18444260:CE-08

EQ-015/S.17.v1 — Ethical Load (Lyapunov potential)

domain: S • tier: Definition • status: current • coq: definition • parents: EQ-015/M.03.v1, EQ-015/S.02.v1

$$V_{A,R}(M) \geq 0$$

Occurrences (1): 18444260:CE-09

EQ-015/S.18.v1 — Causal Memory Constraint

domain: S • tier: Definition • status: current • coq: definition • parents: EQ-015/M.03.v1, EQ-015/S.02.v1

$$\tau'_c(R) > 0$$

Occurrences (1): 18444260:CE-10

EQ-015/S.19.v1 — Spectral Stability Margin

domain: S • tier: Definition • status: current • coq: definition • parents: EQ-015/M.03.v1, EQ-015/S.02.v1

$$\Delta_{spec}(R) > 0$$

Occurrences (1): 18444260:CE-11

EQ-015/S.20.v1 — Causal Ethics Inequality (locked)

domain: S • tier: Definition • status: current • coq: closed • parents: EQ-015/M.03.v1, EQ-015/S.02.v1

$$\begin{aligned} \text{Ethical}(A) &\Leftrightarrow \exists \text{Choice}(A \rightarrow R_A) : d/dt' V_{\{A, R_A\}}(M(t')) \leq 0 \wedge \tau_{-c}'(R_A) > 0 \wedge \\ \Delta_{spec}(R_A) &> 0 \end{aligned}$$

Occurrences (1): 18444260:CE-12

EQ-015/S.21.v1 — Notation: positive-part derivative

domain: S • tier: Definition • status: current • coq: wrapped_related • parents: EQ-015/M.03.v1, EQ-015/S.02.v1

$$\dot{V}^+ := \max(dV/dt', 0)$$

Occurrences (1): 18444260:CE-13

EQ-015/S.22.v1 — Individual Ethics Condition

domain: S • tier: Definition • status: current • coq: wrapped_related • parents: EQ-015/M.03.v1, EQ-015/S.02.v1

$$\text{Eth}_{ind}(A) \Leftrightarrow d/dt' V_{A, R_A} \leq 0 \wedge \Delta_{spec}(R_A) > 0$$

Occurrences (1): 18444260:CE-19

EQ-015/S.23.v1 — Spectral decomposition of M

domain: S • tier: Definition • status: current • coq: wrapped_related • parents: EQ-015/M.03.v1, EQ-015/S.03.v1

$$M(t') = M_0\varphi_0 + \sum_k a_k(t')\varphi_k$$

Occurrences (1): 18444260:CE-14

EQ-015/S.24.v1 — Survival subspace $S_A(\mathbf{R})$

domain: S • tier: Definition • status: current • coq: wrapped_related • parents: EQ-015/M.03.v1, EQ-015/S.03.v1

$$S_A(R) \subseteq \text{span}\{\varphi_k\}$$

Occurrences (1): 18444260:CE-15

EQ-015/S.25.v1 — Survival projection operator

domain: S • tier: Definition • status: current • coq: wrapped_related • parents: EQ-015/M.03.v1, EQ-015/S.03.v1

$$P_{S_A(R)} : M \rightarrow S_A(R)$$

Occurrences (1): 18444260:CE-16

EQ-015/S.26.v1 — Individual survival (limsup>0)

domain: S • tier: Definition • status: current • coq: wrapped_related • parents: EQ-015/M.03.v1, EQ-015/S.03.v1

$$\limsup_{n \rightarrow \infty} \|P_{S_A(R_A)} M(t' + n)\|^2 > 0$$

Occurrences (1): 18444260:CE-17

EQ-015/S.27.v1 — Individual non-survival (lim=0)

domain: S • tier: Definition • status: current • coq: wrapped_related • parents: EQ-015/M.03.v1, EQ-015/S.03.v1

$$\lim_{n \rightarrow \infty} \|P_{S_A(R_A)} M(t' + n)\|^2 = 0$$

Occurrences (1): 18444260:CE-18

EQ-015/S.28.v1 — Collective potential V_G

domain: S • tier: Definition • status: current • coq: wrapped_related • parents: EQ-015/M.03.v1, EQ-015/S.03.v1

$$V_G(M) := \sum_{i=1}^N w_i V_{A_i, R_i}(M), w_i > 0$$

Occurrences (1): 18444260:CE-20

EQ-015/S.29.v1 — Collective ethics condition Eth_{col}

domain: S • tier: Definition • status: current • coq: definition • parents: EQ-015/M.03.v1, EQ-015/S.03.v1

$$Eth_{col}(G) \Leftrightarrow d/dt' V_G(M) \leq 0 \wedge \min_i \Delta_{spec}(R_i) > 0$$

Occurrences (1): 18444260:CE-21

EQ-015/S.30.v1 — Conflict: individual-admissible/collective-destabilizing

domain: S • tier: Definition • status: current • coq: definition • parents: EQ-015/M.03.v1, EQ-015/S.03.v1

$$Conf_{ind \rightarrow col} \Leftrightarrow d/dt' V_{A_i} \leq 0 \wedge d/dt' V_G > 0$$

Occurrences (1): 18444260:CE-22

EQ-015/S.31.v1 — Conflict: collective-admissible/individual-destabilizing

domain: S • tier: *Definition* • status: *current* • coq: *definition* • parents: *EQ-015/M.03.v1, EQ-015/S.03.v1*

$$Conf_{col \rightarrow ind} \Leftrightarrow d/dt' V_G \leq 0 \wedge \exists i : d/dt' V_{A_i} > 0$$

Occurrences (1): 18444260:CE-23

EQ-015/S.32.v1 — Regime incompatibility

domain: S • tier: *Definition* • status: *current* • coq: *wrapped_related* • parents: *EQ-015/M.03.v1, EQ-015/S.03.v1*

$$\Delta_{spec}(R_i \cup R_j) \leq 0$$

Occurrences (1): 18444260:CE-24

EQ-015/S.33.v1 — Structural injustice

domain: S • tier: *Definition* • status: *current* • coq: *closed* • parents: *EQ-015/M.03.v1, EQ-015/S.03.v1*

$$\exists i, j : C_{A_i, R_i} \gg C_{A_j, R_j}$$

Occurrences (1): 18444260:CE-30

EQ-015/S.34.v1 — Non-collapse law 1: objective possibility $\neq$ structural feasibility

domain: S • tier: *Definition* • status: *current* • coq: *closed* • parents: *EQ-015/M.03.v1, EQ-015/S.04.v1*

$$\text{objective possibility} \neq \text{structural feasibility}$$

Occurrences (1): 22357788:CBC-16 (list item 1 of 17)

EQ-015/S.35.v1 — Non-collapse law 2: feasibility $\neq$ live possibility

domain: S • tier: *Definition* • status: *current* • coq: *wrapped_related* • parents: *EQ-015/M.03.v1, EQ-015/S.04.v1*

$$\text{feasibility} \neq \text{live possibility}$$

Occurrences (1): 22357788:CBC-16 (list item 2 of 17)

EQ-015/S.36.v1 — Non-collapse law 3: live possibility $\neq$ stated preference

domain: S • tier: *Definition* • status: *current* • coq: *wrapped_related* • parents: *EQ-015/M.03.v1, EQ-015/S.04.v1*

$$\text{live possibility} \neq \text{stated preference}$$

Occurrences (1): 22357788:CBC-16 (list item 3 of 17)

EQ-015/S.37.v1 — Non-collapse law 4: stated preference $\neq$ free consent

domain: S • tier: *Definition* • status: *current* • coq: *wrapped_related* • parents: *EQ-015/M.03.v1, EQ-015/S.04.v1*

$$\text{stated preference} \neq \text{free consent}$$

Occurrences (1): 22357788:CBC-16 (list item 4 of 17)

EQ-015/S.38.v1 — Non-collapse law 5: choice $\neq$ enactment

domain: S • tier: *Definition* • status: *current* • coq: *wrapped_related* • parents: *EQ-015/M.03.v1, EQ-015/S.04.v1*

$$\text{choice} \neq \text{enactment}$$

Occurrences (1): 22357788:CBC-16 (list item 5 of 17)

EQ-015/S.39.v1 — Non-collapse law 6: enactment $\neq$ evaluator-visible agency

domain: S • tier: Definition • status: current • coq: wrapped_related • parents: EQ-015/M.03.v1, EQ-015/S.04.v1

enactment $\neq$ evaluator-visible agency

Occurrences (1): 22357788:CBC-16 (list item 6 of 17)

EQ-015/S.40.v1 — Non-collapse law 7: option count $\neq$ effective agency

domain: S • tier: Definition • status: current • coq: wrapped_related • parents: EQ-015/M.03.v1, EQ-015/S.04.v1

option count $\neq$ effective agency

Occurrences (1): 22357788:CBC-16 (list item 7 of 17)

EQ-015/S.41.v1 — Non-collapse law 8: accessibility $\neq$ warrant

domain: S • tier: Definition • status: current • coq: wrapped_related • parents: EQ-015/M.03.v1, EQ-015/S.04.v1

accessibility $\neq$ warrant

Occurrences (1): 22357788:CBC-16 (list item 8 of 17)

EQ-015/S.42.v1 — Non-collapse law 9: resonance $\neq$ consent

domain: S • tier: Definition • status: current • coq: wrapped_related • parents: EQ-015/M.03.v1, EQ-015/S.04.v1

resonance $\neq$ consent

Occurrences (1): 22357788:CBC-16 (list item 9 of 17)

EQ-015/S.43.v1 — Non-collapse law 10: repetition $\neq$ legitimacy

domain: S • tier: Definition • status: current • coq: wrapped_related • parents: EQ-015/M.03.v1, EQ-015/S.04.v1

repetition $\neq$ legitimacy

Occurrences (1): 22357788:CBC-16 (list item 10 of 17)

EQ-015/S.44.v1 — Non-collapse law 11: retention $\neq$ improvement

domain: S • tier: Definition • status: current • coq: wrapped_related • parents: EQ-015/M.03.v1, EQ-015/S.04.v1

retention $\neq$ improvement

Occurrences (1): 22357788:CBC-16 (list item 11 of 17)

EQ-015/S.45.v1 — Non-collapse law 12: norm conformity $\neq$ absence of agency

domain: S • tier: Definition • status: current • coq: wrapped_related • parents: EQ-015/M.03.v1, EQ-015/S.04.v1

norm conformity $\neq$ absence of agency

Occurrences (1): 22357788:CBC-16 (list item 12 of 17)

EQ-015/S.46.v1 — Non-collapse law 13: resistance to norms $\neq$ agency by definition

domain: S • tier: Definition • status: current • coq: wrapped_related • parents: EQ-015/M.03.v1, EQ-015/S.04.v1

resistance to norms $\neq$ agency by definition

Occurrences (1): 22357788:CBC-16 (list item 13 of 17)

EQ-015/S.47.v1 — Non-collapse law 14: meaning-shaping $\neq$ domination

domain: S • tier: Definition • status: current • coq: wrapped_related • parents: EQ-015/M.03.v1, EQ-015/S.04.v1

meaning-shaping $\neq$ domination

Occurrences (1): 22357788:CBC-16 (list item 14 of 17)

EQ-015/S.48.v1 — Non-collapse law 15: live-field narrowing $\neq$ structural violence without causal and normative conditions

domain: S • tier: Definition • status: current • coq: wrapped_related • parents: EQ-015/M.03.v1, EQ-015/S.04.v1

live-field narrowing $\neq$ structural violence without causal and normative conditions

Occurrences (1): 22357788:CBC-16 (list item 15 of 17)

EQ-015/S.49.v1 — Non-collapse law 16: AI influence $\neq$ manipulation by definition

domain: S • tier: Definition • status: current • coq: wrapped_related • parents: EQ-015/M.03.v1, EQ-015/S.04.v1

AIinfluence $\neq$ manipulation by definition

Occurrences (1): 22357788:CBC-16 (list item 16 of 17)

EQ-015/S.50.v1 — Non-collapse law 17: coupled performance $\neq$ Human Return

domain: S • tier: Definition • status: current • coq: wrapped_related • parents: EQ-015/M.03.v1, EQ-015/S.04.v1

coupled performance $\neq$ Human Return

Occurrences (1): 22357788:CBC-16 (list item 17 of 17)

EQ-015/S.51.v1 — Witnessed-readout potential $p^*_{\{A,g\}}$

domain: S • tier: Definition • status: current • coq: definition • parents: EQ-015/M.03.v1, EQ-015/S.06.v1

$$p^*_{A,g} = \max_{\pi \in \Pi_{feas_A}(g)} Pr(Rg \cap Dg \cap Xg \cap Fg)$$

Occurrences (1): 22357788:CBC-08

EQ-015/S.52.v1 — [restatement of $p^*_{\{A,g\}}$]

domain: S • tier: Definition • status: current • coq: wrapped_related • parents: EQ-015/M.03.v1, EQ-015/S.06.v1

$p^*_{\{A,g\}}(h,z;T,B,P) := \max_{\pi \in \Pi_{wit_A}(g;h,z,T,B)} Pr^{\pi}_P(Rg \cap Dg \cap Xg \cap Fg)$, with
 $p^*_{\{A,g\}} := 0$ when the set is empty

Occurrences (1): 22361830:(1)

EQ-015/S.53.v1 — p^*_A vector and threshold set C_A^α

domain: S • tier: Definition • status: current • coq: wrapped_related • parents: EQ-015/M.03.v1, EQ-015/S.06.v1

$$p^*_A = (p^*_{A,g})_{g \in G}, C_A^\alpha = \{g : p^*_{A,g} \geq \alpha\}$$

Occurrences (1): 22361830:(2)

EQ-015/S.54.v1 — Aggregate corrigible-agency A_A^{corr}

domain: S • tier: Definition • status: current • coq: wrapped_related • parents: EQ-015/M.03.v1, EQ-015/S.06.v1

$$A_A^{corr}(h, z; T, B, P, w) := \Sigma_g w_g p_{A,g}^* \in [0, 1]$$

Occurrences (1): 22361830:(3)

EQ-015/S.55.v1 — Chain-rule decomposition $p_{A,g}(\pi)=r \cdot d \cdot x \cdot f$

domain: S • tier: Definition • status: current • coq: wrapped_related • parents: EQ-015/M.03.v1, EQ-015/S.06.v1

$$p_{A,g}(\pi) = r \cdot d \cdot x \cdot f, \text{ with } r = Pr(Rg), d = Pr(Dg|Rg), x = Pr(Xg|Rg, Dg), f = Pr(Fg|Rg, Dg, Xg)$$

Occurrences (1): 22361830:unnumbered

EQ-015/S.56.v1 — Layer-2 potential $p^*(2)_{A,g}$

domain: S • tier: Definition • status: current • coq: definition • parents: EQ-015/M.03.v1, EQ-015/S.07.v1

$$p^*(2)_{A,g} := \max_{z \in Jfeas} p_{A,g}^*(h, z; T, B, P)$$

Occurrences (1): 22361830:(4)

EQ-015/S.57.v1 — Recoverable envelope gap $L_A^{recoverable}$

domain: S • tier: Definition • status: current • coq: closed • parents: EQ-015/M.03.v1, EQ-015/S.07.v1

$$L_A^{recoverable} = \max_{z \in Jfeas} A_A^{corr}(h, z) - A_A^{corr}(h, z_0)$$

Occurrences (1): 22361830:(5)

EQ-015/S.58.v1 — Pseudo-peace gate-collapse signature

domain: S • tier: Definition • status: current • coq: closed • parents: EQ-015/M.03.v1, EQ-015/S.10.v1

$$Y_{obs} = calm \wedge p * wit = low \wedge F = blocked$$

Occurrences (1): 22361830:(7)

EQ-015/S.59.v1 — D1: potential vs exercised vs observed

domain: S • tier: Definition • status: current • coq: closed • parents: EQ-015/M.03.v1, EQ-015/S.11.v1

$$Potential \neq exercised \neq observed$$

Occurrences (1): 22361830:D1 (NC-78)

EQ-015/S.60.v1 — D2: declared set vs witnessed set

domain: S • tier: Definition • status: current • coq: wrapped_related • parents: EQ-015/M.03.v1, EQ-015/S.11.v1

$$\text{Declared set} \neq \text{witnessed set}$$

Occurrences (1): 22361830:D2

EQ-015/S.61.v1 — D3: layer 1 vs layer 2

domain: S • tier: Definition • status: current • coq: wrapped_related • parents: EQ-015/M.03.v1, EQ-015/S.11.v1

$$Layer1 \neq layer2$$

Occurrences (1): 22361830:D3

EQ-015/S.62.v1 — D4: task potential vs aggregate potential

domain: S • tier: Definition • status: current • coq: wrapped_related • parents: EQ-015/M.03.v1, EQ-015/S.11.v1

Task potential $\neq$ aggregate potential

Occurrences (1): 22361830:D4

EQ-015/S.63.v1 — D5: feasible vs permitted

domain: S • tier: Definition • status: current • coq: wrapped_related • parents: EQ-015/M.03.v1, EQ-015/S.11.v1

Feasible $\neq$ permitted

Occurrences (1): 22361830:D5

EQ-015/S.64.v1 — D6: diagnosis vs attribution of responsibility

domain: S • tier: Definition • status: current • coq: wrapped_related • parents: EQ-015/M.03.v1, EQ-015/S.11.v1

Diagnosis of compression $\neq$ attribution of responsibility

Occurrences (1): 22361830:D6 (NC-79)

EQ-015/S.65.v1 — D7: recoverable gap vs accumulated loss

domain: S • tier: Definition • status: current • coq: wrapped_related • parents: EQ-015/M.03.v1, EQ-015/S.11.v1

Recoverable gap $\neq$ accumulated loss

Occurrences (1): 22361830:D7

EQ-015/W.01.v1 — industrial and post-labour claim chains

domain: W • tier: Definition • status: not_an_equation • coq: definition • parents: EQ-015/M.02.v1

N_4 split ruling (founder 22:20, 2026-09-06, registry/split_proposal_SW.json): (1) 'human labour $\rightarrow$ production $\rightarrow$ wage $\rightarrow$ claim on output' and (51) 'labour-based claim $\rightarrow$ human/citizen claim on social production' are two distinct causal/conceptual arrow-chains (not equal to one another) with no relational operator ($=$, $\leq$, $\in$, ...) – diagrams of a claim chain, not equations; matches the task's own example 'industrial and post-labour claim chains'.

[print fallback: this entry's LaTeX transcription did not compile as standalone display math; showing the source ascii/latest text instead — the registry, JSON entry and docs site carry the LaTeX transcription unmodified]

human labour $\rightarrow$ production $\rightarrow$ wage $\rightarrow$ claim on output; labour-based claim $\rightarrow$ human/citizen claim on social production

Occurrences (2): 22481924:(1), 22481924:(51)

EQ-015/W.02.v1 — minimal epistemic firewall for machine candidate generation

domain: W • tier: Definition • status: current • coq: definition • parents: EQ-015/M.01.v1

$$\dot{Z}_t = v_t G_t - \delta_Z Z_t, 0 \leq v_t \leq 1$$

Occurrences (1): 22481924:(3)

EQ-015/W.03.v1 — embodied automation and machine productive-capacity block

domain: W • tier: Definition • status: split • coq: definition • parents: EQ-015/M.01.v1

N_4 split (founder ruling 22:20, 2026-09-06): (5) $B^{\sim}RB_t = N^{\sim}RB_t \ q^{\sim}RB_t$, (6) the growth-decomposition identity, (7) the Cobb-Douglas-like M_t definition, (9) $\rho = (\sigma - 1)/\sigma$, (10) the labour-share $s^{\sim}L_t$ definition are five distinct quantities/definitions.

$$B^{\sim}RB_t = N^{\sim}RB_t \ q^{\sim}RB_t; \text{ growth decomposition } \dot{B}^{\sim}RB/B^{\sim}RB = \dot{N}^{\sim}RB/N^{\sim}RB + \dot{q}^{\sim}RB/q^{\sim}RB; \ M_t = (K^{\sim}M_t)^{\sim\kappa} (A^{\sim}AI_t)^{\sim\alpha} (B^{\sim}RB_t)^{\sim\beta}; \ \rho = (\sigma - 1)/\sigma; \ s^{\sim}L_t = (\omega_H \ H_t^{\sim\rho}) / (\omega_H \ H_t^{\sim\rho} + \omega_M \ M_t^{\sim\rho})$$

Occurrences (5): 22481924:(5), 22481924:(6), 22481924:(7), 22481924:(9), 22481924:(10)

EQ-015/W.04.v1 — four-dimensional labour centrality

domain: W • tier: Definition • status: split • coq: definition • parents: EQ-015/M.01.v1

N_4 split (founder ruling 22:20, 2026-09-06): (8) ' $L_t = \langle L^{task}_t, L^{income}_t, L^{bottleneck}_t, L^{bargain}_t \rangle$ ' is the four-dimensional labour-centrality tuple definition; 'Labour-Decentering Proposition [Open]' ('AI and robotics can reduce labour centrality without eliminating labour...') is a narrative Open-tier proposition, not an equation.

$$L_t = \langle L_t^{task}, L_t^{income}, L_t^{bottleneck}, L_t^{bargain} \rangle$$

Occurrences (2): 22481924:(8), 22481924:Labour-Decentering Proposition

EQ-015/W.05.v1 — the claim constitution: from wages to citizen claims

domain: W • tier: Definition • status: split • coq: definition • parents: EQ-015/M.01.v1

N_4 split (founder ruling 22:20, 2026-09-06): (11) q_t definition, (12) Γ_t definition, (13) $q_t^{\min}$ definition, (14) D^{rent} definition, (15) Γ^{net} definition are five distinct quantities chained together in one code.

$$q_t = o_t + \tau_t(1 - o_t); \Gamma_t = s_t^L + q_t(1 - s_t^L); q_t^{\min} = (\bar{\Gamma} - s_t^L)/(1 - s_t^L); D^{rent}, \Gamma^{net}$$

Occurrences (5): 22481924:(11), 22481924:(12), 22481924:(13), 22481924:(14), 22481924:(15)

EQ-015/W.06.v1 — aggregate demand, realization, and profit-wealth feedback

domain: W • tier: Definition • status: split • coq: definition • parents: EQ-015/M.01.v1

N_4 split (founder ruling 22:20, 2026-09-06): (22) AD_t aggregate-demand definition, (23) χ^{dem}_t realization-rate definition, (24) Pi^M_t profit definition are three distinct formulas; (25) and (55) are arrow-chain feedback-loop summaries (' $\Gamma^{eff} Y \rightarrow AD \rightarrow Pi^M \rightarrow W^M_{t+1} \rightarrow M_{t+1}$ ') with no relational operator, restating the loop in prose diagram form rather than as an equation.

$$AD_t = C(\Gamma^{eff}_t Y_t, m_t) + I_t + G_t + NX_t; \chi^{dem}_t = \min\{1, AD_t/Y_t\}; \Pi^M_t = \chi^{dem}_t Y_t - Cost^M_t; \Gamma^{eff} Y \rightarrow AD \rightarrow \Pi^M \rightarrow W^M_{t+1} \rightarrow M_{t+1}$$

Occurrences (5): 22481924:(22), 22481924:(23), 22481924:(24), 22481924:(25), 22481924:(55)

EQ-015/W.07.v1 — ownership stock accumulation and its non-collapse from redistribution

domain: W • tier: Definition • status: split • coq: definition • parents: EQ-015/M.03.v1

N_4 split (founder ruling 22:20, 2026-09-06): (16) ownership-stock recursion $W^M_{i,t+1}$, (17) ownership-share o_t definition, (18) 'current redistribution != future ownership reproduction' (a non-collapse claim) are three distinct formal objects.

$$W^M_{i,t+1} = (1 - \delta_W) W^M_{i,t} + r^M_t W^M_{i,t} + s^{\{M, cap\}}_{i,t} + T^{cap}_{i,t} - Tax^{cap}_{i,t}; \\ o_t = \sum_{i \in B} W^M_{i,t} / \sum_i W^M_{i,t}; \text{current redistribution} \neq \text{future ownership reproduction}$$

Occurrences (3): 22481924:(16), 22481924:(17), 22481924:(18)

EQ-015/W.08.v1 — conversion-gate control, credible exit, and dependency exposure

domain: W • tier: Definition • status: split • coq: definition • parents: EQ-015/M.03.v1

N_4 split (founder ruling 22:20, 2026-09-06): (26) conversion-gate G^{conv} definition, (27) credible-exit X definition, (28) dependency-exposure D definition, (29) 'Concentration != G^{conv} != D ' non-collapse claim, (53) 'concentration !=> dependency !=> agency loss' conditional non-collapse claim – five distinct formal objects.

$$G^{conv}_{j,t}(e) = 1 - [V_t(e|j)/V_t(e)]_+; X_{i,t}(e;j) = \text{clip}(V_{i,t}(e|j)/V_{i,t}(e), 0, 1); D_{i \rightarrow j,t}(g) = \sum_e w_e(g) G^{conv}_{j,t}(e) [1 - X_{i,t}(e;j)]; \text{Concentration} \neq G^{conv} \neq D; \\ \text{concentration} \neq \text{dependency} \neq \text{agency loss (conditional)}$$

Occurrences (5): 22481924:(26), 22481924:(27), 22481924:(28), 22481924:(29), 22481924:(53)

EQ-015/W.09.v1 — relational class position vector

domain: W • tier: Definition • status: current • coq: definition • parents: EQ-015/M.01.v1

$$C_{i,t} = \langle O_{i,t}, G_{i,t}, \Gamma_{i,t}, A_{i,t}^{access}, X_{i,t}, D_{i,t}, R_{i,t}^{rent} \rangle$$

Occurrences (1): 22481924:(30)

EQ-015/W.10.v1 — Human Systemic Position typed audit index

domain: W • tier: Definition • status: split • coq: definition • parents: EQ-015/M.03.v1

N_4 split (founder ruling 22:20, 2026-09-06): (40) $P^{\sim}H_t$ index definition, (41)+(42) its growth-rate decomposition (one continued equation split across two raw labels), (43) ' $Y_dot_t > 0 \Rightarrow P_dot^{\sim}H_t > 0$ ' non-collapse claim, (44) AWA_ t 'Abundance Without Agency' tuple definition are four distinct formal objects; (57) 'growth of machine productive power versus growth of broad human claim, exit, agency, and Human Return' is a narrative comparison with no relational operator.

$$P^{\sim}H_t = (\Gamma^{\sim}eff_t)^{\sim}\theta\Gamma(A^{\sim}corr_H,t)^{\sim}\theta A(\Lambda^{\sim}live_H,t)^{\sim}\theta\Lambda(r^{\sim}H_t)^{\sim}\theta R(S^{\sim}H_t)^{\sim}\theta S(X^{\sim}H_t)^{\sim}\theta X / (1 + D^{\sim}H_t)^{\sim}\theta D; \dot{Y}_t > 0 \not\Rightarrow \dot{P}^{\sim}H_t > 0$$

Occurrences (6): 22481924:(40), 22481924:(41), 22481924:(42), 22481924:(43), 22481924:(44), 22481924:(57)

EQ-015/W.11.v1 — recognized social role/standing dynamic and time-budget identity

domain: W • tier: Definition • status: split • coq: definition • parents: EQ-015/M.01.v1

N_4 split (founder ruling 22:20, 2026-09-06): (36) ' $S_dot^{\sim}H_t = s_1 W_t + s_2 N_t - \delta a_S S^{\sim}H_t$ ' (social-standing dynamic) and (37) ' $1 = l_wage + l_care + l_learn + l_civic + l_leisure$ ' (a time-budget identity) are two distinct equations with different LHS quantities.

$$\dot{S}_t^H = s_1 W_t + s_2 N_t - \delta_S S_t^H; 1 = l_{wage} + l_{care} + l_{learn} + l_{civic} + l_{leisure}$$

Occurrences (2): 22481924:(36), 22481924:(37)

EQ-015/W.12.v1 — social reproduction dynamic and its non-collapse from productive necessity

domain: W • tier: Definition • status: split • coq: definition • parents: EQ-015/M.03.v1

N_4 split (founder ruling 22:20, 2026-09-06): (38) $H^{\sim}cap_t$ human-capability-stock dynamic and (39) 'productive necessity of humans != social necessity of human reproduction' non-collapse claim are two distinct formulas; (56) '{Care,Health,Education} -> $H^{\sim}cap$ -> { $A^{\sim}corr_H,r_H,S_H$ } -> P -> $I_t(t+1)$ ' is an arrow-chain narrative summary, no relational operator.

$$\dot{H}^{\sim}cap_t = f(Care_t, Health_t, Education_t, Nutrition_t, Community_t) - \delta_{HH} H^{\sim}cap_t; \text{productive necessity of humans} \neq \text{social necessity of human reproduction}; \{Care, Health, Education\} \rightarrow H^{\sim}cap \rightarrow \{A^{\sim}corr_H, r_H, S_H\} \rightarrow P \rightarrow I_{t+1}$$

Occurrences (3): 22481924:(38), 22481924:(39), 22481924:(56)

EQ-015/W.13.v1 — normalized decoupling-pressure early-warning diagnostic

domain: W • tier: Dr • status: current • coq: definition • parents: EQ-015/M.01.v1

$$\Omega_t = w_M(g_M - g_H) + w_Dg_D + w_Gg_G - w_qg_q - w_I\Gamma g_I(\Gamma^{\sim}eff) - w_Xg_X - w_Rg_R(r^{\sim}H) - w_Ag_A - w_Hg_H(H^{\sim}cap)$$

Occurrences (1): 22481924:(52)

EQ-015/W.14.v1 — corrigible agency envelope at world-system scale

domain: W • tier: Definition • status: current • coq: definition • parents: EQ-015/M.01.v1

$$A_{H,i,t}^{corr}(g) = \max_{\pi \in \Pi_{i,t}^{live}(g)} Pr^{\pi}(R_g \cap D_g \cap X_g \cap F_g)$$

Occurrences (1): 22481924:(34)

EQ-015/W.15.v1 — Human Return tuple at world-system scale

domain: W • tier: Definition • status: current • coq: definition • parents: EQ-015/M.01.v1

$$R_{H,t}^{return} = \langle C_t, T_t, S_t^{skill}, A_t^{alt} \rangle$$

Occurrences (1): 22481924:(35)

EQ-015/W.16.v1 — Human Conversion Vector and elasticity

domain: W • tier: Definition • status: split • coq: definition • parents: EQ-015/M.01.v1

N_4 split (founder ruling 22:20, 2026-09-06): (1) 'Machine expansion $\neq$ Human expansion' non-collapse claim, (11) C^H_t Human-Conversion-Vector definition, (12) $\eta^{HC}_{(j,t)}$ elasticity definition, (13) sign-constraints on three elasticities are four distinct formulas; 'Proposition 1 [Open]' ('An increase in M_t ... is insufficient to establish human expansion...') and (25)/(26) (a 'versus' comparison and an arrow-chain summary) are narrative, no operator.

Machine expansion $\neq$ Human expansion; $C^H_t = \langle \Gamma^{\text{eff}}_t, X^H_t, \Lambda^{\text{live}}_{\{H,t\}}, A^{\text{corr}}_{\{H,t\}}, R^{\text{route}}_{\{H,t\}}, W^{\text{world}}_{\{H,t\}}, r_{\{H,t\}}, H^{\text{cap}}_t, S_{\{H,t\}} \rangle$; $\eta^{HC}_{\{j,t\}} = \partial \ln C^H_{\{j,t\}} / \partial \ln M_t$

Occurrences (7): 22481926:(1), 22481926:(11), 22481926:(12), 22481926:(13), 22481926:Proposition 1, 22481926:(25), 22481926:(26)

EQ-015/W.17.v1 — high- vs low-conversion interaction forms

domain: W • tier: Definition • status: not_an_equation • coq: definition • parents: EQ-015/M.01.v1

N_4 split ruling (founder 22:20, 2026-09-06, registry/split_proposal_SW.json): (14) 'H problem $\rightarrow$ AI divergence $\rightarrow$ H resistance $\rightarrow$ World test $\rightarrow$ H integration $\rightarrow$ AI removal $\rightarrow$ H return', (15) 'H problem $\rightarrow$ AI answer $\rightarrow$ use $\rightarrow$ dependence', and (24) 'Maximum AI assistance $\rightarrow$ Maximum durable human conversion' are three arrow-chain narratives with no relational operator; 'Proposition 2 [Open]' ('The AI assistance intensity that maximizes current performance need not maximize delayed Human Return...') is a narrative Open-tier proposition – none of the four members is an equation.

H problem $\rightarrow$ AI divergence $\rightarrow$ H resistance $\rightarrow$ World test $\rightarrow$ H integration $\rightarrow$ AI removal $\rightarrow$ H return (high-conversion); H problem $\rightarrow$ AI answer $\rightarrow$ use $\rightarrow$ dependence (low-conversion); Maximum AI assistance $\rightarrow$ Maximum durable human conversion

Occurrences (4): 22481926:(14), 22481926:(15), 22481926:Proposition 2, 22481926:(24)

EQ-015/W.18.v1 — Bad-Mode State Vector

domain: W • tier: Definition • status: current • coq: definition • parents: EQ-015/M.01.v1

$B_t = \langle D_t, \hat{G}^{\text{conv}}_t, C^{\text{info}}_t, 1 - X^H_t, 1 - r_{\{H,t\}}, 1 - \Lambda^{\text{live}}_{\{H,t\}}, 1 - A^{\text{corr}}_{\{H,t\}}, 1 - \Gamma^{\text{eff}}_t, 1 - H^{\text{cap}}_t \rangle$

Occurrences (1): 22481926:(16)

EQ-015/W.19.v1 — Reversibility Window and Urgency Vector

domain: W • tier: Definition • status: split • coq: definition • parents: EQ-015/M.01.v1

N_4 split (founder ruling 22:20, 2026-09-06): (17) reversibility-window W_j definition, (18) Delta $g_{(j,t)}$ definition, (19) urgency $U_{(j,t)}$ definition, (20) urgency vector U_t definition, (21) U^{max}_t definition are five distinct formulas; 'Proposition 3 [Open]' ('the Reversibility Principle') is a narrative Open-tier proposition, no relational operator.

$W_j = \{t: C^{\text{rec}}_{\{j,t\}} \leq C^-_j \wedge \tau^{\text{rec}}_{\{j,t\}} \leq \tau^-_j\}$; $\Delta g_{\{j,t\}} = [g^-_{\{M,t\}} - g^-_{\{Cj,t\}}]_+$; $U_{\{j,t\}} = \Delta g_{\{j,t\}} L_{\{j,t\}} S_{\{j,t\}} \tau^{\text{rec}}_{\{j,t\}}$; $U_t = \langle U_{\{1,t\}}, \dots, U_{\{J,t\}} \rangle$; $U^{\text{max}}_t = \max_j U_{\{j,t\}}$

Occurrences (6): 22481926:(17), 22481926:Proposition 3, 22481926:(18), 22481926:(19), 22481926:(20), 22481926:(21)

EQ-015/W.20.v1 — Robotic-block output $B^R B_t$

domain: W • tier: Definition • status: current • coq: closed • parents: EQ-015/M.01.v1, EQ-015/W.03.v1

$$B^R B_t = N^R B_t q^R B_t$$

Occurrences (1): 22481924:(5)

EQ-015/W.21.v1 — Growth decomposition of $B^R B_t$

domain: W • tier: Definition • status: current • coq: wrapped_related • parents: EQ-015/M.01.v1, EQ-015/W.03.v1

$$B_{dot}^R B_t / B^R B_t = N_{dot}^R B_t / N^R B_t + q_{dot}^R B_t / q^R B_t$$

Occurrences (1): 22481924:(6)

EQ-015/W.22.v1 — Machine output M_t (nested CES/Cobb-Douglas)

domain: W • tier: Definition • status: current • coq: definition • parents: EQ-015/M.01.v1, EQ-015/W.03.v1

$$M_t = (K_t^M)^\kappa (A^A I_t)^\alpha (B^R B_t)^\beta$$

Occurrences (1): 22481924:(7)

EQ-015/W.23.v1 — Elasticity parameter rho

domain: W • tier: Definition • status: current • coq: definition • parents: EQ-015/M.01.v1, EQ-015/W.03.v1

$$\rho = (\sigma - 1) / \sigma$$

Occurrences (1): 22481924:(9)

EQ-015/W.24.v1 — Labour income share s_t^L

domain: W • tier: Definition • status: current • coq: definition • parents: EQ-015/M.01.v1, EQ-015/W.03.v1

$$s_t^L = (\omega_H H_t^\rho) / (\omega_H H_t^\rho + \omega_M M_t^\rho)$$

Occurrences (1): 22481924:(10)

EQ-015/W.25.v1 — Four-dimensional labour centrality tuple

domain: W • tier: Definition • status: current • coq: definition • parents: EQ-015/M.01.v1, EQ-015/W.04.v1

$$L_t = \langle L_t^{task}, L_t^{income}, L_t^{bottleneck}, L_t^{bargain} \rangle$$

Occurrences (1): 22481924:(8)

EQ-015/W.26.v1 — Claim-conversion rate q_t

domain: W • tier: Definition • status: current • coq: closed • parents: EQ-015/M.01.v1, EQ-015/W.05.v1

$$q_t = o_t + \tau_t(1 - o_t) = 1 - (1 - o_t)(1 - \tau_t)$$

Occurrences (1): 22481924:(11)

EQ-015/W.27.v1 — Effective claim Γ_t

domain: W • tier: Definition • status: current • coq: closed • parents: EQ-015/M.01.v1, EQ-015/W.05.v1

$$\Gamma_t = s_t^L + q_t(1 - s_t^L)$$

Occurrences (1): 22481924:(12)

EQ-015/W.28.v1 — Minimum sufficient q_t^{min}

domain: W • tier: Definition • status: current • coq: definition • parents: EQ-015/M.01.v1, EQ-015/W.05.v1

$$q_t^{min} = (\Gamma_{bar} - s_t^L) / (1 - s_t^L)$$

Occurrences (1): 22481924:(13)

EQ-015/W.29.v1 — AI-rent differential D^{rent}

domain: W • tier: Definition • status: current • coq: wrapped_related • parents: EQ-015/M.01.v1, EQ-015/W.05.v1

$$D_{(i,t)}^{rent} = (Rent_{(i,t)}^{(AI,out)} - Rent_{(i,t)}^{(AI,in)})/Y_{(i,t)}$$

Occurrences (1): 22481924:(14)

EQ-015/W.30.v1 — Net effective claim Γ^{net}

domain: W • tier: Definition • status: current • coq: wrapped_related • parents: EQ-015/M.01.v1, EQ-015/W.05.v1

$$\Gamma_{(i,t)}^{net} = \Gamma_{(i,t)} - D_{(i,t)}^{rent}$$

Occurrences (1): 22481924:(15)

EQ-015/W.31.v1 — Aggregate demand AD_t

domain: W • tier: Definition • status: current • coq: definition • parents: EQ-015/M.01.v1, EQ-015/W.06.v1

$$AD_t = C(\Gamma_t^{eff} Y_t, m_t) + I_t + G_t + NX_t$$

Occurrences (1): 22481924:(22)

EQ-015/W.32.v1 — Demand-realization rate χ_t^{dem}

domain: W • tier: Definition • status: current • coq: closed • parents: EQ-015/M.01.v1, EQ-015/W.06.v1

$$\chi_t^{dem} = \min\{1, AD_t/Y_t\}$$

Occurrences (1): 22481924:(23)

EQ-015/W.33.v1 — Machine-sector profit Π_t^M

domain: W • tier: Definition • status: current • coq: definition • parents: EQ-015/M.01.v1, EQ-015/W.06.v1

$$\Pi_t^M = \chi_t^{dem} Y_t - Cost_t^M$$

Occurrences (1): 22481924:(24)

EQ-015/W.34.v1 — Ownership-stock accumulation recursion

domain: W • tier: Definition • status: current • coq: definition • parents: EQ-015/M.03.v1, EQ-015/W.07.v1

$$W_{(i,t+1)}^M = (1 - \delta_W) W_{(i,t)}^M + r_t^M W_{(i,t)}^M + s_{(i,t)}^{(M,cap)} + T_{(i,t)}^{cap} - Tax_{(i,t)}^{cap}$$

Occurrences (1): 22481924:(16)

EQ-015/W.35.v1 — Ownership share o_t

domain: W • tier: Definition • status: current • coq: definition • parents: EQ-015/M.03.v1, EQ-015/W.07.v1

$$o_t = (sum_{iinB} W_{(i,t)}^M) / (sum_i W_{(i,t)}^M)$$

Occurrences (1): 22481924:(17)

EQ-015/W.36.v1 — Non-collapse: redistribution vs ownership reproduction

domain: W • tier: Definition • status: current • coq: closed • parents: EQ-015/M.03.v1, EQ-015/W.07.v1

current redistribution $\neq$ future ownership reproduction

Occurrences (1): 22481924:(18)

EQ-015/W.37.v1 — Conversion-gate control G^{conv}

domain: W • tier: Definition • status: current • coq: definition • parents: EQ-015/M.03.v1, EQ-015/W.08.v1

$$G_{(j,t)}^{conv}(e) = 1 - [V_t(e| - j)/V_t(e)]_+$$

Occurrences (1): 22481924:(26)

EQ-015/W.38.v1 — Credible-exit index X

domain: W • tier: Definition • status: current • coq: wrapped_related • parents: EQ-015/M.03.v1, EQ-015/W.08.v1

$$X_{(i,t)}(e; j) = clip(V_{(i,t)}(e| - j)/V_{(i,t)}(e), 0, 1)$$

Occurrences (1): 22481924:(27)

EQ-015/W.39.v1 — Dependency exposure D

domain: W • tier: Definition • status: current • coq: definition • parents: EQ-015/M.03.v1, EQ-015/W.08.v1

$$D_{(i \rightarrow j, t)}(g) = sum_{(e \in E(g))} w_e(g) G_{(j,t)}^{conv}(e) [1 - X_{(i,t)}(e; j)]$$

Occurrences (1): 22481924:(28)

EQ-015/W.40.v1 — Non-collapse: concentration vs gate vs dependency

domain: W • tier: Definition • status: current • coq: closed • parents: EQ-015/M.03.v1, EQ-015/W.08.v1

$$Concentration \neq G^{conv} \neq D$$

Occurrences (1): 22481924:(29)

EQ-015/W.41.v1 — Conditional non-collapse: concentration/dependency/agency loss

domain: W • tier: Definition • status: current • coq: wrapped_related • parents: EQ-015/M.03.v1, EQ-015/W.08.v1

$$concentration \not\Rightarrow dependency \not\Rightarrow \text{agency loss}$$

Occurrences (1): 22481924:(53)

EQ-015/W.42.v1 — Human Systemic Position index P_t^H

domain: W • tier: Definition • status: current • coq: definition • parents: EQ-015/M.03.v1, EQ-015/W.10.v1

$$P^H_t = [(\Gamma^{\text{eff}}_t)^{\theta_\Gamma} (A^{\text{corr}}(H,t))^{\theta_A} (\Lambda^{\text{live}}(H,t))^{\theta_\Lambda} r^{\theta_R} (S^H_t)^{\theta_S} (X^H_t)^{\theta_X}] / (1 + D^H_t)^{\theta_D}$$

Occurrences (1): 22481924:(40)

EQ-015/W.43.v1 — Growth-rate decomposition of P_t^H (part 1)

domain: W • tier: Definition • status: current • coq: wrapped_related • parents: EQ-015/M.03.v1, EQ-015/W.10.v1

$$P^H_{\text{dot}} / P^H_t = \theta_\Gamma (\Gamma^{\text{dot}}_{\text{eff}} / \Gamma^{\text{eff}}_t) + \theta_A (A^{\text{dot}}_{\text{corr}}(H,t) / A^{\text{corr}}(H,t)) + \theta_R (r^{\text{dot}}_t / r^H_t) + \theta_\Lambda (\Lambda^{\text{dot}}_{\text{live}}(H,t) / \Lambda^{\text{live}}_t)$$

Occurrences (1): 22481924:(41)

EQ-015/W.44.v1 — Growth-rate decomposition of P_t^H (part 2)

domain: W • tier: Definition • status: current • coq: wrapped_related • parents: EQ-015/M.03.v1, EQ-015/W.10.v1

$$\dots + \text{theta_S} (\text{S_dot}^H_t / \text{S}^H_t) + \text{theta_X} (\text{X_dot}^H_t / \text{X}^H_t) - \text{theta_D} (\text{D_dot}^H_t / (1 + \text{D}^H_t))$$

Occurrences (1): 22481924:(42)

EQ-015/W.45.v1 — Non-collapse: output growth vs P_t^H growth

domain: W • tier: Definition • status: current • coq: definition • parents: EQ-015/M.03.v1, EQ-015/W.10.v1

$$\text{Y_dot}_t > 0 \Rightarrow \text{P_dot}^H_t > 0$$

Occurrences (1): 22481924:(43)

EQ-015/W.46.v1 — Abundance-Without-Agency (AWA_t) tuple

domain: W • tier: Definition • status: current • coq: wrapped_related • parents: EQ-015/M.03.v1, EQ-015/W.10.v1

$$\text{AWA}_t = \langle \text{Y}(\text{up}), \text{Gamma}^{\text{eff}} \text{adequate}, \text{O}^H(\text{down}), \text{X}^H(\text{down}), \text{D}^H(\text{up}), \text{Lambda}^{\text{live}}_H(\text{down}), \text{A}^{\text{corr}}_H(\text{down}), \text{r}_H(\text{down}), \text{S}_H(\text{down}) \rangle$$

Occurrences (1): 22481924:(44)

EQ-015/W.47.v1 — Social-standing dynamic S_t^H

domain: W • tier: Definition • status: current • coq: definition • parents: EQ-015/M.01.v1, EQ-015/W.11.v1

$$\text{S_dot}^H_t = \text{s}_1 \text{W}_t + \text{s}_2 \text{N}_t - \text{delta}_S \text{S}^H_t$$

Occurrences (1): 22481924:(36)

EQ-015/W.48.v1 — Time-budget identity

domain: W • tier: Definition • status: current • coq: closed • parents: EQ-015/M.01.v1, EQ-015/W.11.v1

$$1 = l_{\text{wage}} + l_{\text{care}} + l_{\text{learn}} + l_{\text{civic}} + l_{\text{leisure}}$$

Occurrences (1): 22481924:(37)

EQ-015/W.49.v1 — Human-capability-stock dynamic H_t^{cap}

domain: W • tier: Definition • status: current • coq: definition • parents: EQ-015/M.03.v1, EQ-015/W.12.v1

$$\text{H_dot}^{\text{cap}}_t = \text{f}(\text{Care}_t, \text{Health}_t, \text{Education}_t, \text{Nutrition}_t, \text{Community}_t) - \text{delta}_H \text{H}^{\text{cap}}_t$$

Occurrences (1): 22481924:(38)

EQ-015/W.50.v1 — Non-collapse: productive vs social necessity

domain: W • tier: Definition • status: current • coq: definition • parents: EQ-015/M.03.v1, EQ-015/W.12.v1

$$\text{productive necessity of humans} \neq \text{social necessity of human reproduction}$$

Occurrences (1): 22481924:(39)

EQ-015/W.51.v1 — Non-collapse: machine expansion vs human expansion

domain: W • tier: Definition • status: current • coq: definition • parents: EQ-015/M.01.v1, EQ-015/W.16.v1

Machine expansion $\not\approx$ Human expansion

Occurrences (1): 22481926:(1)

EQ-015/W.52.v1 — Human Conversion Vector C_t^H

domain: W • tier: Definition • status: current • coq: definition • parents: EQ-015/M.01.v1, EQ-015/W.16.v1

$C^H_t = \langle \text{Gamma}^{\text{eff}}_t, X^H_t, \text{Lambda}^{\text{live}}_t(H,t), A^{\text{corr}}_t(H,t), R^{\text{route}}_t(H,t), \\ W^{\text{world}}_t(H,t), r_t(H,t), H^{\text{cap}}_t, S_t(H,t) \rangle$

Occurrences (1): 22481926:(11)

EQ-015/W.53.v1 — Conversion elasticity $\eta^{HC}(\mathbf{j},t)$

domain: W • tier: Definition • status: current • coq: definition • parents: EQ-015/M.01.v1, EQ-015/W.16.v1

$$\eta^H C_{(j,t)} = d(\ln C_{(j,t)}^H) / d(\ln M_t)$$

Occurrences (1): 22481926:(12)

EQ-015/W.54.v1 — Elasticity sign constraints

domain: W • tier: Definition • status: current • coq: wrapped_related • parents: EQ-015/M.01.v1, EQ-015/W.16.v1

$$\eta^H C_\Gamma > 0, \eta^H C_{(rH)} < 0, \eta^H C_X < 0$$

Occurrences (1): 22481926:(13)

EQ-015/W.55.v1 — Reversibility window W_j

domain: W • tier: Definition • status: current • coq: definition • parents: EQ-015/M.01.v1, EQ-015/W.19.v1

$W_j = \{ t : C^{\text{rec}}_j(j,t) \leq C_{\text{bar}_j} \text{ AND } \tau^{\text{rec}}_j(j,t) \leq \tau_{\text{bar}_j} \}$

Occurrences (1): 22481926:(17)

EQ-015/W.56.v1 — Capability-conversion gap $\Delta g(\mathbf{j},t)$

domain: W • tier: Definition • status: current • coq: wrapped_related • parents: EQ-015/M.01.v1, EQ-015/W.19.v1

$$\Delta g_{(j,t)} = [g_{(M,t)} - g_{(Cj,t)}]_+$$

Occurrences (1): 22481926:(18)

EQ-015/W.57.v1 — Urgency index $U(\mathbf{j},t)$

domain: W • tier: Definition • status: current • coq: definition • parents: EQ-015/M.01.v1, EQ-015/W.19.v1

$$U_{(j,t)} = \Delta g_{(j,t)} * L_{(j,t)} * S_{(j,t)} * \tau_{(j,t)}^{\text{rec}}$$

Occurrences (1): 22481926:(19)

EQ-015/W.58.v1 — Urgency vector U_t

domain: W • tier: Definition • status: current • coq: wrapped_related • parents: EQ-015/M.01.v1, EQ-015/W.19.v1

$$U_t = \langle U_{(1,t)}, \dots, U_{(J,t)} \rangle$$

Occurrences (1): 22481926:(20)

EQ-015/W.59.v1 — Maximum urgency U_t^{max}

domain: W • tier: *Definition* • status: *current* • coq: *wrapped_related* • parents: *EQ-015/M.01.v1, EQ-015/W.19.v1*

$$U_t^{max} = \max_j U_{(j,t)}$$

Occurrences (1): 22481926:(21)

16 EQ-016 — Regime-split discriminant

domain: — • tier: *Dr* • status: *current* • coq: *root_layer_unwired* • parents: *ModeEquation*

$$\lambda_{\text{c}} = D^2/(4MK)$$

17 EQ-017 — Telegraph/relaxation reduced form

domain: — • tier: *finite_diagnostic* • status: *current* • coq: *root_layer_unwired* • parents: *SpinePDE, Telegraph*

$$\tau_{\text{R}} \dot{\mathbf{I}}_{\text{R}} + \mathbf{L}_{\text{R}} \mathbf{I}_{\text{R}} = \mathbf{S}_{\text{R}} + \eta_{\text{R}}$$

18 EQ-018 — DRL two-field apparatus state tuple

domain: — • tier: *Dr* • status: *current* • coq: *root_layer_unwired* • parents: *LivingGeometry*

$$\mathbb{G}_{\text{n}} = (\mathbf{G}_{\text{n}}, \Lambda_{\text{n}}, \mathcal{T}_{\text{n}}, \Theta_{\text{n}})$$

19 EQ-019 — Reader/record stacked vector

domain: — • tier: *Dr* • status: *current* • coq: *root_layer_unwired* • parents: *EQ-018*

$$\mathbf{X}_{\text{n}}^{\text{A}} = (\Phi_{\text{n}}^{\text{I}}, \Psi_{\text{n}}^{\text{I}})^{\text{T}}$$

20 EQ-020 — Coupled transport operator $\mathbb{G}_{\text{n}}$

domain: — • tier: *Dr* • status: *current* • coq: *root_layer_unwired* • parents: *EQ-019*

$$\mathbb{G}_{\text{n}} = \mathbf{L}_{\{\mathbf{G}_{\text{n}}\}} \otimes \mathbf{I}_{\mathcal{F}} + \mathbf{I}_{\{\mathbf{G}_{\text{n}}\}} \otimes \mathbf{C}_{\mathcal{F}} + \mathbf{C}_{\text{int},\text{n}}$$

21 EQ-021 — Metric-G skew/symmetric decomposition of $\mathbb{G}_{\text{n}}$

domain: — • tier: *Th_coqc* • status: *current* • coq: *root_layer_unwired* • parents: *T1*

$$\mathbb{G}_{\text{n}}^{(+)} = (\mathbb{G}_{\text{n}} + \mathbb{G}_{\text{n}}^{\text{adjG}})/2, \mathbb{G}_{\text{n}}^{(-)} = (\mathbb{G}_{\text{n}} - \mathbb{G}_{\text{n}}^{\text{adjG}})/2 \text{ [retention-metric G-adjoint, not naive transpose]; } \langle \mathbf{z}, \mathbb{G}_{\text{n}}^{(-)} \mathbf{z} \rangle_{\text{G}} = 0$$

22 EQ-022 — Discrete reader/record coupled recurrence

domain: — • tier: *finite_diagnostic* • status: *current* • coq: *root_layer_unwired* • parents: *ReaderRecordEqs*

$$\text{Reader: } \mathbf{M} \delta_{\text{t}}^2 \Phi_{\text{n}} + \mathbf{D} \delta_{\text{t}}^{\text{c}} \Phi_{\text{n}} + \mathbf{K}\mathbf{G}[\Theta_{\text{n}}] \Phi_{\text{n}} + \nabla \mathbf{V}(\Phi_{\text{n}}) - \mathbf{J}_{\text{n}} = \mathcal{R}_{\Phi,\text{n}}; \text{Record: } \mathbf{M} \delta_{\text{t}}^2 \Psi_{\text{n}} - \mathbf{D} \delta_{\text{t}}^{\text{c}} \Psi_{\text{n}} + \mathbf{K}\mathbf{G}[\Theta_{\text{n}}]^{\text{T}} \Psi_{\text{n}} + \nabla^2 \mathbf{V}(\Phi_{\text{n}}) \Psi_{\text{n}} = \mathcal{R}_{\Psi,\text{n}}$$

23 EQ-023 — Potential gradient/Hessian (mother potential)

domain: — • tier: *finite_diagnostic* • status: *current* • coq: *root_layer_unwired* • parents: *ResidualsAndPotentials*

$$\nabla V_i(\Phi) = a_i \Phi_i + b_i \Phi_i^3; (\nabla^2 V)_{ii} = a_i + 3b_i \Phi_i^2$$

24 EQ-024 — Reader one-step implicit solve

domain: — • tier: *finite_diagnostic* • status: *current* • coq: *root_layer_unwired* • parents: *GaussJordanStepper*

$$A_{\Phi} = M/\Delta t^2 + D/2\Delta t; b_{\Phi,n} = \mathcal{R}_{\Phi,n} + J_n - K\mathbb{G}_n\Phi_n - \nabla V(\Phi_n) + (2M/\Delta t^2)\Phi_n + (D/2\Delta t - M/\Delta t^2)\Phi_{n-1}; A_{\Phi}\Phi_{n+1} = b_{\Phi,n}$$

25 EQ-025 — Record one-step implicit solve

domain: — • tier: *finite_diagnostic* • status: *current* • coq: *root_layer_unwired* • parents: *GaussJordanStepper*

$$A_{\Psi} = M/\Delta t^2 - D/2\Delta t; b_{\Psi,n} = \mathcal{R}_{\Psi,n} - K\mathbb{G}_n\hat{T}\Psi_n - \nabla^2 V(\Phi_n)\Psi_n + (2M/\Delta t^2)\Psi_n - (M/\Delta t^2 + D/2\Delta t)\Psi_{n-1}; A_{\Psi}\Psi_{n+1} = b_{\Psi,n}$$

26 EQ-026 — Record/readout/epistemic composite readout

domain: — • tier: *Th_coqc* • status: *current* • coq: *root_layer_unwired* • parents: *Face.10.StrictGap*

$$R_0 = \Gamma_{\{RAR,0\}}(D_0) = \Omega_{A \circ A \circ \Pi \circ T_{\Sigma}}(D_0)$$

27 EQ-027 — N2 — Readout equation

domain: — • tier: *Th_coqc* • status: *current* • coq: *root_layer_unwired* • parents: *EQ-026*

$$M_A[n] = K_A \cdot \theta(E[n]) + \eta_{sel} + \eta_{map} + \eta_{self}$$

28 EQ-028 — N2 epistemic-error non-collapse law

domain: — • tier: *Th_coqc* • status: *current* • coq: *root_layer_unwired* • parents: *EQ-027*

$$M_A[n] \neq \theta(E) \quad \forall n; \epsilon_{tot} = \eta_{sel} + \eta_{map} + \eta_{self} > 0 \quad \forall n$$

Occurrences (1): (26)

29 EQ-029 — bR cross-domain lineage ledger conservation

domain: — • tier: *Dr* • status: *current* • coq: *root_layer_unwired* • parents: *EQ-008*

$$I_Q = I_B + O_C + O_P + O_B$$

30 EQ-030 — Standard Model domain state tuple

domain: — • tier: *Dr* • status: *current* • coq: *root_layer_unwired* • parents: *EQ-008*

$$Z_n = (\Phi_n, \Psi_n, \Theta_n, U_n, \Sigma_n, \mathcal{T}_n, \Lambda_n)$$

31 EQ-031 — Gauge-equivalence readout

domain: — • tier: *Dr* • status: *current* • coq: *root_layer_unwired* • parents: *EQ-030*

$r = 0(X); 0(hX) = 0(X) \forall X \implies h \text{ internal rename}; X \sim X' \iff 0(X) = 0(X'); \text{physical state} = [X]$

32 EQ-032 — No-decoder-recovers-state theorem

domain: — • tier: *Th_cogc* • status: *current* • coq: *root_layer_unwired* • parents: *EQ-031*

$\forall X, R, 0, x1, x2 : x1 \neq x2 \rightarrow 0(x1) = 0(x2) \rightarrow \forall D : D(0(x1)) = x1 \rightarrow D(0(x2)) = x2 \rightarrow \text{False}$

33 EQ-033 — Gauge redundancy forces undecodability

domain: — • tier: *Th_cogc* • status: *current* • coq: *root_layer_unwired* • parents: *EQ-032*

$\forall X, R, 0, h, x : h(x) \neq x \rightarrow 0(h(x)) = 0(x) \rightarrow \forall D : D(0(h(x))) = h(x) \rightarrow D(0(x)) = x \rightarrow \text{False}$

34 EQ-034 — True state exists but no total decoder

domain: — • tier: *Th_cogc* • status: *current* • coq: *root_layer_unwired* • parents: *EQ-032, EQ-033*

$x1 \neq x2 \rightarrow 0(x1) = 0(x2) \rightarrow (\exists x : X, x = x1) \wedge (\exists x : X, x = x2) \wedge \neg(\exists D : R \rightarrow X, D(0(x1)) = x1 \wedge D(0(x2)) = x2)$

35 EQ-035 — Retained (metric-G) inner product

domain: — • tier: *Dr* • status: *current* • coq: *root_layer_unwired* • parents: *EQ-021*

$\langle x, y \rangle_G = x^\dagger G y, G > 0; \|x\|_G^2 = x^\dagger G x \geq 0$

36 EQ-036 — Antisymmetry factor r=-1 (SU(3) closure step 1)

domain: — • tier: *Th_cogc* • status: *current* • coq: *root_layer_unwired* • parents: *EQ-030*

$\Omega(\dots x_i, x_{\{i+1\}} \dots) = r \cdot \Omega(\dots x_{\{i+1\}}, x_i \dots); r^2 = 1 \implies r = \pm 1; r = -1$

37 EQ-037 — Minimal odd arity k=3 (SU(3) closure step 2)

domain: — • tier: *Th_cogc* • status: *current* • coq: *root_layer_unwired* • parents: *EQ-036*

$\Omega(x_1 \dots x_k) = \Omega(x_2 \dots x_k, x_1); (-1)^{\wedge\{k-1\}} = 1 \implies k \text{ odd}; k = 3$

38 EQ-038 — Minimal dimension 3 (SU(3) closure step 3)

domain: — • tier: *Th_cogc* • status: *current* • coq: *root_layer_unwired* • parents: *EQ-037*

$\Omega(x, y, z) = 0 \text{ if } \dim V < 3; \dim V \geq 3; \dim V = 3, V \cong \mathbb{C}^3$

39 EQ-039 — Unitary-group closure (SU(3) closure step 4)

domain: — • tier: Th_cogc • status: current • coq: root_layer_unwired • parents: EQ-038

$$U^\dagger G U = G, \quad G=I \implies U^\dagger U=I \implies U \in U(3)$$

40 EQ-040 — Determinant-one selection giving SU(3) (SU(3) closure step 5)

domain: — • tier: Th_cogc • status: current • coq: root_layer_unwired • parents: EQ-039

$$\Omega(Ux, Uy, Uz) = \det(U) \cdot \Omega(x, y, z); \quad \det U = 1; \quad SU(3) = \{U \in U(3) : \det U = 1\}$$

41 EQ-041 — su(3) dimension count = 8 (SU(3) closure step 6)

domain: — • tier: Th_cogc • status: current • coq: root_layer_unwired • parents: EQ-040

$$9 - 1 = 8 = \dim \mathfrak{su}(3)$$

42 EQ-042 — Admissible gauge-automorphism group and transition relabeling

domain: — • tier: Dr • status: current • coq: root_layer_unwired • parents: EQ-041

$$\mathcal{A} = \{ h : 0h=0, \quad hF=Fh, \quad h^\dagger Gh=G \}; \quad U'_{\{j \leftarrow i\}} = h_{_j} \ U_{\{j \leftarrow i\}} \ h_{_i}^{-1}$$

43 EQ-043 — Loop holonomy and curvature defect

domain: — • tier: Dr • status: current • coq: root_layer_unwired • parents: EQ-042

$$H_C = \prod_{_e \in C} U_e; \quad H_C' = h \ H_C \ h^{-1}; \quad \mathcal{K}_C = H_C - I$$

44 EQ-044 — Unified force residual decomposition

domain: — • tier: Dr • status: current • coq: root_layer_unwired • parents: UnifiedForce_action

$$\mathcal{F}_all = \mathcal{F}_G + \mathcal{F}_EM + \mathcal{F}_W + \mathcal{F}_S + \mathcal{F}_res; \quad \chi^4 = 1 - \|\mathcal{F}_res\|^2 / \|\mathcal{F}_all\|^2$$

45 EQ-045 — Blind-pipeline gauge-algebra dimension result

domain: — • tier: finite_diagnostic • status: current • coq: root_layer_unwired • parents: EQ-041

$$\dim Z(\mathfrak{g}) = 1; \quad \mathfrak{g} \cong \mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3); \quad \dim = 1+3+8 = 12$$

46 EQ-046 — U(1) phase action

domain: — • tier: Dr • status: current • coq: root_layer_unwired • parents: EQ-042

$$x \mapsto e^{i\alpha} x$$

47 EQ-047 — General representation action and invariant tensor

domain: — • tier: Dr • status: current • coq: root_layer_unwired • parents: EQ-042

$$x \mapsto \rho_R(g)x; C_2(R); \mathcal{I} : R_1 \otimes \dots \otimes R_n \rightarrow 1$$

48 EQ-048 — Color-singlet tensor decompositions

domain: — • tier: Dr • status: current • coq: root_layer_unwired • parents: EQ-047

$$3 \otimes 3^- \rightarrow 1 + \dots; 3 \otimes 3 \otimes 3 \rightarrow 1 + \dots \text{ via } \epsilon_{\{abc\}} x^a y^b z^c$$

49 EQ-049 — Triality Z_3 labeling and closure

domain: — • tier: Th_coqc • status: current • coq: root_layer_unwired • parents: EQ-041

$$P/Q \cong Z_3; \tau(p, q) = p + 2q \pmod{3}; \tau(3) = 1, \tau(3^-) = 2, \tau(8) = 0; \tau(R \otimes 8) = \tau(R); 1 + 2 = 0, 1 + 1 + 1 = 0$$

50 EQ-050 — Physical Hilbert subspace at zero constraint

domain: — • tier: Dr • status: current • coq: root_layer_unwired • parents: EQ-042

$$\mathcal{H}_{\text{physical}} \subseteq \mathcal{H}_{\{\tau=0\}}$$

51 EQ-051 — $Z_3/2D$ confinement area law

domain: — • tier: finite_diagnostic • status: current • coq: root_layer_unwired • parents: EQ-049

$$z_e \in Z_3; u_p = \prod z_e \epsilon; S = \kappa \Sigma |u_{p-1}|^2; |\omega - 1|^2 = 3; r = e^{-3\kappa}; q(\kappa) = (1-r)/(1+2r); \langle W(C) \rangle = q^{A(C)}; \sigma = -\log q > 0$$

52 EQ-052 — Retained-information vs surface-entropy inequality

domain: — • tier: finite_diagnostic • status: current • coq: root_layer_unwired • parents: EQ-051

$$N_A \lesssim \mu^A; \Sigma N_A \rho^A \lesssim \Sigma (\mu \rho)^A; \mu \rho < 1 \iff I_{\text{retain}} = -\log \rho > S_{\text{surface}} = \log \mu$$

53 EQ-053 — Character-expansion confinement machinery

domain: — • tier: Th_coqc • status: current • coq: root_layer_unwired • parents: EQ-051

$$S_p(U) = \kappa \|U - I\|_F^2; \|U - I\|^2 = 6 - 2\text{Re Tr } U; K_\kappa(U) = \exp[\kappa(\chi_3 + \chi_3^-)] = \Sigma_R c_R(\kappa) \chi_R; c_R(\kappa) = \int_{\{SU(3)\}} K_\kappa \chi_R^*; dK/d\kappa = (\chi_3 + \chi_3^-)K; c_{R'}(\kappa) = \Sigma_S (N_{\{3S\}}^R + N_{\{3^-S\}}^R) c_S; c_R(0) = \delta_{\{R0\}}; u = c_3/3c_0; v = c_8/8c_0$$

54 EQ-054 — All-order character recursion identity

domain: — • tier: Th_coqc • status: current • coq: root_layer_unwired • parents: EQ-053

$$c_{0'}(\kappa) = 2c_3(\kappa) \text{ [source: InfoAllOrderCharacter.v / all_order_character_v1_0.py -- a different file than EQ-053's SM §11]}$$

55 EQ-055 — Group-averaging projector idempotence/positivity

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *EQ-042*

$$P_e = \int_{\mathcal{G}} \rho_e(h) d\mu(h); P^2 = P; P^\dagger_G = P; \|P_e T\|_G \leq \|T\|_G$$

56 EQ-056 — Reader/record characteristic-root sign split

domain: — • tier: *finite_diagnostic* • status: *current* • coq: *root_layer_unwired* • parents: *EQ-022*

$$\text{reader roots: } M s^2 + D s + K = 0 \implies \text{Re}(s) = -D/(2M); \text{record roots: } M s^2 - D s + K = 0 \\ \implies \text{Re}(s) = +D/(2M)$$

57 EQ-057 — Rational growth ratio dissolving the I1 continuum-completeness injection

domain: — • tier: *finite_diagnostic* • status: *current* • coq: *root_layer_unwired* • parents: *EQ-056*

$$\rho(n) := r^n, r \in \mathbb{Q}, r > 1$$

58 EQ-058 — Mass-ratio geometric-mean fit values (three distinct sectors)

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *EQ-041*

$$r_U = \sqrt{(587.963 \cdot 135.827)} \approx 282.597083; r_D = \sqrt{(20.00 \cdot 44.75)} \approx 29.917803; r_E = \\ \sqrt{(206.85 \cdot 16.81)} \approx 58.970290 \text{ (three distinct numbers, same formula, different inputs --} \\ \text{not an equality)}$$

59 EQ-059 — Golden-ratio identity (proposed use rejected)

domain: — • tier: *Th_coqc* • status: *current* • coq: *root_layer_unwired* • parents: *EQ-058*

$$\phi = (1+\sqrt{5})/2, x^2-x-1 = 0 \text{ (the identity itself is Th_coqc; proposed use as item-1's r is} \\ \text{Open/REJECTED, not adopted)}$$

60 EQ-060 — Regime discriminant Δ^2 and root behavior

domain: — • tier: *finite_diagnostic* • status: *current* • coq: *root_layer_unwired* • parents: *ModeEquation*

$$\Delta^2 = D^2 - 4MK; \lambda_c < 1 : \text{Re}(s) = \mp D/2M \text{ (K-independent); } \lambda_c > 1 : \text{Re}(s) = [D \pm \sqrt{(D^2 - 4MK)}]/2M \\ \text{(K-dependent)}$$

61 EQ-061 — Step-reader/record implicit coefficients

domain: — • tier: *finite_diagnostic* • status: *current* • coq: *root_layer_unwired* • parents: *GaussJordanStepper*

$$\text{step_reader: } A_\Phi = M/dt^2 + D/2dt; \text{step_record: } A_\Psi = M/dt^2 - D/2dt$$

62 EQ-062 — Reader/record envelope negative finding

domain: — • tier: *finite_diagnostic* • status: *current* • coq: *root_layer_unwired* • parents: *EQ-061*

$|\Phi|$ envelope: $1.215839 \rightarrow 0.183275$; $|\Psi|$ envelope: $1.105998 \rightarrow 322.802505$; $\Phi \cdot \Psi$ (n=10) = $-1.188008 \rightarrow \Phi \cdot \Psi$ (n=1999) = -59.738547 (ratio $\approx 50.28\times$) -- negative finding, refutes its own opening hypothesis

63 EQ-065 — Declared branch-tape Π_0 construction

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *EQ-008*

$\Pi_0 = 3\lambda_U + 3\lambda_D + \lambda_E = 6.328453553357985$ (0.0037% error vs the fixture's known-M reconstruction). Branch initial conditions explicitly disclosed as arbitrary/uncalibrated; must not be conflated with the independently-derived real-PDG-mass $\Pi_0 \approx 6.9888$ elsewhere in the domain.

64 EQ-066 — Order-vacuum threshold closure

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *EQ-065*

$\Pi_0 > \alpha_{\text{ord}} \Rightarrow \text{ORDERED_READY}$, $r^* = 3.823356105009073$. Caveat: ORDERED_READY on this stepper is structurally guaranteed regardless of branch-tape data ($\lambda_j \in (0,1]$ forces $\Pi_0 \in (0,7]$, $\alpha_{\text{ord}} = -0.5$ sits below that bound) -- not evidence the branch construction is predictive.

65 EQ-067 — Native vacuum amplitude

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *EQ-066*

$v_{\text{native}} = \sqrt{2 r^*} = 2.7652689218262565$ (no physical unit attached)

66 EQ-069 — RETRACTED — continuous- Θ graph-accumulation approach

domain: — • tier: *RETRACTED* • status: *historical* • coq: *root_layer_unwired* • parents: *EQ-008*

RETRACTED in Readout Genesis: RETRACTED

RETRACTED 2026-07-26 (continuum contamination): $\text{cond}\#(G[\Theta])$ is a smooth bijection $\Theta \leftrightarrow R$ -- a coordinate, not an observable -- so a single continuous knob cannot force generations. Number kept reserved so the stream stays stable; see domains/standards_model/item1_exploration/CONTINUUM_ARC_ERROR_NOTE.md.

67 EQ-070 — RETRACTED — continuous- Θ graph-accumulation approach (duplicate entry)

domain: — • tier: *RETRACTED* • status: *historical* • coq: *root_layer_unwired* • parents: *EQ-069*

RETRACTED in Readout Genesis: RETRACTED

RETRACTED 2026-07-26, same content/reason as EQ-069, number reserved.

68 EQ-071 — RETRACTED — continuous- Θ graph-accumulation approach (duplicate entry)

domain: — • tier: *RETRACTED* • status: *historical* • coq: *root_layer_unwired* • parents: *EQ-069*
RETRACTED in Readout Genesis: RETRACTED

RETRACTED 2026-07-26, same content/reason as EQ-069/EQ-070, number reserved.

69 weld — The one-line master equation (the weld)

domain: — • tier: *Th_coqc* • status: *current* • coq: *root_layer_unwired* • parents: *EQ-008, MQ08-stepper*

$$\delta_R = (a \# b) \vdash [\text{Th_coqc}] \ L_R = D_W - W \vdash [\text{Dr}] \ F \ (\text{MQ.08 stepper}) \equiv \{ q_D : q_D \circ F = F \#_D \circ q_D \}$$

Occurrences (4): (1), (3), (4), (5)

weld/E.01.v1 — Knowledge as stabilized representation: robust to admissible variation

domain: *E* • tier: *Definition* • status: *current* • coq: *definition* • parents: *weld/M.03.v1*

Know_A(W) = 1 iff Dist(R_A[n], R_A^ν[n]) ≤ ϵ_K for all admissible variations ν on window W. Independently echoed as: Knowing = Coupled System + Selection + Retention + Model + Error + Correction.

Occurrences (2): 18925129:(2), 19640361:(13)

weld/E.02.v1 — Possession-Constitution Collapse: a condition on who may possess knowledge is wrongly treated as a condition on what may constitute epistemically significant structure

domain: *E* • tier: *Definition* • status: *current* • coq: *closed* • parents: *weld/M.03.v1*

SC: K(S,p)→Subject(S); the collapse the paper rejects, HSC: Epi(X,p)→Knewer(X,p); Possession-Constitution Collapse named; Knewer fetishism: treating a knower's recognized identity/credentials as possessing epistemic value in itself once the evidential/relational work behind it is abstracted away

Occurrences (6): 22301202:SC, 22301202:HSC, 22301202:Def-1, 22301202:Def-2, 22301202:Prop-1, 22301202:Def-3

weld/E.03.v1 — Knowledge as an auditable bounded status granted to a claim, not inherited from speaker identity

domain: *E* • tier: *Definition* • status: *current* • coq: *definition* • parents: *weld/M.03.v1*

$\sigma_K(p)$ =Admit_E(p|Agent,D,C,O,Access,Language,Tools,Rights,Prov,Ev,Method,Infer,Assumptions,Uncertainty,Scope,Obj,CorrCond,Independence,Lineage); K_p=(p,C_p,P_p,E_p,M_p,I_p,U_p, Ω _p,O_p,L_p, $\sigma_K(p)$); status space $\sigma_K(p) \in \{\text{ADMITTED, LOCAL, TRANSPORTABLE, UNRESOLVED, OBSTRUCTED, RETRACTED, SUPERSEDED}\}$; three levels K_{prop}(p)= $\sigma_K(p) \subset K_{op}(p)=\langle K_{prop}, \text{Task, Completion} \rangle \subset K_{trans}(p)=\langle K_{op}, \Delta\Theta, \Delta I, \Delta U, \Delta \text{Action, Persistence, Transport, Viability} \rangle$

Occurrences (20): 21529456:Def-1, 21529456:(22), 21529456:(23), 21529456:(31), 21529456:(32), 21529456:(56), 21529456:(57), 21529456:(58), 21529456:(59), 21529456:(60), 21529456:(61), 21529456:(62) (+8 more)

weld/E.04.v1 — Practical/operational effectiveness is a separate axis from epistemic status

domain: *E* • tier: *Definition* • status: *current* • coq: *closed* • parents: *weld/M.03.v1*

$$\Pi_{prac}(p) = \text{TestPerformance}(Y, \hat{Y}, \text{intervention}, C, O); \sigma_K(p) \neq \Pi_{prac}(p)$$

Occurrences (1): 21529456:(25)

weld/E.05.v1 — A three/four/five-valued gate outcome or provenance label that never silently converts absence-of-evidence into pass or fail

domain: E • tier: Definition • status: current • coq: definition • parents: weld/M.03.v1

$\chi_G \in \{1, 0, \perp\}$, 1=ADMITTED, 0=OBSTRUCTED, $\perp$ =UNRESOLVED (thirteen admission gates G0-G13); independently, $\chi_t(d) \in \{\text{FORCED, DERIVED, POSITED, BORROWED, OPEN}\}$ as a per-distinction provenance ledger $L_Q, t = \{(d, \chi_t(d))\}$

Occurrences (3): 21529456:(27), 22307148:(9), 22307148:(10)

weld/E.06.v1 — A candidate domain state must be sufficient, and a quotient may not merge states differing on a required future invariant

domain: E • tier: Definition • status: current • coq: closed • parents: weld/M.03.v1

$$Suff_{E,L}(Z_E^{cand}; Q, O, c, T) \in \{1, 0, \perp\}; Inv_E(z) \neq Inv_E(z') \implies q_E(z) \neq q_E(z')$$

Occurrences (2): 21529456:(28), 21529456:(29)

weld/E.07.v1 — The strongest publicly assignable claim tier is bounded by the weakest load-bearing gate

domain: E • tier: Dr • status: current • coq: closed • parents: weld/M.03.v1

$$\tau_{public}(p) \leq \inf_{g \in G_p} \tau(g) (\text{Weakest-link claim ceiling, proved})$$

Occurrences (2): 21529456:(30), 21529456:Prop-2

weld/E.08.v1 — Local knowledge may only be presented as global through a declared, tolerance-bounded commuting-square transport

domain: E • tier: Definition • status: current • coq: closed • parents: weld/M.03.v1

$K_local = K|_{\{\Omega_local\}}$; $T^Y_{ij} \circ K_i \cong K_j \circ T^C_{ij}$, $\epsilon_bridge = d(T^Y_{ij} \circ K_i, K_j \circ T^C_{ij})$; $\Omega(K) = \{(g, d, l, P, O, c) : \text{all required gates pass}\}$

Occurrences (5): 21529456:(33), 21529456:(34), 22307148:(11)-(12), 22307148:(13), 22307148:(14)

weld/E.09.v1 — A raw noisy sensor reading, smoothed by a finite trailing window into a windowed-average readout

domain: E • tier: Definition • status: current • coq: closed • parents: weld/M.01.v1

$S_n : X_n \rightarrow Z_n$; $z_n = \Pi_n S_n(X_n) + \eta_n$, $z_n \in \mathbb{Q}^{\sim\{m_n\}}$; $W_n = \{n-h_n+1, \dots, n\}$;
 $z^{\sim}_n = \sum_{j=0}^{\sim\{h_n-1\}} a_{\{n,j\}} z_{\{n-j\}}$

Occurrences (4): 21425420:(7), 21425420:(8), 21425420:(9), 21425420:(10)

weld/E.10.v1 — Bridge Burden: an inference from source metadata must name its mediating relation or it is pedigree substitution

domain: E • tier: Dr • status: current • coq: closed • parents: weld/M.02.v1

Bridge Burden: any inference from source metadata to a change in epistemic standing must identify the mediating relation that does the work; where none is supplied, the inference is pedigree substitution rather than source-sensitive epistemology.

Occurrences (1): 22301202:Prin-2

weld/E.11.v1 — Effort v0.3 Eq.1: NEW NON-COLLAPSE

domain: E • tier: Dr • status: *current* • coq: *open_prop* • parents: *weld/M.02.v1*

$\backslash\text{uncertainty} \backslash\text{not}\backslash\text{equiv} \backslash\text{stochastic mechanism} \backslash\backslash \backslash\text{stochastic mechanism}$
 $\backslash\text{not}\backslash\text{equiv} \backslash\text{other-agent control} \backslash\backslash \backslash\text{other-agent control} \backslash\text{not}\backslash\text{equiv}$
 $\backslash\text{unresolved residual}$

Occurrences (1): 22622206:eq.01

weld/H.01.v1 — The self as a typed quotient-readout bundling agency, identity, phenomenal ity, ownership, coherence, valence, policy, and lineage

domain: H • tier: *Definition* • status: *current* • coq: *definition* • parents: *weld/M.03.v1*

$S_A[n] = q_self(F^n[\delta_R, T_A, c_{\{0:n\}}]) = \langle A_A[n], \Delta_A[n], H_A[n], Phen_A^{str}[n], P_A^{lived}[n],$
 $Own_A[n], Coh_A[n], Val_A[n], \Pi_A[n], \Lambda_A[n] \rangle \sim self; \text{identity update } I_A[n+1] = q_id(I_A[n] \oplus A_A[n] \oplus P_A^{lived}[n] \oplus \Pi_A^{star}[n] \oplus Res_A[n] \oplus \Delta T_A[n])$

Occurrences (2): 21529456:(47), 21529456:(49)

weld/H.02.v1 — Dynamical, information, and phenomenal horizons — a proposed but only partially-derived weld

domain: H • tier: *Definition* • status: *current* • coq: *definition* • parents: *weld/M.02.v1*

$H_dyn: \Delta_A(\lambda) = D_A^2 - 4M_AK_A\lambda = 0; H_info(A) = \{z \in Z_A : Rec(z) \Rightarrow (E_b < \infty, R_p > 0)\};$
 $H_phen(A) = \{r = A_A^{acc} \Pi_A T_A(\delta_R) : PhenGate(r) = 1\}; H_dyn \dashv DI_K \dashv H_info \dashv IP_K \dashv H_phen$

Occurrences (7): 21529456:(50), 21529456:(51), 21529456:(52), 21529456:(53), 21529456:(54), 21529456:(55), 21425420:(31)

weld/H.03.v1 — Readout-of-readout, finite retention, and a governance state that composes context/lineage access, action formation, interruptibility, and policy revision

domain: H • tier: *finite_diagnostic* • status: *current* • coq: *definition* • parents: *weld/M.03.v1*

$R_A^{\{(1)\}}[n] = 0_A(q_A(Z_A[n])); c_n, T_n); R_A^{\{(2)\}}[n] = 0_A(R_A^{\{(1)\}}[n]; c_n, T_n);$
 $G_A^{MR}[n] = R_{\tau_g}(0_A(0_A(q_A(Z_A[n]))), C_A^{ret}, L_A^{acc}, F_A, \Pi_A); \text{no-free-governance}$
 $\text{law: CurrentReadout+SelectionPriority+SelectionStability+SelfReport} \not\Rightarrow G_A^{MR};$
 $\text{boundedness } B_G = I(R_A^{\{(2)\}}; Z_A) / \tau_c A < \infty, \ker 0_A^{\{(2)\}} \neq \emptyset$

Occurrences (18): 21529456:(67), 21529456:(68), 21529456:(69), 21529456:(71), 21529456:(72), 21529456:(73), 21529456:(74), 21529456:(75), 21529456:(76), 21529456:(77), 21529456:(78), 21529456:(79) (+6 more)

weld/H.04.v1 — The human-domain retained-difference graph and its discrete spine equation

domain: H • tier: *Definition* • status: *current* • coq: *definition* • parents: *weld/M.01.v1*

$G_H[n] = (V_H[n], E_H[n], w_H[n]), w_H[n](e) \in \mathbb{Q}_{>0}, \phi_n \in \mathbb{Q}^{|V_H[n]|}; \mu_H \delta^2 \phi_n + d_H \delta \phi_n$
 $+ \kappa_H L_H[n] \phi_n + \partial_Q V_H(\phi_n) = J_H[n] - \eta_H[n]$

Occurrences (8): 21425420:(1), 21425420:(2), 21425420:(3), 21425420:(4), 21425420:(5), 21425420:(6), 21425420:(11), 21425420:(30)

weld/H.05.v1 — A bounded transport relation must hold between AI-produced and human-produced knowledge objects before either substitutes for the other

domain: H • tier: *Definition* • status: *current* • coq: *definition* • parents: *weld/M.03.v1*

$T_{\{H \leftarrow AI\}} \circ K_{AI} \cong K_{H \rightarrow T_C}$, with explicit defects for semantic loss, source omission, authority laundering, uncertainty compression, and context mismatch

Occurrences (2): 21529456:(88), 21529456:(87)

weld/H.06.v1 — declared defect vector for a domain weld

domain: H • tier: Definition • status: current • coq: definition • parents: weld/M.02.v1

[print fallback: this entry's LaTeX transcription did not compile as standalone display math; showing the source ascii/latest text instead — the registry, JSON entry and docs site carry the LaTeX transcription unmodified]

$$\epsilon_H = \text{Def}(q^{\sim}_H \circ F, F \#_H q^{\sim}_H, 0_H, \text{Inv}_H)$$

Occurrences (2): 22498047:(7), 22424434:(7)

weld/H.07.v1 — session-end retention gate and unaided return battery

domain: H • tier: Definition • status: current • coq: definition • parents: weld/M.01.v1

$$\Delta \Omega^{\sim}_H s = B_s A_s, \text{rank} \ll d_H; \Omega^{\sim}_H \{s+1, 0\} = \Omega^{\sim}_H \{s, 0\} + \eta_s \Delta \Omega^{\sim}_H s; Y^{\sim} \text{return}_{\{s+\Delta\}} \\ = (R_{\text{rec}}, R_{\text{disc}}, T_{\text{new}}, Q_{\text{next}}); \text{RET} = (P^{\sim} \text{post}_{\{H, AI\}} - P^{\sim} \text{pre}_{\{H, AI\}}) - (P^{\sim} \text{post}_{\{H, C\}} - P^{\sim} \text{pre}_{\{H, C\}})$$

Occurrences (9): 22331922:EF-18 (Repair 7), 22331922:EF-19 (Repair 7), 22424434:(36), 22424434:(37), 22639311:Tunnel-v2.0: eq.14, 22550491:Eq. 33, 22550491:Eq. 34, 22550491:Eq. 35 (\label{eq:lowrank}), 22550491:Eq. 36 (\label{eq:gate})

weld/H.08.v1 — effective candidate count as readout-equivalence classes

domain: H • tier: Dr • status: current • coq: definition • parents: weld/M.03.v1

$$D_s^{eff} = |C_s / \sim_R|, d_s = D_s^{eff} / |C_s|; N_{distinct} = |\{C1, ..., Cn\} / \sim_Q|$$

Occurrences (3): 22331922:EF-07 (Repair 3), 22481928:(18), 22639311:Tunnel-v2.0:eq.15

weld/H.09.v1 — Retained experiential reorganization as a layer beneath, not identical to, later demonstrable capability

domain: H • tier: Open • status: current • coq: definition • parents: weld/M.02.v1

retained experiential reorganization $\rightarrow$ possible later CTSA crystallization; retained sensitivity $\rightarrow$ C_return, retained reweighting $\rightarrow$ A_return, retained reorganization $\rightarrow$ Q_next^better

Occurrences (2): 22357744:EMG-25, 22357744:EMG-26

weld/H.10.v1 — human agency, self-context, and external context-frame dominance

domain: H • tier: Definition • status: current • coq: definition • parents: weld/M.01.v1

A:=capacity to decide under context while retaining accountability; S:=(goals, constraints, stakes, role, local evidence, lived situation); F:=externally induced context-frame; AI-Empowerment(H) $\Leftrightarrow$ S preserved $\wedge$ high $\mathcal{E}$; S replaced by language $\Rightarrow$ F dominates $\Rightarrow$ [agency] degrades

Occurrences (7): 18517054:Def-4, 18517054:Def-5, 18517054:Def-6, 18517054:Prop-2, 18517054:Prop-5 (B1), 18517054:(2), 18517054:(3)

weld/H.11.v1 — pre-choice rival-model comparison ladder M0-M4

domain: H • tier: Definition • status: current • coq: definition • parents: weld/M.03.v1

M0=formal option count+stated preference; M1=affordance/capability+cost; M2=constructed -preference/salience; M3=predictive/active-inference; M4=readout-native live-field model

Occurrences (5): 22424434:(44), 22424434:(45), 22424434:(46), 22424434:(47), 22424434:(48)

weld/H.12.v1 — CTSA class-workflow relation and session-boundary architecture

domain: H • tier: Dr • status: *current* • coq: *definition* • parents: *weld/M.01.v1*

$C \rightarrow T \rightarrow \text{Workflow} \rightarrow S \rightarrow A \rightarrow C$; Frozen Human baseline $\rightarrow K_like \rightarrow \text{Difference/Resistance} \rightarrow H \leftrightarrow AI$
 $\rightarrow \text{Retention gate} \rightarrow \text{CTSA Return Profile} \rightarrow \text{Unaided Return Test} \rightarrow \text{Gain/Loss} \rightarrow \text{Regulation} \rightarrow \text{Provenance/Warrant} \rightarrow \text{Epistemic direction}$

Occurrences (2): 22339909:CTSA-03, 22339909:CTSA-05

weld/H.13.v1 — Effort v0.3 Eq.6: NEW PROPOSITION

domain: H • tier: Dr • status: *current* • coq: *open_prop* • parents: *weld/M.01.v1*

$N_{\{ext\}} < \infty$; $N_{\{int\}} = N_{\{ext\}}$

Occurrences (1): 22622206:eq.06

weld/H.14.v1 — Effort v0.3 Eq.20: NEW DOMAIN DEFINITION

domain: H • tier: *Definition* • status: *current* • coq: *definition* • parents: *weld/M.01.v1*, *weld/M.02.v1*

$A_n^Q := q_A(S_n, T_n, c_n; Q)$

Occurrences (1): 22622206:eq.20

weld/H.15.v1 — Effort v0.3 Eq.21: NEW DOMAIN DEFINITION

domain: H • tier: *Definition* • status: *current* • coq: *definition* • parents: *weld*

$\mathcal{I}_n^Q := \mathcal{I}_Q(\mathcal{G}_n, A_n^Q, c_n)$

Occurrences (1): 22622206:eq.21

weld/H.16.v1 — Effort v0.3 Eq.22: NEW DERIVATION

domain: H • tier: Dr • status: *current* • coq: *open_prop* • parents: *weld/M.03.v1*

$\text{SameTrial}_Q(n,m)=1 \iff \text{SID}_Q(n,m) \in \{\text{SAME}_{\text{mech}}, \text{EQUIV}_Q\}, \ A_n^Q \sim_Q A_m^Q, \ \mathcal{I}_n^Q \sim_Q \mathcal{I}_m^Q$

Occurrences (1): 22622206:eq.22

weld/H.17.v1 — Effort v0.3 Eq.23: NEW NON-COLLAPSE

domain: H • tier: Dr • status: *current* • coq: *open_prop* • parents: *weld/M.03.v1*

$\text{same source readout} \not\rightarrow \text{same effective agent} \not\rightarrow \text{same encounter}$

Occurrences (1): 22622206:eq.23

weld/H.18.v1 — Effort v0.3 Eq.24: NEW DOMAIN DEFINITION

domain: H • tier: *Definition* • status: *current* • coq: *definition* • parents: *weld*

$\text{ISW}_Q(\mathcal{G}, A, A'; c) = \text{PASS} \iff d_I^Q(\mathcal{O}_I^Q, \mathcal{I}_Q(\mathcal{G}, A, c), \mathcal{I}_Q(\mathcal{G}, A', c)) > \tau_I$

Occurrences (1): 22622206:eq.24

weld/H.19.v1 — Effort v0.3 Eq.25: NEW DOMAIN DEFINITION

domain: H • tier: Definition • status: current • coq: definition • parents: weld

$$\text{SID_Q}(n, n\{+\}1) = \text{mathsf{SAME}}_{\text{mech}}, \quad A_{n+1}^Q \not\sim_Q A_n^Q, \quad \text{ISW_Q}(\text{mathcal{G}}_n, A_n^Q, A_{n+1}^Q; c_n) = \text{mathsf{PASS}}$$

Occurrences (1): 22622206:eq.25

weld/H.20.v1 — Effort v0.3 Eq.26: NEW PROPOSITION

domain: H • tier: Dr • status: current • coq: open_prop • parents: weld

$$\text{mathfrak{I}}_{n+1}^Q \not\sim_Q \text{mathfrak{I}}_n^Q$$

Occurrences (1): 22622206:eq.26

weld/H.21.v1 — Effort v0.3 Eq.27: NEW PROPOSITION

domain: H • tier: Dr • status: current • coq: open_prop • parents: weld

$$\text{SID_Q}(n, n\{+\}1) = \text{mathsf{SAME}}_{\text{mech}}, \quad A_{n+1}^Q \not\sim_Q A_n^Q, \quad \text{ISW_Q}(\text{mathcal{G}}_n, A_n^Q, A_{n+1}^Q; c_n) = \text{mathsf{FAIL}} \quad ; \rightarrow ; \quad \text{no changed-agent credit for } \text{mathfrak{I}}_n^Q$$

Occurrences (1): 22622206:eq.27

weld/H.22.v1 — Credential is not expertise (methodological non-collapse)

domain: H • tier: Definition • status: current • coq: definition • parents: weld/M.02.v1

$$\text{credential} \neq \text{expertise}$$

weld/H.23.v1 — Project role, expertise type, and AI system are three non-collapsing categories

domain: H • tier: Definition • status: current • coq: definition • parents: weld/M.02.v1

$$\text{project role} \neq \text{expertise type} \neq \text{AI system}$$

weld/H.24.v1 — Three ideal-type expertise states indexed by person, question, domain, time

domain: H • tier: Definition • status: current • coq: definition • parents: weld/M.02.v1

$$\chi_i(Q, D, t) \in \{N, I, C\}$$

weld/H.25.v1 — Expertise-state transition probability conditional on AI assistance, practice, world feedback, resources

domain: H • tier: Open • status: current • coq: open_prop • parents: weld/H.24.v1

$$P \! \! \! \left(\chi_{i,t+1} = b \mid \chi_{i,t} = a, A_t, P_t, W_t, R_t \right)$$

weld/H.26.v1 — H1 - Interactional Compression hypothesis

domain: H • tier: Open • status: current • coq: open_prop • parents: weld/H.25.v1

$$C_{\{N \rightarrow I\}^{\{AI\}}} < C_{\{N \rightarrow I\}^{\{baseline\}}}$$

weld/H.27.v1 — H2 - Contributory Persistence hypothesis

domain: H • tier: *Open* • status: *current* • coq: *open_prop* • parents: *weld/H.25.v1*

$$|\Delta C_{\{I\} \rightarrow C}^{\{AI\}}| < |\Delta C_{\{N\} \rightarrow I}^{\{AI\}}|$$

weld/H.28.v1 — Core Epistemic Registration object (experience-based expert, interactional expert or empty, AI model set)

domain: H • tier: *Definition* • status: *current* • coq: *definition* • parents: *weld/M.02.v1*

$$\mathcal{R}_p = \langle E_p^{\{exp\}}, E_p^{\{int\}}, \mathcal{M}_p^{\{AI\}} \rangle \\ E_p^{\{int\}} = \varnothing \quad \mathcal{M}_p^{\{AI\}} = \{M_1, M_2, \dots, M_k\}$$

weld/H.29.v1 — Live-problem coupled epistemic system research architecture

domain: H • tier: *Definition* • status: *current* • coq: *definition* • parents: *weld/M.02.v1*, *EQ-015/H.24.v1*

$$\begin{aligned} & \text{Live Problem} \rightarrow \text{Core Respondent / Practitioner} + \text{Researcher} \\ & + \text{Interactional Expert if present} + \text{AI Model(s)} \rightarrow \text{Question} \\ & \rightarrow \text{Candidate Set} \rightarrow \text{Discrimination} \rightarrow \\ & \text{World Test} \rightarrow \text{Feedback} \rightarrow \text{Revision}. \end{aligned}$$

weld/H.30.v1 — Core Epistemic Structure of a project

domain: H • tier: *Definition* • status: *current* • coq: *definition* • parents: *weld/H.22.v1*, *weld/H.23.v1*, *weld/H.28.v1*

$$E_p = \langle X_p^{\{exp\}}, X_p^{\{int\}}, \mathcal{M}_p^{\{AI\}} \rangle$$

weld/H.31.v1 — AI model set of a project

domain: H • tier: *Definition* • status: *current* • coq: *definition* • parents: *weld/H.22.v1*, *weld/H.23.v1*, *weld/H.28.v1*

$$\mathcal{M}_p^{\{AI\}} = \{M_1, M_2, \dots, M_k\}$$

weld/H.32.v1 — Interactional-expert slot empty when the role is unheld

domain: H • tier: *Definition* • status: *current* • coq: *definition* • parents: *weld/H.22.v1*, *weld/H.23.v1*, *weld/H.28.v1*

$$X_p^{\{int\}} = \varnothing \quad \text{role unheld, not a zero score}$$

weld/H.33.v1 — Non-collapse: experience-based expertise, interactional expertise, and AI model are distinct

domain: H • tier: *Dr* • status: *current* • coq: *open_prop* • parents: *weld/H.22.v1*, *weld/H.23.v1*, *weld/H.28.v1*

$$\text{Experience-Based Expertise} \neq \text{Interactional Expertise} \neq \text{AI Model}$$

weld/H.34.v1 — Experience-holder decomposition (lived experience, selection, interpretation)

domain: H • tier: *Definition* • status: *current* • coq: *definition* • parents: *weld/H.22.v1*, *weld/H.23.v1*, *weld/H.28.v1*

$$X_p^{\{exp\}} = \langle \text{Exp}, \text{Sel}, \text{Int} \rangle$$

weld/H.35.v1 — Agent-count / provenance-route-count non-collapse

domain: $H \bullet \text{tier: } Dr \bullet \text{status: current} \bullet \text{coq: open_prop} \bullet \text{parents: weld/H.08.v1, A.8/M.02.v1}$

$$N_A(c) \not\equiv N_P(c)$$

Occurrences (1): 22639311:Tunnel-v2.1:eq.5

weld/H.36.v1 — Agreeing agents does not entail independent epistemic routes (network Epistemic Mirror Effect)

domain: $H \bullet \text{tier: } Dr \bullet \text{status: current} \bullet \text{coq: open_prop} \bullet \text{parents: weld/H.08.v1, A.8/M.02.v1}$

$$N_A(c) \uparrow \not\rightarrow N_P(c) \uparrow$$

Occurrences (1): 22639311:Tunnel-v2.1:eq.6

weld/H.37.v1 — Evidence-driven convergence regime

domain: $H \bullet \text{tier: Definition} \bullet \text{status: current} \bullet \text{coq: definition} \bullet \text{parents: weld/H.08.v1, EQ-015/H.14.v1}$

$$\Delta D^{\text{eff}}_{\mathcal{G}} < 0, \Delta R^{\text{ex}}_{\mathcal{G}} > 0 \quad \text{and} \quad \Delta P^{\text{ind}}_{\mathcal{G}} > 0 \quad \text{for} \quad \text{WorldRecord} = \text{Decisive}$$

Occurrences (1): 22639311:Tunnel-v2.1:eq.9

weld/H.38.v1 — AI-side semantic momentum reset across sessions (Fusion/Tunnel scope condition)

domain: $H \bullet \text{tier: Definition} \bullet \text{status: current} \bullet \text{coq: definition} \bullet \text{parents: weld/H.07.v1}$

$$m^{\text{AI}}_{s+1,0} = 0$$

Occurrences (1): 22550491:Eq. 37

weld/M.01.v1 — Root weld: retained distinction forces the graph-Laplacian and the context-sensitive stepper ($\delta_R \vdash L_R \vdash F$)

domain: $M \bullet \text{tier: Th_coqc} \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: weld}$

$$\begin{aligned} \delta_R = (a \# b) \vdash [\text{Th_coqc}] L_R = D_W - W \vdash [Dr] F \text{ (MQ.08 stepper); concretely} \\ S_{n+1} = F(S_n, u_n, c_n, T_n); \text{ independently corroborated by the discrete telegraph form} \\ V_{n+1} = V_n + \Delta\theta \cdot (-\gamma \cdot V_n - D_s \cdot L_R \cdot X_n) \text{ (MQ.08) forced from three postulates including} \\ \text{finite propagation } v = \sqrt{(D/\tau_c)} < \infty, \text{ with universal eigenmode decay } |a_k[n]| \leq |a_k[0]| e^{-\gamma_k n \Delta\theta} \text{ (Impermanence Theorem).} \end{aligned}$$

Occurrences (7): 21529456:(1), 21529456:(3), 21529456:(4), 21529456:(5), 19640361:P3, 19640361:MQ.08, 19640361:Th-5

weld/M.02.v1 — Domain admissibility: a translation q_D commutes with the root dynamics and readout

domain: $M \bullet \text{tier: Definition} \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: weld}$

$$\begin{aligned} q_{\{D,n+1\} \circ F_n} = F_{\{D,n\} \circ q_{\{D,n\}}}; \quad 0_{\{D,n\}} = 0_{\{D,n\} \circ q_{\{D,n\}}}; \text{ equivalently } q_D(F(z, u, c, T)) \\ = F_D(q_D(z), u, c, T) \text{ and } 0_D(z; Q, c) = 0_D(q_D(z); Q, c) \text{ (general); } q_{\{HCA,n+1\} \circ F_n} = \\ F_{\{HCA,n\} \circ q_{\{HCA,n\}}} \text{ (HCA domain instance, READOUT_GENESIS_CORE.md one-line weld)} \end{aligned}$$

Occurrences (8): 21529456:(2), 21529456:(6), 21529456:(7), 22498047:(3), 22424434:(6), 22636999:eq.1, 22639311:Tunnel-v2.0:e q.3, 22550491:Eq. 79

weld/M.03.v1 — No-early-collapse: two states are equivalent only relative to a declared question/reader/context/horizon

domain: $M \bullet \text{tier: Definition} \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: weld}$

$$z_{Q,O,c,L} z' \iff O(F^k z) = O(F^k z') \text{ for all } k \leq L$$

Occurrences (3): 21529456:(8), 22307148:(8), 22639311:Tunnel-v2.0:eq.4

weld/M.04.v1 — A preregistered rival-model ladder any readout-retention theory must beat

domain: M • tier: *Open* • status: *current* • coq: *definition* • parents: *weld/M.03.v1*

M0=arousal/intensity only; M1=familiarity/exposure; M2=prediction/surprise;
M3=language/category construction; M4=memory/schema retrieval; M5=the present
readout-retention model

Occurrences (1): 22357744:EMG-28

weld/M.05.v1 — A problem is a retained residual requiring explanation, repair, or decision; a question selects a contrast from it

domain: M • tier: *Definition* • status: *current* • coq: *definition* • parents: *weld/M.03.v1*

$$r_t = A_t \varepsilon_t - \delta_t, V_t = (1/2) r_t^T W_t r_t; P_t := \text{Retain}_{\Pi_P}(r_t); Q_{\Pi_Q}(P_t) = d_{Q,t}$$

Occurrences (3): 22307148:(3), 22307148:(4), 22307148:(5)

weld/M.06.v1 — Reachability is not accessibility: a weighted, history-shaped access kernel over the semantic/hypothesis graph, distinct from bare graph reachability

domain: M • tier: *Definition* • status: *current* • coq: *closed* • parents: *weld/M.03.v1*

$0 \leq \kappa_{\{A,t,Q\}}(s' | s) \leq 1$; $\mu_{\{A,t\}}(\pi | Q) = \prod \kappa_{\{A,t,Q\}}(s_{\{k+1\}} | s_k)$; $\text{Acc}_{\{A,t\}}(H | Q) = \sup_{\pi} \mu_{\{A,t\}}(\pi | Q)$; $\text{reachable}(H) = 1 \not\Rightarrow \text{Acc}(H | Q) \text{ high}$; restructuring operator B_t :
 $D^{\sim}_{\text{sem}_{\{t-\}} \rightarrow D^{\sim}_{\text{sem}_{\{t+\}}}}$; insight signature $\text{Acc}_{\{t+\}}(H * | Q) \gg \text{Acc}_{\{t-\}}(H * | Q)$

Occurrences (10): 22307148:(15), 22307148:(16), 22307148:(17), 22307148:(23), 22307148:(24), 22307148:(25), 22307148:(26), 22307148:(27), 22307148:(28), 22307148:(29)

weld/M.07.v1 — Discovery time and direction as a first-passage stopping time over a readout-discriminable hypothesis space

domain: M • tier: *Definition* • status: *current* • coq: *closed* • parents: *weld/M.03.v1*

$G^{\sim}K_{\{A,t\}}(Q) = \langle V_{\{A,t\}}, E_{\{A,t\}}, \omega_{\{A,t,Q\}}, \chi_{\{A,t\}} \rangle$; $U_{\{A,t\}}(Q) = \{ [H] \in H^{\sim} \text{disc}_{\{A,t\}}(Q) : W_t(H) \geq w_0, \exists u \in U_{\{D,t\}} \text{ with } \Delta_u(H, H') > 0 \}$; $\tau_U = \inf \{ n \geq 0 : \Gamma_Q(S_n) \in U_{\{A,n\}}(Q) \}$;
 $T_U(G^{\sim}K, Q) = E[\tau_U | G^{\sim}K, Q]$; $\pi^{\sim} \text{first}_{\{G^{\sim}K, Q\}}(C_j) = \Pr[H_{\{\tau_U\}} = C_j | G^{\sim}K, Q]$; $\text{usable} \neq \text{true}$;
central proposal $G^{\sim}K \Rightarrow (L(\tau_U), \pi^{\sim} \text{first})$

Occurrences (19): 22307561:(1), 22307561:(2)-(3), 22307561:(5), 22307561:(6), 22307561:(7), 22307561:(8), 22307561:H1, 22307561:H2, 22307561:(9), 22307561:(10), 22307561:(11), 22307561:(13) (+7 more)

weld/M.08.v1 — A discriminating action, its returned record, and the revision it drives

domain: M • tier: *Definition* • status: *current* • coq: *definition* • parents: *weld/M.03.v1*

discriminating action u^* satisfies $\delta^{\wedge}_i(u^*) \neq \delta^{\wedge}_j(u^*)$; $\delta^*_{\{t+1\}} = 0_D(Z_{\{t+1\}}; u^*)$;
 $r^*_i = \delta^{\wedge}_i(u^*) - \delta^*_{\{t+1\}}$; $(Z_{\{t+1\}}, A_{\{t+1\}}, D^{\sim}_{\text{sem}_{\{t+1\}}}) = \text{Update}(Z_t, A_t, \delta^*_{\{t+1\}}, \Pi_R)$

Occurrences (4): 22307148:(48), 22307148:(49), 22307148:(50), 22307148:(51)

weld/M.09.v1 — ClaimStrength $\leq$ EvidenceStrength invariant

domain: M • tier: *finite_diagnostic* • status: *current* • coq: *closed* • parents: *weld/M.03.v1*

$$\text{ClaimStrength} \leq \text{EvidenceStrength}$$

Occurrences (1): 22163849:(68)

weld/M.10.v1 — conceptual-theorizing and practice-research engine pipelines

domain: *M* • tier: *Definition* • status: *current* • coq: *closed* • parents: *weld/M.02.v1*

Phenomenon→ExistingExplanations→PreciseInadequacy→Mechanism→Boundary→Propositions;
Engine A: Phenomenon→TheoreticalInadequacy→Mechanism→ConceptualContribution; Engine B:
Practice→Observation→Intervention→Evidence→Implementation; Bridge=1⇔Pract
iceΔTheory; ConceptualPaper→CriticalEmpiricalTest→TheoryRevision

Occurrences (5): 22163849:(2), 22163849:(6), 22163849:(7), 22163849:(8), 22163849:(93)

weld/M.11.v1 — K0-K3 knowledge-state staging

domain: *M* • tier: *Definition* • status: *current* • coq: *closed* • parents: *weld/M.03.v1*

K0: private candidate; K1: K0+public timestamp/provenance/provisionality; K2:
K1+substantive external friction; K3: K2+review/certification/empirical constraint;
K0→K1→K2→K3

Occurrences (3): 22163849:(13)-(16), 22163849:(17), 22163849:(97)-(100)

weld/M.12.v1 — scholarly credit velocity, stock pressure, and gradient discipline

domain: *M* • tier: *finite_diagnostic* • status: *current* • coq: *closed* • parents: *weld/M.01.v1*

$\chi_t = \lambda_{\text{mint}} / (\lambda_{\text{conv}} + \epsilon)$; $B_{t+1} = B_t + M_t - \omega X_t$; $M_t \leq \lambda X_t$; $C_{t+1} = (1 - \delta)C_t + G_t + \Phi_t(\text{Stock})I_t P_t^{\{r\}} S_t$; $V_C = dE[C]/dt \approx (\text{CreditCreation} \times \text{Retention} \times \text{Conversion}) / H_{\text{critical}}$;
 $V \propto \prod x_j$; $dV/dx_j = V/x_j$; $\text{Priority}_j = (dV/dx_j) / (\text{MarginalHumanCost}_j + \epsilon)$; $E[C_{t+k}] > C_t$;
 $V_C^* = (\dots) / (H_{\text{critical}} \times (\text{Debt} + \text{Frag} + \text{Bias} + \text{COI} + \epsilon))$

Occurrences (17): 22163849:(40), 22163849:(41), 22163849:(42), 22163849:(43), 22163849:(45), 22163849:(46), 22163849:(51), 22163849:(52), 22163849:(53), 22163849:(54), 22163849:(55), 22163849:(90) (+5 more)

weld/M.13.v1 — A nested ladder of admissible identification sets and the audit that locates a distinction on it

domain: *M* • tier: *Definition* • status: *current* • coq: *closed* • parents: *weld/M.02.v1*

$$A_0 \subseteq A_1 \subseteq \dots \subseteq A_m; l(d_{0_{x'}}) = \min\{j : x' \notin \Gamma_j\}$$

Occurrences (2): 22301318:(15), 22301318:(16)

weld/M.14.v1 — A worked audit of AI-assisted inference showing a hidden extra access route via a retained record

domain: *M* • tier: *Dr* • status: *current* • coq: *closed* • parents: *weld/M.01.v1*

$C_D = G_D \circ K^*$ (worked identity); Retained-record route (proposition): if a supposedly fixed background corpus contains a current-case-specific retained record $D(x)$ varying with the target alternative, available through a route not contained in the stated source K^* , the effective architecture contains additional access not admitting $C = G \circ K^*$ with G independent of that record.

Occurrences (2): 22301318:(25), 22301318:Prop-4

weld/M.15.v1 — Effort v0.3 Eq.9: NEW DOMAIN DEFINITION

domain: *M* • tier: *Definition* • status: *current* • coq: *definition* • parents: *weld/M.02.v1*, *weld/M.03.v1*

$$\mathfrak{M}_Q(r_n) := \{\backslash, m \in \mathfrak{M}_Q^{\text{adm}} : 0_Q^G(m) = r_n \backslash, \backslash\}$$

Occurrences (1): 22622206:eq.09

weld/M.16.v1 — Effort v0.3 Eq.10: NEW DOMAIN DEFINITION

domain: M • tier: Definition • status: current • coq: definition • parents: weld/M.02.v1, weld/M.03.v1

$\text{CSML_Q}(M) = \text{PASS} \iff M \in \text{mathfrak{M}}_Q^{\text{adm}} \wedge \text{Asm}(M, Q) \backslash \text{explicit}$

Occurrences (1): 22622206:eq.10

weld/M.17.v1 — Effort v0.3 Eq.11: NEW DOMAIN DEFINITION

domain: M • tier: Definition • status: current • coq: definition • parents: weld/M.02.v1, weld/M.03.v1

$\text{SAL_Q}^{\text{strong}}(n) = \text{PASS} \iff \text{mathfrak{M}}_Q(r_n) \subseteq \text{mathfrak{M}}_Q^{\text{stoch}}$

Occurrences (1): 22622206:eq.11

weld/M.18.v1 — Effort v0.3 Eq.12: NEW DERIVATION

domain: M • tier: Dr • status: current • coq: open_prop • parents: weld/M.02.v1, weld/M.03.v1

$\text{mathfrak{M}}_Q(r_n) \cap \text{mathfrak{M}}_Q^{\text{stoch}} \neq \text{varnothing}, \text{quad } \text{mathfrak{M}}_Q(r_n) \cap \text{mathfrak{M}}_Q^{\text{nonstoch}} \neq \text{varnothing} \ ; \ ; \Rightarrow \ ; \text{SAL_Q}^{\text{strong}}(n) = \text{HOLD}$

Occurrences (1): 22622206:eq.12

weld/M.19.v1 — Effort v0.3 Eq.13: NEW NON-COLLAPSE

domain: M • tier: Dr • status: current • coq: open_prop • parents: weld/M.02.v1, weld/M.03.v1

$\text{CSML_Q}(M) = \text{PASS} \ ; \ ; \not\Rightarrow \ ; \text{SAL_Q}^{\text{strong}}(n) = \text{PASS}$

Occurrences (1): 22622206:eq.13

weld/M.20.v1 — Effort v0.3 Eq.14: NEW DERIVATION

domain: M • tier: Dr • status: current • coq: open_prop • parents: weld/M.02.v1

$T_{\{R \rightarrow D\}} \circ F_R = F_D \circ T_{\{R \rightarrow D\}}, \text{quad } 0_D \circ T_{\{R \rightarrow D\}} = 0_R, \text{quad } W^D_{\{Q, j\}} \circ T_{\{R \rightarrow D\}} = W^R_{\{Q, j\}} \ ; \ ; \forall j \in J_Q$

Occurrences (1): 22622206:eq.14

weld/M.21.v1 — Effort v0.3 Eq.15: NEW DOMAIN DEFINITION

domain: M • tier: Definition • status: current • coq: definition • parents: weld/M.02.v1

$\text{CDOL_Q}(\text{mathcal{R}} \rightarrow D) = \text{PASS} \iff \exists T_{\{R \rightarrow D\}} : \text{Eq. (14) holds}$

Occurrences (1): 22622206:eq.15

weld/M.22.v1 — Effort v0.3 Eq.16: NEW DOMAIN DEFINITION

domain: M • tier: Definition • status: current • coq: definition • parents: weld/M.02.v1

$\text{ECT_Q}(e_n) \in \{E_0, E_1, E_2, E_3, E_4, \text{HOLD}\}$

Occurrences (1): 22622206:eq.16

weld/M.23.v1 — Effort v0.3 Eq.18: NEW DOMAIN DEFINITION

domain: M • tier: Definition • status: current • coq: definition • parents: weld/M.03.v1

$$\text{SID_Q}(n,m) \in \{\text{SAME}_{\text{mech}}, \text{EQUIV}_{\text{Q}}, \text{DIFF}_{\text{Q}}, \text{HOLD}\}$$

Occurrences (1): 22622206:eq.18

weld/M.24.v1 — Effort v0.3 Eq.19: NEW PROPOSITION

domain: M • tier: Dr • status: current • coq: open_prop • parents: weld/M.03.v1

$$\text{SID_Q}(n,m) = \text{EQUIV}_{\text{Q}} \ ; \ ; \not\rightarrow \ ; \ ; \mathcal{G}_n = \mathcal{G}_m$$

Occurrences (1): 22622206:eq.19

weld/M.25.v1 — Effort v0.3 Eq.33: NEW DERIVATION

domain: M • tier: Dr • status: current • coq: open_prop • parents: weld/M.01.v1, weld/M.02.v1, weld/M.03.v1

$$\text{ERG_D} = \text{PASS} \iff \exists u_a, u_b \in \mathcal{U}_{\text{D}}^{\text{adm}} : d_{\text{Q}}(\text{big}(\text{R_D}(s, u_a), \text{R_D}(s, u_b)) > \tau_E$$

Occurrences (1): 22622206:eq.33

weld/M.26.v1 — Effort v0.3 Eq.34: NEW DOMAIN DEFINITION

domain: M • tier: Definition • status: current • coq: definition • parents: weld/M.01.v1, weld/M.02.v1, weld/M.03.v1

$$\text{CER_Q} = \text{PASS} \iff \text{ERG_D} = \text{PASS} \wedge \text{ID_Q}(u \text{ to } Y) = \text{PASS}$$

Occurrences (1): 22622206:eq.34

weld/M.27.v1 — Effort v0.3 Eq.36: NEW DOMAIN DEFINITION

domain: M • tier: Definition • status: current • coq: definition • parents: weld/M.01.v1

$$\mathbf{C}_n(u) = (\text{big}(\text{C}_n^{\text{int}}(u), \text{C}_n^{\text{ext}}(u), \text{C}_n^{\text{opp}}(u), \text{C}_n^{\text{risk}}(u)))$$

Occurrences (1): 22622206:eq.36

weld/M.28.v1 — Effort v0.3 Eq.37: NEW DERIVATION

domain: M • tier: Dr • status: current • coq: open_prop • parents: weld/M.01.v1

$$\mathcal{U}_{n+1}^{\text{safe}} := \text{Big}\{ u \in \mathcal{U}_{\text{D}}^{\text{adm}} : \mathbf{C}_n(u) \text{ preceq } \mathbf{B}_n, \mathbf{P} \text{ big}(\tau_{\text{D}^c} \leq H \text{ mid } \mathcal{F}_n, u) \text{ big} \leq \alpha \text{ Big} \}$$

Occurrences (1): 22622206:eq.37

weld/M.29.v1 — Effort v0.3 Eq.38: NEW SYNTHESIS

domain: M • tier: Definition • status: current • coq: definition • parents: weld

$$\text{Route}_{\text{Q}}(e_n) = \begin{cases} \mathcal{A}_M, & \text{ECT}_{\text{Q}}(e_n) = E_0, \\ \mathcal{A}_B, & \text{ECT}_{\text{Q}}(e_n) = E_1, \\ \mathcal{A}_P, & \text{ECT}_{\text{Q}}(e_n) = E_2, \\ \mathcal{A}_G, & \text{ECT}_{\text{Q}}(e_n) = E_3, \\ \mathcal{A}_X, & \text{ECT}_{\text{Q}}(e_n) = E_4, \\ \text{HOLD}, & \text{ECT}_{\text{Q}}(e_n) = \text{HOLD}. \end{cases}$$

Occurrences (1): 22622206:eq.38

weld/M.30.v1 — Effort v0.3 Eq.39: NEW SYNTHESIS

domain: M • tier: Definition • status: current • coq: definition • parents: weld/M.01.v1, weld

$$D_{n+1} = \begin{cases} \text{HOLD}, & \text{if } \text{Route}_Q(e_n) = \text{HOLD}, \\ \text{STOP}, & \text{if } \text{U}_{n+1}^{\text{safe}} = \text{varnothing}, \\ \text{otherwise.} & \end{cases}$$

Occurrences (1): 22622206:eq.39

weld/M.31.v1 — Effort v0.3 Eq.40: NEW PROPOSITION

domain: M • tier: Dr • status: current • coq: open_prop • parents: weld/M.01.v1

$$\text{U}_{n+1}^{\text{safe}} = \text{varnothing} \rightarrow \text{STOP}$$

Occurrences (1): 22622206:eq.40

weld/M.32.v1 — Effort v0.3 Eq.41: NEW SYNTHESIS

domain: M • tier: Definition • status: current • coq: definition • parents: weld, EQ-015/M.03.v1, A.5/H.09.v1

\text{Event} \to \text{Readout} \to \text{Conditional/Strong Stochastic License} \to \text{Event-Control Type} \to \text{Witness-Preserving Bridge} \to \text{Interaction Sensitivity} \to \text{Retention/Learning} \to \text{Changed Agent} \to \text{Mathematical Adapter} \to \text{Cost/Risk/Viability} \to \{\text{CONTINUE}, \text{SWITCH}, \text{STOP}, \text{HOLD}\}

Occurrences (1): 22622206:eq.41

weld/S.01.v1 — Finite-Memory Laplacian/Telegraph Generator as the Social-Instability/Peace Spine

domain: S • tier: Definition • status: split • coq: definition • parents: weld/M.01.v1

N_4 split (founder ruling 22:20, 2026-09-06): occurrences (1)-(12) [record 18383439] state a continuum PDE model ($s(x,t) \in R^n$, $V(x,t) = \Phi(s(x,t))$, $\tau \partial_t j + j = -D \nabla s$, ...) that is a DIFFERENT mathematical object from PAR-stepper/L1-L4 [record 22361830] ($LR = DW - W$ (forced Laplacian); $A := LR + \Gamma$; $s[n+1] = \dots$) – two distinct models bundled under one code; the Theorem (No-Go) is a prose theorem statement with proof sketch, not itself an equation.

$L_R = D_W - W$ (forced Laplacian); $A := L_R + \Gamma$; $s[n+1] = s[n] + dt(-A s[n] + J)$. L1 (invariance/recurrence): a bounded amplitude-only intervention cannot move the fixed point $s = A^{-1}J$, and the state returns above threshold after a finite computable number of ticks once the intervention ends. L2 (the bill): holding the state at threshold requires a non-vanishing per-tick force, cumulative cost growing linearly in time held. L3 (operator change): raising dissipation γ_i (an operator-level change, not amplitude suppression) can push the fixed point permanently below threshold at zero ongoing force. L4 (mutation): suppressing one node under permanent resets produces a strictly positive long-run floor in a coupled neighbour node.

Occurrences (18): 18383439:(1), 18383439:(2), 18383439:(3), 18383439:(4), 18383439:(5), 18383439:(6), 18383439:Theorem (No-Go for Elimination by Suppression), 18383439:(7), 18383439:(8), 18383439:(9), 18383439:(10), 18383439:(11) (+6 more)

weld/S.02.v1 — Manifested-Record & Agency-as-Substructure Axioms (Reality-as-Record, Agency-as-Choice, Collective)

domain: S • tier: Ax • status: split • coq: definition • parents: weld/M.02.v1

N_4 split (founder ruling 22:20, 2026-09-06): CE-01 ' $M(t') \in M$ ' (Axiom I), CE-02 ' $A_i(t') \subseteq M(t')$ ' (Axiom II), CE-03 ' $G(t') := \{A_1(t'), \dots, A_N(t')\} \subseteq M(t')$ ' (Axiom IV), CE-04 ' $R \in R_{\text{adm}}(t')$ ' (Admissible Regime Set) are four textually and logically distinct axioms – none is a phi-transform of another – bundled under one code.

$M(t') \in M$ [Axiom I: Reality-as-Record]; $A_i(t') \subseteq M(t')$ [Axiom II: Agency-as-Choice]; $G(t') := \{A_1(t'), \dots, A_N(t')\} \subseteq M(t')$ [Axiom IV: Collective as Coupled Agencies]; $R \in R_{\text{adm}}(t')$ [Admissible Regime Set]

Occurrences (4): 18444260:CE-01, 18444260:CE-02, 18444260:CE-03, 18444260:CE-04

weld/S.03.v1 — Moral Cost Functional, Responsibility, Karma, Cost–Survival Link, and Tragic Regimes

domain: S • tier: Definition • status: split • coq: definition • parents: weld/M.02.v1

N_4 split (founder ruling 22:20, 2026-09-06): CE-25/CE-26 (violation indicators), CE-27 (moral cost functional), CE-28 (responsibility), CE-29 (collective moral cost), CE-31 (karma-as-divergence), CE-32 (cost-survival theorem), CE-33 (zero-cost ethics), CE-34 (constitutional regime selection), plus the choice-gate rule and the Tragic/Tragic-min definitions – twelve distinct formal objects.

$C_{\{A,R\}}[t_1,t_2] := \int_{t_1}^{t_2} [\alpha V^+_{\{A,R\}}(M(t')) + \beta \chi_{\text{causal}}(R) + \gamma \chi_{\text{spec}}(R)] dt'$,
 $\alpha, \beta, \gamma > 0$; $\text{Resp}(A) \equiv C_{\{A,R_A\}}$; Cost-Survival Link: $\lim_{T \rightarrow \infty} C_{\{A,R\}}[0,T] = \infty \Rightarrow$
 $\lim_{n \rightarrow \infty} \|P_{\{S_A\}}M(t'+n)\|^2 = 0$ [Theorem]; choice-gate: $\neg \exists \text{Choice}(A \rightarrow R_A) \Rightarrow$ no
ethics/responsibility/cost attribution; Tragic regime class: every available regime is
either inadmissible or incurs strictly positive unavoidable cost, so ethics reduces to
 $R \uparrow_A \in \text{argmin } C_{\{A,R\}}$ s.t. admissibility (cost-minimization with residue, not pure
pass/fail)

Occurrences (12): 18444260:CE-25, 18444260:CE-26, 18444260:CE-27, 18444260:CE-28, 18444260:CE-29, 18444260:CE-31, 18444260:CE-32, 18444260:CE-33, 18444260:CE-34, 18444260:(CE-choice gate), 18444260:(Tragic), 18444260:(Tragic-min)

weld/S.04.v1 — Belief is a conditioned relation between an agent and a proposition, not the proposition itself; belief scale never self-promotes to epistemic status

domain: S • tier: Definition • status: split • coq: definition • parents: weld/M.03.v1

N_4 split (founder ruling 22:20, 2026-09-06): (17) belief relation, (18) belief-vector definition, (19) update rule, (20) group-stabilization definition are four distinct formulas; Prop-1 ('increasing distribution scale of $\text{Bel}(p)$ does not... increase $\sigma_K(p)$ ') is a proved narrative proposition about them, stated in prose with no standalone relational operator.

$\text{Bel}_{\{i,p,n\}} = \text{Rel}_B(a_{\{i,p\}}, x_{\{i,p\}}, \mu_{\{i,p\}}, E_{\{i,p\}}, M_{\{i,p\}}, \Theta_{\{i,p\}}, I_{\{i,p\}}, S_{\{i,p\}}^{\text{active}}, c_{\{i,p\}})$; canonical belief vector
 $b_{\{i,p,n\}} = (e_{\{i,p\}}, c_{\{i,p\}}, s_{\{i,p\}}, a_{\{i,p\}}, \eta_{\{i,p\}}, g_{\{i,p\}}, r_{\{i,p\}})$; update
 $b_{\{i,p,n+1\}} = U_B(\dots)$; group stabilization $\text{Bel}_{\{g,d,n\}}^{\{1\}}(p) = \text{Stabilize}_B(\{b_{\{i,p,n\}}\}_{i \in G}, C_B)$; Belief-scale nonpromotion (proved): increasing distribution scale of
 $\text{Bel}(p)$ does not by itself raise $\sigma_K(p)$

Occurrences (5): 21529456:(17), 21529456:(18), 21529456:(19), 21529456:(20), 21529456:Prop-1

weld/S.05.v1 — A group's retained state and semantic readout, without positing a group mind

domain: S • tier: Definition • status: split • coq: definition • parents: weld/M.03.v1

N_4 split (founder ruling 22:20, 2026-09-06): (43) group retained-state tuple, (44) group semantic readout, (45) group memory update recursion, (46) group warrant function are four distinct definitions (different LHS quantities: $Z_{\{G,t\}} / S_{\{G,t\}} / m^{\wedge}G / W_{\{G,t\}}$).

$Z_{\{G,t\}} = \langle Z_{\{A_i,t\}} \rangle_{i=1}^n, T_{\{G,t\}}, A_{\{G,t\}}, C_{\{G,t\}} \rangle$; $S_{\{G,t\}} = q_{\{\text{sem}, G, \Omega_{\{G,t\}}\}}(Z_{\{G,t\}})$; $m^{\wedge}G_{\{t+1\}}(e) = \rho_G m^{\wedge}G_t(e) + \sigma_{\{G,t\}}(e)$; $W_{\{G,t\}}(H) = W_G(R_{\{G,\leq t\}}, \text{independence, defects, calibration, objection channels})$

Occurrences (4): 22307148:(43), 22307148:(44), 22307148:(45), 22307148:(46)

weld/S.06.v1 — recoverable live-field gap / power before choice

domain: S • tier: Dr • status: not_an_equation • coq: definition • parents: weld/M.01.v1

N_4 split ruling (founder 22:20, 2026-09-06, registry/split_proposal_SW.json): The only occurrence, 22424434:(28) sec. 9.1 'Power before prohibition', reads verbatim: 'Power may alter agency by altering meaning/accessibility before overt choice' – pure prose, no operator. The CANONICAL.json statement.latest for this code instead reads ' $L^{\wedge} \text{live}_{\{A,g\}} = \max_{\{z \in J_{\text{feas}}\}} D_{\wedge} L(L^{\wedge} z_A(g), L^{\wedge} z_0_A(g))$ ', which is verbatim the formula backing EQ-015/S.08.v1 (occurrence 22357788:CBC-10, 'Recoverable Live-Field Gap'). The statement does not derive from this row's own cited occurrence at all. INTEGRITY FIX: prior statement.latest (" $L^{\wedge} \text{live}_{\{A,g\}} = \max_{\{z \in J_{\text{feas}}\}} D_{\wedge} L(L^{\wedge} z_A(g), L^{\wedge} z_0_A(g))$ ") was misattributed from EQ-015/S.08.v1 and has been replaced with this entry's own occurrence text, quoted above; corrected per N_4 task instruction.

Power may alter agency by altering meaning/accessibility before overt choice

Occurrences (1): 22424434:(28)

weld/S.07.v1 — recommended route for personal/relational dialogue

domain: S • tier: Definition • status: not_an_equation • coq: closed • parents: weld/M.01.v1

N_4 split ruling (founder 22:20, 2026-09-06, registry/split_proposal_SW.json): The single occurrence 22481928:(36) sec. 10.1 reads: 'Experience $\rightarrow$ Interpretations $\rightarrow$ Absent Perspective $\rightarrow$ Observable Evidence $\rightarrow$ Direct Human Conversation.' – a recommended procedural route (arrow chain of stages), no relational operator ($=$, $\leq$, $\in$, ...).

[print fallback: this entry's LaTeX transcription did not compile as standalone display math; showing the source ascii/latest text instead — the registry, JSON entry and docs site carry the LaTeX transcription unmodified]

Experience $\rightarrow$ Interpretations $\rightarrow$ Absent Perspective $\rightarrow$ Observable Evidence $\rightarrow$ Direct Human Conversation

Occurrences (1): 22481928:(36)

weld/S.08.v1 — Operational Linguistic Wisdom's eight Open propositions

domain: S • tier: Open • status: not_an_equation • coq: definition • parents: weld/M.01.v1

N_4 split ruling (founder 22:20, 2026-09-06, registry/split_proposal_SW.json): $P1..P8$ (22456487) are eight distinct narrative propositions for future empirical testing, e.g. $P1$: 'Connectivity quality: Elective Connectivity quality predicts the breadth and safety of linguistic assets available for learning.' – none contains a relational/mathematical operator; this is the paper's own 'Propositions for Future Testing (Open Tier)' list, explicitly named 'eight Open propositions'.

$P1 - P8[Open], \S 7$

Occurrences (8): 22456487:P1, 22456487:P2, 22456487:P3, 22456487:P4, 22456487:P5, 22456487:P6, 22456487:P7, 22456487:P8

weld/S.09.v1 — Live-Field Empirical Programme: eight falsifiable hypotheses (H1-H8)

domain: S • tier: Open • status: not_an_equation • coq: definition • parents: weld/M.01.v1

N_4 split ruling (founder 22:20, 2026-09-06, registry/split_proposal_SW.json): $CBC-H1..CBC-H8$ (sec. 16, 'Empirical Programme: Try to Kill the Theory') are eight falsifiable hypotheses in prose, e.g. $CBC-H1$: 'Live possibility beyond formal option count: the live-field model should predict which options enter deliberation and later action beyond the number of available options.' – no relational operator in any of the eight; the paper's own section explicitly frames these as hypotheses to falsify, not equations.

H1 live possibility beyond option count; H2 meaning access beyond resources; H3 history-shaped re-entry; H4 resonance without consent; H5 structural closure before overt prohibition; H6 evaluator visibility; H7 human-AI live-field carryover; H8 corrigibility predicts durable agency

Occurrences (8): 22357788:CBC-H1, 22357788:CBC-H2, 22357788:CBC-H3, 22357788:CBC-H4, 22357788:CBC-H5, 22357788:CBC-H6, 22357788:CBC-H7, 22357788:CBC-H8

weld/S.10.v1 — state field definition

domain: S • tier: Definition • status: current • coq: not_formalisable • parents: weld/M.01.v1, weld/S.01.v1

$$s(x, t) \in R^n$$

Occurrences (1): 18383439:(1)

weld/S.11.v1 — violence-potential definition

domain: S • tier: Definition • status: current • coq: not_formalisable • parents: weld/M.01.v1, weld/S.01.v1

$$V(x, t) = \Phi(s(x, t))$$

Occurrences (1): 18383439:(2)

weld/S.12.v1 — telegraph flux constitutive relation

domain: S • tier: Definition • status: current • coq: not_formalisable • parents: weld/M.01.v1, weld/S.01.v1

$$\tau \partial t j + j = -D \nabla s$$

Occurrences (1): 18383439:(3)

weld/S.13.v1 — continuity equation

domain: S • tier: Definition • status: current • coq: not_formalisable • parents: weld/M.01.v1, weld/S.01.v1

$$\partial t s = -\nabla \cdot j - \Gamma(s) + S \text{env}$$

Occurrences (1): 18383439:(4)

weld/S.14.v1 — linearized generator

domain: S • tier: Definition • status: current • coq: not_formalisable • parents: weld/M.01.v1, weld/S.01.v1

$$\partial t s = L \tau s$$

Occurrences (1): 18383439:(5)

weld/S.15.v1 — spectral decomposition

domain: S • tier: Definition • status: current • coq: not_formalisable • parents: weld/M.01.v1, weld/S.01.v1

$$s(t) = \sum_k c_k e^{-\lambda_k t} r_k$$

Occurrences (1): 18383439:(6)

weld/S.16.v1 — minimal 2-node matrix model

domain: S • tier: Dr • status: current • coq: definition • parents: weld/M.01.v1, weld/S.01.v1

$$L = [[-\alpha, \varepsilon], [\varepsilon, -\beta]], \alpha, \beta > 0$$

Occurrences (1): 18383439:(7)

weld/S.17.v1 — eigenvalues of minimal model

domain: S • tier: Dr • status: current • coq: not_formalisable • parents: weld/M.01.v1, weld/S.01.v1

$$\lambda \pm = -(\alpha + \beta)/2 \pm \text{sqrt}(((\alpha - \beta)/2)^2 + \varepsilon^2)$$

Occurrences (1): 18383439:(8)

weld/S.18.v1 — control lever: increase dissipation

domain: S • tier: Dr • status: current • coq: definition • parents: weld/M.01.v1, weld/S.01.v1

$$\Gamma \uparrow (\text{increase dissipation})$$

Occurrences (1): 18383439:(9)

weld/S.19.v1 — control lever: reduce environmental load

domain: S • tier: Dr • status: current • coq: definition • parents: weld/M.01.v1, weld/S.01.v1

$$S \text{env} \downarrow (\text{reduce load})$$

Occurrences (1): 18383439:(10)

weld/S.20.v1 — control lever: limit propagation

domain: $S \bullet \text{tier: } Dr \bullet \text{status: current} \bullet \text{coq: definition} \bullet \text{parents: weld/M.01.v1, weld/S.01.v1}$

$$Lij \downarrow (\text{limit propagation})$$

Occurrences (1): 18383439:(11)

weld/S.21.v1 — control lever: shorten memory

domain: $S \bullet \text{tier: } Dr \bullet \text{status: current} \bullet \text{coq: definition} \bullet \text{parents: weld/M.01.v1, weld/S.01.v1}$

$$\tau \downarrow (\text{shorten memory})$$

Occurrences (1): 18383439:(12)

weld/S.22.v1 — discrete forced-Laplacian stepper generator

domain: $S \bullet \text{tier: Definition} \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: weld/M.01.v1, weld/S.01.v1}$

$$LR = DW - W(\text{forced Laplacian}); A := LR + \Gamma; s[n+1] = s[n] + dt(-As[n] + J)$$

Occurrences (1): 22361830:PAR-stepper

weld/S.23.v1 — invariance/recurrence lemma

domain: $S \bullet \text{tier: } Dr \bullet \text{status: current} \bullet \text{coq: definition} \bullet \text{parents: weld/M.01.v1, weld/S.01.v1}$

$s^* = A^{-1}J$ is the unique fixed point independent of u ; $\|s[n] - s^*\| \leq \rho^{n-nB} \|s[nB] - s^*\|$
after the last nonzero tick of a bounded intervention; if $s_i^* > \theta$ there is an explicit
finite N^* after which $s_i[n] > \theta$ for all $n \geq nB + N^*$

Occurrences (1): 22361830:L1

weld/S.24.v1 — force-cost (the bill) lemma

domain: $S \bullet \text{tier: } Dr \bullet \text{status: current} \bullet \text{coq: definition} \bullet \text{parents: weld/M.01.v1, weld/S.01.v1}$

$|u_i[n]| = dt[(\gamma_i + D_i)(s_i^* - \theta) - \sum_{k \neq i} W_{ik}(s_k^* - s_k[n])]$, converging to a rate
 $F_i^\infty > 0$ whenever $s_i^* \geq \theta$, so cumulative force is $F_i^\infty \cdot n + o(n)$

Occurrences (1): 22361830:L2

weld/S.25.v1 — operator-change lemma

domain: $S \bullet \text{tier: } Dr \bullet \text{status: current} \bullet \text{coq: definition} \bullet \text{parents: weld/M.01.v1, weld/S.01.v1}$

$s_i' = s_i^* / (1 + \Delta B_{ii})$ with $B = A^{-1}$ (Sherman-Morrison); any $\theta < s_i^*$ is reached
below with $\Delta > \Delta^* = (s_i^* / \theta - 1) / B_{ii}$, and zero ongoing force

Occurrences (1): 22361830:L3

weld/S.26.v1 — mutation lemma

domain: $S \bullet \text{tier: } Dr \bullet \text{status: current} \bullet \text{coq: definition} \bullet \text{parents: weld/M.01.v1, weld/S.01.v1}$

$s_j \gtrsim (J_j + W_{ij} s_i^-) / (\gamma_j + D_j)$, with s_i^- an explicit increasing function of node i 's
own inflow, under permanent periodic resets of node i

Occurrences (1): 22361830:L4

weld/S.27.v1 — Axiom I: Reality-as-Record

domain: S • tier: Ax • status: current • coq: definition • parents: weld/M.02.v1, weld/S.02.v1

$$M(t') \in M$$

Occurrences (1): 18444260:CE-01

weld/S.28.v1 — Axiom II: Agency-as-Choice

domain: S • tier: Ax • status: current • coq: definition • parents: weld/M.02.v1, weld/S.02.v1

$$A_i(t') \subseteq M(t')$$

Occurrences (1): 18444260:CE-02

weld/S.29.v1 — Axiom IV: Collective as Coupled Agencies

domain: S • tier: Ax • status: current • coq: closed • parents: weld/M.02.v1, weld/S.02.v1

$$G(t') := \{A_1(t'), \dots, A_N(t')\} \subseteq M(t')$$

Occurrences (1): 18444260:CE-03

weld/S.30.v1 — Admissible Regime Set

domain: S • tier: Ax • status: current • coq: open_prop • parents: weld/M.02.v1, weld/S.02.v1

$$R \in R_{adm}(t')$$

Occurrences (1): 18444260:CE-04

weld/S.31.v1 — Causal violation indicator

domain: S • tier: Definition • status: current • coq: definition • parents: weld/M.02.v1, weld/S.03.v1

$$\chi_{causal}(R) := 1[\tau'_c(R) \approx 0]$$

Occurrences (1): 18444260:CE-25

weld/S.32.v1 — Spectral violation indicator

domain: S • tier: Definition • status: current • coq: definition • parents: weld/M.02.v1, weld/S.03.v1

$$\chi_{spec}(R) := 1[\Delta_{spec}(R) \leq 0]$$

Occurrences (1): 18444260:CE-26

weld/S.33.v1 — Moral cost functional

domain: S • tier: Definition • status: current • coq: definition • parents: weld/M.02.v1, weld/S.03.v1

$$C_{\{A,R\}}[t_1, t_2] := \int_{t_1}^{t_2} [\alpha V^+_{\{A,R\}}(M(t')) + \beta \chi_{causal}(R) + \gamma \chi_{spec}(R)] dt', \\ \alpha, \beta, \gamma > 0$$

Occurrences (1): 18444260:CE-27

weld/S.34.v1 — Responsibility definition

domain: S • tier: Definition • status: current • coq: definition • parents: weld/M.02.v1, weld/S.03.v1

$$Resp(A) \equiv C_{A, R_A}$$

Occurrences (1): 18444260:CE-28

weld/S.35.v1 — Collective moral cost

domain: S • tier: *Definition* • status: *current* • coq: *definition* • parents: *weld/M.02.v1, weld/S.03.v1*

$$C_G[t1, t2] := \sum_{i=1}^N w_i C_{A_i, R_i}[t1, t2]$$

Occurrences (1): 18444260:CE-29

weld/S.36.v1 — Karma as cost divergence

domain: S • tier: *Definition* • status: *current* • coq: *not_formalisable* • parents: *weld/M.02.v1, weld/S.03.v1*

$$\lim_{T \rightarrow \infty} C_{A, R}[0, T] = \infty$$

Occurrences (1): 18444260:CE-31

weld/S.37.v1 — Theorem: Cost-Survival Link

domain: S • tier: *Dr* • status: *current* • coq: *definition* • parents: *weld/M.02.v1, weld/S.03.v1*

$$\lim_{T \rightarrow \infty} C_{A, R}[0, T] = \infty \Rightarrow \lim_{n \rightarrow \infty} \|P_{S_A} M(t' + n)\|^2 = 0$$

Occurrences (1): 18444260:CE-32

weld/S.38.v1 — Zero-cost ethics

domain: S • tier: *Definition* • status: *current* • coq: *not_formalisable* • parents: *weld/M.02.v1, weld/S.03.v1*

$$Eth(A) \Leftrightarrow C_{A, R_A}[t, \infty) = 0$$

Occurrences (1): 18444260:CE-33

weld/S.39.v1 — Constitutional regime selection

domain: S • tier: *Definition* • status: *current* • coq: *definition* • parents: *weld/M.02.v1, weld/S.03.v1*

$$R^* \in \operatorname{argmin}_{R \in R_{adm}} C_{system, R, s.t. \tau'_c(R) > 0, \Delta_{spec}(R) > 0}$$

Occurrences (1): 18444260:CE-34

weld/S.40.v1 — Choice-gate rule

domain: S • tier: *Definition* • status: *current* • coq: *closed* • parents: *weld/M.02.v1, weld/S.03.v1*

$\neg \exists \text{ Choice}(A \rightarrow R_A) \Rightarrow \text{no ethics attribution, no responsibility attribution, and no moral cost attribution}$

Occurrences (1): 18444260:(CE-choice gate)

weld/S.41.v1 — Tragic regime class definition

domain: S • tier: *Definition* • status: *current* • coq: *closed* • parents: *weld/M.02.v1, weld/S.03.v1*

$$\forall R \in R_{available}(t') : \neg[\tau'_c(R) > 0 \wedge \Delta_{spec}(R) > 0] \vee C_{A, R}[t', t' + T] > 0, \text{ for all } T > 0$$

Occurrences (1): 18444260:(Tragic)

weld/S.42.v1 — Tragic-min consequence

domain: S • tier: *Definition* • status: *current* • coq: *definition* • parents: *weld/M.02.v1, weld/S.03.v1*

$$R_A^\dagger \in \operatorname{argmin}_{R \in R_{available}(t')} C_{A, R}[t', t' + T] s.t. \tau'_c(R) > 0, \Delta_{spec}(R) > 0$$

Occurrences (1): 18444260:(Tragic-min)

weld/S.43.v1 — Belief relation (relational, multidimensional)

domain: S • tier: *Definition* • status: *current* • coq: *definition* • parents: *weld/M.03.v1, weld/S.04.v1*

$$Bel_{i,p,n} = Rel_B(a_i, p | x_i, \mu_i, E_i, M_i, \Theta_i, I_i, S_i^{active}, c_i)$$

Occurrences (1): 21529456:(17)

weld/S.44.v1 — Canonical belief vector

domain: S • tier: *Definition* • status: *current* • coq: *definition* • parents: *weld/M.03.v1, weld/S.04.v1*

$$b_{i,p,n} = (e_{i,p}, c_{i,p}, s_{i,p}, a_{i,p}, \eta_{i,p}, g_{i,p}, r_{i,p})$$

Occurrences (1): 21529456:(18)

weld/S.45.v1 — Belief update rule

domain: S • tier: *Definition* • status: *current* • coq: *definition* • parents: *weld/M.03.v1, weld/S.04.v1*

$$b_{\{i,p,n+1\}} = U_B(b_{\{i,p,n\}}, Ev_{\{i,p\}}, M_i, \mu_i, E_i, \Theta_i, I_i, Trust_i, Affect_i, Utility_i, Repetition_i, Auth_i, Pow_i, Access_i, Language_i, Algorithm_i, Obj_i, c_i)$$

Occurrences (1): 21529456:(19)

weld/S.46.v1 — Group belief stabilization

domain: S • tier: *Definition* • status: *current* • coq: *closed* • parents: *weld/M.03.v1, weld/S.04.v1*

$$Bel_{g,d,n}^{(l)}(p) = Stabilize_B(\{b_{i,p,n}\}_{i \in G}, C_B)$$

Occurrences (1): 21529456:(20)

weld/S.47.v1 — Group retained-state tuple $Z_{G,t}$

domain: S • tier: *Definition* • status: *current* • coq: *definition* • parents: *weld/M.03.v1, weld/S.05.v1*

$$Z_{G,t} := \langle \{Z_{A_i,t}\}_{i=1}^n, T_{G,t}, A_{G,t}, C_{G,t} \rangle$$

Occurrences (1): 22307148:(43)

weld/S.48.v1 — Group semantic readout $S_{G,t}$

domain: S • tier: *Definition* • status: *current* • coq: *definition* • parents: *weld/M.03.v1, weld/S.05.v1*

$$S_{G,t} := q_{sem,G,\Omega_{G,t}}(Z_{G,t})$$

Occurrences (1): 22307148:(44)

weld/S.49.v1 — Group memory update m^G

domain: S • tier: *Definition* • status: *current* • coq: *closed* • parents: *weld/M.03.v1, weld/S.05.v1*

$$m_{t+1}^G(e) = \rho_G * m_t^G(e) + \sigma_{G,t}(e)$$

Occurrences (1): 22307148:(45)

weld/S.50.v1 — Group warrant function $W_{G,t}$

domain: S • tier: *Definition* • status: *current* • coq: *definition* • parents: *weld/M.03.v1, weld/S.05.v1*

$$W_{G,t}(H) = W_G(R_{G,\leq t}, independence, defects, calibration, objection\ channels)$$

Occurrences (1): 22307148:(46)

weld/W.01.v1 — the political-economy-to-human bridge mechanism

domain: W • tier: Definition • status: current • coq: definition • parents: weld/M.02.v1

$$c_{i,t}^{(PE \rightarrow H)} = B^{(PE \rightarrow H)}(Z_t; i, g)$$

Occurrences (1): 22481924:(31)

weld/W.02.v1 — After Labour's own standalone river summary

domain: W • tier: Definition • status: not_an_equation • coq: definition • parents: weld/M.02.v1

N_4 split ruling (founder 22:20, 2026-09-06, registry/split_proposal_SW.json): The single occurrence 22481924:(54) is one long arrow chain restating the whole paper's causal graph ('candidate generation $\rightarrow$ validated K/A^{AI} $\rightarrow$ robotic embodiment $\rightarrow$ $M_t \rightarrow Y_t \rightarrow \dots \rightarrow$ next system state') with no relational operator anywhere – a narrative summary diagram, matches the task's own example 'standalone river summary'; it is not itself an equation and should point to the paper's own conclusion section rather than hold a Toledo equation code.

candidate generation $\rightarrow$ validated K/A^{AI} $\rightarrow$ robotic embodiment $\rightarrow M_t \rightarrow Y_t \rightarrow L_t \rightarrow s^{\text{L}}_t$
 $\rightarrow \Gamma_t \rightarrow W^{\text{M}}_t \rightarrow o_{\{t+1\}} \rightarrow \Gamma^{\text{eff}}_t \rightarrow AD_t \rightarrow \Pi^{\text{M}}_t \rightarrow W^{\text{M}}_{\{t+1\}} \rightarrow \{G^{\text{conv}}, X, D\} \rightarrow$ human
bridge $\rightarrow \{\Pi^{\text{live}}, A^{\text{corr}}_H, r_H, S_H\} \rightarrow H^{\text{cap}}_t \rightarrow P^{\text{H}}_t \rightarrow P_t \rightarrow$ collective rules $\rightarrow$ next system
state

Occurrences (1): 22481924:(54)

weld/W.03.v1 — Effective candidate-validation workload (New Definition 2)

domain: W • tier: Definition • status: current • coq: definition • parents: weld/M.02.v1

$$\Lambda_t = \sum_{c \in \mathcal{C}_t^{\text{new}}} w(c)$$

weld/W.04.v1 — Validation capacity as a function of contributory/interactional expertise, world access, data, resources, validation-side AI

domain: W • tier: Definition • status: current • coq: definition • parents: weld/W.03.v1

$$\mu_t = V! \left(E_t^{\text{C}}, E_t^{\text{I}}, W_t, D_t, R_t, A_{\{v,t\}} \right)$$

weld/W.05.v1 — Candidate-validation backlog recursion (New Derivation 1)

domain: W • tier: Dr • status: current • coq: open_prop • parents: weld/W.04.v1

$$B_{\{t+1\}} = \max \left\{ 0, (1 - \delta_B) B_t + \Lambda_t - \mu_t \right\}$$

weld/W.06.v1 — Positive-backlog steady state (Proposition 1, Validation-Backlog Equilibrium)

domain: W • tier: Dr • status: current • coq: open_prop • parents: weld/W.05.v1

$$B^* = \frac{\Lambda - \mu}{\delta_B} \quad \text{for } \Lambda > \mu$$

weld/W.07.v1 — AI comparative statics on the backlog and the AI-induced epistemic scarcity shift condition (Proposition 2)

domain: W • tier: Dr • status: current • coq: open_prop • parents: weld/W.06.v1

$$\begin{aligned} \frac{\partial B^*}{\partial A_g} &= \frac{\partial \Lambda}{\partial A_g} \delta_B > 0 \quad \frac{\partial B^*}{\partial A_v} = -\frac{\partial \mu}{\partial A_v} \delta_B < 0 \quad \frac{\partial B^*}{\partial z} = \frac{\partial \Lambda}{\partial z} - \frac{\partial \mu}{\partial z} \delta_B \quad \frac{\partial \Lambda}{\partial z} > \frac{\partial \mu}{\partial z} \delta_B \quad \Rightarrow \quad \frac{\partial B^*}{\partial z} > 0 \end{aligned}$$

weld/W.08.v1 — Validated output value and its non-proportionality to candidate-set size

domain: W • tier: Definition • status: current • coq: definition • parents: weld/M.02.v1

$$Y_{\{K,t\}} = \sum_{c \in \mathcal{P}_t} v(c) \setminus \mathbf{1}[G(c)=1] \setminus \setminus \setminus \mathcal{C}_t^{\text{new}} \uparrow \uparrow \not\rightarrow Y_{\{K,t\}} \uparrow \uparrow \text{proportionally}$$

weld/W.09.v1 — Shadow-value Lagrangian for the validation-constrained institutional optimum (Proposition 3, Bottleneck Revaluation)

domain: W • tier: Dr • status: current • coq: open_prop • parents: weld/W.08.v1

$$\mathcal{L} = U(Y_K) - C(x) + \lambda_V[\Phi(\mu, \mathcal{C}) - Y_K]$$

weld/W.10.v1 — University Pareto-efficient portfolio objective and its constraint set

domain: W • tier: Definition • status: current • coq: definition • parents: weld/W.08.v1

$$\left\{ Y_K, \Delta R_H^{\text{return}}, W, T, N_v \right\} \setminus \setminus \text{Budget} \leq \bar{B}, \setminus \setminus \text{Safety risk} \leq \bar{S}, \setminus \setminus \text{Protocol burden} \leq \bar{U}, \setminus \setminus \text{Provenance and legitimacy gates passed}, \setminus \setminus \text{Open-exploration capacity protected}.$$

weld/W.11.v1 — Effective independent validation capacity

domain: W • tier: Dr • status: current • coq: open_prop • parents: weld/W.04.v1

$$\mu_t^{\text{eff}} = \mu_t(1 - \pi_t^{\text{RET}})$$

Occurrences (1): 22639311:Tunnel-v2.1:eq.23

Part III

APPENDIX C (new analysis, numbered at the end to avoid renumbering EQ-030+; conceptually belongs beside EQ-005/EQ-015, PART IV §IV.1)

70 EQ-063 — Mass-as-readout-of- τ_c bridge

domain: — • tier: finite_diagnostic • status: current • coq: root_layer_unwired • parents: EQ-005, BridgeFormula.MassReadout

$$\tau_c = \hbar / (2mc^2) \text{ [N4 SCALE BUS]}; m = \hbar / (2c^2 \tau_c) \text{ [readout direction -- mass is what } \tau_c \text{ looks like]}. \text{ Unit/calibration bridge only, not a derivation of EQ-015's } M \text{ coefficient.}$$

Part IV

APPENDIX C (2026-07-25 RTM operational-closure arc)

71 EQ-064 — M calibration via method-of-moments EIV/replicate-IV

domain: — • tier: finite_diagnostic • status: current • coq: root_layer_unwired • parents: EQ-063

$M_{\text{hat}} = (a_1^T y_2 + a_2^T y_1) / (2 a_1^T a_2)$ [replicate-IV, preferred]; $M_{\text{joint}} = 1.0004294772248$ (0.042948% error vs known $M_{\text{true}}=1$; 500-seed sweep mean error 0.335552% at $\sigma=1e-5$). Does not close EQ-015's M derivation -- calibration on a declared synthetic tape.

Part V

APPENDIX C (2026-07-25, DRAFT/UNMERGED)

72 EQ-068 — Three negative findings bridging v_{native} to GeV-scale physics

domain: — • tier: finite_diagnostic • status: current • coq: root_layer_unwired • parents: EQ-067

(a) fit $\Lambda_{\text{RD} \rightarrow \text{GeV}} = 246/v_{\text{native}} = 88.96060634765863$; independent Higgs-mass prediction 218.005 GeV vs real PDG 125.20 GeV -- 74.13% error, fails a 5% band. (b) zero-fitted-parameter ratio test -- identical 74.13% error. (c) zero-external-input internal consistency $m_{\text{from}_\tau c_{\text{native}}} := 1/(2 \cdot (M_{\text{joint}}/D))$ vs curvature-mass route -- ratio 0.0612, 94% deviation, not consistent. Open question left unresolved: whether this native-vs-GeV comparison is a valid readout-vs-readout comparison or a category error against a non-readout dimensional constant.

Part VI

THE FORCED SET — I through XXIV

73 I — Forced Set I — retained distinction forces the graph Laplacian

domain: — • tier: Th_coqc • status: current • coq: root_layer_unwired • parents: EQ-001

$L_R = D_W - W$ -- retained distinction forces the graph-Laplacian form; among all candidate operators, only L_R passes every required structural test (InfoRetainedDistinctionForcesLaplacian)

Occurrences (1): (4)

74 II — Forced Set II — action stationarity under node removal

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: I

$\delta S / \delta Z_n |_{\text{removed}} = \delta S / \delta Z_n |_{\text{kept}} + \text{boundary term}$, exactly -- removing a retained node changes the stationary residual by exactly the boundary term -- no free growth (InfoActionStationarity)

75 III — Forced Set III — nonnegative strain / back-reaction

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: I

$\text{strain}(x, e) \geq 0$; $\text{strain}(x, e) = 0 \iff x_u = x_v$ -- strain across any edge is nonnegative, and zero iff there is no retained distinction across it (InfoBackReaction)

76 IV — Forced Set IV — cone inheritance under stepping

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: MQ08-stepper

`step` $\Rightarrow$ same local stencil -- one operator step inherits its own local stencil $\Rightarrow$ the finite causal cone survives composition (InfoConeInheritance)

77 V — Forced Set V — cross-term dominance in a local cell

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: LivingGeometry

`cell_diag` $\perp$ `cell_cross_polarization` -- the cross term in a 2×2 local cell separates cleanly from the diagonal term (InfoCrossTermDominance)

78 VI — Forced Set VI — cubic linearization residual

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: ResidualsAndPotential

`qshift(cubic residual)` ≥ 0 -- the residual linearized from the cubic potential is nonnegative, strictly positive off the fixed point (InfoCubicLinearization)

79 VII — Forced Set VII — cut monotone under edge growth

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: I

cut monotone under edge growth -- screened strain stays nonnegative as the graph grows (InfoCutGrowth)

80 VIII — Forced Set VIII — graph Noether theorem

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: I

graph permutation symmetry $\Rightarrow$ conserved quantity -- a symmetry under graph permutation forces a conserved quantity, Noether, on a graph (InfoGraphNoether)

81 IX — Forced Set IX — frame-covariant causal quadratic form

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: EQ-007

`Q_v = n+.n-` is frame-covariant -- the causal quadratic form built from the cone's edges is covariant under frame change; signature comes from the causal order $\prec$ (InfoLorentz)

82 X — Forced Set X — boost-invariant box quadratic form

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: IX

`box_quad` is boost-invariant ($g^2(1-v^2) = 1$) -- the box quadratic form is unchanged under a boost -- narrow claim only, not the wider "unification" this book refuses (InfoLorentzInvariance)

83 XI — Forced Set XI — memory precedes mass (and its converse)

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *EQ-005, BridgeFormula.Mass Readout*

$\text{rate} \cdot \tau_c = 1$; $\tau_c \text{ equal} \Leftrightarrow \text{dynamics identical}$ -- equal memory (τ_c) forces identical dynamics regardless of how (M,D) split (M,D,dt nonzero) -- memory precedes mass (InfoMemoryBeforeMass); the converse -- one agreeing readout forces shared τ_c , and retains nothing finer than τ_c -- closed 2026-08-29 (InfoSharedReadoutForcesSharedMemory, axiom-free)

84 XII — Forced Set XII — metric is the energy readout

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *Face.8.MetricReadout*

$\text{qform}(L(\text{edges}), x) \equiv \text{energy}(\text{edges}, x)$ -- the metric's Hessian and the energy functional are the same object (InfoMetricIsEnergyReadout)

85 XIII — Forced Set XIII — quantum-relativity dispersion identity survives boost

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *Face.3.Dispersion, X*

$M \cdot \omega^2 = K \cdot \lambda \iff \text{box_quad} = 0$ (survives boost) -- the dispersion identity, and this null-condition survives a boost -- narrow claim only (InfoQuantumRelativityUnification)

86 XIV — Forced Set XIV — torsion seeds the gauge algebra

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *GaugeFaces*

$\text{torsion} \neq 0 \Rightarrow \text{non-abelian group of rank } N$ -- a nonzero torsion witness generates a non-abelian group of rank N -- the gauge-algebra seed (InfoSeedTorsionGroupAndRankN)

87 XV — Forced Set XV — coercive potential, bounded closure

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *TermTier-gradV*

∇V coercive, bounded closure -- the potential's coercivity is bounded -- wshare/wdeg closure (InfoCoercivityBoundedClosure)

88 XVI — Forced Set XVI — Clairaut symmetry of the metric lift

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *Face.8.MetricReadout*

$\partial^2(\text{metric})/\partial x \partial y = \partial^2(\text{metric})/\partial y \partial x$ -- mixed-partial (Clairaut) symmetry of the metric lift (InfoAnalysisLift -- solver arc (private) copy; NOT the causal-quantum-gravity same-named file, which imports Schwarzschild and is refused)

89 XVII — Forced Set XVII — seed readout vanishes iff homogeneous

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *ResidualsAndPotential*

SeedReadout = 0 $\iff \Phi$ homogeneous -- velocity+coupling+damping combine into one seed readout that vanishes iff the field is homogeneous (InfoSeedUnifiedMasterEquation)

90 XVIII — Forced Set XVIII — dispersion relation is scale-gauge invariant

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *Face.3.Dispersion*

dispersion relation is scale-gauge invariant -- the dispersion relation's sign/structure is invariant under a scale-gauge transform (InfoScaleGaugeNonReadout)

91 XIX — Forced Set XIX — selected state minimizes the action

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *DRLAction*

selected state = argmin(action) -- the selected state minimizes the action -- the general variational principle (InfoSeedArgminActionCost)

92 XX — Forced Set XX — closed loop pure gauge iff flat

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *GaugeFaces*

closed loop pure gauge $\iff$ flat -- a connection from any frame difference; genuine curvature is exactly the non-coboundary part (InfoConnectionFromFrame)

93 XXI — Forced Set XXI — discrete Riemann curvature from second differences

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *XX*

flat $\iff$ second difference = 0 -- discrete curvature from second differences (InfoDiscreteRiemannCurvature)

94 XXII — Forced Set XXII — curvature recovered from transport commutator

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *XXI*

curvature recovered from the transport commutator -- a nonvacuous witness recovers curvature from the transport commutator (InfoDiscreteRiemannCommutator)

95 XXIII — Forced Set XXIII — λ_c as the spine's own crossover

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *ModeEquation*

λ_c is the spine's own crossover -- the classical/quantum crossover λ_c -- a distinct object from GR redshift, not a claim of unifying the two (InfoTelegraphHorizonUnification)

96 XXIV — Forced Set XXIV — Jacobi from associativity of composition

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *EQ-042*

$K(X,Y) = XY - YX$ bilinear+antisym.; Jacobi holds -- Jacobi follows from associativity of ordered composition alone -- no Lie algebra imported; the algebraic identity is general, the specific noncommuting witness pair is hand-exhibited (InfoOrderDefectFromComposition)

Part VII

PART I — ROOT AXIOMS: WHAT IS FIRST?

97 RootToTrunk — Root-to-trunk progression diagram

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *EQ-004, EQ-005, EQ-006, EQ-007, EQ-008*

P0: $\exists a,b : a \neq b \rightarrow \text{distinguishability} \rightarrow \text{asymmetry} \rightarrow \text{temporal ordering} \rightarrow$
retention/causal memory $\tau_c > 0 \rightarrow$ finite causal graph $L_R \rightarrow$ ROOT-3: discrete stepper
MQ.08 $\rightarrow$ telegraph coarse-grain $\rightarrow$ UNIVERSAL SPINE PDE

98 InverseArrow — Inverse arrow: domain-discovery engine converges on the same MQ.08/L_R node

domain: — • *tier:* *finite_diagnostic* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *RootToTrunk*

raw transition tape $(N,S) \rightarrow (N',S')$, no domain name, no law handed in $\rightarrow$ domain-discovery engine [law-DSL candidate-basis grammar $\{1, N, S, N^2, N \cdot S, S^2, \dots\}$; identifiability gate $\text{rank}(A) = \# \text{coeff}$ else $\perp = \text{abstain}$; interaction test (mixed 2nd difference $\neq 0$); exact closure over $\mathbb{Q}$ + conservation] $\rightarrow$ minimal exact update law, e.g. $N' = N + N \cdot S/8$, $C = N + S$ conserved -- the SAME MQ.08 / L_R object, recovered from data, never fabricated

Part VIII

PART II — THE UNIVERSAL EQUATION

99 MQ08-stepper — MQ.08 discrete stepper

domain: — • *tier:* *finite_diagnostic* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *EQ-008*

$\gamma = 1/\tau_c$; $D_s = D \cdot \Delta\theta$; $V[n+1] = V[n] + \Delta\theta \cdot (-\gamma \cdot V[n] - D_s \cdot (L_R @ X[n]))$; $X[n+1] = X[n] + \Delta\theta \cdot V[n+1]$

Occurrences (1): (5)

MQ08-stepper/P.01.v1 — discrete ticks and finite causal operator

domain: P • *tier:* *untagged* • *status:* *current* • *coq:* *definition* • *parents:* *MQ08-stepper*

Retention discretizes into ticks $t = n \cdot \Delta\theta$; the finite causal graph and retention operator L_R are built on those ticks, stepped by MQ.08.

100 MQ08-CFL — MQ.08 CFL / stability condition

domain: — • tier: *finite_diagnostic* • status: *current* • coq: *root_layer_unwired* • parents: *MQ08-stepper*

$$\Delta\theta \leq 2 / (\gamma + \text{sqrt}(\gamma^2 + 4 \cdot \lambda_{\text{max}} \cdot D_s))$$

101 Telegraph — Coarse-grain trunk: telegraph equation

domain: — • tier: *finite_diagnostic* • status: *current* • coq: *root_layer_unwired* • parents: *MQ08-stepper*

$$\tau_c \partial_t^2 u + \partial_t u = D \nabla^2 u$$

102 MemoryKernel — Memory-kernel reading of the telegraph equation

domain: — • tier: *finite_diagnostic* • status: *current* • coq: *root_layer_unwired* • parents: *Telegraph*

$$K(\tau) = (1/\tau_c) e^{-(\tau/\tau_c)}; \int_0^\infty \tau K(\tau) d\tau = \tau_c$$

103 SpinePDE — The universal spine PDE (Eq.49)

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *Telegraph*

$$M \partial_t^2 \Phi + D \partial_t \Phi + K \cdot L_R \Phi + \nabla V(\Phi) = J - \eta (\text{memory} \cdot \text{smoother} \cdot \text{graph/operator} \cdot \text{nonlinear} = \text{source} \cdot \text{residual}). \quad \text{Gauge-covariant reading: } \partial_\mu \rightarrow \mathfrak{D}_\mu = \partial_\mu + A_\mu$$

104 GaugeFaces — Four forces as gauge faces

domain: — • tier: *Dr* • status: *current* • coq: *root_layer_unwired* • parents: *SpinePDE*

$$\text{EM} \rightarrow \text{U}(1), A_\mu = \text{photon field}; \text{Weak} \rightarrow \text{SU}(2), W^\pm, Z_0; \text{Strong} \rightarrow \text{SU}(3), \text{gluons}; \text{Gravity} \rightarrow \text{diffeomorphism invariance (readout-facing structure)}$$

105 MasterEqCanon — MQ.08 master equation, Canon §3 state-space form

domain: — • tier: *finite_diagnostic* • status: *current* • coq: *root_layer_unwired* • parents: *SpinePDE*

$$d/dt [X; V] = [[0, I], [-K L_R, -\gamma]] [X; V] + [0; J - \eta]; \text{ expanded: } V[n+1] = V[n] + \Delta\theta(-\gamma \cdot V[n] - D_s \cdot L_R \cdot X[n]); X[n+1] = X[n] + \Delta\theta \cdot V[n+1]$$

106 SignBridge — Continuum limit and provenance: PGFT v0.5 → current form, sign bridge

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *MasterEqCanon*

$$\text{PGFT v0.5 (old): } M \Phi_{tt} + D \Phi_t - K \Delta\Phi + \nabla V(\Phi) = J; \text{ Current form: } M \Phi_{tt} + D \Phi_t + K L_R \Phi + \nabla V(\Phi) = J - \eta; \text{ Sign bridge: } -K \Delta\Phi \leftrightarrow K L_R \Phi \text{ (graph-to-continuum gate declared)}$$

107 TermTier-M — Term-by-term tier: $M \partial^2_t \Phi$ (memory/inertia)

domain: — • *tier:* *finite_diagnostic* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *SpinePDE*

$M \partial^2_t \Phi$ -- memory/inertia. M is POSITED, not derived (8 failed forcing attempts). Mass is a readout of τ_c via $m = \hbar/(2c^2\tau_c)$; τ_c is discrete and logically prior to mass. Only the quantum regime meaningfully exercises M (D/M vs QuTiP residual 7.6×10^{-4}). Turbulence does NOT live in this term.

108 TermTier-D — Term-by-term tier: $D \partial_t \Phi$ (damping/smoothing/drain)

domain: — • *tier:* *finite_diagnostic* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *SpinePDE*

$D \partial_t \Phi$ -- damping/smoothing/drain. Well-behaved as a linear damping term at the telegraph/spine level; deeper identity is $1/\tau_c$ -flavored memory decay. D is NOT the same object as the first-order relaxation constant τ_R that governs RTPE turbulence.

109 TermTier-KLR — Term-by-term tier: $K \cdot L_R \Phi$ (graph/operator transport)

domain: — • *tier:* *Th_cocq* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *SpinePDE*

$K \cdot L_R \Phi$ -- graph/operator transport; principal symbol $\rightarrow$ metric readout. L_R is a full operator on a multimode state, not a scalar $\lambda\phi$ (Scalar-Eigenmode Reduction Error if narrowed). Proposed fix: $L_R = L_R^+(+) + L_R^*(-)$ under retention metric G .

110 TermTier-gradV — Term-by-term tier: $\nabla V(\Phi)$ (nonlinear potential)

domain: — • *tier:* *finite_diagnostic* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *SpinePDE*

$\nabla V(\Phi)$ -- nonlinear potential/cascade/restoring structure. Turbulence correction (2026-07-21): turbulence is NOT carried by the linear $M \partial^2_t \Phi$ term; it lives here, in the paraproduct term $(u \cdot \nabla)u$. The inertial quantity governing turbulence is the first-order relaxation constant τ_R , not M .

111 TermTier-J — Term-by-term tier: J (source/drive)

domain: — • *tier:* *Dr* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *SpinePDE*

J -- source/drive. External forcing/injection, always supplied from outside the trunk -- never derived from M , D , K , L_R , V themselves.

112 TermTier-eta — Term-by-term tier: η (residual/turbulence/irreducible mismatch)

domain: — • *tier:* *finite_diagnostic* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *SpinePDE*

η -- residual/turbulence/irreducible mismatch. The slot that keeps the equation honest about what it does not close. A FAIL-ABLE gate law applies: a Type-P evidence gate needs both a passing and a failing machine-derived control; a claim resting on η alone with no demonstrated failure mode is Type-U convention, not evidence.

113 ModeEquation — Regime split: mode-by-mode damped oscillator

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *SpinePDE*

$$M \ddot{\phi}_k + D \dot{\phi}_k + K \lambda_k \phi_k = 0$$

114 Layer3 — Three stacked layers — Layer 3: LP-NS audit (non-linear paraproduct diagnostic)

domain: — • *tier:* *finite_diagnostic* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *SpinePDE*

Examines the nonlinear paraproduct structure $(u \cdot \nabla)u$ inside ∇V (Littlewood-Paley / Navier-Stokes-flavored audit machinery) to check whether a given solution or regime stays inside admissible bounds. Explicitly a checker, not a native generator of dynamics.

115 LivingGeometry — DRL-Telegraph apparatus: living-geometry recurrence

domain: — • *tier:* *finite_diagnostic* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *SpinePDE*

$$\Theta_{n+1} = A_{\Theta} \Theta_n + B_{\Theta\Phi} \Phi_n + B_{\Theta\Psi} \Psi_n + u_{\Theta,n}; \quad \mathbb{G}[\Theta_n] = \mathbb{G}_0 + \Sigma_a \Theta_n^a \mathbb{G}_a$$

(operator read from the geometry state)

116 MixingCurrent — DRL-Telegraph apparatus: geometric mixing current

domain: — • *tier:* *finite_diagnostic* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *LivingGeometry*

$\Delta\Phi_n = V_{n+1} \Delta\phi_n + (V_{n+1} - V_n) \phi_n$ -- dropping the second term (basis-motion contribution) requires a proof or a stated defect bound; it is not a free simplification.

117 Rgeo — DRL-Telegraph apparatus: geometry-dominance defect bound R_{geo}

domain: — • *tier:* *finite_diagnostic* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *LivingGeometry*

$R_{\text{geo}} = |(V_{n+1} - V_n) \phi_n| / |\Delta\Phi_n|$ -- measures how much of the true step is basis-motion a fixed-basis approximation would discard. When $R_{\text{geo}} \gtrsim 1$ the drop-the-second-term approximation must be auto-rejected, not silently taken (can flip the sign of the change, not merely its magnitude). 2026-07-21 control: $R_{\text{geo}} = 8/\sqrt{53} \approx 1.099$, a 109.89% relative error.

118 DRLAction — DRL-Telegraph apparatus: the DRL discrete action

domain: — • *tier:* *Dr* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *LivingGeometry*

$$S_{\text{DRL}} = \Sigma_n \mathbb{L}^n; \quad \mathbb{L}^n = (1/\Delta t) \Delta\Phi_n^T M_n \Delta\Psi_n + \frac{1}{2} (\Phi_n^T D_n \Delta\Psi_n - \Psi_n^T D_n \Delta\Phi_n) - \Delta t [K_n \Phi_n^T \mathbb{G}_n \Psi_n + \Psi_n^T \nabla V_n(\Phi_n) - J_n^T \Psi_n]$$

119 EtaOmegaMatrices — DRL-Telegraph apparatus: pairing and orientation matrices

domain: — • tier: *Dr* • status: *current* • coq: *root_layer_unwired* • parents: *DRLAction*

$\eta = [[0, 1], [1, 0]]$ (symmetric pairing); $\omega = [[0, 1], [-1, 0]]$ (antisymmetric orientation)

120 ReaderRecordEqs — DRL-Telegraph apparatus: reader and record equations

domain: — • tier: *finite_diagnostic* • status: *current* • coq: *root_layer_unwired* • parents: *EtaOmegaMatrices*

Reader equation: $M \delta_t^2 \Phi_n + D \delta_t^c \Phi_n + K \mathbb{G}[\Theta_n] \Phi_n + \nabla V(\Phi_n) - J_n = \mathcal{R}_{\Phi,n}$.
 Record equation: $M \delta_t^2 \Psi_n - D \delta_t^c \Psi_n + K \mathbb{G}[\Theta_n]^T \Psi_n + \nabla^2 V(\Phi_n) \Psi_n = \mathcal{R}_{\Psi,n}$.
 Note the sign flip on the damping term (+D reader, -D record) and the transpose on the operator's coupling to Ψ (Bateman-doubled structure).

121 ResidualsAndPotential — DRL-Telegraph apparatus: residual decomposition and cubic potential

domain: — • tier: *finite_diagnostic* • status: *current* • coq: *root_layer_unwired* • parents: *ReaderRecordEqs*

$\mathcal{R}_{\Phi} = \mathcal{J}_{\text{cut},\Phi} + \mathcal{J}_{\text{syn}} + \mathcal{J}_{\text{geo}}$; $\mathcal{R}_{\Psi} = \mathcal{J}_{\text{cut},\Psi} + \mathcal{J}_{\Lambda}$; $\nabla V_i(\Phi) = a_i \Phi_i + b_i \Phi_i^3$ (executable cubic potential); $(\nabla^2 V)_{ii} = a_i + 3 b_i \Phi_i^2$

122 GaussJordanStepper — DRL-Telegraph apparatus: explicit Gauss–Jordan stepper with FAIL rule

domain: — • tier: *finite_diagnostic* • status: *current* • coq: *root_layer_unwired* • parents: *ReaderRecordEqs*

$A_{\Phi} = M/\Delta t^2 + D/2\Delta t$; $b_{\Phi,n} = \mathcal{R}_{\Phi,n} + J_n - K \mathbb{G}_n \Phi_n - \nabla V(\Phi_n) + (2M/\Delta t^2)\Phi_n + (D/2\Delta t - M/\Delta t^2)\Phi_{\{n-1\}}$; $A_{\Phi} \Phi_{\{n+1\}} = b_{\Phi,n}$. $A_{\Psi} = M/\Delta t^2 - D/2\Delta t$; $b_{\Psi,n} = \mathcal{R}_{\Psi,n} - K \mathbb{G}_n^T \Psi_n - \nabla^2 V(\Phi_n) \Psi_n + (2M/\Delta t^2)\Psi_n - (M/\Delta t^2 + D/2\Delta t)\Psi_{\{n-1\}}$; $A_{\Psi} \Psi_{\{n+1\}} = b_{\Psi,n}$ -- solved by partial-pivot Gauss–Jordan; the stepper is required to FAIL when a pivot falls below tolerance, no silent regularization.

123 CutCurrent — DRL-Telegraph apparatus: cut current decomposition

domain: — • tier: *finite_diagnostic* • status: *current* • coq: *root_layer_unwired* • parents: *GaussJordanStepper*

$\mathcal{J}_C = \mathcal{J}_{\text{transport}} + \mathcal{J}_{\text{return}} + \mathcal{J}_{\text{readout}} + \mathcal{J}_{\text{tape}}$

124 RTPEOperatorSplit — DRL-Telegraph apparatus: RTPE turbulence operator's own symmetric/skew/nonlinear decomposition

domain: — • tier: *finite_diagnostic* • status: *current* • coq: *root_layer_unwired* • parents: *ResidualsAndPotential*

$\mathcal{L}_R[I, \Theta] \ I = \mathcal{L}_R^{(+)}[\Theta] \ I + \mathcal{L}_R^{(-)}[\Theta] \ I + \mathcal{N}_R[I]; (\mathcal{L}_R^{(+)})^T = \mathcal{L}_R^{(+)} \succeq 0$
 (symmetric part positive-semidefinite); $(\mathcal{L}_R^{(-)})^T = -\mathcal{L}_R^{(-)}$ (skew part antisymmetric)
 ; $(\mathcal{N}_R[I])_i = \sum_j B_{ijk} I_j I_k$ (quadratic nonlinear transfer). Conservative-transfer
 check: $\langle I, \mathcal{N}_R[I] \rangle = 0$; $\langle I, \mathcal{L}_R^{(-)} I \rangle = 0$ (both to be verified, not assumed).

Part IX

PART III — THE TWELVE FACES OF THE SPINE

125 Layer.3 — Stack correction: Layer 3 — LP-NS audit (nonlinear paraproduct diagnostic, a checker not a generator)

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *Layer3*

a CHECKER on candidate solutions, not a native generator

126 Face.1.Eigenmode — Face 1: Eigenmode

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *SpinePDE*

$L_R \phi_k = \lambda_k \phi_k$ -- Meaning: existing structure = retained eigenmode under admissible transport

127 Face.1.CharEq — Face 1: per-mode characteristic equation for mode amplitude a_k

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *Face.1.Eigenmode*

$M s^2 + D s + K \lambda_k = 0$

128 Face.1.Decomposition — Face 1: Scalar-Eigenmode Reduction Error correction — symmetric/skew decomposition of L_R

domain: — • tier: *Dr* • status: *current* • coq: *root_layer_unwired* • parents: *Face.1.Eigenmode, T1*

$L_R = L_R^{(+)} + L_R^{(-)}$, a symmetric part and a skew part under the retention metric G
 [Dr, pending test T1, not yet proven]

129 Face.2 — Face 2: Decay / Impermanence

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *Face.1.Eigenmode*

$|a_k[n]| \leq |a_k[0]| \exp(-\gamma_k n \Delta\theta)$, $\gamma_k > 0$ -- Retained structures are finite and decaying unless driven or repaired.

130 Face.3.CharEq — Face 3: Dispersion Split — per-mode characteristic

domain: — • tier: *Th_cogc* • status: *current* • cog: *root_layer_unwired* • parents: *Face.1.CharEq*

$$M s^2 + D s + K \lambda = 0$$

131 Face.3.CriticalSplit — Face 3: Dispersion Split — critical split (λ_c, k_c)

domain: — • tier: *Th_cogc* • status: *current* • cog: *root_layer_unwired* • parents: *Face.3.CharEq*

$$\lambda_c = D^2 / (4 M K); k_c = 1 / (2 \sqrt{\tau_c D})$$

132 Face.3.Dispersion — Face 3: Dispersion Split — discrete/canon dispersion relation and regime split

domain: — • tier: *Th_cogc* • status: *current* • cog: *root_layer_unwired* • parents: *Face.3.CriticalSplit*

$\lambda \pm(k) = (-1 \pm \sqrt{(1 - 4\tau_c D k^2)}) / (2\tau_c)$; $k < k_c \rightarrow$ overdamped $\rightarrow$ classical-like; $k > k_c \rightarrow$ oscillatory $\rightarrow$ quantum-like. Machine-checked axiom-free: *RDL_SpineStability.split_classical/quantum* ✓

133 Face.4 — Face 4: Stability / Energy

domain: — • tier: *Th_cogc* • status: *current* • cog: *root_layer_unwired* • parents: *Face.2*

$dE/dt = -D \|v\|^2 \leq 0$ -- Energy is monotone non-increasing; strict decay when velocity is nonzero. Machine-checked: *RDL_SpineStability.energy_nonincreasing / strict_decay* ✓

134 Face.5.LightCone — Face 5: Finite-Speed / Relativity — light-cone / finite-front readout

domain: — • tier: *untagged* • status: *current* • cog: *root_layer_unwired* • parents: *EQ-007*

$\|x\| \leq v t$; $v = \sqrt{D / \tau_c}$. Light-cone / finite-front readout. $\tau_c > 0 \rightarrow$ finite propagation guaranteed.

135 Face.5.GraphSpeed — Face 5: graph-native wave speed

domain: — • tier: *Open* • status: *current* • cog: *root_layer_unwired* • parents: *Face.5.LightCone*

$c^2 = K/M$ -- the graph-native wave speed, read off the $M \partial^2_t \Phi$ and $K \cdot L_R \Phi$ terms. Identifying this graph-native c with the measured physical speed of light is *[Open]*.

136 Face.6.MassMemory — Face 6: Mass / Memory — mass-memory formula

domain: — • tier: *Dr* • status: *current* • cog: *root_layer_unwired* • parents: *EQ-005*

$m = \hbar / (2 \tau_c c^2)$ [mass-memory formula; *Dr/readout tier*]. Status: *structural / design-analogy*. Do not treat as a new measured mass prediction.

137 Face.6.ModeMass — Face 6: Mass / Memory — mode-mass formula

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *Face.6.MassMemory*, *Face.1.Eigenmode*

$$m_k^2 = (K + D_s \lambda_k) / \tau_c \text{ [mode-mass formula; structural/readout]}$$

138 Face.7.RootForce — Face 7: Force / Energy / RDU Readback Gate — root-force equation

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *TermTier-gradV*

$$F_{\text{root}} = -K L_R \Phi - \nabla V(\Phi)$$

139 Face.7.Residual — Face 7: residual discipline

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *Face.7.RootForce*

$$\text{spine_residual} = (\text{inertia} + \text{damping} + \text{geometry} + \text{nonlinear}) - (\text{source} - \text{noise})$$

140 Face.7.SINewtonGate — Face 7: SI Newton readback gate

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *Face.7.RootForce*

`force_to_SI_newton(F_root)` iff all of: `dim(F_root) = force`; source certificates present; `dt nonzero` certificate; residual discipline explicit; semantic unit compatibility; claim-tier boundary stated. Verified by: `formal/Quantum_Relativity_Formal_DAG_*.v` + `scripts/stress_master_equations.py`

141 Face.7.Landauer — Face 7: Landauer energy benchmark

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *Face.7.RootForce*

$$E_{\text{min}}(\text{erase 1 bit}) = k_B T \ln 2 \text{ (declared benchmark, not derived from zero)}$$

142 Face.8.MetricReadout — Face 8: Operator-to-Metric Geometry — canonical reading and operational shorthand

domain: — • *tier:* *Th_coqc* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *EQ-008*

`metric` = principal symbol of the second-difference / Laplacian operator; $g^{ij} \approx \frac{1}{2}$ `Hessian(principal symbol)`; `D2dir(Q) = 2h^2.v^THv` (exact, numpy-computable). Machine-checked: `RDL_MetricReadout.metric_form_readout` ✓ axiom-free

143 Face.8.Guardrail — Face 8: operator-to-metric guardrail

domain: — • *tier:* *Th_coqc* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *Face.8.MetricReadout*

`operator` → `metric` ✓; `spectrum alone` → `metric` ✗ (ill-posed)

144 Face.8.CurvatureForce — Face 8: Curvature and force re-stated on this face (companion/response field Ψ)

domain: — • tier: Dr • status: current • coq: root_layer_unwired • parents: Face.8.MetricReadout

Curvature is the record of an order that could not be swapped (τ_c 's non-commutative ordering, read out geometrically as the failure of parallel transport to commute). Force is that same record acting back on the channel that produced it (the $J - \eta$ term read at this face's geometric resolution), proposed to be carried by a companion/response field Ψ (Bateman-doubled / CTP-Keldysh construction). Does not extend RDL_MetricReadout.metric_form_readout's [Th_coqc] certificate.

145 Face.9 — Face 9: CPTP / Quantum-Channel Face — Kraus completeness

domain: — • tier: Th_coqc • status: current • coq: root_layer_unwired • parents: MQ08-stepper

$\sum_j K_j^\dagger K_j = I$ -- Kraus completeness + trace preservation + complete positivity.
Machine-checked in RDL_StarRig* C34-C39 ✓

146 Face.10.StrictGap — Face 10: strict gap (readout-not-truth)

domain: — • tier: Th_coqc • status: current • coq: root_layer_unwired • parents: EQ-002

$M_A[n] \neq \theta(E) \forall n; \epsilon_{\text{tot}} = \eta_{\text{sel}} + \eta_{\text{map}} + \eta_{\text{self}} > 0 \forall n$. Every observer's record is lossy; this is not a defect, it is the definition of finite access. [Th_coqc] grounding added 2026-07-25 (formal/InfoTrueRecordUnreadable_attempt.v, axiom-free, Print Assumptions $\Rightarrow$ Closed for all three lemmas): the true pre-record state continues to EXIST whenever two distinct true states are collapsed by a non-injective readout -- what fails to exist is a single decoder correct on both (no_decoder_recovers_state, true_state_exists_but_no_total_decoder).

147 Face.10.IdentifiabilityGate — Face 10: identifiability gate on the domain-discovery minimal exact law

domain: — • tier: finite_diagnostic • status: current • coq: root_layer_unwired • parents: Face.10.StrictGap

$\text{rank}(A) = (\text{number of candidate coefficients}) \rightarrow \text{law is identifiable -- report it; rank}(A) < (\text{number of candidate coefficients}) \rightarrow \text{UNDERDETERMINED -- abstain}$

Face.10.IdentifiabilityGate/E.01.v1 — Forced-Identification Fraction Φ_{FI}

domain: E • tier: Definition • status: current • coq: closed • parents: Face.10.IdentifiabilityGate

$\Phi_{FI}(q; A1, A2, C, \pi) = 1 - V^*(C \setminus \pi) / V^*(C)$

148 Face.10.bRLedger — Face 10: bR cross-domain lineage ledger conservation

domain: — • tier: finite_diagnostic • status: current • coq: root_layer_unwired • parents: EQ-029

$r_B = E \cdot A_C \cdot A_P \cdot A_B$ (composite lineage across quantum $\rightarrow$ chemical $\rightarrow$ protein $\rightarrow$ biological-transport domain-translations); $I_Q = I_B + O_C + O_P + O_B$ (retained information either survives to the biological end as I_B or is accounted for as an explicit obstruction O at each stage)

149 Face.11.SolveTarget — Face 11: Obstruction / Solve-Target Face

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *SpinePDE*

$O_R(R^{\diamond}) = O_{\{E_R\}}$; $S_R = \|O_R\|^2$. The solve target is the zero-section of an obstruction bundle, not scalar annihilation. Variational lens: minimize $S_R = \|O_R\|^2$.

150 Face.11.FailAbleGate — Face 11: FAIL-ABLE gate law

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *Face.11.SolveTarget*

A Type-P evidence gate -- a genuine, falsifiable pass/fail test of whether some obstruction has actually vanished -- requires both a passing control and a failing control, both machine-derived, not merely asserted. A gate that can only ever pass is a Type-U convention, not evidence the obstruction target has been hit.

151 Face.12 — Face 12: Boundary-Data Face

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *SpinePDE*

Absolute constants = boundary data. Structure can constrain relations / roles / ratios. Numbers are measured, not derived from zero. Boundary data in this project: G , $\hbar$, c , k_B , e , m_e , α , $v_{\text{Higgs}} = 246.22 \text{ GeV}$; $\tau_{\mu} = 2.187 \times 10^{-6} \text{ s}$ (SM-EFF closure, $\S XI$); $\Sigma_Y = 0$ (hypercharge cancellation, $\S XI$).

Part X

PART IV — THE τ_{c} SCALE BUS

152 BridgeFormula.TauC — IV.1 The Universal Bridge Formula — τ_{c} primitive

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *EQ-005*

$\tau_{\text{c}} = \hbar / (2 m c^2)$ [N4 SCALE BUS -- connects every domain by memory time]. τ_{c} is the primitive; mass is a readout of τ_{c} -- founder-locked, 2026-07-21.

153 BridgeFormula.MassReadout — IV.1 The Universal Bridge Formula — mass readout direction

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *BridgeFormula.TauC*

$m = \hbar / (2 c^2 \tau_{\text{c}})$ [readout direction -- mass is what τ_{c} LOOKS LIKE once you insist on asking "how much stuff is this?"]

154 BridgeFormula.MPositedFinding — IV.1 companion finding — M is posited, not derived

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *BridgeFormula.MassReadout*

M is POSITED, not derived -- eight independent attempts to force M out of a deeper generator failed; the only place M's presence is actually exercised is the quantum leaf (V.1), where a discretized D/M comparison against a QuTiP reference gave a residual around 7.6×10^{-4} . τ_{c} is discrete and prior; mass is its readout; M is a positable coefficient that quantum alone currently exercises.

155 Atlas.AlphaQG — IV.2 The Atlas — quantum-gravity diagnostic

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *BridgeFormula.TauC*

$\alpha_{\text{QG}} = G \cdot E^2 / (\hbar c^5) = (E/E_{\text{Planck}})^2$ -- a calculator/diagnostic of how far a given energy E sits from the Planck energy, expressed as a dimensionless ratio; it flags when quantum-gravity effects become non-negligible, it does not by itself derive or solve quantum gravity.

156 Atlas.ScaleTable — IV.2 The Atlas — 85 orders of magnitude scale table

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *BridgeFormula.TauC*

Planck scale $\tau_c \sim 5.4 \times 10^{-44}$ s; Electron $\sim 6.4 \times 10^{-22}$ s; Proton $\sim 3.5 \times 10^{-25}$ s; Quark (top) $\sim 2.5 \times 10^{-27}$ s; Muon $\sim 3.5 \times 10^{-24}$ s; Nuclear $\sim 10^{-23}$ s; Molecular $\sim 10^{-14}$ – 10^{-8} s; Cell $\sim 10^{-3}$ – 10^0 s; Human $\tau_c^H \sim 0.1$ – 3 s; Social $\sim 10^0$ – 10^8 s; Civilizational $\sim 10^9$ – 10^{11} s; Geological $\sim 10^{12}$ – 10^{15} s; Cosmological $\sim 4.35 \times 10^{17}$ s (Hubble time). The atlas contains 37 disciplines, 220 entries (src/anse_spine/tau_c/tau_c_master.py).

157 CommutingSquare.Bridge — IV.5 The Commuting-Square Bridge Criterion — bridge commutation

domain: — • tier: *Th_coqc* • status: *current* • coq: *root_layer_unwired* • parents: *BridgeFormula.TauC*

$T_{\{a \rightarrow b\}} \cdot F\#_a = F\#_b \cdot T_{\{a \rightarrow b\}}$ (readout preserved across the bridge)

158 CommutingSquare.Quotient — IV.5 The Commuting-Square Bridge Criterion — admissibility square, one quotient step

domain: — • tier: *Th_coqc* • status: *current* • coq: *root_layer_unwired* • parents: *CommutingSquare.Bridge*

$q_{\{n+1\}} \cdot F\#_n = F\#_n \cdot q_n$ (the admissibility square, one quotient step). This is exactly the classical Kemeny-Snell lumpability condition, given a machine-checked name: *InfoQuotientCompressionExactness*.

159 Gateway.PrimitiveTuple — IV.6 PGFT-RDU Native-Unit Gateway — native-unit primitive tuple

domain: — • tier: *Dr* • status: *current* • coq: *root_layer_unwired* • parents: *BridgeFormula.TauC*

$R \diamond = (\Gamma, \mathcal{D}, \mathcal{T}, \mathcal{L}, \Pi, \mathcal{A}, \mathbb{G}, \eta)$ -- Γ history, $\mathcal{D}$ retained distinctions, $\mathcal{T}$ tape, $\mathcal{L}$ transport/Laplacian structure, Π projection, $\mathcal{A}$ admissibility, $\mathbb{G}$ the operator (II.8a), η residual

160 Gateway.NativeMasterPDE — IV.6 PGFT-RDU Native-Unit Gateway — native master PDE

domain: — • tier: *Dr* • status: *current* • coq: *root_layer_unwired* • parents: *Gateway.PrimitiveTuple*, *SpinePDE*

$\mathcal{M}_R \partial^2 \Theta \Phi + \mathbb{D}_R \partial \Theta \Phi + \mathbb{K}_R \mathbb{L}_R \Phi + \nabla \Phi \mathbb{V}_R = \mathbb{J}_R - \eta_R$ -- the same trunk as II.3's spine PDE, relabeled in native RD units so SI never has to be assumed foundational.

161 Gateway.GatewayPDE — IV.6 PGFT-RDU Native-Unit Gateway — the gateway PDE (the calculator)

domain: — • *tier:* *finite_diagnostic* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *Gateway.NativeMasterPDE*

$\partial_{\Theta} \phi = -D_R L_R \phi - \mu_R \phi + S_R$ -- a first-order simplification of the native master PDE, VERDICT: PASS on its own acceptance tests; unit ledger requires every term to share $\delta_R^*(e)/\Theta_R$.

162 Gateway.InputAdapter — IV.6 PGFT-RDU Native-Unit Gateway — 14-unit-type input adapter

domain: — • *tier:* *finite_diagnostic* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *Gateway.GatewayPDE*

$I_{\text{nat}} = E / (k_B T)$ where E is energy-equivalent. Supported units: eV, keV, MeV, GeV, TeV, PeV, EeV, J, kg, g, kcal, cal, nat, bit.

163 Gateway.OutputAdapters — IV.6 PGFT-RDU Native-Unit Gateway — output adapters

domain: — • *tier:* *finite_diagnostic* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *Gateway.GatewayPDE*

$E_{\text{out}} = k_B T \times I_{\text{out}}$ (native readout decoded back to energy); $\ell = c\tau$ (readout length); $F_{\lambda} = E/\lambda$ (readout force-at-wavelength). ($\tau = \hbar/2E$, $m = E/c^2$, and $\alpha_{\text{QG}} = G \cdot E^2/(\hbar c^5) = (E/E_{\text{Planck}})^2$ are the same output table's members already carried in IV.1/IV.2.)

164 Gateway.SchwarzschildIdentity — IV.6 PGFT-RDU Native-Unit Gateway — Schwarzschild energy identity

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *Gateway.OutputAdapters*

$F_{\text{Genesis_SchwarzschildEnergy}}: r_s = 2GE/c^4$ [DeclaredFormula] (declared, not derived) -- restated in Coq syntax in *Genesis_Canon_Bridge_v2_4_0*, not a proved theorem, therefore not *Th_coqc*.

165 Gateway.UnruhIdentity — IV.6 PGFT-RDU Native-Unit Gateway — Unruh τ_c identity

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *Gateway.OutputAdapters*

$F_{\text{Genesis_UnruhTauC}}: \tau_c(T_{\text{Unruh}}) = \pi c/a$ [DeclaredFormula] (declared, not derived)

166 Gateway.SemanticLanes — IV.6 PGFT-RDU Native-Unit Gateway — semantic-lane definitions $\tau_c^*(E)$, $\tau_c^*(U)$

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *Gateway.PrimitiveTuple*

$\tau_c^*(E) = \hbar / 2E$ energy / mass-memory lane [DeclaredFormula] (IV.1); $\tau_c^*(U) = \pi c / a_{\text{local Unruh}}$ / acceleration lane [DeclaredFormula] (this gateway)

167 Gateway.OpenBridgeGuard — IV.6 PGFT-RDU Native-Unit Gateway — Schwarzschild/Unruh non-identity guard

domain: — • *tier:* Open • *status:* current • *coq:* root_layer_unwired • *parents:* Gateway.SchwarzschildIdentity, Gateway.UnruhIdentity

$\tau_c^{\wedge}(E) = ? \quad \tau_c^{\wedge}(U) \text{ -- [OpenBridge], NOT an identity. Open until an observer/redshift translation map is supplied and passes the commuting-square test (X.4). Symbol equality is not semantic equality; verdict on any premature unification = INVALID_COMPARISON (equivalently OPEN_MISSING_TRANSLATION), status OPEN, not REFUTED, NOT = FAIL and NOT = PASS.}$

Part XI

PART V — DOMAIN LEAVES

168 V0_DomainDiscoveryEngine — V.0 Domain-discovery engine (finds q_{α} from a raw transition tape)

domain: — • *tier:* finite_diagnostic • *status:* current • *coq:* root_layer_unwired • *parents:* InverseArrow

Given a raw tape of transitions $(N, S) \rightarrow (N', S')$, the engine discovers: (1) how many variables the domain minimally needs via an exact-gate closure test; (2) whether channels interact via a mixed second-difference test; (3) the minimal exact update law over Q that reproduces every observed transition with no residual; (4) conservation laws as the kernel of the discovered update. Adversarial battery: correctly recovered true laws $N \cdot S/8$, $N^2/8$, an affine law, and a three-variable case; never fabricated a plausible-looking wrong law such as $N \cdot S/16$; held out exact on unseen transitions; refused to fit non-polynomial laws.

169 $q_{\text{quantum_osc}}$ — V.1 Quantum — oscillatory modes threshold

domain: — • *tier:* untagged • *status:* current • *coq:* root_layer_unwired • *parents:* V0_DomainDiscoveryEngine, Face.1.Eigenmode

oscillatory modes $k > k_c$

170 $q_{\text{quantum_lambda_c}}$ — V.1 Quantum — dispersion split (critical wavenumber-associated eigenvalue)

domain: — • *tier:* untagged • *status:* current • *coq:* root_layer_unwired • *parents:* Face.3.CriticalSplit

dispersion split $\lambda_{\text{c}} = D^2 / 4MK$

171 $q_{\text{quantum_CPTP}}$ — V.1 Quantum — CPTP channel readout

domain: — • *tier:* untagged • *status:* current • *coq:* root_layer_unwired • *parents:* Face.9

CPTP channel readout $\text{Sum } K\text{-dagger } K = I$

172 q_quantum_Heisenberg — V.1 Quantum — uncertainty guard

domain: — • tier: *Th_coqc* • status: *current* • coq: *root_layer_unwired* • parents: *q_quantum_CPTP*

uncertainty guard $\Delta x * \Delta p \geq \hbar/2$ (Coq: *RDL_Heisenberg*)

173 q_quantum_CHSH — V.1 Quantum — CHSH / Tsirelson bound (quantum-correlation guard)

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *q_quantum_CPTP*

CHSH / Tsirelson bound (quantum-correlation guard)

174 q1_hollow_card_downgrade — V.1 Quantum — 6-of-8 hollow physics-interpretation-card downgrade

domain: — • tier: *Open* • status: *current* • coq: *root_layer_unwired* • parents: *q_quantum_osc*, *q_quantum_lambda_c*, *q_quantum_CPTP*, *q_quantum_Heisenberg*, *q_quantum_CHSH*

A self-audit of the physics interpretation cards attached to the quantum and GR leaves found that 6 of 8 cards were hollow -- labeled with continuum physics names while the actual content was an arithmetic tautology dressed up to look machine-checked. Those 6 cards are downgraded to [Open] as of this edition; the remaining 2 of 8 stand as previously tiered. The CHSH/Tsirelson guard and *RDL_Heisenberg* are unaffected.

175 q_relativity_cone — V.2 Relativity/GR — finite causal cone

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *Face.5.LightCone*

finite cone $\|x\| \leq vt$, $v = \sqrt{D/\tau_c}$

176 q_relativity_metric — V.2 Relativity/GR — metric read from the operator symbol

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *Face.8.MetricReadout*

metric from operator symbol $g^{\{ij\}} = 1/2 \text{ Hess } \sigma_{\{L,R\}}$

177 q_geometry_LR_Laplacian — V.3 Geometry/Manifold — L_R as (minus) the Laplace-Beltrami operator

domain: — • tier: *Th_coqc* • status: *current* • coq: *root_layer_unwired* • parents: *EQ-008*

$L_R \rightarrow -\Delta_g$ (operator-first metric); principal symbol $\rightarrow$ metric

178 q_turbulence_fullPDE — V.4 PDE/Turbulence — full PDE with nonlinear cascade term

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *SpinePDE*

full PDE $M \Phi_{tt} + D \Phi_t + K L_R \Phi + \text{grad}(V) = J - \eta$

179 q_complex_fiedler — V.5 Complex Systems — graph-spectral translation (Fiedler value, criticality)

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: EQ-008

graph spectrum, Fiedler λ_2 ; criticality at λ_c ; synchronization / diffusion readout

180 lambda0_collapse — V.5 Complex Systems — the $\lambda=0$ collapse (kernel of L_R as irreversible-collapse boundary)

domain: — • tier: Dr • status: current • coq: root_layer_unwired • parents: q_complex_fiedler

As λ_2 is driven toward zero, the graph approaches disconnection: the kernel of L_R (the connected-components null space that Fiedler's $\lambda_2 \rightarrow 0$ limit detects) is the point past which the collapse cannot be reversed from inside the graph. At that point the readout genuinely is zero, not merely small -- the zero-side of the same zero-infinity distinction the book insists on for every other non-readout limit; a boundary a system can approach and even name, but never a magnitude the readout itself ever actually reports as a finite number produced by dynamics.

181 q_biology_eigenmode — V.6 Biology — organism as driven self-sustained eigenmode

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: Face.1.Eigenmode

organism = driven self-sustained eigenmode (τ_c = body scale); diffusion, osmotic, Stokes, Nernst, Michaelis-Menten readouts; homeostasis = obstruction reduction; mortality = irreversible decay

182 q_biology_engine_home_leaf — V.6 Biology — domain-discovery engine's home leaf (adversarial battery run as a biological N,S tape)

domain: — • tier: finite_diagnostic • status: current • coq: root_layer_unwired • parents: V0_DomainDiscoveryEngine, q_biology_eigenmode

The V.0 finite_diagnostic engine's adversarial battery (recovering $N \cdot S/8$, $N^2/8$, affine, and 3-variable laws; never fabricating $N \cdot S/16$; exact holdout; refusing non-polynomial fits) was run in the shape of a biological population/nutrient tape (N, S). This is a success at the level of the engine, not yet a biology result -- it has not been run against real chemostat data.

183 q_chemistry_RD — V.7 Chemistry — reaction-diffusion smoother

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: SpinePDE

reaction-diffusion = $\text{grad}(V) + K \cdot \text{grad}^2$ smoother; kinetics and RD patterns; quantum chem binding (τ_c = molecular)

184 q_cognitive_recordlaw — V.8 Cognitive — mind’s record law

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *Face.10.StrictGap*
record law M_A = K_A theta(E) + eta; mutual information / residual; finite-access readout
(eps_tot > 0 always)

185 q_social — V.9 Social — civilization as the spine at social tau_c

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *BridgeFormula.TauC*
civilization = spine at social tau_c; testimony / institutions / repair; diffusion /
agent-network readout

186 AI_ethics_special_case — V.9 Social — AI ethics as a special case of q_social (readout-selection structure, not a universal moral fact)

domain: — • tier: *finite_diagnostic* • status: *current* • coq: *root_layer_unwired* • parents: *q_social*
An AI has no morality of its own. Ethics is a readout-selection structure built from five ingredients: the data retained, the set of answers actually accessible to the system, the selection rule applied over that set, the values of whoever defines the system, and the audit/revision process that can later reopen any of the first four. The goal for an AI: (a) disclose which reading frame it is answering from, (b) preserve every affected human party’s standing ability to correct or object, (c) adapt its reading to local culture without erasing dignity or voice. Corrigibility is restated as boundary-observability via the same J-eta feedback term (Face 7; VIII.5’s Omega_H pipeline).

187 q_finance_RTPE — V.10 Finance — RTPE relaxation equation

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *RTPEOperatorSplit*
RTPE tau_R dI/dt + L_R I = S + eta; OU / mean-reversion face; eta = market turbulence (residual)

188 q_cosmology — V.11 Cosmology — expansion as growth in accessible causal relations

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *EQ-008*
cosmic tau_c; ordering / scale unfolding; expansion = growth in accessible causal relations

189 q_formal — V.12 Formal Math — the leaf that checks the honesty of the other leaves’ translations

domain: — • tier: *Th_coqc* • status: *current* • coq: *root_layer_unwired* • parents: *V0_DomainDiscoveryEngine*
metric readout (Coq: RDL_MetricReadout); CPTP completeness (Coq: RDL_StarRig*); spine stability (Coq: RDL_SpineStability); telescope / no-creation; disclosed Coq axiom tiers (formal/AXIOM_STATUS.md)

q_formal/M.01.v1 — Forman(-Ricci) combinatorial curvature

domain: M • tier: untagged • status: current • coq: wrapped_related • parents: q_formal

Forman(-Ricci) combinatorial curvature

Occurrences (1): ‘InfoDiscreteGraphCurvature.v’

q_formal/M.02.v1 — Ollivier-Ricci curvature

domain: M • tier: untagged • status: current • coq: wrapped_related • parents: q_formal

Ollivier-Ricci curvature

Occurrences (1): ‘SUPPLEMENT.md’ §open, ‘causal-quantum-gravity’ Open Problem 3

q_formal/M.03.v1 — Ricci curvature of graphs (Forman/Ollivier relation)

domain: M • tier: untagged • status: current • coq: wrapped_related • parents: q_formal

Ricci curvature of graphs (Forman/Ollivier relation)

Occurrences (1): ‘mass_note.tex’ §lit

q_formal/M.04.v1 — Rayleigh quotient / theory of sound

domain: M • tier: untagged • status: current • coq: wrapped_related • parents: q_formal

Rayleigh quotient / theory of sound

Occurrences (1): ‘InfoSpectralCeiling_attempt.v’

q_formal/M.05.v1 — Anderson–Morley eigenvalue bound ‘ $\lambda_{\max} \leq \max(\deg u + \deg v)$ ’

domain: M • tier: untagged • status: current • coq: wrapped_related • parents: q_formal

Anderson–Morley eigenvalue bound $\lambda_{\max} \leq \max(\deg u + \deg v)$

Occurrences (1): planned: ‘InfoAndersonMorleyCeiling_attempt.v’ (not yet written as of this entry)

q_formal/M.06.v1 — Algebraic connectivity / Fiedler value

domain: M • tier: untagged • status: current • coq: wrapped_related • parents: q_formal

Algebraic connectivity / Fiedler value

Occurrences (1): ‘mass_note.tex’ §lit (cited only, not yet mechanized)

q_formal/M.07.v1 — Diameter-based algebraic-connectivity floor ‘ $\lambda_2 \geq 4/(nD)$ ’

domain: M • tier: untagged • status: current • coq: wrapped_related • parents: q_formal

Diameter-based algebraic-connectivity floor $\lambda_2 \geq 4/(nD)$

Occurrences (1): (debated, not yet used in any file)

q_formal/M.08.v1 — Spectral graph theory (survey)

domain: M • tier: untagged • status: current • coq: wrapped_related • parents: q_formal

Spectral graph theory (survey)

Occurrences (1): ‘mass_note.tex’ §lit

q_formal/M.09.v1 — Can one hear the shape of a drum?

domain: M • tier: untagged • status: current • coq: wrapped_related • parents: q_formal

Can one hear the shape of a drum?

Occurrences (1): ‘mass_note.tex’ §lit

q_formal/M.10.v1 — Graph Laplacian $\rightarrow$ Laplace–Beltrami convergence

domain: M • tier: untagged • status: current • coq: wrapped_related • parents: q_formal

Graph Laplacian $\rightarrow$ Laplace–Beltrami convergence

Occurrences (1): ‘causal-quantum-gravity/paper/main.tex’ §nd, Open Problem 1

q_formal/M.11.v1 — Crystallographic restriction theorem (only n -fold rotational symmetries with n in $\{1,2,3,4,6\}$ are compatible with a lattice; equivalently $\text{trace} = 2\cos(2\pi k/n) \in \mathbb{Q} \implies n \in \{1,2,3,4,6\}$)

domain: M • tier: untagged • status: current • coq: wrapped_related • parents: q_formal

Crystallographic restriction theorem (only n -fold rotational symmetries with n in $\{1,2,3,4,6\}$ are compatible with a lattice; equivalently $\text{trace} = 2\cos(2\pi k/n) \in \mathbb{Q} \implies n \in \{1,2,3,4,6\}$)

Occurrences (1): ‘InfoModeRotation_attempt.v’ (cited only); ‘InfoReadabilityBoundary_attempt.v’ (period-5 non-example mechanized as one instance)

q_formal/M.12.v1 — Euler–Lagrange first variation

domain: M • tier: untagged • status: current • coq: wrapped_related • parents: q_formal

Euler–Lagrange first variation

Occurrences (1): ‘InfoActionStationarity.v’ (‘mother_step_stationary’)

q_formal/M.13.v1 — Noether’s theorem

domain: M • tier: untagged • status: current • coq: wrapped_related • parents: q_formal

Noether’s theorem

Occurrences (1): ‘InfoGraphNoether.v’

q_formal/M.14.v1 — Courant–Friedrichs–Lewy stability condition

domain: M • tier: untagged • status: current • coq: wrapped_related • parents: q_formal

Courant–Friedrichs–Lewy stability condition

Occurrences (1): ‘InfoSpectralCeiling_attempt.v’ (‘step_ratio_window’)

q_formal/M.15.v1 — Green’s identity

domain: M • tier: untagged • status: current • coq: wrapped_related • parents: q_formal

Green’s identity

Occurrences (1): ‘InfoGraphFluxBalance.v’

q_formal/M.16.v1 — Gauss–Ostrogradsky divergence theorem

domain: M • tier: untagged • status: current • coq: wrapped_related • parents: q_formal

Gauss–Ostrogradsky divergence theorem

Occurrences (1): ‘InfoGraphFluxBalance.v’

q_formal/M.17.v1 — Störmer–Verlet leapfrog integrator

domain: M • tier: untagged • status: current • coq: wrapped_related • parents: q_formal

Störmer–Verlet leapfrog integrator

Occurrences (1): ‘InfoActionStationarity.v’ (the mother-step recurrence)

q_formal/M.18.v1 — Marsden–West discrete mechanics

domain: M • tier: untagged • status: current • coq: wrapped_related • parents: q_formal

Marsden–West discrete mechanics

Occurrences (1): ‘mass_note.tex’ §lit

q_formal/M.19.v1 — Lax–Richtmyer stability equivalence

domain: M • tier: untagged • status: current • coq: wrapped_related • parents: q_formal

Lax–Richtmyer stability equivalence

Occurrences (1): ‘mass_note.tex’ §lit

q_formal/M.20.v1 — Polyak’s heavy-ball momentum method

domain: M • tier: untagged • status: current • coq: wrapped_related • parents: q_formal

Polyak’s heavy-ball momentum method

Occurrences (1): ‘InfoOptimizerWindow_attempt.v’; also ‘InfoAgencyOptimizationRootCoupling_attempt.v’ (PR #48)

q_formal/M.21.v1 — Nesterov’s accelerated gradient method

domain: M • tier: untagged • status: current • coq: wrapped_related • parents: q_formal

Nesterov’s accelerated gradient method

Occurrences (1): ‘mass_note.tex’ §lit

q_formal/M.22.v1 — ODE model of Nesterov’s method

domain: M • tier: untagged • status: current • coq: wrapped_related • parents: q_formal

ODE model of Nesterov’s method

Occurrences (1): ‘mass_note.tex’ §lit

q_formal/M.23.v1 — Integral quadratic constraints for optimizer analysis

domain: M • tier: untagged • status: current • coq: wrapped_related • parents: q_formal

Integral quadratic constraints for optimizer analysis

Occurrences (1): ‘mass_note.tex’ §lit

q_formal/M.24.v1 — Onsager reciprocity relations (symmetric linear-response/susceptibility when circulation vanishes)

domain: M • tier: untagged • status: current • coq: wrapped_related • parents: q_formal

Onsager reciprocity relations (symmetric linear-response/susceptibility when circulation vanishes)

Occurrences (1): ‘domains/standard_model/four_force_circulation_v0_2.py’; ‘formal/InfoFourForceCirculationRecovery_attempt.v’

q_formal/M.25.v1 — Discrete cosine transform (DCT-II) diagonalization of the free/Neumann path-graph Laplacian

domain: M • tier: untagged • status: current • coq: wrapped_related • parents: q_formal

Discrete cosine transform (DCT-II) diagonalization of the free/Neumann path-graph Laplacian

Occurrences (1): ‘engine/graph_structure.py’ (‘path_heat_apply’)

q_formal/M.26.v1 — Discrete sine transform (DST) diagonalization of the pinned/Dirichlet path-graph Laplacian

domain: M • tier: untagged • status: current • coq: wrapped_related • parents: q_formal

Discrete sine transform (DST) diagonalization of the pinned/Dirichlet path-graph Laplacian

Occurrences (1): ‘engine/graph_structure.py’ (registered here; Dirichlet/pinned-boundary route out of scope for this branch — no ‘pins’ support)

q_formal/M.27.v1 — Chebyshev polynomial expansion of a matrix exponential (modified-Bessel coefficients)

domain: M • tier: untagged • status: current • coq: wrapped_related • parents: q_formal

Chebyshev polynomial expansion of a matrix exponential (modified-Bessel coefficients)

Occurrences (1): ‘engine/spectral_routes.py’ (‘ChebyshevHeatCache’)

q_formal/M.28.v1 — Lanczos/Krylov-subspace approximation of matrix functions

domain: M • tier: untagged • status: current • coq: wrapped_related • parents: q_formal

Lanczos/Krylov-subspace approximation of matrix functions

Occurrences (1): ‘engine/spectral_routes.py’ (‘lanczos_expm_action’)

q_formal/M.29.v1 — ‘expm_multiply’ scaling-and-squaring / Krylov baseline algorithm

domain: M • tier: untagged • status: current • coq: wrapped_related • parents: q_formal

expm_multiply scaling-and-squaring / Krylov baseline algorithm

Occurrences (1): ‘engine/router.py’ (baseline route, ‘scipy.sparse.linalg.expm_multiply’)

q_formal/M.30.v1 — Regular-polygon planar central configuration (equal masses, rigid rotation)

domain: M • tier: untagged • status: current • coq: wrapped_related • parents: q_formal

Regular-polygon planar central configuration (equal masses, rigid rotation)

Occurrences (1): ‘engine/router.py’ (‘_regular_polygon_central_configuration’)

q_formal/M.31.v1 — Errors-in-variables (EIV) attenuation: ‘ $M_{\text{hat}}_{\text{OLS}}/M_{\text{true}} \approx \text{Var}(a_{\text{true}})/(\text{Var}(a_{\text{true}})+\text{Var}(\text{noise}))$ ’, method-of-moments correction by subtracting known noise variance from the observed regressor second moment

domain: M • tier: untagged • status: current • coq: wrapped_related • parents: q_formal

Errors-in-variables (EIV) attenuation: $M_{\text{hat}}_{\text{OLS}}/M_{\text{true}} \approx \text{Var}(a_{\text{true}})/(\text{Var}(a_{\text{true}})+\text{Var}(\text{noise}))$, method-of-moments correction by subtracting known noise variance from the observed regressor second moment

Occurrences (1): ‘domains/standard_model/item1_exploration/retained_transition_meter/bias_diagnosis_v1.py’; ‘eiv_corrected_fit_v1.py’; ‘retained_transition_operational_closure/operational_exchange_estimator_v0_1.py’ (‘fit_moment_corrected’)

q_formal/M.32.v1 — Instrumental-variable (IV) estimation via two independent noisy replicate measurements of the same latent regressor, ‘ $M_{\text{hat}} = (a_1 \cdot y_2 + a_2 \cdot y_1) / (2 \cdot a_1 \cdot a_2)$ ’

domain: M • tier: untagged • status: current • coq: wrapped_related • parents: q_formal

Instrumental-variable (IV) estimation via two independent noisy replicate measurements of the same latent regressor, $M_{\text{hat}} = (a_1 \cdot y_2 + a_2 \cdot y_1) / (2 \cdot a_1 \cdot a_2)$

Occurrences (1): ‘domains/standard_model/item1_exploration/retained_transition_operational_closure/operational_exchange_estimator_v0 (‘fit_replicate_iv’)

190 q_formal_commuting_square — V.12 Formal Math — commuting-square identity (InfoQuotientCompressionExactness)

domain: — • tier: Th_coqc • status: current • coq: root_layer_unwired • parents: CommutingSquare.Quotient, q_formal

$q_{\{n+1\}}$. $F_n = F\#_n$. q_n (InfoQuotientCompressionExactness) -- one of q_{formal} ’s Th_coqc entries; the commuting-square criterion of IV.5 gets its teeth here

Occurrences (1): (2)

191 bR_Ledger_r_B — V.13 bR Cross-Domain Lineage Ledger — lineage ratio

domain: — • tier: finite_diagnostic • status: current • coq: root_layer_unwired • parents: EQ-029

$r_B = E \cdot A_C \cdot A_P \cdot A_B$ -- where E is the external SI-decode factor (explicitly external), and A_C , A_P , A_B are the retention-preserving amplitudes attached to the chemical, protein, and biological-transport translation steps respectively.

192 bR_Ledger_qQ_negative_result — V.13 bR Ledger — negative result: a single quantum quotient q_Q does not commute to biology

domain: — • tier: finite_diagnostic • status: current • coq: root_layer_unwired • parents: bR_Ledger_r_B

A single quantum-domain quotient q_Q does not commute, on its own, all the way to a biological question -- attempting to answer a biological-transport question using only q_Q reproduces exactly the kind of commuting-square failure IV.5 predicts for an under-resolved bridge, recorded as a second, independent confirmation of the Scalar-Eigenmode Reduction Error.

193 LR_skew_split — V.13a Scalar-Eigenmode Reduction Error — L_R split under the retention metric G

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: EQ-008

$L_R = L_R^{(+)} + L_R^{(-)}$ (split under the retention metric G , not naive transpose) -- $L_R^{(+)}$ carries the symmetric (diagonal-dominant) part, $L_R^{(-)}$ carries the skew, rotational, cross-channel coupling a scalar reduction throws away.

194 T1 — T1 — metric-G skew-decomposition abstract closure (retention_skew_quadratic_form_vanishes)

domain: — • tier: *Th_coqc* • status: *current* • coq: *root_layer_unwired* • parents: *LR_skew_split*

$z^T (L_R^{\sim}(-)) z = 0$ under the retention inner product, for the G-adjoint (not naive-transpose) skew part -- proven as a general fact about any G-adjoint decomposition, given G invertible. InfoRetentionMetricSkewDecomposition_attempt.v, theorem retention_skew_quadratic_form_vanishes, axiom-free (Print Assumptions: Closed under the global context), fully abstract (no Section/Variable/Hypothesis, explicit forall-premise s). Companion numeric fixture t1_metric_g_skew_verify.py confirms it on a concrete $G \neq I$ case AND confirms the negative control: the naive-transpose split does NOT have this property when $G \neq I$.

195 T1b — T1b — concrete instantiation gap, and T1b milestone #1 (relativity domain, G=I derived)

domain: — • tier: *finite_diagnostic* • status: *current* • coq: *root_layer_unwired* • parents: *T1*

What T1 does NOT yet close: it does not construct the concrete retention metric G for the real G_n operator of II.8a ($L_{\{G_n\}}$ tensor $I_F + I_{\{G_n\}}$ tensor $C_F + C_{int,n}$), confirm G is invertible there, or verify that G_n 's actual G-adjoint matches the assumed form. T1b milestone #1 (2026-07-24, finite_diagnostic, relativity domain, proof of concept): T1b_relativity_instantiation_verify.py closes the first concrete instance of the full chain -- a scalar retained-load functional $F \rightarrow$ Face 8's Hessian-readout (RDL_MetricReadout.v, metric_form_readout, Th_coqc) $\rightarrow$ the retention metric G $\rightarrow$ T1's skew-decomposition -- using only pre-existing, independently verified objects. Result: Face 8 reads $G=I$ exactly off obstruction's Hessian at every sampled base point and step size -- $G=I$ is derived, not assumed, matching the same $G=I$ basis convention SM_INFORMATION_PHILOSOPHY_MASTER.md section 3.2 already uses to derive SU(3). U_C 's skew part then vanishes under this derived G exactly as T1 predicts.

196 T2 — T2 — endogenous state-dependent $L_R[I_R]$ (remaining open two-field-wall gap)

domain: — • tier: *Open* • status: *current* • coq: *root_layer_unwired* • parents: *T1b*

What remains open is the harder, nonlinear case: endogenous state-dependent $L_R[I_R]$, where the operator itself depends on the state it is acting on. Multimode L_R with the G-metric skew split handles the linearized cases of two-field systems without any change to the master PDE; the general nonlinear case is not handled. (Elsewhere, V.19's biology registration reports T2 endogenous $L_R[\Theta]$ CLOSED via operator-memory augmentation: I-only fails, augmented (I,Theta) closes -- a domain-specific instance, while the general nonlinear case named here remains open.)

197 V.14.FailAbleGateLaw — V.14 The FAIL-ABLE Gate Law (Type-P vs Type-U evidence)

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *Face.11.FailAbleGate*

A genuine, Type-P evidence gate for any bridge or reduction claimed in this Part requires both a passing control and a failing control, and both must be machine-derived -- not asserted, not merely imagined as a contrast case. A gate that can only be shown passing, with no demonstrated way for the same machinery to fail on a genuinely different input, is Type-U, a convention dressed as a finding. Where a leaf claims a bridge with only a passing case shown and no demonstrated failing case, that claim is Type-U convention and should be read at Dr tier at best until a failing control is produced.

198 pi_phi_page_curve — V.15 Open bridges — pi/phi retained-history fusion identity (external contribution)

domain: — • *tier:* Open • *status:* current • *coq:* root_layer_unwired • *parents:* BridgeFormula.TauC

tau (x) tau ~ = 1 (+) tau => FPdim(tau) = phi -- closed only under 7 declared gates H1-H7; root-forcing of H1-H7 in any natural domain is untested; a candidate golden-ratio measurement schedule shows opposite results under two different protocols (positive AND negative controls both kept); physical Page-curve/black-hole-unitarity/quantum-advantage claims are explicitly [Open]/not demonstrated. Nothing here enters THE FORCED SET.

199 ChemDomain_ledger — V.16 Registered Domain: Chemistry — conservation ledger

domain: — • *tier:* untagged • *status:* current • *coq:* root_layer_unwired • *parents:* q_chemistry_RD

ledger A(n1-n0)=0 (conservation) -- the free structural carrier N^G under a frozen registry, the exact source-word count homomorphism, the finite permutation-sufficiency gate, conditional decomposition uniqueness, refinement/coarsening commutation, and structural sharing without lineage erasure.

ChemDomain_ledger/C.01.v1 — Amount of substance ‘n = m/M’ (using IUPAC atomic-weight molar masses)

domain: C • *tier:* Definition • *status:* current • *coq:* definition • *parents:* ChemDomain_ledger

Amount of substance $n = m/M$ (using IUPAC atomic-weight molar masses)

Occurrences (1): ‘engine/formulas.py’ (‘moles_from_mass’)

ChemDomain_ledger/C.02.v1 — Percent composition by mass ‘%X = n_X M_X/M_compound * 100’

domain: C • *tier:* Definition • *status:* current • *coq:* definition • *parents:* ChemDomain_ledger

Percent composition by mass $\%X = n_X M_X/M_{\text{compound}} * 100$

Occurrences (1): ‘engine/formulas.py’ (‘percent_composition_by_mass’)

ChemDomain_ledger/C.03.v1 — Mass-mass stoichiometry from a balanced equation

domain: C • *tier:* Definition • *status:* current • *coq:* definition • *parents:* ChemDomain_ledger

Mass-mass stoichiometry from a balanced equation

Occurrences (1): ‘engine/formulas.py’ (‘stoichiometry_mass_mass’)

ChemDomain_ledger/C.04.v1 — Molarity ‘c = n/V’

domain: C • *tier:* Definition • *status:* current • *coq:* definition • *parents:* ChemDomain_ledger

Molarity $c = n/V$

Occurrences (1): ‘engine/formulas.py’ (‘molarity’)

ChemDomain_ledger/C.05.v1 — Dilution ‘C1 V1 = C2 V2’

domain: C • *tier:* Definition • *status:* current • *coq:* definition • *parents:* ChemDomain_ledger

Dilution $C_1 V_1 = C_2 V_2$

Occurrences (1): ‘engine/formulas.py’ (‘dilution’)

ChemDomain_ledger/C.06.v1 — pH scale ‘ $\text{pH} = -\log_{10}([\text{H}^+])$ ’

domain: C • tier: Definition • status: current • coq: definition • parents: ChemDomain_ledger

pH scale $\text{pH} = -\log_{10}([\text{H}^+])$

Occurrences (1): ‘engine/formulas.py’ (‘ph_strong_acid’, ‘ph_weak_acid’)

ChemDomain_ledger/C.07.v1 — Limiting-reagent product yield (compare ‘ n_i/coeff_i ’, take the minimum)

domain: C • tier: Definition • status: current • coq: definition • parents: ChemDomain_ledger

Limiting-reagent product yield (compare n_i/coeff_i , take the minimum)

Occurrences (1): ‘engine/formulas.py’ (‘limiting_reagent_product_moles’)

ChemDomain_ledger/C.08.v1 — Empirical-formula mole ratio from percent composition

domain: C • tier: Definition • status: current • coq: definition • parents: ChemDomain_ledger

Empirical-formula mole ratio from percent composition

Occurrences (1): ‘engine/formulas.py’ (‘empirical_formula_ratio’)

ChemDomain_ledger/C.09.v1 — Molecular-formula multiplier ‘ $k = M_{\text{molecular}}/M_{\text{empirical}}$ ’

domain: C • tier: Definition • status: current • coq: definition • parents: ChemDomain_ledger

Molecular-formula multiplier $k = M_{\text{molecular}}/M_{\text{empirical}}$

Occurrences (1): ‘engine/formulas.py’ (‘molecular_formula_multiplier’)

ChemDomain_ledger/C.10.v1 — Graham’s law of effusion ‘ $\text{rate1}/\text{rate2} = \sqrt{M_2/M_1}$ ’

domain: C • tier: Definition • status: current • coq: definition • parents: ChemDomain_ledger

Graham’s law of effusion $\text{rate1}/\text{rate2} = \sqrt{M_2/M_1}$

Occurrences (1): ‘engine/formulas.py’ (‘grahams_law_effusion_ratio’)

ChemDomain_ledger/C.11.v1 — Hess’s law of constant heat summation

domain: C • tier: Definition • status: current • coq: definition • parents: ChemDomain_ledger

Hess’s law of constant heat summation

Occurrences (1): ‘engine/formulas.py’ (‘hess_law_sum’)

ChemDomain_ledger/C.12.v1 — Reaction enthalpy from average bond dissociation energies ‘ $dH = \text{sum}(\text{broken}) - \text{sum}(\text{formed})$ ’

domain: C • tier: Definition • status: current • coq: definition • parents: ChemDomain_ledger

Reaction enthalpy from average bond dissociation energies $dH = \text{sum}(\text{broken}) - \text{sum}(\text{formed})$

Occurrences (1): ‘engine/formulas.py’ (‘bond_energy_reaction_enthalpy’)

ChemDomain_ledger/C.13.v1 — Law of mass action / equilibrium constant ‘ $K_c = [\text{B}]/[\text{A}]$ ’, ICE-table solution

domain: C • tier: Definition • status: current • coq: definition • parents: ChemDomain_ledger

Law of mass action / equilibrium constant $K_c = [\text{B}]/[\text{A}]$, ICE-table solution

Occurrences (1): ‘engine/formulas.py’ (‘equilibrium_kc_ice’, ‘le_chatelier_new_equilibrium’)

ChemDomain_ledger/C.14.v1 — Le Chatelier's principle (quantitative re-equilibration)

domain: *C* • tier: *untagged* • status: *current* • coq: *not_formalisable* • parents: *ChemDomain_ledger*

Le Chatelier's principle (quantitative re-equilibration)

Occurrences (1): 'engine/formulas.py' ('le_chatelier_new_equilibrium')

ChemDomain_ledger/C.15.v1 — Henderson-Hasselbalch equation ' $\text{pH} = \text{pKa} + \log_{10}([\text{A}^-]/[\text{HA}])$ '

domain: *C* • tier: *Definition* • status: *current* • coq: *definition* • parents: *ChemDomain_ledger*

Henderson-Hasselbalch equation $\text{pH} = \text{pKa} + \log_{10}([\text{A}^-]/[\text{HA}])$

Occurrences (1): 'engine/formulas.py' ('henderson_hasselbalch')

ChemDomain_ledger/C.16.v1 — Acid-base titration equivalence-point stoichiometry ' $\text{C}_{\text{acid}} \text{V}_{\text{acid}} = \text{C}_{\text{base}} \text{V}_{\text{base}}$ '

domain: *C* • tier: *Definition* • status: *current* • coq: *definition* • parents: *ChemDomain_ledger*

Acid-base titration equivalence-point stoichiometry $\text{C}_{\text{acid}} \text{V}_{\text{acid}} = \text{C}_{\text{base}} \text{V}_{\text{base}}$

Occurrences (1): 'engine/formulas.py' ('titration_equivalence_volume')

ChemDomain_ledger/C.17.v1 — Solubility product ' K_{sp} ', molar solubility of a 1:1 salt ' $s = \sqrt{K_{\text{sp}}}$ '

domain: *C* • tier: *Definition* • status: *current* • coq: *definition* • parents: *ChemDomain_ledger*

Solubility product K_{sp} , molar solubility of a 1:1 salt $s = \sqrt{K_{\text{sp}}}$

Occurrences (1): 'engine/formulas.py' ('solubility_from_ksp')

ChemDomain_ledger/C.18.v1 — Standard electrochemical cell potential ' $E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}}$ ', standard reduction potentials

domain: *C* • tier: *Definition* • status: *current* • coq: *definition* • parents: *ChemDomain_ledger*

Standard electrochemical cell potential $E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}}$, standard reduction potentials

Occurrences (1): 'engine/formulas.py' ('standard_cell_potential')

ChemDomain_ledger/C.19.v1 — Faraday's laws of electrolysis ' $m = I t M / (n F)$ '

domain: *C* • tier: *Definition* • status: *current* • coq: *definition* • parents: *ChemDomain_ledger*

Faraday's laws of electrolysis $m = I t M / (n F)$

Occurrences (1): 'engine/formulas.py' ('faraday_electrolysis_mass'); Faraday constant ' $F = N_A e$ ' added to 'CONSTANTS'

ChemDomain_ledger/C.20.v1 — Bimolecular (mixed second-order) rate law ' $\text{rate} = k[\text{A}][\text{B}]$ '

domain: *C* • tier: *Definition* • status: *current* • coq: *definition* • parents: *ChemDomain_ledger*

Bimolecular (mixed second-order) rate law $\text{rate} = k[\text{A}][\text{B}]$

Occurrences (1): 'engine/formulas.py' ('rate_law_second_order_mixed')

ChemDomain_ledger/C.21.v1 — Raoult's law ' $P = x_{\text{solvent}} P_0$ '

domain: *C* • tier: *Definition* • status: *current* • coq: *definition* • parents: *ChemDomain_ledger*

Raoult's law $P = x_{\text{solvent}} P_0$

Occurrences (1): 'engine/formulas.py' ('raoults_law_vapor_pressure')

ChemDomain_ledger/C.22.v1 — Colligative property closed form ' $dT = i K m$ ' (boiling-point elevation / freezing-point depression, one closed form for both via ' $K=K_b$ ' or ' $K=K_f$ ')

domain: C • tier: Definition • status: current • coq: definition • parents: ChemDomain_ledger

Colligative property closed form $dT = i K m$ (boiling-point elevation / freezing-point depression, one closed form for both via $K=K_b$ or $K=K_f$)

Occurrences (1): 'engine/formulas.py' ('colligative_dT')

ChemDomain_ledger/C.23.v1 — Van der Waals real-gas equation of state ' $P = nRT/(V-nb) - a n^2/V^2$ '

domain: C • tier: Definition • status: current • coq: definition • parents: ChemDomain_ledger

Van der Waals real-gas equation of state $P = nRT/(V-nb) - a n^2/V^2$

Occurrences (1): 'engine/formulas.py' ('van_der_waals_pressure')

ChemDomain_ledger/C.24.v1 — Clausius-Clapeyron equation ' $\ln(P_2/P_1) = -(dH_{vap}/R)(1/T_2 - 1/T_1)$ '

domain: C • tier: Definition • status: current • coq: definition • parents: ChemDomain_ledger

Clausius-Clapeyron equation $\ln(P_2/P_1) = -(dH_{vap}/R)(1/T_2 - 1/T_1)$

Occurrences (1): 'engine/formulas.py' ('clausius_clapeyron')

ChemDomain_ledger/C.25.v1 — Gibbs free energy ' $dG = dH - T dS$ '

domain: C • tier: Definition • status: current • coq: definition • parents: ChemDomain_ledger

Gibbs free energy $dG = dH - T dS$

Occurrences (1): 'engine/formulas.py' ('gibbs_free_energy', 'gibbs_to_equilibrium_constant')

ChemDomain_ledger/C.26.v1 — Van 't Hoff equation ' $\ln(K_2/K_1) = -(dH/R)(1/T_2 - 1/T_1)$ '

domain: C • tier: Definition • status: current • coq: definition • parents: ChemDomain_ledger

Van 't Hoff equation $\ln(K_2/K_1) = -(dH/R)(1/T_2 - 1/T_1)$

Occurrences (1): 'engine/formulas.py' ('van_t_hoff_equation')

ChemDomain_ledger/C.27.v1 — Nernst equation ' $E = E_0 - (RT/nF) \ln Q$ '

domain: C • tier: Definition • status: current • coq: definition • parents: ChemDomain_ledger

Nernst equation $E = E_0 - (RT/nF) \ln Q$

Occurrences (1): 'engine/formulas.py' ('nernst_equation')

ChemDomain_ledger/C.28.v1 — Arrhenius equation, two-temperature activation energy ' $E_a = -R \ln(k_2/k_1)/(1/T_2 - 1/T_1)$ '

domain: C • tier: Definition • status: current • coq: definition • parents: ChemDomain_ledger

Arrhenius equation, two-temperature activation energy $E_a = -R \ln(k_2/k_1)/(1/T_2 - 1/T_1)$

Occurrences (1): 'engine/formulas.py' ('arrhenius_activation_energy')

ChemDomain_ledger/C.29.v1 — Zero-order integrated rate law ' $[A] = [A]_0 - k t$ '

domain: C • tier: Definition • status: current • coq: definition • parents: ChemDomain_ledger

Zero-order integrated rate law $[A] = [A]_0 - k t$

Occurrences (1): 'engine/formulas.py' ('integrated_rate_law_zero_order')

ChemDomain_ledger/C.30.v1 — Second-order integrated rate law ‘ $1/[A] = 1/[A]_0 + k t$ ’

domain: C • tier: Definition • status: current • coq: definition • parents: ChemDomain_ledger

Second-order integrated rate law $1/[A] = 1/[A]_0 + k t$

Occurrences (1): ‘engine/formulas.py’ (‘integrated_rate_law_second_order’)

ChemDomain_ledger/C.31.v1 — Beer-Lambert law ‘ $A = \epsilon b c$ ’

domain: C • tier: Definition • status: current • coq: definition • parents: ChemDomain_ledger

Beer-Lambert law $A = \epsilon b c$

Occurrences (1): ‘engine/formulas.py’ (‘beer_lambert_absorbance’)

ChemDomain_ledger/C.32.v1 — Bragg’s law ‘ $n \lambda = 2 d \sin(\theta)$ ’

domain: C • tier: Definition • status: current • coq: definition • parents: ChemDomain_ledger

Bragg’s law $n \lambda = 2 d \sin(\theta)$

Occurrences (1): ‘engine/formulas.py’ (‘bragg_diffraction_angle’)

ChemDomain_ledger/C.33.v1 — Born-Landé lattice energy ‘ $U = -(N_A M z_1 z_2 e^2)/(4 \pi \epsilon_0 r_0)(1-1/n)$ ’

domain: C • tier: Definition • status: current • coq: definition • parents: ChemDomain_ledger

Born-Landé lattice energy $U = -(N_A M z_1 z_2 e^2)/(4 \pi \epsilon_0 r_0)(1-1/n)$

Occurrences (1): ‘engine/formulas.py’ (‘born_landé_lattice_energy’)

ChemDomain_ledger/C.34.v1 — Rigid-rotor rotational constant ‘ $B = h/(8 \pi^2 \mu r^2)$ ’

domain: C • tier: Definition • status: current • coq: definition • parents: ChemDomain_ledger

Rigid-rotor rotational constant $B = h/(8 \pi^2 \mu r^2)$

Occurrences (1): ‘engine/formulas.py’ (‘rigid_rotor_rotational_constant’)

ChemDomain_ledger/C.35.v1 — Debye-Hückel limiting law ‘ $\log_{10}(\gamma) = -A z^2 \sqrt{I}$ ’

domain: C • tier: Definition • status: current • coq: definition • parents: ChemDomain_ledger

Debye-Hückel limiting law $\log_{10}(\gamma) = -A z^2 \sqrt{I}$

Occurrences (1): ‘engine/formulas.py’ (‘debye_huckel_limiting_law’)

ChemDomain_ledger/C.36.v1 — Michaelis-Menten enzyme kinetics ‘ $v = V_{\max} [S]/(K_m + [S])$ ’

domain: C • tier: Definition • status: current • coq: definition • parents: ChemDomain_ledger

Michaelis-Menten enzyme kinetics $v = V_{\max} [S]/(K_m + [S])$

Occurrences (1): ‘engine/formulas.py’ (‘michaelis_menten_rate’)

ChemDomain_ledger/C.37.v1 — Eyring transition-state theory ‘ $k = (k_B T/h) \exp(-dG_{\ddagger}/(R T))$ ’

domain: C • tier: Definition • status: current • coq: definition • parents: ChemDomain_ledger

Eyring transition-state theory $k = (k_B T/h) \exp(-dG_{\ddagger}/(R T))$

Occurrences (1): ‘engine/formulas.py’ (‘eyring_rate_constant’)

ChemDomain_ledger/C.38.v1 — Marcus electron-transfer theory ‘ $k = A \exp(-(dG_0 + \lambda) / (4 \lambda k_B T))$ ’

domain: C • tier: Definition • status: current • coq: definition • parents: ChemDomain_ledger

Marcus electron-transfer theory $k = A \exp(-(dG_0 + \lambda) / (4 \lambda k_B T))$

Occurrences (1): ‘engine/formulas.py’ (‘marcus_electron_transfer_rate’)

ChemDomain_ledger/C.39.v1 — Langmuir adsorption isotherm ‘ $\theta = K P / (1 + K P)$ ’

domain: C • tier: Definition • status: current • coq: definition • parents: ChemDomain_ledger

Langmuir adsorption isotherm $\theta = K P / (1 + K P)$

Occurrences (1): ‘engine/formulas.py’ (‘langmuir_isotherm’)

ChemDomain_ledger/C.40.v1 — Brunauer-Emmett-Teller (BET) multilayer adsorption isotherm ‘ $V/V_m = C x / ((1-x)(1-x+Cx))$ ’

domain: C • tier: Definition • status: current • coq: definition • parents: ChemDomain_ledger

Brunauer-Emmett-Teller (BET) multilayer adsorption isotherm $V/V_m = C x / ((1-x)(1-x+Cx))$

Occurrences (1): ‘engine/formulas.py’ (‘bet_isotherm’)

ChemDomain_ledger/C.41.v1 — Zero-point-energy model of the primary kinetic isotope effect ‘ $k_H/k_D = \exp(hc(w_H - w_D) / (2 k_B T))$ ’

domain: C • tier: Definition • status: current • coq: definition • parents: ChemDomain_ledger

Zero-point-energy model of the primary kinetic isotope effect $k_H/k_D = \exp(hc(w_H - w_D) / (2 k_B T))$

Occurrences (1): ‘engine/formulas.py’ (‘kinetic_isotope_effect_zpe’)

ChemDomain_ledger/C.42.v1 — Stokes-Einstein diffusion coefficient ‘ $D = k_B T / (6 \pi \eta r)$ ’

domain: C • tier: Definition • status: current • coq: definition • parents: ChemDomain_ledger

Stokes-Einstein diffusion coefficient $D = k_B T / (6 \pi \eta r)$

Occurrences (1): ‘engine/formulas.py’ (‘stokes_einstein_diffusion’)

200 RelativityDomain_Qv — V.17 Registered Domain: Relativity — null-coordinate invariant Q_v

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: q_relativity_cone

null coords $n^+/- = vt +/- x$; $Q_v = n^+ \cdot n^- = v^2 t^2 - x^2$ (from the cone’s edges);
gates 01-04, 04 = RD-neutrality ($n^+ n^- = n^+ n^-$)

201 RelativityDomain_lapse_Lorentz — V.17 Registered Domain: Relativity — lapse, Lorentz-form transforms, redshift, horizon

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: RelativityDomain_Qv

$B = N \cdot \text{diag}(\kappa^{-1}, \kappa)$, $N = \sqrt{\det B}$ = lapse; $\Gamma_R = 1/\sqrt{1-\beta^2}$, $x' = \Gamma_R(x - u t)$, $t' = \dots$ (clock/length/simultaneity/composition); redshift $\nu_o = N \cdot \nu_i$; horizon $\Leftrightarrow N = 0$ (rank-loss, finite)

202 RelativityDomain_dynamic_metric_source — V.17 Registered Domain: Relativity — dynamic metric-source law

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *RelativityDomain_lapse_Lorentz*

dynamic metric-source step law: $M_{\text{Theta}} * \delta t^2 \text{Theta} + \text{grad}(U_{\text{Theta}}) + K \cdot \Phi^T (dG / d \text{Theta}) \Psi = 0$ -- with minimal closed state $Z_n = (\Phi_n, \Phi_{\{n-1\}}, \Psi_n, \Psi_{\{n-1\}}, \text{Theta}_n, \text{Theta}_{\{n-1\}})$, exact fixture $\text{Theta}_{\{n+1\}} = 3/8$, and a failing control (dropping the record source Ψ gives residual $1 \neq 0$ -- Ψ is load-bearing).

203 RelativityDomain_calculator_identities — V.17 Registered Domain: Relativity — Schwarzschild/Unruh calculator identities (NOT derived)

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *RelativityDomain_dynamic_metric_source*, *Gateway.SchwarzschildIdentity*

Schwarzschild $r_s = 2GE/c^4$ and Unruh $\tau_c = \pi c / a$ -- derived (they stay [DeclaredFormula] calculator identities, not established as a derivation of real physics).

204 QuantumDomain_osc_split — V.18 Registered Domain: Quantum — oscillatory split (green node)

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *q_quantum_osc*

oscillatory split $\lambda_c = D^2/4MK$ (osc iff $\lambda > \lambda_c$)

205 QuantumDomain_complexification — V.18 Registered Domain: Quantum — complexification gate (i as a readout, with failing control)

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *QuantumDomain_osc_split*

complexification gate: i is a READOUT of a closed real oriented mode-pair $J^2 = -I$, $J^{\dagger} J = -J$ (FAILING control: $J_{\text{bad}}^2 = +I \Rightarrow \text{NO_COMPLEX_QUANTUM}$)

206 QuantumDomain_norm_reversibility — V.18 Registered Domain: Quantum — positive norm and reversible evolution

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *QuantumDomain_complexification*

positive norm $N_Q = \langle \psi, \psi \rangle_G$; reversible $U^{\dagger} U = G$

207 QuantumDomain_Born_target — V.18 Registered Domain: Quantum — Born rule as a target, not an axiom

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *QuantumDomain_norm_reversibility*

The verifier shows refinement-consistency selects the quadratic $w(P_i|\psi) = \langle \psi, P_i \psi \rangle / \langle \psi, \psi \rangle$ (a linear $p=1$ rule fails: $\sqrt{25} \neq \sqrt{9} + \sqrt{16}$) -- but uniqueness across all admissible refinements is [TARGET -- NOT YET DERIVED], so measurement/Born stays RED.

208 BiologyDomain_living_unit — V.19 Registered Domain: Biology — living-unit fixed point (BIO-G3)

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *q_biology_eigenmode*

BIO-G3 living-unit $V_A = \text{Gamma}(V_A)$ under a declared boundary contract; no-free-repair ledger; mortality = irreversible exit from V_A .

BiologyDomain_living_unit/B.01.v1 — Compound microscope total magnification ‘ $M = M_{\text{objective}} * M_{\text{eyepiece}}$ ’

domain: *B* • *tier:* *Definition* • *status:* *current* • *coq:* *definition* • *parents:* *BiologyDomain_living_unit*

Compound microscope total magnification $M = M_{\text{objective}} * M_{\text{eyepiece}}$

Occurrences (1): ‘engine/formulas.py’ (‘microscope_total_magnification’)

BiologyDomain_living_unit/B.02.v1 — Surface-area-to-volume ratio of a cube ‘ $SA/V = 6/\text{side}$ ’

domain: *B* • *tier:* *Definition* • *status:* *current* • *coq:* *definition* • *parents:* *BiologyDomain_living_unit*

Surface-area-to-volume ratio of a cube $SA/V = 6/\text{side}$

Occurrences (1): ‘engine/formulas.py’ (‘surface_area_to_volume_ratio_cube’)

BiologyDomain_living_unit/B.03.v1 — Population density ‘ $\text{density} = \text{count}/\text{area}$ ’

domain: *B* • *tier:* *Definition* • *status:* *current* • *coq:* *definition* • *parents:* *BiologyDomain_living_unit*

Population density $\text{density} = \text{count}/\text{area}$

Occurrences (1): ‘engine/formulas.py’ (‘population_density’)

BiologyDomain_living_unit/B.04.v1 — Body mass index ‘ $\text{BMI} = \text{mass_kg}/\text{height_m}^2$ ’

domain: *B* • *tier:* *Definition* • *status:* *current* • *coq:* *definition* • *parents:* *BiologyDomain_living_unit*

Body mass index $\text{BMI} = \text{mass_kg}/\text{height_m}^2$

Occurrences (1): ‘engine/formulas.py’ (‘body_mass_index’)

BiologyDomain_living_unit/B.05.v1 — Mendelian monohybrid cross recessive-phenotype probability ‘ $P(aa) = p_{\text{maternal}} * p_{\text{paternal}}$ ’

domain: *B* • *tier:* *Definition* • *status:* *current* • *coq:* *definition* • *parents:* *BiologyDomain_living_unit*

Mendelian monohybrid cross recessive-phenotype probability $P(aa) = p_{\text{maternal}} * p_{\text{paternal}}$

Occurrences (1): ‘engine/formulas.py’ (‘monohybrid_cross_recessive_probability’)

BiologyDomain_living_unit/B.06.v1 — Cardiac output ‘ $CO = HR * SV$ ’

domain: B • tier: Definition • status: current • coq: definition • parents: BiologyDomain_living_unit

Cardiac output $CO = HR * SV$

Occurrences (1): ‘engine/formulas.py’ (‘cardiac_output’)

BiologyDomain_living_unit/B.07.v1 — Hardy-Weinberg equilibrium, homozygous-recessive genotype frequency ‘ q^2 ’

domain: B • tier: Definition • status: current • coq: definition • parents: BiologyDomain_living_unit

Hardy-Weinberg equilibrium, homozygous-recessive genotype frequency q^2

Occurrences (1): ‘engine/formulas.py’ (‘hardy_weinberg_genotype_freq’)

BiologyDomain_living_unit/B.08.v1 — Allele frequency from genotype counts ‘ $q = (2 n_{aa} + n_{Aa}) / (2(n_{AA} + n_{Aa} + n_{aa}))$ ’

domain: B • tier: Definition • status: current • coq: definition • parents: BiologyDomain_living_unit

Allele frequency from genotype counts $q = (2 n_{aa} + n_{Aa}) / (2(n_{AA} + n_{Aa} + n_{aa}))$

Occurrences (1): ‘engine/formulas.py’ (‘allele_frequency_from_genotype_counts’)

BiologyDomain_living_unit/B.09.v1 — Chi-square goodness-of-fit statistic ‘ $\chi^2 = \sum((O_i - E_i)^2 / E_i)$ ’

domain: B • tier: Definition • status: current • coq: definition • parents: BiologyDomain_living_unit

Chi-square goodness-of-fit statistic $\chi^2 = \sum((O_i - E_i)^2 / E_i)$

Occurrences (1): ‘engine/formulas.py’ (‘chi_square_goodness_of_fit’)

BiologyDomain_living_unit/B.10.v1 — Exponential (Malthusian) population growth ‘ $N = N_0 \exp(r t)$ ’

domain: B • tier: Definition • status: current • coq: definition • parents: BiologyDomain_living_unit

Exponential (Malthusian) population growth $N = N_0 \exp(r t)$

Occurrences (1): ‘engine/formulas.py’ (‘exponential_population_growth’)

BiologyDomain_living_unit/B.11.v1 — Logistic population growth with carrying capacity ‘ $N = K / (1 + ((K - N_0) / N_0) \exp(-r t))$ ’

domain: B • tier: Definition • status: current • coq: definition • parents: BiologyDomain_living_unit

Logistic population growth with carrying capacity $N = K / (1 + ((K - N_0) / N_0) \exp(-r t))$

Occurrences (1): ‘engine/formulas.py’ (‘logistic_population_growth’)

BiologyDomain_living_unit/B.12.v1 — Lincoln-Petersen mark-recapture population estimator ‘ $N = M C / R$ ’

domain: B • tier: Definition • status: current • coq: definition • parents: BiologyDomain_living_unit

Lincoln-Petersen mark-recapture population estimator $N = M C / R$

Occurrences (1): ‘engine/formulas.py’ (‘lincoln_petersen_population_estimate’)

BiologyDomain_living_unit/B.13.v1 — Plant/cell water potential ‘ $\Psi = \Psi_s + \Psi_p$ ’

domain: B • tier: Definition • status: current • coq: definition • parents: BiologyDomain_living_unit

Plant/cell water potential $\Psi = \Psi_s + \Psi_p$

Occurrences (1): ‘engine/formulas.py’ (‘water_potential’)

BiologyDomain_living_unit/B.14.v1 — Fick’s first law of diffusion ‘ $\text{rate} = D A (dC/dx)$ ’

domain: B • tier: Definition • status: current • coq: definition • parents: BiologyDomain_living_unit

Fick’s first law of diffusion rate $= D A (dC/dx)$

Occurrences (1): ‘engine/formulas.py’ (‘ficks_law_diffusion_rate’)

BiologyDomain_living_unit/B.15.v1 — Respiratory quotient ‘ $RQ = \text{CO}_2\text{produced}/\text{O}_2\text{consumed}$ ’

domain: B • tier: Definition • status: current • coq: definition • parents: BiologyDomain_living_unit

Respiratory quotient $RQ = \text{CO}_2\text{produced}/\text{O}_2\text{consumed}$

Occurrences (1): ‘engine/formulas.py’ (‘respiratory_quotient’)

BiologyDomain_living_unit/B.16.v1 — Goldman-Hodgkin-Katz multi-ion membrane potential ‘ $V_m = (RT/F) \ln((P_K K_{out} + P_{Na} Na_{out} + P_{Cl} Cl_{in}) / (P_K K_{in} + P_{Na} Na_{in} + P_{Cl} Cl_{out}))$ ’

domain: B • tier: Definition • status: current • coq: definition • parents: BiologyDomain_living_unit

Goldman-Hodgkin-Katz multi-ion membrane potential $V_m = (RT/F) \ln((P_K K_{out} + P_{Na} Na_{out} + P_{Cl} Cl_{in}) / (P_K K_{in} + P_{Na} Na_{in} + P_{Cl} Cl_{out}))$

Occurrences (1): ‘engine/formulas.py’ (‘goldman_hodgkin_katz_potential’)

BiologyDomain_living_unit/B.17.v1 — Hill equation cooperative ligand binding ‘ $Y = S^n / (S_0^n + S^n)$ ’ (Vmax-scaled variant also serves the PhD allosteric-enzyme row)

domain: B • tier: Definition • status: current • coq: definition • parents: BiologyDomain_living_unit

Hill equation cooperative ligand binding $Y = S^n / (S_0^n + S^n)$ (Vmax-scaled variant also serves the PhD allosteric-enzyme row)

Occurrences (1): ‘engine/formulas.py’ (‘hill_equation_cooperative_binding’)

BiologyDomain_living_unit/B.18.v1 — Lotka-Volterra predator-prey interior equilibrium ‘ $(x^*, y^*) = (\gamma/\delta, \alpha/\beta)$ ’

domain: B • tier: Definition • status: current • coq: definition • parents: BiologyDomain_living_unit

Lotka-Volterra predator-prey interior equilibrium $(x^*, y^*) = (\gamma/\delta, \alpha/\beta)$

Occurrences (1): ‘engine/formulas.py’ (‘lotka_volterra_equilibrium’)

BiologyDomain_living_unit/B.19.v1 — One-compartment pharmacokinetic clearance ‘ $CL = k V_d$ ’

domain: B • tier: Definition • status: current • coq: definition • parents: BiologyDomain_living_unit

One-compartment pharmacokinetic clearance $CL = k V_d$

Occurrences (1): ‘engine/formulas.py’ (‘pharmacokinetic_clearance’)

BiologyDomain_living_unit/B.20.v1 — One-compartment IV-bolus area under the curve ‘ $AUC = C_0/k$ ’

domain: B • tier: Definition • status: current • coq: definition • parents: BiologyDomain_living_unit

One-compartment IV-bolus area under the curve $AUC = C_0/k$

Occurrences (1): ‘engine/formulas.py’ (‘pharmacokinetic_auc_iv_bolus’)

BiologyDomain_living_unit/B.21.v1 — Mutation-selection balance, recessive allele ‘ $\hat{q} = \sqrt{\mu/s}$ ’

domain: B • tier: Definition • status: current • coq: definition • parents: BiologyDomain_living_unit

Mutation-selection balance, recessive allele $\hat{q} = \sqrt{\mu/s}$

Occurrences (1): ‘engine/formulas.py’ (‘mutation_selection_balance_recessive’)

BiologyDomain_living_unit/B.22.v1 — Kleiber’s law, 3/4-power metabolic scaling ‘ $BMR = a M^{0.75}$ ’

domain: B • tier: Definition • status: current • coq: definition • parents: BiologyDomain_living_unit

Kleiber’s law, 3/4-power metabolic scaling $BMR = a M^{0.75}$

Occurrences (1): ‘engine/formulas.py’ (‘kleibers_law_metabolic_rate’)

BiologyDomain_living_unit/B.23.v1 — Jukes-Cantor phylogenetic distance ‘ $d = -3/4 \ln(1-4p/3)$ ’

domain: B • tier: Definition • status: current • coq: definition • parents: BiologyDomain_living_unit

Jukes-Cantor phylogenetic distance $d = -3/4 \ln(1-4p/3)$

Occurrences (1): ‘engine/formulas.py’ (‘jukes_cantor_distance’)

BiologyDomain_living_unit/B.24.v1 — Moran/Wright-Fisher genic-selection fixation probability ‘ $p_{\text{fix}} = (1 - \exp(-2 N_e s p_0)) / (1 - \exp(-2 N_e s))$ ’

domain: B • tier: Definition • status: current • coq: definition • parents: BiologyDomain_living_unit

Moran/Wright-Fisher genic-selection fixation probability $p_{\text{fix}} = (1 - \exp(-2 N_e s p_0)) / (1 - \exp(-2 N_e s))$

Occurrences (1): ‘engine/formulas.py’ (‘moran_fixation_probability’)

BiologyDomain_living_unit/B.25.v1 — Kingman coalescent expected TMRCA ‘ $E[\text{TMRCA}] = 2(1-1/n) N_e$ ’

domain: B • tier: Definition • status: current • coq: definition • parents: BiologyDomain_living_unit

Kingman coalescent expected TMRCA $E[\text{TMRCA}] = 2(1-1/n) N_e$

Occurrences (1): ‘engine/formulas.py’ (‘coalescent_expected_tmrca’)

BiologyDomain_living_unit/B.26.v1 — Price equation, selection-only term ‘ $\Delta \bar{z} = \text{Cov}(w, z) / \bar{w}$ ’

domain: B • tier: Definition • status: current • coq: definition • parents: BiologyDomain_living_unit

Price equation, selection-only term $\Delta \bar{z} = \text{Cov}(w, z) / \bar{w}$

Occurrences (1): ‘engine/formulas.py’ (‘price_equation_selection_differential’)

BiologyDomain_living_unit/B.27.v1 — Lotka-Volterra interior-equilibrium Jacobian eigenvalue magnitude ‘ $\sqrt{|\text{Im}(\lambda)|} = \sqrt{\alpha \gamma}$ ’

domain: B • tier: Definition • status: current • coq: definition • parents: BiologyDomain_living_unit

Lotka-Volterra interior-equilibrium Jacobian eigenvalue magnitude $\sqrt{|\text{Im}(\lambda)|} = \sqrt{\alpha \gamma}$

Occurrences (1): ‘engine/formulas.py’ (‘lotka_volterra_interior_stability’)

BiologyDomain_living_unit/B.28.v1 — Luria-Delbrück fluctuation-test expected mutant number ‘ $m = N \mu$ ’

domain: B • tier: Definition • status: current • coq: definition • parents: BiologyDomain_living_unit

Luria-Delbrück fluctuation-test expected mutant number $m = N \mu$

Occurrences (1): ‘engine/formulas.py’ (‘luria_delbruck_expected_mutants’)

BiologyDomain_living_unit/B.29.v1 — Karlin-Altschul local-alignment E-value ‘ $E = K m n \exp(-\lambda S)$ ’

domain: B • tier: Definition • status: current • coq: definition • parents: BiologyDomain_living_unit

Karlin-Altschul local-alignment E-value $E = K m n \exp(-\lambda S)$

Occurrences (1): ‘engine/formulas.py’ (‘karlin_altschul_evalu’)

BiologyDomain_living_unit/B.30.v1 — Approaching-sound auditory time-dilation effect sizes (+15% approach / -6% recede)

domain: B • tier: untagged • status: current • coq: not_formalisable • parents: BiologyDomain_living_unit

Approaching-sound auditory time-dilation effect sizes (+15% approach / -6% recede)

Occurrences (1): ‘InfoApproachingWaveLocalTimeDilation_attempt.v’

209 BiologyDomain_heredit — V.19 Registered Domain: Biology — heredity quotient / lineage-count frequency change (BIO-G4)

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: BiologyDomain_living_unit

BIO-G4 heredity $q_H \rightarrow$ lineage counts $N_{\{n+1\}} = B_n N_n \rightarrow$ frequency change [CLOSED count]; worked fixture $B = \text{diag}(2,1)$, $N_0 = (1,1) \rightarrow p_1 = (2/3, 1/3)$, no fitness variable.

210 BiologyDomain_T2_closed — V.19 Registered Domain: Biology — endogenous operator (T2) closure by operator-memory augmentation

domain: — • tier: Open • status: current • coq: root_layer_unwired • parents: T2, BiologyDomain_living_unit

T2 endogenous $L_R[\Theta]$: I-only fails, augmented (I, Θ) closes. [CLOSED]

211 WeldDSL — V.20 THE WELD — master equation and domain-DSL are one object (commuting square)

domain: — • *tier:* *finite_diagnostic* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *weld*, *V0_DomainDiscoveryEngine*

$q_D \circ F = F\#_D \circ q_D$ -- a domain q_D is an admissible SENTENCE of the DSL iff it commutes with the master stepper F . Witnessed by scripts/unification_weld.py (exact-rational, PASS): a quotient q_D that closes the square is admitted as a real sentence; a one-entry perturbation of F breaks the square and the quotient is correctly refused.

Occurrences (1): (2)

212 SM_DAG — V.21 Standard Model — root-native DAG and gauge-as-automorphism ($\text{Aut}(F,O)$)

domain: — • *tier:* *Th_coqc* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *GaugeFaces*, *EQ-042*

$\text{Aut}(F,O) = \{ h : 0 \circ h = 0, h \circ F = F \circ h \}$ -- the internal symmetry, grown from the exact-quotient discipline, never posited. Spine of the DAG: $\delta_R \rightarrow$ typed state $\rightarrow$ ordered paths $\rightarrow$ [R3 readout-preserving automorphism, FIRST GATE] $\rightarrow$ localized transport $\rightarrow$ connection $\rightarrow$ holonomy $\rightarrow$ curvature $\rightarrow$ gauge algebra / quantum completion / scale flow $\rightarrow$ representations $\rightarrow$ chirality/anomaly $\rightarrow$ order-parameter breaking $G \rightarrow H \rightarrow$ mass/coupling structure $\rightarrow$ generations/mixing $\rightarrow$ empirical group-ID $\rightarrow$ bounded SM claim. Verdict: Root-Native Standard-Model Architecture: Substantial Conditional and Node-Level Closure, not a complete result; end-to-end physical SM from the UNRESTRICTED root is still 0% OPEN.

213 UnifiedForce_action — V.22 Unified Force — one action, four projections (master statement)

domain: — • *tier:* *Th_coqc* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *SM_DAG*

$S_{UF} = S_{DRL} + S_{\Theta} + S_U + S_{\Sigma} + S_{\text{cut}/\text{tape}}$, $\delta S_{UF} / \delta Z_n = J_{\{C,n\}[Z_n]}$; $F_D = P_D(-\text{grad}_Z U_{UF} + J_C)$; $P_{\text{geo}} \rightarrow \text{gravity}$. $P_{\text{center}} \rightarrow \text{EM}$. $P_{\text{chiral,broken}} \rightarrow \text{weak}$. $P_{\text{simple,unbroken}} \rightarrow \text{strong}$

214 UnifiedForceClosure_v0.1 — Unified Force Closure v0.1 — internally closed at finite/typed level, chi4 completeness score

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *UnifiedForce_action*

One action, one block master equation, four forces as orthogonal projection readouts $F_{\text{all}} = F_G + F_{EM} + F_W + F_S + F_{\text{res}}$, with a falsifiable completeness score χ_4 ($\chi_4 = 1$ on a strong+weak fixture; $\chi_4 < 1$ when a hidden fifth interaction leaks). Witnessed: automorphism group, connection-law-from-commutation, holonomy invariants, order-defect+Jacobi, covariant Laplacian, the selected-state h/m split -- with six failing controls. Physical four-force unification is OPEN.

215 SM_G0_order_defect — SM-G0/G0.6 — order-defect from ordered composition (borrow #2 discharged)

domain: — • *tier:* *Th_coqc* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *EQ-042*, *UnifiedForceClosure_v0.1*

InfoOrderDefectFromComposition.v (Print Assumptions Closed over Q): ordered composition is plain 2x2 rational matrix product (no non-commuting term written in); associativity is a theorem; $K(X,Y) = XY - YX$ is bilinear + antisymmetric; Jacobi is DERIVED from associativity; non-commutativity is emergent (a witness pair fails to commute); commuting transports imply zero defect (AP20's own control, now derived). $c_self/c_geo = 1$ is not root-derived; the non-abelian input is relocated, not eliminated.

216 FourForceCirculation_v0.2 — Four-Force Circulation v0.2 — directed-response identity and Omega recovery

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: UnifiedForceClosure_v0.1

On a four-sector ring fixture $A = H + \Omega$ (H symmetric reciprocal load, Ω antisymmetric circulation load), $\chi = A^{-1}$: $\text{spec}(H) = \{6,4,4,2\} > 0$, $\text{spec}(A) = \{6,2,4,-i\}$, $\text{Re } \lambda(A) > 0$; exact directed-response identity $\chi - \chi^T = -2 \chi^T \Omega \chi$; $\Omega_{\text{recovered}} = \Omega_{\text{planted}}$ read as the unique antisymmetric part of χ^{-1} (exact by construction), with reciprocal ($\Omega=0 \Rightarrow \chi=\chi^T \Rightarrow \Omega_{\text{rec}}=0$) and missing-edge failing controls. Baseline $\Omega=0$ recovers Onsager reciprocity.

217 ElectroweakDecoder_v0.3 — Unified Force v0.3 — Calibrated Electroweak Decoder (EM $\leftrightarrow$ Weak)

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: FourForceCirculation_v0.2

Structural core (exact): the neutral obstruction $M^2 = (v^2/4) \cdot \text{outer}((g,-g'),(g,-g'))$ is rank-1, so $\det M^2 = 0$ identically, giving exactly one massless + one massive direction -- the photon's masslessness EMERGES: photon $\sim (g',g) = \sin(\theta_W) W^3 + \cos(\theta_W) B$ (eigenvalue 0), $Z \sim (g,-g')$ with $m_Z^2 = (v^2/4)(g^2+g'^2)$, $m_W^2 = g^2 v^2/4$; failing control = a generic rank-2 obstruction has $\det \neq 0$, no massless direction. Calibration (CODATA2022+PDG2025): $v=246.21965$ GeV, $g'/g=0.535802$, $g=0.652824$, $g'=0.349784$; held-out tree-level $M_Z = M_W/\cos(\theta_W) = 91.178$ vs PDG 91.188 (0.011%, radiative layer OPEN).

218 SM_Discovery_Pipeline_v0.4 — SM Discovery Pipeline v0.4 — finite blind discovery of hypercharges and cubic anomaly cancellation

domain: — • tier: Th_coqc • status: current • coq: root_layer_unwired • parents: ElectroweakDecoder_v0.3

BLIND discriminators (group name, reps, charges, Left/Right never inputs) recover on finite fixtures: the algebra (center 1 + ideals $8+3 \Rightarrow u(1)+su(3)+su(2)$; negative control $1+3+3 \Rightarrow \text{FAIL}$); chirality ($\eta_\chi = 1$ chiral vs 0 vectorlike, conditional on a given orientation op); the reps $(3,2)+(1,2)+(3,1)x2+(1,1)$; and the hypercharges $q=1/6$, $l=-1/2$, $u=2/3$, $d=-1/3$, $e=-1$ FORCED by anomaly + coupling closure with the cubic $[U(1)]^3$ anomaly = 0 exact (Th_coqc). A finite radiative log-det engine gives raw curvatures $(r1,r2,r3) = (-153/20,-243/20,-216/5)$ and $\text{Tr } K0^{-1} = 27/10$ -- finite diagnostic, NOT physical beta-functions.

219 ColorSU3_v0.2 — Color=3 and SU(3) from ordered tape (v0.2)

domain: — • tier: Th_coqc • status: current • coq: root_layer_unwired • parents: EQ-041

Order-sensitivity $\Rightarrow$ antisymmetry, and closed-loop start-independence $\Rightarrow (-1)^{\{k-1\}} = 1 \Rightarrow k$ odd (derived, not posited) $\Rightarrow$ minimal $k=3 \Rightarrow \dim V=3$; preserving the load + triple record $\Rightarrow SU(3)$, the common phase $c^3=1 \Rightarrow$ the Z_3 center, $\dim su(3) = 9-1 = 8$; single channels carry no invariant readout, pair and triple do (kinematic color-neutrality).

220 CenterConfinement_v0.3 — Center-sector confinement (v0.3) — Wilson area law

domain: — • tier: Th_coqc • status: current • coq: root_layer_unwired • parents: ColorSU3_v0.2, EQ-051

The root curvature action $\kappa \sum |u_p - 1|^2$ gives $\langle u_p \rangle = (1-r)/(1+2r) = q$, hence $\langle W(C) \rangle = q^{\{A(C)\}}$ and the area law $-\log |\langle W \rangle| = \sigma A$, $\sigma = -\log(q) > 0$, so $V(R) = \sigma R$ -- separation cost grows linearly, derived not assumed (control $\kappa \rightarrow \infty \Rightarrow \sigma \rightarrow 0$). Scope: Z₃-only, 2D, no continuum.

221 ConfinementCertificate_v0.5 — Confinement certificate (v0.5) — computable confinement certificate

domain: — • tier: Th_coqc • status: current • coq: root_layer_unwired • parents: CenterConfinement_v0.3

Derive ρ_t (triality retention = character integral) and $\mu_4 \leq 20e$ (4D plaquette entropy) FROM the action $S_p = \kappa \|U - I\|^2$ (with $\|U - I\|^2 = 6 - 2 \operatorname{Re} \operatorname{Tr} U$ on $SU(3)$), making confinement a computable certificate $c_t = \mu_4 * \rho_t < 1$: rigorous for $0 < \kappa < 0.0020252$ (all reps, $\rho_t \leq e^{\{9\kappa\}} - 1$), candidate $\kappa \sim 0.053$.

222 TrialitySpectralFlow_v0.6 — Triality spectral flow (v0.6) — Block-Scale Existence Theorem

domain: — • tier: Th_coqc • status: current • coq: root_layer_unwired • parents: ConfinementCertificate_v0.5

Serial blocking is convolution $\Rightarrow$ Peter-Weyl $\Rightarrow a_R^{\{m\}} = a_R^m \Rightarrow \rho_t(b) = \rho_t\{b^2\}$, so information cost grows as area $I_t(b) = b^2 * I_t(1)$. Block-Scale Existence Theorem: for any $0 < \rho_t < 1$ and finite μ_4 there is a coarse scale $b_* > \sqrt{\log(\mu_4) / -\log(\rho_t)}$ with $c_t(b_*) < 1$ -- spectral contraction beats surface entropy (analytic PASS: $\kappa=0.01$, $b=2 \Rightarrow c \leq 0.004276$).

223 BlockClosure_v0.7 — b=2 block, first correlated shell (v0.7)

domain: — • tier: Th_coqc • status: current • coq: root_layer_unwired • parents: TrialitySpectralFlow_v0.6

Exact block integral $K_{\{B2\}}$, $c_R^{\{2\}}$, $\rho_t^{\text{full}(2)}$ are defined (finite tensor contraction). First correlated shell (single cube-bumps) is exact: in 4D each plaquette has $2(D-2)=4$ bump directions, minimal surface $A_{\min}=4$, giving 16 single-bumps (area $4 \rightarrow 8$), so $\rho_{\{1, \text{geom}\}}(2) = u^4 (1+16u^4)$ and correlation defect $\epsilon_{\text{geom}\{1\}} = (1/4) \log(1+16u^4) > 0$ -- correlations HELP triality survive, but only at $O(u^4)$. First-shell certificate $20e * u^4 (1+16u^4) < 1 \Leftrightarrow u < 0.34915$ (later shown in v0.9 to be an invalid convergence criterion). A plain Metropolis estimator FAILED (signal $\approx$ noise, FAIL_ESTIMATOR_NOISE).

224 CorrectedCertificate_v0.9 — Corrected certificate (v0.9) — linear convergence criterion, representation-tail closure

domain: — • tier: Th_coqc • status: current • coq: root_layer_unwired • parents: BlockClosure_v0.7

Self-caught correction: v0.7's $20e * u^4 (1+16u^4) < 1 \Leftrightarrow u < 0.34915$ was not a valid convergence criterion -- $u^4 (1+16u^4)$ is the minimal-surface amplitude, not the per-plaquette ratio; entropy grows with area, so the sum $\sum_A N_A u^A$ converges iff $\mu_4 \cdot \hat{u} < 1$ (LINEAR). From $G > 0$ the internal automorphism group is compact, the link intertwiner $P = \int_G \rho(h) dh$ is an orthogonal projector with $\|P\| \leq 1$ (a contraction), so $\hat{u} \leq u/(1-8v)$ for $8v < 1$; corrected window $\kappa \sim 0.05358$.

225 AllOrderCharacter_v1.0 — All-order character closure (v1.0) — exact SU(3) Weyl integral window

domain: — • *tier:* Th_coqc • *status:* current • *coq:* root_layer_unwired • *parents:* CorrectedCertificate_v0.9, E Q-054

u(kappa), v(kappa) closed to all orders from exact SU(3) Weyl integrals (deterministic quadrature, no truncation): $u = c_3/(3 c_0)$, $v = c_8/(8 c_0)$, $\chi_8 = |\chi_3|^2 - 1$ from 3 tensor 3bar = 1 + 8, recursion $c_0' = 2 c_3$; the all-order window is $0 < \kappa < 0.053583974745\dots$, matching the v0.9 low-order series to $\sim 4e-6$ (not a fit). High-precision numerical closure, not a rigorous interval proof.

226 SurfaceAutomaton_v1.1 — Surface automaton for $\mu_4^{\text{admissible}}$ (v1.1) — lower-bound spectral radii

domain: — • *tier:* Th_coqc • *status:* current • *coq:* root_layer_unwired • *parents:* AllOrderCharacter_v1.0

$\mu_4^{\text{admissible}}$ = the Perron growth of a frontier automaton whose state b_s is exactly the minimal sufficient quotient. First real 4D spectral radii (single-sheet sector): canonical connector $z^3 + z^2 + 3z - 1 = 0 \Rightarrow \mu_{\text{can}} = 3.38298$; shortest-paths $2z^3 - z^2 + 4z - 1 = 0 \Rightarrow \mu_{\text{short}} = 3.87513$ (finite strips $H=0..4$ give 1, 2.846, 3.348, 3.555, 3.660, converging). This is a LOWER bound (omits bubbles/handles/branching), giving bracket $3.87513 \leq \mu_4^{\text{admissible}} \leq 54.366$; does not yet replace 20e in the certificate.

227 UpperAutomaton_v1.2 — Upper automaton (v1.2) — entropy ceiling 54 to 7.084

domain: — • *tier:* Th_coqc • *status:* current • *coq:* root_layer_unwired • *parents:* SurfaceAutomaton_v1.1

A 4D edge touches 6 plaquettes (1 in, 5 free x_i in Z_3); Z_3 link closure $\sum x_i = 2 \pmod{3}$ has $3^4=81$ solutions; splitting by the number k of nonzero new plaquettes gives the branching polynomial $B(z) = 5z + 10z^2 + 30z^3 + 25z^4 + 11z^5$ ($\text{Sum}=81$; the $k=2$ coefficient 10 is the Z_3 triple junction $1 \rightarrow 1+1$). A scalar majorant makes $B(z) < 1$ a contraction; critical root $B(z_+) = 1 \Rightarrow z_+ = 0.141161 \Rightarrow \mu^+ \leq 7.084097$ (upper bound, over-counts non-closing surfaces). Squeezes the bracket to $3.87513 \leq \mu_4^{\text{admissible}} \leq 7.08410$ and widens the certificate window to $\kappa < 0.321687$ (about 6x the old 20e window).

228 MassGap_v1.3_v1.4 — Mass gap — finite-transfer gap theorem (v1.3) and universal reflection-positive slab (v1.4)

domain: — • *tier:* Th_coqc • *status:* current • *coq:* root_layer_unwired • *parents:* UpperAutomaton_v1.2

v1.3: given a positive self-adjoint physical transfer operator T_{phys} with a unique vacuum that strictly contracts the nonvacuum sector ($q = \|T_{\text{phys}} P_{\text{perp}}\| < 1$), the gap $\Delta = -(1/a) \log(q) > 0$ and connected gauge-invariant correlators decay exponentially -- with negative controls (massless-diffusion gap closes as $L \rightarrow \infty$; vacuum degeneracy $\Rightarrow \Delta=0$). v1.4: gauge $(U(1)/SU(2)/SU(3)) + \text{scalar} + \text{generic fermionic}$ Fock kernels are Gram-positive $\Rightarrow T_{\text{UF}} \geq 0$, and the same reader $m = -(1/a) \log(\lambda)$ classifies every mass TYPE from the spectral-measure shape; a PDG-2026 unit test shows the extractor reads the input masses back (tests the READER, not a prediction); held-out test: mass ratios $m_i/m_j = \log(\lambda_{i,1})/\log(\lambda_{j,1})$ are lattice-scale independent ($\mu/e = 206.768, \dots$).

229 Hypercharge_Z6_v1.5 — Hypercharge + Z6 global structure — anomaly closure (v1.5)

domain: — • tier: Th_coqc • status: current • coq: root_layer_unwired • parents: SM_Discovery_Pipeline_v0.4

Under the minimal one-generation skeleton (Q, u^c, d^c, L, e^c, H) , interaction closure fixes charges up to two free scales ($u=-q-h$, $d=-q+h$, $l=-3q$, $e=h+3q$); anomaly cancellation forces the ratios: gravitational anomaly $A_{\text{grav}} = 6q+3u+3d+2l+e = h-3q$ and cubic $A_{111} = (h-3q)^3$ factor into the SAME condition $h=3q$ ($A_{111} = (A_{\text{grav}})^3$). At $h=3q$, primitive $q=1$: $y = 6Y = (1, -4, 2, -3, 6, 3) \Rightarrow Y = (1/6, -2/3, 1/3, -1/2, 1, 1/2)$ and $Q_{\text{em}} = T3+Y = (2/3, -1/3, 0, -1)$. Center-lock $2t+3s+y \equiv 0 \pmod{6}$ holds for every record $\Rightarrow$ the invisible common center is exactly Z_6 , so $G_{\text{phys}} = [SU(3) \times SU(2) \times U(1)]/Z_6$. Global $SU(2)$ (Witten): $\# \text{doublets} = 3Q+L = 4$ is even (control: colored-Q-only = 3 odd would FAIL). Negative control: adding a right-handed ν^c makes $A_{\text{grav}} = A_{111} = 0$ for all $(q, h) \Rightarrow Y$ and B-L degenerate, lifted only by the extra self-pair ν^c tensor $\nu^c \rightarrow 1 \Rightarrow 2n=0$.

230 BlindMatterSearch_v1.6 — Blind matter search (v1.6) — forced matter skeleton

domain: — • tier: Th_coqc • status: current • coq: root_layer_unwired • parents: Hypercharge_Z6_v1.5

Enumerate representation multiplicities over the minimal carrier alphabet $\{1, 3, 3\text{bar}\} \times \{1, 2\}$ under gauge-consistency gates (color $[SU(3)]^3$ anomaly, global- $SU(2)$ parity, closure, no-vectorlike, minimality). The unique minimum up to total conjugation is $(3, 2) + 2(3\text{bar}, 1) + (1, 2) + (1, 1)$, $N=5$, $D_{\text{total}}=15$ -- the two colored singlets and the colorless doublet are forced, not listed; names are attached AFTER. Coq proves the unbounded minimality $D_{\text{total}} \geq 15$ (via lia, not just a finite scan).

231 RootNativeChirality_v1.7 — Root-native chirality (v1.7) — chirality grading and exact no-go

domain: — • tier: Th_coqc • status: current • coq: root_layer_unwired • parents: BlindMatterSearch_v1.6

The chirality grading Γ_T GROWS from the ordered triple $(\Gamma^2=I, \Gamma^{\dagger}=G, R \Gamma R = -\Gamma)$, no Γ^5 imported; an exact no-go shows an unbroken tape-reversal makes the two orientation sectors carry equivalent weak reps, so grading alone gives no chiral gauge asymmetry -- it takes an orientation-odd order X_i , and $P_w = (I - X_i \Gamma)/2$ makes $SU(2)$ act on one sector.

232 TapeKineticOperator_v1.8 — Tape kinetic operator (v1.8) — Ginsparg-Wilson relation and no species doubling

domain: — • tier: Th_coqc • status: current • coq: root_layer_unwired • parents: RootNativeChirality_v1.7

A root-native D_T obeys an exact Ginsparg-Wilson relation $\Gamma(I-V) + (I-V)\Gamma = (I-V)\Gamma(I-V)$, so chirality survives at finite spacing without naive anticommutation; the free fixture has no species doubling ($0 < m_0 < 2r \Rightarrow$ one physical zero; $16 \rightarrow 1+15$ at $d=4$) -- the Clifford relation $\{A_\mu, A_\nu\} = 2\delta$ is forced by isotropy, not imported.

233 WhyD4_v1.9 — Why $d=4$ (v1.9) — derived spacetime dimension

domain: — • tier: Th_coqc • status: current • coq: root_layer_unwired • parents: TapeKineticOperator_v1.8

The minimal discrete carrier (orientation bit tensor incidence bit, dim 4) + first-order isotropy + kinetic-closure gate + no-spectator derive d=4: $d \leq 4$ (≤ 3 orthogonal directions in R^3), d even (parity $(-1)^{d-1} = -1$), $d \neq 2$ (incidence spectator), with explicit realization $A_0 = \tau_1$ tensor I , $A_i = \tau_2$ tensor σ_i , and $\Gamma_T = -A_0 A_1 A_2 A_3$.

234 IsotropicFixedPoint_v1.10 — Isotropic fixed point — Lorentz shadow from coarse-graining (v1.10)

domain: — • tier: *Th_coqc* • status: *current* • coq: *root_layer_unwired* • parents: *WhyD4_v1.9*

The finite signed-permutation twirl $\Pi_4(X) = (\text{Tr}(X)/4)$. I (over the order-384 group $B_4 = (Z_2)^4 \text{ semidirect } S_4$) projects onto isotropy; a partial twirl gives the exact contraction $\Delta_{n+1} = (1-\alpha) \Delta_n$. A five-move local mixer has exact spectral radius $\rho_{\text{frame}} = 0.858010754587974\dots < 1$ on the 9-dim traceless space (largest root of $15625 \lambda^6 - 25000 \lambda^5 + 8125 \lambda^4 + 3500 \lambda^3 - 1625 \lambda^2 - 100 \lambda + 59$, rational Bolzano bracket $p(858/1000) < 0 < p(8581/10000)$, $p(1)=584 > 0$). Coupled sectors mix by a doubly-stochastic map, so $c_{\text{gauge}} = c_{\text{matter}} = c_{\text{order}} = c_{\text{tape}} \rightarrow c_*$, giving a 3+1 Lorentz shadow $E^2 = c_*^2 |p|^2 + m_s^2 c_*^4$ (Lorentz universality without mass universality).

235 FrameMixingFromAction_v1.11 — Frame-mixing weights from the action (v1.11) — reflection-positive Gram derivation

domain: — • tier: *Th_coqc* • status: *current* • coq: *root_layer_unwired* • parents: *IsotropicFixedPoint_v1.10*

A primitive frame-rewrite grammar (identity, swap, cycle, cycle^{-1} , sign-flip; order-384 frame group F_4) carries retained-rewrite costs $\epsilon(m)$, giving half-step amplitudes $b_m = e^{-\epsilon(m)/2}$; a hidden-midpoint slab makes the frame transfer a Gram operator $K_{\text{fr}} = B^\dagger B$ (reflection-positive by construction), whose relative-frame expansion yields weights $p(h) = \sum_{\{m^{-1} n = h\}} b_m b_n / (\sum b)^2$ with $p(h) \geq 0$, $\sum p(h) = 1$, $p(h) = p(h^{-1})$. The induced metric channel $M_b(G) = \sum_h p(h) h^T G h$ has only fixed metric the scalar line $R.I_4$ -- isotropy is derived, not assumed. Equal-cost fixture: anisotropy contracts at exact rate $\rho_{\text{aniso}} = 0.7361824549886247\dots < 1$ (largest root of $244140625 \lambda^6 - 371093750 \lambda^5 + 190234375 \lambda^4 - 41812500 \lambda^3 + 4299375 \lambda^2 - 201750 \lambda + 3481$), a stronger contraction than v1.10's 0.8580.

236 OrderHiggsClosure_v1.12 — Minimal Order/Higgs (v1.12) — Higgs rep forced, electroweak mass matrix

domain: — • tier: *Th_coqc* • status: *current* • coq: *root_layer_unwired* • parents: *FrameMixingFromAction_v1.11*

From the blind matter skeleton, the minimal order carrier is forced: a color-singlet, an $SU(2)$ doublet (2 tensor R_2 contains 1 iff $R_2=2$), and hypercharge $y_H=3$ fixed by every matter closure ($1+h-4=0$, $1-h+2=0$, $-3-h+6=0$) -- i.e. $H=(1,2)_{\{1/2\}}$, the Higgs rep derived, not fed. A nonzero order H_* has stabilizer $Q_{\text{res}} = T_3 + Y$ (dim 1), so $4-1=3$ of the four $SU(2) \times U(1)$ generators are broken; the neutral (W^3, B) mass matrix $(v^2/4)[[g^2, -gg'], [-gg', g'^2]]$ has $\det 0$, rank 1 $\Rightarrow$ one massless photon $A \sim (g', g)$ and $m_Z^2 = (g^2 + g'^2) v^2/4$, charged pair gives $m_W^2 = g^2 v^2/4 \Rightarrow m_W = m_Z \cos(\theta)$ and tree-level $\rho=1$; degree-of-freedom count balances $8+4 = 2+9+1$.

237 IntertwinerOrderVacuum_v1.13 — Intertwiner and order-vacuum closure (v1.13) — corrected $\text{Pi}_0 > \alpha$ criterion

domain: — • tier: *Th_coqc* • status: *current* • coq: *root_layer_unwired* • parents: *OrderHiggsClosure_v1.12*

Invariant multiplicities $\text{nu}_U=\text{nu}_D=\text{nu}_E=1$ (each $\text{Hom}_G(\dots,1)$ one-dimensional); fixing the order carrier H , the linear maps $M_j(H)$ satisfy $M_j M_j^\dagger = r \cdot I_{\{d_j\}}$ with rank $(d_U, d_D, d_E) = (3, 3, 1)$ giving 7 closure singular modes and one primitive ordered-tape history per branch. Correction: an earlier exponential ansatz $Z_j(r) = 1 + \text{zeta}_j e^{\{\kappa_j r\}}$ wrongly kept weight $\text{zeta}_j > 0$ at $r=0$ and is superseded by the representation-derived fermionic Fock determinant $Z_j(r) = \det(I+K_j) = (1+\text{lambda}_j r)^{d_j}$. With $V_{\text{eff}}(r) = \alpha r + \beta r^2 - 3 \log(1+\text{lambda}_U r) - 3 \log(1+\text{lambda}_D r) - \log(1+\text{lambda}_E r)$, $V_{\text{eff}}'(0) = \alpha - \text{Pi}_0$, $\text{Pi}_0 = 3 \text{lambda}_U + 3 \text{lambda}_D + \text{lambda}_E$; $\text{Pi}_0 > \alpha \Rightarrow \text{order}$. Convexity $V_{\text{eff}}''(r) > 0$ holds automatically for $\beta > 0$. No-go: $0 < \text{lambda}_j \leq 1 \Rightarrow \text{Pi}_0 \leq 7 \Rightarrow \alpha \geq 7$ forces $r=0$. The gauge-invariant statement is $\langle H^\dagger H \rangle > 0$ (Elitzur); NOT a prediction of the W/Z/Higgs masses.

Part XII

PART V-A — DOMAIN EMERGENCE AND TRANSLATION SUFFICIENCY

238 A.1 — A.1 The root of reality is not a domain

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *EQ-001*

$\delta_R = (a \# b)$ composes two retained differences; the running retained state is the triple $\mathfrak{S}_n = (G_n, \Lambda_n, \mathcal{T}_n)$ -- graph, labels, tape. [[Reality is retained before it is divided into domains]]

Occurrences (2): (1), (3)

239 A.2 — A.2 What a domain is

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *A.1, V0_DomainDiscoveryEngine*

Let $\mathfrak{f}_n$ be a state proposed with translation in mind, F_n its dynamics, Q_α a question, O_α the readout that question demands, and $q_{\{\alpha, n\}}$ the candidate translation map. Then $\mathcal{D}_{\{\alpha, n\}} = \mathfrak{f}_n / \sim_{\{\alpha, n\}}$. [[Domain = minimal sufficient dynamically closed translation]]

240 A.3 — A.3 A domain is relative to its question — but not arbitrary

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *A.2*

$Q_\alpha \neq Q_\beta$ can force $\mathcal{D}_\alpha \neq \mathcal{D}_\beta$. There is no "absolute domain" that exists independently of some readout asking for it; every candidate partition still has to pass the exact quotient gate, and a partition that fails it is not a domain, whatever question motivated proposing it.

241 A.4 — A.4 The Exact Domain Gate

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *A.3*

$$\begin{aligned} q_{\{\alpha, n+1\}} \circ F_n &= F_{\{\alpha, n\}}^{\#} \circ q_{\{\alpha, n\}} \quad 0_{\{\alpha, n\}} = 0_{\{\alpha, n\}}^{\#} \circ q_{\{\alpha, n\}} \\ q_{\{\alpha, n\}}(z) &= q_{\{\alpha, n\}}(z') \implies 0_{\{\alpha, n:n+L\}}(z, u) = 0_{\{\alpha, n:n+L\}}(z', u) \end{aligned}$$

Occurrences (1): (2)

242 A.5 — A.5 State is not domain

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *A.2*

$\mathfrak{S}_n \neq \mathfrak{I}_{\{\alpha, n\}} \neq \mathcal{D}_{\{\alpha, n\}}$ -- retained reality, the representation chosen to work with, and the quotient discovered on top of that representation are three distinct objects.

A.5/E.01.v1 — event occurrence $\neq$ retained record $\neq$ accessible trace; root retention $\neq$ belief/meaning/experience/choice

domain: *E* • *tier:* *Definition* • *status:* *current* • *coq:* *closed* • *parents:* *A.5/M.01.v1*

$$A_n \neq r_n \neq x_n; \text{root retention} \neq | = \text{belief state} \neq | = \text{meaning} \neq | = \text{experience} \neq | = \text{choice}$$

Occurrences (2): 22424434:(8), 22424434:(42)

A.5/E.02.v1 — Candidate-Set Formation & Appraisal Loop (Ψ , Appraise, Gate, cycle)

domain: *E* • *tier:* *Definition* • *status:* *current* • *coq:* *closed* • *parents:* *A.5/M.01.v1*

$$\begin{aligned} P^H_t &\rightarrow Q_{\{H, t\}} \text{ --Decompose_AI} \rightarrow Q^{\sim}_{\{AI\}_{t+1}} \text{ --T_AI} \rightarrow H_{\{t+1\}} \text{ --/}\sim_{\{D, t\}} \rightarrow \\ H^{\sim}_{\{\text{disc}\}_{t+1}} &\text{ --Gate} \rightarrow U_{\{t+1\}} \text{ --Spawn} \rightarrow F_{\{t+1\}} \rightarrow (Q_{\{t+2\}}, H_{\{t+2\}}) \text{ or --u*} \rightarrow \\ \delta_{\{t+1\}} &\text{ --revise} \rightarrow P_{\{t+1\}} \end{aligned}$$

Occurrences (19): 22307564:(1), 22307564:(2), 22307564:(3), 22307564:(4), 22307564:(5), 22307564:(Q5, unlabeled), 22307564:(6), 22308072:(1), 22308072:(2), 22308072:(3), 22308072:(4), 22308072:(5) (+7 more)

A.5/E.03.v1 — A unit of distinction and its provenance path as a labeled graph

domain: *E* • *tier:* *Definition* • *status:* *current* • *coq:* *closed* • *parents:* *A.5/M.01.v1*

$$D_{\Phi}(x_0) = \{d_{0_{x'}} = (x_0, x') : \Phi(x') \neq \Phi(x_0)\}; P(d) = (V_d, E_d, \tau_d)$$

Occurrences (2): 22301318:(3), 22301318:(4)

A.5/E.04.v1 — Role Separation: ten epistemic roles are distinct and must not be identified merely because one agent occupies several

domain: *E* • *tier:* *Dr* • *status:* *current* • *coq:* *closed* • *parents:* *A.5/M.01.v1*

Role Separation: generation, truth, evidential support, reliability, understanding, possession, endorsement, accountability, credibility, and institutional authorization are distinct epistemic roles. No pair may be identified merely because one human agent often occupies both.

Occurrences (1): 22301202:Prin-1

A.5/E.05.v1 — Representationality $\neq$ Selectivity

domain: *E* • *tier:* *Dr* • *status:* *current* • *coq:* *closed* • *parents:* *A.5/M.01.v1*

$$\text{Representationality} \neq \text{Selectivity}$$

Occurrences (1): 22301318:(2)

A.5/E.06.v1 — Meaning Before Naming’s own family of non-collapse chains guarding affective/semantic reorganization

domain: *E* • tier: *Open* • status: *current* • coq: *closed* • parents: *A.5/M.01.v1*

§8: Exposure ≠ felt intensity ≠ retention ≠ improvement. §11: external pattern≠meaning; meaning≠explicit naming; shared stimulus≠shared inner state; affective intensity≠epistemic warrant; accessibility≠truth; resonance≠understanding; retention≠benefit; verbalization≠lossless translation; same surface form≠same reader-relative meaning.

Occurrences (3): 22410666:§8-non-collapse-chain, 22410666:§11-non-collapse-laws, 22410666:H6

A.5/E.07.v1 — Experience Is Meaning-Giving’s own family of non-collapse chains across rhythm, release, and reader-relative meaning

domain: *E* • tier: *Dr* • status: *current* • coq: *closed* • parents: *A.5/M.01.v1*

EMG-15: order≠rhythm, repetition≠rhythm, rhythm≠meaning, rhythm≠retention. EMG-20: release≠transformation, intensity≠truth, shock≠insight, barrier-crossing≠improvement. EMG-27: phenomenon≠experience; event trace≠meaning; meaning-giving≠truth-making; pre-explicit significance≠fully-formed proposition; naming≠passive reporting; resonance≠identity; resonance≠truth; rhythm≠repetition count; rhythm≠meaning; accessibility≠warrant; semantic attraction≠evidence; informational work≠physical energy; shock≠coherent insight; release≠durable transformation; retention≠improvement; Human-AI amplification≠synergy; coupled performance≠Human Return. EMG-29: $\mu^{\text{aff}} \neq \mu^{\text{epi}}$, $\mu^{\text{auto}} \neq \mu^{\text{epi}}$, $\mu^{\text{prag}} \neq \text{truth}$.

Occurrences (4): 22357744:EMG-15, 22357744:EMG-20, 22357744:EMG-27, 22357744:EMG-29

A.5/E.08.v1 — Successful adaptation ≠ mental health, an identity explicitly rejected by the source

domain: *E* • tier: *Dr* • status: *current* • coq: *closed* • parents: *A.5/M.01.v1*

successful adaptation = mental health[explicitlyREJECTED]

Occurrences (1): 21425420:(35)

A.5/E.09.v1 — Knowledge Topology’s own non-collapse pair guarding usability and warrant-decrease from truth

domain: *E* • tier: *untagged* • status: *current* • coq: *closed* • parents: *A.5/M.01.v1*

usable ≠ true; $T_U \text{down} \not\Rightarrow W(H) \text{up} \not\Rightarrow \text{truth}$

Occurrences (2): 22307561:(4), 22307561:(12)

A.5/H.01.v1 — pre-prompt human-state transport $H_t -L_H-> Q_t$, $Q_t \neq H_t$

domain: *H* • tier: *Definition* • status: *current* • coq: *closed* • parents: *A.5/M.01.v1*

$H_t -L_H-> Q_t$, $Q_t \neq H_t$ (Pre-Prompt Human State Principle); session-indexed restatement $H_{\{s,0\}} -L_H-> Q_{\{s,0\}}$

Occurrences (5): 22456487:(1), 22456564:unlabeled (§6, Framing bullet), 22424434:(29), 22481928:(1), 22639311:Tunnel-v2.0:eq .9

A.5/H.02.v1 — Topic Entry Condition and Problem-First default

domain: *H* • tier: *Definition* • status: *current* • coq: *definition* • parents: *A.5/M.01.v1*

$P^{\text{live}}_{\{H,t\}}$ defined; TopicEntry∈{LiveProblem,OpenExploration,RoutineDelegation}; Problem-First⇒{Agenda Ownership,Bounded Relevance,World-Side Testability,Observable Human Return}; Problem-First≠Problem-Only

Occurrences (9): 22481928:(4), 22481928:(5), 22481928:(6), 22481928:(7), 22481928:PFDP, 22481928:(43), 22550491:Eq. 47, 22550491:Eq. 48, 22550491:Eq. 49

A.5/H.03.v1 — Release without destruction: identity-binding can decay to zero while readout stays nonzero

domain: H • tier: Definition • status: current • coq: definition • parents: A.5/M.01.v1

$U_{\{n+1\}} = \Pi_{\{U \geq 0\}}[(I - D_U)U_n + J_{\{\text{reinforce}, n\}} - J_{\{\text{release}, n\}}]$; sufficient condition
 $J_{\text{release}} - J_{\text{reinforce}} \geq \epsilon_{\text{release}} > 0$; release requires $z_n \in V_A$, $0_E(z_n) \neq 0$, $\|U_n\| \rightarrow 0$;
nonbinding mode $v_{\text{free}} \in \ker B_I$ with $0_E(v_{\text{free}}) \neq 0$

Occurrences (4): 21529456:(63), 21529456:(64), 21529456:(65), 21529456:(66)

A.5/H.04.v1 — Human-AI Augmentation/Synergy Ratio and chain-reaction hypotheses

domain: H • tier: Definition • status: current • coq: closed • parents: A.5/M.01.v1

$\Sigma_{\{H+AI\}} = D^{\sim}\{\text{use}\}_{\{H+AI\}}(B, Q) / \max\{D^{\sim}\{\text{use}\}_H(B, Q), D^{\sim}\{\text{use}\}_{\{AI\}}(B, Q), 1\}$;
non-collapse: more AI output $\neq$ more epistemic diversity, more epistemic diversity $\neq$
better warrant

Occurrences (4): 22308072:(11), 22308072:(12), 22308072:(16), 22308072:(17)

A.5/H.05.v1 — choice as downstream of the live set; enactment/observation non-identity

domain: H • tier: Definition • status: current • coq: definition • parents: A.5/M.01.v1

$\pi^{\sim}\text{choice}_{\{A, t\}} \in \Pi^{\sim}\text{live}_{\{A, t\}}(g)$; $\pi^{\sim}\text{act} \neq \pi^{\sim}\text{choice}$ possible, $Y_{\text{obs}} = 0_q(H_{\{0:T\}}) \neq H_{\{0:T\}}$;
possible $\neq$ feasible $\neq$ live $\neq$ chosen $\neq$ enacted $\neq$ observed

Occurrences (3): 22424434:(24), 22424434:(25), 22424434:(26)

A.5/H.06.v1 — changing live-field/accessibility profile after AI encounter

domain: H • tier: Definition • status: current • coq: definition • parents: A.5/M.01.v1

$L_{\{A, t+1\}} \neq L_{\{A, t\}}$ (if a residue is retained); at world-system scale: $\Lambda^{\sim}\text{live}_{\{H, t\}} = \lambda_1$
 $B_t + \lambda_2 X_t + \lambda_3 P^{\sim}\text{plural}_t + \lambda_4 r^{\sim}H_t - \lambda_5 D_t - \lambda_6 C^{\sim}I_t - \delta_{\Lambda} \Lambda^{\sim}\text{live}_{\{H, t\}}$

Occurrences (2): 22424434:(34), 22481924:(33)

A.5/H.07.v1 — Epistemic Fusion v8.1's twelve revised non-collapse laws

domain: H • tier: Definition • status: current • coq: closed • parents: A.5/M.01.v1

AI-first fluency $\neq$ human baseline; explanation $\neq$ verification; resistance quality $\neq$ resist
ance accessibility; uncertainty signal $\neq$ truth; output count $\neq$ epistemic diversity;
trust $\neq$ appropriate reliance; self-confidence $\neq$ competence; amplification $\neq$ warrant;
retention $\neq$ improvement; augmentation $\neq$ synergy; joint gain $\neq$ human return; user
preference $\neq$ epistemic direction

Occurrences (1): 22331922:NCL-14

A.5/H.08.v1 — assisted-performance gain does not entail Human Return gain

domain: H • tier: Definition • status: current • coq: closed • parents: A.5/M.01.v1

$\Delta \text{Performance}_{AI} > 0 \not\Rightarrow \Delta H_{\text{return}} > 0$ (equivalently : Assisted performance $\neq$ Unaided Human Return)

Occurrences (7): 22456564:(4), 22481928:(27), 22498047:(25), 22424434:(40), 22636999:eq.3, 22636999:eq.27, 22639311:Tunnel-
v2.0:eq.17

A.5/H.09.v1 — Exposure $\neq$ Retention $\neq$ Improvement

domain: H • tier: Definition • status: current • coq: closed • parents: A.5/M.01.v1

Exposure $\neq$ Retention $\neq$ Improvement

Occurrences (3): 22331922:EF-17, 22424434:(39), 22550491:Eq. 42

A.5/H.10.v1 — CTSA v9.1's nine preserved non-collapse laws

domain: *H* • tier: *Dr* • status: *current* • coq: *closed* • parents: *A.5/M.01.v1*

AI fluency $\neq$ human baseline; explanation $\neq$ verification; output count $\neq$ epistemic diversity;
exposure $\neq$ retention $\neq$ improvement; trust in AI $\neq$ calibrated confidence in self;
augmentation $\neq$ synergy $\neq$ Human Return; retained gain $\neq$ net expansion; provenance $\neq$ warrant;
persistence $\neq$ truth

Occurrences (1): 22339909:NCL-10

A.5/H.11.v1 — canonical eight-stage Dialogue Conversion Protocol

domain: *H* • tier: *Definition* • status: *current* • coq: *definition* • parents: *A.5/M.01.v1*

Anchor $\rightarrow$ Expand $\rightarrow$ Oppose $\rightarrow$ Discriminate $\rightarrow$ Verify $\rightarrow$ Integrate $\rightarrow$ Remove $\rightarrow$ Return; Q_t-AI $\rightarrow$ {Q_i, K_i}; Rival Generation / Boundary Test; Oppose $\neq$ manufacture

Occurrences (6): 22481928:(11), 22481928:(14), 22481928:(15), 22481928:(16), 22481928:(17), 22481928:(45)

A.5/H.12.v1 — the Verify stage: load-bearing claims and independent routes

domain: *H* • tier: *Definition* • status: *current* • coq: *definition* • parents: *A.5/M.01.v1*

L(c)={Decision,Risk,Money,Health,Legal,Publication,IrreversibleAction}; V={primary source,data,experiment,calculation,expert,independent method,world outcome}; Source $\neq$ AI synthesis $\neq$ Human inference $\neq$ Unverified candidate

Occurrences (3): 22481928:(19), 22481928:(20), 22481928:(21)

A.5/H.13.v1 — event, event-specific context, and translation-as-readout

domain: *H* • tier: *Definition* • status: *current* • coq: *closed* • parents: *A.5/M.01.v1*

E:=event; C_e=(S_e,R_e,A_e, τ_e); T[^]:=I(E|C_{acc}); T[^] $\neq$ E; general: T[^](k)=I(E|C_{acc}[^](k));
epoch 3 (LLM): T[^](3)=I(E|C_{acc}[^](3))+ Δ C via dialogue; E $\rightarrow$ C_{acc}[^](0) $\rightarrow$... $\rightarrow$ C_{acc}[^](3)

Occurrences (7): 18517054:Def-1, 18517054:Def-2, 18517054:Def-3, 18517054:Prop-1, 18517054:Eq-unifying, 18517054:Eq-epoch 3, 18517054:Prop-4

A.5/H.14.v1 — referential grounding, experiential grounding, and embodiment kept distinct

domain: *H* • tier: *Definition* • status: *current* • coq: *closed* • parents: *A.5/M.01.v1*

G(referential grounding); Ge(experiential grounding); Emb(embodiment); G $\neq$ Emb

Occurrences (2): 18517054:Def-7, 18517054:Prop-3

A.5/H.15.v1 — resources $\neq$ access $\neq$ capability $\neq$ realized opportunity

domain: *H* • tier: *Definition* • status: *current* • coq: *closed* • parents: *A.5/M.01.v1*

Resources $\neq$ Access, Access $\neq$ Capability, Capability $\neq$ Realized Opportunity; Access(z) $\neq$ Control(z); Equal AI Access $\neq$ => Equal Capability Conversion

Occurrences (3): 22498047:(9), 22498047:(10), 22498047:(11)

A.5/H.16.v1 — typed barrier ledger and Observed-Difficulty non-collapse

domain: *H* • tier: *Definition* • status: *current* • coq: *definition* • parents: *A.5/M.01.v1*

B[^]_{bar}{i,n} $\subseteq$ {Knowledge,Skill,Language,Tool,ResourceTime,Network,Credential,Permission,Opportunity,Unknown}; Observed Difficulty $\neq$ Skill Deficit

Occurrences (2): 22498047:(12), 22498047:(13)

A.5/H.17.v1 — candidate vs endorsed advancement routes

domain: *H* • tier: *Definition* • status: *current* • coq: *definition* • parents: *A.5/M.01.v1*

$C^{\text{cand}}_{\{i,n\}} = \text{Gen}(P^{\text{live}}, B^{\text{bar}}, K^{\text{life}}, R^{\text{return}})$; $C^{\text{live}}_{\{i,n\}} = \{c \in C^{\text{cand}} : \text{Endorse}_i(c) = 1\}$;
Proactive Suggestion $\neq$ Human Goal Ownership; Capability Advancement $\neq$ AI Goal Authority;
Scaffolding $\neq$ Control

Occurrences (5): 22498047:(15), 22498047:(16), 22498047:(17), 22498047:(18), 22498047:(19)

A.5/H.18.v1 — world-side opportunity readout and its non-collapses

domain: *H* • tier: *Definition* • status: *current* • coq: *definition* • parents: *A.5/M.01.v1*

$\Omega^{\text{real}}_{\{i,n\}} = G_0(R^{\text{return}}_{\{H,i,n\}}, K^{\text{life}}_{\{i,n\}}, \text{Cred}_{\{i,n\}}, \text{Net}_{\{i,n\}}, \text{Perm}_{\{i,n\}}, \text{MarketReadout}_n)$; Credential $\neq$ Capability; Market Legibility $\neq$ Human Worth; Skill $\rightarrow$ Evidence of Skill $\rightarrow$ Recognition $\rightarrow$ Opportunity

Occurrences (4): 22498047:(28), 22498047:(29), 22498047:(30), 22498047:(31)

A.5/H.19.v1 — three-children worked-scenario causal warning and pathway

domain: *H* • tier: *Definition* • status: *current* • coq: *closed* • parents: *A.5/M.01.v1*

Outcome_C – Outcome_A $\neq$ Effect_HCA; Schooling+Human-Owned Inquiry+Supervised Adaptive AI+Projects+Human Return+Opportunity Bridges

Occurrences (2): 22498047:(32), 22498047:(33)

A.5/H.20.v1 — Effort v0.3 Eq.30: NEW DOMAIN DEFINITION

domain: *H* • tier: *Definition* • status: *current* • coq: *definition* • parents: *A.5/H.09.v1*

$LC_Q(n) = \text{PASS} \iff RB_Q(n) = 1 \wedge \Lambda_Q(n) = \text{PASS}$

Occurrences (1): 22622206:eq.30

A.5/H.21.v1 — Effort v0.3 Eq.31: NEW DOMAIN DEFINITION

domain: *H* • tier: *Definition* • status: *current* • coq: *definition* • parents: *A.5/H.09.v1*

$IC_Q(n) = \text{PASS} \iff LC_Q(n) = \text{PASS} \wedge \Gamma_Q(n) > \tau_G$

Occurrences (1): 22622206:eq.31

A.5/H.22.v1 — Effort v0.3 Eq.32: NEW NON-COLLAPSE

domain: *H* • tier: *Dr* • status: *current* • coq: *open_prop* • parents: *A.5/H.09.v1*

$\text{Exposure} \neq \text{Retained Revision} \neq \text{Certified Learning} \neq \text{Improvement}$

Occurrences (1): 22622206:eq.32

A.5/H.23.v1 — Effort v0.3 Eq.35: NEW NON-COLLAPSE

domain: *H* • tier: *Dr* • status: *current* • coq: *open_prop* • parents: *A.5/H.09.v1*

$V_n^{\text{learn}}(u) \not\equiv V_n^{\text{act}}(u)$

Occurrences (1): 22622206:eq.35

A.5/H.24.v1 — Agent accuracy gain does not entail network corrigibility gain

domain: H • tier: Dr • status: *current* • coq: *open_prop* • parents: $A.5/H.08.v1$

$$\Delta \text{Accuracy}_{\{\mathrm{agent}\}} > 0 \not\rightarrow \Delta \text{Corrigibility}_{\{\mathrm{G}\}} > 0$$

Occurrences (1): 22639311:Tunnel-v2.1:eq.15

A.5/M.01.v1 — root state $\neq$ candidate domain representation $\neq$ discovered quotient

domain: M • tier: *Definition* • status: *current* • coq: *closed* • parents: $A.5$

$$S_n \neq Z_{D,n} \neq D_{D,n} (D = HCAorMEMK)$$

Occurrences (2): 22498047:(4), 22424434:(5)

A.5/M.02.v1 — global-Thai dual-track and geographic distribution audit

domain: M • tier: *finite_diagnostic* • status: *current* • coq: *closed* • parents: $A.5/M.01.v1$

$P_{\text{local}} \not\subseteq D_{\text{AI}}$; $\text{MultiAIConsensus} \Rightarrow \text{GeographicCompleteness}$; $\text{DVP} = \text{DVP} + \text{GDA}$; $S_G = L \times G \times M \times B \times F$;
 $\text{ManuscriptStrength} > \text{IdentityDependence}$; $\text{ConversionPlan}_i = \{\text{Global}_i, \text{Thai}_i\}$; $K2, \text{Global} + K2, T$
 $\text{hai} \Rightarrow \text{RouteDiversity} \uparrow$; FrictionValue_e ; $\text{GlobalAsset} \rightarrow \text{ThaiLegibility} \rightarrow \text{ThaiFriction}$

Occurrences (9): 22163849:(69), 22163849:(71), 22163849:(72), 22163849:(75), 22163849:(76), 22163849:(78), 22163849:(79), 22163849:(80), 22163849:(129)

A.5/M.03.v1 — the evidence-registry non-collapse chain (reachability, speed, volume vs warrant/truth)

domain: M • tier: *untagged* • status: *current* • coq: *closed* • parents: $A.5/M.01.v1$

$\text{neighboring evidence} \neq \text{formal-variable validation} \neq \text{truth of the integrated theory}$;
 $\text{reachability} \neq \text{accessibility}$; $\text{speed} \neq \text{quality/warrant}$; $\text{AI output volume} \neq \text{epistemic diversity}$;
 $\text{Attraction} \neq \text{Accessibility} \neq \text{Warrant} \neq \text{Truth}$; $\text{generation speed} \neq \text{hypothesis quality}$

Occurrences (6): 22308066:(unnumbered, Sec.1), 22308066:(1), 22308066:(2), 22308066:(3), 22308066:(unnumbered, Sec.4), 22308066:(unnumbered, Sec.3.4)

A.5/M.04.v1 — knowledge-topology first-passage-time hypothesis

domain: M • tier: *Open* • status: *current* • coq: *definition* • parents: $A.5/M.01.v1$

$$G1K \neq G2K \Rightarrow L(\tau_U | G1K, Q) \neq L(\tau_U | G2K, Q)$$

Occurrences (1): 22308066:(4)

A.5/M.05.v1 — The essential-dependency-set of a distinction, its defeater-partition, and the misrouted-defeat corollary

domain: M • tier: *Definition* • status: *current* • coq: *closed* • parents: $A.5/M.01.v1$

$\text{Ess}(d) = \bigcap_{p \in \Pi(d)} V(p)$; $\Pi_{\{\Delta v\}}(d) = \{p \in \Pi(d) : v \notin V(p)\}$; Corollary (misrouted defeat under silent lift): if $\text{Ess}(d) \Delta \bar{\text{Ess}}(d) \neq \emptyset$, a node-specific defeater aimed at that symmetric difference produces a represented vulnerability that differs from d 's actual dependency vulnerability.

Occurrences (3): 22301318:(17), 22301318:(18), 22301318:Cor-1

A.5/M.06.v1 — Readout Genesis’s own root-level chain of typed non-collapses

domain: M • tier: *finite_diagnostic* • status: *current* • coq: *definition* • parents: *A.5/M.01.v1*

$A \neq x \neq \mu \neq E \neq M \neq \text{Bel} \neq p \neq \sigma_K(p)$; $A_i = A_j \Rightarrow x_i = x_j$; {BeliefStrength, BeliefDistribution, Auth, Pow, Val_E}(p) typed-distinct; StatusOfSpeaker $\neq$ StatusOfClaim, Auth(p) $\neq$ Val_E(p), Pow(p) $\neq$ Val_E(p); $\sigma_K(p) \neq \Pi_{\text{prac}}(p)$; Bel_global(p) $\neq$ K_global(p); $A \neq I_A \neq \text{Phen}_A$ ^str $\neq$ P_A ^lived $\neq$ Own_A $\neq$ Coh_A $\neq$ Val_A; $R_A(1) \neq \text{Claim}_A \neq \text{Identity}_A$, Identity_A $\neq$ ActionCandidate_A $\neq$ CommittedAction_A; CurrentReadout+SelectionPriority+SelectionStability+SelfReport $\Rightarrow$ G_A^MR (no-free-governance)

Occurrences (9): 21529456:(10), 21529456:(12), 21529456:(21), 21529456:(24), 21529456:(26), 21529456:(35), 21529456:(48), 21529456:(70), 21529456:(85)

A.5/M.07.v1 — Credit $\neq$ EpistemicValue

domain: M • tier: *untagged* • status: *current* • coq: *closed* • parents: *A.5/M.01.v1*

Credit $\neq$ EpistemicValue

Occurrences (1): 22163849:(1)

A.5/M.08.v1 — Friction $\neq$ Fellowship

domain: M • tier: *untagged* • status: *current* • coq: *closed* • parents: *A.5/M.01.v1*

Friction $\neq$ Fellowship

Occurrences (1): 22163849:(3)

A.5/M.09.v1 — SelfExperience $\neq$ GeneralEvidence

domain: M • tier: *untagged* • status: *current* • coq: *closed* • parents: *A.5/M.01.v1*

SelfExperience $\neq$ GeneralEvidence

Occurrences (1): 22163849:(9)

A.5/M.10.v1 — PositionalAccess $\neq$ PopulationAuthority

domain: M • tier: *untagged* • status: *current* • coq: *closed* • parents: *A.5/M.01.v1*

PositionalAccess $\neq$ PopulationAuthority

Occurrences (1): 22163849:(11)

A.5/M.11.v1 — CommunityTrust $\neq$ Representativeness

domain: M • tier: *untagged* • status: *current* • coq: *closed* • parents: *A.5/M.01.v1*

CommunityTrust $\neq$ Representativeness

Occurrences (1): 22163849:(12)

A.5/M.12.v1 — DVP $\Rightarrow$ K2

domain: M • tier: *untagged* • status: *current* • coq: *closed* • parents: *A.5/M.01.v1*

DVP $\nRightarrow$ K2

Occurrences (3): 22163849:(18), 22163849:(101), 22307841:(unnumbered, Sec.8.1, disclaimer D-DVP-NOT-K2)

A.5/M.13.v1 — ManyModels $\neq$ Independence

domain: M • tier: *untagged* • status: *current* • coq: *closed* • parents: *A.5/M.01.v1*

ManyModels $\not\Rightarrow$ *Independence*

Occurrences (1): 22163849:(26)

A.5/M.14.v1 — MechanicalValidity $\neq$ SemanticValidity

domain: M • tier: *untagged* • status: *current* • coq: *closed* • parents: *A.5/M.01.v1*

MechanicalValidity $\neq$ *SemanticValidity*

Occurrences (3): 22163849:(28), 22163849:(102), 22307841:(unnumbered, Sec.3.3)

A.5/M.15.v1 — SourceExistence $\neq$ ClaimSupport

domain: M • tier: *untagged* • status: *current* • coq: *closed* • parents: *A.5/M.01.v1*

SourceExistence $\neq$ *ClaimSupport*

Occurrences (2): 22163849:(29), 22163849:(103)

A.5/M.16.v1 — Friendship $\neq$ IndependentEvidence

domain: M • tier: *untagged* • status: *current* • coq: *closed* • parents: *A.5/M.01.v1*

Friendship $\neq$ *IndependentEvidence*

Occurrences (1): 22163849:(31)

A.5/M.17.v1 — Correspondence $\neq$ PeerReview

domain: M • tier: *untagged* • status: *current* • coq: *closed* • parents: *A.5/M.01.v1*

Correspondence $\neq$ *PeerReview*

Occurrences (1): 22163849:(32)

A.5/M.18.v1 — IntellectualAffinity $\neq$ Truth

domain: M • tier: *untagged* • status: *current* • coq: *closed* • parents: *A.5/M.01.v1*

IntellectualAffinity $\neq$ *Truth*

Occurrences (1): 22163849:(33)

A.5/M.19.v1 — ActivationAction $\neq$ CreditEvent

domain: M • tier: *untagged* • status: *current* • coq: *closed* • parents: *A.5/M.01.v1*

ActivationAction $\neq$ *CreditEvent*

Occurrences (1): 22163849:(44)

A.5/M.20.v1 — RawSpeed $\downarrow$ $\neq$ $V_C\downarrow$

domain: M • tier: *untagged* • status: *current* • coq: *closed* • parents: *A.5/M.01.v1*

RawSpeed(down) $\not\Rightarrow$ *V_C (down)*

Occurrences (1): 22163849:(60)

A.5/M.21.v1 — $L_H \neq \text{Truth}, L_V \neq \text{Truth}$

domain: M • tier: *untagged* • status: *current* • coq: *closed* • parents: *A.5/M.01.v1*

$$L_H \neq \text{Truth}, L_V \neq \text{Truth}$$

Occurrences (1): 22163849:(63)

A.5/M.22.v1 — $\text{MultiAIConsensus} \Rightarrow \text{GeographicCompleteness}$

domain: M • tier: *untagged* • status: *current* • coq: *closed* • parents: *A.5/M.01.v1*

$$\text{MultiAIConsensus} \not\Rightarrow \text{GeographicCompleteness}$$

Occurrences (2): 22163849:(70), 22163849:(131)

A.5/M.23.v1 — $\text{DoubleBlind} = \text{Bonus} \neq \text{Requirement}$

domain: M • tier: *untagged* • status: *current* • coq: *closed* • parents: *A.5/M.01.v1*

$$\text{DoubleBlind} = \text{Bonus}; \text{DoubleBlind} \neq \text{Requirement}$$

Occurrences (1): 22163849:(73)-(74)

A.5/M.24.v1 — $\text{K2,Global} \neq \text{K2,Thai}$

domain: M • tier: *untagged* • status: *current* • coq: *closed* • parents: *A.5/M.01.v1*

$$\text{K2,Global} \neq \text{K2,Thai}$$

Occurrences (2): 22163849:(77), 22163849:(130)

A.5/M.25.v1 — $\text{InterventionCreator} \neq \text{SoleEvaluator}$

domain: M • tier: *untagged* • status: *current* • coq: *closed* • parents: *A.5/M.01.v1*

$$\text{InterventionCreator} \neq \text{SoleEvaluator}$$

Occurrences (2): 22163849:(86), 22163849:(127)

A.5/M.26.v1 — $\text{PracticeExperience} \neq \text{PopulationEvidence}$

domain: M • tier: *untagged* • status: *current* • coq: *closed* • parents: *A.5/M.01.v1*

$$\text{PracticeExperience} \neq \text{PopulationEvidence}$$

Occurrences (1): 22163849:(87)

A.5/M.27.v1 — $A_t \neq C_t^{\text{scholarly}}$

domain: M • tier: *untagged* • status: *current* • coq: *closed* • parents: *A.5/M.01.v1*

$$A_t \neq C_t^{\text{scholarly}}$$

Occurrences (1): 22163849:(84)

A.5/M.28.v1 — $M_{\text{attention}} \neq M_{\text{truth}} \neq M_{K2}$

domain: M • tier: *untagged* • status: *current* • coq: *closed* • parents: *A.5/M.01.v1*

$$M_{\text{attention}} \neq M_{\text{truth}}, M_{\text{attention}} \neq M_{K2}$$

Occurrences (1): 22163849:(96)

A.5/M.29.v1 — NoHumanAvailable $\Rightarrow$ ResearchStop

domain: M • tier: *untagged* • status: *current* • coq: *closed* • parents: *A.5/M.01.v1*

$$\text{NoHumanAvailable} \not\Rightarrow \text{ResearchStop}$$

Occurrences (1): 22163849:(104)

A.5/M.30.v1 — Prestige $\Rightarrow$ APCApproval

domain: M • tier: *finite_diagnostic* • status: *current* • coq: *closed* • parents: *A.5/M.01.v1*

$$\text{Prestige} \not\Rightarrow \text{APCApproval}; \text{APCApproval} \Rightarrow \text{FieldFit} + \text{CreditYield} + \text{BudgetFit}$$

Occurrences (1): 22163849:(107)-(108)

A.5/M.31.v1 — DisclosurePenalty $\Rightarrow$ Concealment

domain: M • tier: *finite_diagnostic* • status: *current* • coq: *closed* • parents: *A.5/M.01.v1*

$$\text{DisclosurePenalty} \not\Rightarrow \text{Concealment}; \text{DisclosurePenalty} \Rightarrow \text{BetterProvenance} + \text{BetterHumanDefence} + \text{VenueFit}$$

Occurrences (1): 22163849:(121)-(122)

A.5/M.32.v1 — AIContribution $\neq$ EpistemicResponsibility

domain: M • tier: *untagged* • status: *current* • coq: *closed* • parents: *A.5/M.01.v1*

$$\text{AIContribution} \neq \text{EpistemicResponsibility}$$

Occurrences (2): 22163849:(123), 22307841:(unnumbered, Sec.9 / Appendix disclaimer D-AUTHORSHIP)

A.5/S.01.v1 — Persistence-Control Agency Hierarchy (constrained dynamical system through reflective meta-regulation)

domain: S • tier: *Definition* • status: *split* • coq: *definition* • parents: *A.5/M.01.v1*

N_4 split (founder ruling 22:20, 2026-09-06): Def-1 ($x' = F(x, C)$), Def-2 (viable region membership), Persistence-Regulation-Condition, Constraint-Evolution, State-Constraint-Coupling, Def-3 Agency-Condition, Proto-Agency-Condition, and the L3/L4/L5 constraint-evolution variants are ten distinct definitions forming a level-by-level hierarchy – none is a ϕ -transform of another.

$x' = F(x, C)$; $x(t) \in V$; $d/dt[d(x(t), V)] < 0$ (persistence regulation); $C_{t+1} = G(x_t, C_t)$;
 $\partial C / \partial x \neq 0$ (necessary, not sufficient); Agency-Condition: $\partial T_p / \partial C \cdot \partial C / \partial x > 0$; Proto-Agency:
 $\partial C / \partial x = 0$; L3: $C_{t+1} = G(x_t, C_t, H_t)$; L4: $C_{t+1} = G(x_t, C_t, x^{\wedge}_{t+k})$; L5:
 $G_{t+1} = M(G_t)$

Occurrences (10): 18897585:Def-1, 18897585:Def-2, 18897585:Persistence-Regulation-Condition, 18897585:Constraint-Evolution, 18897585:State-Constraint-Coupling, 18897585:Def-3 (Agency-Condition), 18897585:Proto-Agency-Condition, 18897585:L3-Constraint-Evolution-History, 18897585:L4-Constraint-Evolution-Predictive, 18897585:L5-Meta-Regulation

A.5/S.02.v1 — organizational retention falsifier condition

domain: S • tier: *Open* • status: *current* • coq: *definition* • parents: *A.5/M.01.v1*

$$noH_{org}, n+1 \neq H_{org}, n \text{ attributable to the processing}$$

Occurrences (1): 22456487:unlabeled (§2, falsifier condition)

A.5/S.03.v1 — Six-Level Live-Possibility Non-Collapse Chain (possible $\supseteq$ feasible $\supseteq$ live $\supseteq$ chosen $\supseteq$ enacted $\supseteq$ observed)

domain: S • tier: Definition • status: split • coq: definition • parents: A.5/M.01.v1

N4 split (founder ruling 22:20, 2026-09-06): CBC-01 (containment law + choice membership), CBC-05 (live possibility field definition), CBC-06 (live-set threshold definition), CBC-03 (chosen-vs-enacted-vs-observed distinction), CBC-04 (the six-level non-collapse law itself), CBC-09 (witnessed refinement of the containment) are six distinct formulas; CBC-02 restates part of CBC-01 verbatim ($\pi^{\text{choice}} \in \Pi^{\text{live}}$) and is folded into that same child as a duplicate occurrence.

$\Pi^{\text{live}}_{\{A,t\}}(g) \subseteq \Pi^{\text{feas}}_{\{A,t\}}(g) \subseteq \Pi^{\text{phys}}_t(g); L_{\{A,t\}}(g) = \{(\pi, \lambda^{\text{live}}_{\{A,t\}}(\pi|g)) : \pi \in \Pi^{\text{feas}}_{\{A,t\}}(g)\}; \Pi^{\text{live}}_{\{A,t\}}(g) = \{\pi : \lambda^{\text{live}}_{\{A,t\}}(\pi|g) \geq \tau_{\text{live}}\};$
 $\pi^{\text{choice}}_{\{A,t\}} \in \Pi^{\text{live}}_{\{A,t\}}(g); \pi^{\text{act}} \neq \pi^{\text{choice}}$ possible, $Y_{\text{obs}} = 0_q(H_{\{0:T\}}) \neq H_{\{0:T\}}; \text{possible} \neq \text{feasible} \neq \text{live} \neq \text{chosen} \neq \text{enacted} \neq \text{observed}$

Occurrences (7): 22357788:CBC-01, 22357788:CBC-05, 22357788:CBC-06, 22357788:CBC-02, 22357788:CBC-03, 22357788:CB-C-04, 22357788:CBC-09

A.5/S.04.v1 — Constrained dynamical system

domain: S • tier: Definition • status: current • coq: definition • parents: A.5/M.01.v1, A.5/S.01.v1

$$x' = F(x, C)$$

Occurrences (1): 18897585:Def-1

A.5/S.05.v1 — Viable region membership

domain: S • tier: Definition • status: current • coq: definition • parents: A.5/M.01.v1, A.5/S.01.v1

$x(t) \in V$, where $V \subseteq S$ is the viable region of the state space S , for all times in the interval of observation

Occurrences (1): 18897585:Def-2

A.5/S.06.v1 — Persistence regulation condition

domain: S • tier: Definition • status: current • coq: definition • parents: A.5/M.01.v1, A.5/S.01.v1

$d/dt[d(x(t), V)] < 0$ (in expectation), where $d(x, V)$ is the distance between state x and the viable region V

Occurrences (1): 18897585:Persistence-Regulation-Condition

A.5/S.07.v1 — Constraint evolution law

domain: S • tier: Definition • status: current • coq: definition • parents: A.5/M.01.v1, A.5/S.01.v1

$$C_{t+1} = G(x_t, C_t)$$

Occurrences (1): 18897585:Constraint-Evolution

A.5/S.08.v1 — State-constraint coupling (necessary condition)

domain: S • tier: Definition • status: current • coq: definition • parents: A.5/M.01.v1, A.5/S.01.v1

$$\partial C / \partial x \neq 0$$

Occurrences (1): 18897585:State-Constraint-Coupling

A.5/S.09.v1 — Agency condition

domain: S • tier: Definition • status: current • coq: definition • parents: A.5/M.01.v1, A.5/S.01.v1

$\partial T_p / \partial C \cdot \partial C / \partial x > 0$, where T_p is the expected persistence time of the system within the viable region V

Occurrences (1): 18897585:Def-3 (Agency-Condition)

A.5/S.10.v1 — Proto-agency condition

domain: S • tier: *Definition* • status: *current* • coq: *definition* • parents: $A.5/M.01.v1$, $A.5/S.01.v1$

$$\partial C / \partial x = 0$$

Occurrences (1): 18897585:Proto-Agency-Condition

A.5/S.11.v1 — Level 3: history-dependent constraint evolution

domain: S • tier: *Definition* • status: *current* • coq: *definition* • parents: $A.5/M.01.v1$, $A.5/S.01.v1$

$$C_{t+1} = G(x_t, C_t, H_t), \text{ where } H_t \text{ is the system's interaction history}$$

Occurrences (1): 18897585:L3-Constraint-Evolution-History

A.5/S.12.v1 — Level 4: predictive constraint evolution

domain: S • tier: *Definition* • status: *current* • coq: *definition* • parents: $A.5/M.01.v1$, $A.5/S.01.v1$

$$C_{t+1} = G(x_t, C_t, \hat{x}_{t+k}), \text{ where } \hat{x}_{t+k} \text{ represents predicted future states}$$

Occurrences (1): 18897585:L4-Constraint-Evolution-Predictive

A.5/S.13.v1 — Level 5: meta-regulation

domain: S • tier: *Definition* • status: *current* • coq: *closed* • parents: $A.5/M.01.v1$, $A.5/S.01.v1$

$$G_{t+1} = M(G_t)$$

Occurrences (1): 18897585:L5-Meta-Regulation

A.5/S.14.v1 — Six-level containment + choice membership

domain: S • tier: *Definition* • status: *current* • coq: *definition* • parents: $A.5/M.01.v1$, $A.5/S.03.v1$

$$\Pi \text{live}_{A,t}(g) \subseteq \Pi \text{feas}_{A,t}(g) \subseteq \Pi \text{phys}_t(g), \pi^{\text{choice}} \in \Pi \text{live}_{A,t}(g)$$

Occurrences (1): 22357788:CBC-01

A.5/S.15.v1 — Live possibility field $L_{A,t}(g)$

domain: S • tier: *Definition* • status: *current* • coq: *definition* • parents: $A.5/M.01.v1$, $A.5/S.03.v1$

$$L_{A,t}(g) = \{(\pi, \kappa_{A,t}(\pi|g)) : \pi \in \Pi \text{feas}_{A,t}(g)\}$$

Occurrences (1): 22357788:CBC-05

A.5/S.16.v1 — Live-set threshold definition

domain: S • tier: *Definition* • status: *current* • coq: *definition* • parents: $A.5/M.01.v1$, $A.5/S.03.v1$

$$\Pi \text{live}_{A,t}(g) = \{\pi : \kappa_{A,t}(\pi|g) \geq \tau \text{live}\}$$

Occurrences (1): 22357788:CBC-06

A.5/S.17.v1 — [duplicate of CBC-01: choice membership]

domain: S • tier: *Definition* • status: *current* • coq: *definition* • parents: $A.5/M.01.v1$, $A.5/S.03.v1$

$$\pi_{A,t}^{\text{choice}} \in \Pi \text{live}_{A,t}(g)$$

Occurrences (1): 22357788:CBC-02

A.5/S.18.v1 — Chosen vs enacted vs observed distinction

domain: S • tier: Definition • status: current • coq: definition • parents: A.5/M.01.v1, A.5/S.03.v1

$$\pi^{act} \neq \pi^{\text{choice possible}}; Y_{obs} = Oq(H0 : T); Y_{obs} \neq H0 : T$$

Occurrences (1): 22357788:CBC-03

A.5/S.19.v1 — Six-level non-collapse law

domain: S • tier: Definition • status: current • coq: definition • parents: A.5/M.01.v1, A.5/S.03.v1

$$\text{possible} \neq \text{feasible} \neq \text{live} \neq \text{chosen} \neq \text{enacted} \neq \text{observed}$$

Occurrences (1): 22357788:CBC-04

A.5/S.20.v1 — Witnessed-set refinement of containment

domain: S • tier: Definition • status: current • coq: definition • parents: A.5/M.01.v1, A.5/S.03.v1

$$\Pi \text{live}_{A,t}(g) \subseteq \Pi \text{feas}_A(g; h, z, T, B)$$

Occurrences (1): 22357788:CBC-09

A.5/W.01.v1 — scarce-asset rent burden and effective material claim

domain: W • tier: Definition • status: split • coq: definition • parents: A.5/M.01.v1

N_4 split (founder ruling 22:20, 2026-09-06): (19) $B^{\text{scarce}}_{i,t}$ rent-burden definition, (20) $\Gamma^{\text{eff}}_{i,t}$ effective-claim definition, (21) 'machine abundance $\neq$ low rent burden $\neq$ effective material freedom' (a non-collapse claim) are three distinct formal objects.

$$B^{\text{scarce}}_{i,t} = (R^{\text{house}} + R^{\text{land}} + R^{\text{energy}} + DS) / Y; \Gamma^{\text{eff}}_{i,t} = \max\{0, \Gamma^{\text{net}}_{i,t} \\ \} - B^{\text{scarce}}_{i,t}; \text{machine abundance} \neq \text{low rent burden} \neq \text{effective material freedom}$$

Occurrences (3): 22481924:(19), 22481924:(20), 22481924:(21)

A.5/W.02.v1 — three-channel power vector and recursive political-economy loops

domain: W • tier: Definition • status: split • coq: definition • parents: A.5/M.01.v1

N_4 split (founder ruling 22:20, 2026-09-06): (45) P_t three-channel power vector, (46) P^B_t (bargaining power) proportionality, (47) P^E_t (epistemic power) definition, (48) I_{dot}_t (institutional-change) definition, (50) 'ownership $\neq$ informational power $\neq$ coercive power' non-collapse claim are five distinct formal objects; (49) is a three-strand arrow-chain narrative of the recursive loops with no relational operator.

$$P_t = (P^{\text{econ}}_t, P^{\text{info}}_t, P^{\text{coerc}}_t); P^B_t \propto \Gamma^{\text{eff}}_t A^{\text{corr}}_{\{H,t\}} \wedge \text{live}_{\{H,t\}} X^H_{\text{tr}} H_t; \\ P^E_t = P_E(P_t, \hat{G}^{\text{conv}}, D^H_t, 1 - \Gamma^{\text{eff}}_t); \dot{I}_t = F_I(P^B_t, P^E_t, \text{state capacity}, \text{rules}, \text{shocks}); \\ \text{ownership} \neq \text{informational power} \neq \text{coercive power}$$

Occurrences (6): 22481924:(45), 22481924:(46), 22481924:(47), 22481924:(48), 22481924:(49), 22481924:(50)

A.5/W.03.v1 — Human Conversion Imperative's non-collapse separations (9 mapped; source heading says "ten")

domain: W • tier: Definition • status: split • coq: definition • parents: A.5/M.01.v1

N_4 split (founder ruling 22:20, 2026-09-06): (2)..(10) are nine distinct ' $X \neq Y$ ' non-collapse claims (AI capability $\neq$ validated knowledge; assisted performance $\neq$ human learning; augmentation $\neq$ synergy; formal options $\neq$ live possibilities; access $\neq$ credible exit; income transfer $\neq$ future ownership; material security $\neq$ agency; market concentration $\neq$ domination; productive necessity $\neq$ social necessity) – nine non-equivalent members bundled as one code. INTEGRITY FIX: this entry's own name ("Human Conversion Imperative's ten non-collapse separations") states "ten non-collapse separations", but only 9 member occurrences (22481926:(2), 22481926:(3), 22481926:(4), 22481926:(5), 22481926:(6), 22481926:(7), 22481926:(8), 22481926:(9), 22481926:(10)) are mapped in CANONICAL.json – the tenth separation could not be located in the occurrence set and is not fabricated here; count corrected to 9, honestly, per readout-not-truth.

AI capability≠validated knowledge; assisted performance≠human learning; augmentation≠s
ynergy; formal options≠live possibilities; access≠credible exit; income transfer≠future
ownership; material security≠agency; market concentration≠domination; productive
necessity≠social necessity

Occurrences (9): 22481926:(2), 22481926:(3), 22481926:(4), 22481926:(5), 22481926:(6), 22481926:(7), 22481926:(8), 22481926:(
9), 22481926:(10)

A.5/W.04.v1 — Scarce-asset rent burden B^{scarce}

domain: W • tier: Definition • status: current • coq: definition • parents: A.5/M.01.v1, A.5/W.01.v1

$$B_{(i,t)}^{scarce} = (R_{(i,t)}^{house} + R_{(i,t)}^{land} + R_{(i,t)}^{energy} + DS_{(i,t)})/Y_{(i,t)}$$

Occurrences (1): 22481924:(19)

A.5/W.05.v1 — Effective material claim Γ^{eff}

domain: W • tier: Definition • status: current • coq: definition • parents: A.5/M.01.v1, A.5/W.01.v1

$$\Gamma_{(i,t)}^{eff} = \max\{0, \Gamma_{(i,t)}^{net} - B_{(i,t)}^{scarce}\}$$

Occurrences (1): 22481924:(20)

A.5/W.06.v1 — Non-collapse: abundance vs rent burden vs material freedom

domain: W • tier: Definition • status: current • coq: closed • parents: A.5/M.01.v1, A.5/W.01.v1

machine abundance $\neq$ low rent burden $\neq$ effective material freedom

Occurrences (1): 22481924:(21)

A.5/W.07.v1 — Three-channel power vector P_t

domain: W • tier: Definition • status: current • coq: definition • parents: A.5/M.01.v1, A.5/W.02.v1

$$P_t = \langle P_t^{econ}, P_t^{info}, P_t^{coerc} \rangle$$

Occurrences (1): 22481924:(45)

A.5/W.08.v1 — Bargaining power P_t^B (proportionality)

domain: W • tier: Definition • status: current • coq: definition • parents: A.5/M.01.v1, A.5/W.02.v1

$\hat{P}_t^B$ proportional-to $\hat{\Gamma}_{eff,t}$ $\hat{A}^{corr}_t(H,t)$ $\hat{\Lambda}^{live}_t(H,t)$ $\hat{X}_t^H$ $\hat{r}_t^H$

Occurrences (1): 22481924:(46)

A.5/W.09.v1 — Epistemic power P_t^E

domain: W • tier: Definition • status: current • coq: definition • parents: A.5/M.01.v1, A.5/W.02.v1

$$P_t^E = P_E(P_t, G_{dot}^{conv}, D_t^H, 1 - \Gamma_t^{eff})$$

Occurrences (1): 22481924:(47)

A.5/W.10.v1 — Institutional-change rate I_dot_t

domain: W • tier: Definition • status: current • coq: definition • parents: A.5/M.01.v1, A.5/W.02.v1

$I_dot_t = F_I(\hat{P}_t^B, \hat{P}_t^E, \text{state capacity}, \text{rules}, \text{shocks})$

Occurrences (1): 22481924:(48)

A.5/W.11.v1 — Non-collapse: ownership vs informational vs coercive power

domain: W • tier: Definition • status: current • coq: definition • parents: A.5/M.01.v1, A.5/W.02.v1

ownership $\neq$ informational power $\neq$ coercive power

Occurrences (1): 22481924:(50)

A.5/W.12.v1 — Non-collapse 1: AI capability vs validated knowledge

domain: W • tier: Definition • status: current • coq: closed • parents: A.5/M.01.v1, A.5/W.03.v1

AIcapability $\neq$ validated knowledge

Occurrences (1): 22481926:(2)

A.5/W.13.v1 — Non-collapse 2: assisted performance vs human learning

domain: W • tier: Definition • status: current • coq: closed • parents: A.5/M.01.v1, A.5/W.03.v1

assisted performance $\neq$ human learning

Occurrences (1): 22481926:(3)

A.5/W.14.v1 — Non-collapse 3: augmentation vs synergy

domain: W • tier: Definition • status: current • coq: closed • parents: A.5/M.01.v1, A.5/W.03.v1

augmentation $\neq$ synergy

Occurrences (1): 22481926:(4)

A.5/W.15.v1 — Non-collapse 4: formal options vs live possibilities

domain: W • tier: Definition • status: current • coq: closed • parents: A.5/M.01.v1, A.5/W.03.v1

formal options $\neq$ live possibilities

Occurrences (1): 22481926:(5)

A.5/W.16.v1 — Non-collapse 5: access vs credible exit

domain: W • tier: Definition • status: current • coq: closed • parents: A.5/M.01.v1, A.5/W.03.v1

access $\neq$ credible exit

Occurrences (1): 22481926:(6)

A.5/W.17.v1 — Non-collapse 6: income transfer vs future ownership

domain: W • tier: Definition • status: current • coq: closed • parents: A.5/M.01.v1, A.5/W.03.v1

income transfer $\neq$ future ownership

Occurrences (1): 22481926:(7)

A.5/W.18.v1 — Non-collapse 7: material security vs agency

domain: W • tier: Definition • status: current • coq: closed • parents: A.5/M.01.v1, A.5/W.03.v1

material security $\neq$ agency

Occurrences (1): 22481926:(8)

A.5/W.19.v1 — Non-collapse 8: market concentration vs domination

domain: *W* • tier: *Definition* • status: *current* • coq: *closed* • parents: *A.5/M.01.v1*, *A.5/W.03.v1*

market concentration $\neq$ domination

Occurrences (1): 22481926:(9)

A.5/W.20.v1 — Non-collapse 9: productive necessity vs social necessity

domain: *W* • tier: *Definition* • status: *current* • coq: *closed* • parents: *A.5/M.01.v1*, *A.5/W.03.v1*

productive necessity $\neq$ social necessity

Occurrences (1): 22481926:(10)

243 A.6 — A.6 Sufficiency before discovery

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *A.4*

Sufficiency $\text{Suff}_{\{\alpha, L\}}(\ddagger_{\alpha}^{\text{cand}})$ must be checked before a partition is ever searched for. Mandated order: Retention $\rightarrow$ State Proposal $\rightarrow$ Sufficiency Audit $\rightarrow$ Partition Discovery $\rightarrow$ Exact Quotient. If insufficient: $\ddagger_{\alpha}^{\text{cand}}(k+1) = \ddagger_{\alpha}^{\text{cand}}(k) \boxplus \Delta \ddagger_{\text{missing}}$ -- enlarge the retained state; never compensate by redefining the output's meaning or substituting a target-domain law for the missing retained data.

244 A.7 — A.7 The no-early-collapse law

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *A.6*

[[Refinement cannot recover distinctions absent from its input]] If an encoding E maps $E(z) = E(z')$ while $z \neq z'$, no downstream solver can ever split them back apart. [[compress only after equivalence is audited]]

245 A.8 — A.8 Merging structure must not erase history

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *A.7*

$\gamma \equiv_S \gamma' \not\Rightarrow \gamma =_R \gamma'$ -- sharing a structural node is not the same as being identical at the level of record. Formation and exact recoverability of a signature Σ_{α} : $\Sigma_{\alpha}(\mathfrak{G}) = (Q_{\alpha}(\mathfrak{G}), Q_{\alpha}(\mathfrak{G})) \wedge_{\alpha} \circ \Sigma_{\alpha} = I_{\mathfrak{G}}$

A.8/E.01.v1 — The paper's own governing maxim and its operational checklist form

domain: *E* • tier: *Definition* • status: *current* • coq: *definition* • parents: *A.8/M.01.v1*

No epistemic discrimination without provenance. Operational form: (i) What does the attributed source distinguish? (ii) Which distinction does the claim add? (iii) Which provenance path makes that addition licit?

Occurrences (6): 22301318:(1), 22301318:(26), 22301318:(27), 22301318:(28), 22301318:(29), 22301318:(30)

A.8/E.02.v1 — The Residual Provenance Effect: a measurable shift in credence from provenance alone, after relevant pathways are held fixed

domain: *E* • tier: *Definition* • status: *current* • coq: *closed* • parents: *A.8/M.01.v1*

[print fallback: this entry's LaTeX transcription did not compile as standalone display math; showing the source ascii/latest text instead — the registry, JSON entry and docs site carry the LaTeX transcription unmodified]

(origin, procedure, dependence, checks, answerability) $\rightarrow \Delta \text{Cr}(p)$; $\text{RPE} = \text{Cr}(p|E, R, A, 01) - \text{Cr}(p|E, R, A, 02)$

Occurrences (3): 22301202:(dCr), 22301202:Def-4, 22301202:(RPE)

A.8/E.03.v1 — Provenance may alter epistemic standing only via a specified epistemically relevant condition, never by redescription alone

domain: *E* • tier: *Dr* • status: *current* • coq: *closed* • parents: *A.8/M.01.v1*

Provenance Relevance Constraint: provenance may rationally alter epistemic standing insofar as it changes total evidence, reliability, dependence, assurance, accountability, or another specified epistemically relevant condition; it has no additional free-standing force from redescribing the origin of an otherwise fixed epistemic route.

Occurrences (1): 22301202:Prin-4

A.8/H.01.v1 — the Integrate stage record

domain: *H* • tier: *Definition* • status: *current* • coq: *definition* • parents: *A.8/M.01.v1*

$$I_s = \langle \Delta M_s, E_s^{decisive}, U_s^{remain}, Next_s \rangle$$

Occurrences (1): 22481928:(22)

A.8/H.02.v1 — Expansion and Contraction as alternating modes

domain: *H* • tier: *Definition* • status: *current* • coq: *definition* • parents: *A.8/M.01.v1*

Expansion:=maximize candidate diversity and discriminability; Contraction:=prune by evidence, provenance, stakes, and action relevance

Occurrences (2): 22481928:(31)-(32), 22481928:(33)-(34)

A.8/H.03.v1 — net HCA advancement record and conditional estimand

domain: *H* • tier: *Definition* • status: *current* • coq: *definition* • parents: *A.8/M.01.v1*

$A^{HCA}_i = \langle \text{Gain_CTSA}, \text{Loss}, \text{Transfer}, \text{Ownership}, \text{Burden}, \text{BarrierChange}, \text{OpportunityChange}, \text{Provenance}, \text{Warrant} \rangle$; $ATE_{HCA}(x) = E[Y(1) - Y(0) | K^{life} = x]$

Occurrences (2): 22498047:(36), 22498047:(37)

A.8/H.04.v1 — decisive-record policy in argmax/argmin form

domain: *H* • tier: *Definition* • status: *current* • coq: *definition* • parents: *A.8/M.01.v1*

$u^{diag} = \text{argmax}_u [IG_B(u) - \lambda_C \text{Cost}(u) - \rho \text{Risk}(u)]$; $u^{adv} = \text{argmin}_u E[\dots]$; $\pi \in \text{argmax}_\pi E[\Delta H_s | \pi]$; $\pi^{perf} \in \text{argmax}_\pi E[P^{assist}_s | \pi]$

Occurrences (7): 22498047:(14), 22498047:(20), 22498047:(21), 22481928:(40)-(41), 22481928:(42), 22163849:(81), 22163849:(92)

A.8/M.01.v1 — The historical occurrence itself is not rewritten by later reinterpretation

domain: *M* • tier: *Dr* • status: *current* • coq: *closed* • parents: *A.8*

$$\Delta A_{past} = 0$$

Occurrences (2): 21529456:(16), 22639311:Tunnel-v2.0:eq.5

A.8/M.02.v1 — per-transition provenance/tier/defect/reader/falsifier ledger

domain: *M* • tier: *Definition* • status: *current* • coq: *definition* • parents: *A.8/M.01.v1*

$$\Lambda(e_i) = \langle Prov_i, Tier_i, Def_i, Reader_i, Falsifier_i \rangle$$

Occurrences (3): 22456487:unlabeled (§5.1, provenance ledger), 22424434:(41), 22639311:Tunnel-v2.0:eq.22

A.8/M.03.v1 — provenance-ledger status categories

domain: M • tier: *Definition* • status: *current* • coq: *definition* • parents: *A.8/M.01.v1*

$$Status \in \{Source, AI - Synthesis, Human-Inference, Candidate, Decision\}$$

Occurrences (1): 22481928:(39)

A.8/M.04.v1 — Provenance Relevance Constraint and gated valid credit

domain: M • tier: *finite_diagnostic* • status: *current* • coq: *closed* • parents: *A.8/M.01.v1*

$$PRC(e_i) \in \{0, 1\}; C_t^{valid} = \sum_i w_{ie_i} PRC(e_i)$$

Occurrences (2): 22163849:(49), 22163849:(50)

A.8/M.05.v1 — K2 procurement cost and yield heuristics

domain: M • tier: *finite_diagnostic* • status: *current* • coq: *definition* • parents: *A.8/M.01.v1*

$$Cost_{\{K2, r\}} = (Cash_r + \lambda_{HH_r} + \lambda_{LL_r}) / (D_{rR_rI_r} + \epsilon); ExpectedK2Yield_j = (P(Review_j) Depth_j Fit_j) / (Cash_j + Prep_j + Latency_j + \epsilon); EffectiveK2 = Depth \times Relevance \times Independence$$

Occurrences (3): 22163849:(111), 22163849:(112), 22163849:(113)

A.8/M.06.v1 — institutional starting capital, earned credit, and positional-capital bookkeeping

domain: M • tier: *Definition* • status: *current* • coq: *definition* • parents: *A.8/M.01.v1*

$$K0_start = I0 + R0 + N0 + P0; E_t = f(Q_t, C_t, V_t, U_t, X_t, T_t); PosCap = Access \times Language \times SituatedObservation \times TranslationCapacity \times Trust; NetPractice = P_A - (Co + Bi)$$

Occurrences (4): 22163849:(4), 22163849:(5), 22163849:(10), 22163849:(124)

A.8/M.07.v1 — the Epistemic Isolation Constraint

domain: M • tier: *finite_diagnostic* • status: *current* • coq: *definition* • parents: *A.8/M.01.v1*

$$\mu_{H,f} \approx 0, \mu_A \gg \mu_{H,f}; EIC \Rightarrow BuildSyntheticFormationInfrastructure; AI \text{ speed} \rightarrow synthetic \text{ criticism} \rightarrow K1 \rightarrow human \text{ correction}$$

Occurrences (3): 22163849:(19), 22163849:(20), 22163849:(21)

A.8/M.08.v1 — bottleneck inversion and epistemic debt

domain: M • tier: *finite_diagnostic* • status: *current* • coq: *definition* • parents: *A.8/M.01.v1*

$$\Lambda = \min(\mu_L, \mu_H, \mu_E, \mu_P, \mu_C); V_c = (H \cdot L \cdot T)^{(1/3)}; D_e = A \cdot (1 - V_c); PublicOutputVelocity \leq VerificationCapacity$$

Occurrences (4): 22163849:(22), 22163849:(23), 22163849:(24), 22163849:(25)

A.8/M.09.v1 — association-strength recognition conversion

domain: M • tier: *finite_diagnostic* • status: *current* • coq: *definition* • parents: *A.8/M.01.v1*

$$A_s = AssociationStrength(Author, Problem); A_s(t+1) > A_s(t); Paper1 \rightarrow Theme; Paper2 \rightarrow SameTheme + NewMechanism; Paper3 \rightarrow EmpiricalTest; Paper4 \rightarrow BoundaryExtension$$

Occurrences (3): 22163849:(37), 22163849:(38), 22163849:(39)

A.8/M.10.v1 — concept-cluster compounding and leverage

domain: M • tier: *finite_diagnostic* • status: *current* • coq: *definition* • parents: *A.8/M.01.v1*

FlagshipConcept→Preprint→Conference→Journal→EmpiricalTest→ComparativeExtension→Grant; CreditLeverage_i=($\sum_j$ CreditEvent_{ij})/(CoreIntellectualInvestment_i+ ϵ); PC_i=JournalQuality×ProgrammeFit×ContributionStrength×UptakePotential (superseded)

Occurrences (3): 22163849:(47), 22163849:(48), 22163849:(110)

A.8/M.11.v1 — reactor-control criticality analogy for scholarly output

domain: M • tier: *Definition* • status: *current* • coq: *definition* • parents: *A.8/M.01.v1*

$k_t = \nu_{tf_coh}, tL_g, tp_int, tm_leg, tu_conv, tP_NL, t$; $\beta_D = (\sum w_dE_d) / (\sum w_dE_d + \sum w_pE_p + \epsilon)$; IncreaseMintRate⇒ $\beta_D \geq \beta_{min} \wedge IntegrityClean \wedge XenonLow \wedge ConversionLogOn$; $\rho_R = N^{ind}_{\leq 6m} / N^{cit}_{> 6m}$; $X_t = 1(O_{sub}) + 2(E_{known}) + 3(C_{stale})$; $BR_t = N^{\{ind-reuse\}}_t / N^{\{terminal\}}_t$

Occurrences (6): 22163849:(56), 22163849:(57), 22163849:(58), 22163849:(59), 22163849:(61), 22163849:(62)

A.8/M.12.v1 — a simple residual model of unresolved tension

domain: M • tier: *Definition* • status: *current* • coq: *definition* • parents: *A.8/M.01.v1*

$$r = A\varepsilon - \delta; V = (1/2)r^{TW_r}$$

Occurrences (1): 22163849:(67)

A.8/M.13.v1 — citation, revision, and emergency-shutdown integrity firewall

domain: M • tier: *finite_diagnostic* • status: *current* • coq: *definition* • parents: *A.8/M.01.v1*

AI candidate→OriginalSource→ClaimMatch→VerifiedCitation; Reject→Objecti on→Revision→ ΔQ ; SCRAM=FreezeNewRelease+Correction+ReAudit

Occurrences (3): 22163849:(82), 22163849:(83), 22163849:(88)

A.8/M.14.v1 — the Human Mastery Gate heuristic

domain: M • tier: *finite_diagnostic* • status: *current* • coq: *definition* • parents: *A.8/M.01.v1*

$$H_g = (\text{questions defended without AI})/10$$

Occurrences (1): 22163849:(85)

A.8/M.15.v1 — material feasibility and portfolio capacity constraints

domain: M • tier: *finite_diagnostic* • status: *current* • coq: *closed* • parents: *A.8/M.01.v1*

B_{year}=B_{conference}+B_{ethics}+B_{software}+B_{data}+B_{publication}+B_{travel}; MissingResource⇒ProjectHold; H_{planned}>H_{sustainable}⇒PortfolioShrink; PublicWIP≤3; Priority_i=(I_{in_iE_iF_iY_iS_i})/(D_i+C_i+Frag_i+COI_i+ ϵ)

Occurrences (5): 22163849:(106), 22163849:(109), 22163849:(118), 22163849:(119), 22163849:(120)

A.8/M.16.v1 — discovery-justification separation for embedded practice

domain: M • tier: *finite_diagnostic* • status: *current* • coq: *definition* • parents: *A.8/M.01.v1*

[print fallback: this entry's LaTeX transcription did not compile as standalone display math; showing the source ascii/latest text instead — the registry, JSON entry and docs site carry the LaTeX transcription unmodified]

PracticeObservation→Hypothesis; Claim<-Literature+ExternalEvidence+ComparativeEvidence; ClaimScope≤SamplingScope

Occurrences (3): 22163849:(125), 22163849:(126), 22163849:(128)

A.8/M.17.v1 — The Existence-Attribution-Disclosure norms governing when provenance is adequate

domain: M • tier: Dr • status: *current* • coq: *closed* • parents: $A.8/M.01.v1$

E (provenance existence): $P(d)$ nonempty or the claim is marked provenance-indeterminate.
A (provenance attribution): a credited source S must pass the licensing test for d . D
(provenance disclosure): every non-source node on d 's essential dependency path must remain declared or recoverable.

Occurrences (3): 22301318:Princ-7, 22301318:Princ-8, 22301318:Princ-9

A.8/M.18.v1 — Two named failure modes of provenance licensing: overreach in general, and its silent-omission special case

domain: M • tier: *Definition* • status: *current* • coq: *closed* • parents: $A.8/M.01.v1$

Epistemic overreach: a claim-level distinction lacks an adequate provenance path, is attributed to a node that does not license it, or depends on an essential augmentation concealed in the representation. Silent lift: with $\lambda(d)$ the actual essential dependency set and $\lambda^{\wedge}(d)$ the represented one, a silent lift occurs when an essential node v is omitted, or a source S is credited with d despite failing its licensing condition.

Occurrences (2): 22301318:Def-6, 22301318:Def-7

A.8/M.19.v1 — A worked base-rate audit example (a positive diagnostic test) illustrating the apparatus

domain: M • tier: *Definition* • status: *current* • coq: *closed* • parents: $A.8/M.01.v1$

[print fallback: this entry's LaTeX transcription did not compile as standalone display math; showing the source ascii/latest text instead — the registry, JSON entry and docs site carry the LaTeX transcription unmodified]

$P(D|+) = 0.90(0.01) / [0.90(0.01)+0.09(0.99)] \approx 0.0917$; diagram $(y, K^{\sim}) + \pi + L \rightarrow$
belief/action

Occurrences (2): 22301318:(23), 22301318:(24)

A.8/M.20.v1 — Typed provenance DAG

domain: M • tier: *Definition* • status: *current* • coq: *definition* • parents: $A.8/M.02.v1$

$\backslash \Pi(c) = (V_c, E_c, \backslash \tauau_c)$

Occurrences (1): 22639311:Tunnel-v2.1:eq.7

A.8/M.21.v1 — AI-Independent Consequence Requirement

domain: M • tier: Dr • status: *current* • coq: *open_prop* • parents: $A.8/M.01.v1$, $A.8/M.02.v1$

$H \rightarrow y \quad \text{such that} \quad \text{CausalAncestry}(y) \not\subseteq \mathcal{G}^{\text{recursive}}_{\leq \tauau}$

Occurrences (1): 22639311:Tunnel-v2.1:eq.16

A.8/M.22.v1 — AI-Off World-Closure (AOWC) cycle

domain: M • tier: Dr • status: *current* • coq: *open_prop* • parents: $A.8/M.01.v1$, $EQ-015/H.24.v1$, $A.8/M.13.v1$

Claim $\rightarrow$ Freeze $\rightarrow$ AI-Off $\rightarrow$ World $\rightarrow$ Record $\rightarrow$ Fixed $\rightarrow$ Evaluation $\rightarrow$ Revision

Occurrences (1): 22639311:Tunnel-v2.1:eq.17

A.8/M.23.v1 — AOWC gate conditions

domain: M • tier: Definition • status: current • coq: definition • parents: A.8/M.01.v1, EQ-015/H.24.v1

Freeze(H, τ), \ Predeclare($T, \mathcal{A}_H$), \ AI_{\mathrm{decisive\ execution}}(T)=0, \ AI_{\mathrm{primary\ evaluation}}(T)=0, \ Record(W_T) \ \text{before recursive reinterpretation}, \ T \ \text{allowed to count against} \ H

Occurrences (1): 22639311:Tunnel-v2.1:eq.18

A.8/M.24.v1 — AI-off software closure application

domain: M • tier: Dr • status: current • coq: open_prop • parents: A.8/M.01.v1

Frozen \ Artifact + AI_{\mathrm{runtime}}=0 \rightarrow Test \ Outcome

Occurrences (1): 22639311:Tunnel-v2.1:eq.19

A.8/S.01.v1 — Identifiability Certificate (exact non-identifiability worked example)

domain: S • tier: Dr • status: current • coq: closed • parents: A.8/M.01.v1

P1: $r=d=x=9/10$, channel always open, every reason voiced $\Rightarrow f_{P1}=1$, $p_{P1}=729/1000$. P2: channel open w.p. $1/5$, every open-channel reason voiced $\Rightarrow f_{P2}=1/5$, $p_{P2}=729/5000$. Both induce the identical record $\Pr(\text{complaint})=1/10$, so the record alone supports only the interval $[729/5000, 729/1000]$.

Occurrences (1): 22361830:PAR-cert

246 A.9 — A.9 Meaning is not domain

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: A.5

$\mathcal{S} \neq 0_\alpha(\mathcal{S}) \neq Y_\alpha \neq P_\alpha$ -- a domain is the structure that makes something readable; meaning is what a decoder, context, memory, or criterion does with that readable structure afterward.

247 A.10 — A.10 Number of fields $\neq$ number of domains ★

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: A.2

One field is not one domain; two fields are not two domains. Several fields can be a single domain if the full coupling between them is jointly necessary for one readout: $D_\alpha = (\Phi, \Psi, \chi, \dots) / \sim_\alpha$. "Do we need two fields?" is a sufficiency/closure question (A.6's gate), not a field-counting question; must be settled by whether the candidate state is sufficient (A.6) and the partition closes exactly (A.4), never by counting symbols in the state vector.

248 A.11 — A.11 Domains can change over time

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: A.2

$\mathcal{P}_{\{\alpha, n\}} \neq \mathcal{P}_{\{\alpha, n+1\}}$ -- a partition can develop a defect, a cell splitting $C_a \rightarrow C_{\{a, 1\}} \sqcup C_{\{a, 2\}}$, and when the equivalence that justified a coarse cell breaks, refinement is mandatory, not optional. Coarsening back is permitted only when freshly proven to preserve dynamics, readout, and residue.

249 A.12 — A.12 Cross-domain relationships

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: A.11

$K_{\{\alpha \leftarrow \beta\}}: \mathcal{D}_\beta \times \mathcal{T} \rightarrow \mathcal{D}^\wedge_\alpha$ $\epsilon_{\{\alpha \leftarrow \beta\}} = d_\Delta(\mathcal{D}_\alpha, \mathcal{D}^\wedge_\alpha)$. A bridge between domains need not be exact; when it is not, it must carry a stated, non-zero recovery error $\epsilon_{\{\alpha \leftarrow \beta\}}$, never a silent claim of equivalence. $[[\text{translatable} \neq \text{exactly equivalent}]]$

250 Gate1 — Gate 1 — No-Free-Domain-Law

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: A.4

$\delta_R + \text{Retention} \not\Rightarrow (F, C, V, \theta)_{\text{domain}}$
 $\mathcal{M}_{\text{compatible}} = \{ \mathcal{M} : \mathcal{M} \models \delta_R, \text{Retention}, \mathcal{K} \}$
 If $|\mathcal{M}_{\text{compatible}}| > 1$, no domain-specific law may be declared from retention and the tape alone -- an additional interaction tape, observation, or postulate is required first.

251 Gate2 — Gate 2 — Three-Valued Admissibility

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: Gate1

$C(\xi \mid c, \mathcal{T}) \in \{1, 0, \perp\}$ 1 admitted · 0 obstructed · $\perp$ unresolved
 $\text{Adm}(\xi) = 1 \text{ (T} \wedge \text{I} \wedge \text{B} \wedge [\text{C}=1]) / 0 \text{ (} [\text{C}=0]) / \perp \text{ (} [\text{C}=\perp])$
 $\text{Adm} = 1 \Rightarrow \text{expand}$; $\text{Adm} = 0 \Rightarrow \text{record the obstruction}$; $\text{Adm} = \perp \Rightarrow \text{record it as unresolved and do not guess}$.

Occurrences (1): (27)

252 Gate3 — Gate 3 — Context-Indexed Law

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: Gate2

$F_n = F_n(\mathcal{G}_n, c_n, \mathcal{T}_n)$ $G_n = G_n(G_n, c_n, \mathcal{T}_n)$ $\text{Adm}_n = \text{Adm}_n(\xi; c_n, \mathcal{T}_n)$. Valid in context c does not imply valid in context c' .

253 Gate4 — Gate 4 — State-Sufficiency Gate (three-valued)

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: Gate3, A.6

$\text{Suff}_{\{\alpha, L\}}(\ddagger_\alpha^{\text{cand}}; c, \mathcal{T}) \in \{1, 0, \perp\}$
 $\text{Retention} \rightarrow \text{State Proposal} \rightarrow \text{Sufficiency} \rightarrow \text{Discovery} \rightarrow \text{Quotient}$
 $\ddagger^{\text{cand}}(k+1) = \ddagger^{\text{cand}}(k) \boxplus \Delta \ddagger_{\text{missing}}$ (if insufficient). A threshold or decoder may never be used to compensate for a distinction the state has already lost -- the correction is always to enlarge $\ddagger$, never to patch the readout downstream.

Occurrences (1): (28)

254 Gate5 — Gate 5 — Invariant-Completion Gate

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: Gate4

$D^2(g) = D^2(g')$ but $\chi(g) \neq \chi(g')$ (one invariant set can be insufficient)
 $\text{Inv}_\alpha(z) = (I_1(z), \dots, I_m(z))$
 exact quotient requires: $\text{Inv}_\alpha = \text{Inv}_\alpha^{\text{cand}} \circ q_\alpha$
 $q(z) = q(z')$ but $\text{Inv}_\alpha(z) \neq \text{Inv}_\alpha(z') \Rightarrow \text{refine } C_a \rightarrow C_{\{a,1\}} \sqcup C_{\{a,2\}}$. Apparent symmetry is not valid quotient symmetry.

Occurrences (1): (29)

255 Gate6 — Gate 6 — Query-Relative Symmetry Group

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: Gate5

$\mathcal{H}_\alpha = \{ h : 0_\alpha(hz) = 0_\alpha(z), hF = Fh \}$
 $\mathcal{D}_\alpha = \ddagger / \mathcal{H}_\alpha$. Only transformations that fix this question's readout may be quotiented away.

256 Gate7 — Gate 7 — Calibration Firewall

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: Gate6, EQ-009

$y_\alpha^{\text{known}} = U_\alpha(r_{RD} ; \theta_\alpha, c, C_\alpha^{\text{cal}})$
 $H_{\text{cal}} = \text{Hash}(U_\alpha, \theta_\alpha, \text{training}, \text{units}, \text{protocol})$ (frozen before checking)
 $\text{Checker}(H_{\text{cal}}, Y_{\text{heldout}}) \rightarrow \epsilon_{\text{cal}}$. A native informational functional is not a physical observable until an independently checked calibration has been applied.

257 A13.DefectVector — A.13 The Defect Vector

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: Gate5

$\epsilon_\alpha = (\epsilon_{\text{suff}}, \epsilon_{\text{dyn}}, \epsilon_{\text{read}}, \epsilon_{\text{inv}}, \epsilon_{\text{bridge}}, \epsilon_{\text{cal}})^T$. An exact claim requires $\epsilon_\alpha = 0$ in every component; anything else is approximate and must name which component(s) are non-zero rather than reporting a single blended error term.

258 A13.ArchitectureBox — A.13 The architecture-around-DRL box

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: Gate1, Gate2, Gate3, Gate4, Gate5, Gate6, Gate7

$\delta_R \Rightarrow \mathfrak{S}_n \rightarrow [E_\alpha] \ddagger_\alpha^{\text{cand}} \rightarrow [\text{Suff}_{\{\alpha, L\}}] \ddagger_\alpha \rightarrow [\mathcal{C} \in \{1, 0, \perp\}] \text{Adm}_n \rightarrow [\text{FixRefine}(F_n, 0_\alpha, \text{Inv}_\alpha, c_n)] \mathcal{P}_\alpha \rightarrow [q_\alpha] \mathcal{D}_\alpha$ subject to $q_{\{n+1\}}F_n = F_\#_n q_n$, $0_\alpha = 0_\#_\alpha q_n$, $\text{Inv}_\alpha = \text{Inv}_\#_\alpha q_n \rightarrow [\mathbb{G}_n(G_n, c_n, \mathcal{T}_n)] (\Phi_n, \Psi_n) \rightarrow [S_{\text{DRL}}] r_{RD} \rightarrow [H_{\text{cal}}, U_\alpha]$
 $y_\alpha^{\text{known}} \rightarrow [\text{Checker}] \epsilon_\alpha \rightarrow [\Pi_\alpha] P_\alpha$. S_{DRL} (the DRL-Telegraph recurrence of Part II) appears here exactly once, untouched, as the fixed core the other seven gates are built around.

259 A13.SummaryBox — A.13 Standalone summary box

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: A13.ArchitectureBox

$\delta_R \Rightarrow \mathfrak{S}_n \rightarrow [E_\alpha] \ddagger_\alpha^{\text{cand}} \rightarrow [\text{Suff}] \ddagger_\alpha \rightarrow [\text{FixRefine}] \mathcal{P}_\alpha \rightarrow [q_\alpha] \mathcal{D}_\alpha = \ddagger_\alpha / \mathcal{P}_\alpha$
subject to $q_{\{\alpha, n+1\}}F_n = F_\#_{\{\alpha, n\}} q_{\{\alpha, n\}}$, $0_{\{\alpha, n\}} = 0_\#_{\{\alpha, n\}} q_{\{\alpha, n\}} \rightarrow [\Sigma_\alpha] (0_\alpha, Q_\alpha) \rightarrow [\Pi_\alpha] P_\alpha$. [[Reality is retained before it is translated]] [[Domain = the minimal translation that still preserves the differences that question needs]] [[Meaning arises after readout; it does not sit at the root]] [[Never reduce before proving that what is being merged does not differ with respect to that question]]

Part XIII

PART VI-A — DOMAIN-NEUTRAL EXTRACTION AND THE MAKER–CHECKER EPISTEMIC FIREWALL

260 B.1 — B.1 The Domain-Neutral Problem Object

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *A.2*

$\mathfrak{P}_{RD} = (\Gamma_0, \delta_R, \mathfrak{f}^{\text{cand}}, F, \text{Adm}, \mathcal{B}, \mathcal{T}, \mathcal{Q}, 0, \mathcal{H}, \mathcal{K})$. Γ_0 initial history · δ_R the distinctions retained · $\mathfrak{f}^{\text{cand}}$ candidate state · F transition · Adm which transitions are admissible · $\mathcal{B}$ boundary/cut · $\mathcal{T}$ tape/memory · $\mathcal{Q}$ the question · 0 readout · $\mathcal{H}$ the time/intervention horizon · $\mathcal{K}$ the budget on access, resolution, and reproduction. At this stage there is still no domain name and no target unit.

261 B.2.Step1 — B.2 Step 1 — Phenomenon Capture

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *B.1*

$\Gamma_n = (e_0 \dots e_{\{n-1\}})$; separate what is observed, what changes, what persists, what has left the observation window, what is recorded, and what is simply not yet known. Starting from an assumed mechanism name is forbidden -- that would presuppose the answer.

262 B.2.Step2 — B.2 Step 2 — Domain Vocabulary Stripping

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *B.2.Step1*

Replace domain-specific nouns with their information-theoretic role: particle/molecule/cell/person → retained unit; force/reaction/signal/thought → transition/coupling; energy/concentration/belief → state/readout; environment/solvent/society → boundary/context; genetic history/memory → lineage/tape; measurement/report/behavior → readout/report. This does not assert that stripped phenomena are "the same"; it lets structurally comparable phenomena be compared without being fooled by vocabulary.

263 B.2.Step3 — B.2 Step 3 — Question & Claim Declaration

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *B.2.Step2*

$\mathcal{Q}_\alpha = (\text{target}, \text{horizon}, \text{resolution}, \text{interventions}, \text{tolerance})$, declared before a domain is constructed, not fitted afterward to match a domain already assumed.

264 B.2.Step4 — B.2 Step 4 — Retained-State Proposal

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *B.2.Step3*

$\mathfrak{f}_\alpha^{\text{cand}} = E_\alpha(\mathfrak{S})$. [[candidate representation ≠ discovered domain]]

265 B.2.Step5 — B.2 Step 5 — State-Sufficiency Audit

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *B.2.Step4, A.6*

$\text{Suff}_{\{\alpha, L\}}(\ddagger_{\alpha}^{\text{cand}})$ must check at least: readout sufficiency, dynamical sufficiency, intervention sufficiency, delayed-return/memory sufficiency, and coupling sufficiency. If insufficient: $\ddagger^{\text{cand}}(k+1) = \ddagger^{\text{cand}}(k) \boxplus \Delta \ddagger_{\text{missing}}$ -- never compensated by adding a formula that manufactures the missing answer.

266 B.2.Step6 — B.2 Step 6 — Field & Coupling Discovery

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *B.2.Step5*

The number of fields must be discovered, never fixed in advance: $X_n = (X^{(1)} \dots X^{(m)})^T$, $m_{\alpha} = \min\{m : 0_{\alpha} \text{ and } F \text{ close on } X^{(1:m)}\}$. Forcing several fields down to one merely to match a preferred reduced-form equation is forbidden -- this is the extraction-time twin of A.10.

267 B.2.Step7 — B.2 Step 7 — Automatic Partition Discovery

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *B.2.Step6*

$\mathcal{P}_{\{\alpha, n\}} = \text{FixRefine}(\ddagger_{\{\alpha, n\}}, F_n, 0_{\alpha}, Q_{\alpha})$; states share a cell exactly when their local labels, interaction profiles, transition signatures, and future-readout -under-intervention all agree.

268 B.2.Step8 — B.2 Step 8 — Exact Quotient Audit

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *B.2.Step7*

$q_{\{\alpha, n+1\}} F_n = F_{\ddagger_{\{\alpha, n\}}} q_{\{\alpha, n\}}$ and $0_{\{\alpha, n\}} = 0_{\ddagger_{\{\alpha, n\}}} q_{\{\alpha, n\}}$; when only approximate, the defects ϵ_{dyn} , ϵ_{read} must be declared. Calling an approximate quotient "exact" is a category error, not a rounding choice.

Occurrences (1): (2)

269 B.2.Step9 — B.2 Step 9 — Admissible Synthesis

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *B.2.Step8*

$\text{Adm} = T \wedge I \wedge B \wedge [0_{\rho} = 0]$; this does not enumerate every possible world, and computational nodes may still be shared, but lineage may never be deleted in the process.

270 B.2.Step10 — B.2 Step 10 — Cross-Domain Signature Extraction

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *B.2.Step9*

$\Xi_{\alpha} = (\mathcal{P}_{\alpha}, F_{\ddagger_{\alpha}}, C_{\alpha}, B_{\alpha}, \mathcal{T}_{\alpha}, 0_{\ddagger_{\alpha}})$; a bridge $K_{\{\beta \leftarrow \alpha\}}$: $\mathcal{D}_{\alpha} \rightarrow \mathcal{D}_{\beta}$ is real only where $K F_{\ddagger_{\alpha}} = F_{\ddagger_{\beta}} K$ and $0_{\ddagger_{\beta}} K = D_{\{\beta \leftarrow \alpha\}} 0_{\ddagger_{\alpha}}$, with $D_{\{\beta \leftarrow \alpha\}}$ an honest decoder translating the result, never a bolted-on mechanism engineered to force agreement.

271 B.2a.NormalFormLineage — B.2a Normal-form sharing and the lineage graph

domain: — • *tier:* *finite_diagnostic* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *B.2.Step10*

A candidate $u \star v$ that passes *Adm* is reduced to a normal form and folded into the running synthesis graph and lineage ledger: $g_c = \text{NF}(u \star v)$; $G_{\{n+1\}} = G_n \cup \{g_c\}$, $\Lambda_{\{n+1\}} = \Lambda_n \boxplus \ell_c$ [*finite_diagnostic*]. If that normal form already exists in G_n , the solver shares the structural node -- it does not duplicate it -- but still records the new lineage that reached it: $\gamma \equiv_S \gamma' \not\equiv \gamma =_R \gamma'$ [*Dr*] (structural sharing $\neq$ retained-lineage identity). A candidate that fails *Adm* is never constructed at all, and an obstruction certificate is appended to the tape instead.

272 B.2a.ConservationLedger — B.2a A generic conservation ledger

domain: — • *tier:* *Dr* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *B.2a.NormalFormLineage*, *EQ-029*

Independent of any domain-specific conserved quantity claimed elsewhere in this book (e.g. the *bR* ledger's $I_Q = I_B + O_C + O_P + O_B$, V.13/Face 10), the synthesis-graph state itself obeys a bookkeeping identity between successive steps: $C(\mathcal{S}_{\{n+1\}}) - C(\mathcal{S}_n) = J_n^{\text{in}} - J_n^{\text{out}} + J_n^{\text{created}} - J_n^{\text{obstructed}}$ [*Dr*]. This ledger is architecture, not a physics claim: it reports graph, lineage, tape, currents, and operator powers for a semantic card to audit against its own declared conserved quantity -- it does not, by itself, assert that any particular physical quantity is conserved.

273 B.2a.SynthesisComplexity — B.2a Synthesis complexity

domain: — • *tier:* *finite_diagnostic* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *B.2a.NormalFormLineage*

$D_{\text{syn}}(n) = |\text{NF}_{\text{adm}}(G_n)|$. For occupation normal forms over a partition: $D_{\text{syn}} = \Pi_c (|C_c| + 1)$ [*finite_diagnostic*]. The discovery process reports this count but never enumerates the full lineage space to compute it -- a possible-lineage space of size 2^N is exactly the kind of unrestricted branching the copy licence (I.1a) forbids generating without admissibility. Any polynomial-regime statement about D_{syn} 's growth remains a theorem candidate, not a proven bound.

274 B.3 — B.3 Cross-Domain Equivalence Levels

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *B.2.Step10*

L0 surface analogy · L1 shared variables / relations · L2 shared transition signature · L3 readout-preserving map · L4 dynamically commuting quotient · L5 recoverable bidirectional equivalence. Two equations that merely look alike are L0/L1 at best -- resemblance of form is not shared mechanism.

275 B.4 — B.4 Native-unit and final-unit translation

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *EQ-009*, *EQ-010*, *EQ-011*

$r_{\text{native}} \in \mathbb{U}_{\text{RD}}$; $y_{\alpha}^{\text{known}} = U_{\alpha}(r_{\text{native}}; C_{\alpha}^{\text{cal}})$. U_{α} converts units, names an observable, and applies a declared calibration; it must never introduce a new predictive mechanism in the process, and it must state the range over which it is valid. [[native computation $\neq$ unit conversion $\neq$ semantic report]]

276 B.5 — B.5 The Maker–Checker Epistemic Firewall

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *B.3, B.4*

[[A solver cannot certify its own unrestricted interpretation]] -- a condition on the reliability of inference itself, true regardless of domain. The Maker-Checker firewall governs whether the party making a claim is even structurally capable of being unbiased about it (distinct from VI.7's fail-able gate law, which governs whether any single gate is honest evidence).

Occurrences (1): (88)

277 B.5.Registrar — B.5.2 Registrar/Curator

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *B.5*

$\mathcal{R}_0 = (\mathcal{Q}, \text{allowed inputs, forbidden inputs, metrics, thresholds, failure rules})$. The Registrar fixes the claim boundary, separates inputs a Maker is allowed to see from inputs reserved to the Checker only, states tolerances, records deviations, and freezes a version and hash of everything above -- and never selects a model on the basis of an outcome it has already seen.

278 B.5.Maker — B.5.3 Maker

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *B.5*

$(\mathcal{R}_0, \mathcal{X}_{\text{allowed}}) \rightarrow (\mathcal{P}^\wedge, F^\wedge, \hat{Y}, \Pi_{\text{proof}})$. A Maker must never see: held-out outcomes, checker-only labels, a threshold derived from the answer, a mechanism flag that is itself the answer, or any report produced before freeze. The condition is exact: $\text{Info}(\text{Maker}; Y_{\text{checker}}) = 0$.

279 B.5.PredictionFreeze — B.5.4 Prediction Freeze

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *B.5.Maker*

$H_{\text{freeze}} = \text{Hash}(\mathcal{P}^\wedge, F^\wedge, \hat{Y}, \theta, \text{source}, \mathcal{R}_0)$, covering code, config, parameters, the prediction itself, score orientation, normal form, and dependency versions. A hash does not by itself prove no leakage occurred, but it does let anyone verify that nothing changed after the answer was opened.

280 B.5.Checker — B.5.5 Checker

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *B.5.PredictionFreeze*

$(H_{\text{freeze}}, \hat{Y}, Y_{\text{heldout}}, \mathcal{R}_0) \rightarrow \mathcal{V}$. A Checker verifies the hash, coverage, invariants, any claimed exact-quotient property, baselines and controls, error and uncertainty, and the claim boundary, and rules strictly against the criteria frozen in advance. A Checker must never: swap models, retune parameters, flip a score's sign, cherry-pick favorable cases, or move a threshold -- any of these, once outcomes are visible, after the freeze.

281 B.5.Auditor — B.5.6 Auditor/Reporter

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *B.5.Checker*

$\text{Result} \rightarrow \text{Scope Audit} \rightarrow \text{Bounded Claim}$. [[computed $\neq$ checked $\neq$ interpreted $\neq$ claimed]]

282 B.5.ProtocolDeviation — B.5.7 Protocol Deviation

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *B.5.Auditor*

$\mathcal{T}_{\{n+1\}} = \mathcal{T}_n \boxplus \text{DeviationRecord}$ (append-only): when a deviation was found, what changed, why, whether forbidden data was touched, and whether the claim boundary was affected. A deviation record is never overwritten, only appended.

283 B.6 — B.6 The Refinement Loop

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *B.5.ProtocolDeviation*

{ $\in \{\text{F_state}, \text{F_partition}, \text{F_dynamics}, \text{F_readout}, \text{F_bridge}, \text{F_statistics}, \text{F_leakage}\}$
 { $\rightarrow \text{record} \rightarrow \text{refine hypothesis} \rightarrow \text{new freeze} \rightarrow \text{new untouched check}$. Re-scoring a refined hypothesis against the same previously-seen data and reporting that as an independent confirmation is never permitted -- a refinement earns a fresh freeze and a fresh, untouched check, every time.

284 B.7 — B.7 The full assembled process

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *B.2.Step10, B.6*

$\delta_R \rightarrow \text{Phenomenon Capture} \rightarrow \text{Vocabulary Stripping} \rightarrow \mathcal{Q}_\alpha + \text{Claim Boundary} \rightarrow \mathbb{I}_\alpha^{\text{cand}} \rightarrow \text{Suff}_{\{\alpha, L\}} \rightarrow \text{Field/Coupling Discovery} \rightarrow \mathcal{P}_\alpha = \text{FixRefine} \rightarrow \text{Exact Quotient Audit} \rightarrow \text{Adm} + \text{NF} \rightarrow \Xi_\alpha \rightarrow \mathcal{K}_{\{\beta \leftarrow \alpha\}} + \epsilon_{\text{bridge}} \rightarrow \text{Registrar} \rightarrow \text{Maker} \rightarrow \text{H_freeze} \rightarrow \text{Checker} \rightarrow \text{U}_\alpha \rightarrow \Pi_\alpha \rightarrow \text{Bounded Report} \rightarrow \mathcal{T}_{\{n+1\}}$

285 B.8 — B.8 Cross-reference — Gate 7 of A.13

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *B.7, Gate7*

A.13's Gate 7 (Calibration Firewall) states, at full generality and before any domain is named, the same condition B.4's U_α and this Part's H_freeze/Checker machinery enforce in practice: a native retained-domain readout r_RD is not a physical observable until an independently checked, frozen calibration (H_cal , verified by a Checker who never saw the held-out set beforehand) has been applied to it. B.4/B.5 are Gate 7 operationalized as a procedure; Gate 7 is B.4/B.5 stated once as a principle that holds regardless of which domain's units U_α happens to be converting into.

Part XIV

PART VI — THE EPISTEMIC NUCLEAR CORE

286 N3 — N3 — OBSTRUCTION

domain: — • *tier:* *Th_coqc* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *SpinePDE*

$dE/dt = \langle \partial\Phi, J \rangle - D \|\partial\Phi\|^2 \leq 0, 0 \rightarrow 0$; consistency-seeking; the monotone (Coq-checked)

287 N5 — N5 — INVARIANTS

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *SpinePDE*

anomaly ratios, $2/\alpha^2$, π , ϕ ; the dimensionless readouts that DO close (readout-invariants)

288 N1-Layer3 — N1 Layer 3 — LP-NS audit

domain: — • *tier:* *finite_diagnostic* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *Layer3*
nonlinear paraproduct diagnostic over the $\nabla V / (u \cdot \nabla)u$ term

289 N2-ScalarEigenmode — N2 repair — Scalar-Eigenmode Reduction Error split of L_R

domain: — • *tier:* *Open* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *LR_skew_split, T1*
L_R = L_R^(+) + L_R^(-), split under a retention metric G into symmetric (ordinary damped/relaxation) and antisymmetric (skew, rotational) parts, so a naive scalar/naive-transpose reduction does not silently discard off-diagonal/skew coupling before N2's readout

290 L0 — L0 — NUCLEUS

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *EQ-008*
the 5 equations + the Coq floor (RDL*.v); minimal, domain-independent, machine-checked where stated

291 L1 — L1 — SCALE BUS

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *BridgeFormula.TauC*
the τ_c atlas (37 disciplines) -- connect any domain by memory time [tau_c/tau_c_m aster.py]

292 L2 — L2 — READOUT

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *Face.10.StrictGap*
retrieval = readout · confidence = obstruction · ABSTAIN below ϵ_{tot} [rag_spine]

293 L3 — L3 — ADAPTERS

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *V0_DomainDiscoveryEngine*
per-domain (M, D, K, K_A, L_R); physics / bio / econ / IR / AI plug in identically

294 L4 — L4 — AGENCY

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *L3*
argmin 0 preserving repairability · bend information by choice [ai_agency, info_bending]

295 L5 — L5 — GOVERNANCE

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *L4*
readout-not-truth · bounded-judge · machinic_core (only Coq+lib-verified is core; everything else is diagnostic or open)

296 Re_ep — VI.3 — Re_ep (epistemic Reynolds)

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *SpinePDE, RTPEOperatorSpl*
it

Re_ep = epistemic Reynolds (spread = contestedness / turbulence)

297 F_ep — VI.3 — F_ep (obstruction depth)

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *N3*

F_ep = obstruction depth (strength of best support); "the epistemic reading of N3: how far the best supporting readout is from 0 $\rightarrow$ 0"

298 k_ep — VI.3 — k_ep (consistency coupling)

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *EQ-027*

k_ep = consistency coupling (coherence of the supporting set); "the epistemic reading of N2 through a multimode L_R: how coherently the supporting set of readouts agrees"

299 B1 — VI.4 — B1 Landauer bound

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *N3*

CPU nucleus runs near thermodynamic floor; LLM is $\sim 10^6 \times$ above it.

300 B3 — VI.4 — B3 Regime bound

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *SpinePDE*

stay over-damped ($k < k_c$); no wasted oscillation/retry.

301 B4 — VI.4 — B4 Repair bound

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *L4*

preserve repairability so re-planning is cheap, never a full restart.

302 DomainDiscoveryEngine — VI.5 — Domain-Discovery Engine (L3 made automatic)

domain: — • *tier:* *finite_diagnostic* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *L3*

Given a raw tape of (N,S) state transitions, discovers: (a) how many variables the domain minimally needs -- via an exact-gate closure test; (b) whether channels interact -- via mixed second-difference detection; (c) the minimal exact update law -- solved over $\mathbb{Q}$ (exact rationals, no float drift); (d) conservation laws -- read off the discovered update law.

303 BridgeCommute — VI.6 — Cross-domain bridge commuting condition

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *EQ-063*

$T_{\{a \rightarrow b\}} \cdot F\#_a = F\#_b \cdot T_{\{a \rightarrow b\}}$ (and preserves the readout); a bridge between two domains is real only if the translation commutes

BridgeCommute/H.01.v1 — Closure-meaningful commutation criterion for cross-agent communication

domain: *H* • *tier:* *Definition* • *status:* *current* • *coq:* *definition* • *parents:* *BridgeCommute*

$T_{\{j \rightarrow \text{toi}\}} \cdot F\#_H^j = F\#_H^i \cdot T_{\{j \rightarrow \text{toi}\}}$ (commutation criterion for 'closure-meaningful' cross-agent communication)

304 VerdictGate — VI.3 — Verdict gate (DECIDE / ABSTAIN / ESCALATE)

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *Re_ep, F_ep, k_ep*

DECIDE when F_{ep} high AND Re_{ep} low $\rightarrow$ strong + uncontested; ABSTAIN when F_{ep} low AND k_{ep} low $\rightarrow$ no signal (for free); ESCALATE when Re_{ep} high $\rightarrow$ contested $\rightarrow$ pay the LLM.
Qualitative result: only the contested/under-damped steps pay the large model.

305 AdmissibilitySquare — VI.6 — Admissibility square (Info-QuotientCompressionExactness)

domain: — • *tier:* *Th_coqc* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *BridgeCommute*

$q_{\{n+1\}} \cdot F_n = F\#_n \cdot q_n$; "is not a metaphor for this commuting requirement -- it is the commuting requirement"

306 B2 — VI.4 — B2 Readout bound

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *VerdictGate*

abstain/decide cheaply; only contested steps pay.

307 bRLedger — VI.6 — bR cross-domain lineage ledger

domain: — • *tier:* *finite_diagnostic* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *AdmissibilitySquare, N3*

$r_B = E \cdot A_C \cdot A_P \cdot A_B$; tracing quantum $\rightarrow$ chemical $\rightarrow$ protein $\rightarrow$ biological-transport as a chained readout

308 VI.7.FailAbleGateLaw — VI.7 — The Fail-Able Gate Law

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *AdmissibilitySquare, VerdictGate*

a gate only counts as Type-P (a genuine evidence-bearing predicate) if it has both a machine-derived passing control AND a machine-derived failing control; a gate with only passing cases must be labeled Type-U (a convention or definition dressed up as evidence) until a genuine failing control is produced and correctly rejected

309 ObstructionCertificate — VI.6 — bR ledger obstruction certificate

domain: — • *tier:* *finite_diagnostic* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *bRLedger*, *N3*

$I_Q = I_B + O_C + O_P + O_B$; the information present at the quantum end equals what survives at the biological end plus what was obstructed at each intermediate translation

Part XV

PART VII — UNIT GRAMMAR & NEWTON GATE

310 SI7Vectors — VII.1 — SI 7-Base Dimension Vectors

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *EQ-063*

Dimension vector: [L, M, T, I, Θ , N, J] with L=length(m), M=mass(kg), T=time(s), I=current(A), Θ =temperature(K), N=amount(mol), J=luminosity(cd). Derived examples: force [1,1,-2,0,0,0,0]→Newton; energy [2,1,-2,0,0,0,0]→Joule; power [2,1,-3,0,0,0,0]→Watt; pressure [-1,1,-2,0,0,0,0]→Pascal

311 RDU-tau_c — VII.2 — RDU τ_c readout

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *EQ-063*

τ_c readout → time scale [0,0,1,0,0,0,0], unit: s (the input row; every row beneath it in the RDU table is downstream of τ_c)

312 RDU-m_c — VII.2 — RDU m_c readout

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *EQ-063*

m_c readout → mass [0,1,0,0,0,0,0], unit: kg; $m = \hbar/(2\tau_c c^2)$ -- mass sits in the middle of the table, computed FROM τ_c , not the other way round

313 RDU-ell_c — VII.2 — RDU ℓ_c readout

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *RDU-tau_c*

ℓ_c readout → length [1,0,0,0,0,0,0], unit: m; $\ell = c \cdot \tau_c$

314 RDU-f_c — VII.2 — RDU f_c readout

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *RDU-tau_c*

f_c readout → frequency [0,0,-1,0,0,0,0], unit: Hz; $f = 1/\tau_c$

315 RDU-E_c — VII.2 — RDU E_c readout

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *RDU-tau_c*

E_c readout → energy [2,1,-2,0,0,0,0], unit: J; $E = \hbar/\tau_c$

316 RDU-eps_L — VII.2 — RDU ϵ _L (thermal) readout

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *B1*

ϵ _L readout $\rightarrow$ thermal [2,1,-2,0,0,0,0], unit: J; $\epsilon = k_B T \ln 2$ -- the SI-adapter reading of the Landauer floor (VI.4 bound B1)

317 SemanticLanes — VII.3 — Semantic Lanes

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *SI7Vectors*

Lanes: MECHANICAL (forces, momenta, kinematics), THERMAL (temperature, entropy, heat, Landauer), ELECTROMAGNETIC (field, current, voltage, flux), INFORMATION (bits, entropy, τ_c obstruction), BIOLOGICAL (body-scale τ_c , biochemical rates), FINANCIAL (τ_R , price, volatility)

318 NewtonGate — VII.2 — Newton gate

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *VI.7.FailAbleGateLaw*, *RDU-tau_c*

force_to_SI_newton(F, dim) $\rightarrow$ value iff dim == [1,1,-2,0,0,0,0]; else $\rightarrow$ None (gate closes; readback refused)

319 CompatibilityRule — VII.3 — Lane compatibility rule

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *SemanticLanes*, *VI.7.FailAbleGateLaw*

is_semantic_compatible(a_lane, b_lane) $\rightarrow$ bool; dimension-compatible but lane-incompatible $\rightarrow$ gate closes with warning, not error

320 PythonAPI — VII.4 — Python API (anse_spine.units)

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *SI7Vectors*, *NewtonGate*, *CompatibilityRule*

dimension_of, is_dimension_compatible, readout_rdu, to_newton, validate_unitmap -- five functions; readout_rdu(τ) $\rightarrow$ dict of $\tau/\ell/f/E/m/\epsilon$ values; to_newton(value, dim) $\rightarrow$ float | None; example readout_rdu($\tau=6.4e-22$) $\rightarrow$ {'tau':..., 'length':..., 'energy':...}

Part XVI

PART VIII — HUMAN AGENCY: THE τ_c^H LOOP, THE WAKING SELF, AND Ω_H

321 LayerStack — VIII.0 — Three-layer stack placement (DRL-Telegraph / RTPE / LP-NS)

domain: — • *tier:* *finite_diagnostic* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *SpinePDE*, *RTPEOperatorSplit*

Layer 1 -- DRL-Telegraph root (2nd order, the generator): $M \partial^2 \Phi + D \partial \Phi + K \cdot L_R \cdot \Phi + \nabla V = J - \eta$. Layer 2 -- RTPE turbulence relaxation (1st order, the $M \rightarrow 0$, $V \rightarrow 0$ limit): $\tau_R \dot{I}_R + L_R I_R = S_R + \eta_R$ [PASS_WITH_LIMITS, finite_diagnostic]. Layer 3 -- LP-NS audit (nonlinear paraprodut CHECKER, not a generator): diagnoses $(u \cdot \nabla)u / \nabla V$ for blow-up risk; does not itself produce dynamics. The human memory kernel $K_{\tau H}(s) = (1/\tau_c^H) e^{-s/\tau_c^H}$ introduced in VIII.2 is a first-order exponential relaxation kernel -- structurally the same shape as the RTPE relaxation law, not the second-order Telegraph generator.

322 HumanAgencyTuple — VIII.1 — The Human Agency Tuple A_H

domain: — • tier: Dr • status: current • coq: root_layer_unwired • parents: EQ-063, EQ-027

A_H = (B_H, τ_c^H , L_H, Ω_H , K_H, Φ_H , Π_H , U_H) -- B_H: body-ground (embodied substrate); τ_c^H : human causal-memory timescale (~0.1-3 s species-level); L_H: language/symbol layer (compression + framing); Ω_H : world-model (retained internal map); K_H: framing operator (perception $\rightarrow$ record); Φ_H : phenomenalization function; Π_H : policy function (action selection); U_H: utility / value function.

323 ConnectionRootStream — VIII.2 — Connection to the Root Stream ($\tau_c \rightarrow \tau_c^H$ pipeline + memory kernel)

domain: — • tier: Dr • status: current • coq: root_layer_unwired • parents: EQ-063, EQ-027

τ_c (universal) -- τ_c^H (human canonical readout timescale) |-- memory kernel $K_{\tau H}(s) = (1/\tau_c^H) e^{-s/\tau_c^H}$ |-- readout $M_H[n] = K_H \theta(E[n]) + \eta_H[n]$ |-- phenomenalize $P_H[n] = \Phi_H(M_H[n])$ |-- resonance gain $G_i^H = 1 / (1 + (\omega_i \tau_c^H - 1)^2)$ '-- policy $\pi^* = \text{argmin } E\|Res\|^2 + \lambda C + \rho Risk - \nu Val + \mu RepairLoss$. τ_c^H is the species-level instantiation of the universal causal-memory coordinate τ_c ; channel-specific timescales (τ_{vis} , τ_{aud} , τ_{motor} , τ_{social}) are downstream projections, not separate foundations.

324 Step2Record — VIII.3 Step 2 — RECORD (N2 record law + three-way noise decomposition)

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: EQ-027

$M_H[n] = K_H \cdot \theta(E[n]) + \eta_H[n]$ (always lossy; $\epsilon_{tot} = \eta_{sel} + \eta_{map} + \eta_{self} > 0$). η_{sel} -- selection loss ($K_H \cdot \theta$ admits only part of $E[n]$ past threshold); η_{map} -- mapping loss (K_H 's finite-resolution map folds distinct raw states together); η_{self} -- self-noise (B_H 's own retained history injects noise not originating in $E[n]$).

Occurrences (3): (44), (45), (46)

325 ImaginationTransport — VIII.6a — Imagination as Reversible Transport Across Living Modes ($\lambda \neq 0$)

domain: — • tier: Dr • status: current • coq: root_layer_unwired • parents: LR_skew_split

Imagination is the ability to mix, swap, replace, splice, extend, and transform multiple finite meaning-sets into new forms. The mixing/swapping/splicing operation imagination performs is the same object, algebraically, as the skew part of the multimode operator, $L_R^{\wedge}(-)$, introduced at V.13a and XIV.4 as the proposed $L_R = L_R^{\wedge}(+) + L_R^{\wedge}(-)$ decomposition under the retention metric G. $L_R^{\wedge}(-)$ is exactly reversible and norm-preserving ($z^T L^{\wedge}(-) z = 0$ for the retention metric's inner product), operating only in the living-mode regime $\lambda \neq 0$ (eigenmodes of Face 1 that have not collapsed to the kernel of L_R).

326 Step3Phenomenalize — VIII.3 Step 3 — PHENOMENALIZE

domain: — • tier: Dr • status: current • coq: root_layer_unwired • parents: Step2Record

$P_H[n] = \Phi_H(M_H[n])$ (qualia / subjective-report layer). The record is turned into whatever a subjective report is made of.

Occurrences (1): (54)

327 Step4Resonance — VIII.3 Step 4 — RESONANCE

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: ConnectionRootStream

$G_i^H = 1 / (1 + (\omega_i \tau_c^H - 1)^2)$ (information at matching frequency is amplified; others decay). This is the r0 default gain function (see VIII.5 for the r1 log-normal alternative).

328 Step5Policy — VIII.3 Step 5 — POLICY (optimal action)

domain: — • tier: Dr • status: current • coq: root_layer_unwired • parents: Step4Resonance, L4

$\pi^*[n] = \text{argmin} [E\|Res\|^2 + \lambda C + \rho Risk - \nu Val + \mu RepairLoss]$, where $E\|Res\|^2$: epistemic residual; λC : cost of action; $\rho Risk$: risk of irreversible damage; νVal : value of outcome; $\mu RepairLoss$: loss of repairability. New reading, [Dr], flagged PROPOSED and untested (T1): if the terms are genuinely coupled, Π_H is better modeled as a multimode L_R operator over a policy-state vector, with off-diagonal skew coupling captured by $L_R = L_R^+(+) + L_R^-(-)$ under the retention metric G (not a naive transpose); the scalar form is kept as the working, currently-used approximation.

Occurrences (2): (77), (78)

329 OmegaH — VIII.5 — Ω_H , the Horizon Transduction Operator

domain: — • tier: Dr • status: current • coq: root_layer_unwired • parents: Step4Resonance, AdmissibilitySquare

The resonance equation $\chi_H = \omega_A * \tau_c^H$ must use the agency-side frequency, not the raw physical carrier. WRONG: $\chi_H = \omega_{raw} * \tau_c^H$ (light at 4×10^{14} Hz $\rightarrow$ gain $\approx 0 \rightarrow$ but humans see!). CORRECT: $\chi_H = \omega_A * \tau_c^H$, $\omega_A = \text{Freq}(\Omega_H(R_{raw}))$, $\Omega_H = C_H \circ E_H \circ T_H \circ S_H$. Ω_H pipeline: $R_{raw} \rightarrow S_H$ (select) $\rightarrow T_H$ (transduce) $\rightarrow E_H$ (extract envelope) $\rightarrow C_H$ (compress) $\rightarrow \omega_A$. Gain functions: r0 default $G_{r0} = 1.0 / (1.0 + (\chi - 1.0)**2)$ [at $\chi \rightarrow 0$ yields $G=0.5$, a known asymmetry]; r1 log-normal $G_{r1} = \exp(-\log(\max(\chi, 1e-12))**2/2.0)$ [symmetric suppression both sides].

Occurrences (4): (50), (51), (52), (53)

330 Step6ActRepair — VIII.3 Step 6 — ACT + REPAIR

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: Step5Policy

Action updates world-model Ω_H ; residuals drive repair. Whatever residual Res remains unresolved after the action becomes the input the next repair cycle works against; $dR_H/dt = -\mu \cdot \|Res_H\|^2$ (VIII.4) is the rate law for this repair.

Occurrences (1): (79)

331 CrossDomainLineage — VIII.7 — Cross-Domain Lineage of a Single Percept

domain: — • tier: Dr • status: current • coq: root_layer_unwired • parents: bRLedger, Step2Record

$I_{\text{raw}} = I_A + O_S + O_T + O_E + O_C$, where I_A is the information that survives all the way to ω_A at the policy-facing end of the Ω_H pipeline, and O_S, O_T, O_E, O_C are the named obstruction terms lost at S_H, T_H, E_H, C_H respectively -- exactly parallel in form to $I_Q = I_B + O_C + O_P + O_B$ in the physical (bR) lineage case.

332 FalsifierFhuman — VIII.8 — Falsifier F-human

domain: — • tier: Th_coqc • status: current • coq: root_layer_unwired • parents: VI.7.FailAbleGateLaw, OmegaH, Step5Policy

Claim under test: the τ_c^H loop (VIII.1-VIII.5) with the r_0/r_1 resonance gain and the first-order relaxation memory kernel $K_{\tau H}(s)$ correctly predicts which transduced frequencies ω_A produce strong agency-level response via $\chi_H = \omega_A \cdot \tau_c^H$.
 Passing control: the five worked transduction rows of VIII.5's table each fall in the range giving high resonance gain under $\tau_c^H \approx 0.1\text{--}3$ s. Failing control (any one refutes the section): (1) a genuinely ringing, second-order, M-bearing signature in human working-memory decay data; (2) a domain where the naive $\chi_H = \omega_{\text{raw}} \cdot \tau_c^H$ correctly predicts agency-level response; (3) empirically measured coupling between the five Step-5 policy terms that a scalar argmin genuinely cannot represent even approximately.

333 MeasuringAgency — VIII.4 — Measuring Agency (epistemic bandwidth, repair rate, agency gradient)

domain: — • tier: Dr • status: current • coq: root_layer_unwired • parents: Step2Record, Step5Policy, Step6ActRepair

Epistemic bandwidth: $I(A_H; E) / \tau_c^H$ (bits per unit time). Repair rate: $dR_H/dt = -\mu \cdot \|\text{Res}_H\|^2$. Agency gradient: $\partial \pi^* / \partial \eta_H$ (sensitivity to readout noise). Claim tier: Dr (open theoretical; falsifiable -- §11 of docs/claims/HUMAN_AGENCY_SINGLE_TAU_C.md, and see F-human, VIII.8).

Occurrences (2): (63), (64)

334 SelfClosure — VIII.6 — Self as a Closure Property, Not a Substance

domain: — • tier: Dr • status: current • coq: root_layer_unwired • parents: OmegaH, Step6ActRepair

Self, as used here, is not a substance, a location, or a homunculus. It is the closure property of the τ_c^H loop: the fact that Ω_H both (a) results from the loop's own prior passes and (b) conditions the loop's next pass, at a stable, characteristic causal-memory timescale τ_c^H .

Occurrences (2): (36), (37)

Part XVII

PART IX — THE 42-STEP GENESIS STREAM

335 step1 — Step 1 — Bare distinction (P0)

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *EQ-001*

$\exists a, b : a \neq b$ -- Something is distinguishable from something else.

336 step2 — Step 2 — Discriminator exists

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *EQ-002, step1*

$\exists A : A \text{ discriminates } E_1 \neq E_2$ -- A discriminator exists. The observer relation is introduced.

337 step3 — Step 3 — Asymmetry

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *EQ-003, step2*

$A \rightarrow B \neq B \rightarrow A$ -- Asymmetry. The direction of relation matters.

338 step4 — Step 4 — Time as transition count

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *EQ-004, step3*

$t(s) := \min \# \text{steps}(s_0 \rightarrow s) \in \mathbb{N}$ -- Time = counting admissible transitions. Not a container.

339 step5 — Step 5 — Retention axiom

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *EQ-005, step4*

$\tau_c > 0$ -- Some structure persists between transitions. Memory exists.

340 step6 — Step 6 — Discreteness of time

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *EQ-006, step5*

$t = n \Delta\theta, \Delta\theta > 0$ -- World is discrete before it is continuous.

341 WP.S4.DomainReadableState — Domain-readable state

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S4.StrongReaderEquivalence, WP.S4.LocalReaderEquivalence*

$[Z]_{\sim_D}$ (the equivalence class of Z under $\sim_D$)

342 WP.S5.TransportGaugeTransform — Transport gauge transform

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S5.Transport*

$$U'_{\{j<-i\}} = h_j U_{\{j<-i\}} h_i^{-1}$$

343 WP.S5.TransportMetricCompatibility — Transport metric compatibility

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S5.Transport*, *WP.S3.PositiveMetricPositivity*

$$U_{\{j<-i\}}^{\dagger} G_j U_{\{j<-i\}} = G_i \text{ (equality for isometric transport)}$$

344 WP.S5.Holonomy — Holonomy around a cycle

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S5.Transport*

$$H_C = \text{product}_{\{e \text{ in } C, \text{ ordered}\}} U_e$$

345 WP.S6.ClosureMultiplicity — Closure invariant multiplicity

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S6.InvariantClosureMap*

$\text{nu_alpha} = \dim \text{Hom}_A(R_1(x) \dots (x) R_k, 1)$. If the Hom-space is zero: $\text{closure_status} = \text{OBSTRUCTED}$, $\text{interaction_allowed} = \text{false}$.

346 WP.S7.KKTConditions — KKT conditions for inequality constraints

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S7.ConstrainedEquation*

$$\lambda_b \geq 0, C_b \leq [Z] \leq 0, \lambda_b C_b \leq [Z] = 0$$

347 WP.S7.VariableEquationCoverageGate — Variable equation coverage gate

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S7.ConstrainedEquation*

'If G, Theta, U, Gamma, or Xi is dynamical, it MUST have an equation or update rule. Otherwise it MUST be marked background, externally_prescribed, or derived_cache.'
VariableEquationCoverageGate: for_each_state_slot require_one_of [variational_equation, recurrence, algebraic_constraint, declared_background, derived_cache_with_recompute_rule]; fail_code UNCOVERED_STATE_VARIABLE.

348 WP.S10.NormalizedTransfer — Normalized transfer operator

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S10.ContractionBound*

$T_{RG} = T_+ / \text{zeta}$, $\text{zeta} > 0$, $\text{zeta} \geq \text{lambda_max}(T_+)$. 'The normalization zeta MUST be declared and recorded; it may not be hidden.'

349 WP.S11.EffectivePotentialDiagonal — Effective potential, diagonalized eigenvalue form

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *WP.S11.EffectivePotential*

For eigenvalues λ_μ r with multiplicity d_μ : $V_{\text{eff}}(r) = \alpha r + \beta r^2 - \sum_\mu d_\mu \log(1 + \lambda_\mu r)$

350 step7 — Step 7 — L_R defined as retained transport graph

domain: — • tier: *Definition* • status: *current* • coq: *root_layer_unwired* • parents: *EQ-008, step6*

L_R = retained graph of admissible transport -- The causal graph emerges. What can reach what?

351 WP.S4.DomainDefinition — Domain as sufficient quotient

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *WP.S4.DomainReadableState*

$D_D = Z_D / \sim_D$, where Z_D MUST be sufficient before quotienting.

352 WP.S5.TransportMetricDefect — Transport metric-compatibility defect

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *WP.S5.TransportMetricCompatibility*

$\epsilon_{\{G,U\}} = || U^\dagger G_j U - G_i ||$

353 WP.S5.CurvatureDefect — Curvature defect

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *WP.S5.Holonomy*

$K_C = H_C - I$

354 WP.S6.MultiplicityLedger — Multiplicity ledger

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *WP.S6.ClosureMultiplicity*

MultiplicityLedger: invariant_multiplicity = number_of_independent_invariant_closure_maps; rank_multiplicity = operator_or_matrix_rank_degeneracy; generation_multiplicity = repeated_retained_response_classes; raw_ordering_count = number_of_unquotiented_sequences. 'Do not infer physical channel count from raw ordering count before quotienting by: cyclic start-point equivalence, reversal equivalence when the reader is orientation-insensitive, gauge equivalence, reader equivalence, structural sharing without lineage erasure.'

355 WP.S10.SpectralEquation — Spectral reader eigenequation

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *WP.S10.NormalizedTransfer*

$T_{RG} |\phi_s\rangle = \lambda_s |\phi_s\rangle$

356 WP.S11.EffectivePotentialSlope — Effective potential slope at origin

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S11.EffectivePotentialDiagonal*

$V_{\text{eff}}'(0) = \alpha - \sum_{\mu} d_{\mu} \lambda_{\mu}$

357 WP.S23.FixtureC — Fixture C — Holonomy

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S5.Holonomy*

representation 2x2_real_rotation; cycle [U_2_from_1: rotation_degrees_30, U_3_from_2: rotation_degrees_minus_10, U_1_from_3: rotation_degrees_minus_20]; expected_pass: total_rotation_degrees=0, defect_frobenius_approx=0, status=PASS. failing_control: replace U_1_from_3 with rotation_degrees_minus_15; expected: total_rotation_degrees=5, trace_approx=1.9923893962, determinant_approx=1, defect_frobenius_approx=0.1233742584, status=FAIL_IF_FLATNESS_REQUIRED.

358 step8 — Step 8 — Finite causal propagation speed

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *EQ-007, step5, step7*

$v = \sqrt{(D/\tau_c)} < \infty$ -- Causal propagation is finite-speed.

359 WP.S4.DomainRequiredProperties — Domain definition — required vs not-sufficient properties

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S4.DomainDefinition*

DomainDefinition phrase: 'minimal sufficient dynamically closed translation'.
required_properties: state_sufficiency, readout_factorization, dynamic_commutation, invariant_preservation, declared_context, lineage_preservation. not_sufficient: subject_name, equation_shape_similarity, dimensional_similarity, one_successful_fit, one_passing_control.

360 WP.S4.ExactDomainGate — Exact domain gate — dynamic commutation

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *AdmissibilitySquare, WP.S4.DomainDefinition*

$q_{\{D,n+1\}} \circ F_n = F^{\#}_{\{D,n\}} \circ q_{\{D,n\}}$

Occurrences (1): (6)

361 WP.S5.HolonomyInvariants — Readable holonomy invariants

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S5.CurvatureDefect*

Tr(H_C), det(H_C), spec(H_C). 'Semantic labels such as curvature, reaction-path defect, reasoning drift, or institutional contradiction may be attached only by a validated Domain Card.'

362 WP.S10.SpectralGapFormula — Spectral gap readout

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S10.SpectralEquation*

For $a > 0$ and $0 < \lambda_s \leq 1$: $\epsilon_s = -(1/a) \log(\lambda_s)$

363 WP.S10.SpectralClassification — Spectral classification

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S10.SpectralEquation*

SpectralClassification: *isolated_eigenvalue* → *stable_retained_mode*; *isolated_below_continuum* → *bound_or_composite_mode*; *zero_edge* → *scale_free_or_gapless_mode*; *positive_continuum_edge* → *threshold_persistence*; *no_projected_support* → *no_standalone_readout*.

364 WP.S11.OrderOnsetCriterion — Local order onset criterion

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S11.EffectivePotentialSlope*

$\sum \mu d_\mu \lambda_\mu > \alpha$

365 step9 — Step 9 — Damping definition

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *step5, step8*

$\gamma = 1/\tau_c$ -- Damping = inverse memory time.

366 WP.S4.ReadoutFactorization — Exact domain gate — readout factorization

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S4.ExactDomainGate*

$0_{\{D,n\}} = 0^{\#}_{\{D,n\}} \circ q_{\{D,n\}}$

Occurrences (1): (7)

367 WP.S4.InvariantFactorization — Exact domain gate — invariant factorization

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S4.ExactDomainGate*

$\text{Inv}_{\{D,n\}} = \text{Inv}^{\#}_{\{D,n\}} \circ q_{\{D,n\}}$

368 WP.S10.ResonanceExtension — Resonance extension requirement

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S10.SpectralClassification*

'A resonance as a complex pole does not follow directly from a positive self-adjoint spectrum; it requires a declared extension.' *ResonanceExtension* requires: *resolvent_or_scattering_object*, *analytic_continuation_rule*, *sheet_and_boundary_condition* s; *status_if_absent* NOT_AVAILABLE.

369 WP.S11.OrderSelectionClaimScope — Order selection claim scope

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S11.OrderOnsetCriterion*

```
OrderSelectionClaim: exact_scope [declared_effective_potential, local_slope_at_origin];
not_implied [global_uniqueness, empirical_phase_identity, physical_energy_interpretati
on].
```

370 WP.S13.DomainCard — Executable Domain Card schema

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S4.DomainDefinition, WP.S4.ExactDomainGate*

```
DomainCard schema_version 1.0 with required blocks: identity (domain_id, version,
parent_root_version, status); question (target, horizon, resolution, interventions,
tolerance, context); candidate_state (encoder, slot_activation, fields, memory_slots,
boundary_slots, lineage_slots); carriers; readers; invariants; symmetries; constraints;
dynamics (root_kernel, graph_operator, field_operator, coefficients, currents,
allowed/forbidden reductions, physical_projector, midpoint_amplitude, potential_oper
ator); calibration (native_units, decoder, freeze_hash, held_out_outputs); semantics
(labels ATTACH_AFTER_TRANSLATION); defect_metrics; evidence (positive/negative controls);
provenance (root_derived, architecture_declared, empirical_input, calibration_only,
external_comparison, open_hypothesis); forbidden_imports [domain_final_law_as_root,
target_answer_as_parameter, semantic_name_before_gate, fit_each_output_separately,
physical_unit_without_calibration, checker_outcome_visible_to_maker].
```

371 WP.S17.DefectVector — Canonical defect vector

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S4.ExactDomainGate, WP.S12.ModalReductionDefect*

```
epsilon_D = ( epsilon_suff, epsilon_dyn, epsilon_read, epsilon_inv, epsilon_bridge,
epsilon_cal, epsilon_mode, epsilon_geo, epsilon_solve )^T
```

372 WP.S23.FixtureA — Fixture A — Exact quotient (passing)

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S4.ExactDomainGate*

```
fine_states [0,1,2,3]; quotient {0:A,1:A,2:B,3:B}; fine_dynamics {0:2,1:3,2:2,3:3};
reader quotient_label; expected coarse_dynamics {A:B,B:B}, epsilon_dyn=0, epsilon_read=0,
status=PASS. Reason: fine states 0 and 1 are both in cell A and both transition to B;
states 2 and 3 remain in B.
```

373 WP.S23.FixtureB — Fixture B — Quotient failing control

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S4.ExactDomainGate, XI.6*

```
fine_states [0,1,2,3]; quotient {0:A,1:A,2:B,3:B}; fine_dynamics {0:2,1:0,2:2,3:3};
expected reason "states in A have inconsistent coarse successors", failure_code
LOST_INFORMATION, status FAIL. "This is the failing control for the exact quotient gate."
```

374 step10 — Step 10 — Graph diffusion coefficient

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: step9

$D_s = D \cdot \Delta\theta$ -- Graph diffusion coefficient.

375 WP.S4.DomainGateDefects — Domain gate defects

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: WP.S4.ExactDomainGate, WP.S4.ReadoutFactorization, WP.S4.InvariantFactorization

```
epsilon_dyn = ||q_{n+1} F_n - F^#_n q_n|| ; epsilon_read = ||0_n - 0^#_n q_n|| ;  
epsilon_inv = ||Inv_n - Inv^#_n q_n||. ExactDomainGate classification: exact_when  
sufficiency_status==ADMITTED and all three epsilons ==0; approximate_when sufficiency  
ADMITTED and all required defects within frozen tolerances with at least one nonzero;  
fail_when sufficiency OBSTRUCTED or any defect exceeds frozen tolerance; unresolved_when  
sufficiency UNRESOLVED or norm/tolerance missing or identifiability not decidable.
```

376 WP.S15.CompileDomainAlgorithm — COMPILE_DOMAIN algorithm

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: WP.S13.DomainCard, WP.S14.Gate4

```
FUNCTION COMPILE_DOMAIN(root, raw_tape, domain_card_draft): validate root/philosophy;  
capture phenomenon without mechanism labels; strip domain vocabulary into retained roles;  
freeze question/horizon/interventions/tolerance; build candidate state encoder; run state  
sufficiency audit (enlarge+restart on FAIL, abstain on UNRESOLVED); discover minimal  
field count and coupling structure; discover candidate partition via future readout  
signatures; discover query-relative automorphisms; build quotient q_D; build coarse  
dynamics F_D#; compute epsilon_dyn/read/inv; refine partition until exact/tolerate  
d/impossible; enumerate invariant closure channels; build full operator without premature  
diagonalization; audit modal and moving-basis defects; run native dynamics; build  
positive transfer if requested; calibrate only after native outputs are frozen; run  
independent held-out checker; return DomainRunReport + ClaimLedger.
```

377 WP.S17.DefectRules — Defect classification rules

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: WP.S17.DefectVector

```
DefectRules: exact = 'every required component equals zero in exact arithmetic';  
numerical_pass = 'every required component <= preregistered tolerance'; fail = 'any  
required component > tolerance'; unresolved = 'a required component cannot be  
identified'; report rule = 'never collapse named defects into one scalar without  
retaining the full vector'.
```

378 WP.S20.DomainRunReport — Mandatory Human–AI Run Report schema

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: WP.S13.DomainCard, WP.S17.DefectVector

```
DomainRunReport: document_version, root_hash, domain_card_hash, input_hash,  
code_or_algorithm_hash REQUIRED; philosophy_guards {forbidden_import_check,  
semantic_order_check, lineage_preservation_check} PASS_FAIL; types {state_shapes, units,  
semantic_lanes} REQUIRED; solver {algorithm, time_step, pivot_min, condition_estimate,  
convergence, iterations}; quotient {partition, epsilon_suff, epsilon_dyn, epsilon_read,
```

epsilon_inv} REQUIRED; operator {full_operator_used, epsilon_mode, epsilon_geo, holonomy_defects}; transfer {positivity_certificate, contraction_certificate, eigenvalues, persistence_costs} OPTIONAL; controls {positive, negative, expected_vs_observed} REQUIRED; calibration {used, freeze_hash, held_out_score, epsilon_cal}; final.status one_of [PASS_EXACT, PASS_TOLERANCED, FAIL, UNRESOLVED, PROTOCOL_FAIL, DRIFT]; failure_codes, claim_ids, reproducibility_command.

379 WP.S24.ConcreteReferenceStatus — Concrete reference benchmark scope card

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: WP.S13.DomainCard, WP.S17.DefectVector

ConcreteReferenceStatus: benchmark_id RG-RP4C-001, name "Four-Node Retained Pulse Chain"; purpose [exercise a complete Domain Card; expose a concrete operator and data encoder; calibrate unknown native coefficients without target leakage; evaluate recursively on a frozen held-out segment; require a passing positive control and a failing negative control]; epistemic_scope {data_type: deterministic_synthetic_benchmark_with_frozen_measurement_noise, parameter_fit: finite_numerical_calibration, evidence_type: held_out_synthetic_evidence, external_instrument_calibration: synthetic_independent_fixture, empirical_real_world_validation: false, universal_theory_validation: false}; admissible_claim ["the declared finite pipeline is reproducible", "the declared operator coefficients are identifiable on this training split", "the frozen model predicts the declared held-out segment within preregistered tolerances"]; forbidden_claim ["this benchmark proves a universal law of nature", "synthetic held-out evidence is external empirical validation", "the four-node interpretation is present in the root"].

380 WP.S25.PassFailUnresolvedTable — PASS/FAIL/UNRESOLVED decision table and precedence

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: WP.S17.DefectVector, WP.S18.FailAbleGateLaw

Status precedence is fail-closed: DRIFT or PROTOCOL_FAIL > FAIL > UNRESOLVED > PASS_TOLERANCED > PASS_EXACT. "A higher-precedence status overrides every lower-precedence status." Table rows: All required exact defects are zero -> PASS_EXACT (emit exact-within-scope claim); All required numerical defects within frozen tolerance -> PASS_TOLERANCED (emit numerical/finite claim); A required defect exceeds tolerance -> FAIL (name defect and failure code); Candidate state insufficient -> FAIL_STATE (enlarge state; do not patch output); Identifiability rank deficient -> UNRESOLVED (abstain; acquire more distinctions); Positive control passes but no failing control exists -> TYPE_U_ONLY (at most conditional wording); Maker saw held-out answer -> PROTOCOL_FAIL (invalidate independent evidence claim); Root contains domain-final law -> DRIFT (reject run before calculation); Calibration absent for named physical quantity -> UNCALIBRATED (return native value only); Scalar reduction drops coupling -> REDUCTION_FAIL (use full operator); Projector not Hermitian/idempotent -> FAIL_PROJECTOR (rebuild physical sector); Transfer positive but not contraction-certified -> POSITIVE_ONLY (do not claim nonnegative persistence costs); Resonance requested without analytic extension -> NOT_AVAILABLE (report missing extension).

381 step11 — Step 11 — Velocity update (MQ.08)

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: step9, step10, MQ08-stepper

$V[n+1] = V[n] + \Delta\theta(-\gamma V[n] - D_s L_R X[n])$ -- Velocity update. The engine of change.

Occurrences (1): (5)

382 WP.S4.TranslationFailureTaxonomy — Translation failure taxonomy

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S4.DomainGateDefects*, *X.4b*

TranslationFailure allowed_codes: MISTRANSLATION ('map itself is wrong or ill-posed'); LOST_INFORMATION ('source quotient erased a required distinction'); INSUFFICIENT_RESOLUTION ('retained resolution is too coarse'); TARGET_LACKS_VARIABLES ('target state has no slot for required structure'); NO_CLOSURE ('target has no well-posed update'); CONTEXT_MISMATCH ('law valid in c but applied in c_prime'); CALIBRATION_FAILURE ('native output cannot be independently decoded'); IDENTIFIABILITY_UNRESOLVED ('more than one compatible law remains'). 'A report saying only the reduction fails without a failure code is incomplete.'

383 WP.S22.RG_EXECUTE — Top-level execution pseudocode formal steps

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S15.CompileDomainAlgorithm*, *WP.S21.GlobalRuntimeDAG*

From FUNCTION RG_EXECUTE(root_config, domain_card, observations, checker_bundle): key formal audit/build calls embedded in the control flow -- AUDIT_SUFFICIENCY(candidate_state, domain_card.question) returning status in {OBSTRUCTED, UNRESOLVED, ...}; BUILD_FULL_OPERATOR(graph_operator, field_operator, closures, interaction_operator, geometry=Theta, metric=G); AUDIT_EXACT_QUOTIENT(native_run, partition, readers, invariants, metrics) returning fail/unresolved; native_hash = HASH(native_run, quotient_report, transfer_report); freeze_hash = FREEZE_ALL(root_config, domain_card, native_hash, predictions); status = CLASSIFY(defects, domain_card.defect_metrics); claims = BUILD_CLAIM_LEDGER(status, ALL_PROVENANCE()). Pseudocode control-flow itself is excluded per task scope; only these formal calls/equational content are recorded.

384 WP.S24.DomainCard — 24.1 Concrete Domain Card (RG-RP4C-001)

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S13.DomainCard*, *WP.S24.ConcreteReferenceStatus*

DomainCard schema_version 1.0; identity {domain_id: RG-RP4C-001, version: 1.0.0, parent_root_version: RG-UTW-1.2.0, status: FROZEN_REFERENCE}; question {target: "predict the next seven four-component retained readouts after tape index 10", horizon 7_tape_steps, resolution 1_tape_step, tolerance held_out_rmse_max 0.015, held_out_mae_max 0.012, held_out_max_abs_error_max 0.030, operator_symmetry_max 1.0e-12, operator_row_sum_max 1.0e-12, solve_residual_max 1.0e-12; context topology fixed_four_node_path, boundary closed_endpoints, source zero, potential zero}; candidate_state slot_activation Phi ACTIVE, Psi INACTIVE_WITH_REASON, Theta FIXED_BACKGROUND=four_node_path, Sigma INACTIVE_WITH_REASON, Tape ACTIVE_APPEND_ONLY, G ACTIVE=identity_4, L_cl ACTIVE_EMPTY_CLOSURE_LEDGER, U/Gamma/Xi INACTIVE_WITH_REASON; fields Phi real_vector_length_4; carriers dimensions Phi[4], graph_operator[4,4], retained_metric[4,4], primitive_relations [[0,1],[1,2],[2,3]]; readers node_vector_reader 0_vector(Phi_n)=Phi_n, total_reader 0_total(Phi_n)=sum_i Phi_n[i], equivalence_rule tolerance 1.0e-12; invariants set [L_graph = L_graph^T; L_graph * ones = 0; L_graph positive semidefinite; G = identity_4 positive definite]; dynamics root_kernel finite_DRL_reader_equation_with_Psi_inactive, graph_operator path_laplacian_P4 matrix [[1,-1,0,0],[-1,2,-1,0],[0,-1,2,-1],[0,0,-1,1]], field_operator zero_4x4, coefficients dt=1.0 FROZEN, M=1.0 FROZEN_GAUGE_CHOICE, D=0.348925 TRAINING_CALIBRATED, K=0.221249 TRAINING_CALIBRATED; calibration_decoder equation "Phi_n = (r_n - b) / g", baseline_counts [500,500,500,500], gain_counts_per_native_unit [100,100,100,100], calibration_certificate_sha256 0ab2b3b56d33cd14eefa90e5a79289c3197fbc389706ae5dbbcd992af9b76de2, parameters fixed [M, dt, L_graph, baseline_counts, gain_counts_per_native_unit], fitted [D, K], training_data tape_indices

[0,10], equation_indices [1,9], freeze_hash 88b431490789ad33508efa6f09f7b73fce67624239
09477fd0ef61abc63f89ec, held_out_outputs tape_indices [11,17] visibility FORBIDDEN.

385 WP.S26.AntiDriftRules — Anti-drift rule set (DRIFT_01 through DRIFT_16)

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: WP.S1.ForbiddenOrder, WP.S25.PassFailUnresolvedTable

DriftRules (16 rules, each a prose statement, no formal equations): DRIFT_01_ROOT_IMMUTABILITY "Do not alter root to match one domain answer."; DRIFT_02_NO_SEMANTIC_PREMISE "Attach domain names only after a passed translation."; DRIFT_03_NO_IMPORTED_FINAL_LAW "Imported law must be EXTERNAL_COMPARISON or CALIBRATION_ONLY."; DRIFT_04_NO_FIT_BACK "Held-out target cannot become a parameter, eigenvalue or closure weight."; DRIFT_05_EXACT_SCOPE "Exact must name axioms, architecture, arithmetic and domain scope."; DRIFT_06_FINITE_NOT_CONTINUUM "Finite verifier is not a continuum proof."; DRIFT_07_HESSIAN_NOT_PHYSICAL_POLE "Local curvature is not mass or resonance without spectral calibration."; DRIFT_08_MULTIPLICITY_SEPARATION "Keep invariant, rank, generation and ordering multiplicities distinct."; DRIFT_09_POSITIVE_KERNEL "Cross-midpoint interaction needs Gram or positive-type certificate."; DRIFT_10_NEGATIVE_CONTROL "Every evidentiary gate needs a genuine failing control."; DRIFT_11_SUPERSEDED_FENCED "Old formulas remain only with SUPERSEDED status and replacement pointer."; DRIFT_12_ONE_ROOT_MANY_READOUTS "Domain-specific action requires a declared parent map to S_RG."; DRIFT_13_NO_SILENT_REGULARIZATION "A failing solve must fail visibly."; DRIFT_14_NO_PREMATURE_DIAGONALIZATION "Audit discarded coupling before scalar mode use."; DRIFT_15_CALIBRATION_SEPARATION "Native output, unit conversion and semantic report are distinct stages."; DRIFT_16_PHILOSOPHY_TRACE "Every major implementation gate must cite the constitution principle it enforces." Merged into one row: these are individually enumerated but none contains a distinct formal/mathematical statement beyond prose rules, so recorded as a single combined item per task guidance.

386 step12 — Step 12 — State update (MQ.08)

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: step11

$X[n+1] = X[n] + \Delta\theta V[n+1]$ -- State update. History accretes.

Occurrences (1): (5)

387 WP.S24.RawObservation — Raw observation vector at tape index n

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: WP.S24.DomainCard

$r_n = (r_{\{n,0\}}, r_{\{n,1\}}, r_{\{n,2\}}, r_{\{n,3\}})^T$.

388 WP.S24.MetricAndLaplacian — Retained metric and graph operator (path Laplacian P4)

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: WP.S24.DomainCard

$G = I_4$; $L_{\{P_4\}} = [[1,-1,0,0], [-1,2,-1,0], [0,-1,2,-1], [0,0,-1,1]]$.

389 WP.S28.ReferenceAPI — Minimal implementation interfaces (ReferenceAPI)

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S22.RG_EXECUTE*

ReferenceAPI function signatures (input -> output):
validate_root(root_config) -> guard_report;
encode_state(ordered_tape, encoder) -> candidate_state, provenance;
audit_sufficiency(candidate_state, question) -> ADMITTED_OBSTRUCTED_UNRESOLVED, epsilon_suff;
discover_partition(candidate_state, dynamics, readers, interventions) -> partition, refinement_trace;
audit_quotient(partition, fine_dynamics, coarse_dynamics, readers, invariants) -> epsilon_dyn, epsilon_read, epsilon_inv;
discover_automorphisms(reader, dynamics, G) -> group_generators, certificates;
build_transport(local_frames, automorphisms) -> U;
compute_holonomy(U, ordered_cycles) -> H, K, invariants;
enumerate_closure(representations, automorphism_group) -> intertwiners, multiplicities, obstructions;
build_operator(graph_operator, field_operator, interaction, Theta, G) -> full_operator, symmetric_part, skew_part;
run_drl(state_history, operator, coefficients, currents) -> next_state, residuals, solve_report;
build_transfer(midpoint_amplitude, potential, projector) -> T_positive, positivity, contraction;
spectral_read(T, scale_a) -> spectrum, persistence, support_report;
coarse_grain(state, coarse_map, readers) -> coarse_state, epsilon_cg;
calibrate(native_output, frozen_decoder, training_calibration) -> named_output, epsilon_cal;
check(freeze_hash, held_out, controls) -> checker_report; report(all_reports) -> DomainRunReport, ClaimLedger.

390 step13 — Step 13 — CFL / stability condition

domain: — • *tier:* *Th_coqc* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *step12, MQ08-CFL*

$\Delta\theta \leq 2 / (\gamma + \sqrt{\gamma^2 + 4 \lambda_{\max} D_s})$ -- Sufficient stability gate (conservative). Step too large removes the guarantee -- not proof of breakdown.

391 WP.S24.AffineDecoder — Frozen affine decoder

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S24.RawObservation*

$\Phi_n = (r_n - 500 \cdot 1) / 100.$

392 WP.S24.CanonicalDataHash — canonical_data_sha256 anchor for frozen raw dataset

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S24.RawObservation*

canonical_data_sha256: 43b1aa7081ce1fbf71fa05f45dd35e24b09720479b1afa1f128b93852fbd5231;
hash_scope: UTF-8 bytes from the header row through the final data row, including the final LF and excluding these comment lines. Dataset: 18 rows (t=0..17) of columns t,r0,r1,r2,r3 (four-channel counts), e.g. row t=0: 599.581,500.136,499.999,500.098; row t=17: 524.490,526.098,524.749,524.455.

393 WP.S29.AIContract — AI execution contract

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S1.InformationPhilosophyConstitution, WP.S28.ReferenceAPI*

AIContract: before_run [freeze_root_primitives, state_question_and_tolerance, declare_available_and_forbidden_information, strip_early_domain_semantics]; during_run [preserve_full_operator_until_reduction_is_audited, preserve_lineage_and_tape, return_named_defects, distinguish_zero_obstructed_unresolved, fail_closed_on_singular_or_untyped_operations, separate_native_output_from_calibration]; after_run

[include_positive_and_negative_controls, use_weakest_load_bearing_claim_status, state_finite_continuum_scope, state_simulated_empirical_status, output_reproducibility_record]; forbidden [invent_missing_constants, call_analogy_derivation, call_synthetic_data_empirical_validation, call_machine_checked_structure_physical_truth, hide_nonzero_residual, silently_change_domain_card].

394 step14 — Step 14 — Telegraph equation (DRL-Telegraph root)

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: step13, Telegraph

$\tau_c \partial_t^2 u + \partial_t u = D \nabla^2 u$ -- Coarse-grain shorthand of the stepper. Keeps memory + finite speed. [2026-07-21] identified as the DRL-Telegraph root; the sibling RTPE turbulence relaxation $\tau_R \dot{I}_R + L_R I_R = S_R + \eta_R$ is the $M \rightarrow 0$, $V \rightarrow 0$ limit of this equation, and was found today to PASS_WITH_LIMITS as a controlled reduction; the two, plus the LP-NS nonlinear-paraproduct checker, are stacked layers, not to be merged.

395 step15 — Step 15 — Memory kernel

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: step14, MemoryKernel

$K(\tau) = (1/\tau_c) e^{(-\tau/\tau_c)}$, $\int_0^\infty \tau K(\tau) d\tau = \tau_c$ -- Memory kernel. The past fades exponentially.

396 step16 — Step 16 — Universal spine PDE

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: step15, SpinePDE

$M \partial_t^2 \Phi + D \partial_t \Phi + K L_R \Phi + \nabla V(\Phi) = J - \eta$ -- THE universal spine. One equation for everything. [2026-07-21, central correction of the day] turbulence lives in the nonlinear $\nabla V(\Phi) / (u \cdot \nabla)u$ paraproduct term, not the linear inertial term $M \partial_t^2 \Phi$ (eight forcing attempts on M failed); the inertia relevant to turbulence is τ_R (RTPE), not M; the LP-NS audit is a checker on the $\nabla V(\Phi)$ term, not a generator.

397 WP.S24.CalibrationProtocol — Calibration protocol results (D, K fit)

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: WP.S24.CalibrationDesign, WP.S18.MakerCheckerRoles

CalibrationProtocol: split training_observations [0,10], training_equations [1,9], held_out_observations [11,17]; fixed_before_fit list (dataset_hash, calibration_certificate_hash, split_indices, M, dt, L_graph, zero_source, zero_potential, no_intercept, evaluation_thresholds, positive_control, negative_control); fitted_parameters D estimate 0.348925 (D>=0), K estimate 0.221249 (K>=0); solver ordinary_least_squares_no_intercept, float64, design_rank 2, expected_rank 2, singular_values [1.681640, 0.455975], condition_number 3.688012, residual_variance_estimate 0.0000208442, parameter_standard_error D=0.009894 K=0.003119, approximate_95_percent_interval D=[0.329533,0.368318] K=[0.215135,0.227363]; training_fit equation_rmse 0.004437, equation_max_abs_error 0.010863, status PASS_TOLERANCED; freeze sha256 88b431490789ad33508efa6f09f7b73fce6762423909477fd0ef61abc63f89ec.

398 WP.S24.ConcreteDomainRunReport — Concrete domain run report (RG-RP4C-001-RUN-001)

domain: — • *tier:* *finite_diagnostic* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S20.DomainRunReport*, *WP.S24.ConcreteExecutionDAG*

```
ConcreteDomainRunReport: run_id RG-RP4C-001-RUN-001, root_version RG-UTW-1.2.0,
domain_card_version RG-RP4C-001-1.0.0, dataset_sha256 43b1aa7081ce1fbf71fa05f45dd3
5e24b09720479b1afa1f128b93852fbd5231, calibration_certificate_sha256 0ab2b3b56d33cd
14eefa90e5a79289c3197fbc389706ae5dbbcd992af9b76de2, freeze_hash 88b431490789ad
33508efa6f09f7b73fce6762423909477fd0ef61abc63f89ec; state_dimensions Phi 4, memory_depth
2, active_dynamic_fields 1; operator path_laplacian_P4 shape [4,4], symmetry_defect 0.0,
row_sum_defect 0.0; calibration D_hat 0.348925, K_hat 0.221249, design_rank 2,
condition_number 3.688012, training_equation_rmse 0.004437; held_out horizon 7, rmse
0.004945, mae 0.004030, max_abs_error 0.009580; controls positive_control PASS,
negative_control FAIL_AS_REQUIRED; defects epsilon_suff 0.0, epsilon_dyn 0.0,
epsilon_read 0.0, epsilon_inv 0.0, epsilon_bridge NOT_REQUESTED, epsilon_cal 0.004437,
epsilon_held_out 0.004945; final_status PASS_TOLERANCED; claim tier finite_diagnostic,
statement "The frozen finite benchmark pipeline predicts its seven-step held-out trace
within preregistered tolerances.", exclusions [empirical_real_world_validation,
continuum_claim, universal_domain_claim, unique_operator_derivation].
```

399 WP.S32.OneBoxSummary — One-box summary diagram (boxed equation)

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S31.OfficialStatus*, *WP.S1.InformationPhilosophyConstitution*

```
boxed{ Information Philosophy  $\Downarrow$  delta_R, T, O, C, G  $\Downarrow$  Z_n = (Phi, Psi, Theta, Sigma, T,
G, L^c_l; U, Gamma, Xi)_n  $\Downarrow$  Sufficiency  $\rightarrow$  Equivalence  $\rightarrow$  Automorphism  $\rightarrow$  Transport  $\rightarrow$ 
Closure  $\Downarrow$  P_phys[ delta S_RG + lambda delta C ] = 0  $\Downarrow$  T_+ = P_phys e^{-V/2} A^{\dagger} A
e^{-V/2} P_phys  $\succeq$  0  $\Downarrow$  Quotient + epsilon_D + Calibration Freeze + Independent Checker  $\Downarrow$ 
PASS / FAIL / UNRESOLVED  $\Downarrow$  Bounded Claim }. Summary text: "Differences are retained
before they are named. Structure is checked before it is translated. Meaning follows
readout. Claims are emitted only after computation, defect control, freezing, and
independent checking."
```

400 step17 — Step 17 — Gauge-covariant extension

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *step16*, *GaugeFaces*

$\partial_\mu \rightarrow \mathcal{D}_\mu = \partial_\mu + A_\mu$ -- Gauge-covariant extension. Forces arise from this.

401 WP.S24.HeldOutCheckerReport — Held-out and control evidence report

domain: — • *tier:* *finite_diagnostic* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S24.CalibrationProtocol*, *WP.S18.FailAbleGateLaw*

```
HeldOutCheckerReport: freeze_hash_verified true; recursive_forecast held_out_indices
[11,17], observed_values_used_during_recursion false; metrics rmse 0.004945, mae
0.004030, max_abs_error 0.009580; thresholds rmse_max 0.015, mae_max 0.012,
max_abs_error_max 0.030; positive_control recovered_D 0.350000, recovered_K 0.220000,
status PASS; negative_control (K=0, refit D) refitted_D 0.016637, held_out_rmse 0.123705,
required_rmse_min 0.050, status FAIL_AS_REQUIRED; final_status PASS_TOLERANCED;
claim_status finite_diagnostic; cannot_claim [external_empirical_validation,
universal_domain_law, unique_root_derivation_of_the_operator]. Held-out predicted-vs-observed
table for tape indices 11-17 across Phi_0..Phi_3 given in text.
```

402 step18 — Step 18 — Face 1: eigenmode decomposition

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: step17, Face.1.Eigenmode

$L_R \phi_k = \lambda_k \phi_k$ -- Eigenmodes of the causal graph. Stable structures. [2026-07-21, second major correction] this scalar reading is named the Scalar-Eigenmode Reduction Error: L_R is a full multimode operator; proposed fix $L_R = L_R^{(+)} + L_R^{(-)}$, skew part defined against the retention metric G , [PROPOSED pending test T1]; endogenous state-dependent $L_R[I_R]$ tracked as test T2.

403 WP.S24.ArchivalDisclosureRule — Archival disclosure rule for published held-out data

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: WP.S24.HeldOutCheckerReport

ArchivalDisclosureRule: original_run held_out_status UNSEEN_BEFORE_FREEZE, evidence_status VALID_FOR_RECORDED_VERSION; after_publication held_out_status PUBLIC, evidence_status REPRODUCTION_ONLY; model_revision required [new_domain_card_version, new_freeze_hash, new_hidden_held_out_bundle, repeated_positive_and_negative_controls]. "Because the held-out values are printed in this archival whitepaper, they now support reproduction but no longer constitute untouched evidence for future revisions. Any parameter, operator, encoder, or threshold change MUST use a new unrevealed checker bundle and a new freeze hash."

404 WP.S24.ExternalEvidenceBundle — External evidence adapter contract

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: WP.S24.HeldOutCheckerReport

ExternalEvidenceBundle: bundle_id REQUIRED; source_manifest {producer, acquisition_protocol, acquisition_time, raw_file_sha256, schema, access_or_license_status} REQUIRED; calibration_certificate {certificate_id, calibration_method, valid_range, uncertainty, certificate_sha256} REQUIRED; split_manifest {rule_frozen_before_fit REQUIRED_TRUE, training_ids, held_out_ids_or_commitment_hash, maker_visibility TRAINING_ONLY, checker_visibility HELD_OUT_ONLY}; model_freeze {root_version, domain_card_hash, code_hash, parameter_hash, prediction_hash} REQUIRED; checker {independent_identity_or_process, preregistered_metrics, preregistered_thresholds, positive_control, negative_control} REQUIRED; empirical_claim_gate PASS_when [provenance_complete, no_train_test_leakage, calibration_valid_for_observed_range, held_out_thresholds_pass, positive_control_passes, negative_control_fails_as_required], otherwise FAIL_OR_UNRESOLVED.

405 step19 — Step 19 — Face 2: mode decay

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: step18, Face.2

$|a_k[n]| \leq |a_k[0]| e^{(-\gamma_k n \Delta\theta)}$ -- Modes decay. Impermanence is structural.

406 step20 — Step 20 — Face 3: dispersion split

domain: — • tier: Th_coqc • status: current • coq: root_layer_unwired • parents: step19, Face.3.CriticalSplit

$\lambda_c = D^2/(4MK)$, $k_c = 1/(2\sqrt{(\tau_c D)})$ -- Critical split: below = classical, above = quantum-like.

407 step21 — Step 21 — Face 4: energy monotonicity

domain: — • tier: *Th_coqc* • status: *current* • coq: *root_layer_unwired* • parents: *step20, Face.4*

$dE/dt = -D \|v\|^2 \leq 0$ -- Energy monotone non-increasing. Stability theorem.

408 step22 — Step 22 — Face 5: light-cone

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *step21, Face.5.LightCone, step8*

$\|x\| \leq vt$, $v = \sqrt{D/\tau_c}$ -- Light-cone. Causal horizon.

409 step23 — Step 23 — Face 6: mass-memory

domain: — • tier: *Dr* • status: *current* • coq: *root_layer_unwired* • parents: *step22, Face.6.MassMemory, EQ-063*

$m = \hbar/(2 \tau_c c^2)$ -- Mass = inverse memory. Every particle has a characteristic timescale. [2026-07-21] mass is a readout of τ_c ; τ_c is discrete and prior to mass; M (the inertial coefficient in step 16's spine) is $[Dr]/\text{posited}$, not derived, and is exercised only by the quantum face (D/M vs QuTiP discrepancy 7.6×10^{-4} , *finite_diagnostic*).

410 step24 — Step 24 — Face 7: root force

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *step23, Face.7.RootForce*

$F_{\text{root}} = -K L_R \Phi - \nabla V(\Phi)$ -- Root force = geometric + nonlinear imbalance.

411 step25 — Step 25 — Face 7: Newton readback gate

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *step24, Face.7.SINewtonGate*

`force_to_SI_newton(F_root)` iff gate passes -- Newton readback is admissible only through the unit/source/residual gate.

412 step26 — Step 26 — Face 8: operator→metric

domain: — • tier: *Th_coqc* • status: *current* • coq: *root_layer_unwired* • parents: *step25, Face.8.MetricReadout*

$g^{ij} \approx \frac{1}{2} \text{Hess}(\text{principal symbol of } L_R)$ -- Metric = symbol of the operator. Geometry from dynamics.

413 step27 — Step 27 — Face 9: CPTP completeness

domain: — • tier: *Th_coqc* • status: *current* • coq: *root_layer_unwired* • parents: *step26, Face.9*

$\sum_j K_{j\dagger} K_j = I$ -- CPTP completeness. Quantum channel structure.

414 step28 — Step 28 — Face 10: record law (readout-not-truth)

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *step27, EQ-027*

$M_A[n] = K_A \cdot \theta(E[n]) + \eta_{\text{sel}} + \eta_{\text{map}} + \eta_{\text{self}}, \epsilon_{\text{tot}} > 0$ -- Record law. Every observation is lossy.

415 step29 — Step 29 — Face 11: obstruction

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *step28, Face.11.SolveTarget*

$O_R(R^{\diamond}) = O_{\{E_R\}}, S_R = \|O_R\|^2$ -- Obstruction = residual inconsistency. Solving = zeroing the bundle.

416 step30 — Step 30 — Face 12: boundary discipline

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *step29, Face.12*

absolute constants = boundary data (measured, not derived) -- $G, \hbar, c, k_B, e, m_e, \alpha$ are inputs, not outputs.

417 step31 — Step 31 — Scale bus: τ_c per domain

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *step30, EQ-063*

$\tau_c = \hbar/(2mc^2)$ for each domain's mass/scale -- The bridge connecting all domains. 37 disciplines, 85 orders. [2026-07-21] audited by the bR cross-domain lineage ledger: $r_B = E \cdot A_C \cdot A_P \cdot A_B$; $I_Q = I_B + O_C + O_P + O_B$; a single quantum-domain quotient q_Q does NOT commute directly to a biological-level question -- intermediate chemical/protein layers each contribute their own obstruction term.

418 step32 — Step 32 — One trunk, many leaves (canon scale principle)

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *step31, DomainDiscoveryEngine*

$\forall$ domain d : $PDE(d) = SPINE_PDE$ at $\tau_c(d), L_R(d)$ -- Every domain is the same equation read differently. [2026-07-21] domain-discovery engine result (finite_diagnostic): given a raw (N,S) transition tape, without importing a domain formula in advance, discovers minimal variable count (exact-gate closure test), channel interaction (mixed second differences), minimal exact update law over $\mathbb{Q}$, and conservation laws; recovered $N \cdot S/8, N^2/8$, an affine law, and a genuine 3-variable case; never fabricated a plausible wrong law (e.g. $N \cdot S/16$); holdout data matched exactly; non-polynomial laws correctly refused.

419 step33 — Step 33 — Record genesis (RAR V1.4)

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *step32*

$R_0 = \Omega_A \circ A \circ \Pi \circ T_{\Sigma}(D_0)$ -- How a record is formed from raw domain data D_0 .

Occurrences (1): (67)

420 step34 — Step 34 — Human readout (species-level instantiation)

domain: — • *tier:* *Dr* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *step33, Step2Record*

$M_H[n] = K_H \theta(E[n]) + \eta_H[n]$ -- Human readout. Species-level τ_c^H instantiation.

421 step35 — Step 35 — Optimal policy preserves repairability

domain: — • *tier:* *Dr* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *step34, Step5Policy*

$\pi^* = \operatorname{argmin} [E\|Res\|^2 + \lambda C + \rho Risk - \nu Val + \mu RepairLoss]$ -- Optimal policy preserves repairability.

Occurrences (2): (77), (78)

422 step36 — Step 36 — Three CPU epistemic scalars

domain: — • *tier:* *finite_diagnostic* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *step35, Re_ep, F_ep, k_ep*

Re_ep, F_ep, k_ep (three CPU scalars) -- Any reasoning step reduces to these three numbers.

423 step37 — Step 37 — Verdict gate (DECIDE/ABSTAIN/ESCALATE)

domain: — • *tier:* *Dr* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *step36, VerdictGate*

DECIDE | ABSTAIN | ESCALATE -- Verdict gate over the three scalars. [2026-07-21]
FAIL-ABLE gate law: a genuine Type-P evidence gate needs both a machine-derived passing control and a machine-derived failing control; a gate with only passing or hand-picked-failing examples is Type-U (a convention dressed as evidence, not a validated threshold).

Occurrences (1): (27)

424 step38 — Step 38 — Coq growth ladder

domain: — • *tier:* *Th_coqc* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *step37, step1*

Coq growth ladder: $\forall n, \exists a_{n+1} : a_{n+1} > a_n$

425 step39 — Step 39 — Coq record law

domain: — • *tier:* *Th_coqc* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *step38, step28*

Coq record law: $\forall r \in \text{Records}, \epsilon(r) > 0$

426 step40 — Step 40 — Coq CPTP completeness

domain: — • *tier:* *Th_coqc* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *step39, step27*

Coq CPTP completeness: $\forall K_list, \text{completeness } K_list \rightarrow \sum K \dagger K = I$

427 step41 — Step 41 — Coq operator→metric

domain: — • tier: *Th_coqc* • status: *current* • coq: *root_layer_unwired* • parents: *step40, step26*

Coq operator→metric: $\forall L, \exists g = \text{readout_metric}(L)$

428 step42 — Step 42 — SM-EFF closure (boundary data)

domain: — • tier: *finite_diagnostic* • status: *current* • coq: *root_layer_unwired* • parents: *step41, step30*

$v_{\text{Higgs}} = 246.22 \text{ GeV}$ (CODATA/measured input); $\Sigma_Y = 0$ (hypercharge cancellation -- structural discipline); $\tau_{\mu} = 2.187 \times 10^{-6} \text{ s}$ (muon lifetime -- measured boundary datum).
SM-EFF: standard-model effective readout using declared measured inputs. Not a derivation from zero. Boundary data closes the number, structure closes the roles/relations. There is no step 43 -- the stream is exactly and only 42 steps.

Part XVIII

PART X — FORMAL FLOOR (COQ-CHECKED STRUCTURE)

429 X.1 — X.1 — What Is Machine-Checked (seven readouts of L_R)

domain: — • tier: *Th_coqc* • status: *current* • coq: *root_layer_unwired* • parents: *N3, Face.8.MetricReadout, Face.9*

Machine-checked theorems (axiom-free unless disclosed): growth ladder: $\forall n, \exists a_{n+1} > a_n$; record law: $\forall r, \epsilon(r) > 0$; CPTP completeness: $\text{completeness list} \rightarrow \Sigma K^\dagger K = I$; operator→metric: $\text{principal_symbol } L \rightarrow \text{metric_form}$; energy stability: $dE/dt = -D\|v\|^2$ (energy_nonincreasing / strict_decay); dispersion split: *split_classical / split_quantum*; unit-node grammar: *force readback structure*; domain_rescale_covariant: *constant-coefficient spine (M, D, K, λ held constant) rescales covariantly -- [Th_coqc]*.
Files: *formal/Quantum_Relativity_Formal_DAG_Deep_ConnectedClosure_MassFreeVis ualBH_CognitiveRhythm_NoImpact.v; formal/RDL_SpineStability.v, RDL_MetricReadout.v, RDL_StarRig*.v, RDL_DtN.v.*

430 X.2 — X.2 — The Coq–Physics Boundary

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *X.1*

Machine-checked structure $\neq$ empirical physics truth. Structure can be Coq-verified with disclosed assumptions. Numbers require laboratory measurement. The Coq floor closes STRUCTURE. Measurement closes NUMBERS.

431 X.3 — X.3 — Assumption Ledger (Th_coqc naming corollary)

domain: — • tier: *Th_coqc* • status: *current* • coq: *root_layer_unwired* • parents: *X.2*

The full axiom disclosure lives in *formal/AXIOM_STATUS.md*. Key principle: a claim is *Th_coqc* only when the Coq artifact builds AND assumptions are publicly disclosed.
Corollary (sharpened 2026-07-21): a claim is *Th_coqc* only if the object being named in English matches the object actually quantified over in the .v file -- not a same-named neighbor, not an intended future generalization, not a scalar stand-in for a full operator.

432 X.4a — X.4 — The admissibility square (InfoQuotientCompressionExactness / Kemeny–Snell lumpability)

domain: — • tier: *Th_coqc* • status: *current* • coq: *root_layer_unwired* • parents: *AdmissibilitySquare*

A domain-bridge translation $T_{\{a \rightarrow b\}}$ must commute with the two domains' generators: $T_{\{a \rightarrow b\}} \cdot F\#_a = F\#_b \cdot T_{\{a \rightarrow b\}}$, and must preserve readout. This square -- the admissibility square -- $q_{\{n+1\}} \cdot F_n = F\#_n \cdot q_n$ -- is the machine-checked theorem *InfoQuotientCompressionExactness*, and is the formal restatement of Kemeny–Snell lumpability (a quotient of a Markov chain is itself Markov exactly when this square commutes for the chain's generator), applied to *L_R* instead of a generic Markov generator, as the test for whether a claimed cross-domain reduction is honest or mistranslation.

Occurrences (1): (2)

433 X.4b — X.4 — Failure modes of the admissibility square + the Buckingham trap / correct order

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *X.4a*

FAILURE MODES when $T_{\{a \rightarrow b\}} \cdot F\#_a \neq F\#_b \cdot T_{\{a \rightarrow b\}}$: mistranslation -- the map $T_{\{a \rightarrow b\}}$ itself is wrong/ill-posed; lost information -- q_α discarded exactly the structure *b* needs; insufficient resolution -- *a*'s readout grid is too coarse for *b*'s dynamics; target lacks variables -- *b*'s state space has no slot for what *a* is sending; no closure -- *b* cannot form a well-posed update law from the image. FORBIDDEN (the Buckingham trap): textbook-name $\rightarrow$ stuff-parameters-in $\rightarrow$ declare "derived". CORRECT: Retention $\rightarrow$ Structure $\rightarrow$ Domain-translation $\rightarrow$ Meaning $\rightarrow$ Report.

Part XIX

PART XI — CLAIM DISCIPLINE & READOUT-NOT-TRUTH

434 XI.1a — XI.1 — Claim-Tier Table

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *N3, Face.8.MetricReadout*

Ax = Root axiom / primitive stance ($P0, \tau_c > 0, \Delta\theta > 0$). *Df* = Definition ($\gamma = 1/\tau_c, D_s = D \cdot \Delta\theta$). *Th* = Theorem-level claim in project canon (Asymmetry, temporal ordering). *Th_coqc* = Machine-checked structural theorem (Energy stability, CPTP, metric). *finite_diagnostic* = Computed diagnostic in code units (RDU readout, PGFT v0.6 smoke). *Dr* = Design-analogy / derived reading needing bridge assumptions (Mass-memory, human agency). *Open* = Explicitly open or not closed by current proofs (Quantum gravity, beyond-SM).

435 XI.2a — XI.2 — Readout Invariants (π, ϕ, α)

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *N5*

$\pi = 3.14159\dots$ (readout-invariant: ratio of geometry, not derived from dynamics). $\phi = 1.61803\dots$ (golden ratio: readout-invariant growth ratio). $\alpha = 1/137.036$ (fine-structure: measured boundary datum). These are readout-invariants -- they appear in multiple contexts but are not manufactured by the spine; they are boundary inputs or structural ratios.

436 XI.3a — XI.3 — What This Project Claims (original list)

domain: — • *tier:* *Th_coqc* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *N3, Face.9, Face.8.MetricReadout, EQ-027*

CLAIMS (explicitly): primordial distinguishability as root (Ax); $\tau_c > 0$ as retention axiom (Ax); MQ.08 as fundamental computational equation (Th); universal spine PDE unifies domains structurally (Th); energy monotone non-increasing (Th_coqc); dispersion split exists at λ_c (Th_coqc); metric from operator principal symbol (Th_coqc); CPTP completeness (Th_coqc); every observer's record is lossy (Th, N2); Newton readback admissible through the gate (finite_diagnostic); τ_c atlas spans 37 disciplines, ~85 orders (finite_diagnostic); human agency = τ_c^H instantiation (Dr, open).

437 XI.5-F1 — XI.5 — Falsifier F1 (retention axiom)

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *EQ-005*

F1: Show that $\tau_c = 0$ is consistent with retained structure (falsifies the retention axiom).

438 XI.5-F2 — XI.5 — Falsifier F2 (graph-operator reading)

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *EQ-008*

F2: Show that transport over a finite causal graph is impossible without L_R (falsifies the graph-operator reading).

439 XI.5-F3 — XI.5 — Falsifier F3 (Face 9, CPTP completeness)

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *Face.9*

F3: Show that CPTP completeness fails for admissible channels (falsifies Face 9; Coq refutation would close this).

440 XI.5-F4 — XI.5 — Falsifier F4 (Face 4, energy stability)

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *Face.4*

F4: Show that energy can increase monotonically under pure damping (falsifies Face 4; Coq would catch this).

441 XI.5-F5 — XI.5 — Falsifier F5 (Face 8, operator-to-metric)

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *Face.8.MetricReadout*

F5: Show that operator principal symbol gives no metric readout (falsifies Face 8; Coq refutation would close this).

442 XI.5-FT1 — XI.5 — Falsifier F-T1 (skew-L_R / metric-G algebra, test T1)

domain: — • *tier:* *Dr* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *T1*

F-T1: Test T1 -- construct a two-field system where the claimed skew-L_R / metric-G decomposition is well-defined, and show it FAILS to reproduce known linearized coupling behavior (falsifies the PROPOSED skew-L_R algebra of XIV.3; until T1 runs and passes, the algebra stays PROPOSED).

443 XI.5-FT2 — XI.5 — Falsifier F-T2 (two-field-wall gap, test T2)

domain: — • *tier:* Open • *status:* current • *coq:* root_layer_unwired • *parents:* T2

F-T2: Test T2 -- construct an endogenous, state-dependent L_R[I_R] system (the remaining two-field-wall gap) and show no admissible closure of the master equation reproduces its qualitative behavior (would confirm the wall is real and not merely under-explored; a PASS on T2 instead would shrink the wall further).

444 XI.1b — XI.1 — NEW rule: Th_coqc may never be paired with an unmatched continuum-physics noun (label inflation)

domain: — • *tier:* Th_coqc • *status:* current • *coq:* root_layer_unwired • *parents:* XI.1a

NEW rule, binding after the 2026-07-21 hollow-label audit: Th_coqc may never be paired with a continuum-physics proper noun ("quantum," "relativistic," "the Navier-Stokes equation," "turbulence") unless the .v file's quantified object is itself that continuum object, not a finite/discrete analogue that merely shares its name or its limiting behavior. When a finite analogue is what was actually verified, the correct tag is finite_diagnostic, and the write-up must say so in the same sentence that mentions the physics name. Six of the project's eight physics-interpretation cards were found in violation on 2026-07-21 and downgraded pending re-audit; two survived because their Coq object was the literal finite structure their name claimed to be (energy monotonicity, CPTP completeness).

445 XI.4a — XI.4 — What This Project Does NOT Claim (original list)

domain: — • *tier:* untagged • *status:* current • *coq:* root_layer_unwired • *parents:* XI.3a

NON-CLAIMS (explicitly): derives G , $\hbar$, c , k_B , α from first principles; derives particle masses from zero; proves a new physics theory; solves quantum gravity; provides clinical guidance; provides nuclear authorization; provides orbital control authorization; provides hardware-control authorization; replaces standard cosmological measurement; replaces CODATA, laboratory measurement, or domain-specific safety review.

446 XI.2b — XI.2 — Quantity is not a pre-existing $\mathbb{R}$ (Dr capstone)

domain: — • *tier:* Dr • *status:* current • *coq:* root_layer_unwired • *parents:* XI.4a

There is no $\mathbb{R}$ hiding behind the odometer. What is real is the readout that an agency operator actually returns -- bounded, discrete, at finite resolution h -- never a Platonic magnitude the readout merely approximates from outside. A quantity is a coordinate/projection onto whatever readout space the system can actually give, not a pre-existing continuum value measurement politely uncovers. Two projections can carry identical arithmetic and land on the same numeric value and still be different quantities: a quantity's identity is the role that number plays in the bounded, causal, inferential readout graph it came from, not its momentary number. This is exactly what InfoQuotientCompressionExactness's commuting-square condition is built to make precise: stated over quotient roles, not raw numeric coincidence.

447 XI.3b — XI.3 — Additions carried forward from the 2026-07-21 working session

domain: — • tier: finite_diagnostic • status: current • coq: root_layer_unwired • parents: DomainDiscoveryEngine, bRLedger, VI.7.FailAbleGateLaw

Three governing layers are structurally distinct and must not be merged (Th):
DRL-Telegraph root (2nd order) / RTPE turbulence relaxation (1st order, the $M \rightarrow 0$, $V \rightarrow 0$ limit) / LP-NS paraproduct audit (a CHECKER, not a generator). Turbulence enters through the nonlinear $\nabla V / (u \cdot \nabla)u$ paraproduct term, not the linear $M \cdot a$ inertial term (Th, 2026-07-21 correction). τ_R (first-order relaxation-memory) is the inertia that governs turbulence behavior; M is a posited parameter that quantum alone exercises meaningfully (finite_diagnostic: D/M vs QuTiP agree to $7.6e-4$). Mass is a readout of discrete τ_c via $m = \hbar/(2c^2\tau_c)$; τ_c is discrete and prior to mass (Ax, founder-locked). Multimode L_R with skew decomposition $L_R = L_R^+(+) + L_R^*(-)$ under the retention metric G handles linearized two-field cases without altering the master equation (Dr, PROPOSED, gated on pending test T1). A domain-discovery engine recovers the minimal closed update law, variable count, channel-mixing, and conservation laws from a raw (N,S) transition tape with zero imported domain formulas, and provably refuses to fabricate a wrong law under an adversarial battery (finite_diagnostic). A cross-domain lineage ledger (bR) tracks retained information across a quantum $\rightarrow$ chemical $\rightarrow$ protein $\rightarrow$ biological-transport chain as a multiplicative obstruction account $I_Q = I_B + O_C + O_P + O_B$ (finite_diagnostic architecture; not yet validated against event-resolved real data). A Type-P evidence gate requires BOTH a machine-derived passing control and a machine-derived failing control; a gate with only one side is Type-U convention, not evidence (Th, the FAIL-ABLE gate law).

448 XI.6 — XI.6 — The FAIL-ABLE Gate Law

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: VI.7.FailAbleGateLaw

A gate G is Type-P (evidence) only if BOTH exist and are machine-derived: (a) a PASSING control -- an input on which G is known/constructed to pass; (b) a FAILING control -- an input on which G is known/constructed to fail. If only (a) exists (or only a "should pass" story with no constructed failing case), G is Type-U (convention) -- it documents an agreement, not a discriminating test, and MUST NOT be cited as evidence for a claim. Tier consequence: Type-P gate passing $\rightarrow$ supports Th_coqc / finite_diagnostic tagging; Type-U gate passing $\rightarrow$ supports at most Dr, and must say "convention, not evidence" in the same sentence.

449 XI.4b — XI.4 — Non-claims added from the 2026-07-21 session

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: XI.3b, A13.SummaryBox

Does NOT claim the skew- L_R / metric- G antisymmetry algebra is proven -- it is PROPOSED, gated on test T1, and must not be cited as settled. Does NOT claim the two-field wall (chemostat / Lotka-Volterra / MHD) is fully closed -- the remaining gap (endogenous state-dependent $L_R[I_R]$) is real and gated on test T2. Does NOT claim the domain-discovery engine has been validated on real chemostat or any real biological data -- pilot runs are synthetic/adversarial only; real-data validation is pre-registered future work. Does NOT claim the bR lineage ledger is a first-principles derivation of biological transport from quantum mechanics -- it is a finite_diagnostic accounting architecture over an obstruction ledger, awaiting event-resolved empirical test. Does NOT claim any "machine-checked" physics-interpretation card is checked in the continuum sense unless its Coq object literally is that continuum object.

450 XI.5-Fhuman — XI.5 — Falsifier F-human (τ_c^H / Dr claim)

domain: — • tier: Dr • status: current • coq: root_layer_unwired • parents: FalsifierFhuman

F-human: Measure τ_c^H directly and find it is not in the predicted range for multiple channels; or show the residual model breaks down systematically across populations (falsifies §VIII / Dr claim).

Part XX

PART XII — EXECUTABLE GUARDS

451 Guard1 — Guard 1 — Package health smoke check

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: SpinePDE

```
python -c "from anse_spine import NuclearCore, Domain, Spine; print('anse_spine OK')"
```

452 Guard2 — Guard 2 — PGFT v0.6 reproducible PDE lane smoke

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: SpinePDE

```
python scripts/pgft_v0_6_smoke.py
```

453 Guard4 — Guard 4 — Docs root anchor integrity

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: SpinePDE

```
python scripts/verify_docs_root.py
```

454 Guard6 — Guard 6 — Jupyter notebook smoke (7 cells)

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: SpinePDE

```
jupyter nbconvert -to notebook -execute notebooks/anse_spine_smoke.ipynb
```

455 Guard10 — Guard 10 — Skew-L_R / metric-G well-posedness probe (pre-T1)

domain: — • tier: Dr • status: current • coq: root_layer_unwired • parents: LR_skew_split

Sanity-checks that the PROPOSED $L_R = L_R^{(+)} + L_R^{(-)}$ decomposition under retention metric G is at minimum well-defined (symmetric/antisymmetric split exists and is unique) on the linearized two-field fixtures used so far. This is NOT test T1 itself -- passing this probe does not promote the tier above PROPOSED/Dr. `python scripts/skew_lr_wellposed_probe.py -fixtures chemostat_linearized,lv_linearized`

456 Guard13 — Guard 13 — Schwarzschild/Unruh semantic-lane type-checker

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *Gateway.SchwarzschildIdentity, Gateway.UnruhIdentity*

Refuses any compare(LocalAcceleration, SurfaceGravity) not mediated by a declared, tested ObserverMap. Acceleration readouts must carry mandatory metadata; a bare scalar a is rejected at the gateway: AccelerationReadout = {value, observer_class, normalization_point, local_or_asymptotic, redshift_map, semantic_lane}. PASSING control: a comparison mediated by SurfaceGravity = ObserverMap(LocalAcceleration) with a supplied, checked redshift map -- allowed. FAILING control: the $\tau_c(E)=\tau_c(U) \Rightarrow a=\kappa$ identification with NO ObserverMap -- must return INVALID_COMPARISON/OPEN, and must NOT return FAIL or PASS. `python scripts/bh_gateway_lane_check.py -reject-unmapped-compare -require-metadata observer_class,normalization_point,redshift_map,semantic_lane`

457 Guard15 — Guard 15 — Geometry-dominance diagnostic

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *Rgeo*

The computable defect bound the living-geometry mixing current (II.8a) already demands: for a moving-basis step $\Phi_n=V_n \phi_n$, $R_{\text{geo}} = |(V_{n+1}-V_n)\phi_n| / |\Delta\Phi_n|$; reject a fixed-basis (drop-the-basis-motion-term) approximation when $R_{\text{geo}} \gtrsim 1$. This is a diagnostic ON an existing master-equation term, not an imported rule. PASSING control: $V_{n+1}=V_n$ (no basis motion) $\rightarrow R_{\text{geo}}=0$, fixed-basis allowed. FAILING control: $V_n=I$, $V_{n+1}=3\text{-}4\text{-}5$ rotation, $\phi_n=(2,1) \rightarrow R_{\text{geo}}\approx 1.099 + \text{sign-flip}$; fixed-basis approximation MUST be rejected. `python scripts/geometry_dominance_check.py -reject-above 1.0`

458 Guard3 — Guard 3 — RDU readout suite

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *RDU-tau_c*

`python scripts/run_rdu_readout_suite.py -stress-samples 100 -trial-count 100 -grid-sizes 2`

459 Guard8 — Guard 8 — Domain-discovery adversarial battery

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *DomainDiscoveryEngine*

Must recover $N\cdot S/8$, N^2 , affine, and the 3-variable case exactly (passing controls); must REFUSE $N\cdot S/16$ and any non-polynomial candidate law (failing controls). A battery run that accepts a wrong law is a hard CI failure, not a warning. `python scripts/domain_discovery_battery.py -holdout-exact -refuse-nonpoly`

460 Guard14 — Guard 14 — Domain-discovery identifiability gate

domain: — • *tier:* *finite_diagnostic* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *DomainDiscoveryEngine, A13.SummaryBox*

The discovery engine's concrete realization of the three-valued admissibility gate ($\perp$ = do not guess): before reporting any "minimal exact law", require `rank(design_matrix) == n_candidate_coefficients` over $\mathbb{Q}$, else abstain. NOT a new root rule -- the *finite_diagnostic* instance of the state-sufficiency / $\perp$ -unresolved principle already in A.13/VI, applied to the tool. PASSING control: the $N\cdot S/8$ tape, declared basis $\{1, N, S, N^2, N\cdot S, S^2\}$ (6 rows, rank 6) -- identifiable, law reported. FAILING control: the same basis with any row removed (rank $5 < 6$) -- MUST return UNDERDETERMINED/abstain, NOT a guessed law. `python scripts/discovery_identifiability_gate.py -require-full-rank -abstain-on-deficient`

461 Guard7 — Guard 7 — Label-inflation grep

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: XI.1b

Flags any `Th_coqc` claim paired with a continuum physics proper noun where the `.v` object is a scalar/finite reduction, not the named continuum structure itself. Run before any interpretation card ships. Type-P: passing control = the two surviving cards from the 2026-07-21 audit; failing control = the six retracted cards. `python scripts/label_inflation_guard.py -cards docs/claims/*.md`

462 Guard5 — Guard 5 — Omega_H horizon transduction smoke (27 checks)

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: OmegaH

`python scripts/smoke_omega_h.py`

463 Guard9 — Guard 9 — bR lineage ledger conservation check

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: bRLedger

Verifies $I_Q = I_B + O_C + O_P + O_B$ holds along the quantum→chemical→protein→biological-transport chain for every synthetic fixture; flags any fixture where q_Q alone (skipping the intermediate quotients) spuriously commutes, since that would itself be a reduction-error false positive. `python scripts/br_lineage_conservation_check.py -chain quantum,chemical,protein,biological`

464 Guard11 — Guard 11 — FAIL-ABLE gate audit

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: VI.7.FailAbleGateLaw

Scans the guard suite itself for Type-U gates lacking a constructed failing control, and blocks any PR that adds a new gate without one. `python scripts/failable_gate_audit.py -suite scripts/`

465 Guard12 — Guard 12 — CFL sufficiency-not-necessity control

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: step13

Proves the II.1 / step-13 CFL bound is a CONSERVATIVE SUFFICIENT gate, not a tight instability threshold. On the 4-node fixture $W = \begin{bmatrix} 0 & 1 & .5 & 0 \\ 1 & 0 & 1 & .5 \\ .5 & 1 & 0 & 1 \\ 0 & .5 & 1 & 0 \end{bmatrix}$ ($L_R = D_W - W$, $\lambda_{\max} = 3.618034$, $\gamma = 0.4$, $D = 3$): PASSING control $\Delta\theta = 0.55$ -- VIOLATES the file bound ($\Delta\theta_{\text{CFL}} = 0.4244865$) yet stays stable: per-mode update spectral radius $\rho = 0.883176 < 1$, energy non-increasing, $\|X\|$ decays $3.162 \rightarrow 1.1e-3$ over 50 ticks ($0.55 < \text{the true per-operator boundary } h_{\text{exact}} = 0.6827093$). FAILING control $\Delta\theta = 0.75$ -- past the true spectral threshold: $\rho = 2.610975$, $\|X_{50}\| \approx 2.6e21$, blows up. Together they falsify "exceeding the bound $\Rightarrow$ unconditionally unstable" while leaving "respecting the bound $\Rightarrow$ stable" intact. h_{exact} is a spectral/asymptotic boundary (transient growth as $\Delta\theta \rightarrow h_{\text{exact}}$ via the non-normal update matrix), NOT a safe operating point -- keep a margin. `python scripts/cfl_sufficiency_guard.py -passing 0.55 -failing 0.75 -check energy-non-increasing -check spectral-radius`

Occurrences (1): (4)

466 WP.S9.StabilityBound — Legacy stepper stability bound

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *step13*

$\Delta_{\theta} \leq 2 / (\gamma + \sqrt{\gamma^2 + 4 \lambda_{\max} D_s})$ (sufficient, not necessary)

467 WP.S24.ConcreteReaderRecurrence — Concrete reader recurrence (second-order DRL kernel, RP4C)

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S24.AffineDecoder*, *WP.S24.MetricAndLaplacian*, *WP.S8.ReaderEquation*

$M * \Delta_t^2 \Phi_n + D * \Delta_t^c \Phi_n + K * L_{\{P_4\}} \Phi_n = 0$, with $M=1$, $\Delta_t=1$.
Equivalently: $(M + D/2) \Phi_{n+1} = -K L_{\{P_4\}} \Phi_n + 2M \Phi_n + (D/2 - M) \Phi_{n-1}$.

468 WP.S29.AIResult — AI output template (AIResult)

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S29.AIContract*, *WP.S19.ClaimLedger*

AIResult: `interpretation_order {retention, structure, translation, meaning, report}`
`REQUIRED_SUMMARY each; calculation {inputs REQUIRED_TYPED_LIST, equations_used`
`REQUIRED_LIST, steps REQUIRED_ORDERED_LIST, outputs REQUIRED_TYPED_LIST}; verification`
`{invariants REQUIRED_MAP, defects REQUIRED_VECTOR, positive_control REQUIRED_IF_EV`
`IDENTIARY, negative_control REQUIRED_IF_EVIDENTIARY, checker_status REQUIRED}; claim`
`{status REQUIRED_ENUM, scope REQUIRED_MAP, cannot_claim REQUIRED_LIST, next_bottleneck`
`REQUIRED_OR_NONE}.`

469 WP.S30.HumanChecklist — Human execution checklist

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S29.AIContract*

HumanChecklist: `philosophy ["Did I begin from retained distinctions rather than a domain`
`name?", "Did I separate readout from reality and meaning from phenomenon?"]; model ["Are`
`all variables typed?", "Does every state slot have an update, constraint or background`
`declaration?", "Did I preserve lineage and order?"]; translation ["Was sufficiency`
`checked before quotient?", "Does the square commute?", "Which defect is nonzero?", "Is`
`the symmetry query-relative?"]; computation ["Was the full operator used?", "Was scalar`
`reduction audited?", "Did the solver fail visibly when invalid?", "Is positivity distinct`
`from contraction?"]; evidence ["Is there a failing control?", "Was the prediction`
`frozen?", "Was the checker independent?", "Are calibration and held-out evaluation`
`separate?"]; report ["Is the claim no stronger than its weakest support?", "Are finite,`
`continuum, simulated and empirical scopes explicit?"].`

Part XXI

PART XIII — NEW: PROVEN RESEARCH STREAMS AS READOUTS OF THE ONE TRUNK

470 XIII.1 — XIII.1 — Discrete Calderón / EIT Recovery

domain: — • *tier:* *Th_coqc* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *Face.8.MetricReadout*

This project's discrete version of the Calderón problem asks: given a finite boundary readout (a Dirichlet-to-Neumann map, `RDL_DtN.v` in the formal floor), does it determine a hidden interior "IFF-tree" with no appeal to completeness at infinity? Answer: boundary readout does pin down the interior tree exactly, for the finite graphs tested -- the spine PDE's operator-to-metric face (`F8`, Appendix B) read through the inverse-problem quotient. Tier: the DtN machine-checked structure is `Th_coqc`; the "recovers the interior tree" instance results are `finite_diagnostic` until a broader battery is run.

471 XIII.2 — XIII.2 — Neural-Network-From-Root + Training Convergence

domain: — • tier: finite_diagnostic • status: current • coq: root_layer_unwired • parents: MQ08-stepper

A network whose update rule is derived from the discrete stepper `MQ.08` rather than borrowed from backpropagation folklore trains to convergence over exact rational arithmetic ($\mathbb{Q}$ -closed -- no floating point rounding hides behind the convergence claim). Not a claim that this is a better network; it is a claim that the learning-domain quotient `q_learning` admits a translation from the spine's discrete stepper that commutes with gradient descent's own update structure on the tested fixtures. Tier: `finite_diagnostic`, $\mathbb{Q}$ -closed convergence proven on tested architectures and fixture sizes; general architecture-independent convergence remains Open.

472 XIII.3 — XIII.3 — DEC Toolkit (Discrete Exterior Calculus)

domain: — • tier: Th_coqc • status: current • coq: root_layer_unwired • parents: EQ-008, Face.4

The Discrete Exterior Calculus toolkit -- grad, div, curl, Hodge star, harmonic decomposition, Sobolev bounds, and a discrete maximum principle, all built directly around `L_R` rather than around a continuum manifold. Each operator is a readout of `L_R` under a different classical-calculus quotient: grad and div through the boundary-operator quotient, curl and Hodge through the exterior-algebra quotient, the maximum principle through the monotone-operator quotient that also produces the energy-stability theorem in `X.1`. Tier: the toolkit's structural identities are `Th_coqc`; any specific PDE solved with it inherits `finite_diagnostic` at the instance level.

473 XIII.5 — XIII.5 — Discrete Navier–Stokes Theorems (Stokes limit / advection cubic-form)

domain: — • tier: Th_coqc • status: current • coq: root_layer_unwired • parents: XIII.3, SpinePDE

Stokes-limit theorem: in the regime where the nonlinear advection term is dropped, the discrete spine PDE's solution structure reduces to a discrete Stokes system, machine-checked (`Th_coqc`). Advection cubic-form result: the nonlinear $(\mathbf{u} \cdot \nabla) \mathbf{u}$ term, expressed in the DEC toolkit's discrete calculus, takes a cubic form whose sign properties were checked numerically across tested grid sizes and found consistent with the energy-stability requirement of `X.1` -- `finite_diagnostic`, not yet a general machine-checked theorem, because the check was run on tested instances, not universally quantified and proven in Coq. This pair is the direct formal ancestor of the 2026-07-21 turbulence correction in PART XIV.

474 XIII.4 — XIII.4 — Black-Hole / τ_{c} -of-Agency Knife-Edge

domain: — • tier: Dr • status: current • coq: root_layer_unwired • parents: Face.5.LightCone, OmegaH

The horizon-transduction smoke suite (smoke_omega_h.py, 27 checks) exercises a reading of black-hole-like behavior as an extreme regime of the same spine PDE: a region where the finite cone condition $\|x\| \leq vt$ (Face 5) becomes the effective causal horizon, and where the record law ($\epsilon(r) > 0$) forces every crossing observer's record to be lossy in a way that reads, under the physics quotient, like horizon information loss. The same finite-cone/record-loss machinery, read under the agency quotient instead, produces the τ_c^H human-agency model (HUMAN_AGENCY_SINGLE_TAU_C.md). Both readings sit at the same structural boundary of the spine (the regime where the finite cone shrinks toward degeneracy) and diverge only in which quotient is applied. Neither reading derives the other; they are siblings under one trunk. Tier: firmly Dr for both readings; the underlying finite-cone and record-loss theorems that both depend on are Th_coqc.

Part XXII

PART XIV — NEW: TODAY (2026-07-21) — LIVE STATE

475 XIV.3 — XIV.3 — M Posited; Mass Is a Readout of Discrete τ_c

domain: — • tier: finite_diagnostic • status: current • coq: root_layer_unwired • parents: EQ-063

M, the inertial mass parameter in the trunk equation, is posited, not derived -- eight separate attempts to force M out of more primitive structure this session all failed. Corrected reading, founder-locked: mass is a readout of τ_c , via $m = \hbar / (2 c^2 \tau_c)$, with τ_c itself discrete and prior to mass (Ax, P0). The D/M ratio was checked against QuTiP and agreed to 7.6×10^{-4} (finite_diagnostic), giving some confidence the posited-M reading is numerically consistent in the one regime where M matters, without claiming M has been derived. QSL discriminator: this book's τ_c is NOT the textbook quantum speed limit (QSL) time τ (Mandelstam-Tamm / Margolus-Levitin) -- the two share only an algebraic shadow (both traffic in $\hbar/2E$ -flavor expressions), while τ_c is a discrete, prior, structural retention time and the textbook QSL τ is a derived bound on how fast a quantum state can evolve.

476 XIV.4 — XIV.4 — Scalar-Eigenmode Reduction Error and Skew-L_R

domain: — • tier: Dr • status: current • coq: root_layer_unwired • parents: LR_skew_split, T1

The error: L_R is a full operator acting on a multimode state. A recurring shortcut treated it via its scalar eigenmode reduction, $\lambda\phi$, as if that told the whole story -- it does not, discarding the off-diagonal and skew coupling between modes. The proposed fix: decompose the operator as $L_R = L_R^{(+)} + L_R^{(-)}$ under the retention metric G -- not a naive transpose; G makes "symmetric"/"antisymmetric" mean something physically meaningful in this setting. This metric-G / L_R-antisymmetry algebra is PROPOSED, not proven, and gated on test T1 (XI.5, F-T1). Nothing downstream may cite this decomposition as settled.

477 XIV.5 — XIV.5 — The Two-Field Wall Has Shrunk, Not Vanished

domain: — • tier: Open • status: current • coq: root_layer_unwired • parents: T2, A.10

The "two-field wall": systems like chemostats, Lotka-Volterra predator-prey dynamics, and MHD couple two interacting state fields, which earlier analysis could not fit into

the single-field spine PDE without changing the master equation itself. The multimode-plus-skew-L_R reading (XIV.4) handles the linearized cases of all three without touching the master equation. What remains is a real, named gap: endogenous, state-dependent L_R[I_R] -- cases where the operator itself depends on the state it acts on. This residual gap is gated on test T2 (XI.5, F-T2) and must be reported as an open gap. Cross-reference: A.10 ("number of fields $\neq$ number of domains") is the reason this is not a contradiction of any prior single-field claim.

478 XIV.1 — XIV.1 — Three Stacked Layers — Do Not Merge

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: LayerStack

1. DRL-Telegraph root (2nd order): $M \partial^2 \Phi + D \partial \Phi + K \cdot L_R \cdot \Phi + \nabla V = J - \eta$ -- the trunk equation. 2. RTPE turbulence (1st-order relaxation): $\tau_R \dot{I}_R + L_R I_R = S_R + \eta_R$ -- the $M \rightarrow 0$, $V \rightarrow 0$ LIMIT of layer 1, a genuine reduction, not an independent equation. Status: PASS_WITH_LIMITS. 3. LP-NS audit (nonlinear paraproduct diagnostic): A CHECKER, not a native generator -- audits whether a candidate solution to layers 1/2 is consistent with the nonlinear paraproduct structure; does not itself generate dynamics. Discipline: nothing may cite "the turbulence equation" as if on equal ontological footing with the trunk PDE.

479 XIV.6 — XIV.6 — Domain-Discovery Engine

domain: — • tier: finite_diagnostic • status: current • coq: root_layer_unwired • parents: DomainDiscoveryEngine

Given nothing but a raw tape of (N, S) state transitions -- no imported biology, no imported chemistry, no hand-written domain formula -- the engine discovers, purely from the data: (a) how many variables the domain minimally needs (via an exact-gate closure test); (b) whether channels interact (via a mixed second-difference test); (c) the minimal exact update law over $\mathbb{Q}$ (rational arithmetic -- no floating-point rounding to hide behind); (d) conservation laws. The adversarial battery (FAIL-ABLE gate law, XI.6) includes passing controls (recovers N-S/8, N^2 , an affine law, a genuine 3-variable case) and failing controls it must reject (refuses to fabricate N-S/16, refuses non-polynomial candidate laws). Holdout data is matched exactly, not approximately. Not yet run on real chemostat or any other real biological dataset -- every result so far is synthetic and adversarial-battery-only.

480 XIV.2 — XIV.2 — The Turbulence Correction (supersedes prior reading)

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: XIII.5

Turbulence does not live in the linear M-a inertial term. The corrected reading, following directly from the discrete Navier-Stokes advection-cubic-form result (XIII.5): turbulence lives in the nonlinear $\nabla V / (u \cdot \nabla) u$ paraproduct term. The inertial quantity that actually governs turbulent behavior is τ_R , the first-order relaxation-memory constant from the RTPE limit (XIV.1, layer 2) -- not M.

481 XIV.7 — XIV.7 — The bR Cross-Domain Lineage Ledger

domain: — • tier: finite_diagnostic • status: current • coq: root_layer_unwired • parents: bRLedger

Tracks retained information along a claimed reduction chain quantum $\rightarrow$ chemical $\rightarrow$ protein $\rightarrow$ biological-transport, expressed as $r_B = E \cdot A_C \cdot A_P \cdot A_B$ (a product of an SI-decode factor E, external/measured, and per-stage admissibility factors A_C , A_P , A_B for the chemical, protein, and biological-transport translations). Key diagnostic

result: the single quantum-domain quotient q_Q does not commute for the biological question -- trying to answer a biological-transport question using only the raw quantum quotient, skipping the intermediate chemical and protein translations, fails the admissibility square of X.4. The ledger's obstruction certificates conserve retained lineage as $I_Q = I_B + O_C + O_P + O_B$ -- total quantum-origin information equals what survives to the biological readout (I_B) plus what was lost at each intermediate obstruction (O_C, O_P, O_B). E is treated as external throughout. A real test needs event-resolved empirical data, pre-registered but not yet run.

482 WP.S3.PositiveMetricPositivity — Positive retained metric positivity

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *WP.S3.CanonicalState*

$G_n = G_n^{\dagger}$, $G_n > 0$ (Hermitian, positive definite). PositiveMetricGate checks: hermitian_within_tolerance, smallest_eigenvalue_above_tolerance; pass = both true; fail = any false -> INVALID_STATE_METRIC. 'retained load MUST NOT be called energy until a domain-specific semantic calibration has passed.'

483 WP.S4.LocalReaderEquivalence — Local reader equivalence

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *WP.S4.ReaderFamily*

$x \sim_{\{0,D\}} y \iff 0_{\alpha}(x) = 0_{\alpha}(y)$ for all α

484 WP.S5.AutomorphismGroup — Query-relative automorphism group

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *WP.S4.ReaderFamily*

$A_{\{D,n\}} = \{ h_n : 0_{\{D,n\}} h_n = 0_{\{D,n\}} , h_{\{n+1\}} F_n = F_n h_n , h_n^{\dagger} G_n h_n = G_n \}$. 'This group is relative to the declared question and readers; it is not an absolute symmetry group.'

485 WP.S7.ModularAction — Universal modular action

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *WP.S3.CanonicalState*

$S_{RG} = S_{DRL} + S_{geo} + S_{tr} + S_{int} + S_{ord} + S_{cut} + S_{tape} + S_{read}$

486 WP.S10.PositiveTransferOperator — Positive transfer operator

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *WP.S10.PositiveKernelPositivity*

$T_+ = P_{phys} e^{-V_{\hat{}}/2} K e^{-V_{\hat{}}/2} P_{phys} \geq 0$, $V_{\hat{}} = V_{\hat{}}^{\dagger}$

487 WP.S11.FluctuationDeterminant — Fluctuation determinant

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *WP.S11.FluctuationKernel*

$Z_{\mu}(r) = \det(I + K_{\mu}(r))$

488 WP.S12.ReadoutPreservingCoarseGraining — Readout-preserving coarse-graining condition

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S12.CoarseGrainingMap*
 $0_D^-(b) \circ C_b = 0_D$

489 WP.S12.EffectiveAction — Optional formal effective action

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S12.CoarseGrainingMap*
 $S_{RG}^-(b) = -\log \int \{C_b[Z]=Z^-(b)\} e^{-\{S_{RG}[Z]\}} d\mu[Z]$ ('OPTIONAL_FORMAL unless the Domain Card supplies the measure, domain of integration, finiteness/integrability conditions, and a computable approximation rule.')

490 WP.S12.FixedPoint — Renormalization fixed point

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S12.CoarseGrainingMap*
 $R_b(\theta_*) = \theta_*$

491 WP.S12.ModalReductionGate — Modal reduction admissibility gate

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S12.ModalReductionDefect*, *v011.11*

A scalar or single-mode reduction is admissible only when: (1) the quotient is valid; (2) one mode is sufficient for the declared readout; (3) off-diagonal, skew, and moving-basis currents are below tolerance; (4) geometry is fixed or externally prescribed; (5) no required retained nonlinear transfer exists between modes.
ModalReductionGate: *pass_when* *epsilon_mode* ≤ *preregistered_tolerance* and all five conditions hold; *fail_code* SCALAR_EIGENMODE_REDUCTION_ERROR.

492 WP.S16.BridgeLevels — Bridge evidentiary levels L0-L5

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S16.BridgeError*

BridgeLevels: L0 *surface_analogy*; L1 *shared_variables_or_relations*; L2 *shared_transition_signature*; L3 *readout_preserving_map*; L4 *dynamically_commuting_quotient*; L5 *recoverable_bidirectional_equivalence*. 'Equations that merely look alike may be reported at no more than L0/L1 until stronger gates pass.'

493 WP.S23.FixtureE — Fixture E — Two-node DRL reader step

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S8.DRLStepAlgorithm*

$L = [[1,-1],[-1,1]]$. *drl_two_node_linear*: parameters *M*=1.0, *D*=0.5, *K*=1.0, *dt*=0.1; *operator*=[[1,-1],[-1,1]]; *potential_gradient/hessian* zero; *sources* *J*=[0,0], *R_Phi*=[0,0]; *initial* *Phi*₋₁=[1,-1], *Phi*₀=[1,-1]; *expected* *Phi*₁*_approx*=[0.9804878049,-0.9804878049], *solver_status* PASS. *Reference continuation*: *Phi*₀=[1,-1], *Phi*₁=[0.9804878049,-0.9804878049], *Phi*₂=[0.9427959548,-0.9427959548], *Phi*₃=[0.8885467107,-0.8885467107], *Phi*₄=[0.8196062829,-0.8196062829]. "For the declared discrete energy definition, the implementation SHOULD show a non-increasing trend after a preregistered transient tolerance. This fixture does not support a continuum theorem." *Record-channel*

passing control drl_record_zero_invariant: M=1.0,D=0.5,K=1.0,dt=0.1; Psi_{-1}=[0,0], Psi_0=[0,0], R_Psi=[0,0], potential_hessian zero; expected Psi_1=[0,0], status PASS. Fail-closed record solve control drl_record_singular_step: M=1.0, D=20.0, dt=0.1, derived_A_Psi_scalar=0.0; expected status FAIL, failure_code SINGULAR_RECORD_SOLVE, silent_regularization_allowed false.

494 WP.S27.LegacyMigration — Legacy migration ledger

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: WP.S19.LegacyTierMapping

LegacyMigration: carried_forward_as_normative [commuting_square, failure_taxonomy, state_sufficiency_before_partition, no_early_collapse, maker_checker_firewall, prediction_freeze, fail_able_gate_law, defect_vector, full_multimode_operator, reader_record_pair, living_geometry, moving_basis_current, explicit_fail_closed_linear_solver, modal_reduction_audit, unit_and_semantic_lane_separation, append_only_tape_and_deviation_record]; carried_forward_as_reference_realization [discrete_DRL_stepper, telegraph_like_reduction, two_field_action, finite_fixture_guards]; not_promoted_to_universal_root [tau_c_equals_hbar_over_2mc2, mass_as_universal_memory_identity, four_forces_as_gauge_faces, classical_quantum_identity_from_damping_regime, turbulence_domain_interpretations, Standard_Model_specific_claims, black_hole_specific_claims, biological_or_chemical_semantics, SI_constants, named_physical_equations]; treatment rule "Keep as DomainCard hypothesis, external comparison, calibration or open frontier."

495 XIV.8 — XIV.8 — Consolidated Status Table

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: XIV.6, XIV.7, VI.7.FailAbleGateLaw

Three-layer separation (Telegraph/RTPE/LP-NS): Th, settled this session. Turbulence in ∇V paraproduct, not M: Th, supersedes prior reading. M posited; mass = readout of discrete τ_c : Ax+finite_diagnostic, gate D/M vs QuTiP 7.6e-4, status consistent, not derived. Scalar-eigenmode reduction error identified: Th, settled finding. Skew-L_R / metric-G decomposition: Dr, PROPOSED, gate T1, pending. Two-field wall (linearized cases): Th, shrunk, closed for linearized. Two-field wall (endogenous L_R[I_R]): Open, gate T2, remaining gap, real. Domain-discovery engine: finite_diagnostic, gate adversarial battery (Type-P), passed; real-data validation pending. bR lineage ledger: finite_diagnostic architecture, gate event-resolved test, not yet run. FAIL-ABLE gate law: Th, self-applying, binding, PR #29.

496 WP.S3.PositiveMetricNorm — Retained inner product and norm

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: WP.S3.PositiveMetricPositivity

$$\langle x, y \rangle_{G_n} = x^\dagger G_n y ; ||x||^2_{G_n} = x^\dagger G_n x \geq 0$$

497 WP.S4.StrongReaderEquivalence — Strong (horizon) reader equivalence

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: WP.S4.LocalReaderEquivalence

$$q_D(z) = q_D(z') \Rightarrow 0_{\{D, n:n+L\}}(z, u) = 0_{\{D, n:n+L\}}(z', u) \text{ for all } u \text{ in } U_D$$

Occurrences (1): (8)

498 WP.S5.Transport — Transport map

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *WP.S5.AutomorphismGroup*

$U_{\{j < i\}} : X_i \rightarrow X_j$

499 WP.S6.InvariantClosureMap — Invariant closure map

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *WP.S5.AutomorphismGroup*

$I_{\alpha} : R_1(x) \dots (x) R_k \rightarrow 1$

500 WP.S7.ActionContracts — Action term contracts

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *WP.S7.ModularAction*

ActionContracts: S_DRL consumes [Phi,Psi,G,Tape,declared_coefficients], must be typed/variational-or-explicitly-nonvariational/retained-metric-compatible; S_geo consumes [Theta,Phi,Psi,Sigma,Tape] for relation_update/basis_motion/backreaction; S_tr consumes [U,Theta,G] built from relative_transport/holonomy/retained_mismatch; S_int consumes [closure_maps,Phi,Psi,Sigma,U] restricted to admitted invariant closure channels; S_ord consumes [Sigma,Xi,U,G] for retained_reference/orientation_selection/phase_or_order_transition; S_cut consumes [boundary,cut_currents,closure_ledger] for unresolved_boundary_obligation/information_crossing/interface_cost; S_tape consumes [Tape,Gamma,Xi] for order_sensitivity/reversal_status/sequence_memory; S_read consumes [reader_family,state,coarse_readout] for readout_consistency/microscopic_to_retained_factorization. ActionStatus: universal_modular_interface EXACT_WITHIN_ARCHITECTURE, unique_concrete_action_for_all_domains OPEN, finite_reference_realization_below EXECUTABLE_REFERENCE.

501 WP.S7.ConstrainedEquation — Constrained variational equation

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *WP.S7.ModularAction*

$P_{\text{phys}} [\delta S_{\text{RG}} / \delta Z_n + \sum_a \lambda_a \delta C_a[Z] / \delta Z_n] = 0$,
subject to equality constraints $C_a^* = [Z] = 0$ and inequality constraints $C_b^* = [Z] \leq 0$.

502 WP.S10.ReflectionPositiveTerminology — Reflection-positive terminology requirement

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *WP.S10.PositiveTransferOperator*

'The term reflection-positive is permitted only when the following are supplied: reflection map, positive half-algebra, gluing rule, compatible measure/action. If only $A^\dagger \text{dagger } A$ is established, use the term positive midpoint kernel.'

503 WP.S10.ContractionBound — Contraction certificate — operator bound

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *WP.S10.PositiveTransferOperator*

$0 \leq T_{\text{RG}} \leq I$ (or) $\|T_{\text{RG}}\|_{\text{op}} \leq 1$

504 WP.S11.EffectivePotential — Effective potential

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S11.FluctuationDeterminant, WP.S11.BarePotential*

$V_{\text{eff}}(r) = V_0(r) - \sum_{\mu} \log Z_{\mu}(r)$

505 WP.S12.CoarseGrainingDefect — Coarse-graining defect

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S12.ReadoutPreservingCoarseGraining*

$\epsilon_{\text{cg}} = || O_D^{(b)} C_b - O_D ||$

506 WP.S12.IrrelevantDistinctionCertificate — Irrelevant distinction certificate

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S12.FixedPoint*

$|| \delta \theta_n || \leq \rho^n || \delta \theta_0 ||, \rho < 1$

507 WP.S18.FailAbleGateLaw — Fail-able gate law

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *XI.6*

FailAbleGateLaw: Type_P_evidence_requires: [machine_derived_passing_control, machine_derived_failing_control]; Type_U_convention_when: [only_passing_control_exists, failure_case_cannot_change_gate_output]; tier_consequence: Type_P_pass = "may support finite or machine-checked status within scope", Type_U_pass = "supports at most architecture/conditional wording". "This law tests whether a gate can discriminate; Maker-Checker tests leakage and process bias. Both are required."

508 WP.S23.FixtureD — Fixture D — Positive transfer

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S10.PositiveTransferOperator*

A=[[0.8,0.0],[0.0,0.5]]; V=[[0,0],[0,0]]; P_phys=identity_2; a=1.0; expected K=[[0.64,0.0],[0.0,0.25]], eigenvalues [0.64,0.25], persistence_costs_approx [0.4462871026, 1.3862943611], positivity PASS, contraction PASS. Failing control: supplied_K=[[1.0,2.0],[2.0,1.0]], expected_eigenvalues [3.0,-1.0], expected positivity FAIL, failure_code NON_POSITIVE_KERNEL.

509 WP.S23.FixtureF — Fixture F — Modal reduction

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S12.ModalReductionGate*

modal_reduction_pass: operator=[[2.0,0.0],[0.0,5.0]], projector=[[1,0],[0,0]], expected epsilon_mode=0, status PASS. Failing control modal_reduction_fail: operator=[[2.0,1.0],[-1.0,5.0]], projector=[[1,0],[0,0]], expected epsilon_mode_positive true, failure_code SCALAR_EIGENMODE_REDUCTION_ERROR, status FAIL_IF_SCALAR_REDUCTION_REQUESTED.

Part XXIII

APPENDIX A (v0.11 CARRY-FORWARD) — Turbulence-Integrated equations preserved verbatim

510 v011.1 — APP-v011.1 Hidden-memory kernel + observed/ hidden state split

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *CutCurrent*

$x_{\{0,n+1\}} = F_{\{00,n\}} x_{\{0,n\}} + R_n x_{\{H,n\}} + B_{\{0,n\}} u_n$
 $x_{\{H,n+1\}} = W_n x_{\{0,n\}} + F_{\{H,n\}} x_{\{H,n\}} + B_{\{H,n\}} u_n$
 $K_{\text{mem}}[m] = R_{\{n+m\}} (\prod_{j=1}^{m-1} F_{\{H,n+j\}}) W_n$
 (x_0 = observed sub-state, x_H = hidden sub-state, $K_{\text{mem}}[m]$ = memory kernel at lag m)

511 v011.3 — APP-v011.3 DRL-Telegraph → RTPE conditional bridge

domain: — • *tier:* *Open* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *CommutingSquare.Bridge*

$C_{\text{turb},n+1} \circ F_n = F_{\{R,n\}} \circ C_{\text{turb},n}$ [Df] exact-bridge condition
 $\epsilon_{\text{bridge},n} = C_{\text{turb},n+1}(F_n \circ \mathfrak{S}_n) - F_{\{R,n\}}(C_{\text{turb},n} \circ \mathfrak{S}_n)$ [Df] bridge defect (if only approximate)
 $\eta_{\{R,n\}} = \eta_{\text{unresolved},n} + E_{\text{bridge}}(\epsilon_{\text{bridge},n})$ [Df] defect folded into the RTPE residual
 (That RTPE is an exact reduction of the full DRL-Telegraph architecture is [Open].)

512 v011.5 — APP-v011.5 Readability / identifiability formalism + strict report hierarchy

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *Face.10.StrictGap*

$Y_L = \mathcal{G}_L x_0 + N_L$, $d_{\text{struct}}(L) = \text{rank } \mathcal{G}_L$ [Df]
 $I_{\text{read}} = \mathfrak{J}_{\text{dom}}(Y_L, \mathcal{G}_L, \Sigma_x, \Sigma_N, \alpha, \rho)$ [Df]
 $\Sigma_\alpha(s) = \Sigma_\alpha(s') \Rightarrow s \sim_{\{\Delta,\alpha\}} s'$ [Df]
 $P_{\{\alpha,n\}} \neq Y_{\{\alpha,n\}} \neq 0_\alpha(\mathfrak{S}_n) \neq \mathfrak{S}_n$ [Df] strict report hierarchy

513 v011.7 — APP-v011.7 Total-cost decomposition + replicatio n depth/branching

domain: — • *tier:* *Dr* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *B.2a.SynthesisComplexity*

$c_{\text{URS}} = C_{\text{discover}} + C_{\text{audit}} + C_{\text{normalize}} + C_{\text{frontier}} + C_{\text{lineage}} + C_{\text{tape}} + C_{\text{geometry}}$
 $+ C_{\text{turb}} + C_{\text{recovery}}$
 $r_!(\pi) = \max \# \{ \text{nested licensed replications} \}$, $B_!(\pi) = \prod_{j=1}^{\infty} \{ r_!(\pi) \}^{\kappa_j}$
 Rule [Dr]: intrinsic exponential phase is declared only when BOTH licensed branching and the count of distinguishable admissible normal forms grow exponentially.

514 v011.2 — APP-v011.2 Injectivity laws (output-residue transition, tape-encoder)

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *v011.1*

$(z_n, u_n) \neq (z_{n'}, u_{n'}) \Rightarrow (z_{\{n+1\}}, r_n) \neq (z_{\{n+1\}'}, r_{n'})$ [Ax] no distinction lost at the step
 $E_{\{T,n\}}(r) = E_{\{T,n\}}(r') \Rightarrow r = r'$ [Ax] tape encoder is injective

515 v011.4 — APP-v011.4 LP/NS finite-diagnostic residual (checker only, never native dynamics)

domain: — • tier: *finite_diagnostic* • status: *current* • coq: *root_layer_unwired* • parents: *v011.3*

$H_q^{\text{rem}} = [H_q^{\text{hid}} - C_q^{\text{tr}} - C_q^{\text{p}} - C_q^{\text{inc}} - C_q^{\text{nonalign}} - C_q^{\text{hf}} - C_q^{\text{para}}]_+$
(six correction terms: transport C_q^{tr} , pressure C_q^{p} , incompressibility C_q^{inc} , non-alignment C_q^{nonalign} , high-frequency C_q^{hf} , paraproduct C_q^{para})

516 v011.6 — APP-v011.6 Modal-reduction discarded-coupling audit + admissibility rule

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *XIV.4*

$\epsilon_{\text{mode}} = \|(I-P) \mathbb{G} P\| + \|P \mathbb{G} (I-P)\|$ [Df]
Rule [Ax] (source §13, verbatim): "A scalar eigenmode is admissible only when the discarded coupling is zero or bounded below the preregistered tolerance for the declared readout."

517 v011.8 — APP-v011.8 Theorem-compatibility regimes

domain: — • tier: *Dr* • status: *current* • coq: *root_layer_unwired* • parents: *XIV.5*

Regime 1, fixed linear operator [Dr]: if $\mathbb{G}$, $\mathcal{L}_R$, M , D , K are fixed linear operators, sources are declared and nonlinear currents vanish -- superposition holds for solutions using the same operator trajectory under homogeneous boundary conditions; the zero state is invariant when all sources and currents vanish; rescale covariance holds under the declared weights; an antisymmetric operator sector preserves linearity and contributes no quadratic self-energy.

Regime 2, time-dependent exogenous operator [Dr]: if the operator trajectory is prescribed independently of the state, common-trajectory superposition survives, but the autonomous theorem must be restated.

Regime 3, endogenous geometry or nonlinear transfer [Open]: if $\mathbb{G}_n = \mathbb{G}[\Theta_n]$ with $\Theta_{\{n+1\}} = F_{\Theta}(\Theta_n, X_n)$, or $\mathcal{N}_R \neq 0$, global-solution superposition generally fails -- expected, not a refutation of the linear skeleton (the existing T2 gap / two-field wall).
Honesty line: Regimes 1-2 are elementary linear-algebra facts stated but NOT machine-checked in our Coq floor; they are [Dr] here, not [Th_coqc].

518 v011.11 — APP-v011.11 Five explicit conditions for a valid scalar reduction

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *XIV.4*

The frozen scalar equation $R = M a + D v + K \lambda \phi + V' - j + \eta$ is obtained only when all of the following hold: 1. a valid quotient has already been found; 2. one invariant mode v_a is sufficient for the declared readout; 3. off-diagonal, skew and moving-basis currents vanish in that sector; 4. geometry is fixed or externally prescribed; 5. no retained nonlinear transfer between modes is required. (The scalar equation is a legitimate reduced theorem domain, not the complete ontology of the generalized architecture.)

519 v011.12 — APP-v011.12 15-step runtime protocol

domain: — • tier: Dr • status: current • coq: root_layer_unwired • parents: B.2.Step1, step1

1. Register the domain semantic card without importing its governing equation.
2. Encode observations into RD and declare decoder/calibration separately.
3. Discover the partition and multimode state space.
4. Audit the exact quotient and readout factorization.
5. Build the full operator; do not diagonalize prematurely.
6. Measure symmetric, antisymmetric and discarded modal sectors.
7. Generate only admissible syntheses and append every obstruction certificate.
8. Evolve the DRL-Telegraph reader-record pair.
9. Evolve retained geometry when it is endogenous.
10. Construct the turbulence retained-state readout when requested.
11. Audit the DRL-to-RTPE bridge defect.
12. Apply the LP/NS layer only as a separately labelled checker.
13. Audit conservation, lineage, tape, recovery and identifiability.
14. Decode the final RD report into domain units only after calibration.
15. Compare with preregistered external benchmarks using a checker independent of the maker.

520 v011.13 — APP-v011.13 Named open items

domain: — • tier: Open • status: current • coq: root_layer_unwired • parents: XIV.5

axiom-free Coq lift from scalar rationals to generalized operator modules; exact derivation of RTPE from the full DRL-Telegraph architecture; endogenous living-geometry action with complete backreaction; derivation of nonlinear transfer without importing benchmark laws; continuum-uniform turbulence residual bound; external DNS/experimental validation and component ablation; full chemostat, Lotka-Volterra and MHD derivations from native RD dynamics; a unified hybrid action deriving discovery, admissibility, tape and turbulence together.

521 v011.9 — APP-v011.9 Consolidated master box

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: v011.1, v011.6

$$\begin{aligned} \delta_R: \Gamma \neq \Gamma' &\Rightarrow \mathfrak{R}(\Gamma) \neq \mathfrak{R}(\Gamma') \\ \Gamma \vdash_{\{\alpha, \rho, \kappa\}} \phi, A &\not\Rightarrow A \otimes A, !_{\kappa} A \vdash A^{\otimes m} \\ \mathcal{P}_n &= \text{FixRefine}(\mathcal{M}_n, \mathcal{Q}_\alpha) \\ q_{n+1} F_n &= F_n \# q_n, O_{\{\alpha, n\}} = O_{\{\alpha, n\}}^{\#} q_n \\ \text{Adm} &= T \wedge I \wedge B \wedge [O_\rho = 0] \\ \mathcal{G}_n &= (G_n, \Lambda_n, \mathcal{T}_n, \Theta_n) \\ X_n^A &= (\Phi_n^I, \Psi_n^I)^T, A = (\sigma, I) \\ \mathcal{G}_n &= L_{\{G_n\}} \otimes I + I \otimes C_{\mathcal{F}} + C_{\{\text{int}, n\}} \\ \mathcal{G}_n &= \mathcal{G}_n^{\{(+)\}} + \mathcal{G}_n^{\{(-)\}} \\ \Theta_{n+1} &= F_\Theta(\Theta_n, X_n, \Lambda_n, \mathcal{T}_n, u_n) \\ M \delta_t^2 \Phi + D \delta_t^c \Phi + K \mathbb{G}[\Theta] \Phi + \nabla V - J &= P_\Phi \mathcal{J}_C + \mathcal{J}_{\text{syn}} + \mathcal{J}_{\text{geo}} \\ M \delta_t^2 \Psi - D \delta_t^c \Psi + K \mathbb{G}[\Theta]^T \Psi + \nabla^2 V \Psi &= P_\Psi \mathcal{J}_C + \mathcal{J}_\Lambda \\ I_R &= C_{\text{turb}}(\mathcal{G}, X) \\ \tau_R \delta_t I_R + \mathcal{L}_R[I_R, \Theta] I_R &= S_R + \eta_R \\ \mathcal{L}_R I_R &= \mathcal{L}_R^{\{(+)\}} I_R + \mathcal{L}_R^{\{(-)\}} I_R + \mathcal{N}_R[I_R; \Theta] \\ \mathcal{J}_C &= \mathcal{J}_{\text{transport}} + \mathcal{J}_{\text{return}} + \mathcal{J}_{\text{readout}} + \mathcal{J}_{\text{tape}} \\ \mathcal{T}_{n+1} &= \mathcal{T}_n \boxplus E_T(r_n) \\ \mathcal{G}^{\{\alpha, n\}} &= K_{\{\alpha \leftarrow \beta\}}^{\{[m]\}}(Y_{\{\beta, n-m: n\}}, \mathcal{T}_n) \\ P_{\{\alpha, n\}} &= \Pi_\alpha(Y_{\{\alpha, n\}}, \delta_\alpha, \nu_\alpha, D_\alpha, \theta_\alpha) \end{aligned}$$

(This is a consolidation, not new math -- the whole architecture boxed together for one-point reference, reproduced verbatim from the v0.11 source box.)

Occurrences (1): (2)

522 WP.S1.ForbiddenOrder — Forbidden order

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S1.MandatoryDerivation Order*

Textbook Name -> Import Final Law -> Fit Parameters -> Declare Derived

523 WP.S2.LayerNonBorrowingRules — Layer non-borrowing rules

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S2.ArchitectureLayers*

LayerNonBorrowingRules: 'L0 axiom does not prove L3 domain semantics'; 'L1 mathematical closure does not prove L2 is nature's unique dynamics'; 'L2 finite execution does not prove continuum existence'; 'L3 calibration fit does not equal held-out prediction'; 'L4 clean protocol does not prove the model true'; 'L5 wording MUST use the weakest load-bearing status'.

524 WP.S3.NotRootByDefault — Items not root primitives by default

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S3.RootPrimitives*

NotRootByDefault: spacetime, particle, molecule, gene, neuron, institution, physical_energy, mass, force, probability, continuum, SI_units, domain_final_equation.

525 WP.S3.CanonicalState — Canonical state vector Z_n

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S3.RootPrimitives*

$Z_n = (\text{Phi}_n, \text{Psi}_n, \text{Theta}_n, \text{Sigma}_n, \text{T}_n, \text{G}_n, \text{L}^{\text{cl}}_n ; \text{U}_n, \text{Gamma}_n, \text{Xi}_n)$. Semicolon separates left (primary retained-state slots: Phi proposed_distinction_state, Psi companion_record_response, Theta relational_geometry_translation_state, Sigma retained_order_reference, Tape append_only_history, G positive_retained_metric, L_cl closure_and_lineage_ledger) from right (derived/reified operational slots retained for reproducibility: U transport, Gamma orientation_grading with $\text{Gamma}^{\text{dagger}} = \text{Gamma}$, $\text{Gamma}^2 = \text{I}$, Xi orientation_selecting_order, odd under declared tape reversal; λ_n^a constraint multiplier is auxiliary, not part of L_{cl}).

526 WP.S4.ReaderFamily — Declared reader family

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S3.RootPrimitives*

$\text{O}_D = \{\text{O}_\alpha\}_{\alpha \in \mathbb{R}_D}$

527 WP.S8.OperatorSplitChecks — Operator split required checks

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S8.OperatorSplit*

$(G^{(+)})^{\text{dagger}_G} = G^{(+)}$, $(G^{(-)})^{\text{dagger}_G} = -G^{(-)}$. OperatorSplitStatus: algebraic_split_for_given_G DERIVED_EXACT; universal_physical_interpretation CONDITIONAL; state_dependent_endogenous_operator_closure OPEN_UNLESS_INSTANCE_PROVED.

528 WP.S8.SkewNearZero — Skew-sector near-zero readout condition

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S8.OperatorSplit*
 $\langle z, G^{\sim}(-) z \rangle_G \sim 0$

529 WP.S8.MovingBasisDelta — Moving-basis increment decomposition

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S8.MovingBasis*
 $\Delta \Phi_n = V_{\{n+1\}} \Delta \phi_n + (V_{\{n+1\}} - V_n) \phi_n$

530 WP.S8.DRLStepAlgorithm — DRL_STEP reference runtime algorithm

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S8.LinearSolveGate*, *WP.S8.GeometryDominanceRatio*

```
FUNCTION DRL_STEP(run, state_n_minus_1, state_n): validate dt>0, G positive definite,
provenance/type on every slot; update geometry candidate; build operator; split into
metric-symmetric/skew sectors and check residuals; compute moving-basis current and
R_geo, fail GEOMETRY_REDUCTION_INVALID if fixed-basis requested and R_geo>tolerance;
compute closure channels and append obstruction certificates; assemble R_Phi, R_Psi by
named current channels; build and solve A_Phi*Phi_next=b_Phi and A_Psi*Psi_next=b_Psi via
fail-closed solver; recompute Theta_next/transport U if needed; compute residuals,
constraints, invariants; append to Tape; return FAIL if any hard invariant fails,
UNRESOLVED if any required fact unidentified, else PASS_WITH_OUTPUT.
```

531 WP.S10.PositiveKernelPositivity — Positive kernel positivity

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S10.PositiveKernelDefinition*

$$K = A^{\dagger} A \succeq 0$$

532 WP.S11.FluctuationKernel — Fluctuation kernel

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S11.BarePotential*

$$K_{\mu}(r) = B_{\mu}(r) B_{\mu}(r)^{\dagger}$$

533 WP.S12.CoarseGrainingMap — Coarse-graining map

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *AdmissibilitySquare*

$$Z^{\sim}(b) = C_b[Z]$$

534 WP.S12.ModalReductionDefect — Modal reduction defect

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *v011.6*

$$\epsilon_{\text{mode}} = || (I-P) G P || + || P G (I-P) ||$$

535 WP.S16.BridgeError — Bridge error

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S16.ExactBridgeDynamics*, *WP.S16.BridgeReadout*

$\epsilon_{A \leftarrow B} = d_{\Delta}(D_A, \hat{D}_A)$

Occurrences (1): (34)

536 WP.S19.LegacyTierMapping — Legacy tier mapping

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S19.ClaimLedger*, *XI.1a*

LegacyTierMapping: Ax → ROOT_AXIOM; Df → DEFINITION; Th_coqc → MACHINE_CHECKED_SCOPE;
finite_diagnostic → FINITE_FIXTURE; Dr → CONDITIONAL; Open → OPEN. "A legacy Th
without a proof artifact is not promoted automatically. Map it to CONDITIONAL or
DERIVED_EXACT according to the derivation actually verified."

537 WP.S21.GlobalRuntimeDAG — Global runtime DAG (text form)

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S1.InformationPhilosophyConstitution*, *WP.S3.RootPrimitives*

PHILOSOPHY CONSTITUTION → ROOT PRIMITIVES (delta_R, Tape, Reader, Admissibility, G) →
TYPED RETAINED STATE (Phi, Psi, Theta, Sigma, L_cl) → STATE SUFFICIENCY AUDIT [FAIL →
enlarge state; UNRESOLVED → abstain; PASS →] → READER EQUIVALENCE → QUERY-RELATIVE
AUTOMORPHISM → TRANSPORT U → HOLONOMY H_C + DEFECT K_C → CLOSURE INTERTWINERS → FULL
MULTIMODE OPERATOR [modal defect audit; moving-basis defect audit] → DRL READER-RECORD
STEP → PHYSICAL PROJECTOR → POSITIVE TRANSFER [positivity certificate; contraction
certificate] → SPECTRUM / PERSISTENCE → COARSE-GRAINING → EXACT QUOTIENT AUDIT →
DOMAIN SEMANTICS + CALIBRATION → MAKER FREEZE → INDEPENDENT CHECKER → DEFECT VECTOR
[PASS; FAIL; UNRESOLVED] → BOUNDED CLAIM LEDGER.

538 v011.10 — APP-v011.10 Continuum readout map + validity gate

domain: — • *tier:* *Open* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *step14*

$\Phi_{\epsilon}(t) = C_{\{\alpha, \epsilon\}}(\Phi_0, \Phi_1, \dots)$ [Df]
 $\delta_{t^2} \Phi_n \rightsquigarrow \partial_{t^2} \Phi$, $\delta_{t^c} \Phi_n \rightsquigarrow \partial_t \Phi$ (only when bridge error is controlled) [Open
validity]
(The native order remains discrete; a continuum field is a coarse-grained readout, valid
only when the bridge error is controlled -- otherwise the arrow substitution is
unlicensed. Correct order: retained discrete synthesis → DRL recurrence → continuum
readout.)

539 WP.S9.LegacyStepperRules — Legacy stepper status and forbidden claims

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S9.LegacyStepper*, *WP.S9.StabilityBound*

LegacyStepperRules: status REDUCED_REFERENCE; cfl_bound meaning sufficient_not
_necessary; must_not_claim ['exceeding the bound implies unconditional instability',
'single-field stepper is the complete two-field ontology', 'scalar eigenmode represents
the full operator without discarded-coupling audit'].

540 WP.S24.ConcreteOperatorChecks — Concrete operator invariant checks

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S24.ConcreteReaderRecurrence*

ConcreteOperatorChecks: symmetry expression "norm(L - transpose(L), Frobenius)" expected 0.0; zero_row_sum expression "max_abs(L @ ones_4)" expected 0.0; positive_semidefinite expression "min_eigenvalue(L)" expected_lower_bound -1.0e-12; retained_metric expression "min_eigenvalue(G)" expected_lower_bound 1.0; quotient map identity, epsilon_dyn_expected 0.0, epsilon_read_expected 0.0, epsilon_inv_expected 0.0.

541 WP.S24.CalibrationDesign — Calibration design equations (y_n , X_n , θ OLS)

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S24.ConcreteReaderRecurrence*

For each training equation index $n=1, \dots, 9$: $y_n = -M \delta_t^2 \Phi_n$, $X_n = [\delta_t^c \Phi_n, L_{P_4} \Phi_n]$. After stacking all four components and all training indices: $\theta\text{-hat} = [D\text{-hat}; K\text{-hat}] = (X^T X)^{-1} X^T y$.

542 WP.S24.ConcreteExecutionDAG — 24.6 Concrete execution DAG

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S21.GlobalRuntimeDAG*, *WP.S24.ConcreteReaderRecurrence*

INFORMATION PHILOSOPHY CONSTITUTION -> (forbids semantic and target leakage) RAW
FOUR-CHANNEL COUNTS + DATASET HASH -> (verify order, completeness, lineage) INDEPENDENT
AFFINE CALIBRATION CERTIFICATE -> (decode only by frozen baseline and gain) ENCODED
RETAINED TAPE $\Phi[0:17]$ -> (fixed split before fitting) TRAIN $\Phi[0:10]$ and DECLARED P_4
LAPLACIAN FIXED $M=1$, $dt=1$ -> BUILD LINEAR CALIBRATION SYSTEM $X \theta = y$ -> (rank and conditioning gate) FIT D , K -> (freeze payload and SHA-256) RECURSIVE PREDICTION
 $\Phi_{\text{hat}}[11:17]$ -> (checker reveals held-out values) RMSE + MAE + MAX-ERROR -> POSITIVE
CONTROL (must PASS) / NEGATIVE CONTROL (must FAIL) -> PASS_TOLERANCED / FAIL / UNRESOLVED
-> FINITE, BOUNDED CLAIM ONLY.

543 WP.S31.OfficialStatus — Official status declaration

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S29.AIContract*, *WP.S30.HumanChecklist*

OfficialStatus: name Readout_Genesis_Universal_Technical_Whitepaper, version 1.2.0, language English_only, peer_review_status INTERNAL_TECHNICAL_REVIEW_RESOLVED; closed [philosophy_led_translation_order, anti_drift_constitution, typed_root_and_state_contract, reader_equivalence_contract, exact_quotient_and_defect_contract, automorphism_transport_holonomy_contract, closure_grammar, modular_action_interface, executable_finite_reader_record_reference, positive_transfer_requirements, coarse_graining_and_reduction_gates, domain_compilation_protocol, maker_checker_firewall, claim_and_run_report_schemas, standalone_reference_fixtures, concrete_reference_domain_card, concrete_operator_and_data_encoding, frozen_parameter_calibration, held_out_checker_with_controls]; conditional_or_open [unique_concrete_universal_action, unique_domain_emergence_from_root, generalized_machine_checked_operator_theory, full_endogenous_geometry_backreaction, universal_continuum_limit, universal_empirical_calibration, measurement_and_probability_foundations, spin_statistics, complete_cross_domain_predictions, primitive_constants_from_root]; final_boundary_valid_description ["Universal Research and Derivation Architecture", "Philosophy-Governed Human-AI Executable Specification", "Finite Reference Kernel with Fail-Closed Verification"], invalid_description ["Complete Final Law of Nature", "Empirically Proven Theory of Every Domain"].

Part XXIV

APPENDIX B (EXTERNAL-PAPER INTEGRATION, 2026-07-21) — carried at honest tier, provenance-tagged

544 APP-B.1 — APP-B.1 Conserved pairing charge of the (Φ, Ψ) reader–record system (Paper I)

domain: — • *tier:* *Th_cogc* • *status:* *current* • *cog:* *root_layer_unwired* • *parents:* *EtaOmegaMatrices*

$H = \sum_k (\Phi_k \cdot \pi^* \Psi_k - \Psi_k \cdot \pi^* \Phi_k)$ conserved along the DRL flow
(built from the antisymmetric η/ω pairing already posited in the core.) Machine-checked
here (formal/InfoRetentionLagrangian_attempt.v, PR #185, axiom-free over $\mathbb{Q}$): discrete
Euler-Lagrange equivalence (T1_el_psi_node1/node2, T1_el_phi_node1) and the Legendre
D-cancellation identity (T2_D_cancellation, showing H is D-free), for the 3-ring /
T=3-4-slice scope only. Exact conservation of H along a full trajectory when $D \neq 0$ is NOT
proved (source numerics show $O(\Delta\theta^2)$ charge drift); no generalization off the 3-ring
exists in this repo.

545 WP.S1.InformationPhilosophyConstitution — Information phi- losophy constitution (P1-P10)

domain: — • *tier:* *untagged* • *status:* *current* • *cog:* *root_layer_unwired* • *parents:* *EQ-001*

InformationPhilosophyConstitution: P1 no_built_in_domain_labels ('The root contains no
built-in labels such as physics, chemistry, biology, mind, AI, or society.');

P2
difference_before_entity ('Difference precedes the name of what differs.');

P3
history_before_container_time ('Ordered change precedes time treated as a container.');

P4 retention_before_persistence_semantics ('A state becomes reportable only when some
distinction is retained.');

P5 reader_creates_domain_access_not_root_truth ('A reader
makes structure accessible; it does not retroactively create the root.');

P6
domain_is_translation ('A domain is a minimal sufficient dynamically closed
translation.');

P7 meaning_after_readout ('Meaning arises after readout under context; it
is not stored in the root.');

P8 translation_must_account_for_loss ('Every non-exact
translation must identify where distinctions are lost.');

P9 claim_cannot_borrow_certaint
y ('A claim at one layer may not borrow certainty from another layer.');

P10
report_is_last ('A report is emitted only after retention, structure, translation,
calibration, and checking. '), each with a stated machine_consequence.

546 WP.S6.ThreeValuedAdmissibility — Three-valued admissibil ity

domain: — • *tier:* *untagged* • *status:* *current* • *cog:* *root_layer_unwired* • *parents:* *Gate2*

$C(x \mid c, T)$ in $\{1, 0, \text{perp}\}$. Admissibility: '1' -> status ADMITTED, action construct;
'0' -> status OBSTRUCTED, action append_obstruction_certificate; 'bottom' -> status
UNRESOLVED, action abstain_and_request_more_retained_information. 'UNRESOLVED is not
equivalent to OBSTRUCTED.'

Occurrences (1): (27)

547 WP.S8.FullOperator — Full operator on graph-times-field space

domain: — • *tier:* *untagged* • *status:* *current* • *cog:* *root_layer_unwired* • *parents:* *LR_skew_split*

$G_n = L_{\{\text{graph}, n\}}(x) I_F + I_{\{\text{graph}, n\}}(x) C_F + C_{\{\text{int}, n\}}$

548 WP.S8.LivingGeometry — Living geometry update

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *LivingGeometry*

$$\text{Theta}_{\{n+1\}} = A_{\text{Theta}} \text{Theta}_n + B_{\{\text{Theta}, \text{Phi}\}} \text{Phi}_n + B_{\{\text{Theta}, \text{Psi}\}} \text{Psi}_n + u_{\{\text{Theta}, n\}}$$

549 WP.S8.GeometryDominanceRatio — Geometry-dominance ratio

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *Rgeo*

$R_{\text{geo}} = || (V_{\{n+1\}} - V_n) \text{ phi}_n || / || \Delta \text{Phi}_n ||$. MovingBasisGate: fixed_basis_al
lowed_when $R_{\text{geo}} \leq \text{preregistered_tolerance}$; fixed_basis_rejected_when $R_{\text{geo}} > \text{tolerance}$;
unresolved_when denominator zero and numerator nonzero or basis provenance missing.

550 WP.S8.SecondDifference — Discrete second time-difference operator

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *MQ08-stepper*

$$\text{delta_t}^2 X_n = (X_{\{n+1\}} - 2 X_n + X_{\{n-1\}}) / \Delta t^2$$

551 WP.S8.DiscreteAction — Discrete action / Lagrangian sum

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *DRLAction*

$S_{\text{DRL}} = \sum_n L^n$, with reference (real-coordinate) Lagrangian: $L^n = (1/\Delta t) \Delta \text{Phi}_n^T M_n \Delta \text{Psi}_n + (1/2)(\text{Phi}_n^T D_n \Delta \text{Psi}_n - \text{Psi}_n^T D_n \Delta \text{Phi}_n) - \Delta t [K_n \text{Phi}_n^T G_n \text{Psi}_n + \text{Psi}_n^T \text{grad } V_n(\text{Phi}_n) - J_n^T \text{Psi}_n]$. (Replace transpose by declared Hermitian/metric adjoint in complex coordinates.)

552 WP.S8.ReaderEquation — Reader field equation

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *ReaderRecordEqs*

$$M \text{delta_t}^2 \text{Phi}_n + D \text{delta_t}^c \text{Phi}_n + K G[\text{Theta}_n] \text{Phi}_n + \text{grad } V(\text{Phi}_n) - J_n = R_{\{\text{Phi}, n\}}$$

553 WP.S8.RecordEquation — Record field equation

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *ReaderRecordEqs*

$$M \text{delta_t}^2 \text{Psi}_n - D \text{delta_t}^c \text{Psi}_n + K G[\text{Theta}_n]^{\{\text{dagger}_G\}} \text{Psi}_n + \text{grad}^2 V(\text{Phi}_n) \text{Psi}_n = R_{\{\text{Psi}, n\}}$$

554 WP.S8.ResidualChannels — Residual channel decomposition

domain: — • tier: *untagged* • status: *current* • coq: *root_layer_unwired* • parents: *ResidualsAndPotential*, *CutCurrent*

$$R_{\text{Phi}} = J_{\{\text{cut}, \text{Phi}\}} + J_{\text{syn}} + J_{\text{geo}} ; R_{\text{Psi}} = J_{\{\text{cut}, \text{Psi}\}} + J_{\text{L}} ; \text{cut current: } J_{\text{C}} = J_{\text{transport}} + J_{\text{return}} + J_{\text{readout}} + J_{\text{tape}}$$

555 WP.S8.PhiOneStepSolve — Phi one-step linear solve

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: GaussJordanStepper

```
A_Phi = (M/Delta t^2 + D/(2 Delta t)) I ; b_{Phi,n} = R_{Phi,n} + J_n - K G_n Phi_n -  
grad V(Phi_n) + (2M/Delta t^2) Phi_n + (D/(2 Delta t) - M/Delta t^2) Phi_{n-1} ; A_Phi  
Phi_{n+1} = b_{Phi,n}
```

556 WP.S8.PsiOneStepSolve — Psi one-step linear solve

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: GaussJordanStepper

```
A_Psi = (M/Delta t^2 - D/(2 Delta t)) I ; b_{Psi,n} = R_{Psi,n} - K G_n^{\dagger} Psi_n  
- grad^2 V(Phi_n) Psi_n + (2M/Delta t^2) Psi_n - (M/Delta t^2 + D/(2 Delta t)) Psi_{n-1}  
; A_Psi Psi_{n+1} = b_{Psi,n}
```

557 WP.S9.LegacyStepper — Legacy single-field stepper (reduced reference)

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: MQ08-stepper

```
gamma = 1/tau_c ; D_s = D * Delta_theta ; V[n+1] = V[n] + Delta_theta*(-gamma*V[n] -  
D_s*(L_R @ X[n]) + source[n] - residual[n]) ; X[n+1] = X[n] + Delta_theta*V[n+1]
```

558 WP.S10.PhysicalProjector — Physical projector conditions

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: EQ-050

```
P_phys^{\dagger} = P_phys , P_phys^2 = P_phys. 'Projector construction MUST be traceable to  
a declared redundancy or constraint sector.'
```

559 WP.S11.OrderParameter — Order parameter

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: OrderHiggsClosure_v1.12

```
r = Sigma^{\dagger} G Sigma >= 0
```

560 WP.S12.NoEarlyCollapseLaw — No-early-collapse law

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: A.7

```
If an encoding E satisfies E(z) = E(z') but z != z', no downstream solver can recover the  
distinction from that input alone. NoEarlyCollapse rule: 'compress only after  
equivalence is audited'; correction_if_failed [enlarge_candidate_state, restore_lineag  
e_or_tape, increase_resolution]; forbidden_correction [redefine_output_to_hide_mis  
sing_distinction, import_target_law].
```

561 WP.S14.Gate1 — Gate 1 — No-Free-Domain-Law

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: Gate1

```
delta_R + Retention ⇏ (F, C, V, theta)_domain. 'If more than one compatible model  
remains: status UNRESOLVED, required_next_input [interaction_tape, intervention,  
observation, explicit_postulate].'
```

562 WP.S14.Gate2 — Gate 2 — Three-Valued Admissibility

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *Gate2*

'Return only ADMITTED, OBSTRUCTED, or UNRESOLVED; do not guess.'

Occurrences (1): (27)

563 WP.S14.Gate3 — Gate 3 — Context-Indexed Law

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *Gate3*

$F_n = F_n(Z_n, c_n, T_n)$. 'Validity in c does not imply validity in c_{prime} .'

564 WP.S14.Gate4 — Gate 4 — State Sufficiency

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *Gate4*

$\text{Suff}_{\{D,L\}}(Z_D^{\text{cand}}; c, T)$ in $\{1, 0, \text{perp}\}$. Audit at least: readout sufficiency, dynamic sufficiency, intervention sufficiency, delayed-return/memory sufficiency, coupling sufficiency, geometry/basis sufficiency, boundary sufficiency.

Occurrences (1): (28)

565 WP.S14.Gate5 — Gate 5 — Invariant Completion

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *Gate5*

'If a quotient merges states separated by a required invariant, refine the partition.'

566 WP.S14.Gate6 — Gate 6 — Query-Relative Symmetry

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *Gate6*

$H_D = \{ h : 0_D(h \ z) = 0_D(z) , h \ F = F \ h \}$

567 WP.S14.Gate7 — Gate 7 — Calibration Firewall

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *Gate7*

$y_D^{\text{known}} = U_D(r_{\text{native}}; \theta_D, c, C_D^{\text{cal}})$; $H_{\text{cal}} = \text{Hash}(U_D, \theta_D, \text{training}, \text{units}, \text{protocol})$. 'A native informational value is not a physical, chemical, or biological observable before independent calibration.'

568 WP.S15.FieldCountRule — Minimal field count rule

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *A.10*

$m_D = \min \{ m : 0_D \text{ and } F_D \text{ close on } X^{(1:m)} \}$. 'The number of fields is not the number of domains.'

569 WP.S16.BridgeMap — Cross-domain bridge map

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *CommutingSquare.Bridge*

$K_{\{A \leftarrow B\}} : D_B \times T \rightarrow \hat{\text{hat}}(D)_A$

570 WP.S18.MakerCheckerRoles — Maker-Checker firewall roles

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: B.5.Maker, B.5.Checker

Registrar freezes: question, allowed_inputs, forbidden_inputs, metrics, thresholds, failure_rules, dataset_split, software_versions. Maker may: propose_state, discover_partition, build_operator, calibrate_on_training_only, produce_prediction; must_not_see: held_out_outputs, checker_only_labels, answer_derived_threshold, post_freeze_report. Freeze hash_contents: model, code, configuration, parameters, prediction, score_orientation, normal_form, dependencies. Checker may: verify_hash, open_held_out, evaluate_controls (and further checker duties beyond the excerpted range).

571 WP.S18.ProtocolDeviation — Protocol deviation record update

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: B.5.ProtocolDeviation

$T_{\{n+1\}} = T_n \boxplus \text{DeviationRecord}$. "Deviation records are append-only and MUST NOT be overwritten."

572 B.2 — APP-B.2 Linear-pair honesty caveat (Paper I)

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: APP-B.1

In the linear case the record field decouples: $\Psi \equiv 0 \Rightarrow H \equiv 0$. So the narrative "the record absorbs exactly what the reader loses" is [Open] in the linear regime -- it has content only once nonlinear coupling is retained.

573 APP-B.3 — APP-B.3 Frontier candidates from Paper VI (un-reviewed, same-author)

domain: — • tier: Dr • status: current • coq: root_layer_unwired • parents: APP-B.1

Paper VI ("Universal Readout Closure") states three Gates absent from this core: 1. Self-interaction curvature unit $\rightarrow \times 4$ doubling -- a geometric-unit argument (Weitzenböck-type) yielding a factor-4 doubling. [Open] 2. Spin-statistics exchange sign $(-1)^{2s}$ -- recovered from finite framed dynamics rather than from a relativistic field-theory axiom. [Open] 3. $d = 3$ selector -- a local-directional-feedback criterion selecting three spatial dimensions. [Open] Status: candidate frontier, NOT canon -- same-author, non-peer-reviewed self-closeout using Th (weaker than Th_coqc), disclaims proof-assistant formalization; recorded as [Dr]-declared / [Open] candidates awaiting independent review. (Footnote: Paper III's [Th_sym] (G, Ω) -uniqueness result matches the core's posited η/ω matrices in form but answers a formal-uniqueness question, not the open empirical falsifier T1; noted as [Dr] context only.)

574 WP.S1.MandatoryDerivationOrder — Mandatory derivation order

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: WP.S1.InformationPhilosophyConstitution

Retention $\rightarrow$ Structure $\rightarrow$ Translation $\rightarrow$ Meaning $\rightarrow$ Report

575 WP.S2.ArchitectureLayers — Architecture layers L0-L5

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: WP.S1.InformationPhilosophyConstitution

ArchitectureLayers: L0_ontology {retained_distinction, ordered_history, reader_capability, admissibility, positive_retained_pairing} - 'What is retained, and what remains distinguishable?'; L1_structure {typed_state, equivalence_class, automorphism, transport, holonomy, closure} - 'What structure follows from retention and readout?'; L2_dynamics {action_interface, finite_reference_kernel, constraints, transfer_operator} - 'How does the structure evolve?'; L3_translation {domain_card, sufficiency, quotient, bridge, calibration} - 'How does a domain read the dynamics?'; L4_epistemic {registrar, maker, freeze, checker, controls} - 'Was the claim checked independently?'; L5_report {claim_ledger, scope, defect_vector, reproducibility_record} - 'What may actually be claimed?'

576 WP.S3.RootPrimitives — Root primitives

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: WP.S1.InformationPhilosophyConstitution, EQ-005

RootPrimitives: distinction_record (symbol delta_R, 'record that at least one distinction is retained', primitive:true); ordered_tape (symbol T, 'append-only order of events, rewrites, residues and deviations', primitive:true); reader_family (symbol O, 'declared maps that can distinguish, store or act on retained records', primitive:true); admissibility_language (symbol C, 'rules stating admitted, obstructed or unresolved constructions', primitive:true); positive_retained_pairing (symbol G, 'positive pairing used to measure retained load', primitive:true).

577 WP.S8.MetricAdjoint — Metric-adjoint

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: WP.S8.FullOperator

$A^{\{\dagger_G\}} = G^{-1} A^{\dagger_G} G$

578 WP.S8.OperatorSplit — Metric symmetric/skew split

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: WP.S8.FullOperator

$G^{(+)} = (1/2)(G + G^{\{\dagger_G\}})$; $G^{(-)} = (1/2)(G - G^{\{\dagger_G\}})$

579 WP.S8.OperatorGeometryDependence — Operator dependence on geometry state

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: WP.S8.LivingGeometry

$G[\Theta_n] = G_0 + \sum_a \Theta_n^a G_a$

580 WP.S8.MovingBasis — Moving basis representation

domain: — • tier: untagged • status: current • coq: root_layer_unwired • parents: WP.S8.LivingGeometry

$\Phi_n = V_n \phi_n$

581 WP.S8.CentralDifference — Discrete central time-difference operator

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S8.SecondDifference*

$$\text{delta_t}^c X_n = (X_{\{n+1\}} - X_{\{n-1\}}) / (2 \text{ Delta } t)$$

582 WP.S8.LinearSolveGate — Linear solve gate

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S8.PhiOneStepSolve*, *WP.S8.PsiOneStepSolve*

`LinearSolveGate:` `method reference = partial_pivot_gauss_jordan_or_stable_equivalent;`
`must_report [pivot_min, rank, condition_estimate, solver_residual]; fail_when [pivot_min`
`< pivot_tolerance, rank < expected_rank, solver_residual > solve_tolerance]; must_not`
`[silently regularize, replace singular solve with guessed output].`

583 WP.S10.PhysicalProjectorGate — Physical projector gate

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S10.PhysicalProjector*

`PhysicalProjectorGate checks:` `hermitian, idempotent, removes_only_declared_redundancy,`
`preserves_declared_readout; fail_code INVALID_PHYSICAL_PROJECTOR.`

584 WP.S10.PositiveKernelDefinition — Positive kernel definition

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S10.PhysicalProjector*

$$K(X_+, X_-) = \text{sum}_Y A(Y|X_+) * \text{conj}(A(Y|X_-))$$

585 WP.S11.BarePotential — Bare potential

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S11.OrderParameter*

$$V_0(r) = \alpha r + \beta r^2, \beta > 0$$

586 WP.S16.ExactBridgeDynamics — Exact bridge — dynamics commutation

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S16.BridgeMap*

$$K_{\{A \leftarrow B\}} F_{B\#} = F_{A\#} K_{\{A \leftarrow B\}}$$

587 WP.S16.BridgeReadout — Exact bridge — readout condition

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S16.BridgeMap*

$$O_{A\#} K_{\{A \leftarrow B\}} = D_{\{A \leftarrow B\}} O_{B\#}$$

588 WP.S18.MakerCheckerInfoZero — Maker–Checker mutual information condition

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S18.MakerCheckerRoles*

`Info(Maker; Y_checker) = 0. "This is a protocol condition, not a proof that statistical mutual information is zero merely because a hash exists."`

589 WP.S19.ClaimLedger — Claim ledger schema

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *XI.1a*

`Claim: id REQUIRED, statement REQUIRED; status one_of [ROOT_AXIOM, DEFINITION, DERIVED_EXACT, MACHINE_CHECKED_SCOPE, EXACT_WITHIN_ARCHITECTURE, CONDITIONAL, FINITE_FIXTURE, NUMERICAL, CALIBRATION_ONLY, EMPIRICALLY_HELD_OUT, OPEN, FAILED_CONTROL, SUPERSEDED]; layer one_of [ONTOLOGY, STRUCTURE, DYNAMICS, TRANSLATION, EPISTEMIC, REPORT]; assumptions.explicit REQUIRED; provenance {root_parents, imported_inputs, calibration_inputs, external_comparisons} REQUIRED; derivation {parents, equations, algorithm_version} REQUIRED; verification {proof_or_derivation_artifact REQUIRED_IF_DERIVED_OR_MACHINE_CHECKED, positive_control REQUIRED_IF_EVIDENTIARY, negative_control REQUIRED_IF_EVIDENTIARY, checker_independent REQUIRED_IF_HELD_OUT_OR_EMPIRICAL, defects REQUIRED_IF_COMPUTED_OR_TRANSLATED}; scope {finite_or_continuum, exact_or_toleranced, local_or_global, domain_specific_or_universal, context} REQUIRED; forbidden_upgrade.text REQUIRED.`

590 WP.S23.FixtureG — Fixture G — Geometry dominance

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *WP.S8.GeometryDominanceRatio*

`moving_basis_no_motion: V_n=identity_2, V_next=identity_2; expected R_geo=0, fixed_basis ALLOWED. moving_basis_sign_flip: V_n=identity_2, V_next=rotation_matrix{cos:0.6, sin:0.8}, phi_n=[2.0,1.0], delta_phi_n=[0.5,-0.25]; expected true_delta_Phi=[-1.1,1.45], dropped_basis_term_first_component=0.5, R_geo_approx=1.0988845116, fixed_basis REJECTED.`

Part XXV

Root registry extension (founder ruling 2026-09-07)

591 A2 — A2 - FOLD (the engine)

domain: — • *tier:* *Th_coqc* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *none*

`One generic accumulation over a carrier and operation:  $I_{\oplus}[f](N) = \bigoplus_{k \leq N} f[k]$ ,  $I(Df) = f[N] \boxminus f[0]$ . Changing  $(\oplus, \boxtimes)$  changes the branch of mathematics; the telescoping/FTCC identity is exact.`

A2/M.01.v1 — op_swap (Lemma)

domain: *M* • *tier:* *Th_coqc* • *status:* *current* • *coq:* *mapped_not_wrapped* • *parents:* *A2*

`Lemma op_swap : forall x y z, op x (op y z) = op y (op x z).`

Occurrences (1): ‘formal/IDM_Reduction.v’, ‘op_swap’

A2/M.02.v1 — fold_right_perm (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: A2

```
Theorem fold_right_perm : forall xs ys,
  Permutation xs ys -> fold_right op e xs = fold_right op e ys.
```

Occurrences (1): ‘formal/IDM_Reduction.v’, ‘fold_right_perm’

A2/M.03.v1 — ftcc_Z (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: A2

```
Theorem ftcc_Z : forall (f : nat -> Z) (N : nat),
  fold Z.add 0 (zdelta f) N = f N - f 0%nat.
```

Occurrences (1): ‘formal/IDM_Reduction.v’, ‘ftcc_Z’

A2/M.04.v1 — foldmin_le_init (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: A2

```
Lemma foldmin_le_init : forall v0 f N, fold Z.min v0 f N <= v0.
```

Occurrences (1): ‘formal/IDM_Reduction.v’, ‘foldmin_le_init’

A2/M.05.v1 — foldmin_le_elem (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: A2

```
Lemma foldmin_le_elem : forall v0 f N i,
  (i < N)%nat -> fold Z.min v0 f N <= f i.
```

Occurrences (1): ‘formal/IDM_Reduction.v’, ‘foldmin_le_elem’

A2/M.06.v1 — sum_is_fold (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: A2

```
Theorem sum_is_fold : forall f N, sum_readout f N = fold Z.add 0 f N.
```

Occurrences (1): ‘formal/IDM_Reduction.v’, ‘sum_is_fold’

A2/M.07.v1 — path_is_fold (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: A2

```
Theorem path_is_fold : forall v0 f N, path_accum v0 f N = fold Z.min v0 f N.
```

Occurrences (1): ‘formal/IDM_Reduction.v’, ‘path_is_fold’

A2/M.08.v1 — pivot_preserves (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: A2

```
Theorem pivot_preserves : forall a b c d k x y v w,
  a * x + b * y = v ->
  c * x + d * y = w ->
  (a - k * c) * x + (b - k * d) * y = v - k * w.
```

Occurrences (1): ‘formal/IDM_Reduction.v’, ‘pivot_preserves’

A2/M.09.v1 — fold_ext (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: A2

Lemma fold_ext : forall {A} (op : A -> A -> A) (e : A) (f g : nat -> A) N,
(forall k, f k = g k) -> fold op e f N = fold op e g N.

Occurrences (1): 'formal/IDM_Reduction.v', 'fold_ext'

A2/M.10.v1 — fold_linear (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: A2

Theorem fold_linear : forall (c : Z) (f : nat -> Z) N,
fold Z.add 0 (fun k => c * f k) N = c * fold Z.add 0 f N.

Occurrences (1): 'formal/IDM_Reduction.v', 'fold_linear'

A2/M.11.v1 — fold_add_split (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: A2

Theorem fold_add_split : forall (f g : nat -> Z) N,
fold Z.add 0 (fun k => f k + g k) N = fold Z.add 0 f N + fold Z.add 0 g N.

Occurrences (1): 'formal/IDM_Reduction.v', 'fold_add_split'

A2/M.12.v1 — foldmax_ge_init (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: A2

Lemma foldmax_ge_init : forall v0 f N, v0 <= fold Z.max v0 f N.

Occurrences (1): 'formal/IDM_Reduction.v', 'foldmax_ge_init'

A2/M.13.v1 — foldmax_ge_elem (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: A2

Lemma foldmax_ge_elem : forall v0 f N i,
(i < N)%nat -> f i <= fold Z.max v0 f N.

Occurrences (1): 'formal/IDM_Reduction.v', 'foldmax_ge_elem'

A2/M.14.v1 — foldmax_in (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: A2

Lemma foldmax_in : forall v0 f N,
fold Z.max v0 f N = v0 /\ exists i, (i < N)%nat /\ fold Z.max v0 f N = f i.

Occurrences (1): 'formal/IDM_Reduction.v', 'foldmax_in'

A2/M.15.v1 — sum_list_perm (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: A2

Theorem sum_list_perm : forall xs ys : list Z,
Permutation xs ys -> fold_right Z.add 0 xs = fold_right Z.add 0 ys.

Occurrences (1): 'formal/IDM_Reduction.v', 'sum_list_perm'

A2/M.16.v1 — prod_list_perm (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: A2

```
Theorem prod_list_perm : forall xs ys : list Z,  
Permutation xs ys -> fold_right Z.mul 1 xs = fold_right Z.mul 1 ys.
```

Occurrences (1): ‘formal/IDM_Reduction.v’, ‘prod_list_perm’

A2/M.17.v1 — dot_is_fold (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: A2

```
Theorem dot_is_fold : forall u v N,  
dotf u v N = fold Z.add 0 (fun k => u k * v k) N.
```

Occurrences (1): ‘formal/IDM_Reduction.v’, ‘dot_is_fold’

A2/M.18.v1 — dot_scale (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: A2

```
Theorem dot_scale : forall a u v N, dotf (fun k => a * u k) v N = a * dotf u v N.
```

Occurrences (1): ‘formal/IDM_Reduction.v’, ‘dot_scale’

A2/M.19.v1 — horner_scaled (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: A2

```
Lemma horner_scaled : forall cs x p, p * horner x cs = poly x cs p.
```

Occurrences (1): ‘formal/IDM_Reduction.v’, ‘horner_scaled’

A2/M.20.v1 — horner_is_poly (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: A2

```
Theorem horner_is_poly : forall x cs, horner x cs = poly x cs 1.
```

Occurrences (1): ‘formal/IDM_Reduction.v’, ‘horner_is_poly’

A2/M.21.v1 — foldmin_in (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: A2

```
Lemma foldmin_in : forall v0 f N,  
fold Z.min v0 f N = v0 \/ exists i, (i < N)%nat /\ fold Z.min v0 f N = f i.
```

Occurrences (1): ‘formal/IDM_Reduction.v’, ‘foldmin_in’

A2/M.22.v1 — fold_add_app (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: A2

```
Theorem fold_add_app : forall f m n,  
fold Z.add 0 f (m + n) = fold Z.add 0 f m + fold Z.add 0 (fun k => f (m + k)%nat) n.
```

Occurrences (1): ‘formal/IDM_Reduction.v’, ‘fold_add_app’

A2/M.23.v1 — factorial_is_fold (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: A2

```
Theorem factorial_is_fold : forall N,  
fold Z.mul 1 (fun k => Z.of_nat (S k)) N = Z.of_nat (fact N).
```

Occurrences (1): ‘formal/IDM_Reduction.v’, ‘factorial_is_fold’

A2/M.24.v1 — relax_nonincreasing (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: A2

Lemma relax_nonincreasing : forall d cand : $\mathbb{Z}$, $\mathbb{Z}.min\ d\ cand \leq d$.

Occurrences (1): 'formal/IDM_Reduction.v', 'relax_nonincreasing'

A2/M.25.v1 — relax_idempotent (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: A2

Lemma relax_idempotent : forall d cand : $\mathbb{Z}$, $\mathbb{Z}.min\ (\mathbb{Z}.min\ d\ cand)\ cand = \mathbb{Z}.min\ d\ cand$.

Occurrences (1): 'formal/IDM_Reduction.v', 'relax_idempotent'

A2/M.26.v1 — weak_duality_2 (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: A2

Theorem weak_duality_2 : forall c1 c2 x1 x2 a1 a2 b y,
0 <= x1 -> 0 <= x2 -> 0 <= y ->
c1 <= y * a1 -> c2 <= y * a2 -> (* dual feasibility, per column *)
a1 * x1 + a2 * x2 <= b -> (* primal feasibility: $A\ x \leq b$ *)
c1 * x1 + c2 * x2 <= y * b. (* weak duality: $c \cdot x \leq y \cdot b$ *)

Occurrences (1): 'formal/IDM_Reduction.v', 'weak_duality_2'

A2/M.27.v1 — fold_split_even_odd (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: A2

Theorem fold_split_even_odd : forall f n,
fold Z.add 0 f (2 * n) =
fold Z.add 0 (fun k => f (2 * k)%nat) n + fold Z.add 0 (fun k => f (2 * k + 1)%nat) n.

Occurrences (1): 'formal/IDM_Reduction.v', 'fold_split_even_odd'

A2/M.28.v1 — modpow_is_fold (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: A2

Theorem modpow_is_fold : forall g p N, p <> 0 -> modpow_fold g p N = (g ^ N) mod p.

Occurrences (1): 'formal/IDM_Reduction.v', 'modpow_is_fold'

592 A3 — A3 - DECISION (the search)

domain: — • tier: Th_coqc • status: current • coq: root_layer_unwired • parents: none

Some problems are not accumulations but existence questions: $P(X) \Leftrightarrow \exists w \text{ finite: } \text{check}(X,w) = \top$ (no witness $\Rightarrow$ HOLD). This is also the honesty mechanism: absence of a witness is a refusal, never a guess.

A3/M.01.v1 — witness_sound (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: A3

Theorem witness_sound : forall check bound,
decide check bound = true -> exists w, (w < bound)%nat /\ check w = true.

Occurrences (1): 'formal/IDM_Reduction.v', 'witness_sound'

A3/M.02.v1 — witness_complete (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: A3

```
Theorem witness_complete : forall check bound w,  
(w < bound)%nat -> check w = true -> decide check bound = true.
```

Occurrences (1): ‘formal/IDM_Reduction.v’, ‘witness_complete’

A3/M.03.v1 — decide_dec (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: A3

```
Theorem decide_dec : forall check bound,  
{decide check bound = true} + {decide check bound = false}.
```

Occurrences (1): ‘formal/IDM_Reduction.v’, ‘decide_dec’

A3/M.04.v1 — witness_composite_sound (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: A3

```
Theorem witness_composite_sound : forall n,  
decide (divides_check n) n = true -> composite n.
```

Occurrences (1): ‘formal/IDM_Reduction.v’, ‘witness_composite_sound’

A3/M.05.v1 — composite_has_factor (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: A3

```
Theorem composite_has_factor : forall n,  
composite n -> exists d, Nat.divide d n /\ d <> 1%nat /\ d <> n.
```

Occurrences (1): ‘formal/IDM_Reduction.v’, ‘composite_has_factor’

A3/M.06.v1 — decide_reflect (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: A3

```
Theorem decide_reflect : forall check bound,  
decide check bound = true <-> exists w, (w < bound)%nat /\ check w = true.
```

Occurrences (1): ‘formal/IDM_Reduction.v’, ‘decide_reflect’

A3/M.07.v1 — witness_power_sound (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: A3

```
Theorem witness_power_sound : forall n k,  
decide (fun w => power_check n k w) (S n) = true -> exists w, (w ^ k = n)%nat.
```

Occurrences (1): ‘formal/IDM_Reduction.v’, ‘witness_power_sound’

A3/M.08.v1 — witness_qr_sound (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: A3

```
Theorem witness_qr_sound : forall a p,  
decide (fun w => qr_check a p w) p = true -> exists w, ((w * w) mod p = a mod p)%nat.
```

Occurrences (1): ‘formal/IDM_Reduction.v’, ‘witness_qr_sound’

A3/M.09.v1 — witness_dlog_sound (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: A3

```
Theorem witness_dlog_sound : forall g h p,
  decide (fun w => dlog_check g h p w) p = true -> exists w, ((g ^ w) mod p = h mod p)%nat.
```

Occurrences (1): 'formal/IDM_Reduction.v', 'witness_dlog_sound'

A3/M.10.v1 — sat_reduces_to_decision (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: A3

```
Theorem sat_reduces_to_decision : forall f n,
  decide (fun a => cnf_sat a f) (2 ^ n) = true -> exists a, cnf_sat a f = true.
```

Occurrences (1): 'formal/IDM_Reduction.v', 'sat_reduces_to_decision'

A3/M.11.v1 — sat_model_sound (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: A3

```
Theorem sat_model_sound : forall a f,
  cnf_sat a f = true -> forall cl, In cl f -> clause_sat a cl = true.
```

Occurrences (1): 'formal/IDM_Reduction.v', 'sat_model_sound'

A3/M.12.v1 — tautology_sound (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: A3

```
Theorem tautology_sound : forall check bound,
  all_rows check bound = true -> forall w, (w < bound)%nat -> check w = true.
```

Occurrences (1): 'formal/IDM_Reduction.v', 'tautology_sound'

A3/M.13.v1 — cnf_tautology_sound (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: A3

```
Theorem cnf_tautology_sound : forall f n,
  all_rows (fun a => cnf_sat a f) (2 ^ n) = true -> forall a, (a < 2 ^ n)%nat -> cnf_sat a
  f = true.
```

Occurrences (1): 'formal/IDM_Reduction.v', 'cnf_tautology_sound'

A3/M.14.v1 — witness_crt_sound (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: A3

```
Theorem witness_crt_sound : forall rs ms bound,
  decide (crt_check rs ms) bound = true ->
  exists x, forall r m, In (r, m) (combine rs ms) -> (x mod m = r mod m)%nat.
```

Occurrences (1): 'formal/IDM_Reduction.v', 'witness_crt_sound'

593 CMC — CMC (Causal-Memory Closure) – founder-level bridge axiom (root extension)

domain: — • tier: Ax • status: current • coq: axioms • parents: none

```
Axiom cmc_bridge_axiom : CMC_Bridge_Obligation.
```

CMC/M.01.v1 — decomposed_bridge_obligation (Theorem)

domain: M • tier: Ax • status: current • coq: axioms • parents: CMC

```
Theorem decomposed_bridge_obligation :  
  CMC_Bridge_Obligation.
```

Occurrences (1): 'coq/solver-arc/formal/CMC_Bridge_Decomposition.v', 'decomposed_bridge_obligation'

CMC/M.02.v1 — decomposed_no_refuter (Theorem)

domain: M • tier: Ax • status: current • coq: axioms • parents: CMC

```
Theorem decomposed_no_refuter :  
  forall g : TransportReadout,  
    CMC_Refuter_Burden g -> False.
```

Occurrences (1): 'coq/solver-arc/formal/CMC_Bridge_Decomposition.v', 'decomposed_no_refuter'

CMC/M.03.v1 — closure_free_implies_no_named_closure (Theorem)

domain: M • tier: Th_coqc • status: current • coq: closed • parents: CMC

```
Theorem closure_free_implies_no_named_closure :  
  forall g : TransportReadout,  
    ClosureFree g -> NoNamedClosure g.
```

Occurrences (1): 'coq/solver-arc/formal/CMC_ClosureFree_Exhaustive.v', 'closure_free_implies_no_named_closure'

CMC/M.04.v1 — no_named_closure_implies_closure_free (Theorem)

domain: M • tier: Th_coqc • status: current • coq: closed • parents: CMC

```
Theorem no_named_closure_implies_closure_free :  
  forall g : TransportReadout,  
    NoNamedClosure g -> ClosureFree g.
```

Occurrences (1): 'coq/solver-arc/formal/CMC_ClosureFree_Exhaustive.v', 'no_named_closure_implies_closure_free'

CMC/M.05.v1 — closure_free_iff_no_named_closure (Theorem)

domain: M • tier: Th_coqc • status: current • coq: closed • parents: CMC

```
Theorem closure_free_iff_no_named_closure :  
  forall g : TransportReadout,  
    ClosureFree g <-> NoNamedClosure g.
```

Occurrences (1): 'coq/solver-arc/formal/CMC_ClosureFree_Exhaustive.v', 'closure_free_iff_no_named_closure'

CMC/M.06.v1 — refuter_burden_expands_to_named_absence (Theorem)

domain: M • tier: Th_coqc • status: current • coq: closed • parents: CMC

```
Theorem refuter_burden_expands_to_named_absence :  
  forall g : TransportReadout,  
    CMC_Refuter_Burden g ->  
    CMC_TargetClass g /\ NoNamedClosure g.
```

Occurrences (1): 'coq/solver-arc/formal/CMC_ClosureFree_Exhaustive.v', 'refuter_burden_expands_to_named_absence'

CMC/M.07.v1 — independent_exhaustion_blocks_refuter (Theorem)

domain: M • tier: Th_coqc • status: current • coq: closed • parents: CMC

```
Theorem independent_exhaustion_blocks_refuter :  
forall s : IndependentTransportSignature,  
ClosureExhaustion_independent s ->  
CMC_Refuter_Burden_independent s ->  
False.
```

Occurrences (1): ‘coq/solver-arc/formal/CMC_Independent_Definitions.v’, ‘independent_exhaustion_blocks_refuter’

CMC/M.08.v1 — retained_definition_not_closure_definition (Theorem)

domain: M • tier: Th_coqc • status: current • coq: closed • parents: CMC

```
Theorem retained_definition_not_closure_definition :  
RetainedDiffusive_independent retained_without_closure_signature /\  
ClosureFree_independent retained_without_closure_signature.
```

Occurrences (1): ‘coq/solver-arc/formal/CMC_Independent_Definitions.v’, ‘retained_definition_not_closure_definition’

CMC/M.09.v1 — finite_speed_definition_not_closure_definition (Theorem)

domain: M • tier: Th_coqc • status: current • coq: closed • parents: CMC

```
Theorem finite_speed_definition_not_closure_definition :  
IntrinsicFiniteSpeed_independent finite_speed_without_closure_signature /\  
ClosureFree_independent finite_speed_without_closure_signature.
```

Occurrences (1): ‘coq/solver-arc/formal/CMC_Independent_Definitions.v’, ‘finite_speed_definition_not_closure_definition’

CMC/M.10.v1 — cattaneo_independent_target_has_closure (Theorem)

domain: M • tier: Th_coqc • status: current • coq: closed • parents: CMC

```
Theorem cattaneo_independent_target_has_closure :  
CMC_TargetClass_independent cattaneo_independent_signature /\  
NonzeroClosure_independent cattaneo_independent_signature /\  
~ CMC_Refuter_Burden_independent cattaneo_independent_signature.
```

Occurrences (1): ‘coq/solver-arc/formal/CMC_Independent_Definitions.v’, ‘cattaneo_independent_target_has_closure’

CMC/M.11.v1 — fourier_independent_is_not_target_class (Theorem)

domain: M • tier: Th_coqc • status: current • coq: closed • parents: CMC

```
Theorem fourier_independent_is_not_target_class :  
~ CMC_TargetClass_independent fourier_independent_signature.
```

Occurrences (1): ‘coq/solver-arc/formal/CMC_Independent_Definitions.v’, ‘fourier_independent_is_not_target_class’

CMC/M.12.v1 — class_witness_has_nonzero_closure (Theorem)

domain: M • tier: Th_coqc • status: current • coq: closed • parents: CMC

```
Theorem class_witness_has_nonzero_closure :  
forall (c : WitnessModelClass) (g : TransportReadout),  
ClassWitness c g -> NonzeroClosure g.
```

Occurrences (1): ‘coq/solver-arc/formal/CMC_ModelClass_Witnesses.v’, ‘class_witness_has_nonzero_closure’

CMC/M.13.v1 — class_witness_not_closure_free (Theorem)

domain: M • tier: Th_coqc • status: current • coq: closed • parents: CMC

```
Theorem class_witness_not_closure_free :  
forall (c : WitnessModelClass) (g : TransportReadout),  
ClassWitness c g -> ~ ClosureFree g.
```

Occurrences (1): 'coq/solver-arc/formal/CMC_ModelClass_Witnesses.v', 'class_witness_not_closure_free'

CMC/M.14.v1 — class_witness_blocks_refuter (Theorem)

domain: M • tier: Th_coqc • status: current • coq: closed • parents: CMC

```
Theorem class_witness_blocks_refuter :  
forall (c : WitnessModelClass) (g : TransportReadout),  
ClassWitness c g -> CMC_Refuter_Burden g -> False.
```

Occurrences (1): 'coq/solver-arc/formal/CMC_ModelClass_Witnesses.v', 'class_witness_blocks_refuter'

CMC/M.15.v1 — fourier_memoryless_face_not_target (Theorem)

domain: M • tier: Th_coqc • status: current • coq: closed • parents: CMC

```
Theorem fourier_memoryless_face_not_target :  
forall g : TransportReadout,  
FourierMemorylessFace g -> ~ CMC_TargetClass g.
```

Occurrences (1): 'coq/solver-arc/formal/CMC_ModelClass_Witnesses.v', 'fourier_memoryless_face_not_target'

CMC/M.16.v1 — nonzero_closure_not_closure_free (Theorem)

domain: M • tier: Th_coqc • status: current • coq: closed • parents: CMC

```
Theorem nonzero_closure_not_closure_free :  
forall g : TransportReadout,  
NonzeroClosure g -> ~ ClosureFree g.
```

Occurrences (1): 'coq/solver-arc/formal/CMC_TargetClass_Definitions.v', 'nonzero_closure_not_closure_free'

CMC/M.17.v1 — bridge_obligation_blocks_refuter (Theorem)

domain: M • tier: Th_coqc • status: current • coq: closed • parents: CMC

```
Theorem bridge_obligation_blocks_refuter :  
CMC_Bridge_Obligation ->  
forall g : TransportReadout,  
CMC_Refuter_Burden g -> False.
```

Occurrences (1): 'coq/solver-arc/formal/CMC_TargetClass_Definitions.v', 'bridge_obligation_blocks_refuter'

CMC/M.18.v1 — cmc_no_refuter_under_axioms (Theorem)

domain: M • tier: Ax • status: current • coq: axioms • parents: CMC

```
Theorem cmc_no_refuter_under_axioms :  
forall g : TransportReadout,  
CMC_Refuter_Burden g -> False.
```

Occurrences (1): 'coq/solver-arc/formal/CMC_TargetClass_Definitions.v', 'cmc_no_refuter_under_axioms'

CMC/P.01.v1 — Retention_Lemma_certified (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: CMC}$

```
Theorem Retention_Lemma_certified :  
forall r : CertifiedPhysicalReadout,  
retained_diffusive (certified_gamma r) ->  
exists k : CarrierKind,  
RetentionCarrier k /\ CertifiedCarrierPresent r k.
```

Occurrences (1): ‘coq/solver-arc/formal/CMC_PhysicsClass_Instances.v’, ‘Retention_Lemma_certified’

CMC/P.02.v1 — Finite_Speed_Lemma_certified (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: CMC}$

```
Theorem Finite_Speed_Lemma_certified :  
forall r : CertifiedPhysicalReadout,  
intrinsic_finite_speed (certified_gamma r) ->  
exists k : CarrierKind,  
FiniteSpeedCarrier k /\ CertifiedCarrierPresent r k.
```

Occurrences (1): ‘coq/solver-arc/formal/CMC_PhysicsClass_Instances.v’, ‘Finite_Speed_Lemma_certified’

CMC/P.03.v1 — Closure_Exhaustion_Lemma_certified (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: CMC}$

```
Theorem Closure_Exhaustion_Lemma_certified :  
forall r : CertifiedPhysicalReadout,  
forall kr ks : CarrierKind,  
RetentionCarrier kr ->  
FiniteSpeedCarrier ks ->  
CertifiedCarrierPresent r kr ->  
CertifiedCarrierPresent r ks ->  
NonzeroClosure (certified_gamma r).
```

Occurrences (1): ‘coq/solver-arc/formal/CMC_PhysicsClass_Instances.v’, ‘Closure_Exhaustion_Lemma_certified’

CMC/P.04.v1 — certified_physical_readout_has_nonzero_closure (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: CMC}$

```
Theorem certified_physical_readout_has_nonzero_closure :  
forall r : CertifiedPhysicalReadout,  
NonzeroClosure (certified_gamma r).
```

Occurrences (1): ‘coq/solver-arc/formal/CMC_PhysicsClass_Instances.v’, ‘certified_physical_readout_has_nonzero_closure’

CMC/P.05.v1 — certified_physical_readout_blocks_refuter (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: CMC}$

```
Theorem certified_physical_readout_blocks_refuter :  
forall r : CertifiedPhysicalReadout,  
CMC_Refuter_Burden (certified_gamma r) -> False.
```

Occurrences (1): ‘coq/solver-arc/formal/CMC_PhysicsClass_Instances.v’, ‘certified_physical_readout_blocks_refuter’

CMC/P.06.v1 — cattaneo_telegraph_closure_witness (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: CMC}$

```
Theorem cattaneo_telegraph_closure_witness :  
forall r : CertifiedPhysicalReadout,  
CattaneoTelegraphCertified r ->  
closure_form_present (certified_gamma r) AuxiliaryFluxState.
```

Occurrences (1): ‘coq/solver-arc/formal/CMC_PhysicsClass_Instances.v’, ‘cattaneo_telegraph_closure_witness’

CMC/P.07.v1 — cattaneo_telegraph_blocks_refuter (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: CMC}$

```
Theorem cattaneo_telegraph_blocks_refuter :  
forall r : CertifiedPhysicalReadout,  
CattaneoTelegraphCertified r ->  
CMC_Refuter_Burden (certified_gamma r) -> False.
```

Occurrences (1): 'coq/solver-arc/formal/CMC_PhysicsClass_Instances.v', 'cattaneo_telegraph_blocks_refuter'

CMC/P.08.v1 — kinetic_transport_closure_witness (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: CMC}$

```
Theorem kinetic_transport_closure_witness :  
forall r : CertifiedPhysicalReadout,  
KineticTransportCertified r ->  
closure_form_present (certified_gamma r) KineticHiddenState.
```

Occurrences (1): 'coq/solver-arc/formal/CMC_PhysicsClass_Instances.v', 'kinetic_transport_closure_witness'

CMC/P.09.v1 — kinetic_transport_blocks_refuter (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: CMC}$

```
Theorem kinetic_transport_blocks_refuter :  
forall r : CertifiedPhysicalReadout,  
KineticTransportCertified r ->  
CMC_Refuter_Burden (certified_gamma r) -> False.
```

Occurrences (1): 'coq/solver-arc/formal/CMC_PhysicsClass_Instances.v', 'kinetic_transport_blocks_refuter'

CMC/P.10.v1 — flux_limited_diffusion_closure_witness (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: CMC}$

```
Theorem flux_limited_diffusion_closure_witness :  
forall r : CertifiedPhysicalReadout,  
FluxLimitedDiffusionCertified r ->  
closure_form_present (certified_gamma r) NonlinearSupportGenerator.
```

Occurrences (1): 'coq/solver-arc/formal/CMC_PhysicsClass_Instances.v', 'flux_limited_diffusion_closure_witness'

CMC/P.11.v1 — flux_limited_diffusion_blocks_refuter (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: CMC}$

```
Theorem flux_limited_diffusion_blocks_refuter :  
forall r : CertifiedPhysicalReadout,  
FluxLimitedDiffusionCertified r ->  
CMC_Refuter_Burden (certified_gamma r) -> False.
```

Occurrences (1): 'coq/solver-arc/formal/CMC_PhysicsClass_Instances.v', 'flux_limited_diffusion_blocks_refuter'

CMC/P.12.v1 — wave_bounded_propagator_closure_witness (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: CMC}$

```
Theorem wave_bounded_propagator_closure_witness :  
forall r : CertifiedPhysicalReadout,  
WaveBoundedPropagatorCertified r ->  
closure_form_present (certified_gamma r) BoundedPropagator.
```

Occurrences (1): 'coq/solver-arc/formal/CMC_PhysicsClass_Instances.v', 'wave_bounded_propagator_closure_witness'

CMC/P.13.v1 — wave_bounded_propagator_blocks_refuter (Theorem)

domain: $P \bullet$ tier: Th_coqc • status: current • coq: closed • parents: CMC

```
Theorem wave_bounded_propagator_blocks_refuter :  
forall r : CertifiedPhysicalReadout,  
WaveBoundedPropagatorCertified r ->  
CMC_Refuter_Burden (certified_gamma r) -> False.
```

Occurrences (1): ‘coq/solver-arc/formal/CMC_PhysicsClass_Instances.v’, ‘wave_bounded_propagator_blocks_refuter’

CMC/P.14.v1 — fourier_heat_singular_memoryless_face_not_finite_speed_target (Theorem)

domain: $P \bullet$ tier: Th_coqc • status: current • coq: closed • parents: CMC

```
Theorem fourier_heat_singular_memoryless_face_not_finite_speed_target :  
forall g : TransportReadout,  
FourierHeatSingularMemorylessFace g -> ~ CMC_TargetClass g.
```

Occurrences (1): ‘coq/solver-arc/formal/CMC_PhysicsClass_Instances.v’, ‘fourier_heat_singular_memoryless_face_not_finite_speed_target’

CMC/P.15.v1 — fourier_heat_singular_memoryless_face_not_refuter (Theorem)

domain: $P \bullet$ tier: Th_coqc • status: current • coq: closed • parents: CMC

```
Theorem fourier_heat_singular_memoryless_face_not_refuter :  
forall g : TransportReadout,  
FourierHeatSingularMemorylessFace g -> ~ CMC_Refuter_Burden g.
```

Occurrences (1): ‘coq/solver-arc/formal/CMC_PhysicsClass_Instances.v’, ‘fourier_heat_singular_memoryless_face_not_refuter’

594 D — The naturals D (the retained-difference engine)

domain: — • tier: Th_coqc • status: current • coq: root_layer_unwired • parents: none

The engine ‘D’ is generated by RD1-RD9. ‘D’ is a commutative semiring (Th 2.2); $(D, <)$ is a total, well-ordered order (Th 2.3); $\text{toNat} : D \rightarrow \mathbb{N}$ is a semiring-and-order isomorphism $D \cong \mathbb{N}$ (Th 2.4); $\neg \text{exists } z, 0 < z < \text{succ } 0$ - the discrete floor (Th 2.9, Pr 2.1).

D/M.01.v1 — pigeonhole_bits_needed (Theorem)

domain: $M \bullet$ tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Theorem pigeonhole_bits_needed :  
forall (l : list (list bool)) (rec : list bool -> list bool) (L : nat),  
NoDup l ->  
2 ^ L <= length l ->  
(forall x y, In x l -> In y l -> rec x = rec y -> x = y) ->  
exists x, In x l /\ L <= length (rec x).
```

Occurrences (1): ‘formal/IDM_ApproxCount.v’, ‘pigeonhole_bits_needed’

D/M.02.v1 — weight_cons (Lemma)

domain: $M \bullet$ tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Lemma weight_cons : forall x t, weight (x :: t) = b2n x + weight t.
```

Occurrences (1): ‘formal/IDM_ApproxCount.v’, ‘weight_cons’

D/M.03.v1 — weight_app (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

Lemma weight_app : forall a b, weight (a ++ b) = weight a + weight b.

Occurrences (1): 'formal/IDM_ApproxCount.v', 'weight_app'

D/M.04.v1 — weight_repeat_true (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

Lemma weight_repeat_true : forall k, weight (repeat true k) = k.

Occurrences (1): 'formal/IDM_ApproxCount.v', 'weight_repeat_true'

D/M.05.v1 — weight_repeat_false (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

Lemma weight_repeat_false : forall k, weight (repeat false k) = 0.

Occurrences (1): 'formal/IDM_ApproxCount.v', 'weight_repeat_false'

D/M.06.v1 — fam_elem_weight (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

Lemma fam_elem_weight : forall n r k, weight (fam_elem n r k) = k * (2 * r + 1).

Occurrences (1): 'formal/IDM_ApproxCount.v', 'fam_elem_weight'

D/M.07.v1 — k_bound (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

Lemma k_bound : forall n r k, k <= n / (2 * r + 1) -> k * (2 * r + 1) <= n.

Occurrences (1): 'formal/IDM_ApproxCount.v', 'k_bound'

D/M.08.v1 — fam_elem_length (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

Lemma fam_elem_length : forall n r k, k <= n / (2 * r + 1) -> length (fam_elem n r k) = n.

Occurrences (1): 'formal/IDM_ApproxCount.v', 'fam_elem_length'

D/M.09.v1 — fam_length (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

Lemma fam_length : forall n r, length (fam n r) = S (n / (2 * r + 1)).

Occurrences (1): 'formal/IDM_ApproxCount.v', 'fam_length'

D/M.10.v1 — fam_in (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

Lemma fam_in : forall n r x,
In x (fam n r) -> exists k, k <= n / (2 * r + 1) /\ x = fam_elem n r k.

Occurrences (1): 'formal/IDM_ApproxCount.v', 'fam_in'

D/M.11.v1 — fam_nodup (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

Lemma fam_nodup : forall n r, NoDup (fam n r).

Occurrences (1): 'formal/IDM_ApproxCount.v', 'fam_nodup'

D/M.12.v1 — r_correct_far_apart_False (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Lemma r_correct_far_apart_False :  
forall (val : list bool -> nat) (r : nat) (bs cs : list bool),  
r_correct val r ->  
weight bs + 2 * r < weight cs ->  
val bs = val cs ->  
False.
```

Occurrences (1): 'formal/IDM_ApproxCount.v', 'r_correct_far_apart_False'

D/M.13.v1 — approx_count_deferred_lower_bound (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Theorem approx_count_deferred_lower_bound :  
forall (n r L : nat) (rec : list bool -> list bool) (val : list bool -> nat),  
2 ^ L <= S (n / (2 * r + 1)) ->  
(forall bs cs, rec bs = rec cs -> val bs = val cs) ->  
r_correct val r ->  
exists bs, In bs (fam n r) /\ L <= length (rec bs).
```

Occurrences (1): 'formal/IDM_ApproxCount.v', 'approx_count_deferred_lower_bound'

D/M.14.v1 — b2n_inj (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

Lemma b2n_inj : forall x y : bool, b2n x = b2n y -> x = y.

Occurrences (1): 'formal/IDM_DeclarationBound.v', 'b2n_inj'

D/M.15.v1 — bit_extraction_exact (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Theorem bit_extraction_exact :  
forall (i : nat) (x : bool), b2n x = (i + b2n x) - i.
```

Occurrences (1): 'formal/IDM_DeclarationBound.v', 'bit_extraction_exact'

D/M.16.v1 — profile_injective (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Theorem profile_injective :  
forall (i0 : nat) (bs cs : list bool),  
sturm_profile i0 bs = sturm_profile i0 cs -> bs = cs.
```

Occurrences (1): 'formal/IDM_DeclarationBound.v', 'profile_injective'

D/M.17.v1 — bcube_length (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

Lemma bcube_length : forall n, length (bcube n) = 2 ^ n.

Occurrences (1): 'formal/IDM_DeclarationBound.v', 'bcube_length'

D/M.18.v1 — two_pow_pos (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Lemma two_pow_pos : forall k, 1 <= 2 ^ k.
```

Occurrences (1): ‘formal/IDM_DeclarationBound.v’, ‘two_pow_pos’

D/M.19.v1 — bcube_all_len (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Lemma bcube_all_len : forall n l, In l (bcube n) -> length l = n.
```

Occurrences (1): ‘formal/IDM_DeclarationBound.v’, ‘bcube_all_len’

D/M.20.v1 — bcube_complete (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Lemma bcube_complete : forall l : list bool, In l (bcube (length l)).
```

Occurrences (1): ‘formal/IDM_DeclarationBound.v’, ‘bcube_complete’

D/M.21.v1 — bcube_halves_disjoint (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Lemma bcube_halves_disjoint :  
forall (k : nat) (l : list bool),  
In l (map (cons true) (bcube k)) ->  
In l (map (cons false) (bcube k)) -> False.
```

Occurrences (1): ‘formal/IDM_DeclarationBound.v’, ‘bcube_halves_disjoint’

D/M.22.v1 — NoDup_app_disjoint (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Lemma NoDup_app_disjoint :  
forall (A : Type) (l1 l2 : list A),  
NoDup l1 -> NoDup l2 ->  
(forall x, In x l1 -> In x l2 -> False) ->  
NoDup (l1 ++ l2).
```

Occurrences (1): ‘formal/IDM_DeclarationBound.v’, ‘NoDup_app_disjoint’

D/M.23.v1 — NoDup_map_cons (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Lemma NoDup_map_cons :  
forall (a : bool) (l : list (list bool)),  
NoDup l -> NoDup (map (cons a) l).
```

Occurrences (1): ‘formal/IDM_DeclarationBound.v’, ‘NoDup_map_cons’

D/M.24.v1 — bcube_nodup (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Lemma bcube_nodup : forall n, NoDup (bcube n).
```

Occurrences (1): ‘formal/IDM_DeclarationBound.v’, ‘bcube_nodup’

D/M.25.v1 — short_records_length (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

Lemma short_records_length : forall n, length (short_records n) = $2^n - 1$.

Occurrences (1): 'formal/IDM_DeclarationBound.v', 'short_records_length'

D/M.26.v1 — short_records_complete (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

Lemma short_records_complete :
forall (n : nat) (l : list bool), length l < n -> In l (short_records n).

Occurrences (1): 'formal/IDM_DeclarationBound.v', 'short_records_complete'

D/M.27.v1 — NoDup_map_inj_on (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

Lemma NoDup_map_inj_on :
forall (A B : Type) (f : A -> B) (l : list A),
NoDup l ->
(forall x y, In x l -> In y l -> f x = f y -> x = y) ->
NoDup (map f l).

Occurrences (1): 'formal/IDM_DeclarationBound.v', 'NoDup_map_inj_on'

D/M.28.v1 — deferred_record_bits (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

Theorem deferred_record_bits :
forall (n : nat) (rec : list bool -> list bool),
(forall bs cs, In bs (bcube n) -> In cs (bcube n) -> rec bs = rec cs -> bs = cs) ->
exists bs, In bs (bcube n) /\ n <= length (rec bs).

Occurrences (1): 'formal/IDM_DeclarationBound.v', 'deferred_record_bits'

D/M.29.v1 — declared_forgets_tail (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

Theorem declared_forgets_tail :
forall (j : nat) (bs cs : list bool),
firstn j bs = firstn j cs -> declared_count j bs = declared_count j cs.

Occurrences (1): 'formal/IDM_DeclarationBound.v', 'declared_forgets_tail'

D/M.30.v1 — declaration_separation (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

Theorem declaration_separation :
forall (n : nat),
(forall rec : list bool -> list bool,
(forall bs cs, In bs (bcube n) -> In cs (bcube n) -> rec bs = rec cs -> bs = cs) ->
exists bs, In bs (bcube n) /\ n <= length (rec bs))
/\
(forall (j : nat) (bs cs : list bool),
firstn j bs = firstn j cs -> declared_count j bs = declared_count j cs).

Occurrences (1): 'formal/IDM_DeclarationBound.v', 'declaration_separation'

D/M.31.v1 — qary_symbol_injective (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Theorem qary_symbol_injective :  
forall (q a b : nat),  
a < q -> b < q ->  
(forall r, r < q - 1 -> (a <=? r) = (b <=? r)) ->  
a = b.
```

Occurrences (1): 'formal/IDM_DeclarationBound.v', 'qary_symbol_injective'

D/M.32.v1 — flat_map_cons_length (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Lemma flat_map_cons_length :  
forall (q : nat) (C : list (list nat)) (L : list nat),  
length (flat_map (fun s => map (cons s) C) L) = length L * length C.
```

Occurrences (1): 'formal/IDM_DeclarationBound.v', 'flat_map_cons_length'

D/M.33.v1 — qcube_length (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Lemma qcube_length : forall q n, length (qcube q n) = q ^ n.
```

Occurrences (1): 'formal/IDM_DeclarationBound.v', 'qcube_length'

D/M.34.v1 — qcube_all_len (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Lemma qcube_all_len : forall q n l, In l (qcube q n) -> length l = n.
```

Occurrences (1): 'formal/IDM_DeclarationBound.v', 'qcube_all_len'

D/M.35.v1 — equivariant_stabilizer_containment (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Theorem equivariant_stabilizer_containment :  
forall g x, stabX g x -> stabV g (r x).
```

Occurrences (1): 'formal/IDM_EquivariantReadout.v', 'equivariant_stabilizer_containment'

D/M.36.v1 — faithful_stabilizer_equality (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Theorem faithful_stabilizer_equality :  
forall g x, faithful_on x -> (stabV g (r x) <-> stabX g x).
```

Occurrences (1): 'formal/IDM_EquivariantReadout.v', 'faithful_stabilizer_equality'

D/M.37.v1 — fixed_value_reads_equal (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Theorem fixed_value_reads_equal :  
forall g x, stabV g (r x) -> r (actX g x) = r x.
```

Occurrences (1): 'formal/IDM_EquivariantReadout.v', 'fixed_value_reads_equal'

D/M.38.v1 — nondegenerate_value_moves (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Theorem nondegenerate_value_moves :  
forall g x, r (actX g x) <> r x -> ~ stabV g (r x).
```

Occurrences (1): ‘formal/IDM_EquivariantReadout.v’, ‘nondegenerate_value_moves’

D/M.39.v1 — finite_yoneda (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Theorem finite_yoneda :  
forall (A B : Type) (f g : A -> B) (dom : list A),  
(forall x, In x dom -> f x = g x) ->  
map f dom = map g dom.
```

Occurrences (1): ‘formal/IDM_FiniteWitnesses.v’, ‘finite_yoneda’

D/M.40.v1 — pigeonhole (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Theorem pigeonhole :  
forall (l : list nat) (n : nat),  
(forall x, In x l -> x < n) ->  
NoDup l ->  
length l <= n.
```

Occurrences (1): ‘formal/IDM_FiniteWitnesses.v’, ‘pigeonhole’

D/M.41.v1 — semiring_distrib (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Theorem semiring_distrib :  
forall a b c : nat, a * (b + c) = a * b + a * c.
```

Occurrences (1): ‘formal/IDM_FiniteWitnesses.v’, ‘semiring_distrib’

D/M.42.v1 — no_infinite_readout (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Theorem no_infinite_readout :  
forall (A : Type) (l : list A), exists n : nat, length l = n.
```

Occurrences (1): ‘formal/IDM_FiniteWitnesses2.v’, ‘no_infinite_readout’

D/M.43.v1 — tape_count_succ (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Theorem tape_count_succ :  
forall n : nat, tape_count n < tape_count (S n).
```

Occurrences (1): ‘formal/IDM_FiniteWitnesses2.v’, ‘tape_count_succ’

D/M.44.v1 — tape_no_terminal (Corollary)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Corollary tape_no_terminal :  
forall n : nat, exists m, tape_count n < tape_count m.
```

Occurrences (1): ‘formal/IDM_FiniteWitnesses2.v’, ‘tape_no_terminal’

D/M.45.v1 — same_set_same_size (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Theorem same_set_same_size :  
forall (A : Type) (l l' : list A),  
NoDup l -> NoDup l' ->  
(forall x, In x l <-> In x l') ->  
length l = length l'.
```

Occurrences (1): 'formal/IDM_FiniteWitnesses2.v', 'same_set_same_size'

D/M.46.v1 — measure_additive (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Theorem measure_additive :  
forall (X : Type) (A B : list X), length (A ++ B) = length A + length B.
```

Occurrences (1): 'formal/IDM_FiniteWitnesses3.v', 'measure_additive'

D/M.47.v1 — aut_assoc (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Theorem aut_assoc : forall f g h,  
acts_eq (comp_auto (comp_auto f g) h) (comp_auto f (comp_auto g h)).
```

Occurrences (1): 'formal/IDM_FiniteWitnesses3.v', 'aut_assoc'

D/M.48.v1 — aut_id_left (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Theorem aut_id_left : forall f, acts_eq (comp_auto id_auto f) f.
```

Occurrences (1): 'formal/IDM_FiniteWitnesses3.v', 'aut_id_left'

D/M.49.v1 — aut_id_right (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Theorem aut_id_right : forall f, acts_eq (comp_auto f id_auto) f.
```

Occurrences (1): 'formal/IDM_FiniteWitnesses3.v', 'aut_id_right'

D/M.50.v1 — aut_inv_left (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Theorem aut_inv_left : forall f, acts_eq (comp_auto (inv_auto f) f) id_auto.
```

Occurrences (1): 'formal/IDM_FiniteWitnesses3.v', 'aut_inv_left'

D/M.51.v1 — aut_inv_right (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Theorem aut_inv_right : forall f, acts_eq (comp_auto f (inv_auto f)) id_auto.
```

Occurrences (1): 'formal/IDM_FiniteWitnesses3.v', 'aut_inv_right'

D/M.52.v1 — eval_correct (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Theorem eval_correct :  
  forall f dom env, eval dom f env = true <-> Sat dom f env.
```

Occurrences (1): 'formal/IDM_FirstOrder.v', 'eval_correct'

D/M.53.v1 — sat_fo_decidable (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Theorem sat_fo_decidable :  
  forall f dom env, {Sat dom f env} + {~ Sat dom f env}.
```

Occurrences (1): 'formal/IDM_FirstOrder.v', 'sat_fo_decidable'

D/M.54.v1 — models_decidable (Corollary)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Corollary models_decidable :  
  forall dom f, {Models dom f} + {~ Models dom f}.
```

Occurrences (1): 'formal/IDM_FirstOrder.v', 'models_decidable'

D/M.55.v1 — discrete_floor (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Theorem discrete_floor : ~ (exists z : nat, 0 < z /\ z < S 0).
```

Occurrences (1): 'formal/IDM_Genesis.v', 'discrete_floor'

D/M.56.v1 — no_density_at_root (Corollary)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Corollary no_density_at_root : forall z : nat, 0 < z -> ~ (z < 1).
```

Occurrences (1): 'formal/IDM_Genesis.v', 'no_density_at_root'

D/M.57.v1 — sat_dec (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Theorem sat_dec : forall env f, {sat env f = true} + {sat env f = false}.
```

Occurrences (1): 'formal/IDM_Logic.v', 'sat_dec'

D/M.58.v1 — finite_satisfaction_dec (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Theorem finite_satisfaction_dec :  
  forall (f : form) (envs : list (nat -> bool)),  
  {forall e, In e envs -> sat e f = true} +  
  {exists e, In e envs /\ sat e f = false}.
```

Occurrences (1): 'formal/IDM_Logic.v', 'finite_satisfaction_dec'

D/M.59.v1 — neg_involution (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

Theorem neg_involution : forall v, neg (neg v) = v.

Occurrences (1): 'formal/IDM_ReadoutMinimality.v', 'neg_involution'

D/M.60.v1 — neg_fixed_iff (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

Theorem neg_fixed_iff : forall v, neg v = v <-> (v = Sz \ / v = Sbot).

Occurrences (1): 'formal/IDM_ReadoutMinimality.v', 'neg_fixed_iff'

D/M.61.v1 — neg_moves_iff (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

Theorem neg_moves_iff : forall v, neg v <> v <-> (v = Sp \ / v = Sm).

Occurrences (1): 'formal/IDM_ReadoutMinimality.v', 'neg_moves_iff'

D/M.62.v1 — readout_of_selfneg_is_neutral (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

Lemma readout_of_selfneg_is_neutral : neg (r x0) = r x0.

Occurrences (1): 'formal/IDM_ReadoutMinimality.v', 'readout_of_selfneg_is_neutral'

D/M.63.v1 — minimal_three_values (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

Theorem minimal_three_values :
exists a b c : A,
r a <> r b /\ r a <> r c /\ r b <> r c.

Occurrences (1): 'formal/IDM_ReadoutMinimality.v', 'minimal_three_values'

D/M.64.v1 — third_value_is_neutral (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

Theorem third_value_is_neutral : r x0 = Sz \ / r x0 = Sbot.

Occurrences (1): 'formal/IDM_ReadoutMinimality.v', 'third_value_is_neutral'

D/M.65.v1 — neutral_distinct_from_bottom (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

Theorem neutral_distinct_from_bottom : Sz <> Sbot.

Occurrences (1): 'formal/IDM_ReadoutMinimality.v', 'neutral_distinct_from_bottom'

D/M.66.v1 — two_distinct_neutrals (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

Theorem two_distinct_neutrals :
neg Sz = Sz /\ neg Sbot = Sbot /\ Sz <> Sbot.

Occurrences (1): 'formal/IDM_ReadoutMinimality.v', 'two_distinct_neutrals'

D/M.67.v1 — bottom_is_least (Theorem)

domain: M • tier: Th_cocq • status: current • coq: mapped_not_wrapped • parents: D

Theorem bottom_is_least : forall v, sqle Sbot v.

Occurrences (1): ‘formal/IDM_ReadoutMinimality.v’, ‘bottom_is_least’

D/M.68.v1 — bottom_unique (Theorem)

domain: M • tier: Th_cocq • status: current • coq: mapped_not_wrapped • parents: D

Theorem bottom_unique : forall x, (forall v, sqle x v) -> x = Sbot.

Occurrences (1): ‘formal/IDM_ReadoutMinimality.v’, ‘bottom_unique’

D/M.69.v1 — neutral_is_not_below_sign (Theorem)

domain: M • tier: Th_cocq • status: current • coq: mapped_not_wrapped • parents: D

Theorem neutral_is_not_below_sign : ~ sqle Sz Sp.

Occurrences (1): ‘formal/IDM_ReadoutMinimality.v’, ‘neutral_is_not_below_sign’

D/M.70.v1 — neutral_is_not_bottom (Theorem)

domain: M • tier: Th_cocq • status: current • coq: mapped_not_wrapped • parents: D

Theorem neutral_is_not_bottom : Sz <> Sbot /\ ~ (forall v, sqle Sz v).

Occurrences (1): ‘formal/IDM_ReadoutMinimality.v’, ‘neutral_is_not_bottom’

D/M.71.v1 — classify_bot_iff (Theorem)

domain: M • tier: Th_cocq • status: current • coq: mapped_not_wrapped • parents: D

Theorem classify_bot_iff :
forall floor v,
classify floor v = Sbot <-> (~ floor == 0 /\ - floor <= v /\ v <= floor).

Occurrences (1): ‘formal/IDM_ResolvedCount.v’, ‘classify_bot_iff’

D/M.72.v1 — bot_needs_positive_resolution (Theorem)

domain: M • tier: Th_cocq • status: current • coq: mapped_not_wrapped • parents: D

Theorem bot_needs_positive_resolution :
forall floor v, 0 <= floor -> classify floor v = Sbot -> 0 < floor.

Occurrences (1): ‘formal/IDM_ResolvedCount.v’, ‘bot_needs_positive_resolution’

D/M.73.v1 — classify_zero_iff (Theorem)

domain: M • tier: Th_cocq • status: current • coq: mapped_not_wrapped • parents: D

Theorem classify_zero_iff :
forall floor v, classify floor v = Sz <-> (floor == 0 /\ v == 0).

Occurrences (1): ‘formal/IDM_ResolvedCount.v’, ‘classify_zero_iff’

D/M.74.v1 — classify_plus_sound (Theorem)

domain: M • tier: Th_cocq • status: current • coq: mapped_not_wrapped • parents: D

Theorem classify_plus_sound :
forall floor v, 0 <= floor -> classify floor v = Sp -> 0 < v.

Occurrences (1): ‘formal/IDM_ResolvedCount.v’, ‘classify_plus_sound’

D/M.75.v1 — classify_minus_sound (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Theorem classify_minus_sound :  
forall floor v, 0 <= floor -> classify floor v = Sm -> v < 0.
```

Occurrences (1): ‘formal/IDM_ResolvedCount.v’, ‘classify_minus_sound’

D/M.76.v1 — classify_not_bot_is_determinate (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Theorem classify_not_bot_is_determinate :  
forall floor v, classify floor v <> Sbot ->  
classify floor v = Sp \ / classify floor v = Sm \ / classify floor v = Sz.
```

Occurrences (1): ‘formal/IDM_ResolvedCount.v’, ‘classify_not_bot_is_determinate’

D/M.77.v1 — bot_monotone_in_floor (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Theorem bot_monotone_in_floor :  
forall f1 f2 v, 0 <= f1 -> f1 <= f2 ->  
classify f1 v = Sbot -> classify f2 v = Sbot.
```

Occurrences (1): ‘formal/IDM_ResolvedCount.v’, ‘bot_monotone_in_floor’

D/M.78.v1 — count_if_split (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Lemma count_if_split :  
forall (p q r : Q -> bool) vs,  
(forall v, ((if p v then 1 else 0) = (if q v then 1 else 0) + (if r v then 1 else  
0))%nat) ->  
count_if p vs = (count_if q vs + count_if r vs)%nat.
```

Occurrences (1): ‘formal/IDM_ResolvedCount.v’, ‘count_if_split’

D/M.79.v1 — signedfloor_is_certain_plus_unresolved (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Theorem signedfloor_is_certain_plus_unresolved :  
forall floor vs,  
signed_floor_below floor vs = (certain_below floor vs + unresolved floor vs)%nat.
```

Occurrences (1): ‘formal/IDM_ResolvedCount.v’, ‘signedfloor_is_certain_plus_unresolved’

D/M.80.v1 — certain_le_signedfloor (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: D

```
Theorem certain_le_signedfloor :  
forall floor vs, (certain_below floor vs <= signed_floor_below floor vs)%nat.
```

Occurrences (1): ‘formal/IDM_ResolvedCount.v’, ‘certain_le_signedfloor’

595 δ_R — The primitive — a retained difference exists

domain: — • tier: Th_coqc • status: current • coq: root_layer_unwired • parents: EQ-001

δ_R - a retained difference exists. Ontologically primitive; nothing is assumed below it.

delta_R/M.01.v1 — sing_eq_iff (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: delta_R

Lemma sing_eq_iff : forall a c, seteq (sing a) (sing c) <-> a = c.

Occurrences (1): 'formal/IDM_FiniteWitnesses.v', 'sing_eq_iff'

delta_R/M.02.v1 — kuratowski_pair_inj (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: delta_R

Theorem kuratowski_pair_inj :
forall a b c d : nat,
seteq (fst (kpair a b)) (fst (kpair c d)) ->
seteq (snd (kpair a b)) (snd (kpair c d)) ->
a = c /\ b = d.

Occurrences (1): 'formal/IDM_FiniteWitnesses.v', 'kuratowski_pair_inj'

delta_R/M.03.v1 — primordial_difference_exists (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: delta_R

Theorem primordial_difference_exists : exists a b : nat, a <> b.

Occurrences (1): 'formal/IDM_Genesis.v', 'primordial_difference_exists'

delta_R/M.04.v1 — rdl_non_explosion (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: delta_R

Theorem rdl_non_explosion :
exists vp vq : V4,
designated (conj4 vp (neg4 vp)) = true /\ designated vq = false.

Occurrences (1): 'formal/IDM_Logic.v', 'rdl_non_explosion'

delta_R/M.05.v1 — classical_would_explode (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: delta_R

Theorem classical_would_explode :
forall x : V4, x = VT \/ x = VF ->
designated (conj4 x (neg4 x)) = false.

Occurrences (1): 'formal/IDM_Logic.v', 'classical_would_explode'

delta_R/M.06.v1 — lookup_graph_in (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: delta_R

Lemma lookup_graph_in :
forall f dom a, In a dom -> lookup (graph f dom) a = Some (f a).

Occurrences (1): 'formal/IDM_SetsFunctions.v', 'lookup_graph_in'

delta_R/M.07.v1 — graph_total (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: delta_R

Theorem graph_total :
forall f dom a, In a dom -> exists b, maps_to (graph f dom) a b.

Occurrences (1): 'formal/IDM_SetsFunctions.v', 'graph_total'

delta_R/M.08.v1 — graph_single_valued (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: delta_R

```
Theorem graph_single_valued :  
forall f dom a b1 b2,  
maps_to (graph f dom) a b1 -> maps_to (graph f dom) a b2 -> b1 = b2.
```

Occurrences (1): 'formal/IDM_SetsFunctions.v', 'graph_single_valued'

delta_R/M.09.v1 — graph_faithful (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: delta_R

```
Theorem graph_faithful :  
forall f g dom, graph f dom = graph g dom -> forall a, In a dom -> f a = g a.
```

Occurrences (1): 'formal/IDM_SetsFunctions.v', 'graph_faithful'

delta_R/M.10.v1 — graph_faithful_converse (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: delta_R

```
Theorem graph_faithful_converse :  
forall f g dom, (forall a, In a dom -> f a = g a) -> graph f dom = graph g dom.
```

Occurrences (1): 'formal/IDM_SetsFunctions.v', 'graph_faithful_converse'

delta_R/M.11.v1 — functional_relation_has_map (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: delta_R

```
Theorem functional_relation_has_map :  
forall R dom, is_function_on R dom ->  
exists f : A -> B, forall a, In a dom -> maps_to R a (f a).
```

Occurrences (1): 'formal/IDM_SetsFunctions.v', 'functional_relation_has_map'

596 Keystone — The keystone — $B(\Phi, \Phi) = I(\Phi)$

domain: — • tier: Th_coqc • status: current • coq: root_layer_unwired • parents: EQ-008

Def 5.1 (retained-information density): $I(\Phi) :=$ the distinguishability of neighbouring retained states of the field Φ on the graph. Th 5.1 (keystone, 'Th_coqc', witness 'formal/IDM_Keystone.v: keystone_B_eq_I'): $B(\Phi, \Phi) = I(\Phi)$: in the edge assembly of $L_R = D_W - W$ this is the identity $\Phi^T L_R \Phi = \sum_{\text{edges}} w(\Phi_i - \Phi_j)^2$, proved axiom-free over $\mathbb{Q}$.

Keystone/M.01.v1 — cauchy_schwarz_2 (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: Keystone

```
Theorem cauchy_schwarz_2 :  
forall a b c d : Q,  
(a*c + b*d) * (a*c + b*d) <= (a*a + b*b) * (c*c + d*d).
```

Occurrences (1): 'formal/IDM_FiniteWitnesses3.v', 'cauchy_schwarz_2'

Keystone/M.02.v1 — keystone_edge (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: Keystone

```
Lemma keystone_edge : forall phi e, B_edge phi e == I_edge phi e.
```

Occurrences (1): 'formal/IDM_Keystone.v', 'keystone_edge'

Keystone/M.03.v1 — keystone_B_eq_I (Theorem)

domain: M • tier: Th_coqc • status: *current* • coq: *mapped_not_wrapped* • parents: *Keystone*

```
Theorem keystone_B_eq_I :  
  forall (phi : Phi) (g : list edge), B_form phi g == I_form phi g.
```

Occurrences (1): 'formal/IDM_Keystone.v', 'keystone_B_eq_I'

Keystone/M.04.v1 — Qsq_nonneg (Lemma)

domain: M • tier: Th_coqc • status: *current* • coq: *mapped_not_wrapped* • parents: *Keystone*

```
Lemma Qsq_nonneg : forall x : Q, 0 <= x * x.
```

Occurrences (1): 'formal/IDM_Keystone.v', 'Qsq_nonneg'

Keystone/M.05.v1 — Qadd_nonneg (Lemma)

domain: M • tier: Th_coqc • status: *current* • coq: *mapped_not_wrapped* • parents: *Keystone*

```
Lemma Qadd_nonneg : forall a b : Q, 0 <= a -> 0 <= b -> 0 <= a + b.
```

Occurrences (1): 'formal/IDM_Keystone.v', 'Qadd_nonneg'

Keystone/M.06.v1 — I_edge_nonneg (Lemma)

domain: M • tier: Th_coqc • status: *current* • coq: *mapped_not_wrapped* • parents: *Keystone*

```
Lemma I_edge_nonneg :  
  forall phi e, (let '(_,_,w) := e in 0 <= w) -> 0 <= I_edge phi e.
```

Occurrences (1): 'formal/IDM_Keystone.v', 'I_edge_nonneg'

Keystone/M.07.v1 — keystone_nonneg (Theorem)

domain: M • tier: Th_coqc • status: *current* • coq: *mapped_not_wrapped* • parents: *Keystone*

```
Theorem keystone_nonneg :  
  forall (phi : Phi) (g : list edge),  
  (forall e, In e g -> let '(_,_,w) := e in 0 <= w) ->  
  0 <= B_form phi g.
```

Occurrences (1): 'formal/IDM_Keystone.v', 'keystone_nonneg'

Keystone/M.08.v1 — relaxation_dissipation (Theorem)

domain: M • tier: Th_coqc • status: *current* • coq: *mapped_not_wrapped* • parents: *Keystone*

```
Theorem relaxation_dissipation :  
  forall (phi : Phi) (g : list edge) (tau : Q),  
  0 < tau ->  
  (forall e, In e g -> let '(_,_,w) := e in 0 <= w) ->  
  - ((2 # 1) / tau) * B_form phi g <= 0.
```

Occurrences (1): 'formal/IDM_Keystone.v', 'relaxation_dissipation'

Keystone/P.01.v1 — Qsq_nonneg (Lemma)

domain: P • tier: Th_coqc • status: *current* • coq: *mapped_not_wrapped* • parents: *Keystone*

```
Lemma Qsq_nonneg : forall x : Q, 0 <= x * x.
```

Occurrences (1): 'formal/IDM_Hilbert.v', 'Qsq_nonneg'

Keystone/P.02.v1 — Sum_scale (Lemma)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: mapped_not_wrapped} \bullet \text{parents: Keystone}$

Lemma Sum_scale : forall n c f, Sum n (fun k => c * f k) == c * Sum n f.

Occurrences (1): 'formal/IDM_Hilbert.v', 'Sum_scale'

Keystone/P.03.v1 — Sum_nonneg (Lemma)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: mapped_not_wrapped} \bullet \text{parents: Keystone}$

Lemma Sum_nonneg : forall n f, (forall k, 0 <= f k) -> 0 <= Sum n f.

Occurrences (1): 'formal/IDM_Hilbert.v', 'Sum_nonneg'

Keystone/P.04.v1 — inner_sym (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: mapped_not_wrapped} \bullet \text{parents: Keystone}$

Theorem inner_sym : forall n u v, inner n u v == inner n v u.

Occurrences (1): 'formal/IDM_Hilbert.v', 'inner_sym'

Keystone/P.05.v1 — inner_linear_l (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: mapped_not_wrapped} \bullet \text{parents: Keystone}$

Theorem inner_linear_l : forall n c u1 u2 v,
inner n (fun k => c * u1 k + u2 k) v == c * inner n u1 v + inner n u2 v.

Occurrences (1): 'formal/IDM_Hilbert.v', 'inner_linear_l'

Keystone/P.06.v1 — inner_pos (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: mapped_not_wrapped} \bullet \text{parents: Keystone}$

Theorem inner_pos : forall n v, 0 <= inner n v v.

Occurrences (1): 'formal/IDM_Hilbert.v', 'inner_pos'

Keystone/P.07.v1 — parallelogram Law (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: mapped_not_wrapped} \bullet \text{parents: Keystone}$

Theorem parallelogram_law : forall n u v,
inner n (fun k => u k + v k) (fun k => u k + v k)
+ inner n (fun k => u k - v k) (fun k => u k - v k)
== 2 * inner n u u + 2 * inner n v v.

Occurrences (1): 'formal/IDM_Hilbert.v', 'parallelogram_law'

Keystone/P.08.v1 — pythagoras_orthogonal (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: mapped_not_wrapped} \bullet \text{parents: Keystone}$

Theorem pythagoras_orthogonal : forall n u v,
inner n u v == 0 ->
inner n (fun k => u k + v k) (fun k => u k + v k) == inner n u u + inner n v v.

Occurrences (1): 'formal/IDM_Hilbert.v', 'pythagoras_orthogonal'

Keystone/P.09.v1 — cauchy_schwarz_2 (Theorem)

domain: P • tier: Th_coqc • status: *current* • coq: *mapped_not_wrapped* • parents: *Keystone*

```
Theorem cauchy_schwarz_2 : forall u1 u2 v1 v2 : Q,  
(u1 * v1 + u2 * v2) * (u1 * v1 + u2 * v2)  
<= (u1 * u1 + u2 * u2) * (v1 * v1 + v2 * v2).
```

Occurrences (1): 'formal/IDM_Hilbert.v', 'cauchy_schwarz_2'

Keystone/P.10.v1 — adjoint_involutive (Theorem)

domain: P • tier: Th_coqc • status: *current* • coq: *mapped_not_wrapped* • parents: *Keystone*

```
Theorem adjoint_involutive : forall n A, meq n (adjoint (adjoint A)) A.
```

Occurrences (1): 'formal/IDM_Hilbert.v', 'adjoint_involutive'

Keystone/P.11.v1 — adjoint_of_product (Theorem)

domain: P • tier: Th_coqc • status: *current* • coq: *mapped_not_wrapped* • parents: *Keystone*

```
Theorem adjoint_of_product : forall n A B,  
meq n (adjoint (mmul n A B)) (mmul n (adjoint B) (adjoint A)).
```

Occurrences (1): 'formal/IDM_Hilbert.v', 'adjoint_of_product'

Keystone/P.12.v1 — hermitian_2x2_discriminant_nonneg (Theorem)

domain: P • tier: Th_coqc • status: *current* • coq: *mapped_not_wrapped* • parents: *Keystone*

```
Theorem hermitian_2x2_discriminant_nonneg : forall a b c : Q,  
0 <= (a - c) * (a - c) + 4 * (b * b).
```

Occurrences (1): 'formal/IDM_Hilbert.v', 'hermitian_2x2_discriminant_nonneg'

Keystone/P.13.v1 — hermitian_2x2_gap_is_discriminant (Theorem)

domain: P • tier: Th_coqc • status: *current* • coq: *mapped_not_wrapped* • parents: *Keystone*

```
Theorem hermitian_2x2_gap_is_discriminant : forall a b c lam1 lam2 : Q,  
lam1 + lam2 == a + c ->  
lam1 * lam2 == a * c - b * b ->  
(lam1 - lam2) * (lam1 - lam2) == (a - c) * (a - c) + 4 * (b * b).
```

Occurrences (1): 'formal/IDM_Hilbert.v', 'hermitian_2x2_gap_is_discriminant'

Keystone/P.14.v1 — projection_idempotent (Theorem)

domain: P • tier: Th_coqc • status: *current* • coq: *mapped_not_wrapped* • parents: *Keystone*

```
Theorem projection_idempotent : forall n P, is_projection n P -> meq n (mmul n P P) P.
```

Occurrences (1): 'formal/IDM_Hilbert.v', 'projection_idempotent'

Keystone/P.15.v1 — projection_self_adjoint (Theorem)

domain: P • tier: Th_coqc • status: *current* • coq: *mapped_not_wrapped* • parents: *Keystone*

```
Theorem projection_self_adjoint : forall n P, is_projection n P -> meq n (adjoint P) P.
```

Occurrences (1): 'formal/IDM_Hilbert.v', 'projection_self_adjoint'

Keystone/P.16.v1 — Qsq_nonneg (Lemma)

domain: P • tier: Th_coqc • status: *current* • coq: *mapped_not_wrapped* • parents: *Keystone*

Lemma Qsq_nonneg : forall x : Q, 0 <= x * x.

Occurrences (1): 'formal/IDM_HilbertReadout.v', 'Qsq_nonneg'

Keystone/P.17.v1 — Qadd_nonneg (Lemma)

domain: P • tier: Th_coqc • status: *current* • coq: *mapped_not_wrapped* • parents: *Keystone*

Lemma Qadd_nonneg : forall a b : Q, 0 <= a -> 0 <= b -> 0 <= a + b.

Occurrences (1): 'formal/IDM_HilbertReadout.v', 'Qadd_nonneg'

Keystone/P.18.v1 — partial_energy_nonneg (Theorem)

domain: P • tier: Th_coqc • status: *current* • coq: *mapped_not_wrapped* • parents: *Keystone*

Theorem partial_energy_nonneg : forall xs : list Q, 0 <= partial_energy xs.

Occurrences (1): 'formal/IDM_HilbertReadout.v', 'partial_energy_nonneg'

Keystone/P.19.v1 — partial_energy_app (Theorem)

domain: P • tier: Th_coqc • status: *current* • coq: *mapped_not_wrapped* • parents: *Keystone*

Theorem partial_energy_app :
forall xs ys : list Q,
partial_energy (xs ++ ys) == partial_energy xs + partial_energy ys.

Occurrences (1): 'formal/IDM_HilbertReadout.v', 'partial_energy_app'

Keystone/P.20.v1 — partial_energy_monotone (Theorem)

domain: P • tier: Th_coqc • status: *current* • coq: *mapped_not_wrapped* • parents: *Keystone*

Theorem partial_energy_monotone :
forall (xs : list Q) (x : Q),
partial_energy xs <= partial_energy (xs ++ [x]).

Occurrences (1): 'formal/IDM_HilbertReadout.v', 'partial_energy_monotone'

Keystone/P.21.v1 — weighted_energy_nonneg (Theorem)

domain: P • tier: Th_coqc • status: *current* • coq: *mapped_not_wrapped* • parents: *Keystone*

Theorem weighted_energy_nonneg :
forall wxs : list (Q * Q),
Forall (fun p => 0 <= fst p) wxs ->
0 <= weighted_energy wxs.

Occurrences (1): 'formal/IDM_HilbertReadout.v', 'weighted_energy_nonneg'

Keystone/P.22.v1 — weighted_energy_app (Theorem)

domain: P • tier: Th_coqc • status: *current* • coq: *mapped_not_wrapped* • parents: *Keystone*

Theorem weighted_energy_app :
forall a b : list (Q * Q),
weighted_energy (a ++ b) == weighted_energy a + weighted_energy b.

Occurrences (1): 'formal/IDM_HilbertReadout.v', 'weighted_energy_app'

597 L_R — The relation graph and the operator L_R

domain: — • tier: Th_coqc • status: current • coq: root_layer_unwired • parents: EQ-008

Def 4.3: the retained relation graph; the interval count $N(x,y) = \text{toNat } y - \text{toNat } x - 1$. Th 4.3 (Laplacian mechanics): L_R (=lap/B) is symmetric (B_sym), PSD (B_self_nonneg), its kernel $\supseteq$ constants (lap_kills_constants), linear (lap_linear), equals div-grad (lap_is_div_grad), and satisfies summation-by-parts (sbp).

L_R/M.01.v1 — handshake_lemma (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: L_R

Theorem handshake_lemma :
forall edges : list (nat * nat),
length (endpoints edges) = 2 * length edges.

Occurrences (1): ‘formal/IDM_FiniteWitnesses.v’, ‘handshake_lemma’

L_R/M.02.v1 — no_fibonacci_integer_dim (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: L_R

Theorem no_fibonacci_integer_dim : forall d : Z, d * d <> 1 + d.

Occurrences (1): ‘formal/IDM_FiniteWitnesses3.v’, ‘no_fibonacci_integer_dim’

L_R/M.03.v1 — orient_swap_bc (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: L_R

Lemma orient_swap_bc : forall ax ay bx by_ cx cy,
Orient ax ay cx cy bx by_ == - Orient ax ay bx by_ cx cy.

Occurrences (1): ‘formal/IDM_Geometry.v’, ‘orient_swap_bc’

L_R/M.04.v1 — orient_swap_ab (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: L_R

Lemma orient_swap_ab : forall ax ay bx by_ cx cy,
Orient bx by_ ax ay cx cy == - Orient ax ay bx by_ cx cy.

Occurrences (1): ‘formal/IDM_Geometry.v’, ‘orient_swap_ab’

L_R/M.05.v1 — orient_cyclic (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: L_R

Lemma orient_cyclic : forall ax ay bx by_ cx cy,
Orient ax ay bx by_ cx cy == Orient bx by_ cx cy ax ay.

Occurrences (1): ‘formal/IDM_Geometry.v’, ‘orient_cyclic’

L_R/M.06.v1 — orient_coincident_ab (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: L_R

Lemma orient_coincident_ab : forall ax ay cx cy,
Orient ax ay ax ay cx cy == 0.

Occurrences (1): ‘formal/IDM_Geometry.v’, ‘orient_coincident_ab’

L_R/M.07.v1 — orient_coincident_ac (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: L_R

```
Lemma orient_coincident_ac : forall ax ay bx by_,
  Orient ax ay bx by_ ax ay == 0.
```

Occurrences (1): ‘formal/IDM_Geometry.v’, ‘orient_coincident_ac’

L_R/M.08.v1 — orient_coincident_bc (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: L_R

```
Lemma orient_coincident_bc : forall ax ay bx by_,
  Orient ax ay bx by_ bx by_ == 0.
```

Occurrences (1): ‘formal/IDM_Geometry.v’, ‘orient_coincident_bc’

L_R/M.09.v1 — orient_translation (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: L_R

```
Lemma orient_translation : forall ax ay bx by_ cx cy tx ty,
  Orient (ax + tx) (ay + ty) (bx + tx) (by_ + ty) (cx + tx) (cy + ty)
  == Orient ax ay bx by_ cx cy.
```

Occurrences (1): ‘formal/IDM_Geometry.v’, ‘orient_translation’

L_R/M.10.v1 — orient_scale (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: L_R

```
Lemma orient_scale : forall s ax ay bx by_ cx cy,
  Orient (s*ax) (s*ay) (s*bx) (s*by_) (s*cx) (s*cy)
  == (s * s) * Orient ax ay bx by_ cx cy.
```

Occurrences (1): ‘formal/IDM_Geometry.v’, ‘orient_scale’

L_R/M.11.v1 — orient_collinear_mid (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: L_R

```
Lemma orient_collinear_mid : forall ax ay bx by_,
  Orient ax ay bx by_ ((ax + bx) / (2#1)) ((ay + by_) / (2#1)) == 0.
```

Occurrences (1): ‘formal/IDM_Geometry.v’, ‘orient_collinear_mid’

L_R/M.12.v1 — orient_collinear_aff (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: L_R

```
Lemma orient_collinear_aff : forall ax ay bx by_ t,
  Orient ax ay bx by_ (ax + t*(bx - ax)) (ay + t*(by_ - ay)) == 0.
```

Occurrences (1): ‘formal/IDM_Geometry.v’, ‘orient_collinear_aff’

L_R/M.13.v1 — repeated_event_zero (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: L_R

```
Theorem repeated_event_zero : forall C : Q, C == - C -> C == 0.
```

Occurrences (1): ‘formal/IDM_Harvest.v’, ‘repeated_event_zero’

L_R/M.14.v1 — odd_from_cyclic_closure (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: L_R

```
Theorem odd_from_cyclic_closure :  
  forall k : nat, (k >= 1)%nat -> Nat.Even (k - 1) -> Nat.Odd k.
```

Occurrences (1): ‘formal/IDM_Harvest.v’, ‘odd_from_cyclic_closure’

L_R/M.15.v1 — least_nontrivial_odd_is_three (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: L_R

```
Theorem least_nontrivial_odd_is_three :  
  forall k : nat, (k > 1)%nat -> Nat.Odd k -> (k >= 3)%nat.
```

Occurrences (1): ‘formal/IDM_Harvest.v’, ‘least_nontrivial_odd_is_three’

L_R/M.16.v1 — sym_skew_reconstruct (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: L_R

```
Theorem sym_skew_reconstruct :  
  forall (A : M) i j, A i j == sympart A i j + skewpart A i j.
```

Occurrences (1): ‘formal/IDM_Harvest.v’, ‘sym_skew_reconstruct’

L_R/M.17.v1 — sympart_self_adjoint (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: L_R

```
Theorem sympart_self_adjoint :  
  forall (A : M) i j, sympart A i j == sympart A j i.
```

Occurrences (1): ‘formal/IDM_Harvest.v’, ‘sympart_self_adjoint’

L_R/M.18.v1 — skew_antisym (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: L_R

```
Theorem skew_antisym :  
  forall (A : M) i j, skewpart A i j == - skewpart A j i.
```

Occurrences (1): ‘formal/IDM_Harvest.v’, ‘skew_antisym’

L_R/M.19.v1 — skew_diag_zero (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: L_R

```
Theorem skew_diag_zero :  
  forall (A : M) i, skewpart A i i == 0.
```

Occurrences (1): ‘formal/IDM_Harvest.v’, ‘skew_diag_zero’

L_R/M.20.v1 — laplacian_symmetric (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: L_R

```
Theorem laplacian_symmetric : forall i j, Lap i j == transpose Lap i j.
```

Occurrences (1): ‘formal/IDM_Matrix.v’, ‘laplacian_symmetric’

L_R/M.21.v1 — laplacian_rowsum_zero (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: L_R

Theorem laplacian_rowsum_zero : forall i, (i < n)%nat -> Sum n (fun j => Lap i j) == 0.

Occurrences (1): ‘formal/IDM_Matrix.v’, ‘laplacian_rowsum_zero’

L_R/M.22.v1 — laplacian_ones_in_kernel (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: L_R

Theorem laplacian_ones_in_kernel :
forall i, (i < n)%nat -> Sum n (fun j => Lap i j * 1) == 0.

Occurrences (1): ‘formal/IDM_Matrix.v’, ‘laplacian_ones_in_kernel’

L_R/M.23.v1 — schur_congruence_00 (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: L_R

Theorem schur_congruence_00 : MtDM 0 0 == a.

Occurrences (1): ‘formal/IDM_Schur.v’, ‘schur_congruence_00’

L_R/M.24.v1 — schur_congruence_01 (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: L_R

Theorem schur_congruence_01 : MtDM 0 (S 0) == b.

Occurrences (1): ‘formal/IDM_Schur.v’, ‘schur_congruence_01’

L_R/M.25.v1 — schur_congruence_10 (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: L_R

Theorem schur_congruence_10 : MtDM (S 0) 0 == b.

Occurrences (1): ‘formal/IDM_Schur.v’, ‘schur_congruence_10’

L_R/M.26.v1 — schur_congruence_11 (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: L_R

Theorem schur_congruence_11 : MtDM (S 0) (S 0) == c.

Occurrences (1): ‘formal/IDM_Schur.v’, ‘schur_congruence_11’

L_R/M.27.v1 — diag_inertia_additive (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: L_R

Theorem diag_inertia_additive :
forall p q, diag_neg_count p q = (negb_pivot p + negb_pivot q)%nat.

Occurrences (1): ‘formal/IDM_Schur.v’, ‘diag_inertia_additive’

L_R/M.28.v1 — schur_pivots_are_boundary_and_complement (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: L_R

Theorem schur_pivots_are_boundary_and_complement :
forall a b c : Q, ~ a == 0 ->
diag_neg_count a (schur a b c) = (negb_pivot a + negb_pivot (schur a b c))%nat.

Occurrences (1): ‘formal/IDM_Schur.v’, ‘schur_pivots_are_boundary_and_complement’

598 Q — The rationals Q (field of fractions of Z)

domain: — • *tier:* *Th_coqc* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *none*

Def 3.3: $Q := (Z \times Z_{\neq 0}) / \sim$; value homomorphism into the standard Q. Th 3.4: $(Q, +, \cdot)$ is a field: *qadd_neg*, *qmul_assoc*, *qmul_distrib_l*, and the multiplicative inverse *qmul_inv*.

Q/M.01.v1 — Qsq_nonneg3 (Lemma)

domain: *M* • *tier:* *Th_coqc* • *status:* *current* • *coq:* *mapped_not_wrapped* • *parents:* *Q*

Lemma *Qsq_nonneg3* : forall x : Q, 0 <= x * x.

Occurrences (1): ‘formal/IDM_FiniteWitnesses3.v’, ‘Qsq_nonneg3’

Q/M.02.v1 — Sum_plus (Lemma)

domain: *M* • *tier:* *Th_coqc* • *status:* *current* • *coq:* *mapped_not_wrapped* • *parents:* *Q*

Lemma *Sum_plus* : forall n f g, Sum n (fun k => f k + g k) == Sum n f + Sum n g.

Occurrences (1): ‘formal/IDM_Matrix.v’, ‘Sum_plus’

Q/M.03.v1 — Sum_opp (Lemma)

domain: *M* • *tier:* *Th_coqc* • *status:* *current* • *coq:* *mapped_not_wrapped* • *parents:* *Q*

Lemma *Sum_opp* : forall n f, Sum n (fun k => - f k) == - Sum n f.

Occurrences (1): ‘formal/IDM_Matrix.v’, ‘Sum_opp’

Q/M.04.v1 — Sum_zero (Lemma)

domain: *M* • *tier:* *Th_coqc* • *status:* *current* • *coq:* *mapped_not_wrapped* • *parents:* *Q*

Lemma *Sum_zero* : forall n, Sum n (fun _ => 0) == 0.

Occurrences (1): ‘formal/IDM_Matrix.v’, ‘Sum_zero’

Q/M.05.v1 — Sum_ext (Lemma)

domain: *M* • *tier:* *Th_coqc* • *status:* *current* • *coq:* *mapped_not_wrapped* • *parents:* *Q*

Lemma *Sum_ext* : forall n f g, (forall k, f k == g k) -> Sum n f == Sum n g.

Occurrences (1): ‘formal/IDM_Matrix.v’, ‘Sum_ext’

Q/M.06.v1 — Sum_ext_lt (Lemma)

domain: *M* • *tier:* *Th_coqc* • *status:* *current* • *coq:* *mapped_not_wrapped* • *parents:* *Q*

Lemma *Sum_ext_lt* : forall n f g, (forall k, (k < n)%nat -> f k == g k) -> Sum n f == Sum n g.

Occurrences (1): ‘formal/IDM_Matrix.v’, ‘Sum_ext_lt’

Q/M.07.v1 — Sum_delta (Lemma)

domain: *M* • *tier:* *Th_coqc* • *status:* *current* • *coq:* *mapped_not_wrapped* • *parents:* *Q*

Lemma *Sum_delta* : forall n i f,
(i < n)%nat -> Sum n (fun k => if Nat.eqb i k then f k else 0) == f i.

Occurrences (1): ‘formal/IDM_Matrix.v’, ‘Sum_delta’

Q/M.08.v1 — madd_comm (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: Q

Theorem madd_comm : forall n A B, meq n (madd A B) (madd B A).

Occurrences (1): ‘formal/IDM_Matrix.v’, ‘madd_comm’

Q/M.09.v1 — madd_assoc (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: Q

Theorem madd_assoc : forall n A B C, meq n (madd (madd A B) C) (madd A (madd B C)).

Occurrences (1): ‘formal/IDM_Matrix.v’, ‘madd_assoc’

Q/M.10.v1 — transpose_involutive (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: Q

Theorem transpose_involutive : forall n A, meq n (transpose (transpose A)) A.

Occurrences (1): ‘formal/IDM_Matrix.v’, ‘transpose_involutive’

Q/M.11.v1 — transpose_mmul (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: Q

Theorem transpose_mmul : forall n A B, meq n (transpose (mmul n A B)) (mmul n (transpose B) (transpose A)).

Occurrences (1): ‘formal/IDM_Matrix.v’, ‘transpose_mmul’

Q/M.12.v1 — mid_left (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: Q

Theorem mid_left : forall n A i j, (i < n)%nat -> mmul n mid A i j == A i j.

Occurrences (1): ‘formal/IDM_Matrix.v’, ‘mid_left’

Q/M.13.v1 — twirl_image_scalar (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: Q

Theorem twirl_image_scalar :
forall n A, exists c, forall i j, twirl n A i j = scalarM c i j.

Occurrences (1): ‘formal/IDM_Matrix.v’, ‘twirl_image_scalar’

Q/M.14.v1 — Sum_const (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: Q

Lemma Sum_const : forall n c, Sum n (fun _ => c) == inject_Z (Z.of_nat n) * c.

Occurrences (1): ‘formal/IDM_Matrix.v’, ‘Sum_const’

Q/M.15.v1 — trace_scalarM (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: Q

Lemma trace_scalarM : forall n c, trace n (scalarM c) == inject_Z (Z.of_nat n) * c.

Occurrences (1): ‘formal/IDM_Matrix.v’, ‘trace_scalarM’

Q/M.16.v1 — inject_nat_nonzero (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: Q

Lemma inject_nat_nonzero : forall n, (n <> 0)%nat -> ~ inject_Z (Z.of_nat n) == 0.

Occurrences (1): ‘formal/IDM_Matrix.v’, ‘inject_nat_nonzero’

Q/M.17.v1 — trace_twirl (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: Q

Lemma trace_twirl : forall n A, (n <> 0)%nat -> trace n (twirl n A) == trace n A.

Occurrences (1): ‘formal/IDM_Matrix.v’, ‘trace_twirl’

Q/M.18.v1 — twirl_idempotent (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: Q

Theorem twirl_idempotent :
forall n A i j, (n <> 0)%nat -> twirl n (twirl n A) i j == twirl n A i j.

Occurrences (1): ‘formal/IDM_Matrix.v’, ‘twirl_idempotent’

Q/M.19.v1 — scalar_line_one_dim (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: Q

Theorem scalar_line_one_dim :
forall c d, scalarM c 0%nat 0%nat = scalarM d 0%nat 0%nat -> forall i j, scalarM c i j = scalarM d i j.

Occurrences (1): ‘formal/IDM_Matrix.v’, ‘scalar_line_one_dim’

599 R — The reals R – the continuum AS a readout

domain: — • tier: Th_coqc • status: current • coq: root_layer_unwired • parents: none

Def 3.4: $R :=$ Bishop regular Cauchy sequences of Q : $f:N^+ \rightarrow Q$ with $|f\ n - f\ m| \leq 1/n + 1/m$. Equality Req (an equivalence). Injection inj_Q : $Q \hookrightarrow R$. Th 3.5 (ordered field up to Req). Th 3.6 (Cauchy-completeness, $R_complete$): every real Cauchy sequence has an explicitly constructed limit.

R/M.01.v1 — apriori_multiplicative_contracts (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: R

Theorem apriori_multiplicative_contracts :
forall (rho : Q) (m s : nat -> Q),
(forall k, Qabs (m k) <= rho) ->
(forall k, s (S k) == m k * s k) ->
(forall k, Qabs (s (S k)) <= rho * Qabs (s k)).

Occurrences (1): ‘formal/IDM_Apriori.v’, ‘apriori_multiplicative_contracts’

R/M.02.v1 — apriori_geometric_contracts (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: R

Theorem apriori_geometric_contracts :
forall (rho a q : Q) (s : nat -> Q),
Qabs q <= rho ->
(forall k, s k == a * qpow q k) ->
(forall k, Qabs (s (S k)) <= rho * Qabs (s k)).

Occurrences (1): ‘formal/IDM_Apriori.v’, ‘apriori_geometric_contracts’

R/M.03.v1 — `apriori_stable` (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: R

```
Theorem apriori_stable :  
forall (rho : Q) (m s : nat -> Q) (N M : nat),  
rho <= 1 ->  
(forall k, Qabs (m k) <= rho) ->  
(forall k, s (S k) == m k * s k) ->  
(1 - rho) * Qabs (tailsum s N M) <= Qabs (s N).
```

Occurrences (1): ‘formal/IDM_Apriori.v’, ‘apriori_stable’

R/M.04.v1 — `half_nonneg` (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: R

```
Lemma half_nonneg : 0 <= (1 # 2).
```

Occurrences (1): ‘formal/IDM_Apriori.v’, ‘half_nonneg’

R/M.05.v1 — `richardson_ratio_nonneg` (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: R

```
Lemma richardson_ratio_nonneg : forall p, 0 <= richardson_ratio p.
```

Occurrences (1): ‘formal/IDM_Apriori.v’, ‘richardson_ratio_nonneg’

R/M.06.v1 — `richardson_ratio_le_one` (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: R

```
Lemma richardson_ratio_le_one : forall p, richardson_ratio p <= 1.
```

Occurrences (1): ‘formal/IDM_Apriori.v’, ‘richardson_ratio_le_one’

R/M.07.v1 — `richardson_apriori_contracts` (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: R

```
Theorem richardson_apriori_contracts :  
forall (p : nat) (a : Q) (s : nat -> Q),  
(forall k, s k == a * qpow (richardson_ratio p) k) ->  
(forall k, Qabs (s (S k)) <= richardson_ratio p * Qabs (s k)).
```

Occurrences (1): ‘formal/IDM_Apriori.v’, ‘richardson_apriori_contracts’

R/M.08.v1 — `richardson_apriori_stable` (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: R

```
Theorem richardson_apriori_stable :  
forall (p : nat) (a : Q) (s : nat -> Q) (N M : nat),  
(forall k, s k == a * qpow (richardson_ratio p) k) ->  
(1 - richardson_ratio p) * Qabs (tailsum s N M) <= Qabs (s N).
```

Occurrences (1): ‘formal/IDM_Apriori.v’, ‘richardson_apriori_stable’

R/M.09.v1 — `FTCC_exact` (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: R

```
Theorem FTCC_exact :  
forall (f : seqf) (N : nat), agg f N == f N - f 0%nat.
```

Occurrences (1): ‘formal/IDM_Bridge.v’, ‘FTCC_exact’

R/M.10.v1 — FTCC_eps_exact (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: R

```
Theorem FTCC_eps_exact :  
forall (f : seqf) (eps : Q) (N : nat),  
~ eps == 0 -> Ieps (Deps f eps) eps N == f N - f 0%nat.
```

Occurrences (1): 'formal/IDM_Bridge.v', 'FTCC_eps_exact'

R/M.11.v1 — bridge_faithful_exact_core (Corollary)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: R

```
Corollary bridge_faithful_exact_core :  
forall f N, agg f N == f N - f 0%nat.
```

Occurrences (1): 'formal/IDM_Bridge.v', 'bridge_faithful_exact_core'

R/M.12.v1 — geom_certified_identity (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: R

```
Theorem geom_certified_identity : forall (r : Q) (n : nat),  
(1 - r) * geom_sum r n == 1 - qpow r n.
```

Occurrences (1): 'formal/IDM_Certified.v', 'geom_certified_identity'

R/M.13.v1 — geom_certified_defect (Corollary)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: R

```
Corollary geom_certified_defect : forall (r : Q) (n : nat),  
1 - (1 - r) * geom_sum r n == qpow r n.
```

Occurrences (1): 'formal/IDM_Certified.v', 'geom_certified_defect'

R/M.14.v1 — geom_majorant_tail (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: R

```
Theorem geom_majorant_tail : forall (rho : Q) (t : nat -> Q),  
(forall k, 0 <= t k) ->  
(forall k, t (S k) <= rho * t k) ->  
forall (M N : nat), (1 - rho) * tailsum t N M <= t N.
```

Occurrences (1): 'formal/IDM_Certified.v', 'geom_majorant_tail'

R/M.15.v1 — one_le_Sk (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: R

```
Lemma one_le_Sk : forall k, 1 <= inject_Z (Z.of_nat (S k)).
```

Occurrences (1): 'formal/IDM_Certified.v', 'one_le_Sk'

R/M.16.v1 — q01 (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: R

```
Lemma q01 : (0 < 1)%Q.
```

Occurrences (1): 'formal/IDM_Certified.v', 'q01'

R/M.17.v1 — q01_le (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: R

Lemma q01_le : (0 <= 1)%Q.

Occurrences (1): ‘formal/IDM_Certified.v’, ‘q01_le’

R/M.18.v1 — pos_Sk (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: R

Lemma pos_Sk : forall k, 0 < inject_Z (Z.of_nat (S k)).

Occurrences (1): ‘formal/IDM_Certified.v’, ‘pos_Sk’

R/M.19.v1 — div_le_self (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: R

Lemma div_le_self : forall a d, 0 <= a -> 1 <= d -> a / d <= a.

Occurrences (1): ‘formal/IDM_Certified.v’, ‘div_le_self’

R/M.20.v1 — exp_term_nonneg (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: R

Lemma exp_term_nonneg : forall x k, 0 <= x -> 0 <= exp_term x k.

Occurrences (1): ‘formal/IDM_Certified.v’, ‘exp_term_nonneg’

R/M.21.v1 — exp_term_ratio (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: R

Lemma exp_term_ratio : forall x k, 0 <= x -> exp_term x (S k) <= x * exp_term x k.

Occurrences (1): ‘formal/IDM_Certified.v’, ‘exp_term_ratio’

R/M.22.v1 — exp_tail_certified (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: R

Theorem exp_tail_certified : forall (x : Q) (N M : nat),
0 <= x ->
(1 - x) * tailsum (exp_term x) N M <= exp_term x N.

Occurrences (1): ‘formal/IDM_Certified.v’, ‘exp_tail_certified’

R/M.23.v1 — two_nonneg (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: R

Lemma two_nonneg : (0 <= 2)%Q.

Occurrences (1): ‘formal/IDM_Certified.v’, ‘two_nonneg’

R/M.24.v1 — sq_error_propagation (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: R

Theorem sq_error_propagation : forall (p v e : Q),
Qabs (p - v) <= e ->
Qabs (p * p - v * v) <= (2 * Qabs v + e) * e.

Occurrences (1): ‘formal/IDM_Certified.v’, ‘sq_error_propagation’

R/M.25.v1 — mono_step (Lemma)

domain: M • tier: Th_coqc • status: *current* • coq: *mapped_not_wrapped* • parents: R

Lemma mono_step : forall b1 b2 e, b1 <= b2 -> 0 <= e -> (2 * b1 + e) * e <= (2 * b2 + e) * e.

Occurrences (1): 'formal/IDM_Certified.v', 'mono_step'

R/M.26.v1 — valbound_nonneg (Lemma)

domain: M • tier: Th_coqc • status: *current* • coq: *mapped_not_wrapped* • parents: R

Lemma valbound_nonneg : forall a m, 0 <= a -> 0 <= valbound a m.

Occurrences (1): 'formal/IDM_Certified.v', 'valbound_nonneg'

R/M.27.v1 — errbound_nonneg (Lemma)

domain: M • tier: Th_coqc • status: *current* • coq: *mapped_not_wrapped* • parents: R

Lemma errbound_nonneg : forall a e m, 0 <= a -> 0 <= e -> 0 <= errbound a e m.

Occurrences (1): 'formal/IDM_Certified.v', 'errbound_nonneg'

R/M.28.v1 — iter_sq_valbound (Lemma)

domain: M • tier: Th_coqc • status: *current* • coq: *mapped_not_wrapped* • parents: R

Lemma iter_sq_valbound : forall a v m, Qabs v <= a -> Qabs (iter_sq v m) <= valbound a m.

Occurrences (1): 'formal/IDM_Certified.v', 'iter_sq_valbound'

R/M.29.v1 — iter_sq_certified (Theorem)

domain: M • tier: Th_coqc • status: *current* • coq: *mapped_not_wrapped* • parents: R

Theorem iter_sq_certified : forall (a e p v : Q) (m : nat),
Qabs v <= a -> Qabs (p - v) <= e ->
Qabs (iter_sq p m - iter_sq v m) <= errbound a e m.

Occurrences (1): 'formal/IDM_Certified.v', 'iter_sq_certified'

R/M.30.v1 — Qmult_le_l_nonneg (Lemma)

domain: M • tier: Th_coqc • status: *current* • coq: *mapped_not_wrapped* • parents: R

Lemma Qmult_le_l_nonneg : forall a b c, 0 <= c -> a <= b -> c * a <= c * b.

Occurrences (1): 'formal/IDM_Certified.v', 'Qmult_le_l_nonneg'

R/M.31.v1 — abs_tailsum_le (Lemma)

domain: M • tier: Th_coqc • status: *current* • coq: *mapped_not_wrapped* • parents: R

Lemma abs_tailsum_le : forall (s : nat -> Q) (N M : nat),
Qabs (tailsum s N M) <= tailsum (fun k => Qabs (s k)) N M.

Occurrences (1): 'formal/IDM_Certified.v', 'abs_tailsum_le'

R/M.32.v1 — refine_stable (Theorem)

domain: M • tier: Th_coqc • status: *current* • coq: *mapped_not_wrapped* • parents: R

```
Theorem refine_stable : forall (rho : Q) (s : nat -> Q) (N M : nat),
rho <= 1 ->
(forall k, Qabs (s (S k)) <= rho * Qabs (s k)) ->
(1 - rho) * Qabs (tailsum s N M) <= Qabs (s N).
```

Occurrences (1): ‘formal/IDM_Certified.v’, ‘refine_stable’

R/M.33.v1 — radd_at (Theorem)

domain: M • tier: Th_coqc • status: *current* • coq: *mapped_not_wrapped* • parents: R

```
Theorem radd_at : forall g h n, radd g h n = g n + h n.
```

Occurrences (1): ‘formal/IDM_Continuum.v’, ‘radd_at’

R/M.34.v1 — rmul_at (Theorem)

domain: M • tier: Th_coqc • status: *current* • coq: *mapped_not_wrapped* • parents: R

```
Theorem rmul_at : forall g h n, rmul g h n = g n * h n.
```

Occurrences (1): ‘formal/IDM_Continuum.v’, ‘rmul_at’

R/M.35.v1 — radd_comm (Theorem)

domain: M • tier: Th_coqc • status: *current* • coq: *mapped_not_wrapped* • parents: R

```
Theorem radd_comm : forall g h n, radd g h n == radd h g n.
```

Occurrences (1): ‘formal/IDM_Continuum.v’, ‘radd_comm’

R/M.36.v1 — const_gap_zero (Theorem)

domain: M • tier: Th_coqc • status: *current* • coq: *mapped_not_wrapped* • parents: R

```
Theorem const_gap_zero : forall q n, gap (rconst q) n == 0.
```

Occurrences (1): ‘formal/IDM_Continuum.v’, ‘const_gap_zero’

R/M.37.v1 — gap_subadditive (Theorem)

domain: M • tier: Th_coqc • status: *current* • coq: *mapped_not_wrapped* • parents: R

```
Theorem gap_subadditive :
forall g h n, gap (radd g h) n <= gap g n + gap h n.
```

Occurrences (1): ‘formal/IDM_Continuum.v’, ‘gap_subadditive’

R/P.01.v1 — s_val (Lemma)

domain: P • tier: Th_coqc • status: *current* • coq: *mapped_not_wrapped* • parents: R

```
Lemma s_val : s = 15999999999.
```

Occurrences (1): ‘case_studies/double_pendulum_readout/ShakeIrrational.v’, ‘s_val’

R/P.02.v1 — squeeze (Lemma)

domain: P • tier: Th_coqc • status: *current* • coq: *mapped_not_wrapped* • parents: R

```
Lemma squeeze : s * s < delta_num < (s + 1) * (s + 1).
```

Occurrences (1): ‘case_studies/double_pendulum_readout/ShakeIrrational.v’, ‘squeeze’

R/P.03.v1 — delta_num_not_perfect_square (Theorem)

domain: P • tier: Th_coqc • status: *current* • coq: *mapped_not_wrapped* • parents: R

Theorem delta_num_not_perfect_square : forall m : $\mathbb{Z}$, $m * m <> \text{delta_num}$.

Occurrences (1): ‘case_studies/double_pendulum_readout/ShakeIrrational.v’, ‘delta_num_not_perfect_square’

600 RD1 — Axiom of Retained Difference RD1

domain: — • tier: Ax • status: *current* • coq: *root_layer_unwired* • parents: *none*

RD1 there is a ground ‘0’ (no prior distinction).

601 RD2 — Axiom of Retained Difference RD2

domain: — • tier: Ax • status: *current* • coq: *root_layer_unwired* • parents: *none*

RD2 a successor ‘succ’ (retain one more distinction).

602 RD3 — Axiom of Retained Difference RD3

domain: — • tier: Ax • status: *current* • coq: *root_layer_unwired* • parents: *none*

RD3 ‘succ $n \neq 0$ ’ (a retained tick is never the ground). (witness ‘RD3_succ_ne_zero’)

RD3/M.01.v1 — succ_ground_distinct (Theorem)

domain: M • tier: Th_coqc • status: *current* • coq: *mapped_not_wrapped* • parents: $RD3$

Theorem succ_ground_distinct : $S\ 0 <> 0$.

Occurrences (1): ‘formal/IDM_Genesis.v’, ‘succ_ground_distinct’

603 RD4 — Axiom of Retained Difference RD4

domain: — • tier: Ax • status: *current* • coq: *root_layer_unwired* • parents: *none*

RD4 ‘succ’ is injective (distinct histories stay distinct).

604 RD5 — Axiom of Retained Difference RD5

domain: — • tier: Ax • status: *current* • coq: *root_layer_unwired* • parents: *none*

RD5 induction: a property holding at ‘0’ and closed under ‘succ’ holds throughout.

605 RD6 — Axiom of Retained Difference RD6-RD9

domain: — • tier: Ax • status: *current* • coq: *root_layer_unwired* • parents: *none*

the recursion clauses defining ‘ $\oplus$ ’ (retained addition) and ‘ $\otimes$ ’ (retained composition) and the order ‘ $\prec$ ’ from ‘succ’.

606 RD7 — Axiom of Retained Difference RD7 (bundled with RD6-RD9)

domain: — • *tier:* *Ax* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *none*

RD6-RD9 the recursion clauses defining ‘ $\oplus$ ’ (retained addition) and ‘ $\otimes$ ’ (retained composition) and the order ‘ $\prec$ ’ from ‘succ’. (the treatise bundles RD6, RD7, RD8, RD9 under one heading bullet - no separate text distinguishes RD7 individually; not invented here.)

607 RD8 — Axiom of Retained Difference RD8 (bundled with RD6-RD9)

domain: — • *tier:* *Ax* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *none*

RD6-RD9 the recursion clauses defining ‘ $\oplus$ ’ (retained addition) and ‘ $\otimes$ ’ (retained composition) and the order ‘ $\prec$ ’ from ‘succ’. (the treatise bundles RD6, RD7, RD8, RD9 under one heading bullet - no separate text distinguishes RD8 individually; not invented here.)

608 RD9 — Axiom of Retained Difference RD9 (bundled with RD6-RD9)

domain: — • *tier:* *Ax* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *none*

RD6-RD9 the recursion clauses defining ‘ $\oplus$ ’ (retained addition) and ‘ $\otimes$ ’ (retained composition) and the order ‘ $\prec$ ’ from ‘succ’. (the treatise bundles RD6, RD7, RD8, RD9 under one heading bullet - no separate text distinguishes RD9 individually; not invented here.)

609 Theta — Theta – living/relational-geometry root state (root extension)

domain: — • *tier:* *untagged* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *EQ-008, EQ-022*

$G[\text{Theta}] = G_0 + \text{Sum}_a \text{Theta}^a G_a$

Theta/P.01.v1 — real_quartet_no_cp_readout (Theorem)

domain: *P* • *tier:* *Th_coqc* • *status:* *current* • *coq:* *closed* • *parents:* *Theta*

Theorem real_quartet_no_cp_readout :
forall a b c d : Q,
(cim (Cmul (Cmul (mkC a 0) (mkC b 0))
(Cmul (Cconj (mkC c 0)) (Cconj (mkC d 0))))) == 0)%Q.

Occurrences (1): ‘coq/readout_genesis/formal/InfoThetaCPSquareObstruction_attempt.v’, ‘real_quartet_no_cp_readout’

Theta/P.02.v1 — theta_census_exists_3 (Theorem)

domain: *P* • *tier:* *Th_coqc* • *status:* *current* • *coq:* *closed* • *parents:* *Theta*

Theorem theta_census_exists_3 :
exists w01 w02 w12 : Q,
(0 <= w01)%Q /\ (0 <= w02)%Q /\ (0 <= w12)%Q /\
(L00 == w01 + w02)%Q /\ (L01 == - w01)%Q /\ (L02 == - w02)%Q /\
(L11 == w01 + w12)%Q /\ (L12 == - w12)%Q /\ (L22 == w02 + w12)%Q.

Occurrences (1): ‘coq/readout_genesis/formal/InfoThetaEdgeCensus_attempt.v’, ‘theta_census_exists_3’

Theta/P.03.v1 — theta_census_unique_3 (Theorem)

domain: P • tier: Th_coqc • status: current • coq: closed • parents: Θ

```
Theorem theta_census_unique_3 :  
forall w01 w02 w12 v01 v02 v12 : Q,  
(L01 == - w01)%Q -> (L02 == - w02)%Q -> (L12 == - w12)%Q ->  
(L01 == - v01)%Q -> (L02 == - v02)%Q -> (L12 == - v12)%Q ->  
(w01 == v01)%Q /\ (w02 == v02)%Q /\ (w12 == v12)%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaEdgeCensus_attempt.v', 'theta_census_unique_3'

Theta/P.04.v1 — edge_generators_independent_3 (Theorem)

domain: P • tier: Th_coqc • status: current • coq: closed • parents: Θ

```
Theorem edge_generators_independent_3 :  
forall a b c : Q,  
(a + b == 0)%Q -> (- a == 0)%Q -> (- b == 0)%Q ->  
(a + c == 0)%Q -> (- c == 0)%Q -> (b + c == 0)%Q ->  
(a == 0)%Q /\ (b == 0)%Q /\ (c == 0)%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaEdgeCensus_attempt.v', 'edge_generators_independent_3'

Theta/P.05.v1 — fixed_point_balance_law (Theorem)

domain: P • tier: Th_coqc • status: current • coq: closed • parents: Θ

```
Theorem fixed_point_balance_law :  
(p0*p0*p0*s0 + p1*p1*p1*s1 + p2*p2*p2*s2 == 0)%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaFixedPointBalance_attempt.v', 'fixed_point_balance_law'

Theta/P.06.v1 — mirror_symmetry_is_dead (Theorem)

domain: P • tier: Th_coqc • status: current • coq: closed • parents: Θ

```
Theorem mirror_symmetry_is_dead :  
forall a b gp p : Q,  
~ (b == 0)%Q ->  
(gp + a*p + b*(p*p*p) == 0)%Q -> (* reader, Psi=-Phi *)  
((- gp) + (a + (3#1)*b*(p*p))*(- p) == 0)%Q -> (* record, Psi=-Phi *)  
(p == 0)%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaFixedPointBalance_attempt.v', 'mirror_symmetry_is_dead'

Theta/P.07.v1 — agreement_is_dead (Theorem)

domain: P • tier: Th_coqc • status: current • coq: closed • parents: Θ

```
Theorem agreement_is_dead :  
forall a b p : Q,  
~ (a == 0)%Q ->  
(a*p + b*(p*p*p) == 0)%Q -> (* reader, G = 0 *)  
((a + (3#1)*b*(p*p))*p == 0)%Q -> (* record, Psi=Phi, G = 0 *)  
(p == 0)%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaFixedPointBalance_attempt.v', 'agreement_is_dead'

Theta/P.08.v1 — locus_Z1Z3_J_square (Theorem)

domain: P • tier: Th_coqc • status: current • coq: closed • parents: Θ

```
Theorem locus_Z1Z3_J_square :  
forall Phi0 Phi1 Phi2 Phi3 Psi0 Psi1 Psi2 Psi3 : Q,  
(Phi1 == Phi3)%Q -> (Psi1 == Psi3)%Q ->  
((Phi0*Psi1 - Phi1*Psi0) * (Phi1*Psi2 - Phi2*Psi1)  
* (Phi2*Psi3 - Phi3*Psi2) * (Phi3*Psi0 - Phi0*Psi3)  
== ((Phi0*Psi1 - Phi1*Psi0) * (Phi1*Psi2 - Phi2*Psi1))  
* ((Phi0*Psi1 - Phi1*Psi0) * (Phi1*Psi2 - Phi2*Psi1)))%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaLivingOrientationSign_attempt.v', 'locus_Z1Z3_J_square'

Theta/P.09.v1 — qsquare_nonneg (Lemma)

domain: P • tier: Th_coqc • status: current • coq: closed • parents: Θ

```
Lemma qsquare_nonneg : forall x : Q, (0 <= x*x)%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaLivingOrientationSign_attempt.v', 'qsquare_nonneg'

Theta/P.10.v1 — locus_Z1Z3_J_nonneg (Corollary)

domain: P • tier: Th_coqc • status: current • coq: closed • parents: Θ

```
Corollary locus_Z1Z3_J_nonneg :  
forall Phi0 Phi1 Phi2 Phi3 Psi0 Psi1 Psi2 Psi3 : Q,  
(Phi1 == Phi3)%Q -> (Psi1 == Psi3)%Q ->  
(0 <= (Phi0*Psi1 - Phi1*Psi0) * (Phi1*Psi2 - Phi2*Psi1)  
* (Phi2*Psi3 - Phi3*Psi2) * (Phi3*Psi0 - Phi0*Psi3))%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaLivingOrientationSign_attempt.v', 'locus_Z1Z3_J_nonneg'

Theta/P.11.v1 — qsquare_pos_of_nonzero (Lemma)

domain: P • tier: Th_coqc • status: current • coq: closed • parents: Θ

```
Lemma qsquare_pos_of_nonzero : forall x : Q, ~ (x == 0)%Q -> (0 < x*x)%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaLivingOrientationSign_attempt.v', 'qsquare_pos_of_nonzero'

Theta/P.12.v1 — locus_Z1Z3_J_pos (Corollary)

domain: P • tier: Th_coqc • status: current • coq: closed • parents: Θ

```
Corollary locus_Z1Z3_J_pos :  
forall Phi0 Phi1 Phi2 Phi3 Psi0 Psi1 Psi2 Psi3 : Q,  
(Phi1 == Phi3)%Q -> (Psi1 == Psi3)%Q ->  
~ ((Phi0*Psi1 - Phi1*Psi0) == 0)%Q ->  
~ ((Phi1*Psi2 - Phi2*Psi1) == 0)%Q ->  
(0 < (Phi0*Psi1 - Phi1*Psi0) * (Phi1*Psi2 - Phi2*Psi1)  
* (Phi2*Psi3 - Phi3*Psi2) * (Phi3*Psi0 - Phi0*Psi3))%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaLivingOrientationSign_attempt.v', 'locus_Z1Z3_J_pos'

Theta/P.13.v1 — locus_Z0Z2_J_square (Theorem)

domain: P • tier: Th_coqc • status: current • coq: closed • parents: Θ

```
Theorem locus_Z0Z2_J_square :  
forall Phi0 Phi1 Phi2 Phi3 Psi0 Psi1 Psi2 Psi3 : Q,  
(Phi0 == Phi2)%Q -> (Psi0 == Psi2)%Q ->  
((Phi0*Psi1 - Phi1*Psi0) * (Phi1*Psi2 - Phi2*Psi1)  
* (Phi2*Psi3 - Phi3*Psi2) * (Phi3*Psi0 - Phi0*Psi3)  
== ((Phi0*Psi1 - Phi1*Psi0) * (Phi2*Psi3 - Phi3*Psi2))  
* ((Phi0*Psi1 - Phi1*Psi0) * (Phi2*Psi3 - Phi3*Psi2)))%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaLivingOrientationSign_attempt.v', 'locus_Z0Z2_J_square'

Theta/P.14.v1 — locus_Z0Z2_J_nonneg (Corollary)

domain: P • tier: Th_coqc • status: current • coq: closed • parents: $Theta$

```
Corollary locus_Z0Z2_J_nonneg :
forall Phi0 Phi1 Phi2 Phi3 Psi0 Psi1 Psi2 Psi3 : Q,
(Phi0 == Phi2)%Q -> (Psi0 == Psi2)%Q ->
(0 <= (Phi0*Psi1 - Phi1*Psi0) * (Phi1*Psi2 - Phi2*Psi1)
 * (Phi2*Psi3 - Phi3*Psi2) * (Phi3*Psi0 - Phi0*Psi3))%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaLivingOrientationSign_attempt.v', 'locus_Z0Z2_J_nonneg'

Theta/P.15.v1 — locus_Z0Z2_J_pos (Corollary)

domain: P • tier: Th_coqc • status: current • coq: closed • parents: $Theta$

```
Corollary locus_Z0Z2_J_pos :
forall Phi0 Phi1 Phi2 Phi3 Psi0 Psi1 Psi2 Psi3 : Q,
(Phi0 == Phi2)%Q -> (Psi0 == Psi2)%Q ->
~ ((Phi0*Psi1 - Phi1*Psi0) == 0)%Q ->
~ ((Phi2*Psi3 - Phi3*Psi2) == 0)%Q ->
(0 < (Phi0*Psi1 - Phi1*Psi0) * (Phi1*Psi2 - Phi2*Psi1)
 * (Phi2*Psi3 - Phi3*Psi2) * (Phi3*Psi0 - Phi0*Psi3))%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaLivingOrientationSign_attempt.v', 'locus_Z0Z2_J_pos'

Theta/P.16.v1 — diffR_taylor_identity (Theorem)

domain: P • tier: Th_coqc • status: current • coq: closed • parents: $Theta$

```
Theorem diffR_taylor_identity :
forall Phi0 Phi2 Phi3 Psi0 Psi2 Psi3 dPhi dPsi : Q,
(
(* diffR, exactly, with Phi1:=Phi3+dPhi, Psi1:=Psi3+dPsi substituted *)
dPhi*dPhi*dPhi + (-1#1)*dPhi
+ Psi0*dPhi*dPhi + Psi2*dPhi*dPhi + (-2#1)*Psi3*dPhi*dPhi
+ (-2#1)*dPsi*Phi3*Phi3 + (-2#1)*dPsi*dPhi*dPhi
+ (3#1)*Phi3*dPhi*dPhi + (3#1)*dPhi*Phi3*Phi3
+ (-4#1)*Phi3*Psi3*dPhi + (-4#1)*Phi3*dPhi*dPsi
+ (-3#1)*Phi0*Psi0*dPhi + (-3#1)*Phi2*Psi2*dPhi
+ (2#1)*Phi0*Phi3*dPsi + (2#1)*Phi0*Psi3*dPhi + (2#1)*Phi0*dPhi*dPsi
+ (2#1)*Phi2*Phi3*dPsi + (2#1)*Phi2*Psi3*dPhi + (2#1)*Phi2*dPhi*dPsi
+ (2#1)*Phi3*Psi0*dPhi + (2#1)*Phi3*Psi2*dPhi
==
(* D*dPhi + E*dPsi *)
( ((-1#1) + (3#1)*Phi3*Phi3 + (-4#1)*Phi3*Psi3
+ (-3#1)*Phi0*Psi0 + (-3#1)*Phi2*Psi2
+ (2#1)*Phi0*Psi3 + (2#1)*Phi2*Psi3
+ (2#1)*Phi3*Psi0 + (2#1)*Phi3*Psi2) * dPhi )
+ ( ((-2#1)*Phi3*Phi3 + (2#1)*Phi0*Phi3 + (2#1)*Phi2*Phi3) * dPsi )
+
(* HOT_R, the quadratic+cubic remainder *)
( (1#1)*dPhi*dPhi*dPhi + (1#1)*Psi0*dPhi*dPhi + (1#1)*Psi2*dPhi*dPhi
+ (-2#1)*Psi3*dPhi*dPhi + (-2#1)*dPsi*dPhi*dPhi + (3#1)*Phi3*dPhi*dPhi
+ (-4#1)*Phi3*dPhi*dPsi + (2#1)*Phi0*dPhi*dPsi + (2#1)*Phi2*dPhi*dPsi )
)%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaLivingOrientationSign_attempt.v', 'diffR_taylor_identity'

Theta/P.17.v1 — diffRec_taylor_identity (Theorem)

domain: P • tier: Th_coqc • status: current • coq: closed • parents: $Theta$

```
Theorem diffRec_taylor_identity :
forall Phi0 Phi2 Phi3 Psi0 Psi2 Psi3 dPhi dPsi : Q,
(
```

```

(* diffRec, exactly, with Phi1:=Phi3+dPhi, Psi1:=Psi3+dPsi substituted *)
(-1#1)*dPsi
+ Phi0*dPsi*dPsi + Phi2*dPsi*dPsi + (-2#1)*Phi3*dPsi*dPsi
+ (-2#1)*dPhi*Psi3*Psi3 + (-2#1)*dPhi*dPsi*dPsi
+ (3#1)*Psi3*dPhi*dPhi + (3#1)*dPsi*Phi3*Phi3 + (3#1)*dPsi*dPhi*dPhi
+ (-4#1)*Phi3*Psi3*dPsi + (-4#1)*Psi3*dPhi*dPsi
+ (-3#1)*Phi0*Psi0*dPsi + (-3#1)*Phi2*Psi2*dPsi
+ (2#1)*Phi0*Psi3*dPsi + (2#1)*Phi2*Psi3*dPsi
+ (2#1)*Phi3*Psi0*dPsi + (2#1)*Phi3*Psi2*dPsi
+ (2#1)*Psi0*Psi3*dPhi + (2#1)*Psi0*dPhi*dPsi
+ (2#1)*Psi2*Psi3*dPhi + (2#1)*Psi2*dPhi*dPsi
+ (6#1)*Phi3*Psi3*dPhi + (6#1)*Phi3*dPhi*dPsi
==
(* F*dPhi + D*dPsi *)
( (((-2#1)*Psi3*Psi3 + (2#1)*Psi0*Psi3 + (2#1)*Psi2*Psi3
+ (6#1)*Phi3*Psi3) * dPhi) )
+ ( (((-1#1) + (3#1)*Phi3*Phi3 + (-4#1)*Phi3*Psi3
+ (-3#1)*Phi0*Psi0 + (-3#1)*Phi2*Psi2
+ (2#1)*Phi0*Psi3 + (2#1)*Phi2*Psi3
+ (2#1)*Phi3*Psi0 + (2#1)*Phi3*Psi2) * dPsi) )
+
(* HOT_Rec, the quadratic+cubic remainder *)
( (1#1)*Phi0*dPsi*dPsi + (1#1)*Phi2*dPsi*dPsi + (-2#1)*Phi3*dPsi*dPsi
+ (-2#1)*dPhi*dPsi*dPsi + (3#1)*Psi3*dPhi*dPhi + (3#1)*dPsi*dPhi*dPhi
+ (-4#1)*Psi3*dPhi*dPsi + (2#1)*Psi0*dPhi*dPsi + (2#1)*Psi2*dPhi*dPsi
+ (6#1)*Phi3*dPhi*dPsi )
)%Q.

```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaLivingOrientationSign_attempt.v', 'diffRec_taylor_identity'

Theta/P.18.v1 — linear_diff_system_nondegenerate (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

```

Theorem linear_diff_system_nondegenerate :
forall D E F dPhi dPsi : Q,
(D*dPhi + E*dPsi == 0)%Q ->
(F*dPhi + D*dPsi == 0)%Q ->
~ (D*D == E*F)%Q ->
(dPhi == 0)%Q /\ (dPsi == 0)%Q.

```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaLivingOrientationSign_attempt.v', 'linear_diff_system_nondegenerate'

Theta/P.19.v1 — edge_locus_kills_support (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

```

Theorem edge_locus_kills_support :
forall Phi_i Phi_j Psi_i Psi_j : Q,
(Psi_i == Psi_j)%Q -> (Psi_i == Psi_j)%Q ->
((Phi_i - Phi_j) * (Psi_i - Psi_j) == 0)%Q.

```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaLivingOrientationSign_attempt.v', 'edge_locus_kills_support'

Theta/P.20.v1 — edge_locus_incompatible_with_strict_support_rule (Corollary)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

```

Corollary edge_locus_incompatible_with_strict_support_rule :
forall Phi_i Phi_j Psi_i Psi_j : Q,
(Psi_i == Psi_j)%Q -> (Psi_i == Psi_j)%Q ->
((Phi_i - Phi_j) * (Psi_i - Psi_j) < 0)%Q ->
False.

```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaLivingOrientationSign_attempt.v', 'edge_locus_incompatible_with_strict_support_rule'

Theta/P.21.v1 — c4_edge01_locus_excluded (Corollary)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

```
Corollary c4_edge01_locus_excluded :  
forall Phi0 Phi1 Psi0 Psi1 : Q,  
(Phi0 == Phi1)%Q -> (Psi0 == Psi1)%Q ->  
((Phi0 - Phi1) * (Psi0 - Psi1) < 0)%Q -> False.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaLivingOrientationSign_attempt.v', 'c4_edge01_locus_excluded'

Theta/P.22.v1 — c4_edge12_locus_excluded (Corollary)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

```
Corollary c4_edge12_locus_excluded :  
forall Phi1 Phi2 Psi1 Psi2 : Q,  
(Phi1 == Phi2)%Q -> (Psi1 == Psi2)%Q ->  
((Phi1 - Phi2) * (Psi1 - Psi2) < 0)%Q -> False.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaLivingOrientationSign_attempt.v', 'c4_edge12_locus_excluded'

Theta/P.23.v1 — c4_edge23_locus_excluded (Corollary)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

```
Corollary c4_edge23_locus_excluded :  
forall Phi2 Phi3 Psi2 Psi3 : Q,  
(Phi2 == Phi3)%Q -> (Psi2 == Psi3)%Q ->  
((Phi2 - Phi3) * (Psi2 - Psi3) < 0)%Q -> False.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaLivingOrientationSign_attempt.v', 'c4_edge23_locus_excluded'

Theta/P.24.v1 — c4_edge03_locus_excluded (Corollary)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

```
Corollary c4_edge03_locus_excluded :  
forall Phi0 Phi3 Psi0 Psi3 : Q,  
(Phi0 == Phi3)%Q -> (Psi0 == Psi3)%Q ->  
((Phi0 - Phi3) * (Psi0 - Psi3) < 0)%Q -> False.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaLivingOrientationSign_attempt.v', 'c4_edge03_locus_excluded'

Theta/P.25.v1 — c4_reflection_01_23_excluded (Corollary)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

```
Corollary c4_reflection_01_23_excluded :  
forall Phi0 Phi1 Phi2 Phi3 Psi0 Psi1 Psi2 Psi3 : Q,  
(Phi0 == Phi1)%Q -> (Psi0 == Psi1)%Q ->  
(Phi2 == Phi3)%Q -> (Psi2 == Psi3)%Q ->  
((Phi0 - Phi1) * (Psi0 - Psi1) < 0)%Q -> False.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaLivingOrientationSign_attempt.v', 'c4_reflection_01_23_excluded'

Theta/P.26.v1 — c4_reflection_12_30_excluded (Corollary)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

```
Corollary c4_reflection_12_30_excluded :  
forall Phi0 Phi1 Phi2 Phi3 Psi0 Psi1 Psi2 Psi3 : Q,  
(Phi1 == Phi2)%Q -> (Psi1 == Psi2)%Q ->  
(Phi3 == Phi0)%Q -> (Psi3 == Psi0)%Q ->  
((Phi1 - Phi2) * (Psi1 - Psi2) < 0)%Q -> False.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaLivingOrientationSign_attempt.v', 'c4_reflection_12_30_excluded'

Theta/P.27.v1 — n2_empty_support_dead (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

```
Theorem n2_empty_support_dead :  
forall p s : Q,  
(- p + p*p*p == 0)%Q ->  
(((- (1#1) + (3#1)*(p*p)) * s == 0)%Q ->  
(s == 0)%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaMinimalLiving_attempt.v', 'n2_empty_support_dead'

Theta/P.28.v1 — red_rsum (Lemma)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

```
Lemma red_rsum :  
((4#1) * ((- D*D*E - p0 + p0*p0*p0) + (D*D*E - p1 + p1*p1*p1))  
== U * (U*U + (3#1)*(D*D) - (4#1)))%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaMinimalLiving_attempt.v', 'red_rsum'

Theta/P.29.v1 — red_rdiff (Lemma)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

```
Lemma red_rdiff :  
((4#1) * ((- D*D*E - p0 + p0*p0*p0) - (D*D*E - p1 + p1*p1*p1))  
== D * (- (8#1)*(D*E) - (4#1) + (3#1)*(U*U + D*D))%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaMinimalLiving_attempt.v', 'red_rdiff'

Theta/P.30.v1 — red_csum (Lemma)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

```
Lemma red_csum :  
((4#1) * ((- D*E*E + (- (1#1) + (3#1)*(p0*p0))*s0)  
+ (D*E*E + (- (1#1) + (3#1)*(p1*p1))*s1))  
== - (4#1)*T + (3#1)*T*(U*U + D*D) + (6#1)*(U*(D*E)))%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaMinimalLiving_attempt.v', 'red_csum'

Theta/P.31.v1 — red_cdiff (Lemma)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

```
Lemma red_cdiff :  
((4#1) * ((- D*E*E + (- (1#1) + (3#1)*(p0*p0))*s0)  
- (D*E*E + (- (1#1) + (3#1)*(p1*p1))*s1))  
== - (8#1)*(D*E)*E - (4#1)*E + (6#1)*(U*(D*T)) + (3#1)*E*(U*U + D*D))%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaMinimalLiving_attempt.v', 'red_cdiff'

Theta/P.32.v1 — n2_edge_present_dead (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

```
Theorem n2_edge_present_dead :  
forall U D T E : Q,  
(U * (U*U + (3#1)*(D*D) - (4#1))) == 0)%Q ->  
(D * (- (8#1)*(D*E) - (4#1) + (3#1)*(U*U + D*D)) == 0)%Q ->  
(- (4#1)*T + (3#1)*T*(U*U + D*D) + (6#1)*(U*(D*E))) == 0)%Q ->  
(- (8#1)*(D*E)*E - (4#1)*E + (6#1)*(U*(D*T)) + (3#1)*E*(U*U + D*D) == 0)%Q ->  
~ (D == 0)%Q -> ~ (E == 0)%Q -> False.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaMinimalLiving_attempt.v', 'n2_edge_present_dead'

Theta/P.33.v1 — n2_no_living_fixed_point (Theorem)

domain: $P \bullet$ tier: Th_coqc • status: current • coq: closed • parents: Θ

```
Theorem n2_no_living_fixed_point :  
forall p0 p1 s0 s1 w : Q,  
( (w == 0)%Q /\ (0 <= (p0-p1)*(s0-s1))%Q )  
\ / ( (w == - ((p0-p1)*(s0-s1)))%Q /\ ((p0-p1)*(s0-s1) < 0)%Q ) ->  
(w*(p0-p1) - p0 + p0*p0*p0 == 0)%Q ->  
(- (w*(p0-p1)) - p1 + p1*p1*p1 == 0)%Q ->  
(w*(s0-s1) + (- (1#1) + (3#1)*(p0*p0))*s0 == 0)%Q ->  
(- (w*(s0-s1)) + (- (1#1) + (3#1)*(p1*p1))*s1 == 0)%Q ->  
(s0 == 0)%Q /\ (s1 == 0)%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaMinimalLiving_attempt.v', 'n2_no_living_fixed_point'

Theta/P.34.v1 — qsign_exists (Lemma)

domain: $P \bullet$ tier: Th_coqc • status: current • coq: closed • parents: Θ

```
Lemma qsign_exists : forall x : Q, exists e : Q, (e*e == 1)%Q /\ (0 <= e*x)%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaOrientedSkewObstruction_attempt.v', 'qsign_exists'

Theta/P.35.v1 — cyclic_product_switching_invariant_triangle (Theorem)

domain: $P \bullet$ tier: Th_coqc • status: current • coq: closed • parents: Θ

```
Theorem cyclic_product_switching_invariant_triangle :  
forall a01 a12 a20 eps0 eps1 eps2 : Q,  
(eps0*eps0 == 1)%Q -> (eps1*eps1 == 1)%Q -> (eps2*eps2 == 1)%Q ->  
((eps0*eps1*a01) * (eps1*eps2*a12) * (eps2*eps0*a20)  
== a01*a12*a20)%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaOrientedSkewObstruction_attempt.v', 'cyclic_product_switching_invariant_triangle'

Theta/P.36.v1 — cyclic_product_switching_invariant_C4 (Theorem)

domain: $P \bullet$ tier: Th_coqc • status: current • coq: closed • parents: Θ

```
Theorem cyclic_product_switching_invariant_C4 :  
forall a01 a12 a23 a30 eps0 eps1 eps2 eps3 : Q,  
(eps0*eps0 == 1)%Q -> (eps1*eps1 == 1)%Q ->  
(eps2*eps2 == 1)%Q -> (eps3*eps3 == 1)%Q ->  
((eps0*eps1*a01) * (eps1*eps2*a12) * (eps2*eps3*a23) * (eps3*eps0*a30)  
== a01*a12*a23*a30)%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaOrientedSkewObstruction_attempt.v', 'cyclic_product_switching_invariant_C4'

Theta/P.37.v1 — tree_gauge_fixable_P3 (Theorem)

domain: $P \bullet$ tier: Th_coqc • status: current • coq: closed • parents: Θ

```
Theorem tree_gauge_fixable_P3 :  
forall a01 a12 : Q,  
exists eps0 eps1 eps2 : Q,  
(eps0*eps0 == 1)%Q /\ (eps1*eps1 == 1)%Q /\ (eps2*eps2 == 1)%Q /\  
(0 <= eps0*eps1*a01)%Q /\ (0 <= eps1*eps2*a12)%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaOrientedSkewObstruction_attempt.v', 'tree_gauge_fixable_P3'

Theta/P.38.v1 — tree_gauge_fixable_star3 (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

```
Theorem tree_gauge_fixable_star3 :  
forall a01 a02 : Q,  
exists eps0 eps1 eps2 : Q,  
(eps0*eps0 == 1)%Q /\ (eps1*eps1 == 1)%Q /\ (eps2*eps2 == 1)%Q /\  
(0 <= eps0*eps1*a01)%Q /\ (0 <= eps0*eps2*a02)%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaOrientedSkewObstruction_attempt.v', 'tree_gauge_fixable_star3'

Theta/P.39.v1 — tree_gauge_fixable_P4 (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

```
Theorem tree_gauge_fixable_P4 :  
forall a01 a12 a23 : Q,  
exists eps0 eps1 eps2 eps3 : Q,  
(eps0*eps0 == 1)%Q /\ (eps1*eps1 == 1)%Q /\  
(eps2*eps2 == 1)%Q /\ (eps3*eps3 == 1)%Q /\  
(0 <= eps0*eps1*a01)%Q /\ (0 <= eps1*eps2*a12)%Q /\  
(0 <= eps2*eps3*a23)%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaOrientedSkewObstruction_attempt.v', 'tree_gauge_fixable_P4'

Theta/P.40.v1 — tree_gauge_fixable_star4 (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

```
Theorem tree_gauge_fixable_star4 :  
forall a01 a02 a03 : Q,  
exists eps0 eps1 eps2 eps3 : Q,  
(eps0*eps0 == 1)%Q /\ (eps1*eps1 == 1)%Q /\  
(eps2*eps2 == 1)%Q /\ (eps3*eps3 == 1)%Q /\  
(0 <= eps0*eps1*a01)%Q /\ (0 <= eps0*eps2*a02)%Q /\  
(0 <= eps0*eps3*a03)%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaOrientedSkewObstruction_attempt.v', 'tree_gauge_fixable_star4'

Theta/P.41.v1 — quartet_rescale_invariant (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

```
Theorem quartet_rescale_invariant :  
forall alpha0 alpha1 beta1 beta2 v01 v12 v02 v11 : Cq,  
(cre alpha0 * cre alpha0 + cim alpha0 * cim alpha0 == 1)%Q ->  
(cre alpha1 * cre alpha1 + cim alpha1 * cim alpha1 == 1)%Q ->  
(cre beta1 * cre beta1 + cim beta1 * cim beta1 == 1)%Q ->  
(cre beta2 * cre beta2 + cim beta2 * cim beta2 == 1)%Q ->  
(quartetJraw (Cmul (Cmul alpha0 v01) beta1)  
(Cmul (Cmul alpha1 v12) beta2)  
(Cmul (Cmul alpha0 v02) beta2)  
(Cmul (Cmul alpha1 v11) beta1)  
== quartetJraw v01 v12 v02 v11)%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaQuartetSquareObstruction_attempt.v', 'quartet_rescale_invariant'

Theta/P.42.v1 — quartet_real_scalar_rescale (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

```
Theorem quartet_real_scalar_rescale :  
forall lam : Q, forall v01 v12 v02 v11 : Cq,  
(quartetJraw (Cmul (mkC lam 0) v01) (Cmul (mkC lam 0) v12)  
(Cmul (mkC lam 0) v02) (Cmul (mkC lam 0) v11)  
== lam*lam*lam*lam * quartetJraw v01 v12 v02 v11)%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaQuartetSquareObstruction_attempt.v', 'quartet_real_scalar_rescale'

Theta/P.43.v1 — family_B_K3_exact (Theorem)

domain: $P \bullet$ tier: Th_coqc • status: current • coq: closed • parents: Θ

```
Theorem family_B_K3_exact :  
forall a1 a2 a3 : Q,  
(cim (Cmul (Cmul (zB a1) (zB a2)) (Cconj (zB a3)))) == a1*a2*a3)%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaQuartetSquareObstruction_attempt.v', 'family_B_K3_exact'

Theta/P.44.v1 — family_B_C4_vanishes (Theorem)

domain: $P \bullet$ tier: Th_coqc • status: current • coq: closed • parents: Θ

```
Theorem family_B_C4_vanishes :  
forall a01 a12 a23 a30 : Q,  
(cim (Cmul (Cmul (zB a01) (zB a23))  
(Cmul (Cconj (zB a30)) (Cconj (zB a12))))  
== 0)%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaQuartetSquareObstruction_attempt.v', 'family_B_C4_vanishes'

Theta/P.45.v1 — family_C_rank1_vanishes (Theorem)

domain: $P \bullet$ tier: Th_coqc • status: current • coq: closed • parents: Θ

```
Theorem family_C_rank1_vanishes :  
forall Za Zb Zc Zd : Cq,  
(cim (Cmul (Cmul (Nij Za Zb) (Nij Zc Zd))  
(Cmul (Cconj (Nij Za Zd)) (Cconj (Nij Zc Zb))))  
== 0)%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaQuartetSquareObstruction_attempt.v', 'family_C_rank1_vanishes'

Theta/P.46.v1 — family_A_not_switching_invariant (Theorem)

domain: $P \bullet$ tier: Th_coqc • status: current • coq: closed • parents: Θ

```
Theorem family_A_not_switching_invariant :  
exists eps0 eps1 eps2 eps3 : Q,  
(eps0*eps0 == 1)%Q /\ (eps1*eps1 == 1)%Q /\  
(eps2*eps2 == 1)%Q /\ (eps3*eps3 == 1)%Q /\  
Qeq_bool (PA 1 2 3 4 5 (eps2*eps3*6) 7 (eps3*eps0*8))  
(PA 1 2 3 4 5 6 7 8) = false.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaQuartetSquareObstruction_attempt.v', 'family_A_not_switching_invariant'

Theta/P.47.v1 — family_A_witness_original_value (Theorem)

domain: $P \bullet$ tier: Th_coqc • status: current • coq: closed • parents: Θ

```
Theorem family_A_witness_original_value :  
(PA 1 2 3 4 5 6 7 8 == 188)%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaQuartetSquareObstruction_attempt.v', 'family_A_witness_original_value'

Theta/P.48.v1 — family_A_witness_switched_value (Theorem)

domain: $P \bullet$ tier: Th_coqc • status: current • coq: closed • parents: Θ

```
Theorem family_A_witness_switched_value :  
(PA 1 2 3 4 5 (-6) 7 (-8) == 144)%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaQuartetSquareObstruction_attempt.v', 'family_A_witness_switched_value'

Theta/P.49.v1 — skew_source_switching_covariant (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

```
Theorem skew_source_switching_covariant :  
forall Phi_i Phi_j Psi_i Psi_j Di Dj : Q,  
(Di*Di == 1)%Q -> (Dj*Dj == 1)%Q ->  
((Di*Phi_i)*(Dj*Psi_j) - (Dj*Phi_j)*(Di*Psi_i))  
== Di*Dj*(Phi_i*Psi_j - Phi_j*Psi_i))%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaQuartetSquareObstruction_attempt.v', 'skew_source_switching_covariant'

Theta/P.50.v1 — symmetric_source_switched_value (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

```
Theorem symmetric_source_switched_value :  
(sQ_switched 1 (-1) 1 2 3 4 == 21)%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaQuartetSquareObstruction_attempt.v', 'symmetric_source_switched_value'

Theta/P.51.v1 — symmetric_source_naive_covariant_prediction (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

```
Theorem symmetric_source_naive_covariant_prediction :  
(1 * (-1) * sQ 1 2 3 4 == -1)%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaQuartetSquareObstruction_attempt.v', 'symmetric_source_naive_covariant_prediction'

Theta/P.52.v1 — symmetric_source_not_switching_covariant (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

```
Theorem symmetric_source_not_switching_covariant :  
~ (sQ_switched 1 (-1) 1 2 3 4 == 1 * (-1) * sQ 1 2 3 4)%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaQuartetSquareObstruction_attempt.v', 'symmetric_source_not_switching_covariant'

Theta/P.53.v1 — c4_reversal_is_gauge_witness (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

```
Theorem c4_reversal_is_gauge_witness :  
exists eps0 eps1 eps2 eps3 : Q,  
(eps0*eps0 == 1)%Q /\ (eps1*eps1 == 1)%Q /\  
(eps2*eps2 == 1)%Q /\ (eps3*eps3 == 1)%Q /\  
(eps0*eps1 == -1)%Q /\ (eps1*eps2 == -1)%Q /\  
(eps2*eps3 == -1)%Q /\ (eps3*eps0 == -1)%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaQuartetSquareObstruction_attempt.v', 'c4_reversal_is_gauge_witness'

Theta/P.54.v1 — k3_no_uniform_reversal_gauge (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

```
Theorem k3_no_uniform_reversal_gauge :  
forall eps0 eps1 eps2 : Q,  
(eps0*eps0 == 1)%Q -> (eps1*eps1 == 1)%Q -> (eps2*eps2 == 1)%Q ->  
~ ((eps0*eps1 == -1)%Q /\ (eps1*eps2 == -1)%Q /\ (eps2*eps0 == -1)%Q).
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaQuartetSquareObstruction_attempt.v', 'k3_no_uniform_reversal_gauge'

Theta/P.55.v1 — k3_invariance_forces_uniform (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

```
Theorem k3_invariance_forces_uniform :  
forall w01 w02 w12 : Q,  
(- w02 == - w12)%Q -> (* invariance under transposition (0 1) *)  
(- w01 == - w02)%Q -> (* invariance under transposition (1 2) *)  
(w01 == w02)%Q /\ (w02 == w12)%Q /\ (w01 == w12)%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaSectorSpectrum_attempt.v', 'k3_invariance_forces_uniform'

Theta/P.56.v1 — p3_invariance_forces_uniform (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

```
Theorem p3_invariance_forces_uniform :  
forall w01 w12 : Q,  
(- w01 == - w12)%Q -> (* invariance under the end swap (0 2) *)  
(w01 == w12)%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaSectorSpectrum_attempt.v', 'p3_invariance_forces_uniform'

Theta/P.57.v1 — k3_spectrum_zero (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

```
Theorem k3_spectrum_zero :  
forall w : Q,  
(2*w*1 + (-w)*1 + (-w)*1 == 0*1)%Q /\  
((-w)*1 + 2*w*1 + (-w)*1 == 0*1)%Q /\  
((-w)*1 + (-w)*1 + 2*w*1 == 0*1)%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaSectorSpectrum_attempt.v', 'k3_spectrum_zero'

Theta/P.58.v1 — k3_spectrum_top_a (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

```
Theorem k3_spectrum_top_a : (* eigenvector (1,-1,0), eigenvalue 3w *)  
forall w : Q,  
(2*w*1 + (-w)*(-1) + (-w)*0 == 3*w*1)%Q /\  
((-w)*1 + 2*w*(-1) + (-w)*0 == 3*w*(-1))%Q /\  
((-w)*1 + (-w)*(-1) + 2*w*0 == 3*w*0)%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaSectorSpectrum_attempt.v', 'k3_spectrum_top_a'

Theta/P.59.v1 — k3_spectrum_top_b (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

```
Theorem k3_spectrum_top_b : (* eigenvector (1,0,-1), SAME eigenvalue 3w *)  
forall w : Q,  
(2*w*1 + (-w)*0 + (-w)*(-1) == 3*w*1)%Q /\  
((-w)*1 + 2*w*0 + (-w)*(-1) == 3*w*0)%Q /\  
((-w)*1 + (-w)*0 + 2*w*(-1) == 3*w*(-1))%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaSectorSpectrum_attempt.v', 'k3_spectrum_top_b'

Theta/P.60.v1 — p3_spectrum_zero (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

```
Theorem p3_spectrum_zero :  
forall w : Q,  
(w*1 + (-w)*1 + 0*1 == 0*1)%Q /\  
((-w)*1 + 2*w*1 + (-w)*1 == 0*1)%Q /\  
(0*1 + (-w)*1 + w*1 == 0*1)%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaSectorSpectrum_attempt.v', 'p3_spectrum_zero'

Theta/P.61.v1 — p3_spectrum_mid (Theorem)

domain: P • tier: Th_coqc • status: current • coq: closed • parents: $Theta$

```
Theorem p3_spectrum_mid : (* eigenvector (1,0,-1), eigenvalue w *)
forall w : Q,
(w*1 + (-w)*0 + 0*(-1) == w*1)%Q /\
((-w)*1 + 2*w*0 + (-w)*(-1) == w*0)%Q /\
(0*1 + (-w)*0 + w*(-1) == w*(-1))%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaSectorSpectrum_attempt.v', 'p3_spectrum_mid'

Theta/P.62.v1 — p3_spectrum_top (Theorem)

domain: P • tier: Th_coqc • status: current • coq: closed • parents: $Theta$

```
Theorem p3_spectrum_top : (* eigenvector (1,-2,1), eigenvalue 3w *)
forall w : Q,
(w*1 + (-w)*(-2) + 0*1 == 3*w*1)%Q /\
((-w)*1 + 2*w*(-2) + (-w)*1 == 3*w*(-2))%Q /\
(0*1 + (-w)*(-2) + w*1 == 3*w*1)%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaSectorSpectrum_attempt.v', 'p3_spectrum_top'

Theta/P.63.v1 — p3_levels_distinct (Theorem)

domain: P • tier: Th_coqc • status: current • coq: closed • parents: $Theta$

```
Theorem p3_levels_distinct :
forall w : Q, (0 < w)%Q ->
~ (0 == w)%Q /\ ~ (0 == 3*w)%Q /\ ~ (w == 3*w)%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaSectorSpectrum_attempt.v', 'p3_levels_distinct'

Theta/P.64.v1 — edge_source_bilinear_01 (Theorem)

domain: P • tier: Th_coqc • status: current • coq: closed • parents: $Theta$

```
Theorem edge_source_bilinear_01 :
forall p0 p1 s0 s1 : Q,
(p0*s0 - p0*s1 - p1*s0 + p1*s1 == (p0 - p1) * (s0 - s1))%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaTopologyReadout_attempt.v', 'edge_source_bilinear_01'

Theta/P.65.v1 — edge_source_bilinear_02 (Theorem)

domain: P • tier: Th_coqc • status: current • coq: closed • parents: $Theta$

```
Theorem edge_source_bilinear_02 :
forall p0 p2 s0 s2 : Q,
(p0*s0 - p0*s2 - p2*s0 + p2*s2 == (p0 - p2) * (s0 - s2))%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaTopologyReadout_attempt.v', 'edge_source_bilinear_02'

Theta/P.66.v1 — edge_source_bilinear_12 (Theorem)

domain: P • tier: Th_coqc • status: current • coq: closed • parents: $Theta$

```
Theorem edge_source_bilinear_12 :
forall p1 p2 s1 s2 : Q,
(p1*s1 - p1*s2 - p2*s1 + p2*s2 == (p1 - p2) * (s1 - s2))%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaTopologyReadout_attempt.v', 'edge_source_bilinear_12'

Theta/P.67.v1 — Qmult_pos_pos (Lemma)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

Lemma Qmult_pos_pos : forall a b : Q, (0 < a)%Q -> (0 < b)%Q -> (0 < a * b)%Q.

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaTopologyReadout_attempt.v', 'Qmult_pos_pos'

Theta/P.68.v1 — Qsquare_nonneg (Lemma)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

Lemma Qsquare_nonneg : forall d : Q, (0 <= d * d)%Q.

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaTopologyReadout_attempt.v', 'Qsquare_nonneg'

Theta/P.69.v1 — clipped_stationarity_absent (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

Theorem clipped_stationarity_absent :
forall mu K s w : Q,
(0 < mu)%Q -> (0 < K)%Q -> (0 <= s)%Q -> (0 <= w)%Q ->
(0 <= mu * w * w + (2#1) * K * s * w)%Q.

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaTopologyReadout_attempt.v', 'clipped_stationarity_absent'

Theta/P.70.v1 — clipped_stationarity_present (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

Theorem clipped_stationarity_present :
forall mu K s w wstar : Q,
(0 < mu)%Q -> (0 < K)%Q -> (s < 0)%Q ->
(mu * wstar == - (K * s))%Q ->
(0 < wstar)%Q /\
(mu * wstar * wstar + (2#1) * K * s * wstar
<= mu * w * w + (2#1) * K * s * w)%Q.

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaTopologyReadout_attempt.v', 'clipped_stationarity_present'

Theta/P.71.v1 — discordance_complement_flip (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

Theorem discordance_complement_flip :
forall pi pj si sj : Q,
((pi - pj) * ((- si) - (- sj)) == - ((pi - pj) * (si - sj)))%Q.

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaTopologyReadout_attempt.v', 'discordance_complement_flip'

Theta/P.72.v1 — witness_total_disorder_K3 (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

Theorem witness_total_disorder_K3 :
((1 - 2) * (3 - 2) < 0)%Q /\
((1 - 3) * (3 - 1) < 0)%Q /\
((2 - 3) * (2 - 1) < 0)%Q.

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaTopologyReadout_attempt.v', 'witness_total_disorder_K3'

Theta/P.73.v1 — witness_partial_disorder_P3 (Theorem)

domain: P • tier: Th_coqc • status: current • coq: closed • parents: Θ

```
Theorem witness_partial_disorder_P3 :  
((3 - 1) * ((-2) - (-1)) < 0)%Q /\  
(0 < (3 - 2) * ((-2) - (-5)))%Q /\  
((1 - 2) * ((-1) - (-5)) < 0)%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoThetaTopologyReadout_attempt.v', 'witness_partial_disorder_P3'

Theta/P.74.v1 — quartet_decomp (Lemma)

domain: P • tier: Th_coqc • status: current • coq: closed • parents: Θ

```
Lemma quartet_decomp :  
forall a b c d e f g h : Q,  
(cim (Cmul (mkC a b) (mkC g h)) (Cmul (Cconj (mkC c d)) (Cconj (mkC e f))))  
== (a*c + b*d + (e*g + f*h)) * (e*h - f*g)  
+ ((b*c - a*d) + (f*g - e*h)) * (e*g + f*h)%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoCPEquivariantGenerationBound_attempt.v', 'quartet_decomp'

Theta/P.75.v1 — two_gen_quartet_im_vanishes (Theorem)

domain: P • tier: Th_coqc • status: current • coq: closed • parents: Θ

```
Theorem two_gen_quartet_im_vanishes :  
forall a b c d e f g h : Q,  
(a*c + b*d + (e*g + f*h) == 0)%Q ->  
((b*c - a*d) + (f*g - e*h) == 0)%Q ->  
(cim (Cmul (Cmul (mkC a b) (mkC g h)) (Cmul (Cconj (mkC c d)) (Cconj (mkC e f))))  
== 0)%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoCPEquivariantGenerationBound_attempt.v', 'two_gen_quartet_im_vanishes'

Theta/P.76.v1 — quartet_conj_flips_sign (Theorem)

domain: P • tier: Th_coqc • status: current • coq: closed • parents: Θ

```
Theorem quartet_conj_flips_sign :  
forall z1 z2 z3 z4 : Cq,  
(cim (Cmul (Cmul (Cconj z1) (Cconj z2))  
(Cmul (Cconj (Cconj z3)) (Cconj (Cconj z4)))))  
== - cim (Cmul (Cmul z1 z2) (Cmul (Cconj z3) (Cconj z4)))))%Q.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoCPEquivariantGenerationBound_attempt.v', 'quartet_conj_flips_sign'

Theta/P.77.v1 — witness_unitary (Theorem)

domain: P • tier: Th_coqc • status: current • coq: closed • parents: Θ

```
Theorem witness_unitary : unitary3 V3w = true.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoCPEquivariantGenerationBound_attempt.v', 'witness_unitary'

Theta/P.78.v1 — witness_J_value (Theorem)

domain: P • tier: Th_coqc • status: current • coq: closed • parents: Θ

```
Theorem witness_J_value : Qeq_bool (quartetJ V3w) (110592 # 4151485) = true.
```

Occurrences (1): 'coq/readout_genesis/formal/InfoCPEquivariantGenerationBound_attempt.v', 'witness_J_value'

Theta/P.79.v1 — witness_J_nonzero (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

Theorem witness_J_nonzero : Qeq_bool (quartetJ V3w) 0 = false.

Occurrences (1): 'coq/readout_genesis/formal/InfoCPEquivariantGenerationBound_attempt.v', 'witness_J_nonzero'

Theta/P.80.v1 — witness_cp_unitary (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

Theorem witness_cp_unitary : unitary3 (conj3 V3w) = true.

Occurrences (1): 'coq/readout_genesis/formal/InfoCPEquivariantGenerationBound_attempt.v', 'witness_cp_unitary'

Theta/P.81.v1 — witness_cp_flips (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

Theorem witness_cp_flips : Qeq_bool (quartetJ (conj3 V3w)) (- (110592 # 4151485)) = true.

Occurrences (1): 'coq/readout_genesis/formal/InfoCPEquivariantGenerationBound_attempt.v', 'witness_cp_flips'

Theta/P.82.v1 — neutral_cp_fixed (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

Theorem neutral_cp_fixed : meq3 (conj3 I3) I3 = true.

Occurrences (1): 'coq/readout_genesis/formal/InfoCPEquivariantGenerationBound_attempt.v', 'neutral_cp_fixed'

Theta/P.83.v1 — neutral_unitary (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

Theorem neutral_unitary : unitary3 I3 = true.

Occurrences (1): 'coq/readout_genesis/formal/InfoCPEquivariantGenerationBound_attempt.v', 'neutral_unitary'

Theta/P.84.v1 — neutral_J_zero (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

Theorem neutral_J_zero : Qeq_bool (quartetJ I3) 0 = true.

Occurrences (1): 'coq/readout_genesis/formal/InfoCPEquivariantGenerationBound_attempt.v', 'neutral_J_zero'

Theta/P.85.v1 — three_values_realized (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

Theorem three_values_realized :
sgnQ (quartetJ V3w) = SPlus
/\ sgnQ (quartetJ (conj3 V3w)) = SMinus
/\ sgnQ (quartetJ I3) = SZero.

Occurrences (1): 'coq/readout_genesis/formal/InfoCPEquivariantGenerationBound_attempt.v', 'three_values_realized'

Theta/P.86.v1 — values_pairwise_distinct (Theorem)

domain: $P \bullet \text{tier: } Th_coqc \bullet \text{status: current} \bullet \text{coq: closed} \bullet \text{parents: Theta}$

Theorem values_pairwise_distinct :
SPlus <> SMinus /\ SPlus <> SZero /\ SMinus <> SZero.

Occurrences (1): 'coq/readout_genesis/formal/InfoCPEquivariantGenerationBound_attempt.v', 'values_pairwise_distinct'

610 Z — The integers Z (Grothendieck completion of D)

domain: — • *tier:* *Th_coqc* • *status:* *current* • *coq:* *root_layer_unwired* • *parents:* *none*

Def 3.1: $Z := (D \times D) / \sim$ with $(a, b) \sim (c, d) \Leftrightarrow a \oplus d = c \oplus b$; value homomorphism to the standard Z. Th 3.1: $(Z, +, \cdot, 0, 1)$ is a commutative ring.

Z/M.01.v1 — delta_sum (Theorem)

domain: *M* • *tier:* *Th_coqc* • *status:* *current* • *coq:* *mapped_not_wrapped* • *parents:* *Z*

Theorem delta_sum : forall f g n, Delta (fun k => f k + g k) n == Delta f n + Delta g n.

Occurrences (1): ‘formal/IDM_Calculus.v’, ‘delta_sum’

Z/M.02.v1 — delta_scalar (Theorem)

domain: *M* • *tier:* *Th_coqc* • *status:* *current* • *coq:* *mapped_not_wrapped* • *parents:* *Z*

Theorem delta_scalar : forall (c : Q) f n, Delta (fun k => c * f k) n == c * Delta f n.

Occurrences (1): ‘formal/IDM_Calculus.v’, ‘delta_scalar’

Z/M.03.v1 — delta_product (Theorem)

domain: *M* • *tier:* *Th_coqc* • *status:* *current* • *coq:* *mapped_not_wrapped* • *parents:* *Z*

Theorem delta_product :
forall f g n, Delta (fun k => f k * g k) n == f (S n) * Delta g n + g n * Delta f n.

Occurrences (1): ‘formal/IDM_Calculus.v’, ‘delta_product’

Z/M.04.v1 — Deps_sum (Theorem)

domain: *M* • *tier:* *Th_coqc* • *status:* *current* • *coq:* *mapped_not_wrapped* • *parents:* *Z*

Theorem Deps_sum :
forall f g eps n, ~ eps == 0 ->
Deps (fun k => f k + g k) eps n == Deps f eps n + Deps g eps n.

Occurrences (1): ‘formal/IDM_Calculus.v’, ‘Deps_sum’

Z/M.05.v1 — Deps_product (Theorem)

domain: *M* • *tier:* *Th_coqc* • *status:* *current* • *coq:* *mapped_not_wrapped* • *parents:* *Z*

Theorem Deps_product :
forall f g eps n, ~ eps == 0 ->
Deps (fun k => f k * g k) eps n == f (S n) * Deps g eps n + g n * Deps f eps n.

Occurrences (1): ‘formal/IDM_Calculus.v’, ‘Deps_product’

Z/M.06.v1 — FTCC_telescope (Theorem)

domain: *M* • *tier:* *Th_coqc* • *status:* *current* • *coq:* *mapped_not_wrapped* • *parents:* *Z*

Theorem FTCC_telescope : forall f N, Agg f N == f N - f 0%nat.

Occurrences (1): ‘formal/IDM_Calculus.v’, ‘FTCC_telescope’

Z/M.07.v1 — PSum_ext (Lemma)

domain: *M* • *tier:* *Th_coqc* • *status:* *current* • *coq:* *mapped_not_wrapped* • *parents:* *Z*

Lemma PSum_ext : forall f g N, (forall n, f n == g n) -> PSum f N == PSum g N.

Occurrences (1): ‘formal/IDM_Calculus.v’, ‘PSum_ext’

Z/M.08.v1 — PSum_delta_telescope (Lemma)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: Z

Lemma PSum_delta_telescope : forall h N, PSum (fun k => Delta h k) N == h N - h 0%nat.

Occurrences (1): ‘formal/IDM_Calculus.v’, ‘PSum_delta_telescope’

Z/M.09.v1 — summation_by_parts (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: Z

Theorem summation_by_parts :
forall f g N,
PSum (fun n => f (S n) * Delta g n + g n * Delta f n) N
== f N * g N - f 0%nat * g 0%nat.

Occurrences (1): ‘formal/IDM_Calculus.v’, ‘summation_by_parts’

Z/M.10.v1 — lagrange_order_div (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: Z

Theorem lagrange_order_div :
forall g n : nat, 0 < n -> Nat.divide (n / Nat.gcd g n) n.

Occurrences (1): ‘formal/IDM_FiniteWitnesses2.v’, ‘lagrange_order_div’

Z/M.11.v1 — ring_distrib_Z (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: Z

Theorem ring_distrib_Z :
forall a b c : Z, a * (b + c) = a * b + a * c /\ a * (b - c) = a * b - a * c.

Occurrences (1): ‘formal/IDM_FiniteWitnesses3.v’, ‘ring_distrib_Z’

Z/M.12.v1 — tmin_assoc (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: Z

Theorem tmin_assoc : forall a b c, tmin a (tmin b c) = tmin (tmin a b) c.

Occurrences (1): ‘formal/IDM_Tropical.v’, ‘tmin_assoc’

Z/M.13.v1 — tmin_comm (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: Z

Theorem tmin_comm : forall a b, tmin a b = tmin b a.

Occurrences (1): ‘formal/IDM_Tropical.v’, ‘tmin_comm’

Z/M.14.v1 — tmin_idem (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: Z

Theorem tmin_idem : forall a, tmin a a = a.

Occurrences (1): ‘formal/IDM_Tropical.v’, ‘tmin_idem’

Z/M.15.v1 — tmax_assoc (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: Z

Theorem tmax_assoc : forall a b c, tmax a (tmax b c) = tmax (tmax a b) c.

Occurrences (1): ‘formal/IDM_Tropical.v’, ‘tmax_assoc’

Z/M.16.v1 — tmax_comm (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: Z

Theorem tmax_comm : forall a b, tmax a b = tmax b a.

Occurrences (1): ‘formal/IDM_Tropical.v’, ‘tmax_comm’

Z/M.17.v1 — tmax_idem (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: Z

Theorem tmax_idem : forall a, tmax a a = a.

Occurrences (1): ‘formal/IDM_Tropical.v’, ‘tmax_idem’

Z/M.18.v1 — tadd_assoc (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: Z

Theorem tadd_assoc : forall a b c, tadd a (tadd b c) = tadd (tadd a b) c.

Occurrences (1): ‘formal/IDM_Tropical.v’, ‘tadd_assoc’

Z/M.19.v1 — tadd_comm (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: Z

Theorem tadd_comm : forall a b, tadd a b = tadd b a.

Occurrences (1): ‘formal/IDM_Tropical.v’, ‘tadd_comm’

Z/M.20.v1 — tadd_0_l (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: Z

Theorem tadd_0_l : forall a, tadd 0 a = a.

Occurrences (1): ‘formal/IDM_Tropical.v’, ‘tadd_0_l’

Z/M.21.v1 — minplus_distrib (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: Z

Theorem minplus_distrib : forall a b c, tadd a (tmin b c) = tmin (tadd a b) (tadd a c).

Occurrences (1): ‘formal/IDM_Tropical.v’, ‘minplus_distrib’

Z/M.22.v1 — maxplus_distrib (Theorem)

domain: M • tier: Th_coqc • status: current • coq: mapped_not_wrapped • parents: Z

Theorem maxplus_distrib : forall a b c, tadd a (tmax b c) = tmax (tadd a b) (tadd a c).

Occurrences (1): ‘formal/IDM_Tropical.v’, ‘maxplus_distrib’

Z/M.23.v1 — bottleneck_distrib (Theorem)

domain: M • tier: Th_coqc • status: $current$ • coq: $mapped_not_wrapped$ • parents: Z

Theorem bottleneck_distrib : forall a b c, tmin a (tmax b c) = tmax (tmin a b) (tmin a c).

Occurrences (1): 'formal/IDM_Tropical.v', 'bottleneck_distrib'

Part XXVI

End matter

611 Status counts at this build

Computed at this build over 1877 entries.

By status

status	count
current	1783
unverified	52
split	30
not_an_equation	9
historical	3

By tier

tier	count
untagged	615
Th_coqc	501
Definition	432
Dr	128
finite_diagnostic	111
Open	53
Ax	34
RETRACTED	3

By coq_status

coq_status	count
root_layer_unwired	609
definition	381
closed	277
mapped_not_wrapped	274
wrapped_related	210
not_formalisable	70
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